

Macroeconomía Dinámica

Introducción: Parte 1

La Economía de Robinson Crusoe

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Robinson Crusoe

- Robinson Crusoe debe elegir cuanto trabajar y cuanto consumir.
- No hay mercados ni comercio.
- Abstracción para entender a la economía como el resultado de la agregación de las decisiones individuales.
- Modelo contiene la esencia de los problemas de decisión que aparecen en economías más complejas.
- Conceptos fundamentales: **efectos sustitución y riqueza** ante cambios en las oportunidades.

Definición de una economía

Una economía es un conjunto de agentes tomando decisiones sujeto a restricciones.

En particular, una economía está definida por:

1. Las **preferencias** y objetivos de los agentes económicos
2. Las **restricciones** que enfrentan los agentes en la toma de decisiones:
 - a) Restricciones presupuestarias
 - b) Restricciones tecnológicas
 - c) Dotaciones iniciales
3. Condiciones de **consistencia agregada**
 - a) Consumo agregado = producción agregada

La economía de Robinson Crusoe

- Preferencias de Robinson Crusoe:

$$u(c, h)$$

donde c es consumo y h es ocio.

- Consumo y ocio son bienes *normales*.

- Restricciones en la toma de decisiones:

- Tecnología: función de producción $y = f(l)$
- Restricción sobre la dotación de tiempo: $h + l = 1$

- Condición de consistencia agregada $y = c$.

Supuestos sobre la tecnología

- Función de producción transforma trabajo, l , en bienes de consumo y :

$$y = f(l); \quad (f(l) = Al^\alpha \text{ donde } 0 < \alpha < 1)$$

- Propiedades de la función de producción:

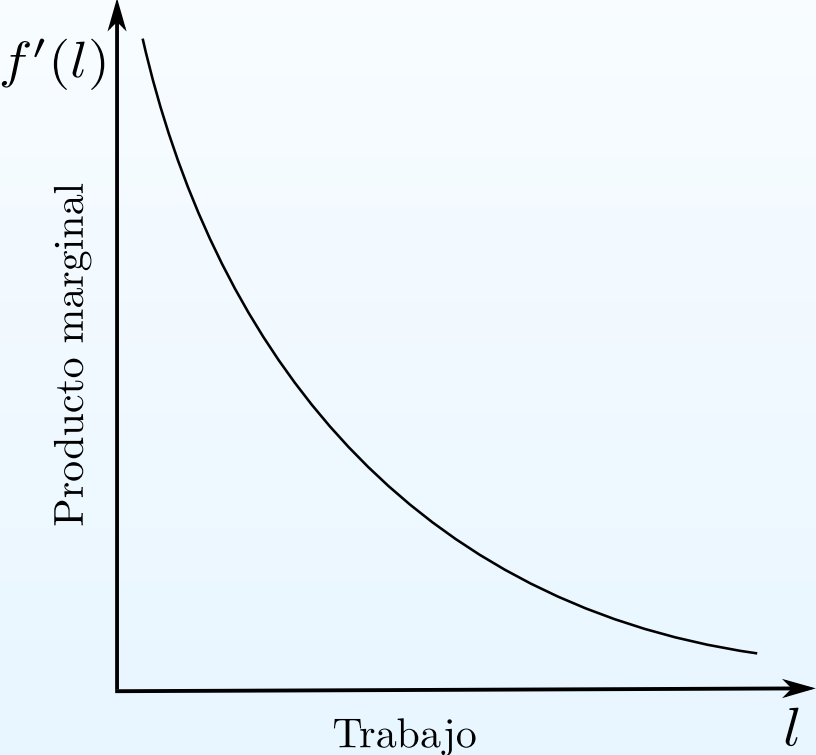
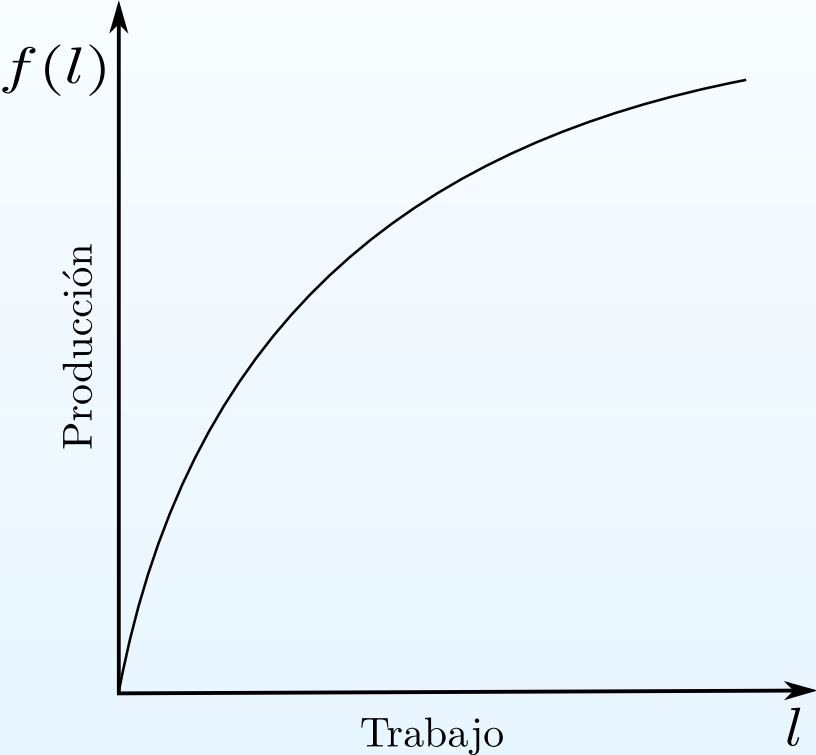
- Productividad marginal del trabajo positiva: incremento del producto al incrementar el trabajo en 1 unidad,

$$f'(l) > 0; \quad (f'(l) = \alpha Al^{\alpha-1} > 0)$$

- Ley de rendimientos decrecientes: la PML disminuye a medida que el trabajo aumenta,

$$f''(l) < 0; \quad (f''(l) = -(1 - \alpha)\alpha Al^{\alpha-2} < 0).$$

Función de producción y productividad marginal del trabajo



Avance tecnológico y aumentos de la productividad

- ¿Cómo modelamos un avance tecnológico o suba de la productividad?

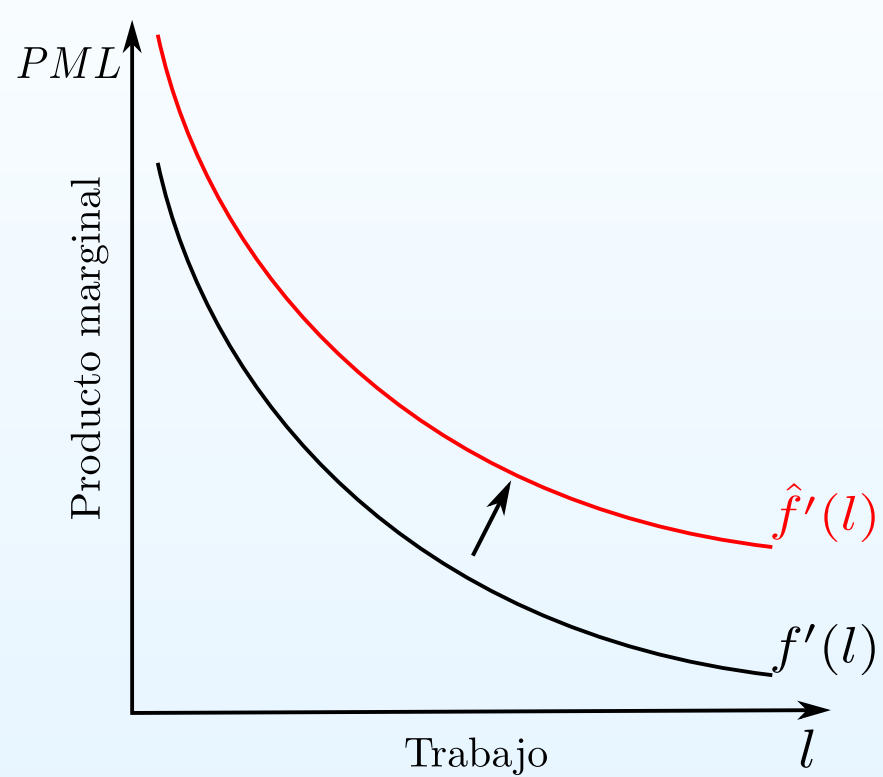
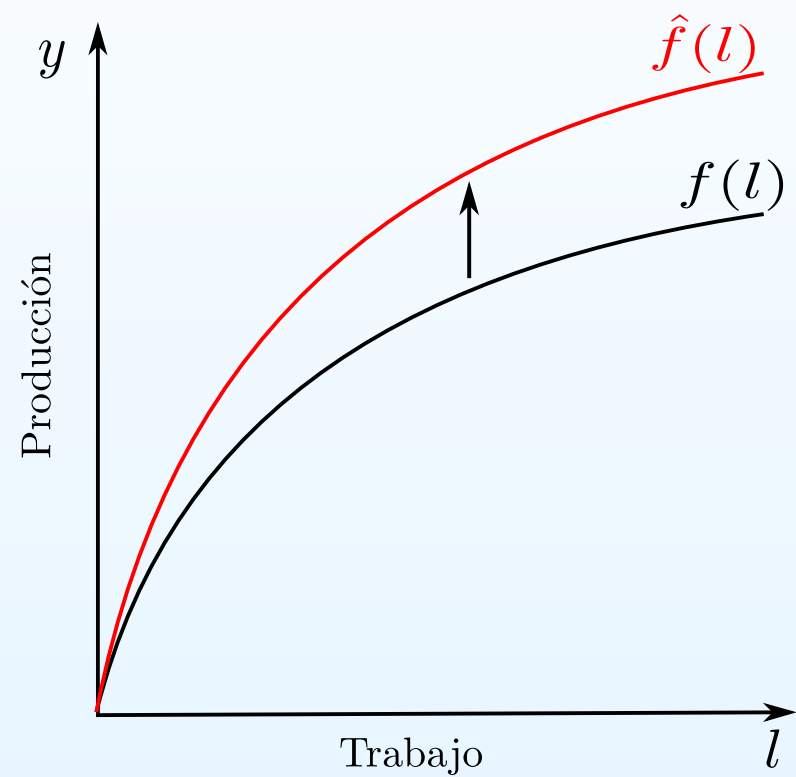
- Robinson Crusoe puede producir más bienes con la misma cantidad de trabajo:

$$\hat{f}(l) > f(l).$$

- ¿Qué pasa con la productividad marginal del trabajo cuando hay un avance tecnológico?

- En general, puede subir o bajar.
- Caso relevante desde el punto de vista empírico: aumenta para cada nivel de l .
- ¿Cómo cambia la PML cuando cambia A en el caso Cobb-Douglas $f(l) = Al^\alpha$?

Avance tecnológico y aumentos de la productividad



Preferencias entre consumo y ocio

- Robinson Crusoe obtiene utilidad de consumir *ocio* y bienes de *consumo*.
 - Equivalentemente, trabajar genera *desutilidad*.
- Sin embargo, trabajar le permite a Robinson producir bienes de *consumo*
- Robinson elige cuanto tiempo dedica a *trabajar* y cuanto tiempo dedica a disfrutar del *ocio*
- Ejemplos de elección de fuerza laboral del hogar:
 - Trabajo part-time o full-time
 - Horas extra
 - Participación de hombres y mujeres en la fuerza de trabajo

Supuestos sobre las preferencias

- Función de utilidad de Robinson Crusoe

$$u(c, h),$$

donde c es consumo y h es ocio.

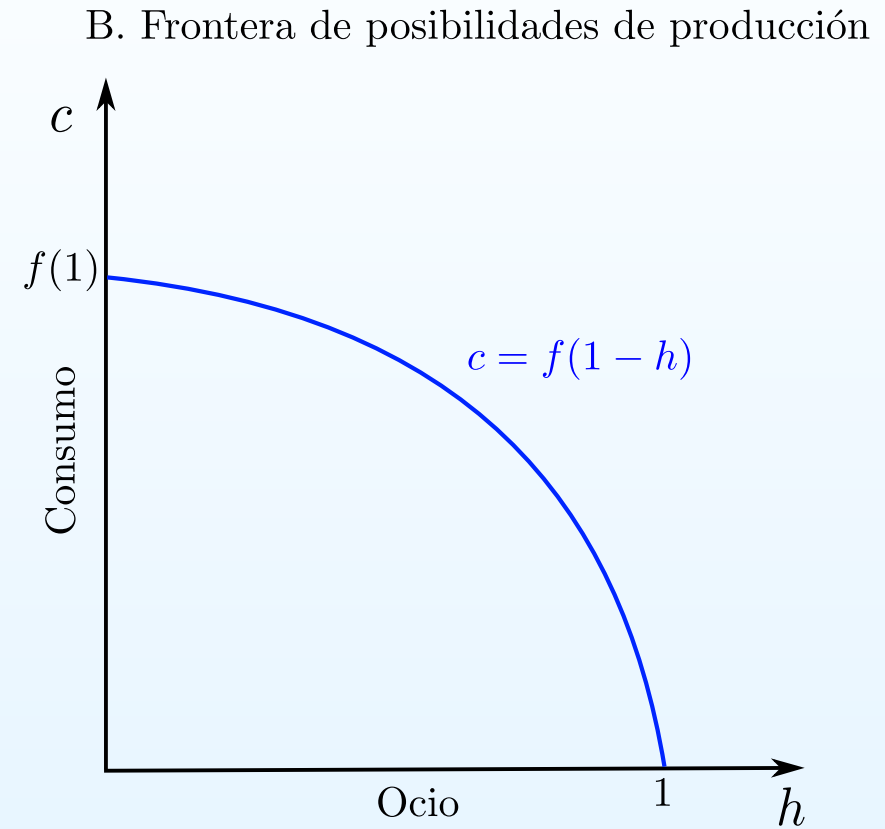
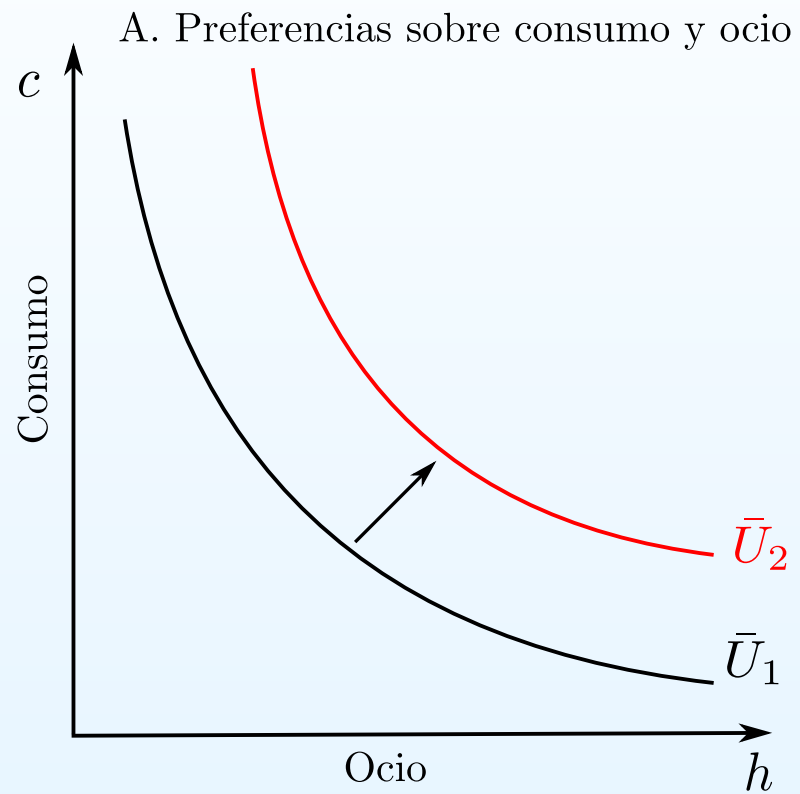
- Como Robinson valora el consumo y el ocio,

$$u_c(c, h) > 0$$

$$u_h(c, h) > 0.$$

- *Curvas de indiferencia:* pares de consumo y ocio que dejan a Robinson con el mismo nivel de utilidad (indiferente).
- *Pendiente de la curva de indiferencia:* *tasa marginal de sustitución* entre consumo y ocio. ¿Qué nos dice esta pendiente?
- *Curvas de indiferencia convexas.* ¿Interpretación económica?

Preferencias entre consumo/ocio y posibilidades de producción



Supuestos sobre las preferencias

- Pendiente de la curva de indiferencia negativa:

$$\left. \frac{dc}{dh} \right|_{\text{dado un nivel de utilidad } \bar{U}} = - \frac{u_h(c, h)}{u_c(c, h)} < 0$$

- Curva de indiferencia convexa: tasa marginal de sustitución es decreciente
 - A mayor nivel de consumo, el consumidor está dispuesto a resignar cada vez menos ocio para aumentar el consumo en una unidad adicional

Ejemplo Cobb-Douglas

Función de utilidad

$$u(c, h) = c^\phi h^{1-\phi}$$

donde $0 < \phi < 1$.

Entonces,

$$\begin{aligned} u_c(c, h) &= \phi c^{\phi-1} h^\phi \\ u_h(c, h) &= (1 - \phi) c^\phi h^{-\phi} \\ \frac{u_h(c, h)}{u_c(c, h)} &= \frac{1 - \phi}{\phi} \frac{c}{h} \end{aligned}$$

Curvas de indiferencia:

$$\bar{U} = c^\phi h^{1-\phi} \Rightarrow c = \left(\frac{\bar{U}}{h^{1-\phi}} \right)^{\frac{1}{\phi}}.$$

El problema de Robinson Crusoe

$$\max_{c,h,l} u(c,h) \quad \text{sujeto a} \quad \begin{cases} c = f(l) \\ l = 1 - h \end{cases}$$

- Se puede reescribir como

$$\max_{c,h} u(c,h) \quad \text{sujeto a} \quad c = f(1 - h)$$

- Lagrangiano (λ es el multiplicador de Lagrange)

$$\max_{c,h,\lambda} u(c,h) - \lambda[c - f(1 - h)]$$

- Condiciones de primer orden

$$\begin{aligned} \frac{dL}{dc} = 0 &\Rightarrow u_c(c,h) = \lambda \\ \frac{dL}{dh} = 0 &\Rightarrow u_h(c,h) = \lambda f'(1 - h) \\ \frac{dL}{d\lambda} = 0 &\Rightarrow c = f(1 - h) \end{aligned}$$

El problema de Robinson Crusoe

- Solución: sistema de 2 ecuaciones en 2 incógnitas:

$$\begin{aligned}\frac{u_h(c, h)}{u_c(c, h)} &= f'(1 - h) \\ c &= f(1 - h)\end{aligned}$$

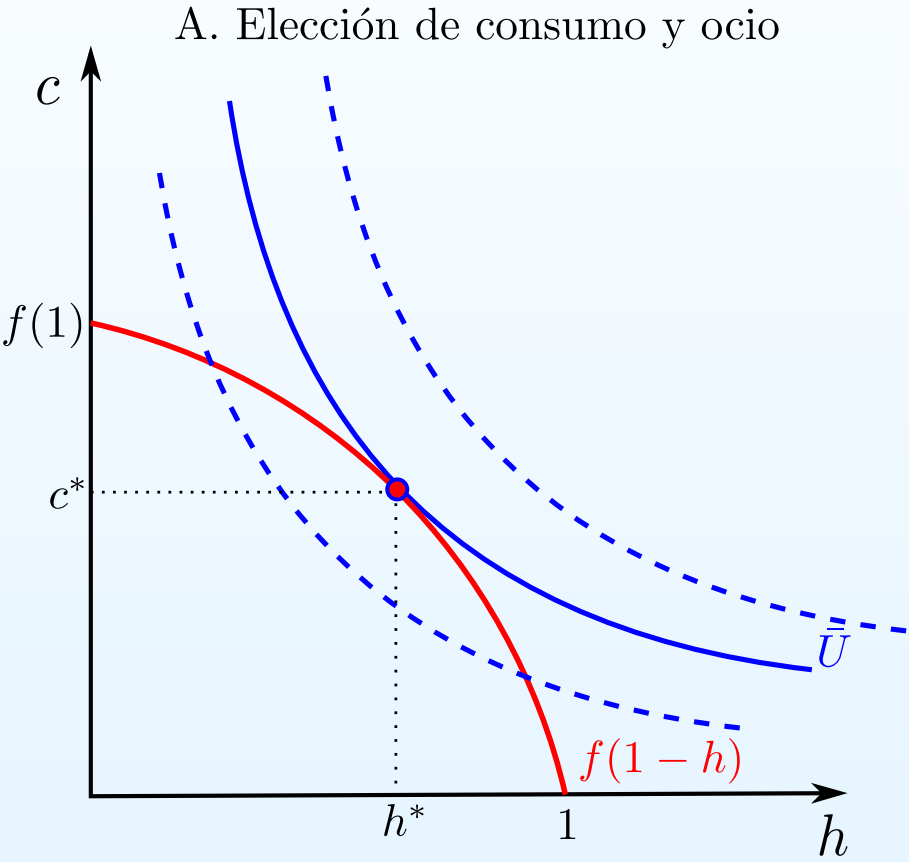
- Interpretación 1 de la primera ecuación:

- El lado izquierdo nos dice la tasa a la cual el consumidor **está dispuesto** a cambiar ocio por bienes de consumo.
- El lado derecho nos dice la tasa a la cual la tecnología **le permite** cambiar ocio (trabajo) por bienes de consumo.

- Interpretación 2 de la primera ecuación: costo marginal = beneficio marginal,

$$u_h(c, h) = f'(1 - h)u_c(c, h).$$

El problema de Robinson Crusoe



Forma equivalente: preferencias entre consumo y trabajo (Barro)

- Barro escribe la función de utilidad en términos de trabajo
- ¿Cómo? Usando la restricción de tiempo $l + h = 1 \Rightarrow h = 1 - l$,

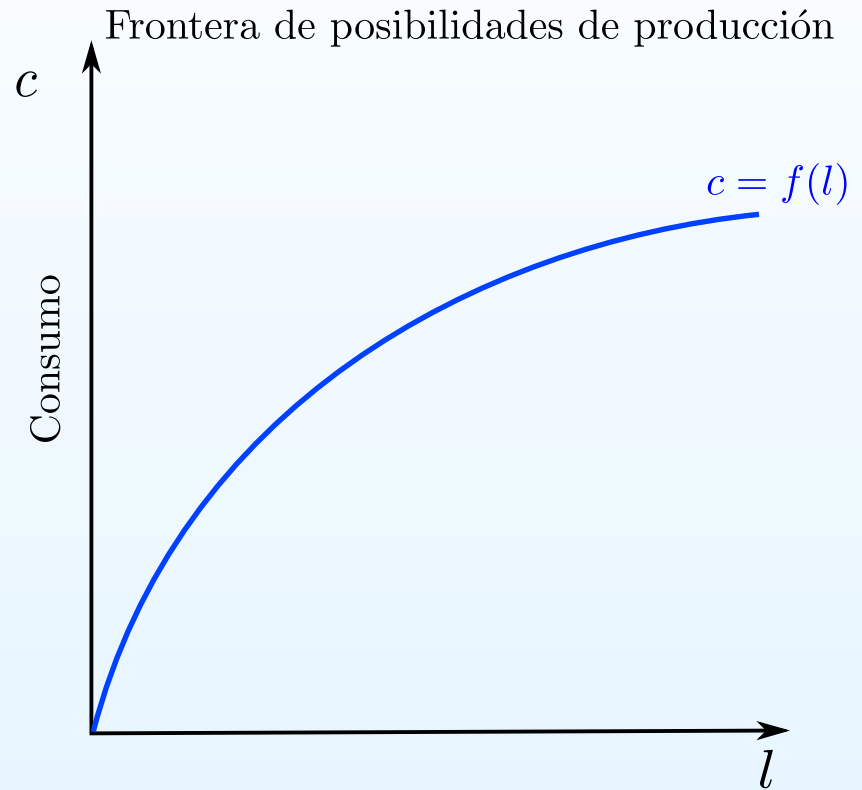
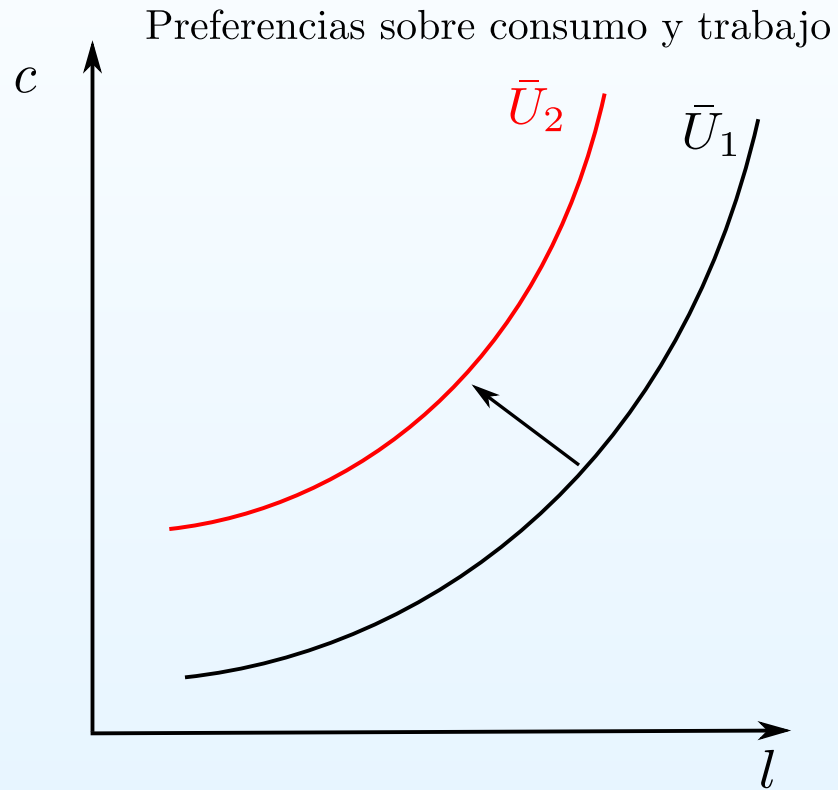
$$u(c, 1 - l)$$

- Propiedades de $u(c, h)$ implican

$$\begin{aligned}\frac{\partial u(c, 1 - l)}{\partial c} &= u_c(c, 1 - l) > 0 \\ \frac{\partial u(c, 1 - l)}{\partial l} &= -u_h(c, 1 - l) < 0\end{aligned}$$

- Curva de indiferencia con pendiente positiva: si Robinson Crusoe trabaja más debe ser compensado con más consumo para permanecer indiferente.
- Curva de indiferencia convexa: compensación requerida en términos de consumo aumenta cuando aumenta el trabajo.

Preferencias sobre consumo/trabajo y posibilidades de producción



El problema de Robinson Crusoe en términos de trabajo

$$\max_{c,l} u(c, 1 - l) \text{ sujeto a } c = f(l)$$

- Podemos reemplazar la restricción y resolver

$$\max_l u(f(l), 1 - l)$$

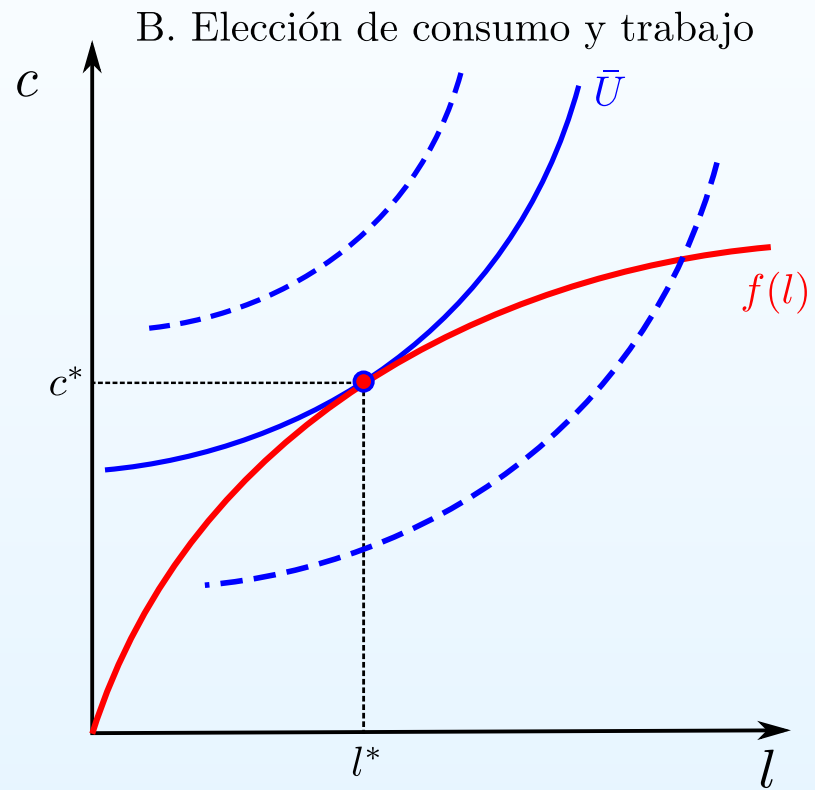
- Condición de primer orden

$$u_c(f(l), 1 - l)f'(l) - u_h(f(l), 1 - l) = 0$$

- Igual que antes:

$$\begin{aligned} \frac{u_h(c, 1 - l)}{u_c(c, 1 - l)} &= f'(l) \\ c &= f(l) \end{aligned}$$

El problema de Robinson Crusoe en términos de trabajo



Ejemplo Cobb Douglas

$$\begin{aligned}u(c, h) &= c^\phi h^{1-\phi} \\ f(l) &= Al^\alpha.\end{aligned}$$

- Tasa marginal de sustitución entre ocio y consumo:

$$\frac{u_h(c, 1-l)}{u_c(c, 1-l)} = \frac{1-\phi}{\phi} \frac{c}{1-l}$$

- Productividad marginal del trabajo

$$f'(l) = \alpha Al^{\alpha-1}$$

- Igualando las dos condiciones y usando $c = Al^\alpha$,

$$\frac{1-\phi}{\phi} \frac{Al^\alpha}{1-l} = \alpha Al^{\alpha-1}$$

Ejemplo Cobb Douglas

- Trabajo óptimo

$$l^* = \frac{\phi\alpha}{1 - \phi + \phi\alpha}$$

- Consumo óptimo

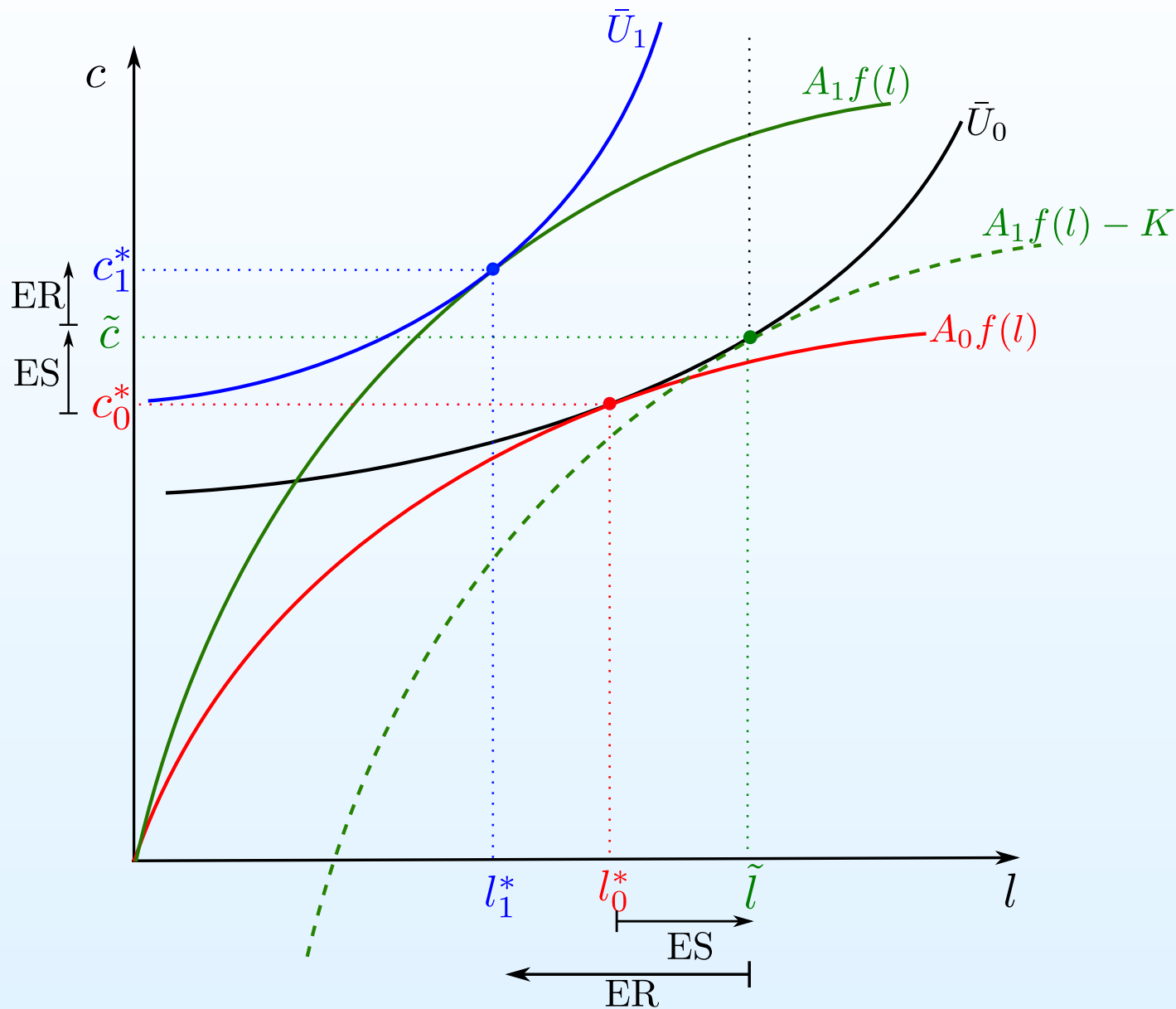
$$c^* = A \left(\frac{\phi\alpha}{1 - \phi + \phi\alpha} \right)^\alpha$$

- Note que l^* no depende de A . ¿Interpretación económica?

Cambio tecnológico: suba en la productividad

- ¿Cómo reacciona Robinson Crusoe ante un aumento de la productividad?
- ¿Qué es un cambio en la productividad?
 - Sequías, aparición de la electricidad, internet, leyes, etc.
- Función de producción:
$$y = Af(l)$$
- Avance tecnológico es una suba de A .

Cambio tecnológico: suba en la productividad ($A_1 > A_0$)



Efecto riqueza

- Robinson Crusoe puede producir más bienes con la misma cantidad de trabajo: efecto riqueza positivo
- **Efecto riqueza puro**: función de producción se desplaza en forma paralela hacia arriba (PML no cambia)
- ¿Cual es la reacción óptima ante este cambio?
 - Supondremos que **el ocio y el consumo son bienes normales**
 - Entonces es óptimo aumentar el consumo y el ocio.

Efecto sustitución

- El aumento en tecnología incluye también una rotación de la función de producción (sube la PML).
- Se modifica la tasa a la cual Robinson Crusoe puede cambiar bienes por ocio.
- El cambio en la pendiente genera un efecto sustitución.
- (Hicks), trasladamos la función de producción paralelamente hacia abajo de manera de dejar al consumidor con el nivel de utilidad original: elegimos una constante K tal que la curva $c = f(l) - K$ sea tangente a la curva de indiferencia original.
 - Consumo sube
 - Trabajo sube (ocio baja)

Combinando los efectos sustitución e ingreso

- Para el consumo, los efectos riqueza y sustitución van en la misma dirección
- Para el trabajo, los efectos riqueza y sustitución van en direcciones opuestas:
 - **Efecto sustitución:** sube la productividad marginal del trabajo, el ocio se hace más caro relativo al consumo y Robinson Crusoe trabaja más.
 - **Efecto riqueza:** Robinson Crusoe es más rico, por lo que quiere consumir más ocio y trabajar menos.
 - El efecto total sobre el trabajo depende del efecto dominante.

Macroeconomía Dinámica

Introducción: Parte 2

El Mercado de Trabajo

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El mercado de trabajo

- La economía de Robinson Crusoe no distingue entre firmas, trabajadores ni consumidores.
- Hoy agregamos (descentralizamos) los mercados de trabajo y de bienes:
 - Empresas demandan trabajo de los hogares y producen bienes que venden a los consumidores.
 - Hogares ofrecen trabajo a las empresas y compran bienes para consumir
 - Hogares son dueños de las empresas, por lo que

$\text{Ingreso del hogar} = \text{ingreso laboral} + \text{beneficios de las empresas (ingreso no laboral)}$

- Plan:
 1. Problema de las empresas
 2. Problema de los hogares
 3. Condiciones de consistencia agregada.

El problema de la empresas

- Firms demandan trabajo y producen bienes de consumo usando la tecnología $y = Af(l)$.
- Firms eligen la demanda de trabajo l^d para maximizar sus beneficios.

$$\max_{l^d} Af(l^d) - wl^d$$

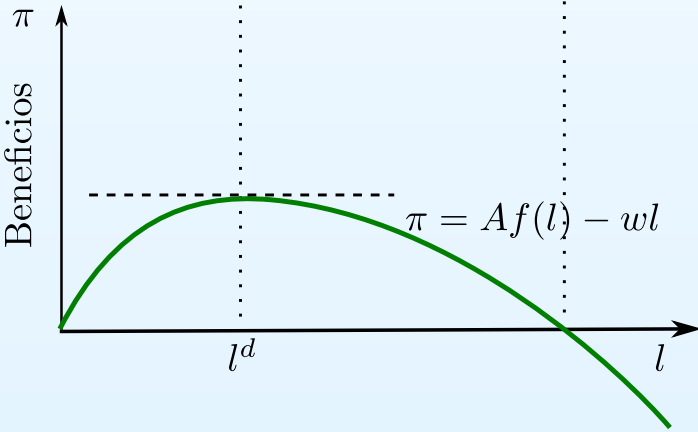
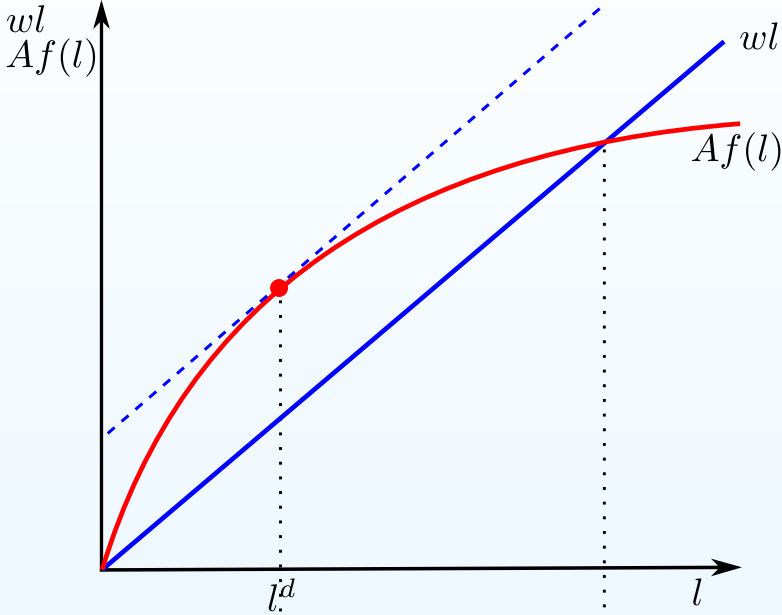
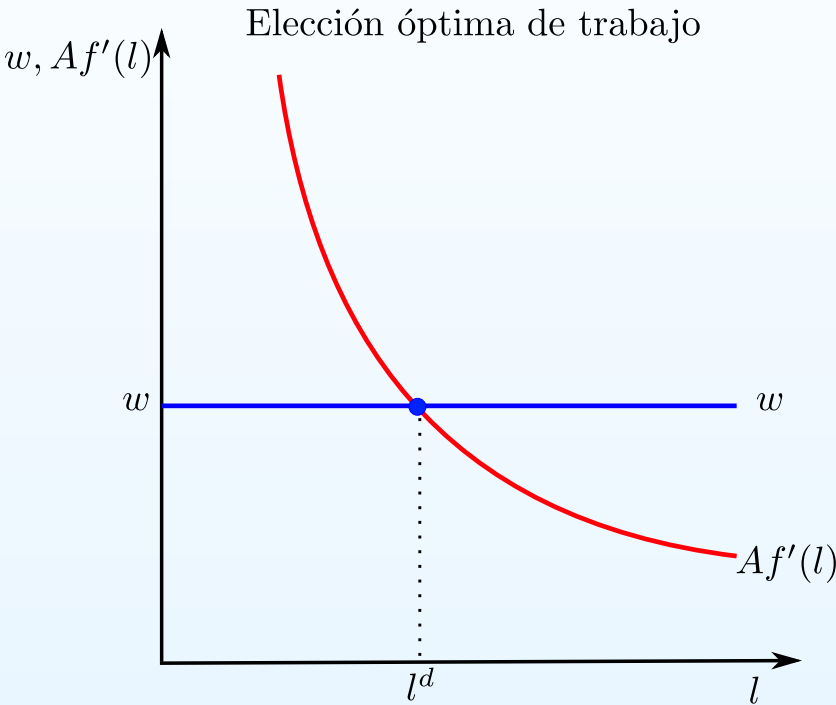
donde normalizamos el precio del bien a $p = 1$ y w es el *salario real* (en término de bienes de consumo)

- Condición de primer orden (determina la demanda de trabajo):

$$Af'(l^d) = w \Rightarrow l^d(\bar{w}, \bar{A})$$

- Empresas demandan trabajo hasta que (el valor) de la PML se iguala al salario real.
- Demanda de trabajo cae cuando sube el salario real y aumenta cuando sube la productividad.

El problema de la empresas



El problema de la empresas

- Oferta de bienes:

$$y^s = Af(l^d(w, A)) = y^s(\bar{w}, \bar{A})$$

- Beneficios:

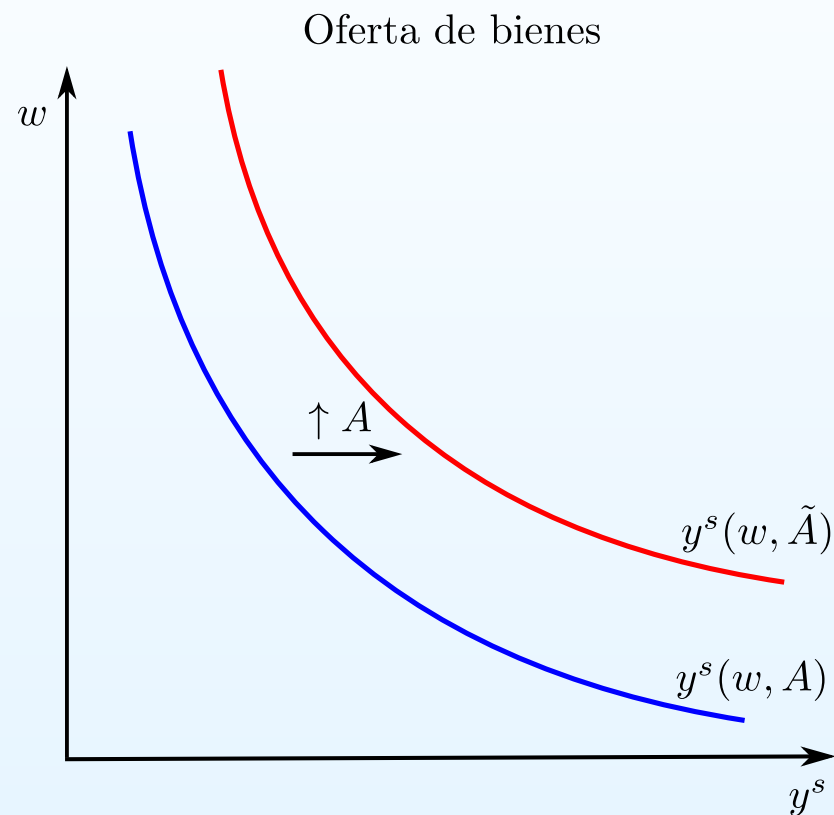
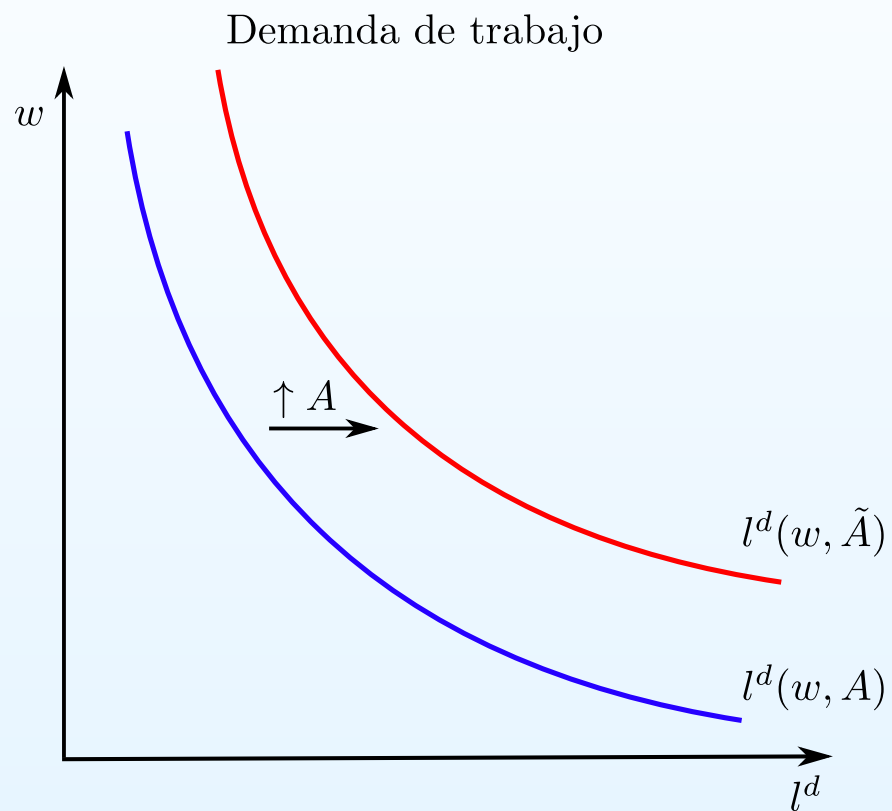
$$\pi = Af(l^d(w, A)) - wl^d(w, A) = \pi(\bar{w}, \bar{A})$$

Teorema de la envolvente:

$$\begin{aligned}\frac{\partial \pi}{\partial w} &= Af'(l^d(w, A)) \frac{\partial l^d}{\partial w} - w \frac{\partial l^d}{\partial w} - l^d(w, A) \\ &= \underbrace{\left[Af'(l^d(w, A)) - w \right]}_{=0 \text{ (CPO)}} \frac{\partial l^d}{\partial w} - l^d(w, A) < 0\end{aligned}$$

$$\begin{aligned}\frac{\partial \pi}{\partial A} &= f(l^d(w, A)) + Af'(l^d(w, A)) \frac{\partial l^d}{\partial A} - w \frac{\partial l^d}{\partial A} \\ &= f(l^d(w, A)) + \underbrace{\left[Af'(l^d(w, A)) - w \right]}_{=0 \text{ (CPO)}} \frac{\partial l^d}{\partial A} > 0\end{aligned}$$

El problema de la empresas



El problema de los hogares

- Hogares compran bienes de consumo y “alquilan” su trabajo a las empresas al salario real w .
- Hogares reciben los beneficios de las empresas (*que toman como dados al tomar sus decisiones*)

$$\max_{c, l^s} u(c, 1 - l^s)$$

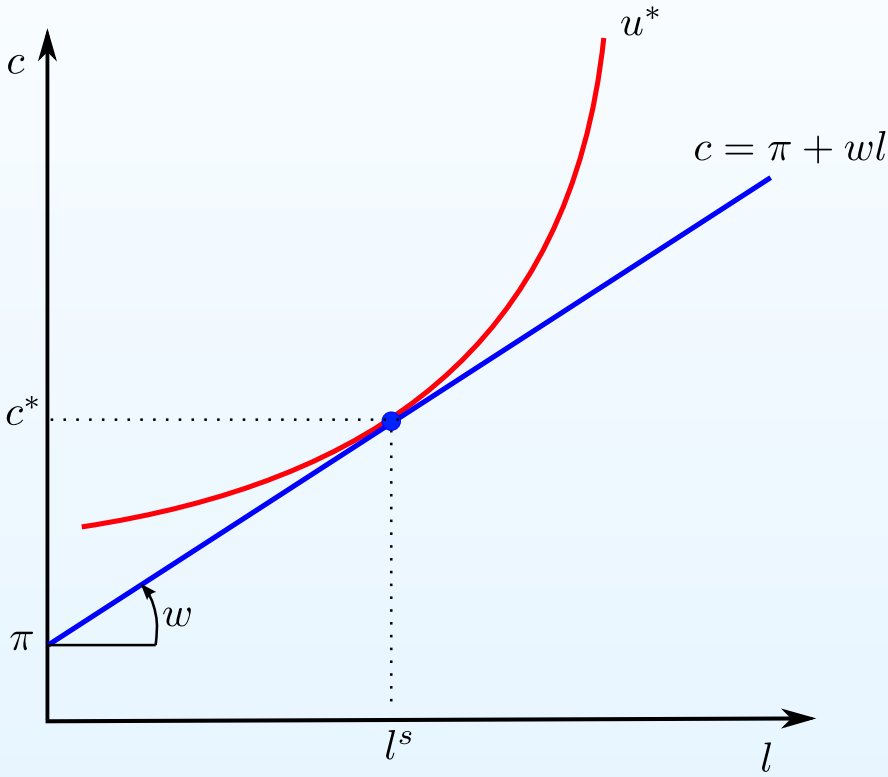
$$\text{sujeto a } c = wl^s + \pi.$$

- CPO:

$$\frac{u_h(c, 1 - l^s)}{u_c(c, 1 - l^s)} = w$$

- Tasa marginal de sustitución entre ocio y consumo = salario real.

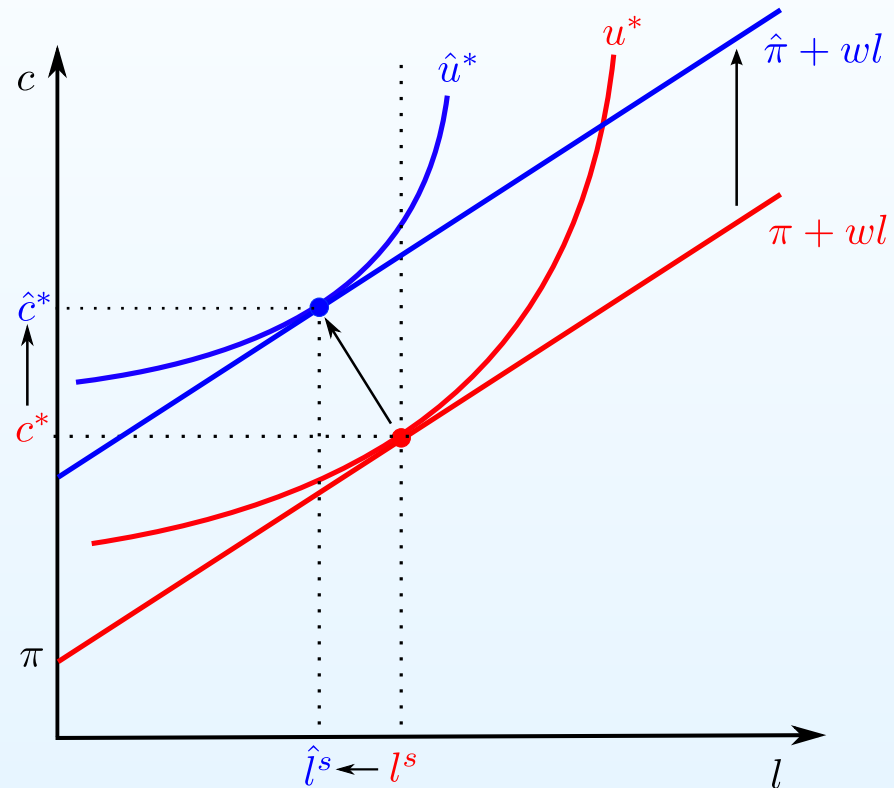
El problema de los hogares



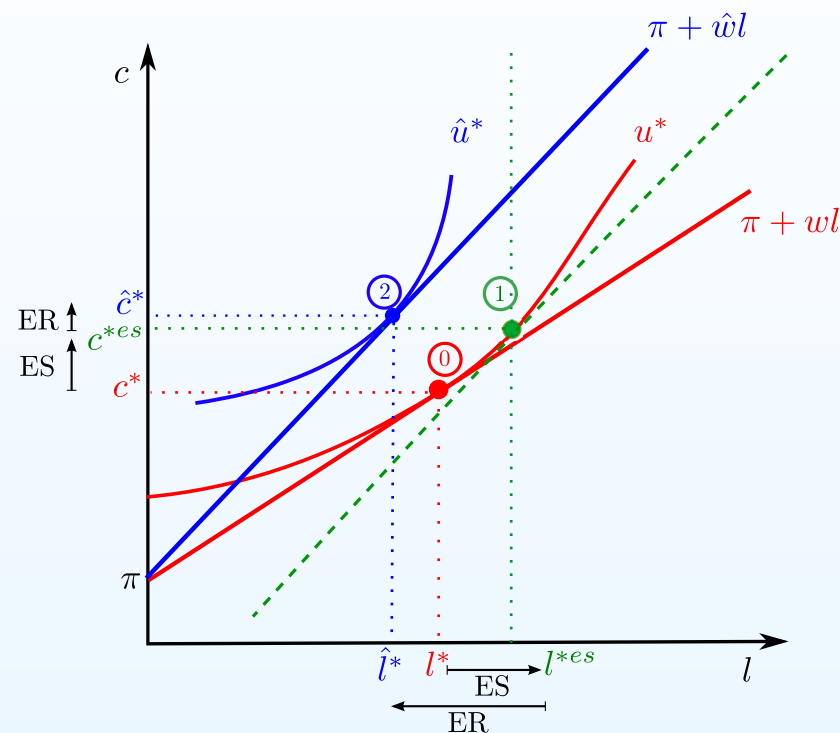
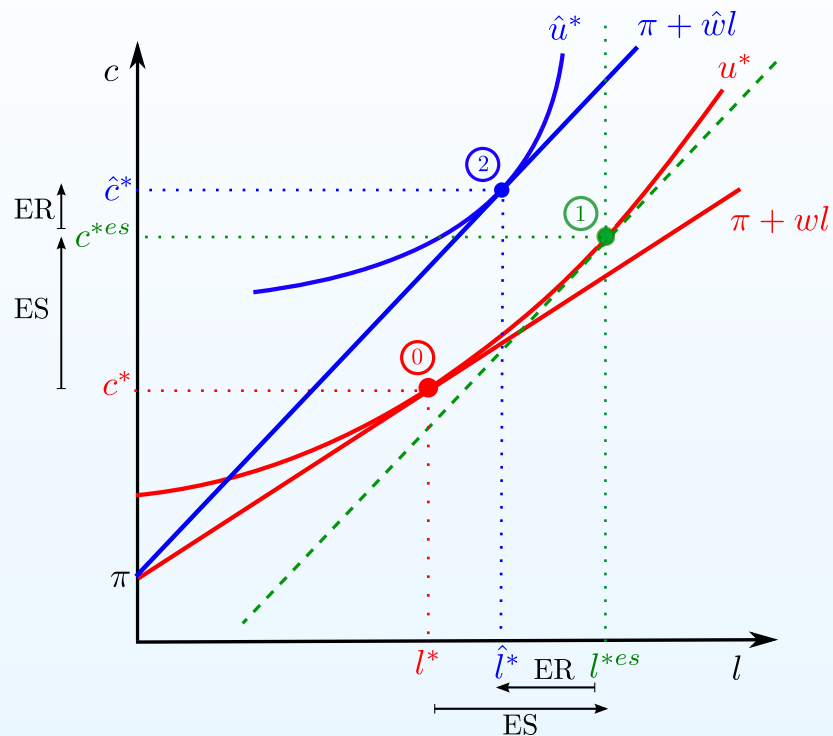
Aumento en los beneficios de las firmas π

- Consumo y ocio son bienes normales

Suba en beneficios $\hat{\pi} > \pi$ (efecto riqueza puro)



Aumento en el salario real w (con π fijo)



- Figura izquierda: domina el efecto sustitución y el trabajo aumenta
- Figura derecha: domina el efecto riqueza y el trabajo disminuye

El problema de los hogares

- Oferta de trabajo y la demanda de bienes

$$l^s = \tilde{l}^s(\overset{?}{w}, \bar{\pi})$$

$$c = \tilde{c}^d(\overset{+}{w}, \overset{+}{\pi})$$

- Efectos sustitución y riqueza:

$$\uparrow \pi \Rightarrow \left\{ \begin{array}{ll} \text{ES :} & \text{no hay} \\ \text{ER (bs. normales):} & \downarrow l^s, \uparrow c \end{array} \right.$$

$$\uparrow w \Rightarrow \left\{ \begin{array}{ll} \text{Efecto sustitución:} & \uparrow l^s, \uparrow c \\ \text{Efecto riqueza (bs. normales):} & \downarrow l^s, \uparrow c \end{array} \right.$$

El problema de los hogares

- Pero los hogares son los dueños de las empresas. Por lo tanto, del problema de las empresas,

$$\pi = \pi(\bar{w}, \bar{A}).$$

- Reemplazando en la oferta de trabajo y demanda de bienes:

$$\begin{aligned} l^s &= \tilde{l}^s(w, \pi(w, A)) = \tilde{l}^s(\bar{w}, \bar{A}) \\ c &= \tilde{c}^d(w, \pi(w, A)) = \tilde{c}^d(\bar{w}, \bar{A}) \end{aligned}$$

- ¿Por qué una suba de w implica un aumento en la oferta de trabajo y del consumo?
- Hay dos efectos riqueza que se cancelan (aproximadamente)

$$\uparrow w \Rightarrow \left\{ \begin{array}{ll} \text{ES:} & \uparrow l^s, \uparrow c \\ \text{ER1 } (\uparrow wl): & \downarrow l^s, \uparrow c \\ \text{ER2 } (\downarrow \pi(w, A)): & \uparrow l^s, \downarrow c \end{array} \right.$$

El problema de los hogares

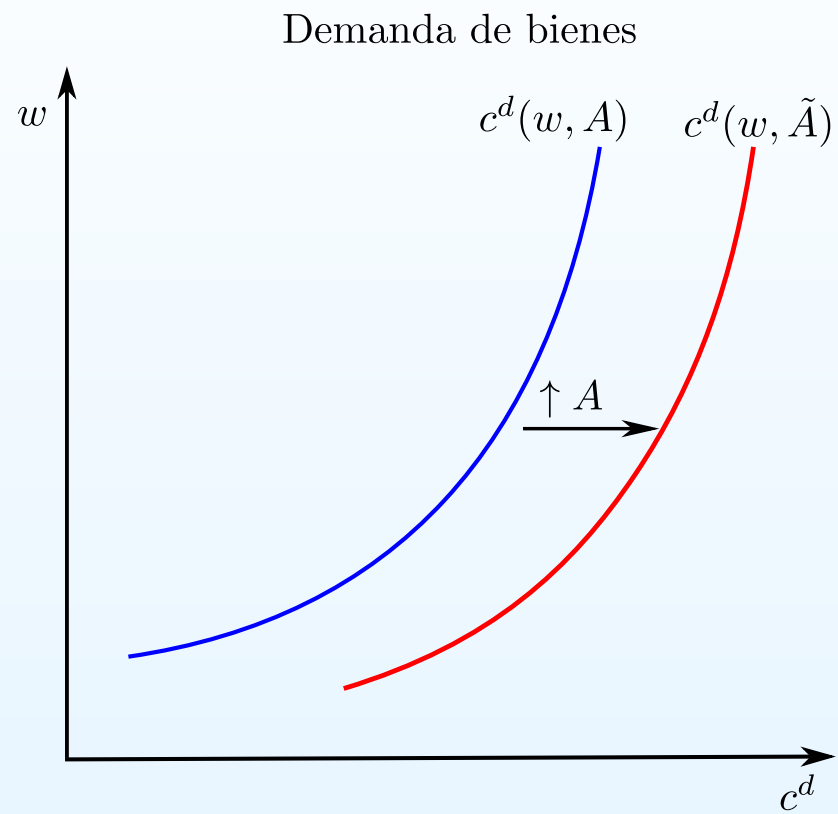
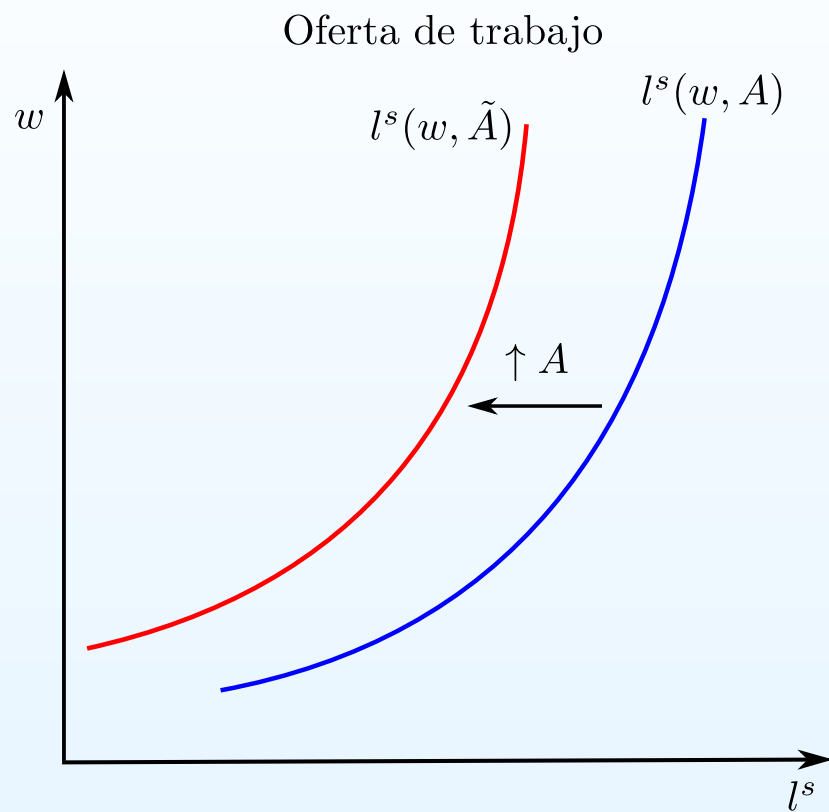
- Con consumidores iguales, ER1 se cancela exactamente con ER2 (efecto de equilibrio general)

- Impacto directo wl^s se cancela con los menores beneficios $\pi = Af(l^d) - wl^d$:
- Restricción presupuestaria del hogar representativo

$$\begin{aligned}c &= wl^s + \pi \\ &= wl^s + Af(l^d) - wl^d.\end{aligned}$$

- ¿Consumidores heterogéneos? En el agregado tienden a cancelarse.
- Por lo tanto, *en el modelo de equilibrio general*, la oferta de trabajo como función del salario real w tendrá siempre pendiente positiva.

Los hogares: oferta de trabajo y demanda de bienes



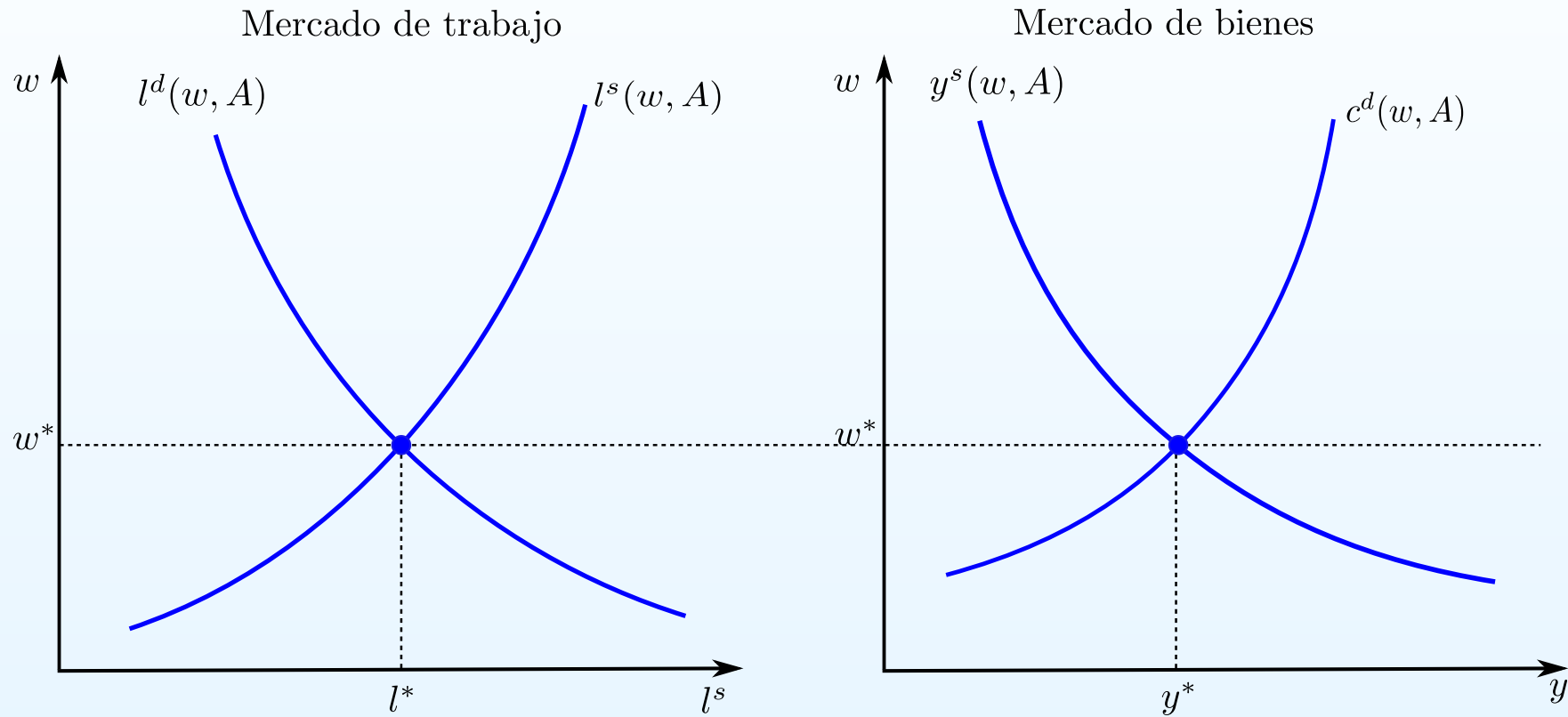
Condiciones de equilibrio (consistencia agregada)

- Los mercados de bienes y de trabajo se deben vaciar

$$\begin{aligned}l^s(\bar{w}^+, \bar{A}) &= l^d(\bar{w}, \bar{A}^+) \\ y^s(\bar{w}, \bar{A}^+) &= c^d(\bar{w}^+, \bar{A}).\end{aligned}$$

- ¡Dos ecuaciones pero una sola incógnita: w ! ¿Qué está pasando?...
- La “**Ley de Walras**”:
 - Si hay N mercados operando simultáneamente y $N - 1$ están en equilibrio, entonces el mercado N -ésimo también estará en equilibrio.
- ¿Qué condición usamos para determinar el salario de equilibrio? Cualquiera, pero es *conveniente* usar la del mercado de trabajo.

Equilibrio



- Por la Ley de Walras, el mismo salario w^* vacía ambos mercados

Ley de Walras

- Si hay N mercados operando simultáneamente y $N - 1$ están en equilibrio, entonces el mercado N -ésimo también estará en equilibrio.
- En nuestro modelo, hay $N=2$ mercados: el mercado de bienes y el mercado de trabajo.
- Usamos la restricción presupuestaria del hogar y los beneficios de la empresa:

$$\begin{aligned}c^d &= wl^s + \pi \\&= wl^s + Af(l^d) - wl^d \\&= w(l^s - l^d) + y^s.\end{aligned}$$

o bien

$$(c^d - y^s) + w(l^d - l^s) = 0.$$

- Si el salario es tal que el mercado de trabajo está en equilibrio, $l^s = l^d$, entonces el mercado de bienes también estará en equilibrio: $c^d = y^s$.
- Si el salario es tal que mercado de bienes está en equilibrio, $c^d = y^s$, entonces el mercado de trabajo también estará en equilibrio: $l^d = l^s$.

Relación con el modelo de Robinson Crusoe. Equilibrio satisface:

Empresas y hogares maximizan

$$\begin{aligned} Af'(l^d) &= w \\ \frac{u_h(c, 1 - l^s)}{u_c(c, 1 - l^s)} &= w \\ c &= Af(l^d) - wl^d + wl^s \end{aligned}$$

Los mercados de trabajo y de bienes están en equilibrio

$$\begin{aligned} l^s &= l^d \\ c &= Af(l^d) \end{aligned}$$

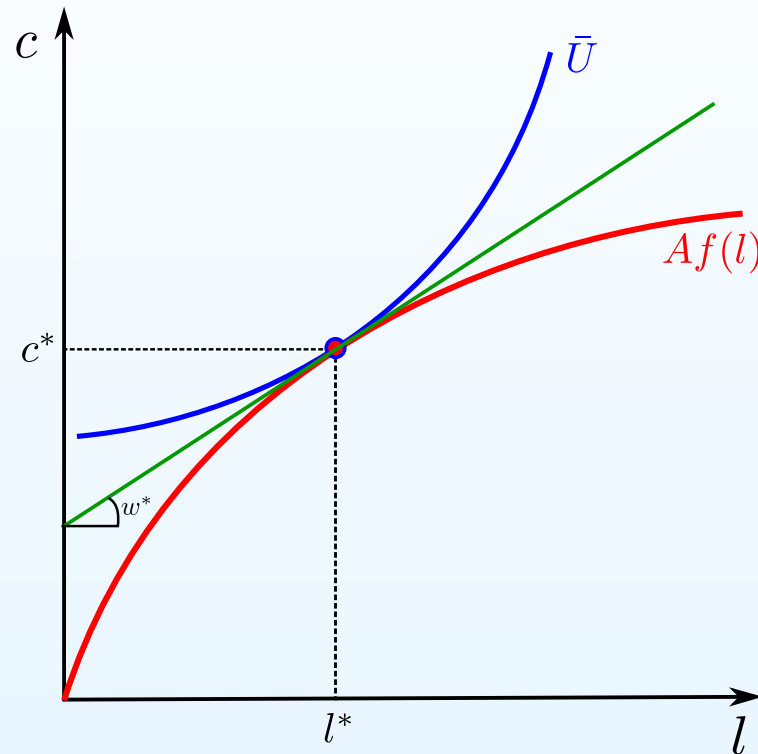
Haciendo los reemplazos pertinentes encontramos

$$\begin{aligned} \frac{u_h(c^*, 1 - l^*)}{u_c(c^*, 1 - l^*)} &= Af'(l^*) \\ c^* &= Af(l^*) \end{aligned}$$

Estas son las mismas condiciones que encontramos en el modelo centralizado de Robinson Crusoe!

Relación con modelo de Robinson Crusoe

- Mismo problema, misma solución (más el salario de equilibrio)



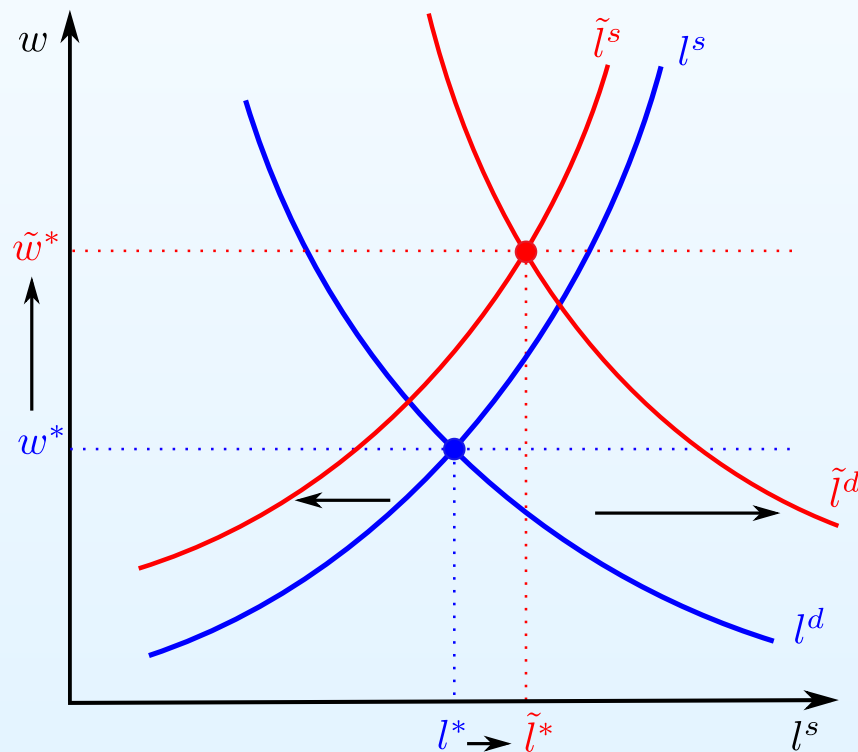
- **Estrategia:** ignoramos el mercado de trabajo. Si nos interesa encontrar el salario, usamos que en equilibrio el salario satisface

$$w = Af'(l^*) = \frac{u_h(c^*, 1 - l^*)}{u_c(c^*, 1 - l^*)}.$$

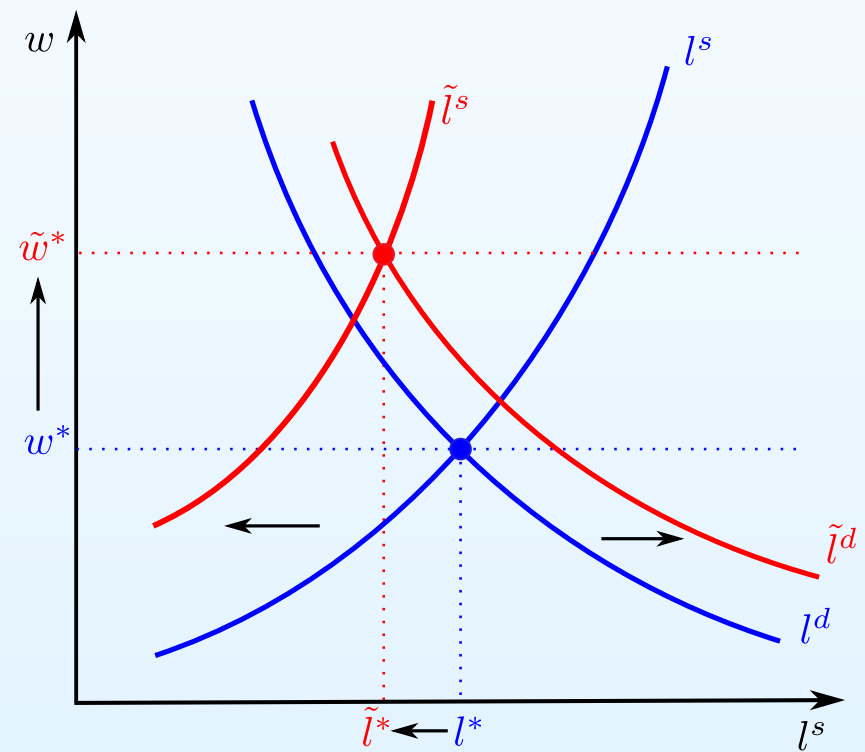
Estática comparativa: sube la productividad A

- $\uparrow A$ implica $\uparrow w^*$, $\uparrow c^*$, y un efecto ambiguo sobre l^* .
- Pero esto ya lo sabíamos del modelo de Robinson Crusoe: efectos sustitución y riqueza van en dirección opuesta en relación al trabajo.

A. Domina efecto sustitución

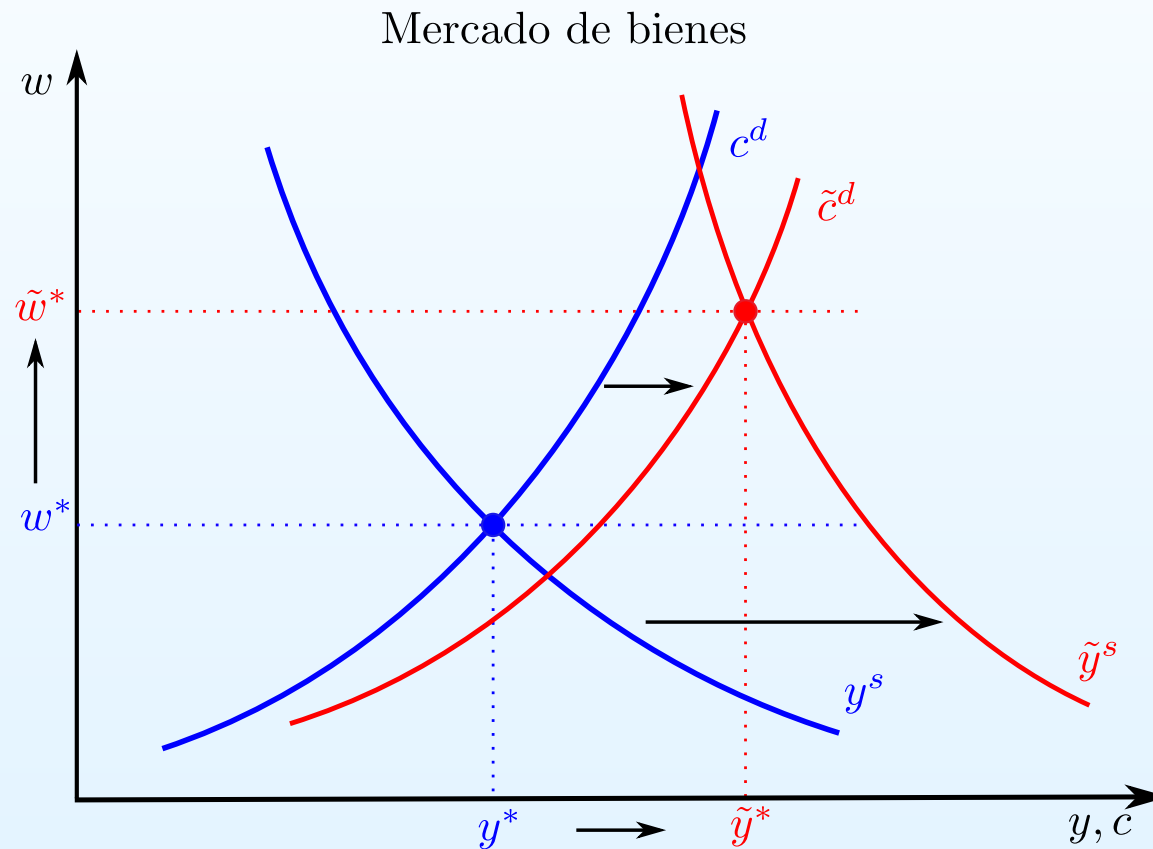


B. Domina efecto ingreso



Estática comparativa: sube la productividad A

- No hay ambigüedad: aumenta el producto.
- Sin embargo, en esta figura no es obvio que debería aumentar el salario. Por ello es conveniente ver el equilibrio en el mercado de trabajo.



Macroeconomía Dinámica

Introducción: Parte 3

Elección Intertemporal de Consumo y Mercado de Crédito

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Elección intertemporal del consumo

- **Hasta hoy:** Robinson Crusoe nace, trabaja, consume y muere en el mismo período.
- **Desde hoy:** modelo dinámico (dos períodos).
- Hogares reciben un ingreso exógeno en cada período.
- Consumidores deciden cuanto consumir hoy, cuanto ahorrar y cuanto consumir en el futuro.
- Rol de los mercados de crédito.
- Comparación con la teoría keynesiana del consumo:
 - Propensión marginal al consumo ante cambios temporarios y permanentes del ingreso.
- **Evidencia empírica:** estímulo al gasto durante la gran recesión en EEUU.

Preferencias

- Preferencias sobre consumo presente y futuro, c_1 y c_2

- Función de utilidad

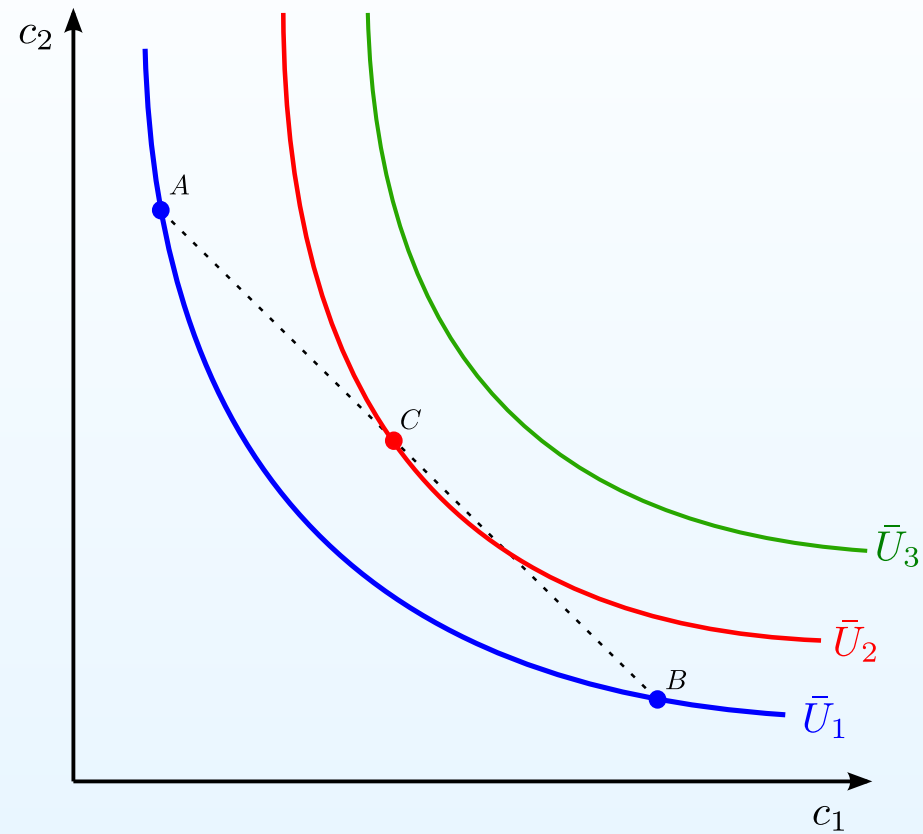
$$U(c_1, c_2) = u(c_1) + \beta u(c_2).$$

- β es el factor de descuento. Usualmente suponemos $0 < \beta < 1$. Interpretación?

- Suponemos

- $u'(c) > 0 \Rightarrow c_1$ y c_2 son bienes normales (separabilidad $\implies$ normalidad).
- $u''(c) < 0$: preferencias por patrones de consumo “suaves” a través del tiempo.

Preferencias



- **Curvas de indiferencia convexas:** preferencias por patrones suaves de consumo. Canasta $C = \frac{1}{2} A + \frac{1}{2} B$ da más utilidad que A o B .

Restricciones presupuestarias flujo

- Familias reciben ingresos exógenos de y_1 e y_2 en los períodos 1 y 2.
- Mercado de crédito: hogares pueden ahorrar o endeudarse en bonos b_t
- Activos pagan tasa de interés real r_t .
- Restricciones presupuestaria flujo:

$$c_1 + b_1 = y_1 + (1 + r_0)b_0$$

$$c_2 + b_2 = y_2 + (1 + r_1)b_1$$

- Los consumidores no pueden morir endeudados (¿por qué?):

$$b_2 \geq 0.$$

Ahorro

- El ahorro es la parte del ingreso total que no se consume:

$$s_t = \underbrace{y_t + r_{t-1}b_{t-1}}_{\text{ingreso total en } t} - \underbrace{c_t}_{\text{consumo en } t}$$

- De la restricción presupuestaria flujo,

$$c_t + b_t = y_t + (1 + r_{t-1})b_{t-1}$$

o bien,

$$b_t = \underbrace{y_t + r_{t-1}b_{t-1} - c_t}_{s_t} + b_{t-1},$$

por lo que

$$s_t = b_t - b_{t-1}.$$

- El ahorro es igual al cambio en el stock de activos

El problema del consumidor

$$\max_{c_1, c_2, b_1, b_2} u(c_1) + \beta u(c_2) \text{ sujeto a}$$

$$c_1 + b_1 = y_1 + (1 + r_0)b_0$$

$$c_2 + b_2 = y_2 + (1 + r_1)b_1$$

$$b_2 \geq 0.$$

- Es óptimo elegir $b_2 = 0$ (¿por qué?)
- Supongamos también $b_0 = 0$ (no es esencial)

El problema del consumidor

$$\max_{c_1, c_2, b_1} u(c_1) + \beta u(c_2) \text{ sujeto a}$$

$$c_1 + b_1 = y_1$$

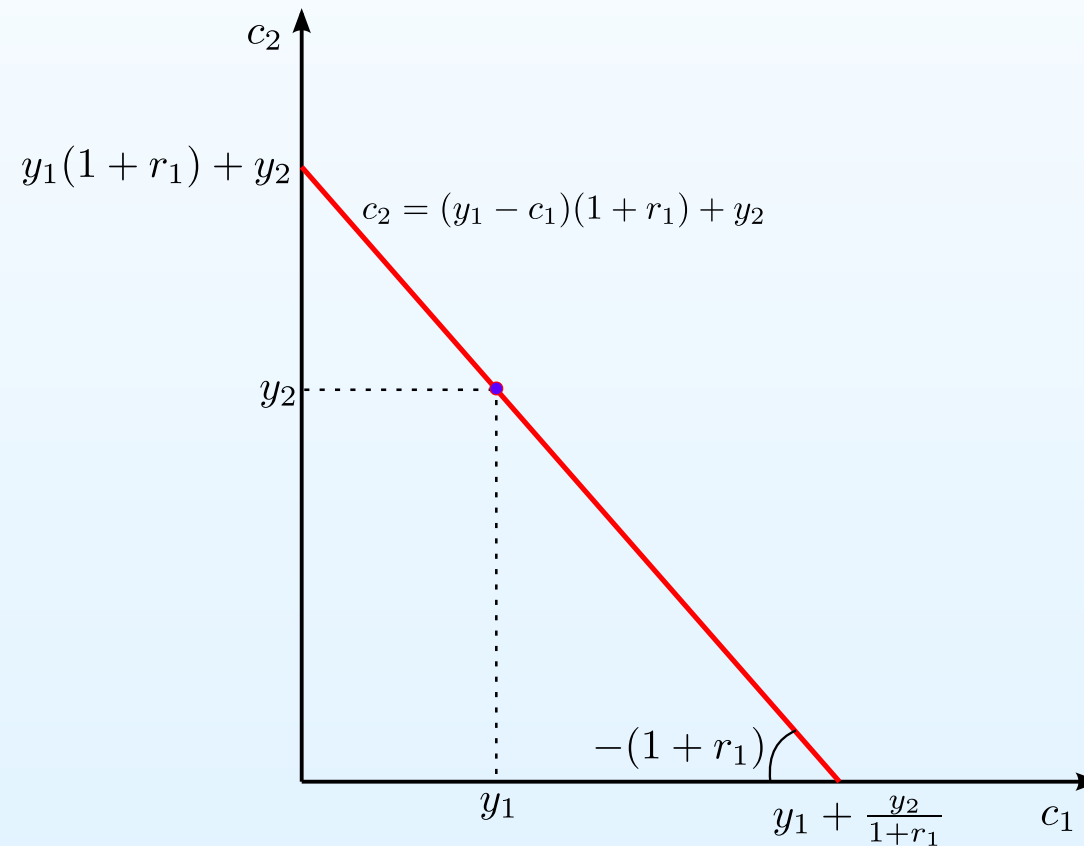
$$c_2 = y_2 + (1 + r_1) b_1.$$

- Tres formas equivalentes de resolver el problema
 1. Construir una única restricción: la **restricción presupuestaria intertemporal**
 2. Asignar un multiplicador a cada restricción presupuestaria flujo
 3. Reemplazar las restricciones en la función objetivo y elegir solo b_1 .

La restricción presupuestaria intertemporal

- Combino restricciones flujo y obtengo la **restricción presupuestaria intertemporal**:

$$c_1 + \frac{c_2}{1 + r_1} = y_1 + \frac{y_2}{1 + r_1}$$



La restricción presupuestaria intertemporal

- La *restricción presupuestaria intertemporal* surge de la posibilidad de endeudarse o ahorrar la cantidad que se desee a la tasa de interés r_1 .
- Implicancias:
 1. Con mercados de crédito podemos pensar que el consumidor enfrenta una sola restricción presupuestaria.
 2. El valor presente del consumo es igual al valor presente del ingreso.
 3. Precio relativo de c_2 en términos de c_1 es $\frac{1}{1+r_1}$:
 - a) Puedo transformar $\frac{1}{1+r_1}$ bienes hoy en $\frac{1}{1+r_1} \times (1 + r_1)$ bienes mañana.
 4. Una suba en r_1 significa que el consumo futuro se hace más barato relativo al consumo presente.

El problema del consumidor

$$\max_{c_1, c_2} u(c_1) + \beta u(c_2) \quad \text{sujeto a} \quad c_1 + \frac{c_2}{1 + r_1} = y_1 + \frac{y_2}{1 + r_1}$$

- Lagrangiano

$$L = u(c_1) + \beta u(c_2) - \lambda \left[c_1 + \frac{c_2}{1 + r_1} - y_1 - \frac{y_2}{1 + r_1} \right]$$

- Condiciones de primer orden

$$\frac{\partial L}{\partial c_1} = 0 \Rightarrow u'(c_1) = \lambda \quad (1)$$

$$\frac{\partial L}{\partial c_2} = 0 \Rightarrow \beta u'(c_2) = \frac{\lambda}{1 + r_1} \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow c_1 + \frac{c_2}{1 + r_1} = y_1 + \frac{y_2}{1 + r_1} \quad (3)$$

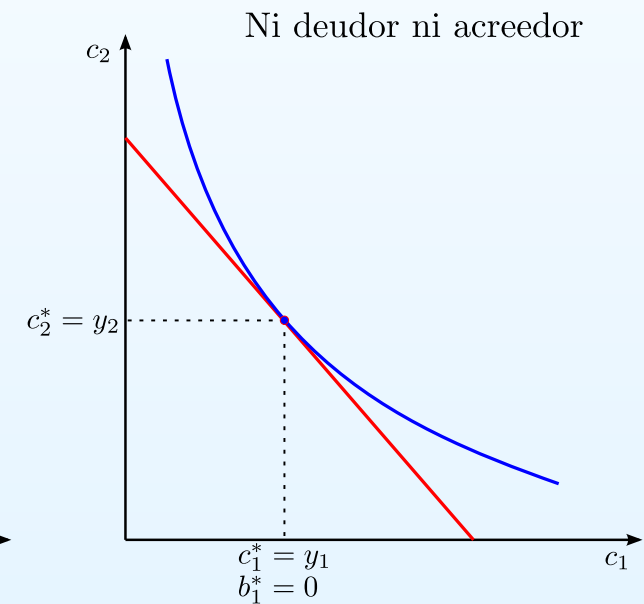
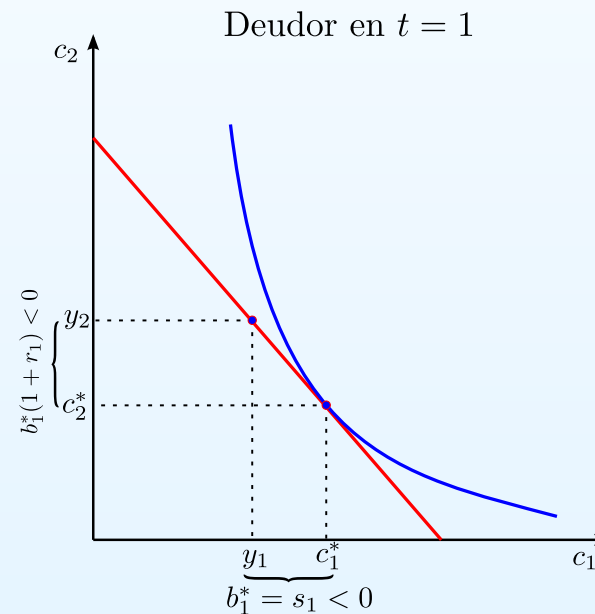
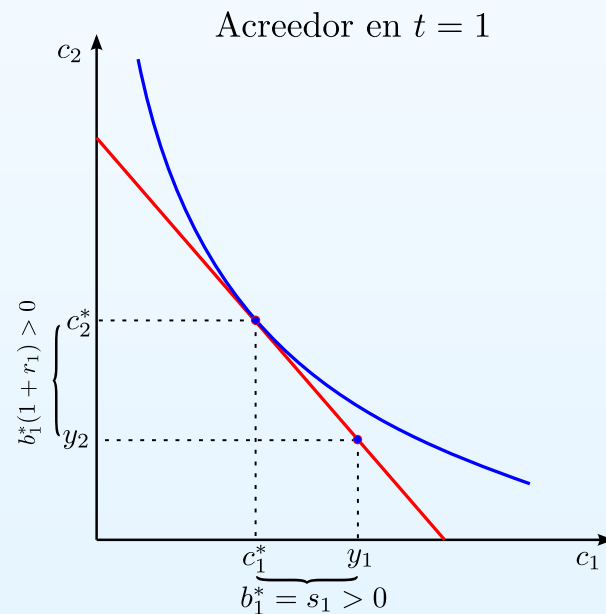
- Sustituyendo (1) en (2) obtenemos la **Ecuación de Euler**

$$u'(c_1) = (1 + r_1)\beta u'(c_2). \quad (4)$$

El problema del consumidor: demandas de consumo

Sistema de dos ecuaciones en dos incógnitas, c_1 y c_2 :

$$\frac{\beta u'(c_2)}{u'(c_1)} = \frac{1}{1 + r_1}$$
$$c_1 + \frac{c_2}{1 + r_1} = y_1 + \frac{y_2}{1 + r_1}$$



- Mercado de crédito aumenta el bienestar de los consumidores

El problema de los hogares

- Demandas de consumo dependen de la tasa de interés y del valor presente del ingreso

$$c_1^* = c_1^d \left(r_1, y_1 + \frac{y_2}{1 + r_1} \right) \quad (5)$$

$$c_2^* = c_2^d \left(r_1, y_1 + \frac{y_2}{1 + r_1} \right) \quad (6)$$

- y_1 e y_2 no aparecen como argumentos independientes
- Mercados de crédito permiten cambiar el patrón temporal de consumo
- Demanda de activos b_1

$$b_1^* = b_1^d(r_1, y_1, y_2) = y_1 - c_1^d \left(r_1, y_1 + \frac{y_2}{1 + r_1} \right).$$

- Demanda de bonos sí dependen de y_1 además del valor presente del ingreso. Intuición.

Ejemplo: Función de utilidad logarítmica

$$u(c) = \log c \quad \Rightarrow \quad u'(c) = \frac{1}{c}.$$

- Ecuación de Euler $u'(c_1) = (1 + r_1)\beta u'(c_2)$ es

$$\frac{1}{c_1} = (1 + r_1) \beta \frac{1}{c_2}$$

o bien

$$c_2 = \beta (1 + r_1) c_1$$

- Resolviendo,

$$c_1 = \frac{1}{1 + \beta} \left[y_1 + \frac{y_2}{1 + r_1} \right] \quad (7)$$

$$c_2 = \frac{\beta(1 + r_1)}{1 + \beta} \left[y_1 + \frac{y_2}{1 + r_1} \right] \quad (8)$$

$$b_1^d = \frac{\beta}{1 + \beta} \left[y_1 - \frac{y_2}{\beta(1 + r_1)} \right] \quad (9)$$

Forma equivalente de resolver el problema del consumidor

- Lagrangiano con restricciones presupuestarias flujo:

$$L = u(c_1) + \beta u(c_2) - \lambda_1 [c_1 + b_1 - y_1] - \lambda_2 [c_2 - y_2 - (1 + r_1)b_1]$$

- Condiciones de primer orden

$$\frac{\partial L}{\partial c_1} = 0 \Rightarrow u'(c_1) = \lambda_1$$

$$\frac{\partial L}{\partial c_2} = 0 \Rightarrow \beta u'(c_2) = \lambda_2$$

$$\frac{\partial L}{\partial b_1} = 0 \Rightarrow \lambda_1 = \lambda_2(1 + r_1)$$

$$\frac{\partial L}{\partial \lambda_1} = 0 \Rightarrow c_1 + b_1 = y_1$$

$$\frac{\partial L}{\partial \lambda_2} = 0 \Rightarrow c_2 = y_2 - (1 + r_1)b_1$$

Forma equivalente de resolver el problema del consumidor

- Ecuaciones se reducen a

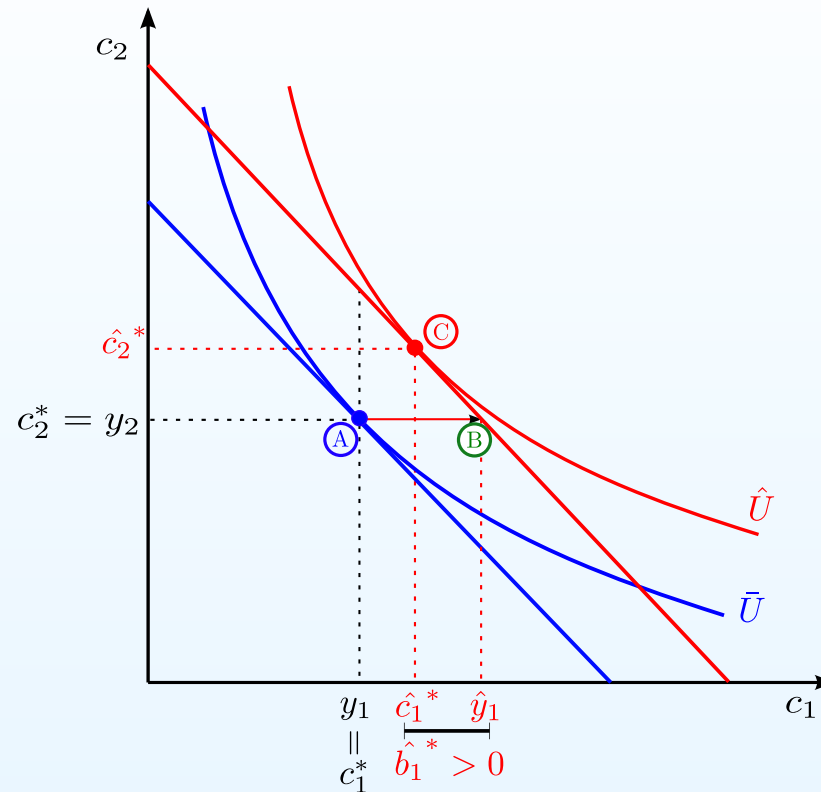
$$u'(c_1) = (1 + r_1)\beta u'(c_2)$$
$$c_1 + \frac{c_2}{1 + r_1} = y_1 + \frac{y_2}{1 + r_1},$$

- Misma solución que encontramos antes.

Análisis de cambios en los ingresos y de la tasa de interés

- Queremos entender la función de demanda de consumo de los hogares.
- Impacto sobre el patrón de consumo intertemporal de los siguientes cambios:
 - *Cambio temporario del ingreso*: sube y_1 e y_2 no cambia
 - *Cambio del ingreso futuro*: y_1 no cambia e y_2 sube
 - *Cambio permanente del ingreso*: suben y_1 e y_2 en la misma cantidad
 - *Aumento de la tasa de interés real* r_1
- Supondremos que antes del cambio el consumidor no era ni deudor ni acreedor.

Suba temporaria del ingreso corriente: $\uparrow y_1, \bar{y}_2$



- **Efecto riqueza:** c_1 y c_2 bienes normales, $\uparrow c_1, \uparrow c_2$
- **Aumento temporario del ingreso:** $0 < \Delta c_1 < \Delta y_1 \Rightarrow \hat{b}_1^* > 0, \quad 0 = \Delta y_2 < \Delta c_2,$
- **Efecto sustitución:** no hay porque no hay cambio en precios relativos

Suba temporaria del ingreso corriente. Ejemplo utilidad logarítmica

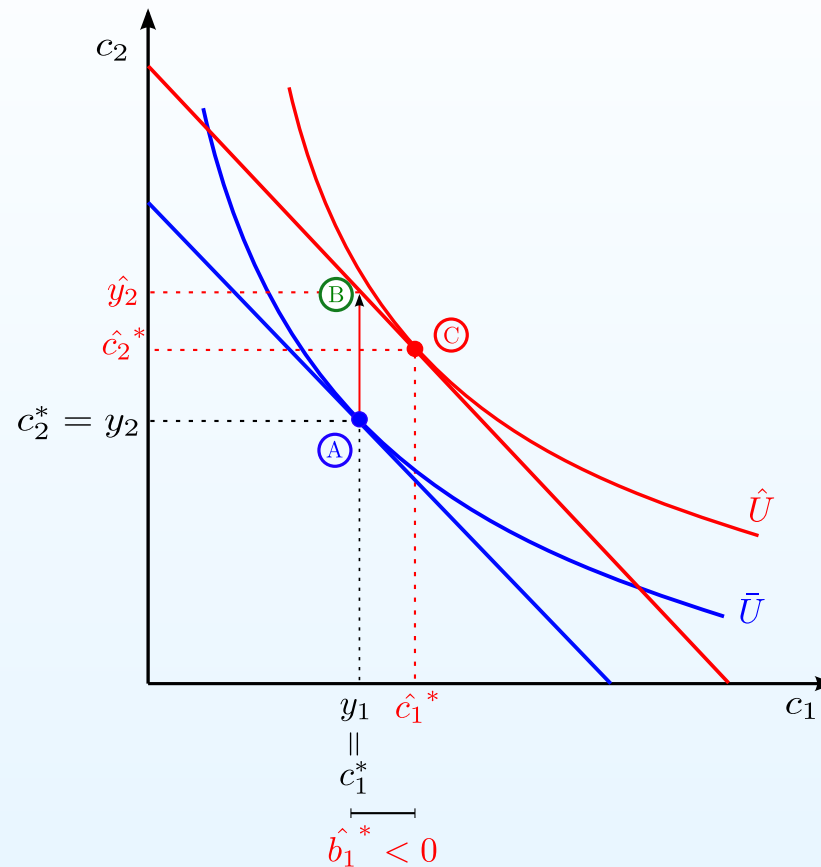
- Usando que $\Delta y_2 = 0$ y $\Delta y_1 > 0$, las demandas (7), (8), y (9) implican

$$\Delta c_1^* = \frac{1}{1 + \beta} \left[\Delta y_1 + \frac{\Delta y_2}{1 + r_1} \right] = \frac{\Delta y_1}{1 + \beta} < \Delta y_1$$

$$\Delta c_2^* = \frac{\beta(1 + r_1)}{1 + \beta} \left[\Delta y_1 + \frac{\Delta y_2}{1 + r_1} \right] = \frac{\beta(1 + r_1)}{1 + \beta} \Delta y_1 > 0$$

$$\Delta b_1^* = \frac{\beta}{1 + \beta} \left[\Delta y_1 - \frac{\Delta y_2}{\beta(1 + r_1)} \right] = \frac{\beta \Delta y_1}{1 + \beta} > 0.$$

Sube del ingreso futuro: $\bar{y}_1, \uparrow y_2$



- **Efecto riqueza:** c_1 y c_2 bienes normales, $\uparrow c_1, \uparrow c_2$
- **Aumento esperado del ingreso futuro:** $0 = \Delta y_1 < \Delta c_1 \Rightarrow \hat{b}_1^* < 0, \quad 0 < \Delta c_2 < \Delta y_2$.
- **Efecto sustitución:** no hay porque no hay cambio en precios relativos

Suba esperada del ingreso futuro. Ejemplo utilidad logarítmica

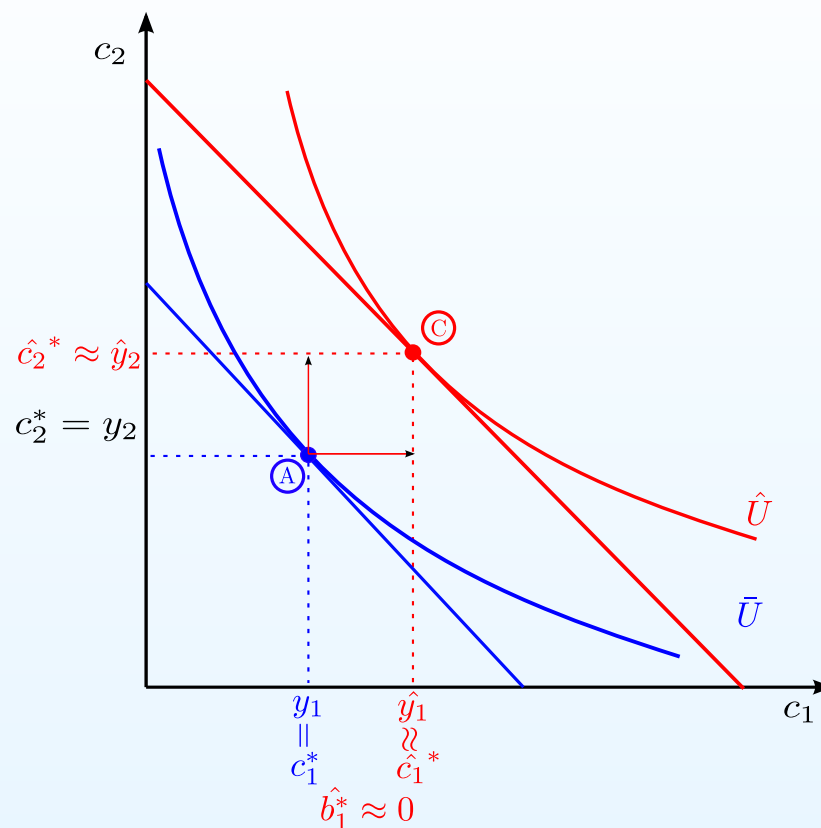
- Usando que $\Delta y_2 > 0$ y $\Delta y_1 = 0$, las demandas (7), (8), y (9) implican

$$\Delta c_1^* = \frac{1}{1 + \beta} \left[\Delta y_1 + \frac{\Delta y_2}{1 + r_1} \right] = \frac{1}{1 + \beta} \frac{\Delta y_2}{1 + r_1} > 0$$

$$\Delta c_2^* = \frac{\beta(1 + r_1)}{1 + \beta} \left[\Delta y_1 + \frac{\Delta y_2}{1 + r_1} \right] = \frac{\Delta y_2}{1 + \beta} > 0$$

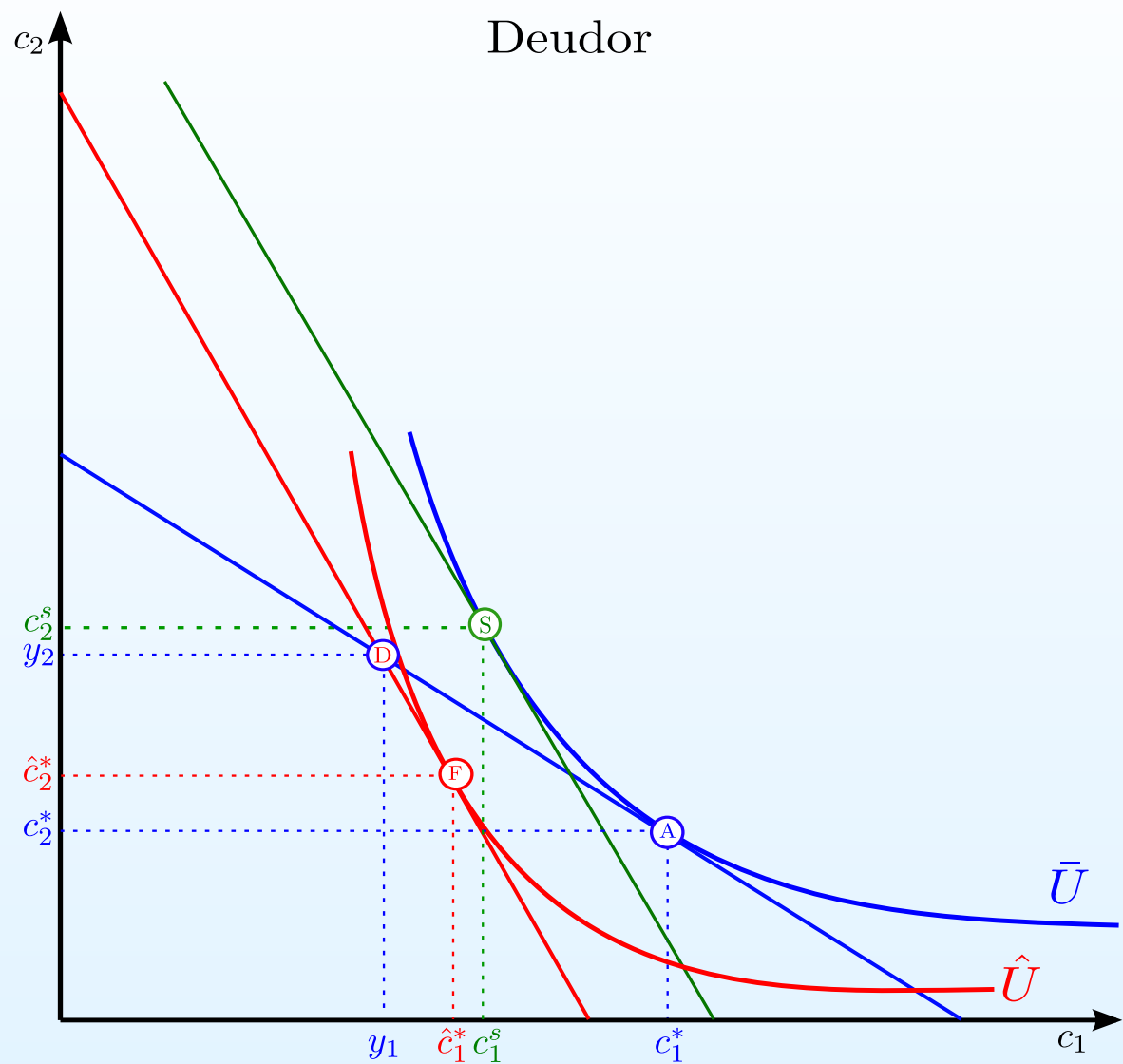
$$\Delta b_1^* = \frac{\beta}{1 + \beta} \left[\Delta y_1 - \frac{\Delta y_2}{\beta(1 + r_1)} \right] = \frac{-\Delta y_2}{(1 + \beta)(1 + r_1)} < 0.$$

Suba permanente en el ingreso: $\uparrow y_1, \uparrow y_2$ en igual cantidad, $\Delta y_1 = \Delta y_2$

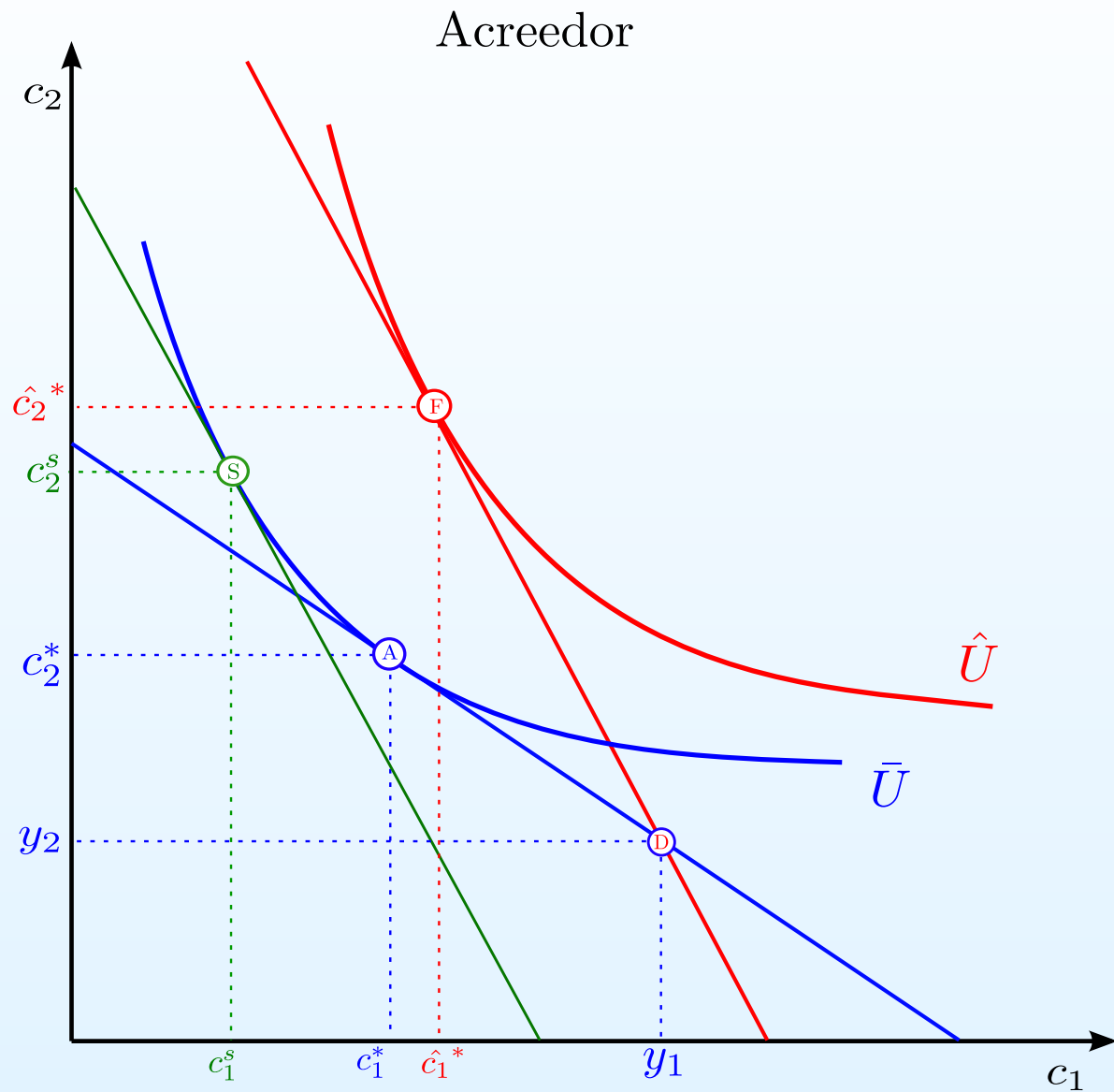


- **Efecto riqueza:** c_1 y c_2 bienes normales: $\uparrow c_1, \uparrow c_2$
- **Aumento permanente del ingreso** $\Delta y_1 = \Delta y_2$ $\Delta c_1 \approx \Delta y_1$ $\Delta c_2 \approx \Delta y_2$ $\Delta b_1 \approx 0$.
- **Efecto sustitución:** no hay porque no hay cambio en precios relativos.

Efecto de una suba en la tasa de interés real: deudor



Efecto de una suba en la tasa de interés real: acreedor



Efecto de una suba en la tasa de interés real: resumen

- Suba en la tasa de interés tiene un efecto sustitución y un efecto riqueza
- **Efecto sustitución:** $\uparrow r_1$ implica que c_2 se hace más barato relativo a c_1 : $\downarrow c_1, \uparrow c_2$
- **Efecto riqueza (deudor):** $\uparrow r_1$ implica que aumenta el costo financiero de la deuda, por lo que la riqueza disminuye. Como c_1 y c_2 son bienes normales, $\downarrow c_1, \downarrow c_2$
- **Efecto riqueza (acreedor):** $\uparrow r_1$ implica que aumenta su ingreso financiero, por lo que la riqueza sube. Como c_1 y c_2 son bienes normales, $\uparrow c_1, \uparrow c_2$

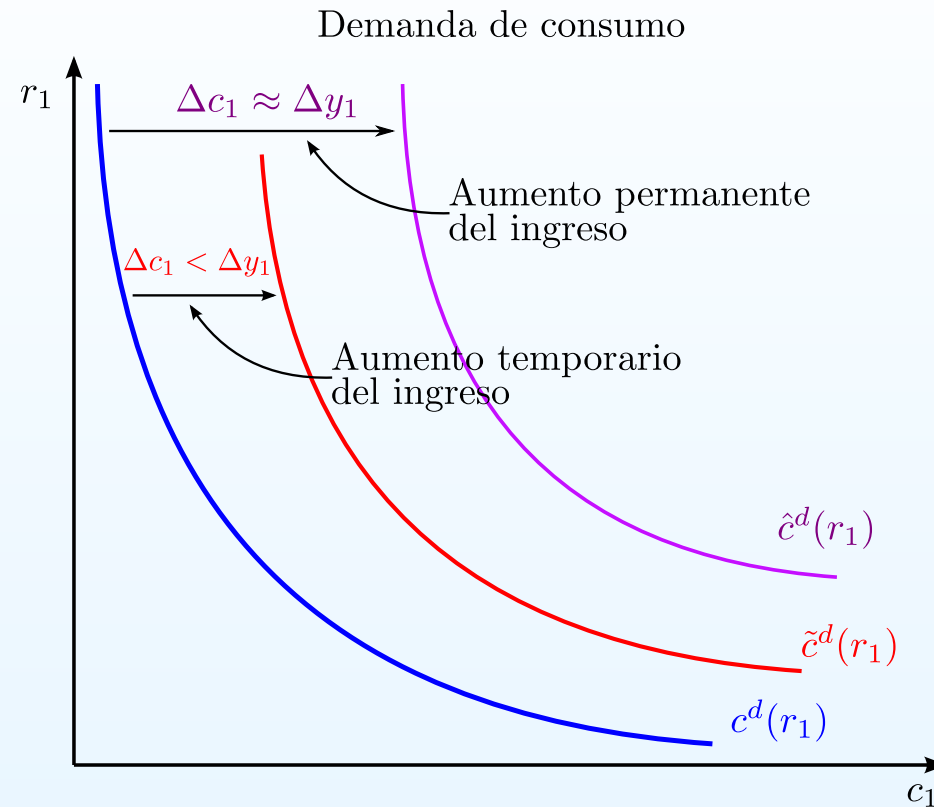
En resumen

	Deudor	Acreedor
Efecto sustitución	$\downarrow c_1, \uparrow c_2$	$\downarrow c_1, \uparrow c_2$
Efecto riqueza	$\downarrow c_1, \downarrow c_2$	$\uparrow c_1, \uparrow c_2$
Efecto total	$\downarrow c_1, c_2?$	$c_1?, \uparrow c_2$

La demanda del consumidor representativo de la economía

- Efecto final depende de si el consumidor es acreedor o deudor.
- Si consideramos una economía cerrada y nos interesa el comportamiento agregado:
 - Si alguien debe 100 pesos hay otra persona que es acreedora de esos 100 pesos.
 - El consumidor “promedio” no es ni deudor ni acreedor.
- Por lo tanto, para el consumidor representativo *de una economía cerrada* un cambio en la tasa de interés no tiene efecto riqueza.

La demanda del consumidor representativo de la economía



- **Importante:** en el caso del consumidor representativo, la pendiente negativa de la demanda de consumo surge del efecto sustitución intertemporal del consumo ante cambios en la tasa de interés.

La propensión marginal al consumo y la función de consumo keynesiana

- La propensión marginal al consumo (PMC) mide el cambio en el consumo corriente cuando sube el ingreso corriente en una unidad:

$$PMC = \frac{dc_t}{dy_t}$$

- Keynes argumentó que:
 1. el consumo depende principalmente del ingreso corriente disponible
 2. cuando el ingreso aumenta en \$1, el consumo aumenta en menos de \$1, y
 3. cuando aumenta su ingreso, el consumidor ahorra una proporción mayor del mismo.

La propensión marginal al consumo y la función de consumo Keynesiana

- Función de consumo Keynesiana que satisface 1, 2 y 3:

$$c_t = A + By_t$$

- A : componente autónomo del consumo
- B : propensión marginal al consumo

- Requisitos para que se cumplan 1. 2. y 3.

$$A > 0 \text{ y } 0 < B < 1.$$

- Se estima que la PMC es alrededor de 0.7-0.8: de cada peso que ganan los consumidores asignan entre 70 y 80 centavos al consumo y entre 30 y 20 centavos al ahorro.

La propensión marginal al consumo y la función de consumo Keynesiana

- Teoría de consumo keynesiana no diferencia entre cambios permanentes y transitorios del ingreso.
 - Cualquier aumento del ingreso se traslada en la proporción $B \approx 0,8$ al consumo.
- La teoría del consumo que analizamos arriba es distinta a la keynesiana:
 - La diferencia en la respuesta del consumo ante cambios transitorios versus permanentes del ingreso es un componente fundamental de la teoría.
 - Nuestro modelo predice que la PMC ante un cambio permanente en el ingreso es cercana a 1, mientras que la PMC de un cambio temporario es mucho menor (más extremo en modelos de muchos períodos)

¿Cuándo es la PMC igual a 1 ante un cambio permanente del ingreso?

- Ecuación de Euler

$$u'(c_1) = (1 + r_1)\beta u'(c_2)$$

- Si $\beta(1 + r_1) = 1$, entonces

$$u'(c_1) = u'(c_2) \Rightarrow c_1 = c_2.$$

- En este caso, la PMC ante cambios permanentes en el ingreso es exactamente igual a 1. (Prueba).
- Estimaciones de β implican $\beta(1 + r_1) \approx 1$, por lo que la PMC ante cambios permanentes en el ingreso es aproximadamente 1.

Macroeconomía Dinámica

Introducción: Parte 4

Elección Intertemporal de Consumo y Trabajo

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Elección intertemporal de consumo y trabajo

- Hoy juntamos el modelo de consumo intertemporal con el modelo de trabajo endógeno.
- El consumidor opera su propia función de producción.
- Equivalente si hay un mercado de trabajo.
- Consumidores deciden
 - cuánto consumir y trabajar hoy,
 - cuánto ahorrar y
 - cuánto consumir y trabajar en el futuro.

El problema del consumidor

- Función de utilidad

$$U = u(c_1) + v(1 - l_1) + \beta [u(c_2) + v(1 - l_2)].$$

- Las restricciones presupuestarias flujo son

$$\begin{aligned}c_1 + b_1 &= A_1 f(l_1) \\c_2 &= A_2 f(l_2) + b_1(1 + r_1).\end{aligned}$$

- Consumidor maximiza su utilidad sujeto a las restricciones presupuestarias
- Conveniente trabajar con la restricción presupuestaria intertemporal:

$$c_1 + \frac{c_2}{1 + r_1} = A_1 f(l_1) + \frac{A_2 f(l_2)}{1 + r_1}$$

El problema del consumidor

- Lagrangiano,

$$\max_{c_1, c_2, l_1, l_2} u(c_1) + v(1 - l_1) + \beta [u(c_2) + v(1 - l_2)] - \lambda \left[c_1 + \frac{c_2}{1 + r_1} - A_1 f(l_1) - \frac{A_2 f(l_2)}{1 + r_1} \right],$$

- Condiciones de primer orden:

$$\frac{\partial L}{\partial c_1} = 0 \Rightarrow u'(c_1) = \lambda \quad (1)$$

$$\frac{\partial L}{\partial l_1} = 0 \Rightarrow v'(1 - l_1) = \lambda A_1 f'(l_1) \quad (2)$$

$$\frac{\partial L}{\partial c_2} = 0 \Rightarrow \beta u'(c_2) = \lambda \frac{1}{1 + r_1} \quad (3)$$

$$\frac{\partial L}{\partial l_2} = 0 \Rightarrow \beta v'(1 - l_2) = \lambda \frac{A_2 f'(l_2)}{1 + r_1} \quad (4)$$

El problema del consumidor

- Ecuaciones (1) y (2) implican

$$\frac{v'(1 - l_1)}{u'(c_1)} = A_1 f'(l_1).$$

- Interpretación económica: $v'(1 - l_1) = A_1 f'(l_1)u'(c_1)$.

- En el segundo periodo,

$$\frac{v'(1 - l_2)}{u'(c_2)} = A_2 f'(l_2).$$

- Ecuación de Euler del consumo

$$u'(c_1) = \beta (1 + r_1) u'(c_2)$$

Solución del problema del consumidor:

- Solución: sistema de 4 ecuaciones con 4 incógnitas:

$$\frac{v'(1 - l_1)}{u'(c_1)} = A_1 f'(l_1)$$

$$\frac{v'(1 - l_2)}{u'(c_2)} = A_2 f'(l_2)$$

$$u'(c_1) = \beta(1 + r_1) u'(c_2)$$

$$c_1 + \frac{c_2}{1 + r_1} = A_1 f(l_1) + \frac{A_2 f(l_2)}{1 + r_1}$$

- Dadas las demandas de consumo c_1^d y c_2^d , y las ofertas de trabajo l_1^s y l_2^s podemos obtener:

- Demanda de bonos

$$b_1^d = A_1 f(l_1^s) - c_1^d$$

- Oferta de bienes

$$y_t^s = A_t f(l_t^s)$$

La ecuación de Euler para el trabajo

- Combino (2) y (4)

$$v'(1 - l_1) = \frac{A_1 f'(l_1)}{A_2 f'(l_2)} (1 + r_1) \beta v'(1 - l_2). \quad (5)$$

- **Importante:** (5) es redundante; implícita en las otras ecuaciones.
- Pero (5) nos permite entender la sustitución intertemporal del trabajo.
- Este efecto es fundamental para entender la curva de oferta agregada de bienes.
- Interpretación: aumento marginal de l_1 ,
 - **Costo marginal:** $v'(1 - l_1)$
 - **Beneficio marginal:** $\frac{A_1 f'(l_1)}{A_2 f'(l_2)} (1 + r_1) \beta v'(1 - l_2)$.

La sustitución intertemporal del trabajo

Sustitución intertemporal del trabajo: cambio en l_1 vs l_2 cuando cambian la tasa de interés r_1 y las productividades A_1 y A_2 .

Análisis gráfico:

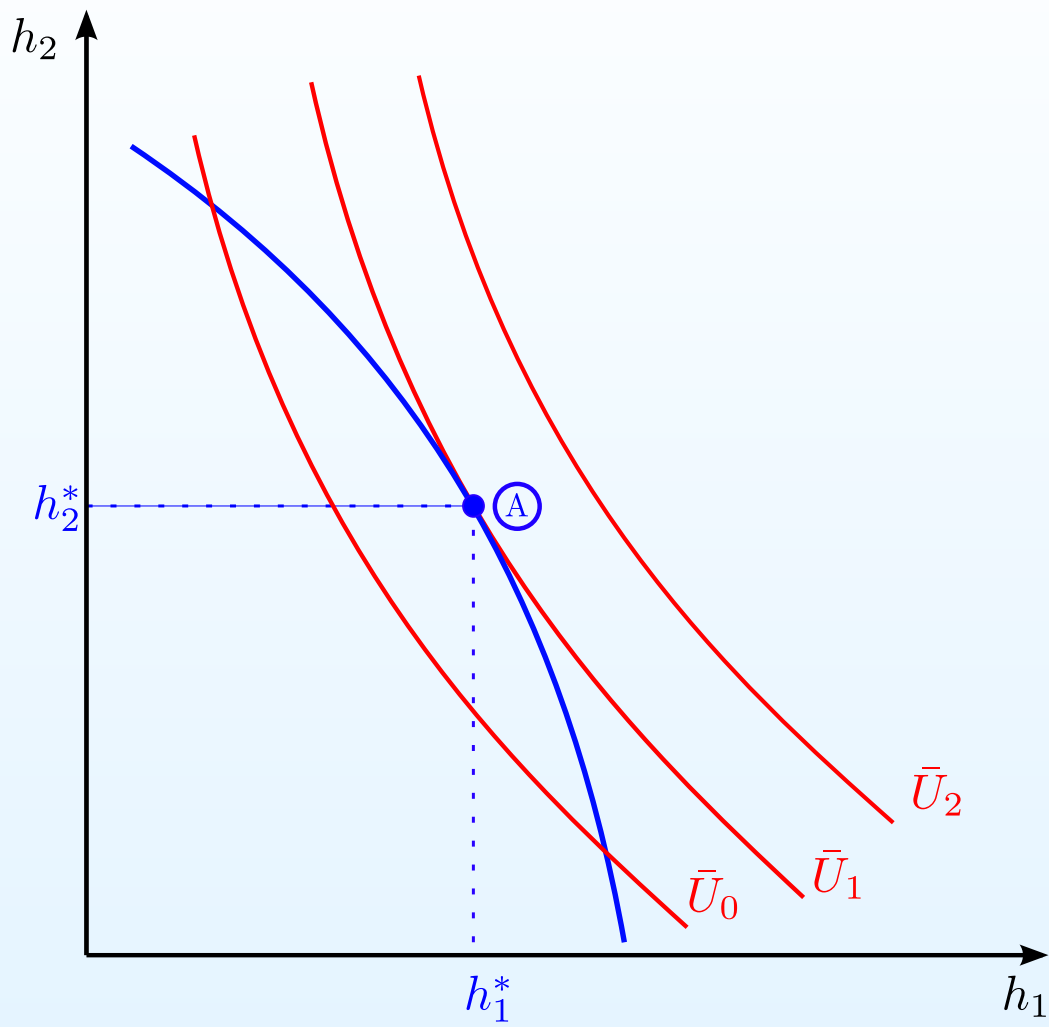
1. *Curvas de indiferencia* entre ocio futuro y ocio presente (c_1 y c_2 dados): valoración subjetiva del ocio futuro en términos de ocio presente
2. *Frontera de posibilidades de producción intertemporal del ocio*: Tasa a la cual la tecnología permite transformar ocio presente por ocio futuro.

Curvas de indiferencia entre h_1 y h_2 : (dados c_1 y c_2)

$$\bar{U} = v(h_1) + \beta v(h_2) + \text{términos relacionados al consumo}$$

- Curvas de indiferencia son decrecientes y convexas.

La sustitución intertemporal del trabajo



La sustitución intertemporal del trabajo

Frontera de posibilidades de producción: Escribo restricción presupuestaria intertemporal dado un valor presente del consumo $\bar{C}$

$$\bar{C} = A_1 f(1 - h_1) + \frac{A_2 f(1 - h_2)}{1 + r_1} \quad (6)$$

- Par de ocios h_1 y h_2 que generan un valor presente del ingreso de $\bar{C}$.
- Forma de esta frontera: h_2 como una función de h_1 .
- La frontera es decreciente y cóncava.

La sustitución intertemporal del trabajo

- La frontera es decreciente: derivó (6) con respecto a h_1 usando $h_2(h_1)$:

$$0 = -A_1 f'(1 - h_1) - \frac{A_2 f'(1 - h_2)}{1 + r_1} \frac{dh_2}{dh_1} \quad (7)$$

o bien,

$$\frac{dh_2}{dh_1} = - (1 + r_1) \frac{A_1 f'(1 - h_1)}{A_2 f'(1 - h_2)} < 0$$

- Intuición: si aumento h_1 y debo financiar $\bar{C}$, deberé reducir h_2 .

La sustitución intertemporal del trabajo

- La frontera es cóncava: derivo (7) una vez más

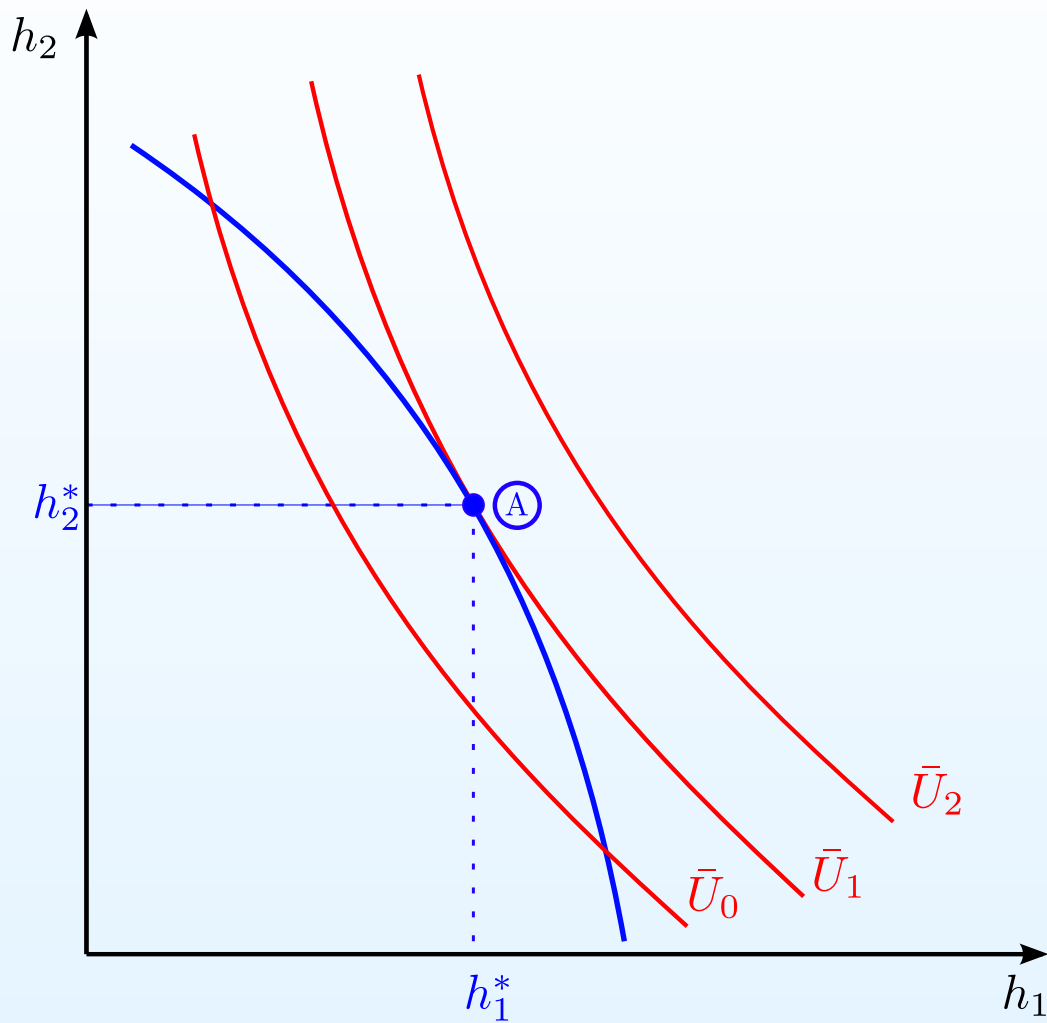
$$0 = A_1 f''(1 - h_1) + \frac{A_2 f''(1 - h_2)}{1 + r_1} \left(\frac{dh_2}{dh_1} \right)^2 - \frac{A_2 f'(1 - h_2)}{1 + r_1} \frac{d^2 h_2}{d(h_1)^2}.$$

O bien,

$$\frac{d^2 h_2}{d(h_1)^2} = \frac{A_1 f''(1 - h_1) + \frac{A_2 f''(1 - h_2)}{1 + r_1} \left(\frac{dh_2}{dh_1} \right)^2}{\frac{A_2 f'(1 - h_2)}{1 + r_1}} < 0.$$

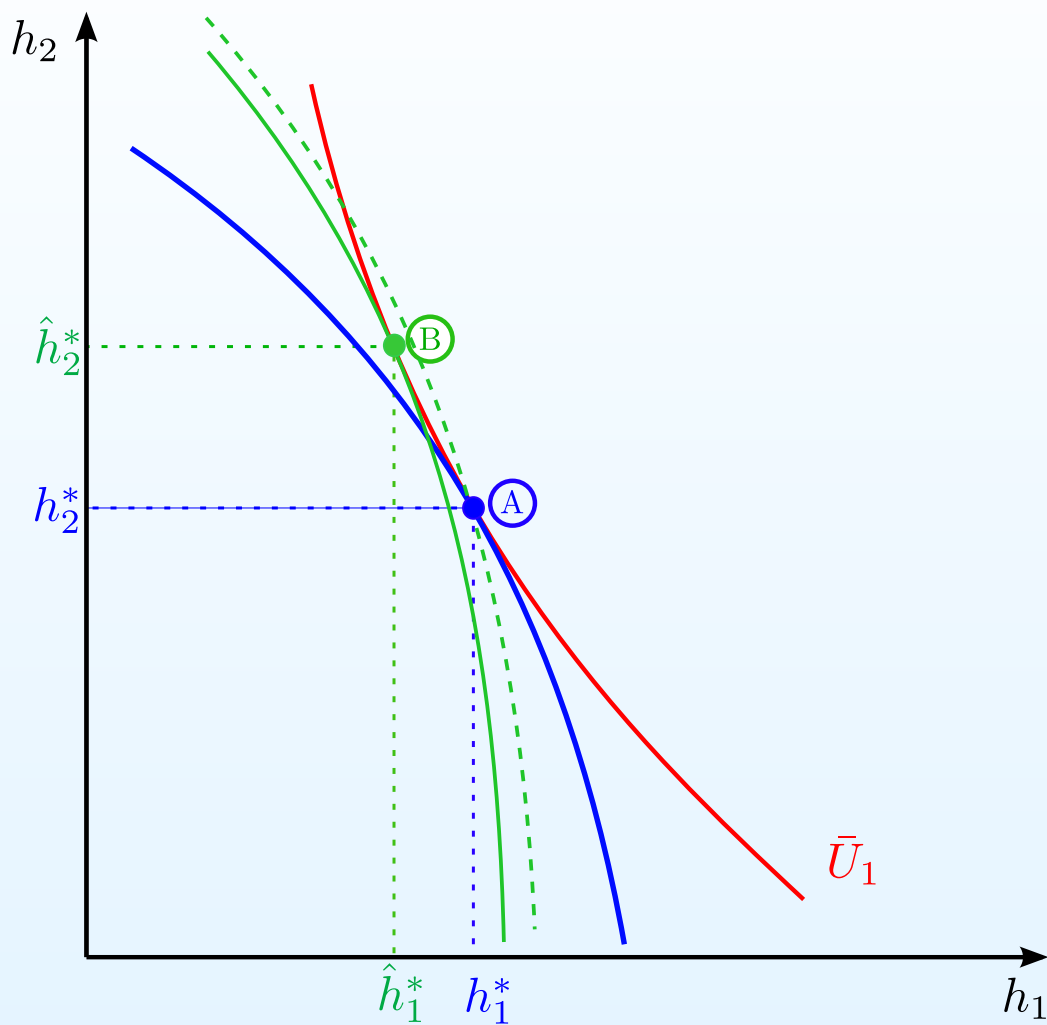
- Intuición: retornos marginales decrecientes del trabajo.

La sustitución intertemporal del trabajo



- Punto A: representación gráfica de la E. de Euler del trabajo.

La sustitución intertemporal del trabajo: aumenta A_1 con A_2 y r_1 fijos



Efecto sustitución intertemporal de $\uparrow A_1 \Rightarrow \uparrow l_1 (\downarrow h_1)$ y $\downarrow l_2 (\uparrow h_2)$. (Intuición).

La sustitución intertemporal del trabajo: suben A_1 y A_2 en igual proporción

- Pendiente de la frontera:

$$\frac{dh_2}{dh_1} = - (1 + r_1) \frac{A_1}{A_2} \frac{f'(1 - h_1)}{f'(1 - h_2)}$$

- En la canasta original (h_1^*, h_2^*) , la pendiente no cambia.
- El efecto sustitución es nulo ante suba permanente de la productividad en la misma proporción
- Recuerde que el consumidor es compensado para que la nueva frontera sea tangente a la curva de indiferencia original.

La sustitución intertemporal del trabajo: aumenta r_1

- Similar al caso de una suba temporaria en la productividad A_1 .
- El efecto sustitución intertemporal de $\uparrow r_1 \Rightarrow \uparrow l_1 (\downarrow h_1)$ y $\downarrow l_2 (\uparrow h_2)$.
- Las notas de clase contienen también un análisis de lo anterior un poco más formal.

Estática Comparativa

- Consideramos aumentos en las productividades A_1 y A_2 , y en la tasa de interés r_1 .
- Nos enfocamos en el periodo $t = 1$.
- *En este análisis usamos todo lo que aprendimos hasta ahora.*

Suba temporaria en la productividad: $\uparrow A_1$ pero A_2 y r_1 no cambian

- **Impacto:** Dados $l_1 = \bar{l}_1$ y $l_2 = \bar{l}_2$, $\uparrow y_1 = \uparrow A_1 f(\bar{l}_1)$. Entonces hay un efecto riqueza positivo.
- **ER positivo:** $\uparrow c_1$, $\uparrow c_2$ $\downarrow l_1$ y $\downarrow l_2$ (bs normales). La caída de l_1 implica $\downarrow y_1$. Sin embargo, la caída de y_1 es menor que la suba del producto debido al efecto directo (impacto) del aumento de A_1 . De otro modo no sería posible aumentar los consumos.
- **ES en el periodo 1:** $\uparrow l_1 \Rightarrow \uparrow y_1$ y $\uparrow c_1$ (sube el precio del ocio relativo al consumo)
- **ES en el periodo 2:** No hay porque no cambia A_2 .
- **ES intertemporal de trabajo:** El aumento de A_1 implica $\uparrow l_1 \Rightarrow \uparrow y_1$ (también $\downarrow l_2$ y $\downarrow y_2$).
- **ES intertemporal de consumo:** No hay.

Efecto total: $\uparrow y_1$, $\uparrow c_1$ pero l_1 puede subir o bajar.

Suba permanente en la productividad: $\uparrow A_1$ y $\uparrow A_2$ en igual proporción

- **Impacto:** Dados $l_1 = \bar{l}_1$ y $l_2 = \bar{l}_2$, $\uparrow y_1 = \uparrow A_1 f(\bar{l}_1)$ y $\uparrow y_2 = \uparrow A_2 f(\bar{l}_2)$. Efecto riqueza positivo.
- **ER positivo:** $\uparrow c_1, \uparrow c_2 \downarrow l_1$ y $\downarrow l_2$ (bs normales). La caída de l_1 implica $\downarrow y_1$. Sin embargo, la caída de y_1 es menor que la suba del producto debido al efecto directo (impacto) del aumento de A_1 . De otro modo no sería posible aumentar los consumos.
- **ES en el periodo 1:** $\uparrow l_1 \Rightarrow \uparrow y_1$ y $\uparrow c_1$ (sube el precio del ocio relativo al consumo)
- **ES en el periodo 2:** $\uparrow l_2 \Rightarrow \uparrow y_2$ y $\uparrow c_2$ (sube el precio del ocio relativo al consumo)
- **ES intertemporal de trabajo:** No hay porque A_1/A_2 no cambia.
- **ES intertemporal de consumo:** No hay.

Efecto total: $\uparrow y_1, \uparrow c_1$ pero l_1 puede subir o bajar.

Suba esperada en la productividad: $\uparrow A_2$ y A_1 no cambia

- **Impacto:** Dado $l_2 = \bar{l}_2$, $\uparrow y_2 = \uparrow A_2 f(\bar{l}_2)$. Entonces hay un efecto riqueza positivo.
- **ER positivo:** $\uparrow c_1, \uparrow c_2 \downarrow l_1$ y $\downarrow l_2$. Cae la producción hoy, $\downarrow y_1$.
- **ES en el periodo 1:** No hay
- **ES en el periodo 2:** $\uparrow l_2 \Rightarrow \uparrow y_2$ y $\uparrow c_2$
- **ES intertemporal de trabajo:** Cae A_1/A_2 . Ocio futuro se hace más caro que el ocio presente: $\downarrow l_1 \Rightarrow \downarrow y_1$.
- **ES intertemporal de consumo:** No hay.

Efecto total: $\downarrow y_1, \uparrow c_1, \downarrow l_1$.

Suba en la tasa de interés r_1

- **Impacto:** No hay.
- **Efecto riqueza:** Depende de si el consumidor es deudor o acreedor

$$\text{Efecto riqueza} = \begin{cases} \text{Deudor } (b_1 < 0) & : \downarrow c_1, \downarrow c_2, \uparrow l_1, \uparrow l_2, \uparrow y_1 \\ \text{Acreedor } (b_1 > 0) & : \uparrow c_1, \uparrow c_2, \downarrow l_1, \downarrow l_2, \downarrow y_1 \\ b_1 = 0 & : \text{No hay} \end{cases}$$

Esto es idéntico a lo que vimos la clase pasada.

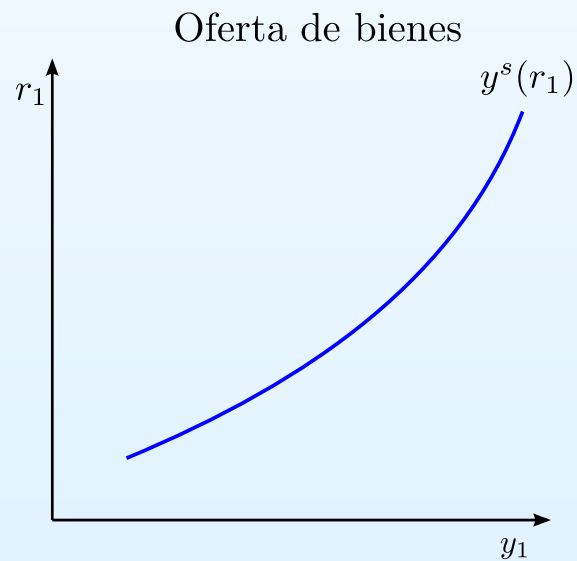
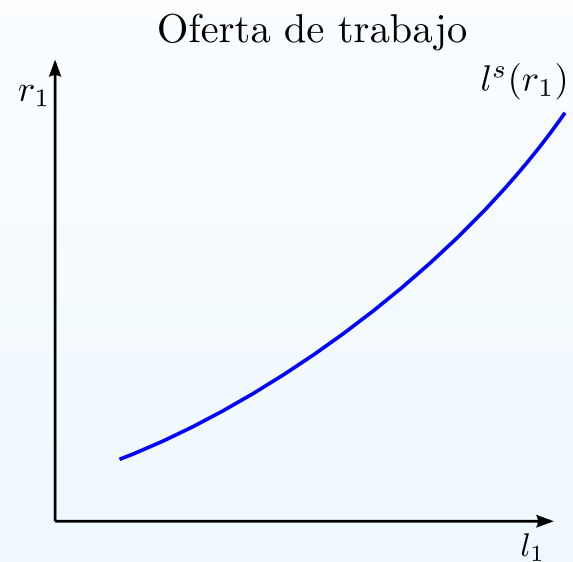
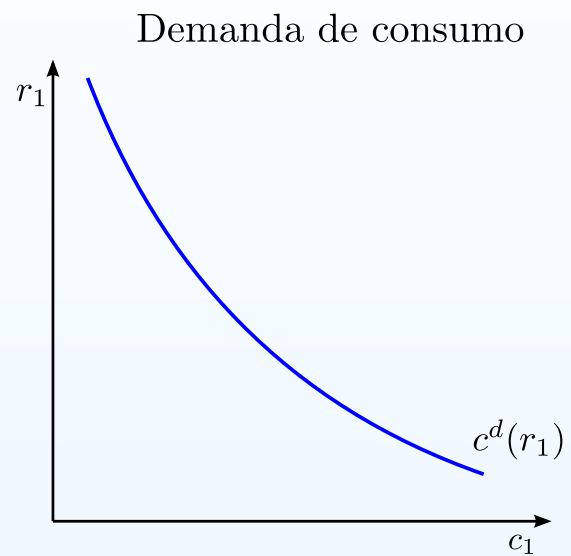
- **ES en el periodo 1:** No hay
- **ES en el periodo 2:** No hay.
- **ES intertemporal del trabajo:** $\uparrow l_1$ y $\downarrow l_2$. Por lo tanto, $\uparrow y_1$ y $\downarrow y_2$.
- **ES intertemporal del consumo:** $\downarrow c_1$ y $\uparrow c_2$ porque c_2 se hace más barato relativo a c_1 .
- En una economía cerrada, el consumidor representativo no es ni deudor ni acreedor:

Efecto total (economía cerrada) : $\uparrow l_1, \uparrow y_1, \downarrow c_1$.

Resumen estática comparativa: efectos en el primer periodo

Shock/variables	c_1	y_1	l_1
Aumento temporario de productividad ($\uparrow A_1$)	$\uparrow$	$\uparrow$	?
Aumento permanente de productividad (A_1/A_2 no cambia)	$\uparrow$	$\uparrow$	?
Aumento esperado de productividad ($\uparrow A_2$)	$\uparrow$	$\downarrow$	$\downarrow$
Aumento de tasa de interés real ($\uparrow r_1$) – economía cerrada	$\downarrow$	$\uparrow$	$\uparrow$

Ofertas y demandas del consumidor representativo



Macroeconomía Dinámica

Introducción: Parte 5

Equilibrio General en el Modelo de dos Períodos

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Elección intertemporal de consumo y trabajo

- Hasta hoy: problema de decisión de los consumidores, trabajadores y empresas.
- Hoy estudiaremos el equilibrio general: determinación de los precios y las cantidades de equilibrio.
- El equilibrio general consiste en encontrar la tasa de interés real que vacíe los mercados de bienes y de crédito.
- El mercado de trabajo estará implícito en el análisis.

Agregación de las demandas y ofertas individuales

- Buscamos las ofertas y demandas agregadas de la economía
- Supongamos que hay $i = 1, 2, \dots, N$ hogares (N grande).
- Cada familia resuelve su problema de decisión:
 - demanda de consumo,
 - oferta de trabajo y
 - oferta de bienes.
- Restricciones presupuestaria flujo del hogar i :

$$c_1^i + b_1^i = y_1^i$$

$$c_2^i = y_2^i + b_1^i (1 + r_1)$$

El problema del consumidor

- Sumamos a través de hogares

$$\sum_{i=1}^N c_1^i + \sum_{i=1}^N b_1^i = \sum_{i=1}^N y_1^i$$

$$\sum_{i=1}^N c_2^i = \sum_{i=1}^N y_2^i + \sum_{i=1}^N b_1^i (1 + r_1)$$

- Restricciones presupuestarias flujo agregadas:

$$C_1 + B_1 = Y_1$$

$$C_2 = Y_2 + B_1 (1 + r_1)$$

- Restricción presupuestaria intertemporal agregada:

$$C_1 + \frac{C_2}{1 + r_1} = Y_1 + \frac{Y_2}{1 + r_1}$$

Consistencia agregada y la Ley de Walras

- Mercados:

1. Bienes en los periodos 1 y 2
2. Bonos en el primer periodo

- Condiciones de equilibrio en el primer periodo:

$$\begin{aligned}C_1^d &= Y_1^s \\ B_1^d &= 0\end{aligned}$$

- Condiciones de equilibrio en el segundo periodo:

$$C_2^d = Y_2^s.$$

Ley de Walras en el modelo dinámico

- **Enunciado usual:** *si hay J mercados y $J - 1$ de ellos están en equilibrio, entonces el mercado restante estará en equilibrio.*
- **Modelos dinámicos:** ley de Walras se cumple de manera secuencial para cada periodo $t = 1, 2$
 - Primer periodo: mercado de bienes y mercado de crédito.
 - Segundo periodo: mercado de bienes
- **Prueba de la ley de Walras en el primer periodo:**

$$(C_1^d - Y_1^s) + B_1^d = 0$$

$$Y_1^s = C_1^d \Rightarrow B_1^d = 0.$$

$$B_1^d = 0 \Rightarrow Y_1^s = C_1^d.$$

Ley de Walras en el modelo dinámico

- Restricción presupuestaria flujo agregada en el segundo periodo:

$$C_2^d - Y_2^s = B_1^d (1 + r_1) .$$

- Dado que hubo equilibrio en el primer periodo ($B_1^d = 0$) entonces

$$C_2^d = Y_2^s .$$

Por lo tanto, si nos aseguramos que se cumple la condición de consistencia agregada en el mercado de bienes del primer periodo, entonces se cumplirán las condiciones de consistencia agregada tanto en el mercado de bonos como en el mercado de bienes del segundo periodo.

- Nos enfocaremos entonces en el mercado de bienes en $t = 1$.
 - Encontramos un caso donde al asegurarnos que se cumple el equilibrio en un mercado, entonces sabremos que los otros dos mercados estarán en equilibrio.
- ¿Contradicción a la Ley de Walras?

La demanda agregada de bienes

- Demanda individual de consumo: $c_1 = c_1^{id}(r_1, A_1, A_2)$.
- Por lo tanto, demanda agregada es también una función de (r_1, A_1, A_2) :

$$C_1^d = \sum_{i=1}^N c_1^{id}(r_1, A_1, A_2),$$

o bien

$$C_1 = C_1^d(r_1, A_1, A_2)$$

- De las demandas individuales:

$$\uparrow A_1 \Rightarrow \uparrow c_1 \text{ (bajo normalidad)}$$

$$\uparrow A_2 \Rightarrow \uparrow c_1 \text{ (bajo normalidad)}$$

- Entonces C_1^d aumenta cuando sube A_1 o A_2 .

La demanda agregada de bienes

- De las demandas individuales:

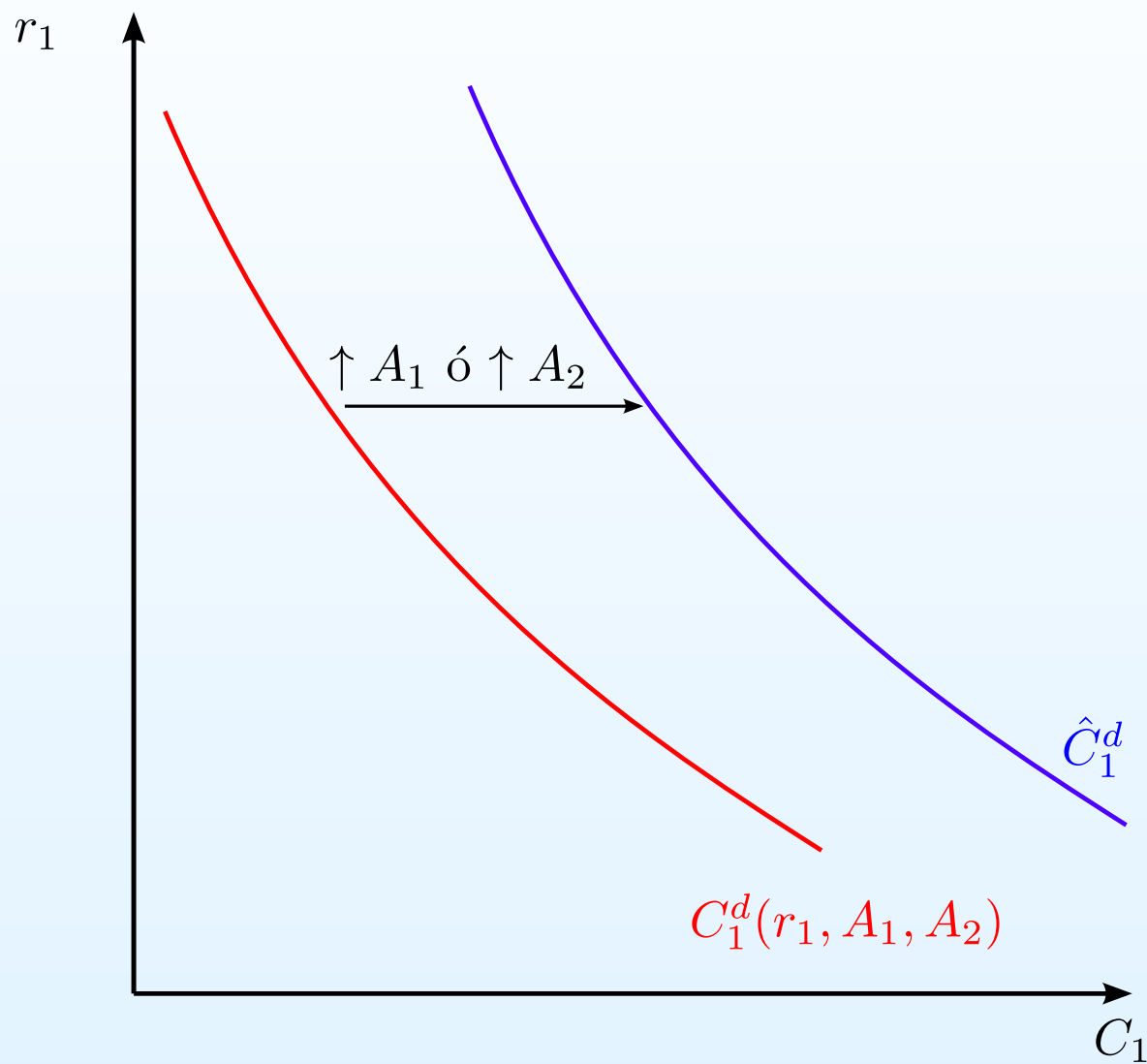
$$\uparrow r_1 \Rightarrow \left\{ \begin{array}{ll} \text{Efecto sustitución:} & \downarrow c_1 \\ & (c_2 \text{ más barato que } c_1) \\ \text{Efecto riqueza:} & \left\{ \begin{array}{ll} \downarrow c_1 & \text{si es deudor} \\ \uparrow c_1 & \text{si es acreedor} \\ = & \text{si no es acreedor ni deudor} \end{array} \right. \end{array} \right.$$

¿Efecto riqueza de una suba en la tasa de interés sobre la demanda agregada?

- El consumidor representativo no es deudor ni acreedor, o
- Efecto riqueza de acreedores y deudores se cancela aproximadamente.
- Para que C_1^d sea creciente en r_1 , ER de acreedores debe ser mayor que la suma del ER de los deudores *más* el ES de *todos* los consumidores.
- Por lo tanto, demanda agregada satisface

$$C_1 = C_1^d(\bar{r}_1, \bar{A}_1, \bar{A}_2).$$

La demanda agregada de bienes



La oferta agregada de bienes

- Oferta individual de bienes: $y_1 = y_1^{is}(r_1, A_1, A_2)$.
- Por lo tanto, la oferta agregada es una función de (r_1, A_1, A_2) :

$$Y_1^s = \sum_{i=1}^N y_1^{is}(r_1, A_1, A_2),$$

o bien

$$Y_1 = Y_1^s(r_1, A_1, A_2)$$

- ¿Cómo cambia la oferta agregada cuando cambian A_1 , A_2 y r_1 ?

La oferta agregada de bienes

Aumento de A_1 con A_2 y r_1 fijos

- **Impacto:** dado $\bar{l}_1$, sube la cantidad de bienes producidas, $\uparrow y_1 = \uparrow A_1 f(\bar{l}_1) \implies ER$ positivo
- **ER positivo:** $\downarrow l_1 \implies \downarrow y_1$. Sin embargo, el ER es menor al efecto directo de la mayor productividad.
- **ES intratemporal:** $\uparrow l_1 \implies \uparrow y_1$.
- **ES intertemporal del trabajo:** $\uparrow l_1 \implies \uparrow y_1$ (ocio hoy más caro relativo a ocio futuro)
- Por lo tanto, $\uparrow A_1 \implies \uparrow Y_1^s$.

La oferta agregada de bienes

Aumento de A_2 con A_1 y r_1 fijos

- **Impacto:** dados $\bar{l}_1$ y $\bar{l}_2$, $\uparrow y_2 = \uparrow A_2 f(\bar{l}_2) \implies ER$ positivo.
- **ER riqueza positivo:** $\downarrow l_1 \Rightarrow \downarrow y_1$.
- **ES intratemporal:** no hay.
- **ES intertemporal del trabajo:** $\downarrow l_1 \Rightarrow \downarrow y_1$ (ocio hoy más barato relativo a ocio futuro)
- Por lo tanto, $\uparrow A_2 \Rightarrow \downarrow Y_1^s$.

La oferta agregada de bienes

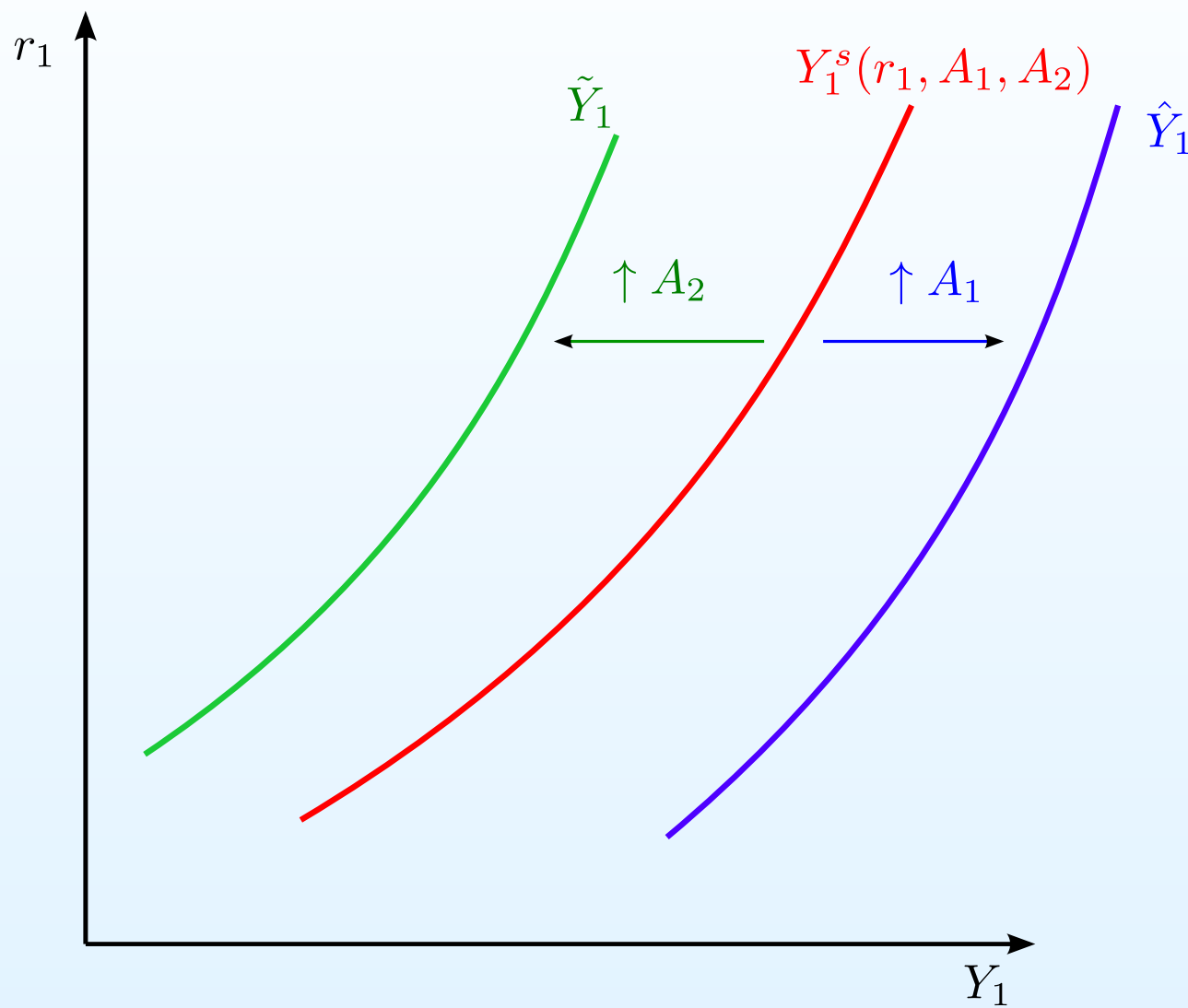
Aumento de la tasa de interés r_1 con A_1 y A_2 fijos

- **ER nulo:** (Consumidor representativo o se cancelan entre deudores y acreedores)
- **ES intratemporal:** no hay.
- **ES intertemporal del trabajo:** $\uparrow l_1 \Rightarrow \uparrow y_1$ (ocio hoy más caro relativo a ocio futuro)
- Por lo tanto, $\uparrow r_1 \Rightarrow \uparrow Y_1^s$.

Por lo tanto, la oferta agregada de bienes satisface

$$Y_1 = Y_1^s(r_1^+, \bar{A}_1^+, \bar{A}_2^-).$$

La oferta agregada de bienes



La demanda neta de bonos agregada

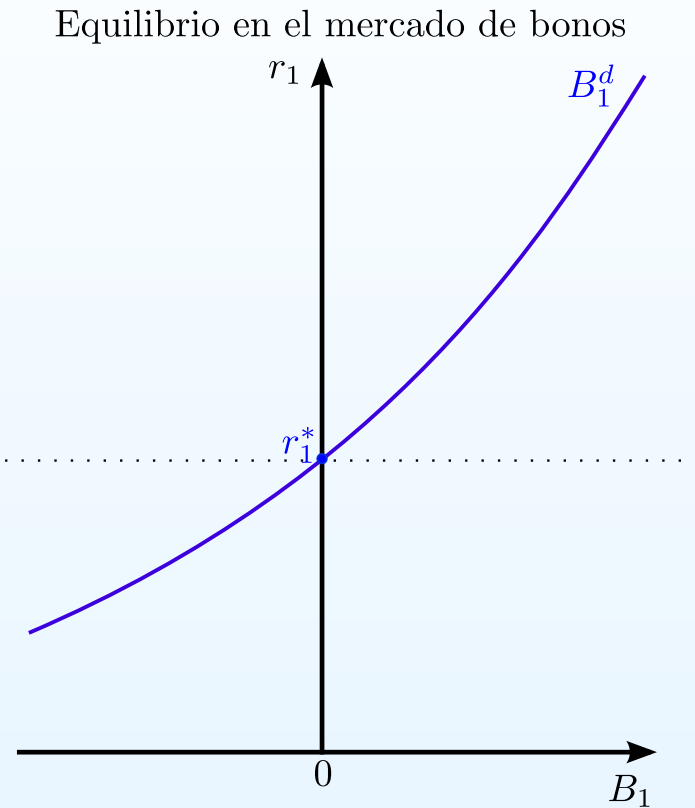
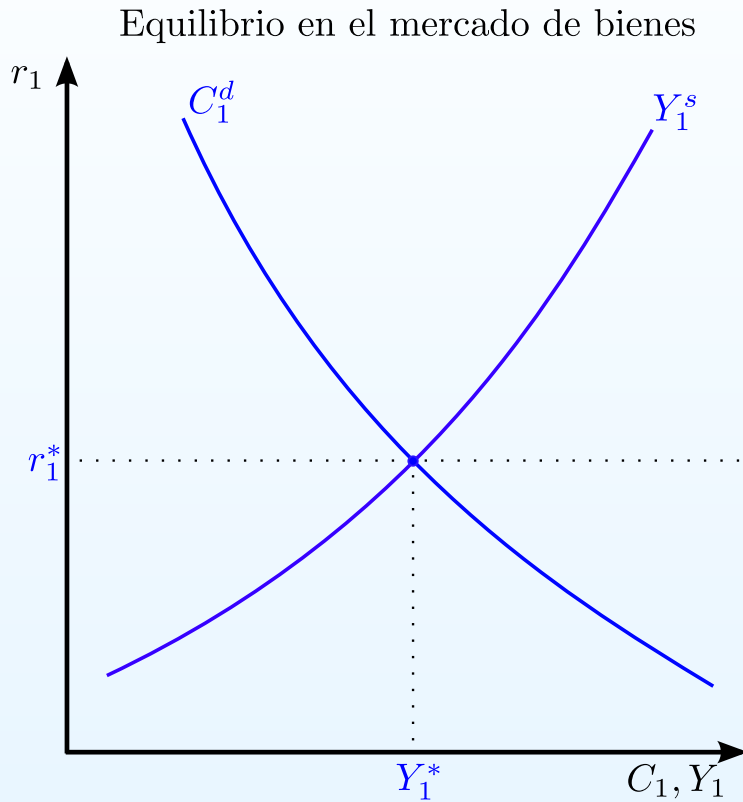
- De la restricción presupuestaria flujo, $b_1 = y_1 - c_1$.
- Sumando para todos los consumidores

$$B_1^d = Y_1^s(r_1^+, \bar{A}_1, \bar{A}_2) - C_1^d(r_1^-, \bar{A}_1, \bar{A}_2).$$

- Demanda agregada de bonos creciente en r_1 y decreciente A_2 .
- ¿Qué ocurre cuando aumenta A_1 (A_2 fijo)? Por un lado, sube la oferta agregada pero por el otro, sube el consumo agregado.
- Aumento temporario implica que $\Delta C_1 < \Delta Y_1$ (suavización intertemporal del consumo).
- Por lo tanto,

$$B_1^d = B_1^d(r_1^+, \bar{A}_1, \bar{A}_2)$$

Equilibrio en el mercado de bienes y de bonos



- Ley de Walras: ambos mercados se vacían a la misma tasa de interés r_1^* .

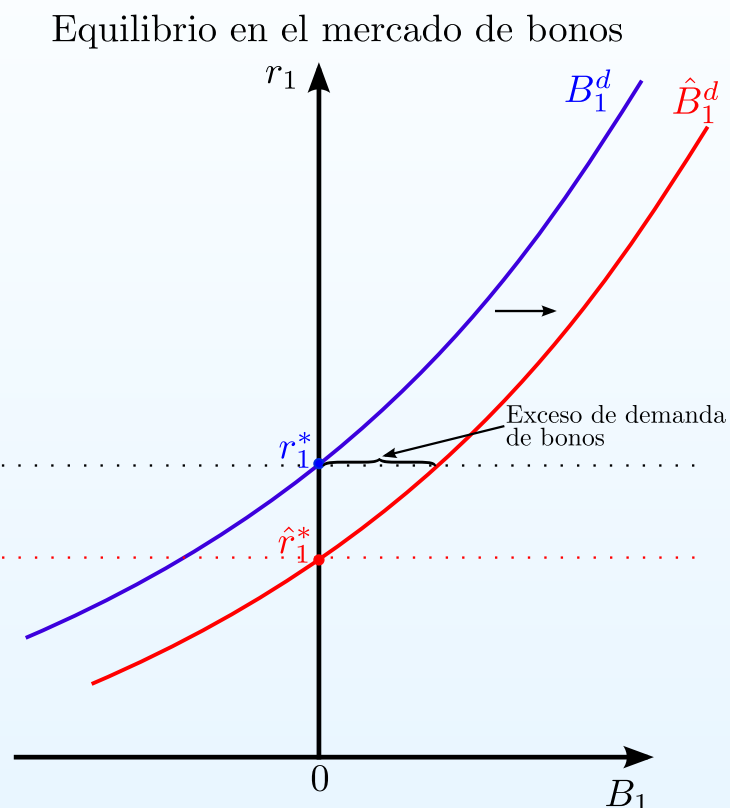
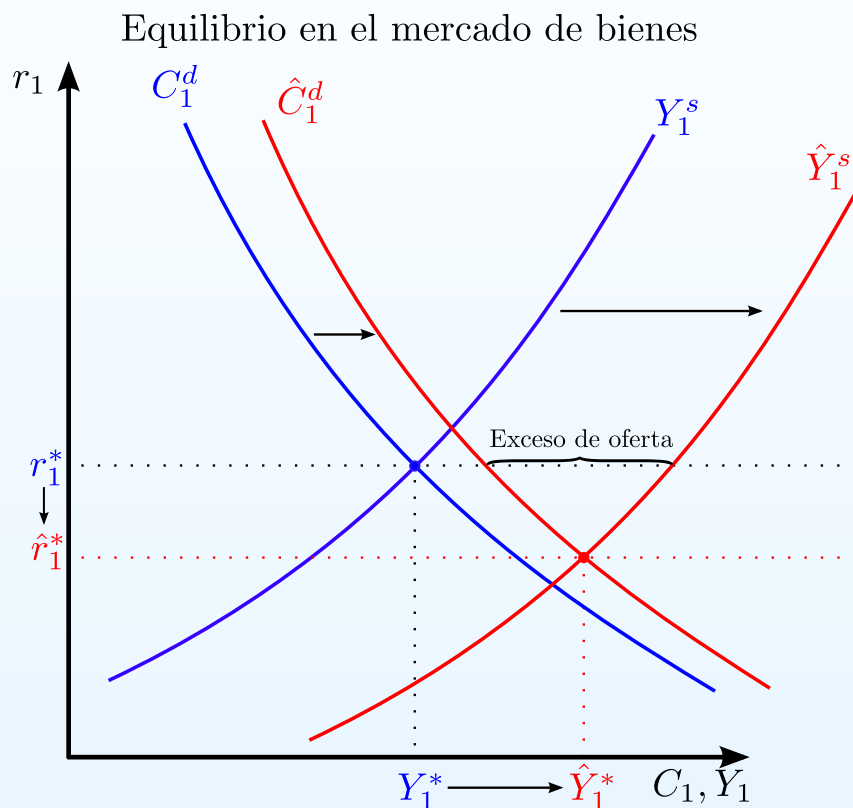
Estática comparativa

- Determinamos el equilibrio general:
 - Encontramos el precio de equilibrio r_1^* y la cantidad de equilibrio Y_1^* .
 - Ley de Walras garantiza que el mercado de bonos y el de bienes mañana estarán en equilibrio.
- Analizaremos cómo cambia la tasa de interés y las cantidades de equilibrio ante cambios en las productividades A_1 y A_2 .
- Consideraremos cambios temporarios, permanentes y esperados de la productividad.
- Idea: entender como distintos shocks pueden afectar la actividad agregada de una economía.

Suba temporaria de la productividad: sube A_1 y A_2 no cambia

- *Desplazamientos de las curvas de oferta y demanda dada una tasa de interés r_1 :*
 - **Impacto:** Dado $\bar{l}_1$, $\uparrow y_1 = \uparrow A_1 f(\bar{l}_1)$.
 - **ER positivo:** $\uparrow c_1$, $\uparrow c_2$, $\downarrow l_1$ y $\downarrow l_2$. La caída de l_1 implica $\downarrow y_1$, aunque menor que el impacto. Efecto riqueza no es muy grande por aumento temporario: $\Delta c_1 < \Delta y_1$.
 - **ES intratemporal en $t = 1$:** $\uparrow c_1$ y $\uparrow l_1 \Rightarrow \uparrow y_1$
 - **ES intertemporal del trabajo:** $\uparrow l_1$ y $\downarrow l_2$. Por lo tanto, $\uparrow y_1$.
- **Resumen:** C_1^d e Y_1^s se trasladan hacia la derecha, pero C_1^d se traslada menos que Y_1^s porque el aumento de productividad es temporario.

Suba temporaria de la productividad: sube A_1 y A_2 no cambia



- En r_1^* : exceso de oferta de bienes y exceso de demanda de bonos. Cae r_1 para equilibrar los mercados.
- Intuición relacionada a la escasez relativa de bienes.

Suba temporaria de la productividad: sube A_1 y A_2 no cambia

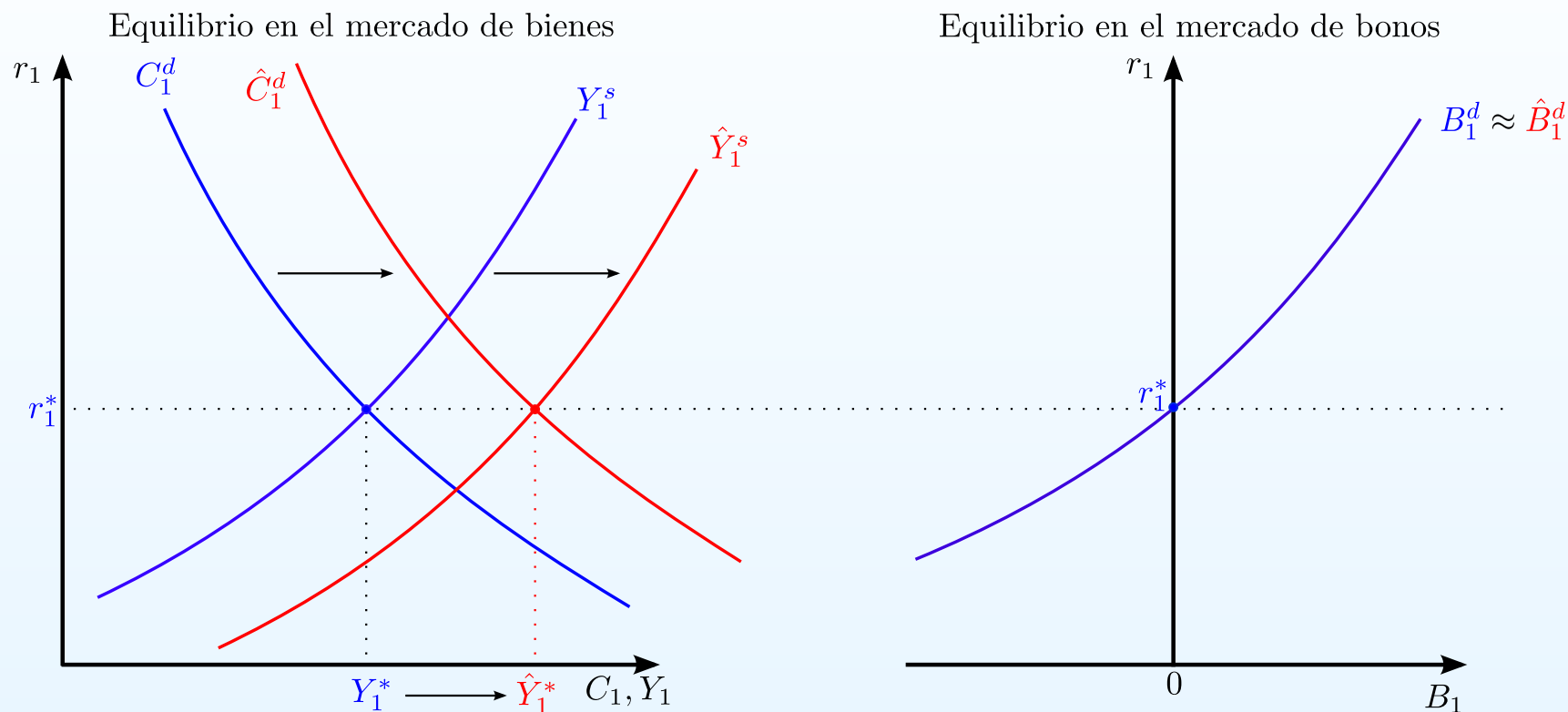
- Desplazamientos *sobre* las curvas de oferta y demanda reflejan los efectos sustitución intertemporal del trabajo y del consumo.
- Para comparar con los datos:
 1. Tasa de interés es *contracíclica*: sube el producto y cae la tasa de interés. En los datos es levemente *procíclica*.
 2. El cambio del trabajo es ambiguo. En los datos es procíclico.

$$\Delta L_1 = \left\{ \begin{array}{ll} \text{Efecto riqueza} & :\downarrow L_1 \\ \text{Efecto sustitución intratemporal} & :\uparrow L_1 \\ \text{Efecto sustitución intertemporal de } \uparrow A_1 & :\uparrow L_1 \\ \text{Efecto sustitución intertemporal de } \downarrow r_1 & :\downarrow L_1 \end{array} \right.$$

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción

- *Desplazamientos de las curvas de oferta y demanda dada una tasa de interés r_1 :*
 - **Impacto:** Dados $\bar{l}_1$ y $\bar{l}_2$, $\uparrow y_1 = \uparrow A_1 f(\bar{l}_1)$ y $\uparrow y_2 = \uparrow A_2 f(\bar{l}_2)$ (en igual proporción)
 - **ER positivo:** $\uparrow c_1, \uparrow c_2, \downarrow l_1$ y $\downarrow l_2$. La caída de l_1 implica $\downarrow y_1$, aunque menor que el impacto, lo que implica $\Delta y_1 > 0$. Efecto riqueza es grande por el aumento permanente: $\Delta c_1 \approx \Delta y_1$.
 - **ES intratemporal en $t = 1$:** $\uparrow c_1$ y $\uparrow l_1 \Rightarrow \uparrow y_1$
 - **ES intertemporal del trabajo:** No hay (A_1/A_2 no cambia).
- **Resumen:** C_1^d e Y_1^s se trasladan hacia la derecha en aproximadamente la misma cantidad por el aumento permanente de la productividad.

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción



- En r_1^* no hay (aprox.) ningún exceso de oferta o demanda.
- La tasa de interés no cambia y el producto aumenta.

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción

- Para comparar con los datos:

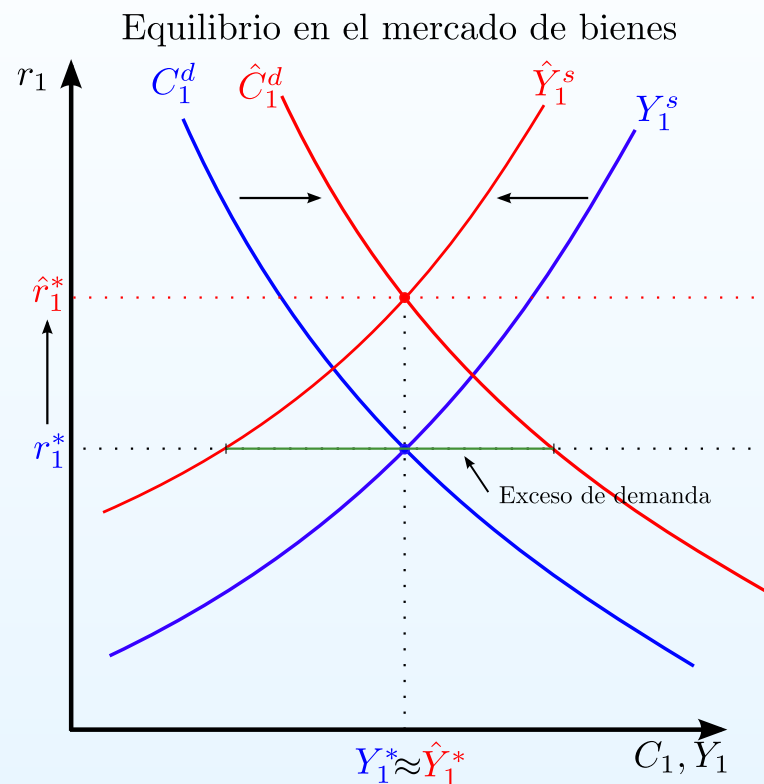
1. La tasa de interés es acíclica: sube el producto Y_1 y la tasa de interés r_1 no cambia.
2. El cambio en el trabajo L_1 es ambiguo:

$$\Delta L_1 = \left\{ \begin{array}{ll} \text{Efecto riqueza} & :\downarrow L_1 \\ \text{Efecto sustitución intratemporal} & :\uparrow L_1 \end{array} \right.$$

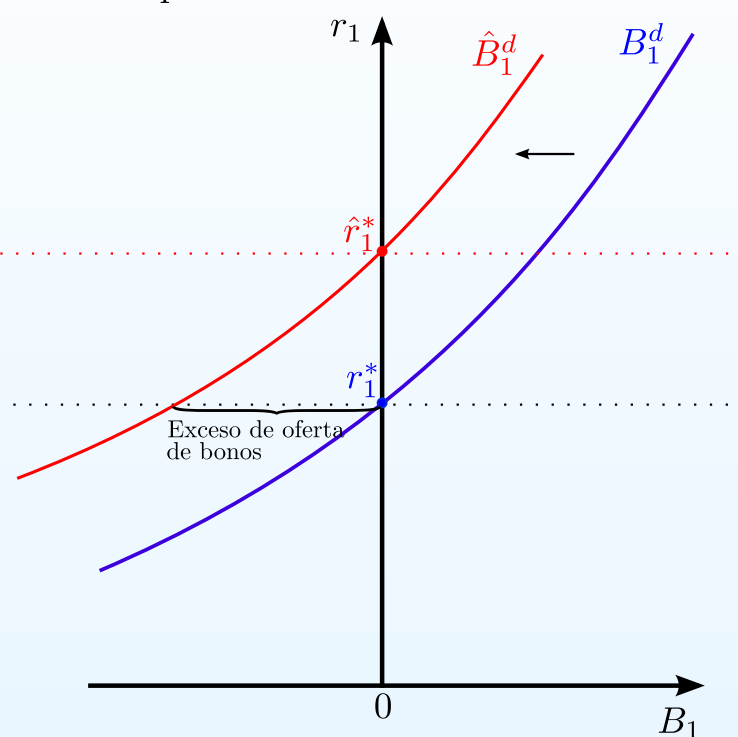
Suba esperada en la productividad: sube A_2 y A_1 no cambia

- *Desplazamientos de las curvas de oferta y demanda dada una tasa de interés r_1 :*
 - **Impacto:** Dado $\bar{l}_2$, $\uparrow y_2 = \uparrow A_2 f(\bar{l}_2)$
 - **ER positivo:** $\uparrow c_1$, $\uparrow c_2$, $\downarrow l_1$ y $\downarrow l_2$. La caída de l_1 implica $\downarrow y_1$.
 - **ES intratemporal:** no hay
 - **ES intertemporal del trabajo:** El aumento en A_2 implica $\downarrow l_1$, $\uparrow l_2$, por lo que $\downarrow y_1$.
- **Resumen:** C_1^d se traslada hacia la derecha e Y_1^s se traslada hacia la izquierda.

Suba esperada en la productividad: sube A_2 y A_1 no cambia



Equilibrio en el mercado de bonos



- En r_1^* : exceso de demanda de bienes y exceso de oferta de bonos. Sube r_1 para equilibrar los mercados.
- En la figura el producto no cambia.

Suba esperada en la productividad: sube A_2 y A_1 no cambia

- Caso general: el cambio del producto y del empleo es ambiguo.
- Si todos son iguales no hay ambigüedad: ni el trabajo ni el producto cambian en $t = 1$; solo sube r_1 .
- *Prueba:* en equilibrio se cumple

$$\frac{v'(1 - L_1)}{u'(C_1)} = A_1 f'(L_1)$$

$$C_1 = A_1 f(L_1).$$

- Ecuaciones determinan C_1 y L_1 (e Y_1) como función de A_1 (no depende de A_2).
- *Intuición:* como nadie puede endeudarse y las posibilidades de producción no cambiaron en $t = 1$, las cantidades de equilibrio tampoco cambiarán.

Macroeconomía Dinámica

Introducción: Parte 6

Capital, Inversión y Ciclos Reales

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Modelo de ciclos reales

- Hoy agregamos un componente de la demanda agregada: la inversión
- Extenderemos el modelo dinámico agregando capital en la función de producción.
- Estudiaremos los determinantes de la inversión.
- Como este modelo no contiene dinero, se lo conoce como el modelo de los ciclos reales.

La función de producción con capital y trabajo

- Bienes de consumo y de inversión se producen con trabajo y capital.
- Función de producción:

$$y_t = A_t f(k_{t-1}, l_t)$$

- k_{t-1} es elegido en $t - 1$ y está disponible para producir (instalado) en t .
- Capital inicial k_0 está dado.

La función de producción con capital y trabajo

- Productividad marginal positiva:

$$f_k(k_{t-1}, l_t) > 0 \quad f_l(k_{t-1}, l_t) > 0$$

- Ley de rendimientos decrecientes:

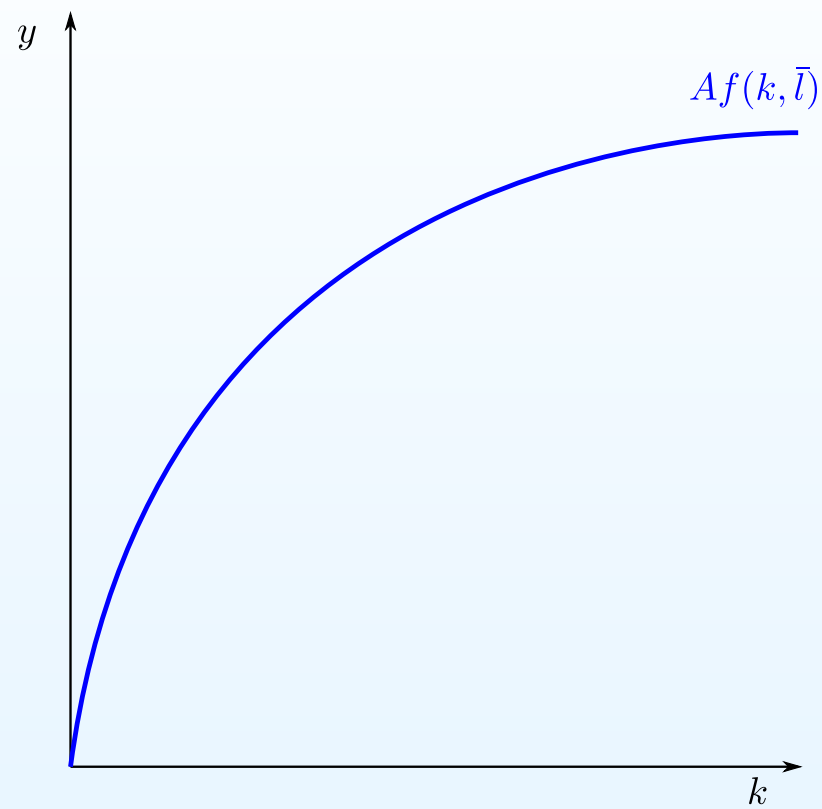
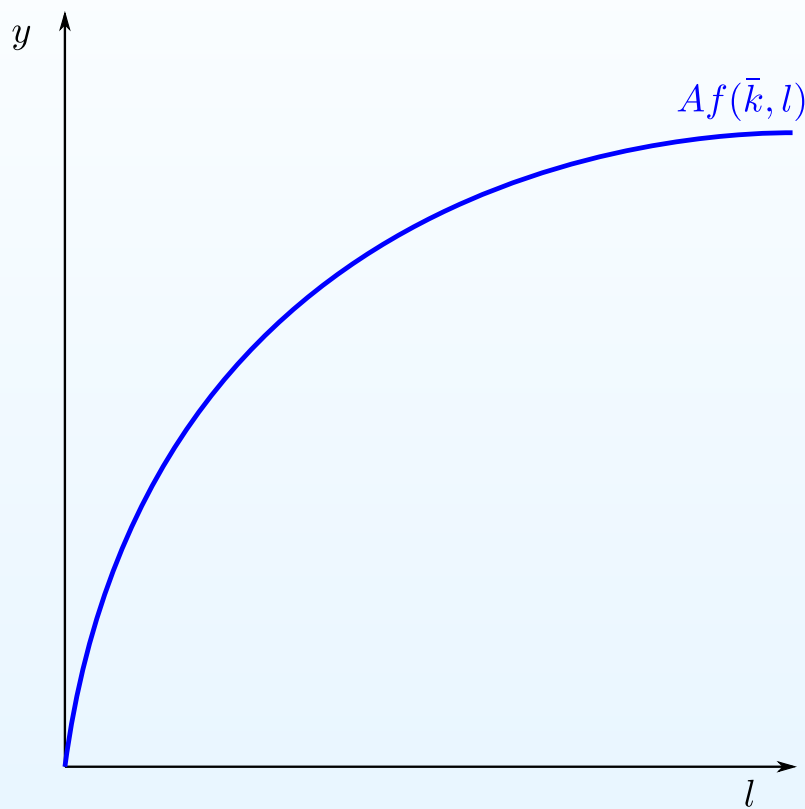
$$f_{kk}(k_{t-1}, l_t) < 0 \quad f_{ll}(k_{t-1}, l_t) < 0$$

- Además es usual suponer

$$f_{kl}(k_{t-1}, l_t) > 0.$$

- Se cumple necesariamente con retornos constantes a escala

La función de producción con capital y trabajo



La ley de acumulación de capital

- El capital se produce con inversión de acuerdo a

$$k_t = k_{t-1} + i_t - \delta k_{t-1}$$

o bien

$$k_t = (1 - \delta)k_{t-1} + i_t$$

- Podemos reescribir la ecuación anterior como

$$\underbrace{i_t}_{\text{inversión bruta}} = \underbrace{(k_t - k_{t-1})}_{\text{inversión neta}} + \underbrace{\delta k_{t-1}}_{\text{depreciación}}.$$

El problema del consumidor/productor

$$\max_{c_1, c_2, l_1, l_2, i_1, i_2, k_1, k_2, b_1, b_2} u(c_1) + v(1 - l_1) + \beta[u(c_2) + v(1 - l_2)]$$

sujeto a

$$c_1 + b_1 + i_1 = A_1 f(k_0, l_1) \quad (1)$$

$$c_2 + b_2 + i_2 = A_2 f(k_1, l_2) + (1 + r_1)b_1 \quad (2)$$

$$k_1 = (1 - \delta)k_0 + i_1 \quad (3)$$

$$k_2 = (1 - \delta)k_1 + i_2 \quad (4)$$

- Stock de capital inicial es k_0 .
- Stock inicial de bonos es cero, $b_0 = 0$.
- Consumidor elige morir sin activos $\Rightarrow b_2 = 0$ y $k_2 = 0$.

El problema del consumidor/productor

- Reducimos el número de restricciones:
- Reemplazamos (3) en (1), y (4) en (2):

$$\begin{aligned}c_1 + b_1 + k_1 &= A_1 f(k_0, l_1) + (1 - \delta)k_0 \\c_2 &= A_2 f(k_1, l_2) + (1 - \delta)k_1 + (1 + r_1)b_1.\end{aligned}$$

- Restricción presupuestaria intertemporal:

$$c_1 + \frac{c_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1}k_1 = A_1 f(k_0, l_1) + \frac{A_2 f(k_1, l_2)}{1 + r_1} + (1 - \delta)k_0.$$

El problema del consumidor/productor

Lagrangiano:

$$\begin{aligned} \max_{c_1, c_2, l_1, l_2, k_1} L = & u(c_1) + v(1 - l_1) + \beta[u(c_2) + v(1 - l_2)] \\ & - \lambda \left[c_1 + \frac{c_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} k_1 - A_1 f(k_0, l_1) - \frac{A_2 f(k_1, l_2)}{1 + r_1} - (1 - \delta)k_0 \right] \end{aligned}$$

Condiciones de primer orden

$$\frac{\partial L}{\partial c_1} = 0 \Rightarrow u'(c_1) = \lambda \quad (5)$$

$$\frac{\partial L}{\partial l_1} = 0 \Rightarrow v'(1 - l_1) = \lambda A_1 f_l(k_0, l_1) \quad (6)$$

$$\frac{\partial L}{\partial c_2} = 0 \Rightarrow \beta u'(c_2) = \lambda \frac{1}{1 + r_1} \quad (7)$$

$$\frac{\partial L}{\partial l_2} = 0 \Rightarrow \beta v'(1 - l_2) = \lambda \frac{A_2 f_l(k_1, l_2)}{1 + r_1} \quad (8)$$

$$\frac{\partial L}{\partial k_1} = 0 \Rightarrow r_1 = A_2 f_k(k_1, l_2) - \delta \quad (9)$$

más la restricción presupuestaria intertemporal.

El problema del consumidor/productor

- Condiciones (5)–(8) son las mismas que en el modelo sin capital.
- Ecuación (9) determina la elección óptima de capital.
- El retorno de un bono debe igualarse a la PMK neta de depreciación
- Intución: ausencia de arbitraje entre bonos y capital

El problema del consumidor/productor

$$\frac{v'(1 - l_1)}{u'(c_1)} = A_1 f_l(k_0, l_1)$$

$$\frac{v'(1 - l_2)}{u'(c_2)} = A_2 f_l(k_1, l_2)$$

$$u'(c_1) = (1 + r_1)\beta u'(c_2)$$

$$r_1 = A_2 f_k(k_1, l_2) - \delta$$

$$c_1 + \frac{c_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} k_1 = A_1 f(k_0, l_1) + \frac{A_2 f(k_1, l_2)}{1 + r_1} + (1 - \delta)k_0.$$

- 5 ecuaciones y 5 incógnitas: c_1 , l_1 , c_2 , l_2 y k_1 .
- Solución: demandas de consumo, c_1 y c_2 ; ofertas de trabajo l_1 y l_2 ; demanda de capital k_1 como funciones de (r_1, A_1, A_2)

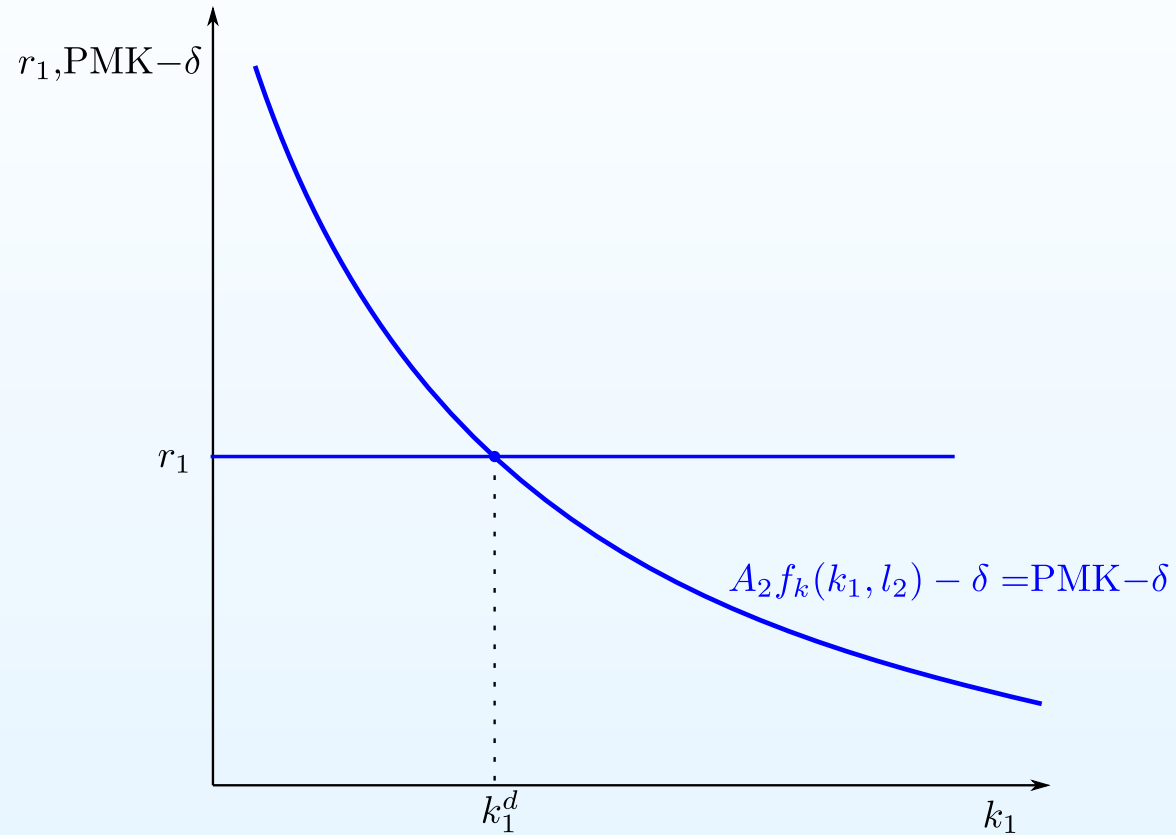
La demanda condicional de capital e inversión

- Demanda condicional de capital (condicional en l_2): valor del capital que satisface la CPO

$$r_1 = A_2 f_k(k_1, l_2) - \delta$$

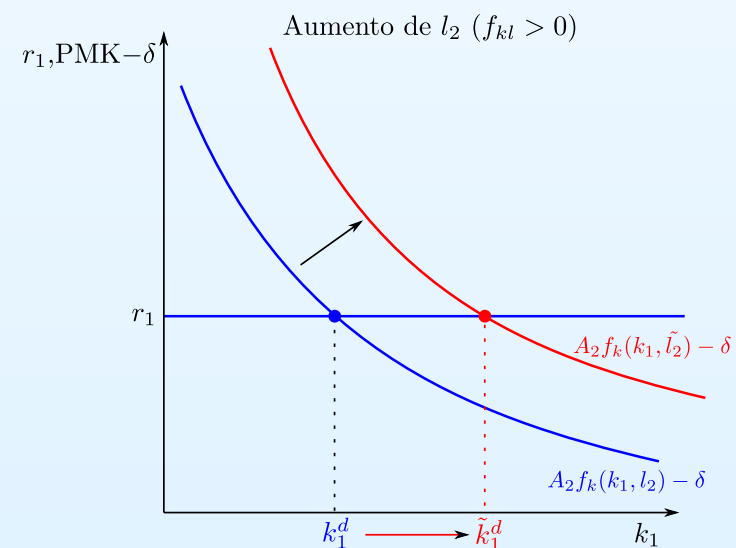
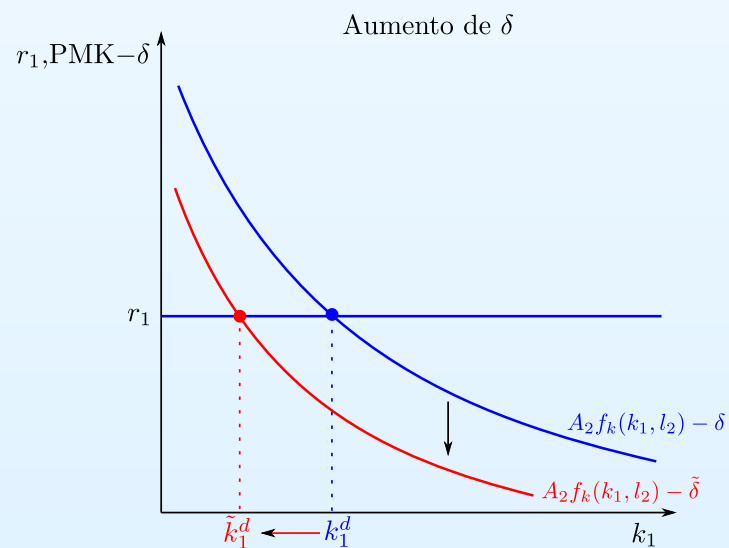
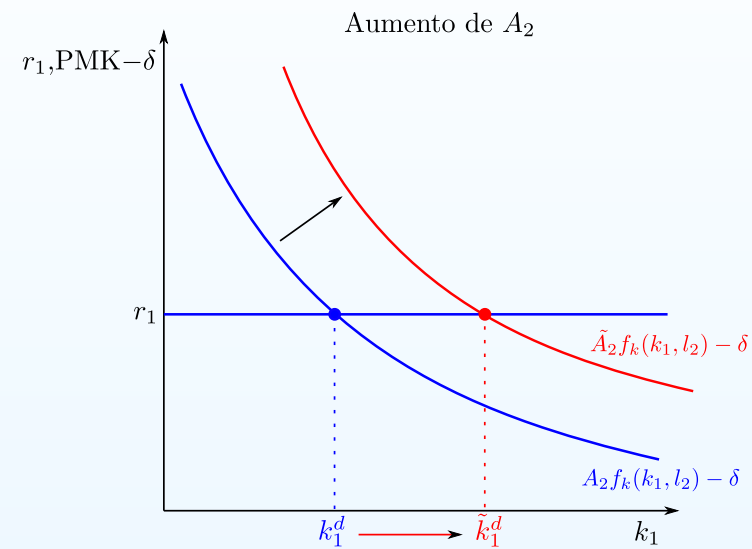
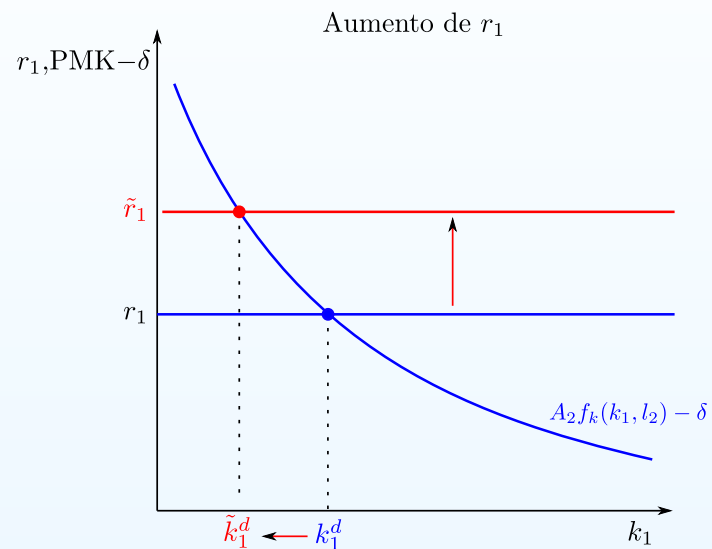
- Note que $A_2 f_k(k_1, l_2) - \delta$ es decreciente en k_1 debido a $f_{kk} < 0$
- Supondremos $f_{kl} > 0$: PMK aumenta cuando aumenta l_2 .

La demanda condicional de capital e inversión



- Demanda condicional de capital k_1^d es una función de r_1 , A_2 y l_2 .
- En particular, no depende de A_1 .

Las demandas condicionales de capital e inversión



Las demandas condicionales de capital e inversión

- Resumiendo,

$$k_1 = k_1^d(\bar{r}_1, \bar{A}_2, \bar{\delta}, \bar{l}_2)$$

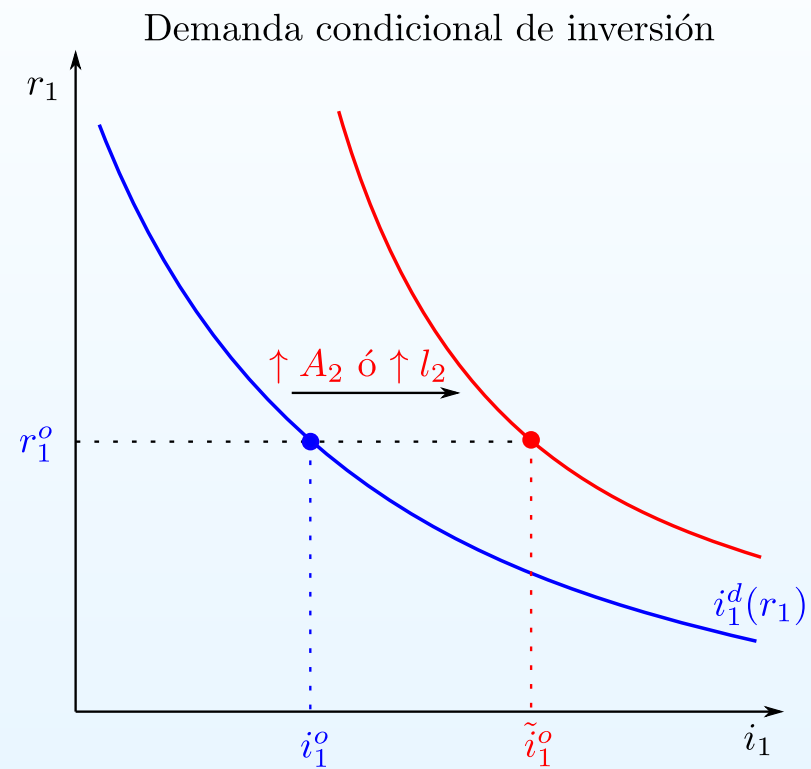
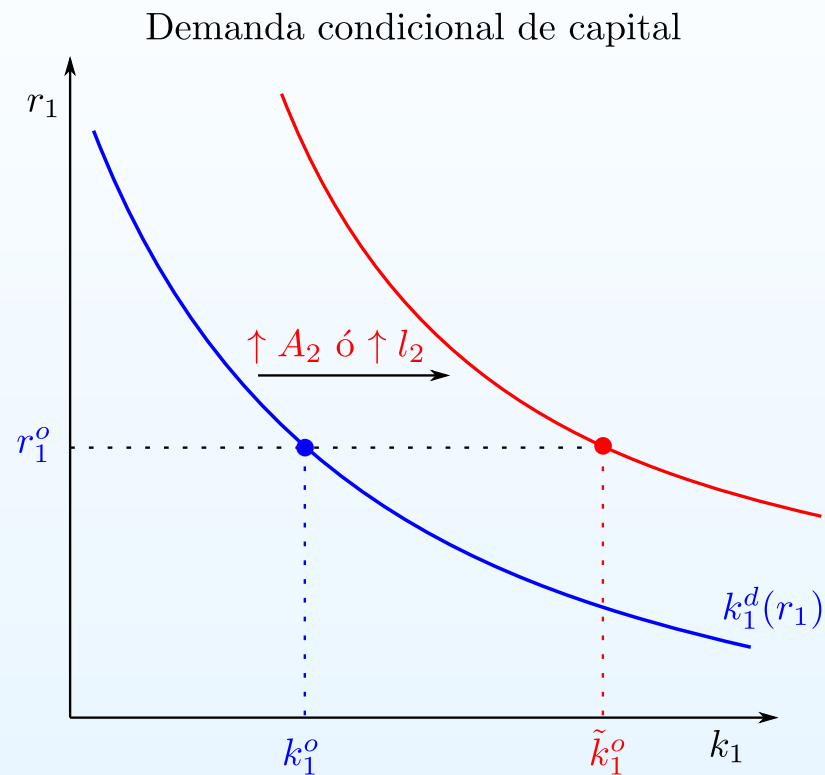
- Demanda condicional de inversión sale de

$$i_1 = k_1 - (1 - \delta)k_0$$

o bien

$$i_1 = i_1^d(\bar{r}_1, \bar{A}_2, \bar{l}_2, \bar{k}_0, \bar{\delta}) = k_1^d(r_1, A_2, \delta, l_2) - (1 - \delta)k_0$$

Las demandas condicionales de capital e inversión



- ¿Inversión en $t = 2$?

$$i_2 = k_2 - (1 - \delta)k_1 \Rightarrow k_2 = 0 \Rightarrow i_2 = -(1 - \delta)k_1.$$

Agregación, Equilibrio y Ley de Walras

- Sumo restricciones presupuestarias flujo de las familias

$$\begin{aligned}C_1^d + I_1^d + B_1^d &= Y_1^s \\C_2^d + I_2^d &= Y_2^s + (1 + r_1)B_1^d\end{aligned}$$

- Inversión satisface $I_1 = K_1 - (1 - \delta)K_0$ e $I_2 = -(1 - \delta)K_1$

- Demanda agregada:

$$Y_t^d = C_t^d + I_t^d$$

- Entonces,

$$\begin{aligned}(Y_1^d - Y_1^s) + B_1^d &= 0, \\(Y_2^d - Y_2^s) &= (1 + r_1)B_1^d.\end{aligned}$$

- De aquí deducimos la Ley de Walras.

Agregación, Equilibrio y Ley de Walras

- Equilibrio en el mercado de bienes:

$$Y_1^d = Y_1^s \quad \text{en } t = 1$$

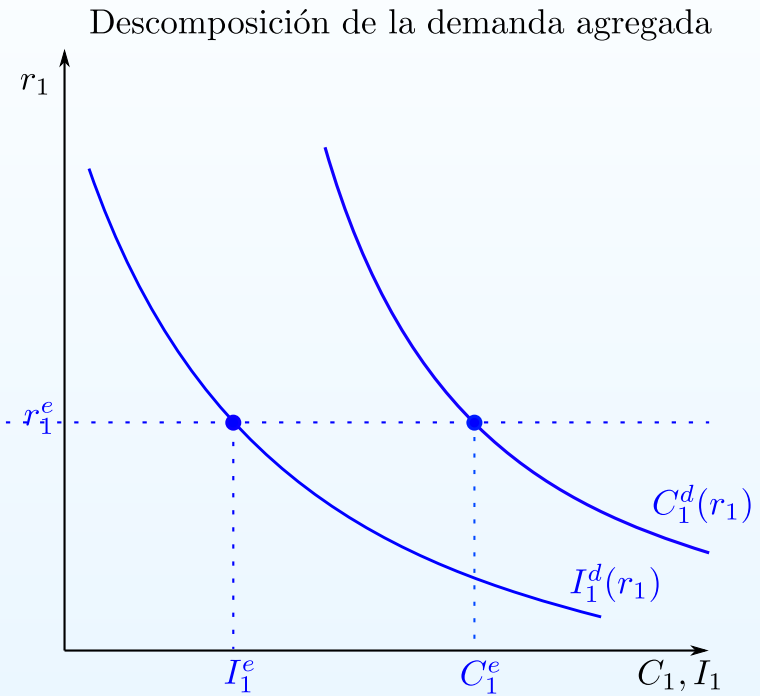
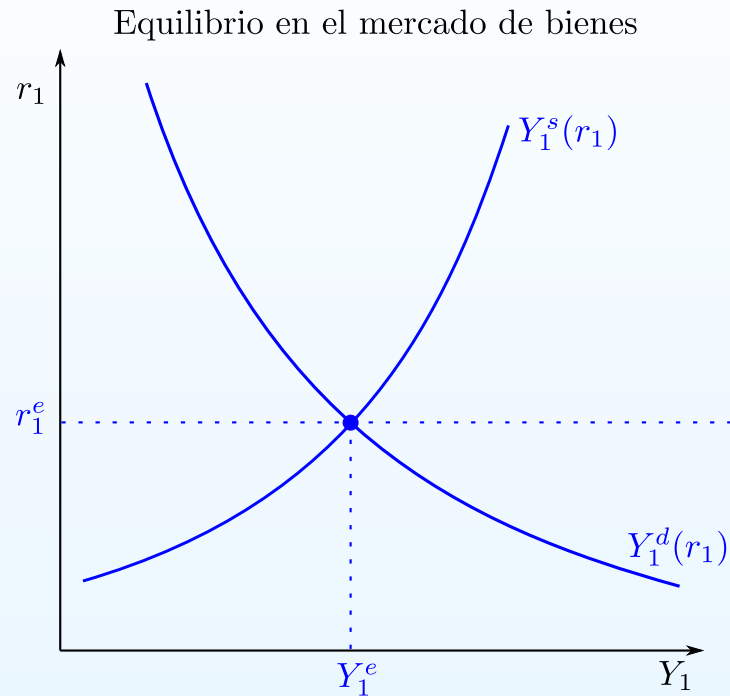
$$Y_2^d = Y_2^s \quad \text{en } t = 2$$

- Equilibrio en el mercado de bonos:

$$B_1^d = 0.$$

- Por Ley de Walras nos enfocamos en el mercado de bienes de $t = 1$.

Equilibrio en el mercado de bienes



- Demanda agregada: demanda de consumo más demanda de inversión,

$$Y_1^d(r_1, \dots) = C_1^d(r_1, \dots) + I_1^d(r_1, \dots)$$

Estática comparativa

- Equilibrio general:
 - Encontramos el precio de equilibrio r_1^e y las cantidades de equilibrio Y_1^e , C_1^e e I_1^e .
 - Ley de Walras garantiza que el mercado de bonos y el de bienes mañana también estarán en equilibrio.
- Analizaremos cómo cambia la tasa de interés y las cantidades de equilibrio ante cambios en las productividades A_1 y A_2 .
- Cambios temporarios, permanentes y esperados de la productividad.

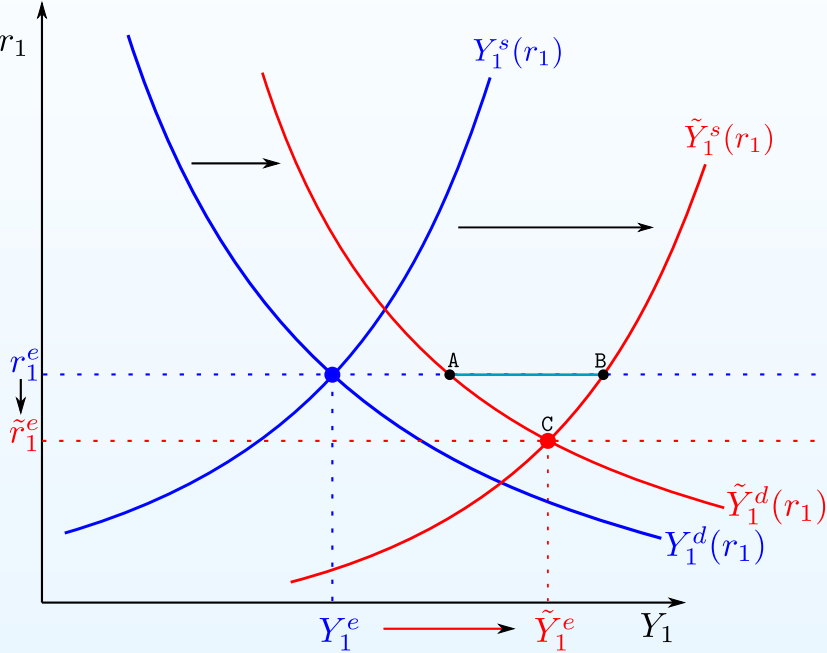
Suba temporaria de la productividad: sube A_1 y A_2 no cambia

Desplazamientos de las curvas dada una tasa de interés r_1 ,

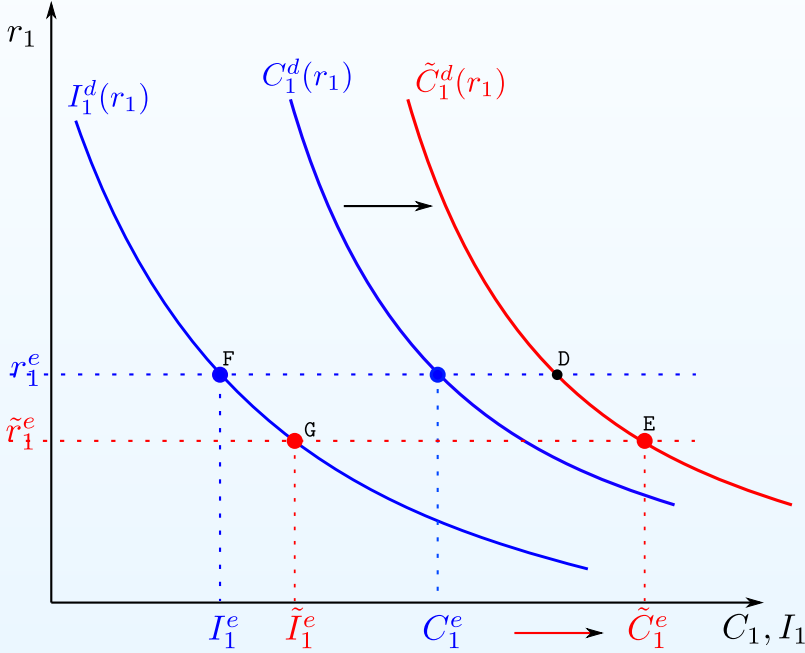
1. **Impacto:** Dados $\bar{L}_1$ y K_0 , $\uparrow Y_1 = \uparrow A_1 f(K_0, \bar{L}_1) \Rightarrow$ efecto riqueza positivo.
2. **ER positivo:** $\uparrow C_1, \downarrow L_1 \Rightarrow \downarrow Y_1$ (también $\downarrow L_2 \Rightarrow \downarrow Y_2$). Pero la caída de L_1 no domina efecto directo de A_1 .
3. **ES intratemporal en el período 1:** $\uparrow C_1, \uparrow L_1 \Rightarrow \uparrow Y_1$.
4. **ES intertemporal del trabajo:** $(\uparrow A_1/A_2)$ implica $\uparrow L_1 \Rightarrow \uparrow Y_1$ (también $\downarrow L_2 \Rightarrow \downarrow Y_2$)
5. **Demanda de inversión:** $\uparrow A_1$ no afecta PMK de mañana. Ignorando el cambio de L_2 , entonces $I_1^d(r)$ no cambia.
 - $\Delta Y_1^s(r_1) > 0$
 - $\Delta C_1^d(r_1) > 0$. Pero shock transitorio: $0 < \Delta C_1^d(r_1) < \Delta Y_1^s(r_1)$.
 - $\Delta I^d(r_1) \approx 0$ (si el impacto de L_2 en I_1 es pequeño)
 - Entonces, $0 < \Delta Y_1^d(r_1) < \Delta Y_1^s(r_1)$.

Suba temporaria de la productividad: sube A_1 y A_2 no cambia

Equilibrio en el mercado de bienes



Descomposición de la demanda agregada



Suba temporaria de la productividad: sube A_1 y A_2 no cambia

- Exceso de oferta de bienes: $\downarrow r_1 \implies \uparrow C_1, \uparrow I_1$ y $\downarrow Y_1$ eliminando el exceso de oferta.
- Mecanismo que opera:
 1. **ES intertemporal del consumo:** $\downarrow r_1^e$ implica $\uparrow C_1$ (y $\downarrow C_2$). Movimiento sobre la curva de demanda de consumo desde D hasta E.
 2. **Cae del retorno de los bonos:** $\downarrow r_1^e \implies$ aumenta la inversión en capital fijo. Movimiento sobre la curva de demanda de inversión desde F hasta G.
 3. **ES intertemporal del trabajo:** $\downarrow r_1^e$ implica $\downarrow L_1$ (y $\uparrow L_2$). $\downarrow L_1 \implies \downarrow Y_1$. Movimiento sobre la curva de oferta de bienes desde B hasta C.

Suba temporaria de la productividad: sube A_1 y A_2 no cambia

Conclusión:

- Aumenta Y_1
- Aumenta C_1 : consumo es procíclico como en los datos
- Aumenta I_1 : inversión es procíclica como en los datos
- Cae r_1 : tasa de interés es contracíclica (en los datos es levemente procíclica).

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción

Desplazamientos de las curvas dada una tasa de interés r_1

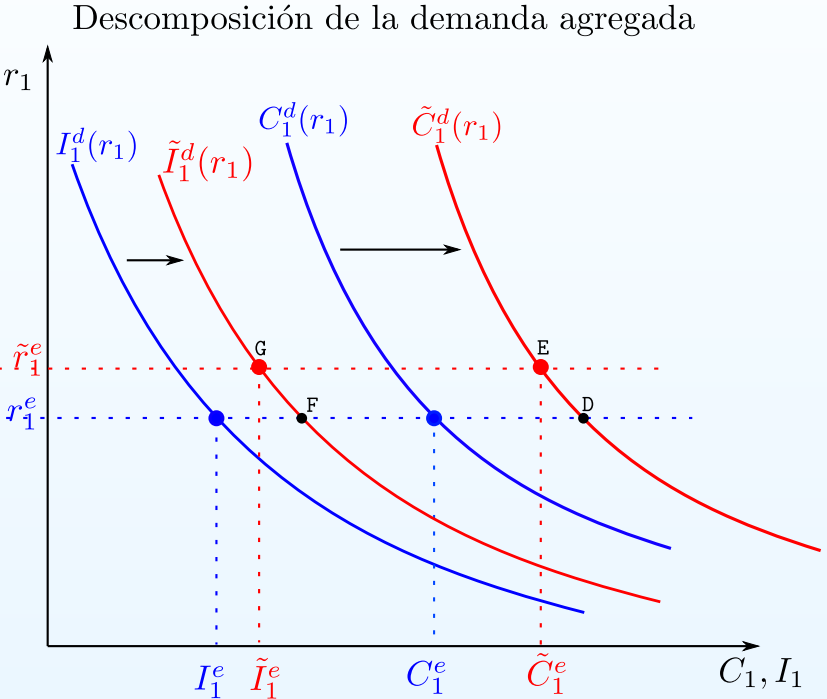
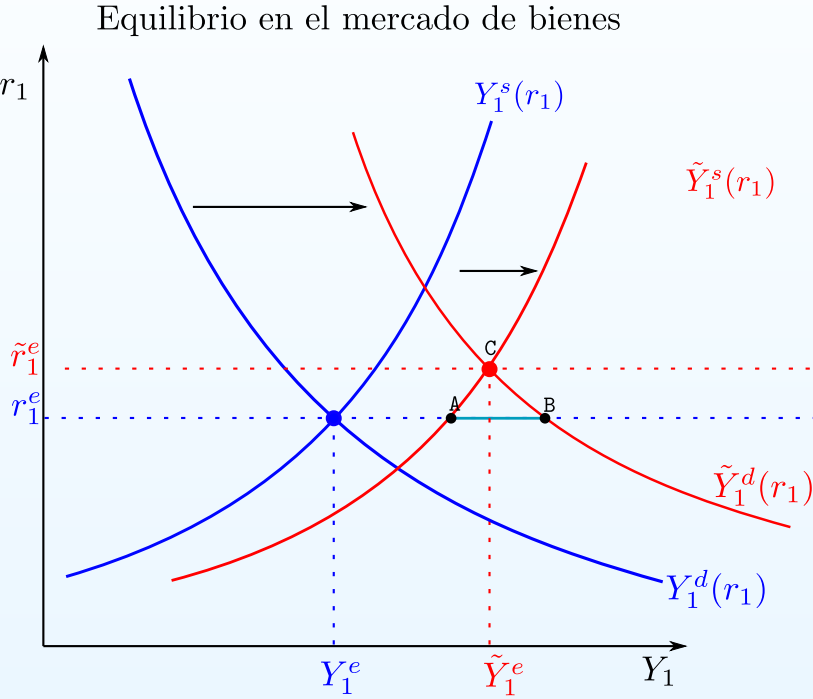
1. **Impacto:** Dados $\bar{L}_1, \bar{L}_2, K_0$ y K_1 , $\uparrow Y_1 = \uparrow A_1 f(K_0, \bar{L}_1)$ y $\uparrow Y_2 = \uparrow A_2 f(K_1, \bar{L}_2)$. Efecto riqueza positivo—más fuerte que en el caso anterior.
2. **ER positivo:** $\uparrow C_1, \downarrow L_1 \Rightarrow \downarrow Y_1$ (también $\downarrow L_2 \Rightarrow \downarrow Y_2$). Pero la caída de L_1 no domina efecto directo de la mayor productividad.
3. **ES intratemporal en el período 1:** $\uparrow C_1, \uparrow L_1 \Rightarrow \uparrow Y_1$.
4. **ES intratemporal en el período 2:** $\uparrow C_2, \uparrow L_2 \Rightarrow \uparrow Y_2$.
5. **ES intertemporal del trabajo:** No hay (A_2/A_1 no cambia).
6. **Demanda de inversión:** $\uparrow PMK_2 = \uparrow A_2 f_k(K_1, L_2) \Rightarrow \uparrow I_1$.

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción

Por lo tanto,

- $\Delta Y_1^s(r_1) > 0$
- $\Delta C_1^d(r_1) > 0$. Shock permanente implica $\Delta C_1^d(r_1) \approx \Delta Y_1^s(r_1)$.
- $\Delta I^d(r_1) > 0$.
- Entonces, $\Delta Y^d(r_1) > \Delta Y^s(r_1) > 0$

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción



Suba permanente de la productividad: suben A_1 y A_2 en igual proporción

- Exceso de demanda de bienes: $\uparrow r_1 \implies \downarrow C_1, \downarrow I_1$ y $\uparrow Y_1$ eliminando el exceso de demanda.
- Mecanismo que opera:
 1. **ES intertemporal del consumo:** $\uparrow r_1^e$ implica $\downarrow C_1$. Movimiento *sobre la curva de demanda de consumo* desde D hasta E. El nuevo nivel de consumo será mayor al original.
 2. **ES intertemporal del trabajo:** $\uparrow r_1^e$ implica $\uparrow L_1 \implies \uparrow Y_1$. Movimiento *sobre la curva de oferta de bienes* desde A hasta C.
 3. **Sube del retorno de los bonos:** $\uparrow r_1^e \implies$ aumenta retorno de ahorrar en bonos. Cae la inversión en capital fijo. Movimiento *sobre la curva de demanda de inversión* desde F hasta G.

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción

Efecto final sobre la inversión: ¿sube o baja?

- a. Sube por el aumento de la productividad futura del capital
- b. Cae por el aumento de la tasa de interés real de equilibrio.

¿Qué efecto domina?

1. Como $\tilde{Y}_1^e > Y_1^e$, entonces debe ser cierto que $C_1 + I_1$ sube,

$$0 < \Delta Y_1^e = \Delta C_1^e + \Delta I_1^e. \quad (10)$$

2. ¿Cómo se distribuye la suba del producto entre consumo e inversión?

a) Cambio permanente de la productividad implica $\Delta C_1(r_1) \approx \Delta Y_1(r_1)$ (para cada r_1)

b) Pero $\uparrow r_1^e \Rightarrow \downarrow C_1$ y $\uparrow Y_1$. Por lo tanto $\Delta C_1^e < \Delta Y_1^e$.

3. Por lo tanto (10) implica $\Delta I_1^e > 0$.

Suba permanente de la productividad: suben A_1 y A_2 en igual proporción

Conclusión:

- Aumenta Y_1
- Aumenta C_1 : consumo es procíclico como en los datos
- Aumenta I_1 : inversión es procíclica como en los datos
- Aumenta r_1 : tasa de interés es procíclica (en los datos es levemente procíclica).

Observaciones

■ *Tasas de interés:*

- Shock temporario: tasa de interés contracíclica
- Shock permanente: tasa de interés es procíclica.

■ *Nueva herramienta de ahorro:*

- Modelo sin capital: ahorro agregado es siempre cero.
- Modelo con capital: familias pueden ahorrar en capital fijo

■ *¿Mayor o menor inversión con la persistencia del shock?:*

- Shock permanente: invertir hoy porque productividad del capital futura es alta
- Shock temporario: invertir para suavizar consumo (con mayor producción futura)

Suba esperada de la productividad: A_1 no cambia y sube A_2

Desplazamientos de las curvas dada una tasa de interés r_1

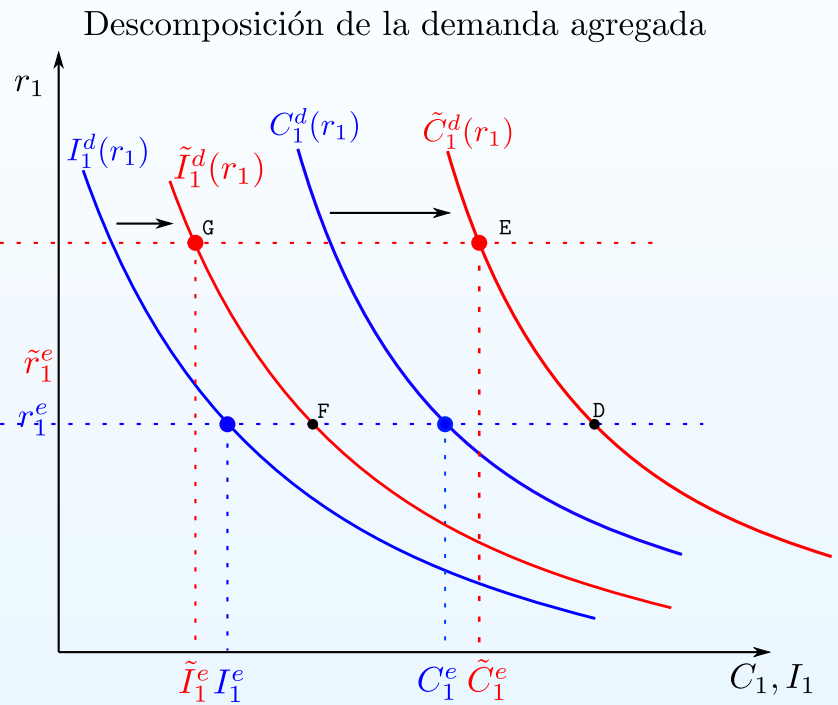
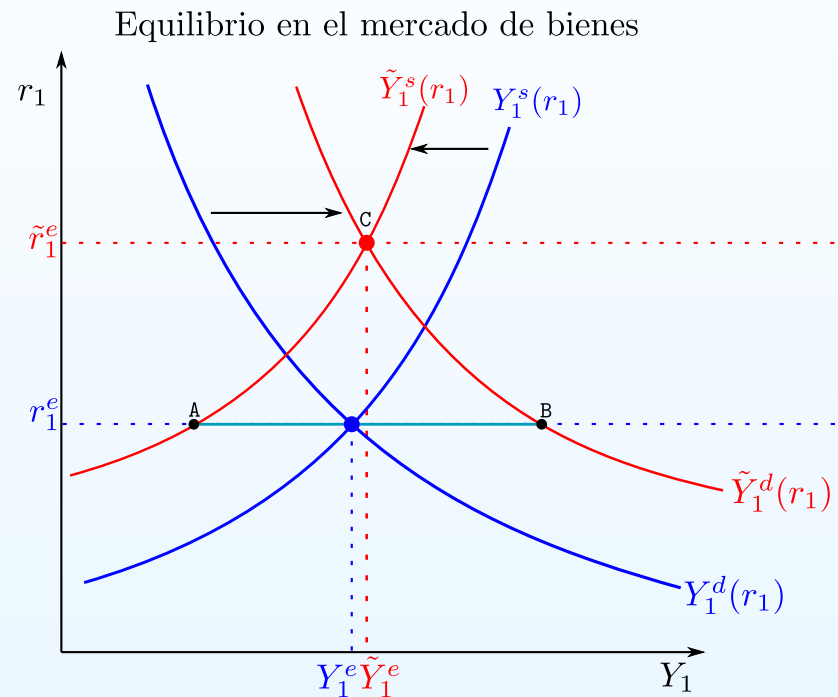
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2. **ER positivo:** $\uparrow C_1, \downarrow L_1 \Rightarrow \downarrow Y_1$ (también $\uparrow C_2, \downarrow L_2 \Rightarrow \downarrow Y_2$).
3. **ES intratemporal en el período 1:** No hay
4. **ES intratemporal en el período 2:** $\uparrow C_2, \uparrow L_2 \Rightarrow \uparrow Y_2$.
5. **ES intertemporal del trabajo:** Cae A_1/A_2 , por lo que el ocio futuro se hace más caro que el ocio presente: $\downarrow L_1 \Rightarrow \downarrow Y_1$ y $\uparrow L_2$.
6. **Demanda de inversión:** $\uparrow PMK_2 = \uparrow A_2 f_k(K_1, L_2) \Rightarrow \uparrow I_1$.

Suba esperada de la productividad: A_1 no cambia y sube A_2

Por lo tanto,

- $\Delta Y_1^s(r_1) < 0$
- $\Delta C_1^d(r_1) > 0$.
- $\Delta I^d(r_1) > 0$.
- Entonces, $\Delta Y^d(r_1) > 0$

Suba esperada de la productividad: A_1 no cambia y sube A_2



Suba esperada de la productividad: A_1 no cambia y sube A_2

- Exceso de demanda de bienes: $\uparrow r_1 \implies \downarrow C_1, \downarrow I_1$ y $\uparrow Y_1$ eliminando el exceso de demanda.
- Mecanismo que opera:
 1. **ES intertemporal del consumo:** $\uparrow r_1^e$ implica $\downarrow C_1$. Movimiento sobre la curva de demanda de consumo desde D hasta E.
 2. **ES intertemporal del trabajo:** $\uparrow r_1^e$ implica $\uparrow L_1 \implies \uparrow Y_1$. Movimiento sobre la curva de oferta de bienes desde A hasta C.
 3. **Sube del retorno de los bonos:** $\uparrow r_1^e \implies$ aumenta retorno de ahorrar en bonos. Cae la inversión en capital fijo. Movimiento sobre la curva de demanda de inversión desde F hasta G.
- En resumen, sube la tasa de interés de equilibrio y hay un efecto ambiguo sobre las cantidades del primer período.

Suba esperada de la productividad: A_1 no cambia y sube A_2

- Sin embargo, podemos demostrar lo siguiente en $t = 1$: independientemente de si Y_1^e sube o baja, inversión y trabajo son procíclicos y consumo es contracíclico.
- Como lo vemos? De la condición intratemporal

$$\frac{v'(1 - L_1)}{u'(C_1)} = A_1 f_l(K_0, L_1) \Rightarrow \frac{v'(1 - L_1)}{A_1 f_l(K_0, L_1)} = u'(C_1).$$

Por lo tanto, si L_1^e sube, C_1^e baja y si L_1^e baja, C_1^e sube.

- Pero si L_1^e sube, entonces Y_1^e sube. De aquí sale que consumo es contracíclico y trabajo es procíclico.
- De consistencia agregada en $t = 1$ sabemos

$$I_1 = Y_1 - C_1.$$

Por lo tanto, si Y_1^e sube, $Y_1^e - C_1^e$ sube, y I_1^e sube. De aquí sale que la inversión es procíclica.

Macroeconomía Dinámica

Introducción: Parte 7a

Sector Público

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Introducción

- Hoy introducimos el sector público en el modelo de equilibrio general.
- El gobierno consume bienes y cobra impuestos.
- Los impuestos son de *suma fija*.
 - ¿Son de suma fija el IVA o el impuesto a las ganancias?
- *Equivalencia Ricardiana*: resultado de irrelevancia del perfil temporal de los impuestos
- Consumidor representativo

Sector público

- Gobierno consume G_1 y G_2 , cobra impuestos T_1 y T_2 , y puede ahorrar en activos B_1^g en el periodo 1.
- Ingreso corriente del gobierno: impuestos T_1 y T_2 .
- Restricciones presupuestarias flujo del sector público:

$$G_1 + B_1^g = T_1 \quad (1)$$

$$G_2 = T_2 + B_1^g (1 + r_1) \quad (2)$$

- Restricción presupuestaria intertemporal del gobierno

$$G_1 + \frac{G_2}{1 + r_1} = T_1 + \frac{T_2}{1 + r_1}$$

Hogares

- Función de utilidad

$$\begin{aligned} U &= u(c_1) + v(1 - l_1) + w(g_1) + \beta [u(c_2) + v(1 - l_2) + w(g_2)] \\ &= u(c_1) + v(1 - l_1) + \beta [u(c_2) + v(1 - l_2)] + w(g_1) + \beta w(g_2) \end{aligned}$$

- Gasto público puede generar utilidad, pero ingresa de manera separable.

- $w(g_1) + \beta w(g_2)$ es una constante: la podemos ignorar en el problema de decisión de los hogares

- Este término puede ser importante si analizamos la elección óptima de gasto público. Pero eso lo estudiarán en Finanzas Públicas

Hogares

- Problema de los hogares

$$\max_{c_1, l_1, c_2, l_2, b_1^p, i_1, i_2, k_1} u(c_1) + v(1 - l_1) + \beta [u(c_2) + v(1 - l_2)] \quad \text{sujeto a}$$

$$c_1 + i_1 + b_1^p = A_1 f(k_0, l_1) - T_1 \quad (3)$$

$$c_2 + i_2 = A_2 f(k_1, l_2) - T_2 + b_1^p (1 + r_1) \quad (4)$$

$$k_1 = (1 - \delta)k_0 + i_1 \quad (5)$$

$$i_2 = -(1 - \delta)k_1.$$

- Restricción presupuestaria intertemporal:

$$c_1 + \frac{c_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} k_1 = A_1 f(k_0, l_1) + \frac{A_2 f(k_1, l_2)}{1 + r_1} + (1 - \delta)k_0 - \left(T_1 + \frac{T_2}{1 + r_1} \right)$$

Hogares

Lagrangiano

$$L = u(c_1) + v(1 - l_1) + \beta[u(c_2) + v(1 - l_2)] \\ - \lambda \left[c_1 + \frac{c_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} k_1 - A_1 f(k_0, l_1) - \frac{A_2 f(k_1, l_2)}{1 + r_1} - (1 - \delta)k_0 + \left(T_1 + \frac{T_2}{1 + r_1} \right) \right]$$

Condiciones de primer orden son iguales a las del modelo sin sector público:

$$\frac{\partial L}{\partial c_1} = 0 \Rightarrow u'(c_1) = \lambda \quad (6)$$

$$\frac{\partial L}{\partial l_1} = 0 \Rightarrow v'(1 - l_1) = \lambda A_1 f_l(k_0, l_1) \quad (7)$$

$$\frac{\partial L}{\partial c_2} = 0 \Rightarrow \beta u'(c_2) = \lambda \frac{1}{1 + r_1} \quad (8)$$

$$\frac{\partial L}{\partial l_2} = 0 \Rightarrow \beta v'(1 - l_2) = \lambda \frac{A_2 f_l(k_1, l_2)}{1 + r_1} \quad (9)$$

$$\frac{\partial L}{\partial k_1} = 0 \Rightarrow r_1 = A_2 f_k(k_1, l_2) - \delta \quad (10)$$

Hogares

Condiciones de optimalidad:

$$\frac{v'(1 - l_1)}{u'(c_1)} = A_1 f_l(k_0, l_1)$$

$$\frac{v'(1 - l_2)}{u'(c_2)} = A_2 f_l(k_1, l_2)$$

$$u'(c_1) = (1 + r_1)\beta u'(c_2)$$

$$r_1 = A_2 f_k(k_1, l_2) - \delta$$

$$c_1 + \frac{c_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} k_1 = A_1 f(k_0, l_1) + \frac{A_2 f(k_1, l_2)}{1 + r_1} + (1 - \delta)k_0 - \left(T_1 + \frac{T_2}{1 + r_1}\right)$$

Además, la ecuación de Euler del trabajo

$$v'(1 - l_1) = \frac{A_1 f_l(k_0, l_1)}{A_2 f_l(k_1, l_2)} (1 + r_1) \beta v'(1 - l_2)$$

Agregación

- Restricción presupuestaria intertemporal del sector privado

$$C_1 + \frac{C_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} K_1 = A_1 f(K_0, L_1) + \frac{A_2 f(K_1, L_2)}{1 + r_1} + (1 - \delta) K_0 - \left(T_1 + \frac{T_2}{1 + r_1} \right)$$

- Restricción presupuestaria intertemporal del gobierno:

$$G_1 + \frac{G_2}{1 + r_1} = T_1 + \frac{T_2}{1 + r_1}$$

- Restricción presupuestaria intertemporal agregada:

$$C_1 + \frac{C_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} K_1 = A_1 f(K_0, L_1) + \frac{A_2 f(K_1, L_2)}{1 + r_1} + (1 - \delta) K_0 - \left(G_1 + \frac{G_2}{1 + r_1} \right).$$

Agregación

- Evaluamos las CPO en las decisiones del consumidor representativo:

$$\frac{v'(1 - L_1)}{u'(C_1)} = A_1 f_l(K_0, L_1)$$

$$\frac{v'(1 - L_2)}{u'(C_2)} = A_2 f_l(K_1, L_2)$$

$$u'(C_1) = (1 + r_1)\beta u'(C_2)$$

$$r_1 = A_2 f_k(K_1, L_2) - \delta$$

$$C_1 + \frac{C_2}{1 + r_1} + \frac{r_1 + \delta}{1 + r_1} K_1 = A_1 f(K_0, L_1) + \frac{A_2 f(K_1, L_2)}{1 + r_1} + (1 - \delta)K_0 - \left(G_1 + \frac{G_2}{1 + r_1} \right)$$

Agregación

- Solución: demandas y ofertas agregadas de la economía

$$C_1 = C_1^d \left(r_1, A_1, A_2, G_1 + \frac{G_2}{1 + r_1} \right)$$

$$C_2 = C_2^d \left(r_1, A_1, A_2, G_1 + \frac{G_2}{1 + r_1} \right)$$

$$I_1 = I_1^d(r_1, A_2, L_2)$$

$$L_1 = L_1^s \left(r_1, A_1, A_2, G_1 + \frac{G_2}{1 + r_1} \right)$$

$$L_2 = L_2^s \left(r_1, A_1, A_2, G_1 + \frac{G_2}{1 + r_1} \right)$$

$$Y_1 = Y_1^s \left(r_1, A_1, A_2, G_1 + \frac{G_2}{1 + r_1} \right)$$

$$Y_2 = Y_2^s \left(r_1, A_1, A_2, G_1 + \frac{G_2}{1 + r_1} \right)$$

Importante

1. *Restricción presupuestaria intertemporal implica que oferta agregada y demanda de consumo dependen del valor presente descontado del gasto público, $G_1 + \frac{G_2}{1+r_1}$, y no de la distribución particular de G_1 y G_2 .*
2. *Reinterpreto el modelo como sigue:*

- a) Consumidor cuyo ingreso en el periodo t viene dado por

$$\begin{aligned}\hat{Y}_t &= Y_t^s - G_t \\ &= A_t F(K_{t-1}, L_t) - G_t\end{aligned}$$

- b) $\hat{Y}_t$ es el ingreso disponible del sector privado neto del gasto público
- c) Puedo interpretar shocks al gasto público como shocks al nivel de la producción.

Condiciones de consistencia agregada

- Mercado de bienes en $t = 1$

$$Y_1^d = Y_1^s$$

- Mercado de bonos en $t = 1$

$$B_1^p + B_1^g = 0$$

- Mercado de bienes en $t = 2$

$$Y_2^d = Y_2^s$$

Ley de Walras

- Restricciones presupuestarias flujo en $t = 1$ de los hogares y del gobierno:

$$\begin{aligned}C_1 + I_1 + B_1^p &= Y_1 - T_1 \\G_1 + B_1^g &= T_1\end{aligned}$$

- Sumo,

$$\underbrace{(C_1^d + I_1^d + G_1^d - Y_1^s)}_{=Y_1^d(r_1)} + (B_1^p + B_1^g) = 0$$

o bien

$$(Y_1^d - Y_1^s) + B_1^d = 0 \tag{11}$$

- Si el mercado de bienes está en equilibrio, el mercado de bonos está en equilibrio y viceversa.

Ley de Walras

- Restricciones presupuestarias flujo en $t = 2$ de los hogares y del gobierno:

$$\begin{aligned}C_2 + I_2 &= Y_2 - T_2 + B_1^p(1 + r_1) \\G_2 &= T_2 + B_1^g(1 + r_1)\end{aligned}$$

- Sumo,

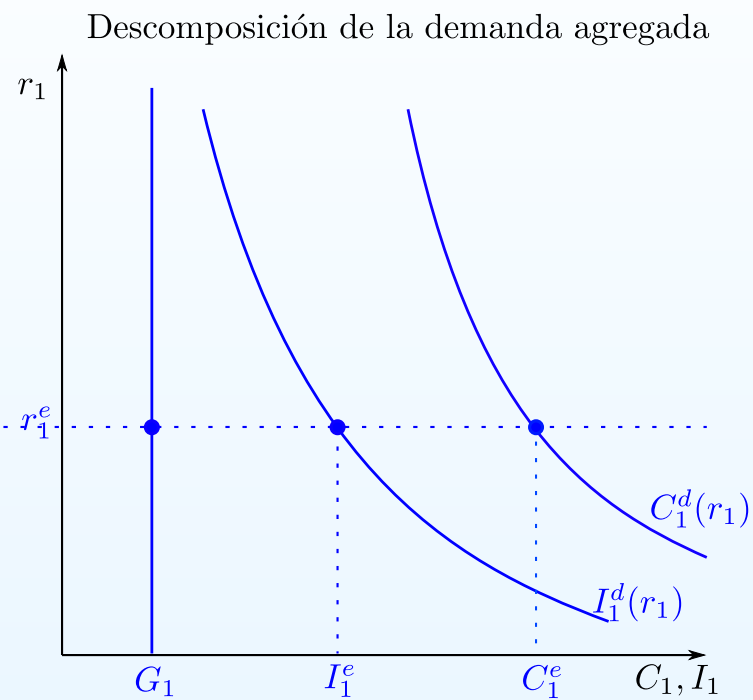
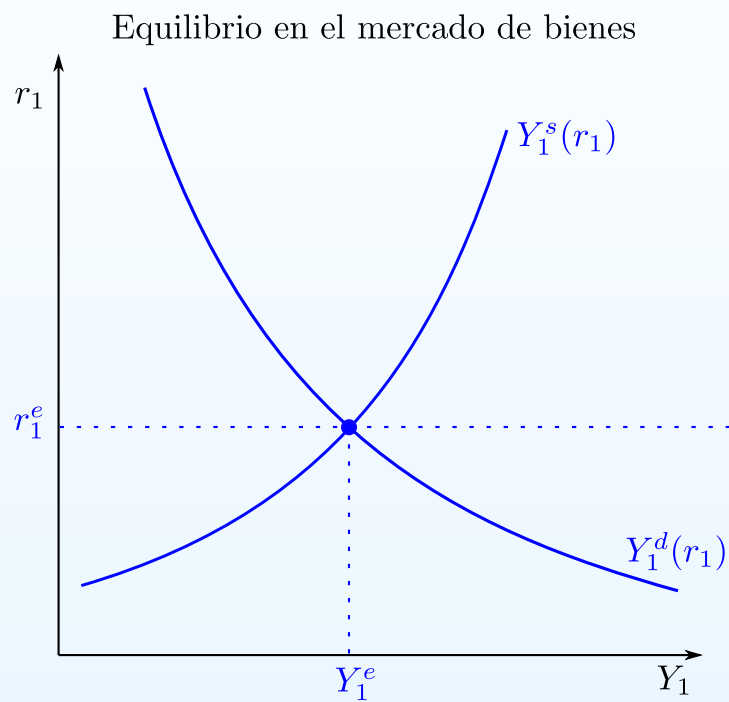
$$\underbrace{(C_2^d + I_2^d + G_2^d - Y_2^s)}_{=Y_2^d(r_1)} = (B_1^g + B_1^p)(1 + r_1)$$

o bien

$$(Y_2^d - Y_2^s) = B_1^d(1 + r_1) \tag{12}$$

- Si el mercado de bonos en $t = 1$ estuvo en equilibrio, el mercado de bienes en $t = 2$ estará en equilibrio

Equilibrio en el mercado de bienes



Equivalencia Ricardiana

Ejemplo: Dados G_1 y G_2 considere una disminución de T_1 compensada por una suba de T_2 .

Oferta agregada $Y_1^s(r_1)$:

- No cambia el valor descontado del gasto público ni las productividades $\Rightarrow L_1^s$ no cambia, $\Delta L_1^s(r_1) = 0$.
- Por lo tanto, tampoco cambia la curva de oferta agregada $\Rightarrow \Delta Y_1^s(r_1) = 0$.

Demanda agregada $Y_1^d(r_1) = C_1^d(r_1) + I_1^d(r_1) + G_1$:

- $\Delta C_1^d(r_1) = 0$ porque A_1 , A_2 , G_1 y G_2 no cambian.
- $\Delta I_1^d(r_1) = 0$ porque ni A_2 ni L_2 cambian
- $\Delta G_1 = 0$
- Entonces la demanda agregada no cambia $\Delta Y_1^d(r_1) = 0$

Por lo tanto, ni la tasa de interés ni las cantidades de equilibrio cambian cuando bajan los impuestos T_1 .

Equivalencia Ricardiana

- Pero sí cambia la distribución de activos entre el sector público y el sector privado:
- Restricción presupuestaria flujo del gobierno (y $\Delta G_1 = 0$)

$$\Delta G_1 + \Delta B_1^g = \Delta T_1 \Rightarrow \Delta B_1^g = \Delta T_1 < 0.$$

El gobierno se está endeudando

- Equilibrio mercado de bonos

$$\Delta B_1^p + \Delta B_1^g = 0 \Rightarrow \Delta B_1^p = -\Delta B_1^g > 0$$

- Sector privado compra los bonos emitidos por el sector público.

Equivalencia Ricardiana

La proposición general:

Equivalencia Ricardiana: *Suponga un cambio en la estructura temporal de los impuestos de suma fija tal que el valor presente de los impuestos no cambia. El resto de la economía es idéntico. Entonces, ni los precios ni la asignación de consumo, trabajo, inversión y producción de equilibrio se modificarán. El único cambio que habrá es una redistribución de activos entre el sector público y el sector privado.*

Comentarios:

1. Supuestos clave para obtener la Equivalencia Ricardiana:
 - a) Existen mercados de crédito donde hogares pueden ahorrar o endeudarse
 - b) Consumidores miran el futuro al tomar sus decisiones e internalizan la restricción presupuestaria del gobierno
 - c) Impuestos de suma fija
2. Teorema supone que G_1 y G_2 no cambian. *Cambios en el gasto público sí generan cambios en los precios y cantidades de equilibrio.*

Shocks al gasto público

- Consideramos el impacto de cambios en el gasto público en la asignación y precios de equilibrio.
 - Aumentos temporarios,
 - Aumentos permanentes
 - Aumentos esperados.
- Veremos que *el multiplicador fiscal es menor que 1.*
- Modelo keynesiano: multiplicador fiscal puede ser (es usualmente) mayor que 1

Suba temporaria del gasto público: $\uparrow G_1$ y $\bar{G}_2$ no cambia

Desplazamientos de las curvas dada una tasa de interés r_1 ,

1. **Impacto:** Suba de $G_1 \Rightarrow$ sube valor presente descontado del gasto $\uparrow \left(G_1 + \frac{G_2}{1+r_1} \right)$. Sube el valor presente descontado de los impuestos: efecto riqueza negativo.
2. **Efecto riqueza negativo:** $\downarrow C_1, \downarrow C_2, \uparrow L_1, \uparrow L_2 \Rightarrow \uparrow Y_1$ y $\uparrow Y_2$.
3. **Efectos sustitución:** no hay.
4. **Demanda de inversión:** no cambia (ignorando el efecto de L_2)

Por lo tanto,

- El punto 2 implica $\Delta Y_1^s(r_1) > 0$
- El punto 4 implica $\Delta I^d(r_1) \approx 0$.
- Mostraremos que se cumplen las siguientes desigualdades:

$$\Delta G_1 > \Delta Y_1^d(r_1) > \Delta Y_1^s(r_1) > 0.$$

Suba temporaria del gasto público: $\uparrow G_1$ y $\bar{G}_2$ no cambia

1. Cae la curva de ingreso disponible $\hat{Y}_1(r_1) = Y_1^s(r_1) - G_1$ cuando sube G_1 . Usando $\Delta I_1 \approx 0$,

$$\Delta C_1^d(r_1) + \frac{\Delta C_2^d(r_1)}{1 + r_1} = \Delta Y_1^s(r_1) - \Delta G_1 + \frac{\Delta Y_2^s(r_1) - \Delta G_2}{1 + r_1} = \Delta \hat{Y}_1(r_1) + \frac{\Delta Y_2^s(r_1)}{1 + r_1}$$

- Lado izquierdo es negativo porque $\Delta C_1^d < 0$ y $\Delta C_2^d < 0$.
- Del lado derecho $\Delta G_2 = 0$ y $\Delta Y_2^s > 0$. Por lo tanto, necesariamente $\Delta \hat{Y}_1(r_1) < 0$.

2. La demanda agregada aumenta, $\Delta Y_1^d(r_1) > 0$. Usando $\Delta I_1 \approx 0$ y $\Delta G_2 = 0$,

$$\Delta C_1^d(r_1) + \Delta G_1 + \frac{\Delta C_2^d(r_1)}{1 + r_1} = \Delta Y_1^s(r_1) + \frac{\Delta Y_2^s(r_1)}{1 + r_1} > 0$$

- Como $\Delta C_2^d(r_1) < 0$ entonces $\Delta Y_1^d(r_1) = \Delta C_1^d(r_1) + \Delta G_1 > 0$
- Por lo tanto el consumo cae menos que lo que sube el gasto (intuición shock temporario)

Suba temporaria del gasto público: $\uparrow G_1$ y $\bar{G}_2$ no cambia

3. *Demanda agregada aumenta menos que el gasto público, $\Delta G_1 > \Delta Y_1^d(r_1)$:*

$$\Delta Y_1^d(r_1) = \Delta C_1^d(r_1) + \underbrace{\Delta I_1^d(r_1)}_{\approx 0} + \Delta G_1 > 0$$

pero como $|\Delta C_1^d| < \Delta G_1$,

$$\Delta G_1 > \Delta Y_1^d(r_1) > 0.$$

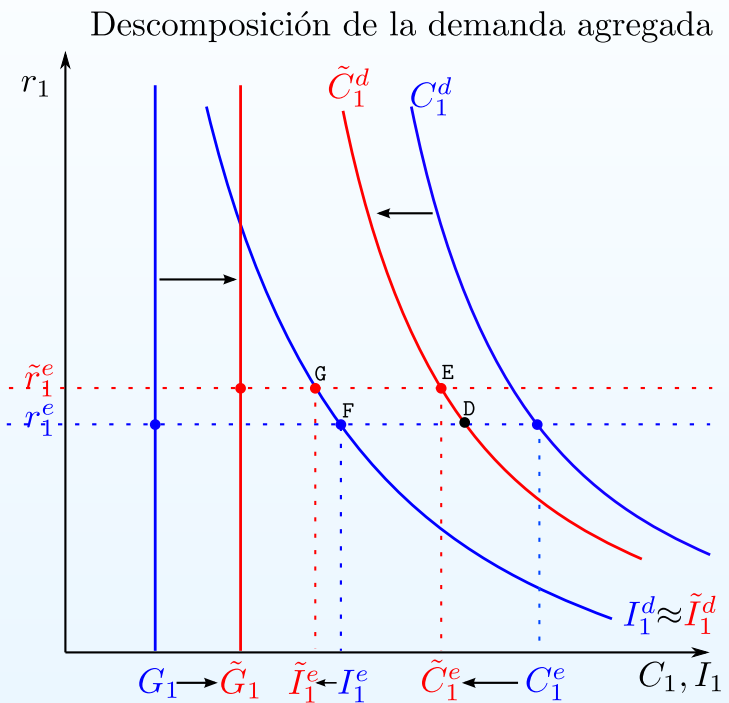
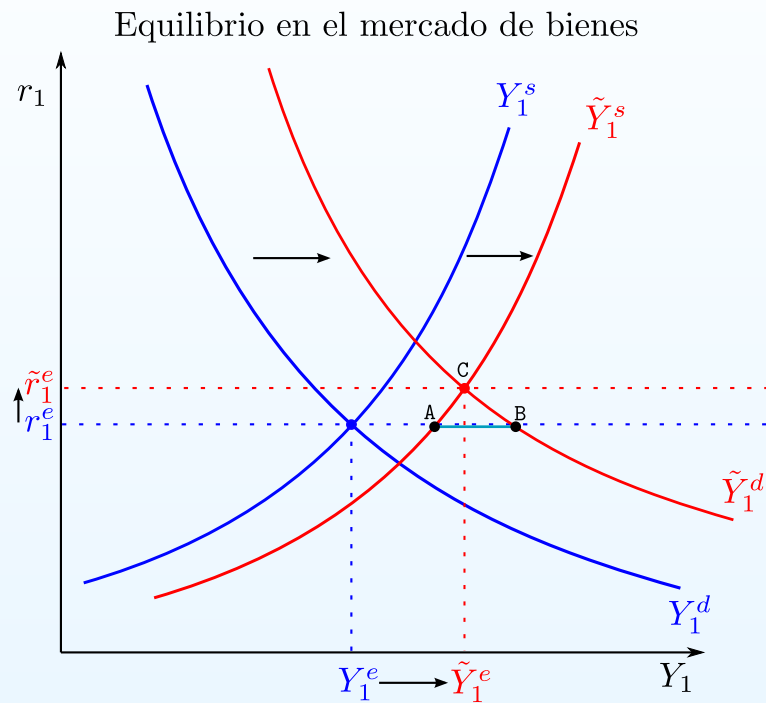
4. *Demanda agregada aumenta más que la oferta agregada, $\Delta Y_1^d(r_1) > \Delta Y_1^s(r_1)$:*

$$\underbrace{\Delta C_1(r_1) + \Delta G_1}_{1 = \Delta Y_1^d \text{ porque } \Delta I_1 \approx 0} - \Delta Y_1^s(r_1) = \frac{\Delta Y_2^s(r_1) - \Delta C_2^d(r_1)}{1 + r_1}$$
$$\Delta Y_1^d(r_1) - \Delta Y_1^s(r_1) = \frac{\overbrace{\Delta Y_2^s(r_1)}^{>0} - \overbrace{\Delta C_2^d(r_1)}^{<0}}{1 + r_1} > 0$$

Conclusión,

$$\Delta Y_1^d(r_1) > \Delta Y_1^s(r_1).$$

Suba temporaria del gasto público: $\uparrow G_1$ y $\bar{G}_2$ no cambia



Suba temporaria del gasto público: $\uparrow G_1$ y $\bar{G}_2$ no cambia

- Exceso de demanda: sube tasa de interés. Luego, cae el consumo, cae la inversión y aumenta el producto.
- Mecanismo:
 1. **ES intertemporal del consumo:** $\uparrow r_1^e \implies \downarrow C_1$ (y $\uparrow C_2$). Movimiento *sobre la curva de demanda de consumo* desde el punto D hasta el punto E.
 2. **ES intertemporal del trabajo:** $\uparrow r_1^e \implies \uparrow L_1 \Rightarrow \uparrow Y_1$. Movimiento *sobre la curva de oferta de bienes* $\tilde{Y}_1^s(r_1)$ desde el punto A hasta el punto C.
 3. **Aumenta del retorno de los bonos:** $\uparrow r_1^e$ aumenta el retorno de ahorrar en bonos. Cae la inversión hasta que los retornos se igualan. Movimiento *sobre la curva de demanda de inversión* desde el punto F hasta el punto G.

Suba temporaria del gasto público: $\uparrow G_1$ y $\bar{G}_2$ no cambia

- Efecto final:
 - Aumenta el producto Y_1
 - Cae el consumo privado C_1
 - Cae la inversión privada I_1
 - Aumenta la tasa de interés r_1 .
- Mecanismo por el cual un aumento del gasto público genera un aumento del producto:
 - Consumidores trabajan más por el efecto riqueza negativo y por la suba en la tasa de interés real.
 - Pero el aumento del gasto público genera un crowding-out del consumo y la inversión
- Por lo tanto, multiplicador fiscal es menor que 1.
- En el modelo keynesiano el multiplicador fiscal es (usualmente) mayor que 1.

Suba temporaria del gasto público: financiamiento del gasto

- Dos casos:
 1. Financio el gasto con impuestos corrientes
 2. Financio el gasto con un déficit fiscal (deuda)
- **Equivalencia Ricardiana:** ni la tasa de interés ni las cantidades de equilibrio dependen de la manera en que se financie el aumento del gasto público.

Suba temporaria del gasto público: financiamiento del gasto

Caso 1: financio el aumento de G_1 subiendo los impuestos hoy: $\Delta G_1 = \Delta T_1$

- Demanda de bonos del gobierno:

$$\Delta B_1^g = \Delta T_1 - \Delta G_1 = 0.$$

- Demanda de bonos de los hogares

$$\begin{aligned}\Delta B_1^p(r_1) &= \Delta Y_1^s(r_1) - \underbrace{\Delta T_1}_{=\Delta G_1} - \Delta C_1^d(r_1) - \underbrace{\Delta I_1(r_1)}_{=0} \\ &= \Delta Y_1^s(r_1) - (\Delta G_1 + \Delta C_1^d(r_1)) \\ &= \Delta Y_1^s(r_1) - \Delta Y_1^d(r_1) < 0.\end{aligned}$$

- Demanda agregada de bonos

$$\Delta B_1^d(r_1) = \Delta B_1^g + \Delta B_1^p(r_1) = \Delta Y_1^s(r_1) - \Delta Y_1^d(r_1) < 0.$$

- En equilibrio los consumidores no pueden endeudarse (más que antes): sube la tasa de interés.

Suba temporaria del gasto público: financiamiento del gasto

Caso 2: financio el aumento de G_1 con deuda

- Demanda de bonos del gobierno:

$$\Delta B_1^g = -\Delta G_1 < 0.$$

- Demanda de bonos de los hogares

$$\begin{aligned}\Delta B_1^p(r_1) &= \Delta Y_1^s(r_1) - \Delta T_1 - \Delta C_1^d(r_1) - \Delta I_1^d(r_1) \\ &= \Delta Y_1^s(r_1) - 0 - \Delta C_1^d(r_1) - 0 \\ &= \underbrace{\Delta Y_1^s(r_1)}_{>0} - \underbrace{\Delta C_1^d(r_1)}_{<0} > 0.\end{aligned}$$

- Demanda agregada de bonos

$$\begin{aligned}\Delta B_1^d &= \Delta B_1^p(r_1) + \Delta B_1^g \\ &= \Delta Y_1^s(r_1) - \Delta Y_1^d(r_1) < 0.\end{aligned}$$

- Igual que caso 1, pero ahora el sector privado le presta al sector público.

Aumento permanente del gasto público: $\Delta G_1 = \Delta G_2 > 0$

Desplazamientos de las curvas dada una tasa de interés r_1 ,

1. **Impacto:** Sube el valor presente descontado del gasto $\uparrow \left(G_1 + \frac{G_2}{1+r_1} \right)$. De la RPI, sube el valor presente descontado de los impuestos y eso genera un efecto riqueza negativo.
2. **Efecto riqueza negativo:** $\downarrow C_1, \downarrow C_2, \uparrow L_1, \uparrow L_2$. El aumento de L_1 y L_2 genera una suba del producto en ambos periodos $\uparrow Y_1$ y $\uparrow Y_2$.
3. **Efectos sustitución:** no hay.
4. **Demanda de inversión:** No hay (ignorando el efecto de L_2). (Si el efecto indirecto a través de L_2 es importante, la demanda de inversión se traslada hacia la derecha.)

Por lo tanto,

- El punto 2 implica que la curva de oferta agregada se traslada hacia la derecha, $\Delta Y^s(r_1) > 0$.
- El punto 4 implica que la curva de demanda de inversión no cambia, $\Delta I^d(r_1) \approx 0$.
- Argumentaremos que $\Delta Y_1^d(r_1) \approx \Delta Y_1^s(r_1)$

Aumento permanente del gasto público: $\Delta G_1 = \Delta G_2 > 0$

- Del punto 2, $\Delta C_1^d(r_1) < 0$.
- Como el aumento del gasto público es permanente, el *ingreso disponible* $\tilde{Y}_t(r_1) = Y_t^s(r_1) - G_t$ sufre una caída permanente.

- Esto implica

$$\Delta C^d(r_1) \approx \hat{Y}_1(r_1) = \Delta Y^s(r_1) - \Delta G_1,$$

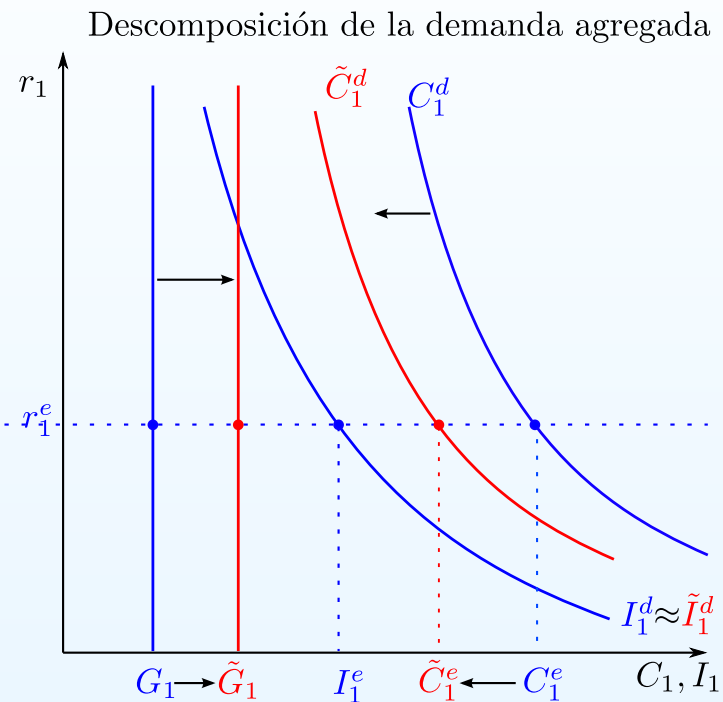
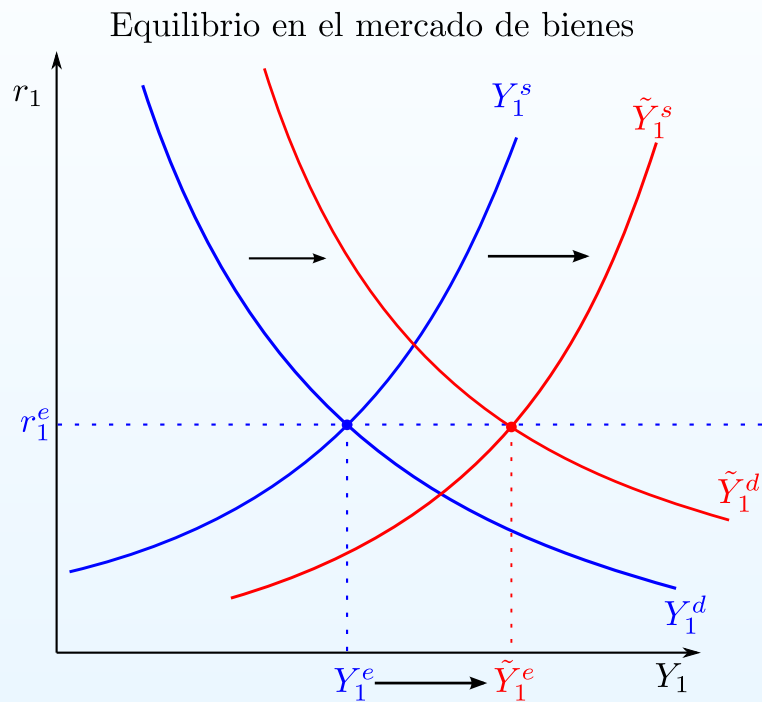
o bien,

$$\Delta C^d(r_1) + \Delta G_1 \approx \Delta Y^s(r_1).$$

- Por lo tanto, la demanda agregada se desplaza hacia la derecha en aproximadamente la misma cantidad que la oferta agregada:

$$\Delta Y_1^d(r_1) = \Delta C_1^d(r_1) + \underbrace{\Delta I_1^d(r_1)}_{\approx 0} + \Delta G_1 = \Delta C_1^d(r_1) + \Delta G_1 \approx \Delta Y_1^s(r_1).$$

Aumento permanente del gasto público: $\Delta G_1 = \Delta G_2 > 0$



Aumento permanente del gasto público: $\Delta G_1 = \Delta G_2 > 0$

- Efecto final del aumento permanente del gasto público:
 - Aumenta el producto Y_1
 - Cae el consumo privado C_1
 - La inversión privada I_1 no cambia
 - La tasa de interés real r_1 no cambia.
- El aumento permanente del gasto público solo genera un crowding-out del consumo privado pero no de la inversión ya que no cambia la tasa de interés de equilibrio.
- El aumento del producto sigue siendo menor que el aumento del gasto público—crowding-out del consumo
- El aumento permanente del gasto público también genera un *multiplicador fiscal menor a 1*.

Aumento esperado del gasto público: $\bar{G}_1$ y $\uparrow G_2$

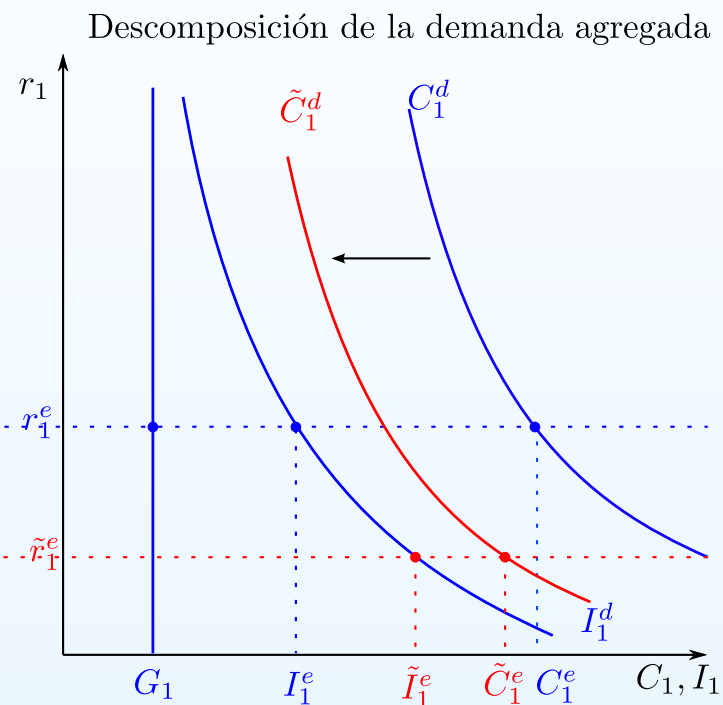
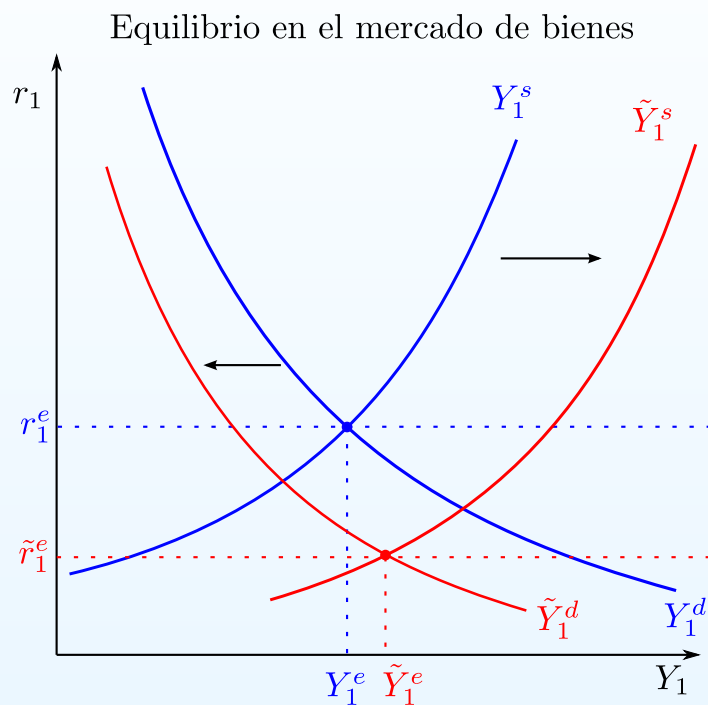
Desplazamientos de las curvas dada una tasa de interés r_1 ,

1. **Impacto:** Sube el valor presente descontado del gasto $\uparrow \left(G_1 + \frac{G_2}{1+r_1} \right)$, lo que genera un efecto riqueza negativo.
2. **Efecto riqueza negativo:** $\downarrow C_1, \downarrow C_2, \uparrow L_1, \uparrow L_2 \implies \uparrow Y_1$.
3. **Efectos sustitución:** no hay.
4. **Demanda de inversión:** no hay (ignorando el efecto de L_2)

Por lo tanto,

- El punto 1 implica que la curva de oferta agregada se traslada hacia la derecha, $\Delta Y^s(r_1) > 0$.
- El punto 4 implica que la curva de demanda de inversión se mantiene aproximadamente en el mismo lugar, $\Delta I^d(r_1) \approx 0$.
- El punto 2 implica que la curva de demanda de consumo $C_1^d(r_1)$ se desplaza hacia la izquierda.
- Por lo tanto, la demanda agregada $Y_1^d(r_1)$ se desplaza hacia la izquierda en la cantidad ΔC_1^d porque ni G_1 ni I_1 cambian.

Aumento esperado del gasto público: $\bar{G}_1$ y $\uparrow G_2$



- En principio, uno podría pensar que el producto puede subir o bajar.
- Sin embargo, mostraremos que el producto y la inversión aumentan, y el consumo cae.

Aumento esperado del gasto público: $\bar{G}_1$ y $\uparrow G_2$

- De la condición intratemporal

$$\frac{v'(1 - L_1)}{u'(C_1)} = A_1 f_l(K_0, L_1) \Rightarrow \frac{v'(1 - L_1)}{A_1 f_l(K_0, L_1)} = u'(C_1).$$

vemos que C_1 y L_1 se mueven en dirección opuesta. Por lo tanto, C_1 e Y_1 se mueven en dirección opuesta.

- Por la caída de la tasa de interés, la inversión I_1 sube.
- Probaremos por contradicción que C_1^e cae. Suponga, por el contrario, que el consumo hoy C_1^e sube. Entonces, de consistencia agregada en $t = 1$,

$$\Delta Y_1^e = \Delta C_1^e + \Delta I_1^e + \underbrace{\Delta G_1}_{=0} = \Delta C_1^e + \Delta I_1^e > 0$$

por lo que el producto sube $\Delta Y_1^e > 0$. Este resultado contradice que consumo y producto se mueven en dirección opuesta.

- Por lo tanto, necesariamente tenemos que $\Delta Y_1^e > 0$, $\Delta I_1^e > 0$, $\Delta C_1^e < 0$.

Aumento esperado del gasto público: $\bar{G}_1$ y $\uparrow G_2$

- El efecto final del aumento esperado del gasto público es el siguiente:
 - Cae la tasa de interés de equilibrio r_1
 - Aumenta la inversión hoy I_1^e
 - Aumenta el producto hoy Y_1^e
 - Cae el consumo hoy C_1^e

Aumento esperado del gasto público: financiación del gasto

Caso 1: el gobierno financia el aumento de G_2 aumentando los impuestos mañana

- Demanda de bonos del gobierno:

$$\underbrace{\Delta G_1}_{=0} + \Delta B_1^g = \underbrace{\Delta T_1}_{=0} \Rightarrow \Delta B_1^g = 0.$$

- Demanda de bonos de los hogares

$$\underbrace{\Delta C_1^d}_{<0} + \Delta B_1^p = \underbrace{\Delta Y_1^s}_{>0} - \underbrace{\Delta T_1}_{=0} \Rightarrow \Delta B_1^p = \Delta Y_1^s - \Delta C_1^d > 0$$

- Demanda agregada de bonos

$$\Delta B_1^d(r_1) = \Delta B_1^g + \Delta B_1^p(r_1) = \Delta Y_1^s(r_1) - \Delta C_1^d(r_1) > 0.$$

- Demanda agregada de bonos aumenta para cada r_1 (consumidores quieren ahorrar). Pero como en equilibrio los hogares no pueden ahorrar más que antes (porque $\Delta B^g = 0$), cae la tasa de interés de equilibrio para desincentivar el ahorro.

Aumento esperado del gasto público: financiación del gasto

Caso 2: el gobierno financia el aumento de G_2 aumentando los impuestos hoy

- Demanda de bonos del gobierno:

$$\Delta B_1^g = \Delta T_1 = \frac{\Delta G_2}{1+r} > 0.$$

- Demanda de bonos de los hogares

$$\begin{aligned}\Delta B_1^p &= \Delta Y_1^s - \Delta C_1^d - \Delta T_1 \\ &= \Delta Y_1^s - \Delta C_1^d - \Delta B_1^g\end{aligned}$$

- Demanda agregada de bonos

$$\Delta B_1^d(r_1) = \Delta B_1^g + \Delta B_1^p(r_1) = \Delta Y_1^s(r_1) - \Delta C_1^d(r_1) > 0.$$

- Aumento de la demanda agregada de bonos es igual que en el Caso 1 (equivalencia ricardiana). La diferencia es que ahora parte del ahorro lo hace el gobierno.

Macroeconomía Dinámica

Introducción: Parte 7b

Impuestos Distorsivos

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Introducción

- Impacto de impuestos distorsivos.
- Economía de dotaciones con impuestos al consumo.
- Único rol del gobierno: cobra impuestos y los devuelve como transferencia de suma fija.
- El gobierno no consume, no ahorra ni se endeuda.
- No se cumple la equivalencia Ricardiana

Hogares

- Función de utilidad

$$u(C_1) + \beta u(C_2).$$

- Ingreso exógeno de Y_1 e Y_2 , transferencias de suma fija V_1 y V_2 , e impuestos T_1 y T_2 .

- Restricciones presupuestarias flujo

$$C_1 + B_1 = Y_1 - T_1 + V_1 \tag{1}$$

$$C_2 = Y_2 - T_2 + V_2 + B_1(1 + r_1). \tag{2}$$

- Impuestos al consumo

$$T_1 = \tau_1 C_1 \tag{3}$$

$$T_2 = \tau_2 C_2. \tag{4}$$

Hogares

- Sustituyo (3) y (4) en (1) y (2),

$$(1 + \tau_1) C_1 + B_1 = Y_1 + V_1$$

$$(1 + \tau_2) C_2 = Y_2 + V_2 + B_1 (1 + r_1).$$

- Restricción presupuestaria intertemporal

$$(1 + \tau_1) C_1 + \frac{(1 + \tau_2) C_2}{1 + r_1} = Y_1 + V_1 + \frac{Y_2 + V_2}{1 + r_1} \quad (5)$$

- Impuestos afectan el precio relativo del consumo *que enfrenta el consumidor*

$$C_1 + \left(\frac{1 + \tau_2}{1 + \tau_1} \right) \frac{1}{1 + r_1} C_2 = \frac{Y_1 + V_1}{1 + \tau_1} + \frac{Y_2 + V_2}{(1 + \tau_1)(1 + r_1)}.$$

$$\frac{P_2}{P_1} = \left(\frac{1 + \tau_2}{1 + \tau_1} \right) \frac{1}{1 + r_1}$$

Hogares

- Problema del consumidor

$$L = u(C_1) + \beta u(C_2) - \lambda \left[(1 + \tau_1) C_1 + \frac{(1 + \tau_2) C_2}{1 + r_1} - (Y_1 + V_1) - \frac{Y_2 + V_2}{1 + r_1} \right].$$

- Condiciones de primer orden

$$u'(C_1) = \lambda (1 + \tau_1)$$

$$\beta u'(C_2) = \lambda \frac{(1 + \tau_2)}{1 + r_1}.$$

- Ecuación de Euler:

$$u'(C_1) = \left(\frac{1 + \tau_1}{1 + \tau_2} \right) (1 + r_1) \beta u'(C_2) \quad (6)$$

- Sin impuestos:

$$u'(C_1) = \beta (1 + r_1) u'(C_2)$$

Hogares

Ejemplo con utilidad logarítmica: $u(C) = \log(C) \Rightarrow u'(C) = 1/C$

- Ecuación de Euler

$$\frac{1}{C_1} = \left(\frac{1 + \tau_1}{1 + \tau_2} \right) (1 + r_1) \beta \frac{1}{C_2}$$

- Despejando C_2 ,

$$C_2 = \left(\frac{1 + \tau_1}{1 + \tau_2} \right) (1 + r_1) \beta C_1. \quad (7)$$

Efecto sustitución intertemporal de cambio en los impuestos

- Aumento de τ_1 es equivalente a un aumento de r_1 : el consumo presente se vuelve más caro que el consumo futuro.
- Un aumento de τ_2 implica que el consumo presente se vuelve más barato que el consumo futuro.
- Por lo tanto:
 1. Un aumento de r_1 o de τ_1 implican un *efecto sustitución intertemporal* por el cual C_1 cae y C_2 aumenta
 2. Un aumento de τ_2 implica un *efecto sustitución intertemporal* por el cual C_1 aumenta y C_2 cae.
- Efecto riqueza?
 - En equilibrio se cancela porque el gobierno devuelve los impuestos como transferencias

Sector público

- Gobierno cobra impuestos y los devuelve inmediatamente como transferencias de suma fija.
- El gobierno no se endeuda ni ahorra activos.
 - Podemos extender el modelo
- Restricciones presupuestarias flujo del gobierno

$$V_1 = \tau_1 C_1$$

$$V_2 = \tau_2 C_2$$

Equilibrio

Equilibrio competitivo: asignación C_1^e, C_2^e y tasa de interés r_1^e tales que

1. Los hogares optimizan
2. Se cumplen las restricciones presupuestarias del gobierno
3. Se vacían los mercados
 - **Mercados:** bienes en los períodos 1 y 2 y bonos en el período 1

$$C_1 = Y_1 \tag{8}$$

$$C_2 = Y_2 \tag{9}$$

$$B_1 = 0. \tag{10}$$

- **Ley de Walras:** mercado de bienes del primer período.

Equilibrio

- Sustituyo restricciones presupuestarias del gobierno en RPI del consumidor

$$(1 + \tau_1) C_1 + \frac{(1 + \tau_2) C_2}{1 + r_1} = Y_1 + \tau_1 C_1 + \frac{Y_2 + \tau_2 C_2}{1 + r_1}$$

- *Restricción presupuestaria intertemporal agregada:*

$$C_1 + \frac{C_2}{1 + r_1} = Y_1 + \frac{Y_2}{1 + r_1}. \quad (11)$$

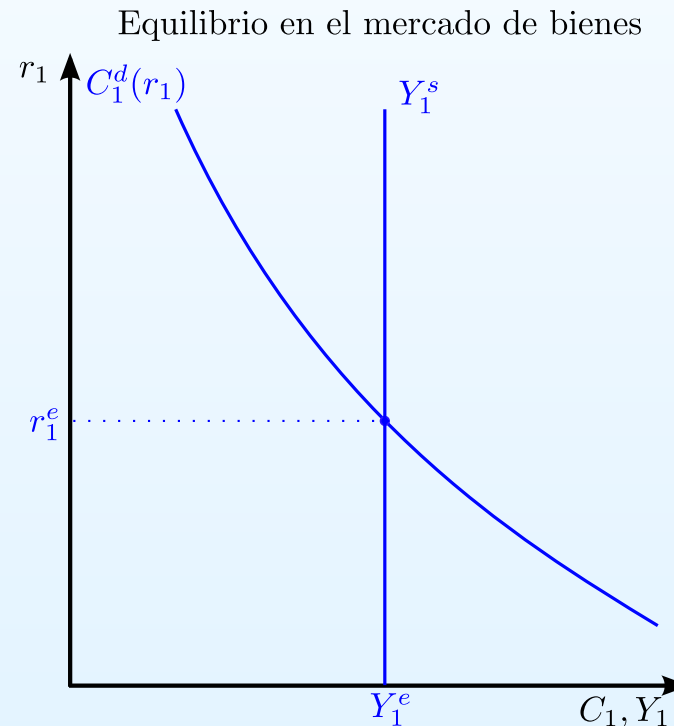
- Implicancia: **en equilibrio los impuestos no tienen efecto riqueza**
 - Cada peso que se recauda es devuelto al consumidor como una transferencia de suma fija.
 - Efecto de equilibrio general: el consumidor individual no tiene en cuenta que su pago particular le será devuelto
 - Transferencias dependen de las decisiones de consumo agregadas de toda la economía.

Equilibrio: demandas y ofertas agregadas

- Demanda agregada de consumo C_1^d cambia con la tasa de interés y los impuestos *únicamente debido a efectos sustitución intertemporal*:

$$C_1 = C_1^d \left(\bar{r}_1, \bar{\tau}_1, \tau_2^+ \right) \quad (12)$$

- La oferta agregada de bienes Y_1^s es constante.



Ejemplo con utilidad logarítmica

- Ecuación de Euler

$$C_2 = \left(\frac{1 + \tau_1}{1 + \tau_2} \right) (1 + r_1) \beta C_1.$$

Reemplazo en RPI agregada

$$C_1 + \frac{C_1 \left(\frac{1 + \tau_1}{1 + \tau_2} \right) \beta (1 + r_1)}{1 + r_1} = Y_1 + \frac{Y_2}{1 + r_1}.$$

- Demanda agregada C_1

$$C_1^d(r_1, \tau_1, \tau_2) = \frac{Y_1 + \frac{Y_2}{1 + r_1}}{1 + \beta \left(\frac{1 + \tau_1}{1 + \tau_2} \right)} \quad (13)$$

- Tasa de interés de equilibrio surge de imponer $C_1^d = Y_1$:

$$r_1^e = \frac{1}{\beta} \frac{Y_2}{Y_1} \left(\frac{1 + \tau_2}{1 + \tau_1} \right) - 1. \quad (14)$$

Estática comparativa

Consideraremos

- Aumento temporarios de los impuestos al consumo.
- Aumento permanente de los impuestos al consumo, comenzando con el mismo nivel de impuestos en ambos períodos.
- Aumento esperado de los impuestos al consumo.

Aumento temporario de los impuestos: $\uparrow \tau_1, \bar{\tau}_2$

Desplazamientos de las curvas dada la tasa de interés r_1 :

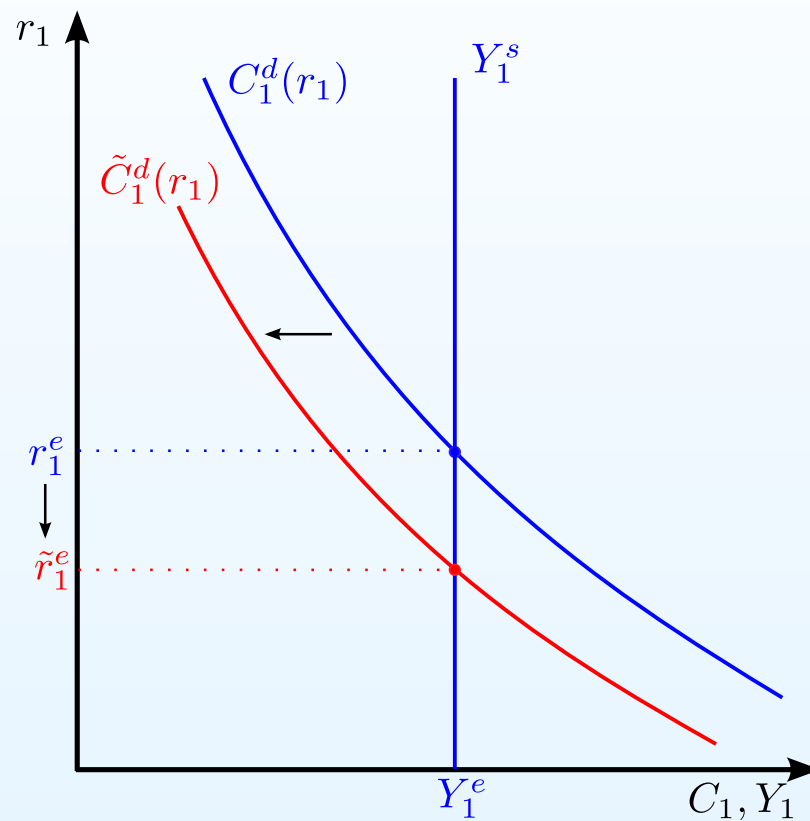
- **Efecto riqueza:** no hay (impuestos se cancelan con transferencias)
- **Efecto sustitución intertemporal:** $\uparrow \tau_1$ implica $\downarrow C_1$ y $\uparrow C_2$: consumo presente se hace más caro relativo al consumo futuro.

Conclusión:

1. Demanda de bienes se desplaza a la izquierda.
2. La oferta de bienes no cambia porque es exógena.

Aumento temporario de los impuestos: $\uparrow \tau_1, \bar{\tau}_2$

Equilibrio en el mercado de bienes



Aumento temporario de los impuestos: $\uparrow \tau_1, \bar{\tau}_2$

- A la tasa de interés original hay un exceso de oferta de bienes: cae la tasa de interés real r_1^e
- Intuición:
 1. $\uparrow \tau_1$ aumenta el precio relativo del consumo presente.
 2. Aumentar la demanda de bonos para aumentar el consumo futuro.
 3. Como todos aumentan su demanda de bonos, pero en equilibrio la demanda neta de bonos debe ser cero, debe caer la tasa de interés para que los consumidores *no quieran* ahorrar.
 4. Al caer r_1^e aumenta el precio del consumo futuro reduciendo así la demanda de bonos.

Aumento permanente de los impuestos: $\tau_1 = \tau_2 = \tau$ y $\uparrow \tau$

Desplazamientos de las curvas dada la tasa de interés r_1 :

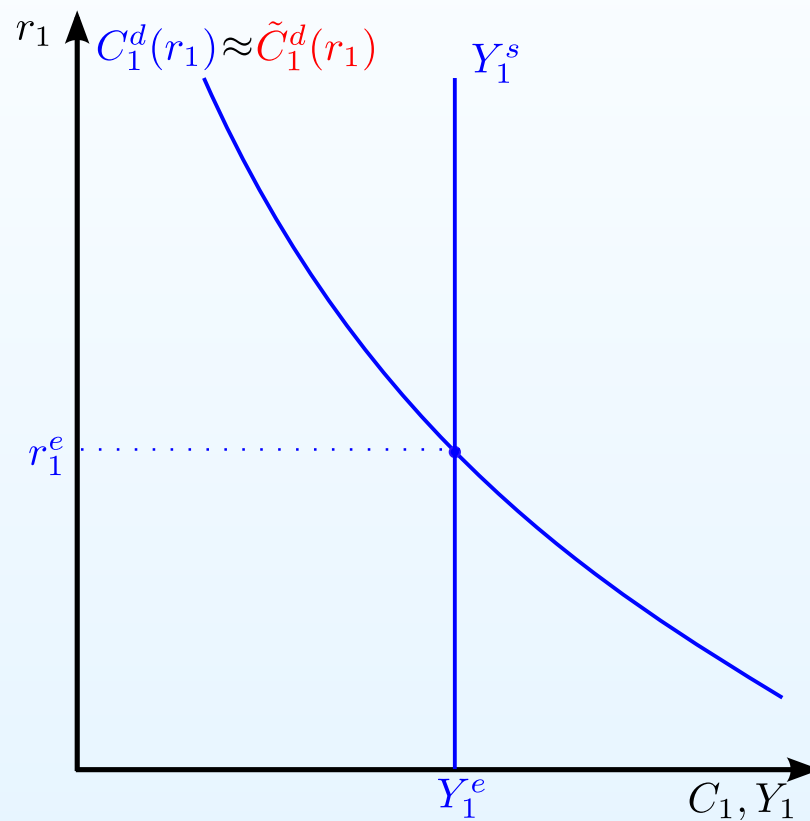
- **Efecto riqueza:** no hay (impuestos se cancelan con transferencias)
- **Efecto sustitución intertemporal:** no hay porque no cambia el precio relativo entre consumo presente y futuro.

Conclusión:

1. La demanda de consumo no cambia.
2. La oferta de bienes no cambia porque es exógena.

Aumento permanente de los impuestos: $\tau_1 = \tau_2 = \tau$ y $\uparrow \tau$

Equilibrio en el mercado de bienes



Aumento permanente de los impuestos: $\tau_1 = \tau_2 = \tau$ y $\uparrow \tau$

- La tasa de interés de equilibrio no cambia
- Intuición
 - Un impuesto al consumo constante (de cualquier nivel) no afecta el precio relativo del consumo
 - Como la recaudación impositiva vuelve a los hogares como una transferencia de suma fija, un impuesto constante es equivalente a un impuesto de suma fija.
- Este resultado depende fuertemente de que la producción sea exógena.
- Con producción endógena, impuestos al consumo (constantes o no) generan distorsiones en la oferta de trabajo.
 - Impuesto al consumo introduce una brecha entre TMS y PML

Suba esperada de los impuestos: $\bar{\tau}_1$ y $\uparrow \tau_2$

Desplazamientos de las curvas dada la tasa de interés r_1 :

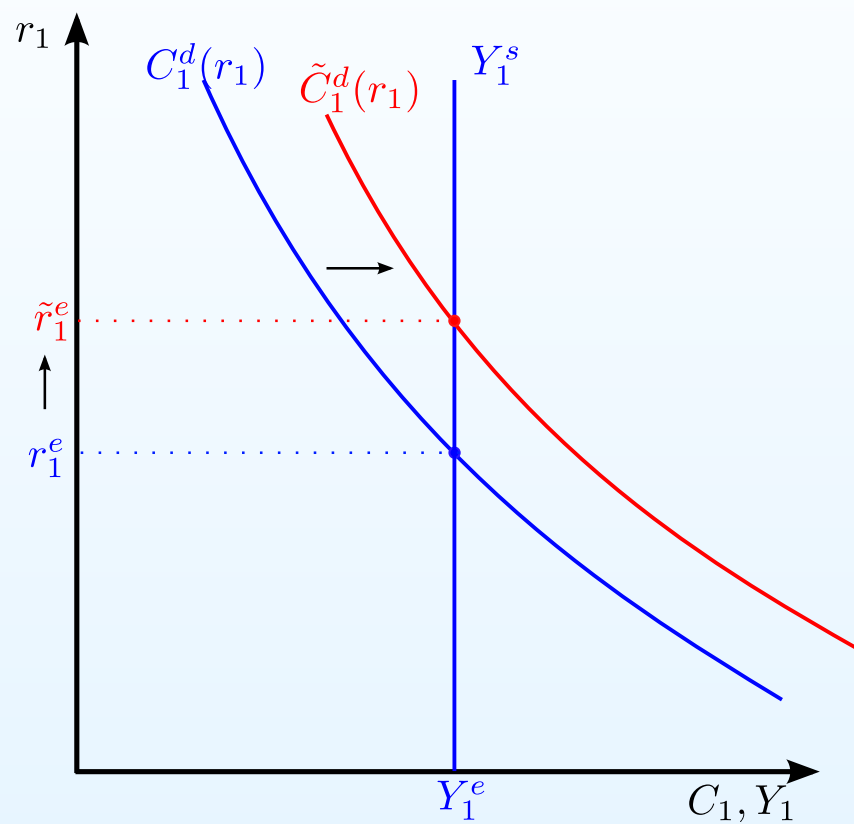
- **Efecto riqueza:** no hay (impuestos se cancelan con transferencias)
- **Efecto sustitución intertemporal:** $\uparrow \tau_2$ implica $\uparrow C_1$ y $\downarrow C_2$: consumo presente se hace más barato relativo al consumo futuro.

Conclusión:

1. Demanda de bienes se desplaza a la derecha.
2. La oferta de bienes no cambia porque es exógena.

Suba esperada de los impuestos: $\bar{\tau}_1$ y $\uparrow \tau_2$

Equilibrio en el mercado de bienes



Suba esperada de los impuestos: $\bar{\tau}_1$ y $\uparrow \tau_2$

- A la tasa de interés original hay un exceso de demanda de bienes: aumenta la tasa de interés real r_1^e
- Intuición:
 1. $\uparrow \tau_2$ aumenta el precio relativo del consumo futuro.
 2. Aumenta la oferta de bonos (el deseo de endeudarse) para aumentar el consumo presente.
 3. Como todos quieren endeudarse, pero en equilibrio la demanda neta de bonos debe ser cero, debe subir la tasa de interés para que los *hogares no quieran* endeudarse.
 4. Al aumentar r_1^e aumenta el precio del consumo presente reduciendo así la oferta de bonos.

Recursive Problems and Dynamic Programming

A General Intertemporal Problem

Consider the following problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^T} U(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T)$$

s.t.

$$F_0(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T) = 0$$

$$F_1(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T) = 0$$

.....

$$F_T(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T) = 0$$

$$k_0 \text{ given}$$

where $c_t \in \mathbb{R}$, $k_t \in \mathbb{R}$ (disregard non-negativity constraints, just for simplicity).

The solution of the problem will be of the form:

$$\begin{aligned}
 c_0 &= H_0(k_0) \\
 &\dots\dots\dots \\
 c_T &= H_T(k_0) \\
 \\
 k_1 &= J_1(k_0) \\
 &\dots\dots\dots \\
 k_{T+1} &= J_{T+1}(k_0)
 \end{aligned}$$

Assume the problem is "nicely behaved" and form the Lagrangian:

$$\begin{aligned}
 \mathcal{L} &= U(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T) \\
 &+ \lambda_0 F_0(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T) \\
 &+ \lambda_1 F_1(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T) \\
 &+ \dots \\
 &+ \lambda_T F_T(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T)
 \end{aligned}$$

First-Order Conditions (FOC)

Notation: $k \equiv (k_0, k_1, \dots, k_{T+1})$, $c \equiv (c_0, c_1, \dots, c_T)$

$$\frac{\partial \mathcal{L}}{\partial c_0} = 0 \Rightarrow \frac{\partial U(k, c)}{\partial c_0} + \sum_{s=0}^T \lambda_s \frac{\partial F_s(k, c)}{\partial c_0} = 0$$

.....

$$\frac{\partial \mathcal{L}}{\partial c_T} = 0 \Rightarrow \frac{\partial U(k, c)}{\partial c_T} + \sum_{s=0}^T \lambda_s \frac{\partial F_s(k, c)}{\partial c_T} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_1} = 0 \Rightarrow \frac{\partial U(k, c)}{\partial k_1} + \sum_{s=0}^T \lambda_s \frac{\partial F_s(k, c)}{\partial k_1} = 0$$

.....

$$\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow \frac{\partial U(k, c)}{\partial k_{T+1}} + \sum_{s=0}^T \lambda_s \frac{\partial F_s(k, c)}{\partial k_{T+1}} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_0} = 0 \Rightarrow F_0(k, c) = 0$$

.....

$$\frac{\partial \mathcal{L}}{\partial \lambda_T} = 0 \Rightarrow F_T(k, c) = 0$$

We have a system of $3(T + 1)$ equations to solve for $3(T + 1)$ unknowns, $c_0, \dots, c_T, k_1, \dots, k_{T+1}, \lambda_0, \dots, \lambda_T$, in terms of k_0 .

Notice that, in general, the system has to be solved *simultaneously* for the $3(T + 1)$ unknowns.

Special Case: A Recursive Problem

Assume:

$$U(k, c) = u_0(k_0, c_0) + \dots + u_T(k_T, c_T) + S(k_{T+1})$$

$$F_t(k, c) = G_t(k_t, c_t) - k_{t+1}, \quad t \in \{0, \dots, T\}$$

With these assumptions, our problem becomes:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^T} u_0(k_0, c_0) + \dots + u_T(k_T, c_T) + S(k_{T+1})$$

s.t.

$$k_1 = G_0(k_0, c_0)$$

$$k_2 = G_1(k_1, c_1)$$

.....

$$k_{T+1} = G_T(k_T, c_T)$$

$$k_0 \text{ given}$$

$c_0, c_1, \dots, c_T$: control variables

$k_0, k_1, \dots, k_{T+1}$: state variables

$u_t(k_t, c_t)$: one-period return function at t

$S(k_{T+1})$: scrap value

$k_{t+1} = G_t(k_t, c_t)$: transition equation at t

The special time-separable structure of this problem implies that, for given k_τ, c_τ influences the future values of the state variable $\{k_{t+1}\}_{t=\tau}^T$ and the current and future returns $\{u_t(k_t, c_t)\}_{t=\tau}^T$ but it does not influence their past values, $\{k_{t+1}\}_{t=0}^{\tau-1}$ and $\{u_t(k_t, c_t)\}_{t=0}^{\tau-1}$. We have:

$$\begin{aligned} k_{\tau+1} &= G_\tau(k_\tau, c_\tau) \\ k_{\tau+2} &= G_{\tau+1}(k_{\tau+1}, c_{\tau+1}) \\ &= G_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}) \\ k_{\tau+3} &= G_{\tau+2}(k_{\tau+2}, c_{\tau+2}) \\ &= G_{\tau+2}(G_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}), c_{\tau+2}) \\ &\dots \end{aligned}$$

and

$$\begin{aligned} u_\tau(k_\tau, c_\tau) \\ u_{\tau+1}(k_{\tau+1}, c_{\tau+1}) &= u_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}) \\ u_{\tau+2}(k_{\tau+2}, c_{\tau+2}) &= u_{\tau+2}(G_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}), c_{\tau+2}) \\ &\dots \end{aligned}$$

Therefore, k_τ constitutes a complete description of the current position of the system (past values $\{c_t, k_t\}_{t=0}^{\tau-1}$ add no information beyond that contained in k_τ ; this is why we call k_τ the state variable at time τ). We show below that this key property gives the problem a recursive structure.

The Lagrangian is:

$$\begin{aligned}\mathcal{L} = & u_0(k_0, c_0) + u_1(k_1, c_1) + \dots + u_T(k_T, c_T) + S(k_{T+1}) \\ & + \lambda_0[G_0(k_0, c_0) - k_1] + \lambda_1[G_1(k_1, c_1) - k_2] + \dots \\ & + \lambda_{T-1}[G_{T-1}(k_{T-1}, c_{T-1}) - k_T] + \lambda_T[G_T(k_T, c_T) - k_{T+1}]\end{aligned}$$

First-Order Conditions (FOC)

$$\frac{\partial \mathcal{L}}{\partial c_0} = 0 \Rightarrow \frac{\partial u_0(k_0, c_0)}{\partial c_0} + \lambda_0 \frac{\partial G_0(k_0, c_0)}{\partial c_0} = 0$$

$$\dots\dots\dots \frac{\partial \mathcal{L}}{\partial c_T} = 0 \Rightarrow \frac{\partial u_T(k_T, c_T)}{\partial c_T} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial c_T} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_1} = 0 \Rightarrow \frac{\partial u_1(k_1, c_1)}{\partial k_1} - \lambda_0 + \lambda_1 \frac{\partial G_1(k_1, c_1)}{\partial k_1} = 0$$

$$\dots\dots\dots \frac{\partial \mathcal{L}}{\partial k_T} = 0 \Rightarrow \frac{\partial u_T(k_T, c_T)}{\partial k_T} - \lambda_{T-1} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial k_T} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow S'(k_{T+1}) - \lambda_T = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_0} = 0 \Rightarrow k_1 = G_0(k_0, c_0)$$

$$\dots\dots\dots \frac{\partial \mathcal{L}}{\partial \lambda_T} = 0 \Rightarrow k_{T+1} = G_T(k_T, c_T)$$

Now we'll show that the system of first-order conditions can be solved recursively. A *recursive solution* takes the form:

$$c_t = h_t(k_t)$$

$$k_{t+1} = g_t(k_t) \equiv G_t(k_t, h_t(k_t))$$

where the functions $h_t(\cdot)$ are called *policy functions* (also called, *optimal feedback rules*).

Iterations over the functions above, starting from the initial value of k_0 at $t = 0$, give the solution we are looking for:

$$c_0 = H_0(k_0) \equiv h_0(k_0)$$

$$k_1 = J_1(k_0) \equiv G_0(k_0, H_0(k_0))$$

$$c_1 = H_1(k_0) \equiv h_1(J_1(k_0))$$

$$k_2 = J_2(k_0) \equiv G_1(J_1(k_0), H_1(k_0))$$

.....

$$c_T = H_T(k_0) \equiv h_T(J_T(k_0))$$

$$k_{T+1} = J_{T+1}(k_0) \equiv G_T(J_T(k_0), H_T(k_0))$$

Start at $t = T$.

The FOC for period T are:

$$\frac{\partial u_T(k_T, c_T)}{\partial c_T} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial c_T} = 0$$

$$\lambda_T = S'(k_{T+1})$$

$$k_{T+1} = G_T(k_T, c_T)$$

Notice that we have a system of three equations that can be solved for the three unknowns, c_T , k_{T+1} , and λ_T , in terms of k_T . We get:

$$c_T = h_T(k_T)$$

$$k_{T+1} = g_T(k_T) \equiv G_T(k_T, h_T(k_T))$$

$$\lambda_T = \ell_T(k_T)$$

Now go one step back to $t = T - 1$.

The FOC for period $T - 1$ are:

$$\frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} = 0$$

$$\frac{\partial u_T(k_T, c_T)}{\partial k_T} - \lambda_{T-1} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial k_T} = 0$$

$$k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

We know from our period- T FOC that $c_T = h_T(k_T)$ and $\lambda_T = \ell_T(k_T)$. Hence, we can write:

$$\frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} = 0$$

$$\frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} - \lambda_{T-1} + \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} = 0$$

$$k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

This is a system of three equations that can be solved for the three unknowns, c_{T-1} , k_T , and λ_{T-1} , in terms of k_{T-1} . We get:

$$c_{T-1} = h_{T-1}(k_{T-1})$$

$$k_T = g_{T-1}(k_{T-1}) \equiv G_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))$$

$$\lambda_{T-1} = \ell_{T-1}(k_{T-1})$$

Now go one step back to $t = T - 2$.

The FOC for period $T - 2$ are:

$$\frac{\partial u_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} + \lambda_{T-2} \frac{\partial G_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} = 0$$

$$\frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial k_{T-1}} - \lambda_{T-2} + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial k_{T-1}} = 0$$

$$k_{T-1} = G_{T-2}(k_{T-2}, c_{T-2})$$

We know from the FOC for period $T - 1$ that $c_{T-1} = h_{T-1}(k_{T-1})$ and $\lambda_{T-1} = \ell_{T-1}(k_{T-1})$. Hence, we can write:

$$\frac{\partial u_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} + \lambda_{T-2} \frac{\partial G_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} = 0$$

$$\frac{\partial u_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))}{\partial k_{T-1}} - \lambda_{T-2}$$

$$+ \ell_{T-1}(k_{T-1}) \frac{\partial G_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))}{\partial k_{T-1}} = 0$$

$$k_{T-1} = G_{T-2}(k_{T-2}, c_{T-2})$$

This is a system of three equations that can be solved for the three unknowns, c_{T-2} , k_{T-1} , and λ_{T-2} , in terms of k_{T-2} . We get:

$$c_{T-2} = h_{T-2}(k_{T-2})$$

$$k_{T-1} = g_{T-2}(k_{T-2}) \equiv G_{T-2}(k_{T-2}, h_{T-2}(k_{T-2}))$$

$$\lambda_{T-2} = \ell_{T-2}(k_{T-2})$$

Keep going until $t = 0$

The FOC for period 0 are:

$$\frac{\partial u_0(k_0, c_0)}{\partial c_0} + \lambda_0 \frac{\partial G_0(k_0, c_0)}{\partial c_0} = 0$$

$$\frac{\partial u_1(k_1, c_1)}{\partial k_1} - \lambda_0 + \lambda_1 \frac{\partial G_1(k_1, c_1)}{\partial k_1} = 0$$

$$k_1 = G_0(k_0, c_0)$$

We know from the FOC for period 1 that $c_1 = h_1(k_1)$ and $\lambda_1 = \ell_1(k_1)$. Hence, we can write:

$$\frac{\partial u_0(k_0, c_0)}{\partial c_0} + \lambda_0 \frac{\partial G_0(k_0, c_0)}{\partial c_0} = 0$$

$$\frac{\partial u_1(k_1, h_1(k_1))}{\partial k_1} - \lambda_0 + \ell_1(k_1) \frac{\partial G_1(k_1, h_1(k_1))}{\partial k_1} = 0$$

$$k_1 = G_0(k_0, c_0)$$

This is a system of three equations that can be solved for the three unknowns, c_0 , k_1 , and λ_0 , in terms of k_0 . We get:

$$c_0 = h_0(k_0)$$

$$k_1 = g_0(k_0) \equiv G_0(k_0, h_0(k_0))$$

$$\lambda_0 = \ell_0(k_0)$$

and we are done.

Bellman Equations (BE)

Define the following sequence of *one-period* optimization problems, for $t \in \{0, 1, \dots, T\}$:

$$\begin{aligned} W_t(k_t) \equiv \max_{c_t, k_{t+1}} & \quad u_t(k_t, c_t) + W_{t+1}(k_{t+1}) \\ \text{s.t.} & \\ & k_{t+1} = G_t(k_t, c_t) \\ & k_t \text{ given} \end{aligned}$$

with

$$W_{T+1}(k_{T+1}) = S(k_{T+1})$$

We'll show that the solution to this sequence of one-period problems coincides with the solution to our original problem.

Consider period $t = T$.

Then:

$$\begin{aligned} W_T(k_T) \equiv \max_{c_T, k_{T+1}} \quad & u_T(k_T, c_T) + S(k_{T+1}) \\ \text{s.t.} \quad & k_{T+1} = G_T(k_T, c_T) \\ & k_T \text{ given} \end{aligned}$$

The Lagrangian for this problem is:

$$\mathcal{L} = u_T(k_T, c_T) + S(k_{T+1}) + \lambda_T [G_T(k_T, c_T) - k_{T+1}]$$

FOC

$$\frac{\partial \mathcal{L}}{\partial c_T} = 0 \Rightarrow \frac{\partial u_T(k_T, c_T)}{\partial c_T} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial c_T} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow \lambda_T = S'(k_{T+1})$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_T} = 0 \Rightarrow k_{T+1} = G_T(k_T, c_T)$$

Notice that this system coincides exactly with the period- T FOC of our original problem. Hence, we get:

$$c_T = h_T(k_T)$$

$$k_{T+1} = g_T(k_T) \equiv G_T(k_T, h_T(k_T))$$

$$\lambda_T = \ell_T(k_T) \equiv S'(G_T(k_T, h_T(k_T)))$$

Substituting into the objective function we get:

$$\begin{aligned} W_T(k_T) &= u_T(k_T, h_T(k_T)) + S(g_T(k_T)) \\ &= u_T(k_T, h_T(k_T)) + S(G_T(k_T, h_T(k_T))) \end{aligned}$$

Now consider period $t = T - 1$.

Then:

$$\begin{aligned}
 W_{T-1}(k_{T-1}) \equiv \max_{c_{T-1}, k_T} & \quad u_{T-1}(k_{T-1}, c_{T-1}) + W_T(k_T) \\
 \text{s.t.} & \quad k_T = G_{T-1}(k_{T-1}, c_{T-1}) \\
 & \quad k_{T-1} \text{ given}
 \end{aligned}$$

The Lagrangian for this problem is:

$$\mathcal{L} = u_{T-1}(k_{T-1}, c_{T-1}) + W_T(k_T) + \lambda_{T-1} [G_{T-1}(k_{T-1}, c_{T-1}) - k_T]$$

FOC

$$\begin{aligned}
 \frac{\partial \mathcal{L}}{\partial c_{T-1}} = 0 \Rightarrow & \quad \frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} \\
 & \quad + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} = 0
 \end{aligned}$$

$$\frac{\partial \mathcal{L}}{\partial k_T} = 0 \Rightarrow \quad W'_T(k_T) - \lambda_{T-1} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_{T-1}} = 0 \Rightarrow \quad k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

We know from our period- T problem that

$$W_T(k_T) = u_T(k_T, h_T(k_T)) + S(G_T(k_T, h_T(k_T)))$$

Then:

$$\begin{aligned}
W'_T(k_T) &= \\
&= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} + \frac{\partial u_T(k_T, h_T(k_T))}{\partial c_T} h'_T(k_T) \\
&+ S'(G_T(k_T, h_T(k_T))) \times \left[\frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} + \frac{\partial G_T(k_T, h_T(k_T))}{\partial c_T} h'_T(k_T) \right] \\
&= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} \\
&+ \left[\frac{\partial u_T(k_T, h_T(k_T))}{\partial c_T} + S'(G_T(k_T, h_T(k_T))) \frac{\partial G_T(k_T, h_T(k_T))}{\partial c_T} \right] h'_T(k_T) \\
&+ S'(G_T(k_T, h_T(k_T))) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} \\
&= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} + \left[\frac{\partial u_T(k_T, h_T(k_T))}{\partial c_T} + \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial c_T} \right] h'_T(k_T) \\
&+ \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} \quad \text{the term in } [\bullet] \text{ equals 0 by FOC of period } T \\
&= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} + \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T}
\end{aligned}$$

Then, the period- $T - 1$ FOC become:

$$\frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} = 0$$

$$\frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} + \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} - \lambda_{T-1} = 0$$

$$k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

But this system coincides exactly with the period- $T - 1$ FOC of our original problem. Hence:

$$c_{T-1} = h_{T-1}(k_{T-1})$$

$$k_T = g_{T-1}(k_{T-1}) \equiv G_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))$$

$$\lambda_{T-1} = \ell_{T-1}(k_{T-1})$$

Substituting into the objective function we get:

$$\begin{aligned} W_{T-1}(k_{T-1}) &= u_{T-1}(k_{T-1}, h_{T-1}(k_{T-1})) + W_T(g_{T-1}(k_{T-1})) \\ &= u_{T-1}(k_{T-1}, h_{T-1}(k_{T-1})) + W_T(G_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))) \end{aligned}$$

We can keep going backward in time and show that, in each period, the solution coincides with the solution to our original problem.

Bottom Line: we can solve our original problem by solving the sequence of one-period problems defined by the Bellman Equations.

Another perspective on the recursive nature of the problem can be obtained from the following observations.

First, notice that:

$$\begin{aligned}
W_0(k_0) &= \max_{c_0, k_1} \{u_0(k_0, c_0) + W_1(k_1)\} \\
&= \max_{c_0, k_1} \{u_0(k_0, c_0) + \max_{c_1, k_2} \{u_1(k_1, c_1) + W_2(k_2)\}\} \\
&= \max_{c_0, k_1} \{u_0(k_0, c_0) + \max_{c_1, k_2} \{u_1(k_1, c_1) \\
&\quad + \max_{c_2, k_3} \{u_2(k_2, c_2) + W_3(k_3)\}\}\} \\
&= \max_{c_0, k_1} \{u_0(k_0, c_0) + \max_{c_1, k_2} \{u_1(k_1, c_1) \\
&\quad + \max_{c_2, k_3} \{u_2(k_2, c_2) + \max_{c_3, k_4} \{u_3(k_3, c_3) + W_4(k_4)\}\}\}\} \\
&= \max_{c_0, k_1} \{u_0(k_0, c_0) + \max_{c_1, k_2} \{u_1(k_1, c_1) \\
&\quad + \max_{c_2, k_3} \{u_2(k_2, c_2) + \max_{c_3, k_4} \{u_3(k_3, c_3) \\
&\quad + \dots + \max_{c_T, k_{T+1}} \{u_T(k_T, c_T) + S(k_{T+1})\}\dots\}\}\}\}
\end{aligned}$$

where, in each step, the maximization is understood to be subject to $k_{t+1} = G_t(k_t, c_t)$, with k_t given.

Second, we know the solutions to our two problems coincide. Then:

$$W_0(k_0) = \max_{\{c_t, k_{t+1}\}_{t=0}^T} \{u_0(k_0, c_0) + \dots + u_T(k_T, c_T) + S(k_{T+1})\}$$

where the maximization is subject to $k_{t+1} = G_t(k_t, c_t)$ for all $t \in \{0, 1, \dots\}$, with k_0 given.

Combining the two results:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^T} \{u_0(k_0, c_0) + \dots + u_T(k_T, c_T) + S(k_{T+1})\} =$$

$$\max_{c_0, k_1} \{u_0(k_0, c_0) + \max_{c_1, k_2} \{u_1(k_1, c_1) + \max_{c_2, k_3} \{u_2(k_2, c_2) +$$

$$\max_{c_3, k_4} \{u_3(k_3, c_3) + \dots + \max_{c_T, k_{T+1}} \{u_T(k_T, c_T) + S(k_{T+1})\} \dots\}\}\}$$

The expression above shows that, for our problem, it is legitimate to cascade the maximization operator. That is, the large optimization problem of the left-hand side can be broken up into $T + 1$ smaller problems. First, the problem in the innermost brackets is solved and the optimal values of c_T and k_{T+1} are obtained for any value of k_T : $c_T = h_T(k_T)$ and $k_{T+1} = g_T(k_T)$. From these we obtain the optimized value $W_T(k_T)$. Then the problem in the second innermost brackets is solved, the optimizers being $c_{T-1} = h_{T-1}(k_{T-1})$ and $k_T = g_{T-1}(k_{T-1})$, with optimized value $W_{T-1}(k_{T-1})$. This process continues until the problem in the outermost brackets is reached.

The preceding argument implies the following principle, usually called the "Bellman's Principle of Optimality": An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

Bellman's Principle of Optimality (BPO)

Suppose the policy functions $\{c_t = h_t(k_t)\}_{t=0}^T$ solve

$$\max_{\{c_t, k_{t+1}\}_{t=0}^T} u_0(k_0, c_0) + \dots + u_T(k_T, c_T) + S(k_{T+1})$$

$$\text{s.t.} \quad k_{t+1} = G_t(k_t, c_t) \quad t \in \{0, 1, \dots, T\}$$

$$k_0 \text{ given}$$

Then $\{c_t = h_t(k_t)\}_{t=\tau}^T$ solve

$$\max_{\{c_t, k_{t+1}\}_{t=\tau}^T} u_\tau(k_\tau, c_\tau) + \dots + u_T(k_T, c_T) + S(k_{T+1})$$

$$\text{s.t.} \quad k_{t+1} = G_t(k_t, c_t) \quad t \in \{\tau, \tau + 1, \dots, T\}$$

$$k_\tau \text{ given}$$

for any $\tau \in \{0, 1, \dots, T\}$.

Remark. BPO implies that, as time advances, there is no incentive to depart from the original plan. Hence, optimal policies have a *self-enforcing* character. Optimal policies that satisfy BPO are said to be *time consistent*. This property is a consequence of the recursive character of the problem and will not characterize the solutions to more general problems.

Particular Case

$$u_t(k_t, c_t) = \beta^t u(k_t, c_t) \quad \text{with } \beta \in (0, 1)$$

Then, BE becomes:

$$W_t(k_t) \equiv \max_{c_t, k_{t+1}} \beta^t u(k_t, c_t) + W_{t+1}(k_{t+1})$$

$$\text{s.t.} \quad k_{t+1} = G_t(k_t, c_t)$$

$$k_t \text{ given}$$

Define the *current* value function:

$$V_t(k_t) \equiv \frac{W_t(k_t)}{\beta^t}$$

Remark. For $t = 0$, $V_0(k_0) = W_0(k_0)$

Then:

$$\begin{aligned} V_t(k_t) &= \frac{W_t(k_t)}{\beta^t} \\ &= \frac{\max_{c_t, k_{t+1}} \beta^t u(k_t, c_t) + W_{t+1}(k_{t+1})}{\beta^t} \\ &= \max_{c_t, k_{t+1}} \frac{\beta^t u(k_t, c_t) + W_{t+1}(k_{t+1})}{\beta^t} \\ &= \max_{c_t, k_{t+1}} u(k_t, c_t) + \frac{W_{t+1}(k_{t+1})}{\beta^t} \\ &= \max_{c_t, k_{t+1}} u(k_t, c_t) + \beta \frac{W_{t+1}(k_{t+1})}{\beta^{t+1}} \\ V_t(k_t) &= \max_{c_t, k_{t+1}} u(k_t, c_t) + \beta V_{t+1}(k_{t+1}) \end{aligned}$$

Up to now, we've been dealing with the following problem:

$$\begin{aligned}
& \max_{\{c_t, k_{t+1}\}_{t=0}^T} \sum_{t=0}^T u_t(k_t, c_t) + S(k_{T+1}) \\
& \text{s.t.} \quad k_{t+1} = G_t(k_t, c_t) \quad t \in \{0, 1, \dots, T\} \\
& \quad k_0 \text{ given}
\end{aligned}$$

In general, the solution to this problem gives a different policy function $c_t = h_t(k_t)$ to be used at each date $t \in \{0, 1, \dots, T\}$. This is a consequence of two features of the problem: the fact that the horizon is finite and the fact that the functions $u_t(\cdot)$ and $G_t(\cdot)$ have been permitted to depend on time in arbitrary ways.

We would like to discover contexts in which the same policy function $c_t = h(k_t)$ is used for each period t . These are called *time-invariant* or *stationary* policies. The previous remark suggests that we consider an *infinite-horizon* setup, and that we restrict the time dependence of $u_t(\cdot)$ and $G_t(\cdot)$ appropriately.

In addition to an infinite horizon, we'll assume:

$$u_t(k_t, c_t) = \beta^t u(k_t, c_t) \quad \beta \in (0, 1)$$

$$G_t(k_t, c_t) = G(k_t, c_t)$$

An Infinite-Horizon Problem

Consider the problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(k_t, c_t) = u(k_0, c_0) + \beta u(k_1, c_1) + \beta^2 u(k_2, c_2) \dots$$

$$\text{s.t.} \quad k_{t+1} = G(k_t, c_t) \quad t \in \{0, 1, \dots\}$$

$$k_0 \text{ given}$$

Notation: $\sum_{t=0}^{\infty} \beta^t u(k_t, c_t)$ denotes $\lim_{T \rightarrow \infty} \sum_{t=0}^T \beta^t u(k_t, c_t)$.

Define the period-1 continuation problem:

$$\max_{\{c_t, k_{t+1}\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} u(k_t, c_t) = u(k_1, c_1) + \beta u(k_2, c_2) + \beta^2 u(k_3, c_3) \dots$$

$$\text{s.t.} \quad k_{t+1} = G(k_t, c_t) \quad t \in \{1, 2, \dots\}$$

$$k_1 \text{ given}$$

Notice that both problems have exactly the same form, except for the initial value of the state variable (the time invariance of the functions $u(\cdot)$ and $G(\cdot)$ is crucial for this). Moreover, this is true for arbitrary periods, t and $t + s$.

The property described above suggests that the solution to the original problem will be a time-invariant policy function

$$c_t = h(k_t)$$

and that the value function for the original problem and any of the continuation problems will be the same. Hence, for any $t \in \{0, 1, \dots\}$, the Bellman Equation will be:

$$V(k_t) \equiv \max_{c_t, k_{t+1}} u(k_t, c_t) + \beta V(k_{t+1})$$

$$\text{s.t. } k_{t+1} = G(k_t, c_t)$$

$$k_t \text{ given}$$

Remark. To emphasize the irrelevance of the time subscript, some authors use c for the control variable, k for the current-period state and k' for its next-period value.

Substituting the constraint into the objective:

$$V(k_t) = \max_{c_t} u(k_t, c_t) + \beta V(G(k_t, c_t))$$

This is a *functional equation* (an equation in which the unknown is a function) to be solved for the function $V(\cdot)$. Notice that in the process of solving for $V(\cdot)$ we will be jointly solving for the policy function $h(\cdot)$.

There are different methods for solving the BE. We describe two of them.

1. *Value-function Iteration*

This method proceeds by constructing a sequence of value functions and associated policy functions. The sequence is created by iterating on

$$V_{j+1}(k) = \max_c u(k, c) + \beta V_j(G(k, c))$$

starting from some function V_0 (usually, $V_0 = 0$), and continuing until V_j has converged (*i.e.*, until $V_{j+1} = V_j$).

Remark. To show that this method works, one defines an operator T by $(TV)(k) = \max_c u(k, c) + \beta V(G(k, c))$. This operator maps V_j into $V_{j+1} : V_{j+1} = TV_j$. Under certain conditions, this operator is a contraction. Then, the contraction-mapping theorem ensures that T has a

unique fixed point, $V = TV$. Moreover, the fixed point can be calculated by successive iterations on $V_{j+1} = TV_j$ starting with some function V_0 (usually $V_0 = 0$).

2. *Guess and Verify*

This method involves guessing a solution V to the BE and verifying whether the guess is correct. The method of undetermined coefficients is used to verify the guess.

The Euler Equation

The previous solution methods require that we find the value function V . We show now that it may be possible to solve the problem without this knowledge.

Consider the Bellman Equation for the infinite-horizon problem:

$$V(k_t) = \max_{c_t} u(k_t, c_t) + \beta V(G(k_t, c_t))$$

If V were known (and differentiable), we could solve the problem in the right-hand side using standard calculus methods. The FOC for an interior solution is:

$$u_c(k_t, c_t) + \beta V'(G(k_t, c_t))G_c(k_t, c_t) = 0$$

The equation above defines, implicitly, the policy function:

$$c_t = h(k_t)$$

Substituting into the objective we get:

$$V(k_t) = u(k_t, h(k_t)) + \beta V(G(k_t, h(k_t)))$$

Differentiating with respect to k_t we get:

$$\begin{aligned} V'(k_t) &= \\ &= u_k(k_t, h(k_t)) + u_c(k_t, h(k_t))h'(k_t) \\ &\quad + \beta V'(G(k_t, h(k_t))) [G_k(k_t, h(k_t)) + G_c(k_t, h(k_t))h'(k_t)] \\ &= u_k(k_t, h(k_t)) + \beta V'(G(k_t, h(k_t))) G_k(k_t, h(k_t)) \\ &\quad + \underbrace{[u_c(k_t, h(k_t)) + \beta V'(G(k_t, h(k_t))) G_c(k_t, h(k_t))]}_{=0 \text{ by FOC}} h'(k_t) \end{aligned}$$

Then:

$$V'(k_t) = u_k(k_t, h(k_t)) + \beta V'(G(k_t, h(k_t))) G_k(k_t, h(k_t))$$

This is the so-called **Benveniste-Scheinkman (BS)** formula, or Envelope Condition (since it can be obtained through a direct application of the Envelope Theorem).

In many problems, there is no unique way of defining states and controls, and several alternative definitions lead to the same solution of the problem. Suppose we manage to choose the state and control variables in such a way that the transition equation does not depend on k_t , so $k_{t+1} = G(c_t)$ [see the appendix for the more general case]. Then, $G_k = 0$, and the BS formula becomes:

$$\begin{aligned} V'(k_t) &= u_k(k_t, h(k_t)) \\ &= u_k(k_t, c_t) \end{aligned}$$

Shifting the expression above one period ahead we get:

$$V'(k_{t+1}) = u_k(k_{t+1}, c_{t+1})$$

From the FOC:

$$u_c(k_t, c_t) + \beta V'(k_{t+1})G_c(k_t, c_t) = 0$$

Combining the two expressions above we obtain the *Euler Equation*:

$$u_c(k_t, c_t) + \beta u_k(k_{t+1}, c_{t+1})G_c(k_t, c_t) = 0$$

From $k_{t+1} = G(c_t)$ we get (assuming G can be inverted): $c_t = G^{-1}(k_{t+1}) \equiv m(k_{t+1})$. Then, the Euler Equation becomes:

$$u_c(k_t, m(k_{t+1})) + \beta u_k(k_{t+1}, m(k_{t+2})) G_c(k_t, m(k_{t+1})) = 0$$

which is a second-order difference equation in k . This equation can be solved using two boundary conditions: the initial condition k_0 , and an appropriate transversality condition (TVC).

Remark 1. Notice that we've managed to eliminate the value function V from our problem. The cost of doing this is that now we need to solve a nonlinear second-order difference equation. Moreover, in order to solve this equation we need to derive the appropriate TVC that will serve as our second boundary condition (more on this next class).

Remark 2. The Guess-Verify method can sometimes be used to solve the Euler Equation. Instead of guessing the form of the value function, we guess the form of the policy function $h(\bullet)$, and check that the Euler Equation is satisfied.

Remark 3. The solution to the Euler Equation will be an *open-loop* solution, since it will express k_t as a function of time only. Solutions of the form $c_t = h(k_t)$ and $k_{t+1} = g(k_t)$, like the ones we found earlier, are usually called *closed-loop* solutions.

Stochastic Dynamic Programming

Consider the (stationary) infinite-horizon stochastic problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(k_t, c_t)$$

$$\text{s.t.} \quad k_{t+1} = G(k_t, c_t, \varepsilon_t) \quad t \in \{0, 1, \dots\}$$

$$k_0 \text{ given}$$

where $\{\varepsilon_t\}_{t=0}^{\infty}$ is a sequence of random variables with conditional distribution function $f(\varepsilon_{t+1}|\varepsilon_t)$ for all t , and $\mathbb{E}_t x$ denotes the mathematical expectation of the random variable x conditional on information available at time t .

It is assumed that ε_t becomes known at t before c_t is chosen at t . The value of k_t is also known before c_t is chosen. This implies, through the transition equation, that k_{t+1} is known at t . Remark: In some specifications, the transition equation is of the form $k_{t+1} = G(k_t, c_t, \varepsilon_{t+1})$. In this case, ε_{t+1} and k_{t+1} are unknown at t . In other specifications, a stochastic shock is allowed to affect the return function: $u(k_t, c_t, \varepsilon_t)$.

The stochastic problem described above continues to have a recursive structure, which follows from the additive separability of the objective function in pairs (k_t, c_t) , and from the particular form assumed for the transition equation. This implies that dynamic programming methods remain appropriate.

The Bellman Equation is:

$$V(k_t, \varepsilon_t) = \max_{c_t} u(k_t, c_t) + \beta \mathbb{E}_t \{V(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1})\}$$

where we have used the transition equation to eliminate k_{t+1} . Notice that the state vector is now (k_t, ε_t) . The expectation is (assuming the shock is a continuous random variable):

$$\mathbb{E}_t \{V(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1})\} = \int V(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1}) f(\varepsilon_{t+1} | \varepsilon_t) d\varepsilon_{t+1}$$

By solving Bellman's functional equation we'll find the value function V , and *contingent plans* of the form:

$$c_t = h(k_t, \varepsilon_t)$$

and

$$k_{t+1} = g(k_t, \varepsilon_t) \equiv G(k_t, h(k_t, \varepsilon_t), \varepsilon_t)$$

The solution V of the Bellman Equation can be found by iterating on

$$V_{j+1}(k_t, \varepsilon_t) = \max_{c_t} u(k_t, c_t) + \beta \mathbb{E}_t \left\{ V_j(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1}) \right\}$$

starting from an appropriate function V_0 (e.g., $V_0 = 0$).

For some cases it is also possible to use the Guess-Verify method.

We can also obtain the *stochastic Euler Equation*. The FOC give:

$$u_c(k_t, c_t) + \beta \mathbb{E}_t \{ V_k(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1}) G_c(k_t, c_t, \varepsilon_t) \} = 0$$

Then:

$$u_c(k_t, c_t) + \beta \mathbb{E}_t \{ V_k(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1}) \} G_c(k_t, c_t, \varepsilon_t) = 0$$

The equation above defines, implicitly, the contingency plan:

$$c_t = h(k_t, \varepsilon_t)$$

Substituting into the objective we get:

$$V(k_t, \varepsilon_t) = u(k_t, h(k_t, \varepsilon_t)) + \beta \mathbb{E}_t \{ V(G(k_t, h(k_t, \varepsilon_t), \varepsilon_t), \varepsilon_{t+1}) \}$$

Differentiating with respect to k_t we get:

$$\begin{aligned}
V_k(k_t, \varepsilon_t) &= u_k(k_t, h(k_t, \varepsilon_t)) + u_c(k_t, h(k_t, \varepsilon_t)) h_k(k_t, \varepsilon_t) \\
&+ \beta \mathbb{E}_t \left\{ \begin{aligned} &V_k(G(k_t, h(k_t, \varepsilon_t), \varepsilon_t), \varepsilon_{t+1}) \times \\ &\times [G_k(k_t, h(k_t, \varepsilon_t), \varepsilon_t) + G_c(k_t, h(k_t, \varepsilon_t), \varepsilon_t) h_k(k_t, \varepsilon_t)] \end{aligned} \right\} \\
&= u_k(k_t, h(k_t, \varepsilon_t)) \\
&+ \beta \mathbb{E}_t \{ V_k(G(k_t, h(k_t, \varepsilon_t), \varepsilon_t), \varepsilon_{t+1}) \} G_k(k_t, h(k_t, \varepsilon_t), \varepsilon_t) \\
&+ \left[\begin{aligned} &u_c(k_t, h(k_t, \varepsilon_t)) \\ &+ \beta \mathbb{E}_t \{ V_k(G(k_t, h(k_t, \varepsilon_t), \varepsilon_t), \varepsilon_{t+1}) \} G_c(k_t, h(k_t, \varepsilon_t)) \end{aligned} \right] h_k(k_t, \varepsilon_t)
\end{aligned}$$

where the last term in the right-hand side of the second equality drops because of the FOC.

Then:

$$\begin{aligned}
V_k(k_t, \varepsilon_t) &= u_k(k_t, h(k_t, \varepsilon_t)) \\
&+ \beta \mathbb{E}_t \{ V_k(G(k_t, h(k_t, \varepsilon_t), \varepsilon_t), \varepsilon_{t+1}) \} G_k(k_t, h(k_t, \varepsilon_t), \varepsilon_t)
\end{aligned}$$

This is the **Benveniste-Scheinkman (BS)** formula (or Envelope Condition) for the stochastic case.

Suppose we manage to choose the state and control variables in such a way that the transition equation does not depend on k_t , so $k_{t+1} = G(c_t, \varepsilon_t)$. Then, $G_k = 0$ and the BS formula becomes:

$$\begin{aligned} V_k(k_t, \varepsilon_t) &= u_k(k_t, h(k_t, \varepsilon_t)) \\ &= u_k(k_t, c_t) \end{aligned}$$

Shifting the expression above one period ahead we get:

$$V_k(k_{t+1}, \varepsilon_{t+1}) = u_k(k_{t+1}, c_{t+1})$$

From the FOC:

$$u_c(k_t, c_t) + \beta \mathbb{E}_t \{V_k(k_{t+1}, \varepsilon_{t+1})\} G_c(k_t, c_t, \varepsilon_t) = 0$$

Combining the two expressions above we obtain the *stochastic Euler Equation*:

$$u_c(k_t, c_t) + \beta \mathbb{E}_t \{u_k(k_{t+1}, c_{t+1})\} G_c(k_t, c_t, \varepsilon_t) = 0$$

Appendix: The Euler Equation Again

The FOC gives:

$$u_c(k_t, c_t) + \beta V'(k_{t+1})G_c(k_t, c_t) = 0$$

The BS formula gives:

$$V'(k_t) = u_k(k_t, c_t) + \beta V'(k_{t+1})G_k(k_t, c_t)$$

From the expression above we get:

$$V'(k_{t+1}) = \frac{V'(k_t) - u_k(k_t, c_t)}{\beta G_k(k_t, c_t)}$$

Substituting into the FOC we get

$$u_c(k_t, c_t) + \beta \frac{V'(k_t) - u_k(k_t, c_t)}{\beta G_k(k_t, c_t)} G_c(k_t, c_t) = 0$$

which gives:

$$V'(k_t) = u_k(k_t, c_t) - u_c(k_t, c_t) \frac{G_k(k_t, c_t)}{G_c(k_t, c_t)}$$

Shifting the expression above one period ahead we get:

$$V'(k_{t+1}) = u_k(k_{t+1}, c_{t+1}) - u_c(k_{t+1}, c_{t+1}) \frac{G_k(k_{t+1}, c_{t+1})}{G_c(k_{t+1}, c_{t+1})}$$

Substituting the previous expression into the FOC we obtain the Euler Equation:

$$u_c(k_t, c_t) + \beta \left[u_k(k_{t+1}, c_{t+1}) - u_c(k_{t+1}, c_{t+1}) \frac{G_k(k_{t+1}, c_{t+1})}{G_c(k_{t+1}, c_{t+1})} \right] G_c(k_t, c_t) = 0$$

If we rewrite the transition equation $k_{t+1} = G(k_t, c_t)$ as $c_t = m(k_t, k_{t+1})$, the Euler Equation becomes:

$$0 = u_c(k_t, m(k_t, k_{t+1})) + \beta \left[u_k(k_{t+1}, m(k_{t+1}, k_{t+2})) - u_c(k_{t+1}, m(k_{t+1}, k_{t+2})) \frac{G_k(k_{t+1}, m(k_{t+1}, k_{t+2}))}{G_c(k_{t+1}, m(k_{t+1}, k_{t+2}))} \right] G_c(k_t, m(k_t, k_{t+1}))$$

which is a second-order difference equation in k . This equation can be solved using two boundary conditions: the initial condition k_0 , and an appropriate transversality condition (TVC).

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A Simple Example

Choose c_0 , c_1 , x_1 , and x_2 , in order to solve:

$$\begin{aligned} \max \quad & \ln c_0 + \ln c_1 + \ln x_2 \\ \text{s.t.} \quad & x_1 = x_0 - c_0 \\ & x_2 = x_1 - c_1 \\ & x_0 > 0 \text{ given} \end{aligned}$$

Lagrange

$$\mathcal{L} = \ln c_0 + \ln c_1 + \ln x_2 + \lambda_1(x_0 - c_0 - x_1) + \lambda_2(x_1 - c_1 - x_2)$$

FOC

$$c_0 : \frac{1}{c_0} = \lambda_1$$

$$c_1 : \frac{1}{c_1} = \lambda_2$$

$$x_1 : \lambda_1 = \lambda_2$$

$$x_2 : \frac{1}{x_2} = \lambda_2$$

+ constraints

From the expressions above we get:

$$\frac{1}{c_0} = \frac{1}{c_1} = \frac{1}{x_2}$$

Then:

$$x_2 = c_1 = c_0$$

Substituting into the second constraint: $c_0 = x_1 - c_0 \Rightarrow$

$$x_1 = 2c_0$$

Substituting into the first constraint: $2c_0 = x_0 - c_0 \Rightarrow$

$$3c_0 = x_0$$

$$c_0^* = \frac{1}{3}x_0$$

Then: $x_2 = c_1 = c_0 \Rightarrow$

$$c_1^* = \frac{1}{3}x_0$$

$$x_2^* = \frac{1}{3}x_0$$

Then: $x_1 = 2c_0 \Rightarrow$

$$x_1^* = \frac{2}{3}x_0$$

Dynamic Programming

At $t = 1$:

$$W_1(x_1) \equiv \max_{c_1, x_2} \ln c_1 + W_2(x_2)$$

$$\text{s.t.} \quad x_2 = x_1 - c_1$$

$$x_1 \text{ given}$$

with $W_2(x_2) = \ln x_2$.

Then:

$$W_1(x_1) \equiv \max_{c_1, x_2} \ln c_1 + \ln x_2$$

$$\text{s.t.} \quad x_2 = x_1 - c_1$$

$$x_1 \text{ given}$$

Then:

$$W_1(x_1) \equiv \max_{c_1} \ln c_1 + \ln(x_1 - c_1)$$

FOC:

$$\frac{1}{c_1} = \frac{1}{x_1 - c_1}$$

Then: $x_1 - c_1 = c_1 \Rightarrow$

$$c_1 = \frac{1}{2}x_1$$

Substituting into the constraint: $x_2 = x_1 - c_1 \Rightarrow x_2 = x_1 - \frac{1}{2}x_1 \Rightarrow$

$$x_2 = \frac{1}{2}x_1$$

Substituting into the objective: $W_1(x_1) = \ln(\frac{1}{2}x_1) + \ln(\frac{1}{2}x_1) \Rightarrow W_1(x_1) = 2\ln(\frac{1}{2}x_1) \Rightarrow$

$$W_1(x_1) = 2\ln(\frac{1}{2}) + 2\ln x_1$$

At $t = 0$

$$W_0(x_0) \equiv \max_{c_0, x_1} \ln c_0 + W_1(x_1)$$

$$\text{s.t.} \quad x_1 = x_0 - c_0$$

$$x_0 \text{ given}$$

where $W_1(x_1)$ was found in the previous step.

Then:

$$W_0(x_0) \equiv \max_{c_0, x_1} \ln c_0 + 2 \ln\left(\frac{1}{2}\right) + 2 \ln x_1$$

$$\text{s.t.} \quad x_1 = x_0 - c_0$$

$$x_0 \text{ given}$$

Then:

$$W_0(x_0) \equiv \max_{c_0} \ln c_0 + 2 \ln\left(\frac{1}{2}\right) + 2 \ln(x_0 - c_0)$$

FOC:

$$\frac{1}{c_0} = \frac{2}{x_0 - c_0}$$

Then: $x_0 - c_0 = 2c_0 \Rightarrow 3c_0 = x_0 \Rightarrow$

$$c_0^* = \frac{1}{3}x_0$$

Substituting into the constraint: $x_1 = x_0 - c_0 \Rightarrow x_1 = x_0 - \frac{1}{3}x_0 \Rightarrow$

$$x_1^* = \frac{2}{3}x_0$$

Substituting the expression above into $c_1 = \frac{1}{2}x_1$ we get:
 $c_1 = \frac{1}{2} \cdot \frac{2}{3}x_0 \Rightarrow$

$$c_1^* = \frac{1}{3}x_0$$

Substituting $x_1 = \frac{2}{3}x_0$ into $x_2 = \frac{1}{2}x_1$ we get:

$$x_2^* = \frac{1}{3}x_0$$

Notice that both methods give the same solution.

Neoclassical Growth Model

$$Y_t = F(K_t, A_t L_t) \quad \text{production function}$$

$$I_t = K_{t+1} - (1 - \delta)K_t \quad \text{gross investment}$$

$$C_t + I_t \leq Y_t \quad \text{resource constraint}$$

$$L_t = (1 + n)^t L_0, \quad n \geq 0 \quad \text{labor = population}$$

$$A_t = (1 + g)^t A_0, \quad g \geq 0 \quad \text{technology (labor augmenting)}$$

$$\hat{U} = \sum_{t=0}^{\infty} \hat{\beta}^t u(\hat{c}_t) \frac{L_t}{H} \quad \text{rep. household's utility}$$

where H = number of households, $\frac{L_t}{H}$ = number of members of the rep. household, $\hat{c}_t = \frac{C_t}{L_t}$ = consumption per capita, and $\hat{\beta} \equiv \frac{1}{1+\rho}$ = discount factor, with $\rho > 0$ (ρ is the discount rate).

For any variable X , define:

$$x_t \equiv \frac{X_t}{A_t L_t}$$

By CRS: $\lambda Y_t = F(\lambda K_t, \lambda A_t L_t) \quad \forall \lambda > 0$. Choosing $\lambda = \frac{1}{A_t L_t}$: $\frac{Y_t}{A_t L_t} = F(\frac{K_t}{A_t L_t}, 1)$. Then:

$$y_t = f(k_t)$$

where $f(k_t) \equiv F(k_t, 1)$.

We can rewrite the resource constraint as follows:

$$C_t + K_{t+1} - (1 - \delta)K_t \leq Y_t$$

Dividing by $A_t L_t$:

$$\frac{C_t}{A_t L_t} + \frac{K_{t+1}}{A_{t+1} L_{t+1}} \frac{A_{t+1} L_{t+1}}{A_t L_t} - (1 - \delta) \frac{K_t}{A_t L_t} \leq \frac{Y_t}{A_t L_t}$$

Then:

$$c_t + k_{t+1}(1 + g)(1 + n) - (1 - \delta)k_t \leq f(k_t)$$

We can rewrite the utility function as follows:

$$\begin{aligned}
\hat{U} &= \sum_{t=0}^{\infty} \hat{\beta}^t u(\hat{c}_t) \frac{L_t}{H} \\
&= \sum_{t=0}^{\infty} \hat{\beta}^t u\left(\frac{C_t}{L_t}\right) \frac{L_t}{H} \\
&= \sum_{t=0}^{\infty} \hat{\beta}^t u\left(\frac{C_t A_t}{L_t A_t}\right) \frac{L_t}{H} \\
&= \sum_{t=0}^{\infty} \left(\frac{1}{1+\rho}\right)^t u(c_t A_t) \frac{L_0(1+n)^t}{H} \\
&= \frac{L_0}{H} \sum_{t=0}^{\infty} \left(\frac{1+n}{1+\rho}\right)^t u(c_t A_t) \\
&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t u(c_t A_t)
\end{aligned}$$

where $\beta \equiv \frac{1+n}{1+\rho}$. Assume $\rho > n$, so $\beta \in (0, 1)$.

Particular Case: Log Utility, Cobb-Douglas Technology, 100% Depreciation

Assume:

$$u(\hat{c}_t) = \ln \hat{c}_t$$

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha} \quad \alpha \in (0, 1)$$

$$K_{t+1} = I_t \quad (i.e., \delta = 1)$$

Then:

$$\begin{aligned} y_t &\equiv \frac{Y_t}{A_t L_t} \\ &= \frac{K_t^\alpha (A_t L_t)^{1-\alpha}}{A_t L_t} \\ &= \frac{K_t^\alpha}{(A_t L_t)^\alpha} \\ &= \left(\frac{K_t}{A_t L_t} \right)^\alpha \\ &= k_t^\alpha \end{aligned}$$

Then:

$$c_t + (1 + g)(1 + n)k_{t+1} \leq k_t^\alpha$$

Also

$$\begin{aligned}\hat{U} &= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln(\hat{c}_t) \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln(c_t A_t) \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t (\ln A_t + \ln c_t) \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln A_t + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t \\&= \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t [\ln A_0 + t \ln(1 + g)] + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t \\&= \frac{L_0}{H} \ln A_0 \sum_{t=0}^{\infty} \beta^t + \frac{L_0}{H} \ln(1 + g) \sum_{t=0}^{\infty} \beta^t t + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t \\&= \frac{L_0 \ln A_0}{H} \frac{1}{1-\beta} + \frac{L_0 \ln(1+g)\beta}{H(1-\beta)^2} + \frac{L_0}{H} \sum_{t=0}^{\infty} \beta^t \ln c_t\end{aligned}$$

Then:

$$\hat{U} = \text{constant}_1 + \text{constant}_2 \sum_{t=0}^{\infty} \beta^t \ln c_t$$

Since $\text{constant}_2 > 0$ we can work with:

$$U \equiv \sum_{t=0}^{\infty} \beta^t \ln c_t$$

The Planner's Problem

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln c_t$$

$$\text{s.t.} \quad c_t + (1+n)(1+g)k_{t+1} \leq k_t^\alpha$$

$$k_0 > 0 \text{ given}$$

Remark 1. It's clear that the constraint will be binding at the optimum.

Remark 2. We didn't include non-negativity constraints because our assumptions ensure that consumption and capital will be strictly positive at the optimum.

Bellman Equation

For any period t :

$$V(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta V(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_t \text{ given}$$

Solution Method: Value-function Iteration

1. Computation of V_1

Setting $V_0 \equiv 0$ we get:

$$V_1(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_{t+1} \geq 0$$

$$k_t \text{ given}$$

Solution:

$$k_{t+1} = 0$$

$$c_t = k_t^\alpha$$

$$V_1(k_t) = \alpha \ln k_t$$

Remark. We need to impose the constraint $k_{t+1} \geq 0$ explicitly, otherwise, with $V_0 \equiv 0$, the problem has no solution.

2. Computation of V_2

$$V_2(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta V_1(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_t \text{ given}$$

From the first iteration we know $V_1(k_{t+1}) = \alpha \ln k_{t+1}$.
Then, using the constraint to eliminate k_{t+1} :

$$V_2(k_t) \equiv \max_{c_t} \ln c_t + \beta \alpha \ln \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)$$

FOC:

$$\frac{1}{c_t} + \alpha \beta \frac{1}{\frac{k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \alpha \beta} k_t^\alpha$$

Then:

$$k_{t+1} = \frac{k_t^\alpha - \frac{1}{1+\alpha\beta} k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\alpha \beta}{1 + \alpha \beta} k_t^\alpha$$

Then:

$$\begin{aligned} V_2(k_t) &= \ln\left(\frac{1}{1+\alpha\beta} k_t^\alpha\right) + \alpha\beta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\alpha\beta}{1+\alpha\beta} k_t^\alpha\right) \\ &= \ln\left(\frac{1}{1+\alpha\beta}\right) + \alpha \ln k_t + \alpha\beta \ln\left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)}\right) + \beta\alpha^2 \ln k_t \\ &= \left[\ln\left(\frac{1}{1+\alpha\beta}\right) + \alpha\beta \ln\left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)}\right) \right] + \alpha(1+\alpha\beta) \ln k_t \\ &= C + \alpha(1+\alpha\beta) \ln k_t \end{aligned}$$

3. Computation of V_3

$$V_3(k_t) \equiv \max_{c_t, k_t} \ln c_t + \beta V_2(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

k_t given

Using our result for $V_2(k_{t+1})$:

$$V_3(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta[C + \alpha(1 + \alpha\beta) \ln k_{t+1}]$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

k_t given

Using the constraint to eliminate k_{t+1} :

$$V_3(k_t) \equiv \max_{c_t} \ln c_t + \beta C + \alpha\beta(1 + \alpha\beta) \ln \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)$$

FOC:

$$\frac{1}{c_t} + \alpha\beta(1 + \alpha\beta) \frac{1}{\frac{k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \alpha\beta + (\alpha\beta)^2} k_t^\alpha$$

Then:

$$k_{t+1} = \frac{k_t^\alpha - \frac{1}{1+\alpha\beta+(\alpha\beta)^2}k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\alpha\beta(1+\alpha\beta)}{1+\alpha\beta+(\alpha\beta)^2} k_t^\alpha$$

Then:

$$V_3(k_t) = \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}k_t^\alpha\right) + \beta \left[C + \alpha(1+\alpha\beta) \ln\left(\frac{1}{(1+n)(1+g)} \frac{\alpha\beta+(\alpha\beta)^2}{1+\alpha\beta+(\alpha\beta)^2} k_t^\alpha\right) \right]$$

$$V_3(k_t) = \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}\right) + \alpha \ln k_t + \beta C + \alpha\beta(1+\alpha\beta) \ln\left(\frac{1}{(1+n)(1+g)} \frac{\alpha\beta+(\alpha\beta)^2}{1+\alpha\beta+(\alpha\beta)^2}\right) + \alpha\beta(1+\alpha\beta)\alpha \ln k_t$$

$$V_3(k_t) = \left[\ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}\right) + \beta C + \alpha\beta(1+\alpha\beta) \ln\left(\frac{\alpha\beta(1+\alpha\beta)}{(1+n)(1+g)[1+\alpha\beta+(\alpha\beta)^2]}\right) + \alpha[1+\alpha\beta+(\alpha\beta)^2] \ln k_t \right]$$

$$\begin{aligned}
V_3(k_t) = & \left[\ln \left(\frac{1}{1 + \alpha\beta + (\alpha\beta)^2} \right) + \beta \ln \left(\frac{1}{1 + \alpha\beta} \right) + \alpha\beta^2 \ln \left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)} \right) \right. \\
& \left. + \alpha\beta(1 + \alpha\beta) \ln \left(\frac{\alpha\beta(1 + \alpha\beta)}{(1+n)(1+g)[1 + \alpha\beta + (\alpha\beta)^2]} \right) \right] \\
& + \alpha[1 + \alpha\beta + (\alpha\beta)^2] \ln k_t
\end{aligned}$$

Then:

$$V_3(k_t) = D + \alpha[1 + \alpha\beta + (\alpha\beta)^2] \ln k_t$$

4. Computation of V_4

Using the same procedure, we get:

$$c_t = \frac{1}{1 + \alpha\beta + (\alpha\beta)^2 + (\alpha\beta)^3} k_t^\alpha$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\alpha\beta[1 + \alpha\beta + (\alpha\beta)^2]}{1 + \alpha\beta + (\alpha\beta)^2 + (\alpha\beta)^3} k_t^\alpha$$

and

$$\begin{aligned}
V_4(k_t) = & \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2+(\alpha\beta)^3}\right) + \beta \ln\left(\frac{1}{1+\alpha\beta+(\alpha\beta)^2}\right) \\
& + \beta^2 \ln\left(\frac{1}{1+\alpha\beta}\right) + \alpha\beta^3 \ln\left(\frac{\alpha\beta}{(1+n)(1+g)(1+\alpha\beta)}\right) \\
& + \alpha\beta^2(1+\alpha\beta) \ln\left(\frac{\alpha\beta(1+\alpha\beta)}{(1+n)(1+g)[1+\alpha\beta+(\alpha\beta)^2]}\right) \\
& + \alpha\beta[1+\alpha\beta+(\alpha\beta)^2] \ln\left(\frac{\alpha\beta[1+\alpha\beta+(\alpha\beta)^2]}{(1+n)(1+g)[1+\alpha\beta+(\alpha\beta)^2+(\alpha\beta)^3]}\right) \\
& + \alpha[1+\alpha\beta+(\alpha\beta)^2+(\alpha\beta)^3] \ln k_t
\end{aligned}$$

$$V_4(k_t) = E + \alpha[1 + \alpha\beta + (\alpha\beta)^2 + (\alpha\beta)^3] \ln k_t$$

5. Computation of V_j

From the previous steps we recognize the following pattern:

$$c_t = \frac{1}{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1}} k_t^\alpha$$

$$k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} \frac{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-2}}{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1}} k_t^\alpha$$

and

$$V_j(k_t) = e_j + d_j \ln k_t$$

where:

$$d_j = \alpha[1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1}]$$

$$e_j = a_j + b_j$$

$$a_j = \sum_{s=0}^{j-2} \beta^s \ln \left(\frac{1}{1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1-s}} \right)$$

$$b_j = \alpha\beta \sum_{s=0}^{j-2} \beta^s \left[\begin{array}{l} [1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-2-s}] \\ \times \ln \left(\frac{\alpha\beta[1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-2-s}]}{(1+n)(1+g)[1 + \alpha\beta + (\alpha\beta)^2 + \dots + (\alpha\beta)^{j-1-s}]} \right) \end{array} \right]$$

6. Convergence

Letting $j \rightarrow \infty$ we obtain:

$$c_t = (1 - \alpha\beta) k_t^\alpha$$

$$k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} k_t^\alpha$$

$$V(k_t) = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln \left(\frac{\alpha\beta}{(1+n)(1+g)} \right) \right] + \frac{\alpha}{1-\alpha\beta} \ln k_t$$

Notice that:

$$c_t = (1 - \alpha\beta) y_t$$

$$C_t = (1 - \alpha\beta) Y_t$$

Hence, the saving rate is constant:

$$\begin{aligned} s &= \alpha\beta \\ &= \frac{\alpha(1+n)}{1+\rho} \end{aligned}$$

Remark. The model looks like a Solow Model, except that now we know the determinants of the saving rate.

Solution Method: Guess and Verify

Guess:

$$V^G(k_t) = \Omega + \Theta \ln k_t$$

where Ω and Θ are parameters to be determined.

Now solve:

$$\tilde{V}(k_t) \equiv \max_{c_t, k_{t+1}} \ln c_t + \beta V^G(k_{t+1})$$

$$\text{s.t.} \quad (1+n)(1+g)k_{t+1} = k_t^\alpha - c_t$$

$$k_t \text{ given}$$

The idea is to find $\tilde{V}(k_t)$ and then look for the values of Ω and Θ that make $\tilde{V}(k_t) = V^G(k_t) \forall k_t$.

We can rewrite the problem above as follows:

$$\tilde{V}(k_t) = \max_{c_t} \ln c_t + \beta \left[\Omega + \Theta \ln \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right) \right]$$

FOC:

$$\frac{1}{c_t} + \beta\Theta \frac{1}{\frac{k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \beta\Theta} k_t^\alpha$$

Then:

$$k_{t+1} = \frac{k_t^\alpha - \frac{1}{1+\beta\Theta} k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1 + \beta\Theta} k_t^\alpha$$

Then:

$$\tilde{V}(k_t) = \ln\left(\frac{1}{1+\beta\Theta} k_t^\alpha\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta} k_t^\alpha\right) \Rightarrow$$

$$\tilde{V}(k_t) = \left[\ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right) \right] + \alpha(1+\beta\Theta) \ln k_t$$

Then, $\tilde{V}(k_t) = V^G(k_t) \forall k_t$ requires:

$$\Omega = \ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right)$$

$$\Theta = \alpha(1+\beta\Theta)$$

From the second equation:

$$\Theta = \frac{\alpha}{1 - \alpha\beta}$$

Substit. into the first:

$$\Omega = \frac{1}{1 - \beta} \left[\ln(1 - \alpha\beta) + \frac{\alpha\beta}{1 - \alpha\beta} \ln\left(\frac{\alpha\beta}{(1+n)(1+g)}\right) \right]$$

Then:

$$V(k_t) = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln\left(\frac{\alpha\beta}{(1+n)(1+g)}\right) \right] + \frac{\alpha}{1-\alpha\beta} \ln k_t$$

which coincides with the value function we found through value-function iteration.

Also, $c_t = \frac{1}{1+\beta\Theta} k_t^\alpha$ and $k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta} k_t^\alpha$
become:

$$c_t = (1 - \alpha\beta) k_t^\alpha$$

and

$$k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} k_t^\alpha$$

which coincide with the expressions we found through
value-function iteration.

Solution Method: FOC + B&S (Envelope) $\Rightarrow$ Euler

$$V(k_t) \equiv \max_{c_t, k_{t+1}} u(c_t) + \beta V(k_{t+1})$$

$$\text{s.t.} \quad c_t = f(k_t) - (1+n)(1+g)k_{t+1}$$

k_t given

Then:

$$V(k_t) \equiv \max_{k_{t+1}} u(f(k_t) - (1+n)(1+g)k_{t+1}) + \beta V(k_{t+1})$$

Remark. By using the constraint to eliminate c_t we have a problem where the state variable in period t is k_t and the control variable is k_{t+1} . The transition equation is simply $k_{t+1} = k_{t+1}$ (that is, the control variable in period t fully determines the value of the state in $t + 1$). This is simpler than the alternative of using the constraint to replace k_{t+1} in $V(k_{t+1})$ because in the former case the transition equation for period t does not depend on the state at t , which makes the application of the envelope condition (the B&S formula) simpler (we discussed this during the first lecture).

FOC

$$-u'(f(k_t) - (1+n)(1+g)k_{t+1})(1+n)(1+g) + \beta V'(k_{t+1}) = 0$$

The expression above implicitly defines:

$$k_{t+1} = h(k_t)$$

Substituting $k_{t+1} = h(k_t)$ into the objective:

$$V(k_t) = u(f(k_t) - (1+n)(1+g)h(k_t)) + \beta V(h(k_t))$$

Differentiating:

$$\begin{aligned} V'(k_t) = & u'(f(k_t) - (1+n)(1+g)h(k_t)) [f'(k_t) - (1+n)(1+g)h'(k_t)] \\ & + \beta V'(h(k_t))h'(k_t) \end{aligned}$$

$$\begin{aligned} V'(k_t) = & u'(f(k_t) - (1+n)(1+g)h(k_t))f'(k_t) \\ & + \left[\begin{array}{l} -u'(f(k_t) - (1+n)(1+g)h(k_t))(1+n)(1+g) \\ + \beta V'(h(k_t)) \end{array} \right] h'(k_t) \end{aligned}$$

The second term in the r-h-s is zero by FOC. Then:

$$V'(k_t) = u'(f(k_t) - (1+n)(1+g)h(k_t)) f'(k_t)$$

Shifting the expression above one period ahead we get (B&S/Envelope Condition):

$$V'(k_{t+1}) = u'(f(k_{t+1}) - (1+n)(1+g)h(k_{t+1})) f'(k_{t+1})$$

Substituting into FOC, and using $h(k_{t+1}) = k_{t+2}$:

$$\begin{aligned} & -u'(f(k_t) - (1+n)(1+g)k_{t+1}) (1+n)(1+g) \\ & + \beta u'(f(k_{t+1}) - (1+n)(1+g)k_{t+2}) f'(k_{t+1}) = 0 \end{aligned}$$

This is the Euler equation. It's a second-order difference equation in k .

The two boundary conditions are:

$$k_0 = \bar{k}_0 \quad (\text{initial condition})$$

and

$$\lim_{t \rightarrow \infty} \beta^t u' (f(k_t) - (1 + n)(1 + g)k_{t+1}) k_{t+1} = 0 \quad (\text{TVC})$$

Remark 1. Read the note on the transversality condition (TVC).

Remark 2. We can write the Euler equation and the TVC as follows:

$$-u'(c_t)(1 + n)(1 + g) + \beta u'(c_{t+1}) f'(k_{t+1}) = 0$$

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0$$

For our particular case, the Euler Equation gives:

$$c_{t+1} = \frac{\alpha\beta}{(1 + n)(1 + g)} k_{t+1}^{\alpha-1} c_t$$

Remark 3. Check that our previous solution satisfies the Euler equation and the TVC.

Steady State

From $k_{t+1} = \frac{\alpha\beta}{(1+n)(1+g)} k_t^\alpha$ and $k_t = k_{t+1}$:

$$\begin{aligned} k_{ss} &= \left[\frac{\alpha\beta}{(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}} \\ &= \left[\frac{\alpha}{(1+\rho)(1+g)} \right]^{\frac{1}{1-\alpha}} \quad \text{since } \beta \equiv \frac{1+n}{1+\rho} \end{aligned}$$

From the policy function:

$$c_{ss} = (1 - \alpha\beta) k_{ss}^\alpha$$

From the production function:

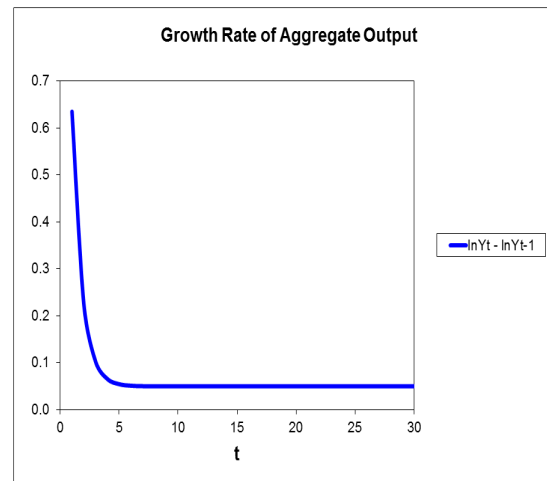
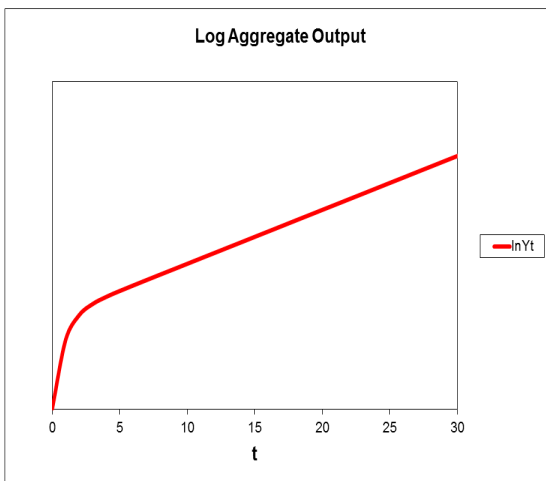
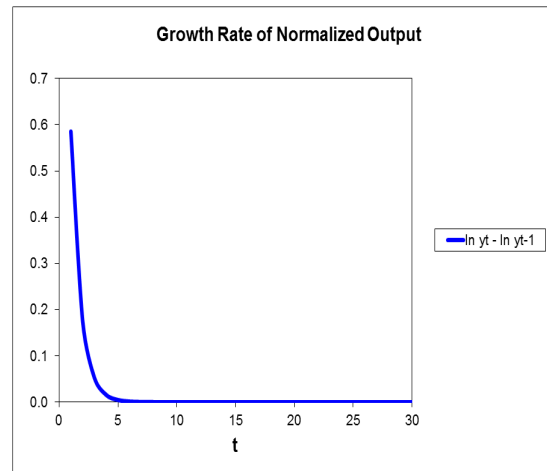
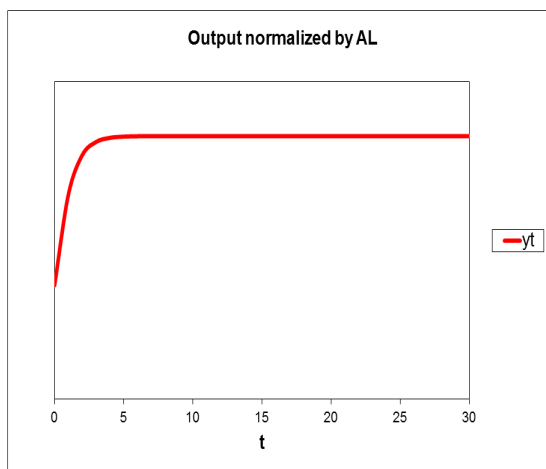
$$y_{ss} = k_{ss}^\alpha$$

From $k_t = \frac{K_t}{A_t L_t}$, $c_t = \frac{C_t}{A_t L_t}$, and $y_t = \frac{Y_t}{A_t L_t}$ we conclude that, in the s-s, the (gross) growth rate of K , C , and Y is $(1+g)(1+n)$ [balanced growth].

From $\hat{c}_t = \frac{C_t}{L_t}$ and $\hat{y}_t = \frac{Y_t}{L_t}$ we conclude that, in s-s, the (gross) growth rate of $\hat{c}_t$ and $\hat{y}_t$ is $1+g$.

Plots

Suppose: $\alpha = 0.3$, $\rho = 0.05$, $n = 0.03$, $g = 0.02$, $k_0 = 0.01$, $L_0 = 1$, $A_0 = 4$. These imply: $k_{ss} = 0.162$ and $y_{ss} = 0.58$.



Decentralized Version

Producer

$$\max_{K_t, L_t} K_t^\alpha (A_t L_t)^{1-\alpha} - w_t L_t - R_t K_t$$

FOC give:

$$R_t = \alpha k_t^{\alpha-1}$$

$$w_t = (1 - \alpha) A_t k_t^\alpha$$

Substituting these expressions into the objective function we see that profits are zero.

Household

Just for simplicity, normalize $H = 1$ so we have a unique household of size L_t at t .

$$\max_{\{C_t, I_t, K_{t+1}, B_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \hat{\beta}^t \ln \left(\frac{C_t}{L_t} \right) L_t$$

$$\text{s.t.} \quad C_t + I_t + B_t = w_t L_t + R_t K_t + (1 + r_{t-1}) B_{t-1}$$

$$I_t = K_{t+1} \quad (\text{since } \delta = 1)$$

$$K_0 > 0, B_{-1} = 0 \text{ given}$$

where we have imposed zero profits in the income side.

Since $L_t = L_0(1 + n)^t$ we can solve:

$$\max_{\{C_t, K_{t+1}, B_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln \left(\frac{C_t}{L_t} \right)$$

$$\text{s.t.} \quad C_t + K_{t+1} + B_t = w_t L_t + R_t K_t + (1 + r_{t-1}) B_{t-1}$$

$$K_0 > 0, B_{-1} = 0 \text{ given}$$

where $\beta \equiv \frac{1+n}{1+\rho} \in (0, 1)$.

The Lagrangian is:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \ln\left(\frac{C_t}{L_t}\right) + \sum_{t=0}^{\infty} \lambda_t [w_t L_t + R_t K_t + (1+r_{t-1})B_{t-1} - C_t - K_{t+1} - B_t]$$

FOC

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow \beta^t \frac{1}{C_t} \frac{1}{L_t} = \lambda_t \Rightarrow \frac{\beta^t}{C_t} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial B_t} = 0 \Rightarrow \lambda_t = \lambda_{t+1}(1 + r_t)$$

$$\frac{\partial \mathcal{L}}{\partial K_{t+1}} = 0 \Rightarrow \lambda_t = \lambda_{t+1} R_{t+1}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow C_t + K_{t+1} + B_t = w_t L_t + R_t K_t + (1+r_{t-1})B_{t-1}$$

From the first two expressions we get: $\frac{\beta^t}{C_t} = \frac{\beta^{t+1}}{C_{t+1}}(1+r_t)$.
Then:

$$C_{t+1} = \beta(1 + r_t)C_t$$

From the second and third expressions we get:

$$R_{t+1} = 1 + r_t$$

Combining the two expressions above we get:

$$C_{t+1} = \beta R_{t+1} C_t$$

Dividing both sides by $A_{t+1}L_{t+1} = (1+g)A_t(1+n)L_t$ we get: $\frac{C_{t+1}}{A_{t+1}L_{t+1}} = \beta R_{t+1} \frac{C_t}{(1+g)A_t(1+n)L_t}$. Then:

$$c_{t+1} = \frac{\beta c_t}{(1+g)(1+n)} R_{t+1}$$

From the FOC for the firm we get: $R_{t+1} = \alpha k_{t+1}^{\alpha-1}$. Substituting into the expression above we get:

$$c_{t+1} = \frac{\alpha\beta}{(1+g)(1+n)} k_{t+1}^{\alpha-1} c_t$$

which coincides with the Euler Equation we derived from the Planner's problem.

Market clearing in the credit market requires $B_t = 0 \forall t$ (closed economy). Imposing $B_t = 0 \forall t$ into the constraint we get: $C_t + K_{t+1} = w_t L_t + R_t K_t$. Since profits are zero, this becomes: $C_t + K_{t+1} = Y_t$. Dividing by $A_t L_t$ and rearranging appropriately we get:

$$c_t + (1 + g)(1 + n)k_{t+1} = k_t^\alpha$$

which coincides with the constraint faced by the Planner.

Hence, the equilibrium allocations of the decentralized competitive economy coincide with the ones chosen by the planner.

Stochastic Neoclassical Growth Model

(Brock and Mirman 1972)

Now assume:

$$Y_t = \theta_t F(K_t, A_t L_t)$$

where

$\{\theta_t\}_{t=0}^{\infty}$ is a sequence of i.i.d. random variables

By CRS:

$$y_t = \theta_t f(k_t)$$

Assume:

$$y_t = \theta_t k_t^{\alpha}$$

and

$$\ln \theta_t \sim \mathcal{N}(0, \sigma^2)$$

The planner chooses contingent plans for consumption and capital in order to solve:

$$\begin{aligned} \max \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln c_t \\ \text{s.t.} \quad & c_t + (1+n)(1+g)k_{t+1} = \theta_t k_t^\alpha \\ & k_0 > 0, \theta_0 > 0 \text{ given} \end{aligned}$$

where $\mathbb{E}_0$ is the conditional expectation as of time $t = 0$.

At time t the planner knows the realization of θ_t (and earlier) but not the future realizations. We define the state of the system as (k_t, θ_t) . Therefore, the BE is:

$$\begin{aligned} V(k_t, \theta_t) \equiv \max_{c_t, k_{t+1}} \quad & \ln c_t + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1}) \\ \text{s.t.} \quad & (1+n)(1+g)k_{t+1} = \theta_t k_t^\alpha - c_t \\ & k_t, \theta_t \text{ given} \end{aligned}$$

Using the constraint to eliminate k_{t+1} :

$$V(k_t, \theta_t) \equiv \max_{c_t} \ln c_t + \beta \mathbb{E}_t V \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)}, \theta_{t+1} \right)$$

Remark. Since c_t , k_t and θ_t are known at t , and the θ'_s are i.i.d., $\mathbb{E}_t$ runs over the distribution of θ_{t+1} .

We solve the problem using the Guess-Verify Method.

Guess:

$$V^G(k_t, \theta_t) = \Omega + \Theta \ln k_t + \Phi \ln \theta_t$$

Then:

$$\tilde{V}(k_t, \theta_t) = \max_{c_t} \ln c_t + \beta \mathbb{E}_t \left[\Omega + \Theta \ln \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)} \right) + \Phi \ln \theta_{t+1} \right]$$

$$\tilde{V}(k_t, \theta_t) = \max_{c_t} \ln c_t + \beta \Omega + \beta \Theta \ln \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)} \right) + \beta \Phi \mathbb{E}_t \ln \theta_{t+1}$$

$$\tilde{V}(k_t, \theta_t) = \max_{c_t} \ln c_t + \beta \Omega + \beta \Theta \ln \left(\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)} \right)$$

FOC:

$$\frac{1}{c_t} + \beta \Theta \frac{1}{\frac{\theta_t k_t^\alpha - c_t}{(1+n)(1+g)}} \frac{-1}{(1+n)(1+g)} = 0 \Rightarrow$$

$$c_t = \frac{1}{1 + \beta \Theta} \theta_t k_t^\alpha$$

Then:

$$k_{t+1} = \frac{\theta_t k_t^\alpha - \frac{1}{1 + \beta \Theta} \theta_t k_t^\alpha}{(1+n)(1+g)} \Rightarrow$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{\beta \Theta}{1 + \beta \Theta} \theta_t k_t^\alpha$$

Then:

$$\begin{aligned} \tilde{V}(k_t, \theta_t) = & \ln \left(\frac{1}{1 + \beta \Theta} \theta_t k_t^\alpha \right) \\ & + \beta \mathbb{E}_t \left[\Omega + \Theta \ln \left(\frac{1}{(1+n)(1+g)} \frac{\beta \Theta}{1 + \beta \Theta} \theta_t k_t^\alpha \right) + \Phi \ln \theta_{t+1} \right] \end{aligned}$$

$$\begin{aligned}\tilde{V}(k_t, \theta_t) = & \left[\ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right) \right] \\ & + \alpha(1+\beta\Theta) \ln k_t + (1+\beta\Theta) \ln \theta_t\end{aligned}$$

Therefore, $\tilde{V}(k_t, \theta_t) = V^G(k_t, \theta_t) \forall (k_t, \theta_t)$ requires:

$$\Omega = \ln\left(\frac{1}{1+\beta\Theta}\right) + \beta\Omega + \beta\Theta \ln\left(\frac{1}{(1+n)(1+g)} \frac{\beta\Theta}{1+\beta\Theta}\right)$$

$$\Theta = \alpha(1+\beta\Theta)$$

$$\Phi = 1 + \beta\Theta$$

From the second equation:

$$\Theta = \frac{\alpha}{1 - \alpha\beta}$$

Substituting into the third equation:

$$\Phi = \frac{1}{1 - \alpha\beta}$$

Substit. $\Theta = \frac{\alpha}{1-\alpha\beta}$ into the first equation:

$$\Omega = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln \left(\frac{\alpha\beta}{(1+n)(1+g)} \right) \right]$$

Then:

$$V(k_t) = \frac{1}{1-\beta} \left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln \left(\frac{\alpha\beta}{(1+n)(1+g)} \right) \right] + \frac{\alpha}{1-\alpha\beta} \ln k_t + \frac{1}{1-\alpha\beta} \ln \theta_t$$

and

$$c_t = (1-\alpha\beta)\theta_t k_t^\alpha$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \alpha\beta\theta_t k_t^\alpha$$

Nonstochastic Steady State

Suppose $\ln \theta_t$ equals its mean in every period: $\ln \theta_t = 0$
 $\forall t$. Hence: $\theta_t = 1 \forall t$.

Then, the evolution of the capital stock is:

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \alpha \beta k_t^\alpha$$

Setting $k_{t+1} = k_t$ we get:

$$k_{nss} = \left[\frac{\alpha \beta}{(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}}$$

where the subscript nss stands for "nonstochastic steady state".

Then:

$$c_{nss} = (1 - \alpha \beta) k_{nss}^\alpha$$

$$y_{nss} = k_{nss}^\alpha$$

Limiting (Ergodic) Distribution

From $k_{t+1} = \frac{1}{(1+n)(1+g)} \alpha \beta \theta_t k_t^\alpha$ we get:

$$\begin{aligned}\ln k_{t+1} &= \ln \left(\frac{\alpha \beta}{(1+n)(1+g)} \right) + \alpha \ln k_t + \ln \theta_t \\ &= \psi + \alpha \ln k_t + \ln \theta_t\end{aligned}$$

Then:

$$\ln k_1 = \psi + \alpha \ln k_0 + \ln \theta_0$$

$$\begin{aligned}\ln k_2 &= \psi + \alpha \ln k_1 + \ln \theta_1 \\ &= \psi + \alpha(\psi + \alpha \ln k_0 + \ln \theta_0) + \ln \theta_1 \\ &= \psi(1 + \alpha) + \alpha^2 \ln k_0 + \alpha \ln \theta_0 + \ln \theta_1\end{aligned}$$

$$\begin{aligned}\ln k_3 &= \psi + \alpha \ln k_2 + \ln \theta_2 \\ &= \psi + \alpha[\psi(1 + \alpha) + \alpha^2 \ln k_0 + \alpha \ln \theta_0 + \ln \theta_1] + \ln \theta_2 \\ &= \psi(1 + \alpha + \alpha^2) + \alpha^3 \ln k_0 + \alpha^2 \ln \theta_0 + \alpha \ln \theta_1 + \ln \theta_2\end{aligned}$$

$$\begin{aligned}
\ln k_t &= \psi(1 + \alpha + \alpha^2 + \dots + \alpha^{t-1}) + \alpha^t \ln k_0 \\
&\quad + \alpha^{t-1} \ln \theta_0 + \alpha^{t-2} \ln \theta_1 + \dots + \alpha \ln \theta_{t-2} + \ln \theta_{t-1} \\
&= \psi \sum_{s=0}^{t-1} \alpha^s + \alpha^t \ln k_0 + \sum_{s=0}^{t-1} \alpha^s \ln \theta_{t-1-s} \\
&= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \sum_{s=0}^{t-1} \alpha^s \ln \theta_{t-1-s}
\end{aligned}$$

Notice that, given k_0 and $\ln \theta_0$, $\ln k_t$ is a linear combination of independent normally distributed random variables [$\ln \theta_t \sim \mathcal{N}(0, \sigma^2)$]. Therefore, given k_0 and $\ln \theta_0$, $\ln k_t$ is normally distributed. We can calculate its mean and variance. The mean is

$$\begin{aligned}
\mathbb{E}_0 \ln k_t &= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 \\
&\quad + \sum_{s=0}^{t-2} \alpha^s \mathbb{E}_0 \ln \theta_{t-1-s} \\
&= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 + \sum_{s=0}^{t-1} \alpha^s \times 0 \\
&= \psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0
\end{aligned}$$

The variance is:

$$\begin{aligned}
 Var_0\{\ln k_t\} &= Var_0\left\{\psi\frac{1-\alpha^t}{1-\alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 + \sum_{s=0}^{t-2} \alpha^s \ln \theta_{t-1-s}\right\} \\
 &= Var\left\{\sum_{s=0}^{t-2} \alpha^s \ln \theta_{t-1-s}\right\} \\
 &= \sum_{s=0}^{t-2} (\alpha^s)^2 Var\{\ln \theta_{t-1-s}\} \\
 &= \sum_{s=0}^{t-2} \alpha^{2s} \sigma^2 \\
 &= \sigma^2 \sum_{s=0}^{t-2} (\alpha^2)^s
 \end{aligned}$$

Then:

$$Var_0\{\ln k_t\} = \sigma^2 \frac{1 - (\alpha^2)^{t-1}}{1 - \alpha^2}$$

Taking limits for $t \rightarrow \infty$:

$$\begin{aligned}
\lim_{t \rightarrow \infty} \mathbb{E}_0 \ln k_t &= \lim_{t \rightarrow \infty} \left[\psi \frac{1 - \alpha^t}{1 - \alpha} + \alpha^t \ln k_0 + \alpha^{t-1} \ln \theta_0 \right] \\
&= \psi \lim_{t \rightarrow \infty} \frac{1 - \alpha^t}{1 - \alpha} + \ln k_0 \lim_{t \rightarrow \infty} \alpha^t \\
&\quad + \ln \theta_0 \lim_{t \rightarrow \infty} \alpha^{t-1} \\
&= \frac{\psi}{1 - \alpha} \quad [\alpha \in 0, 1)]
\end{aligned}$$

$$\begin{aligned}
\lim_{t \rightarrow \infty} Var_0 \{ \ln k_t \} &= \lim_{t \rightarrow \infty} \left[\sigma^2 \frac{1 - (\alpha^2)^{t-1}}{1 - \alpha^2} \right] \\
&= \frac{\sigma^2}{1 - \alpha^2} \lim_{t \rightarrow \infty} \left[1 - (\alpha^2)^{t-1} \right] \\
&= \frac{\sigma^2}{1 - \alpha^2}
\end{aligned}$$

From $y_t = \theta_t k_t^\alpha$:

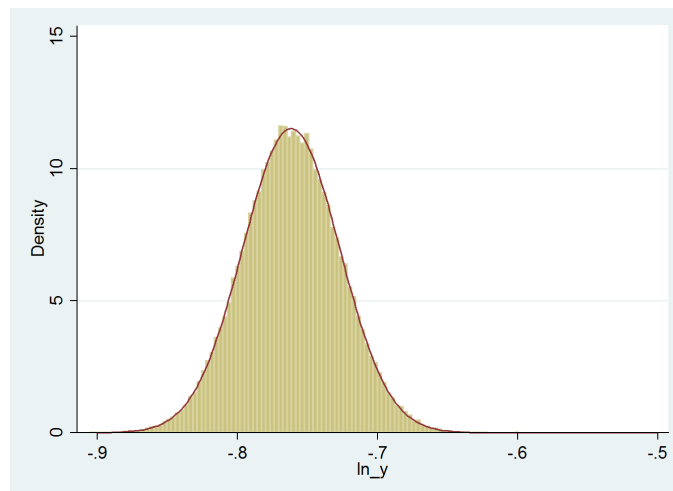
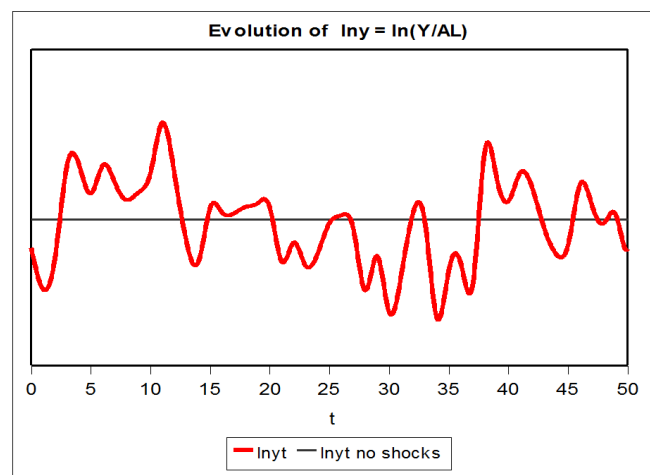
$$\ln y_t = \ln \theta_t + \alpha \ln k_t$$

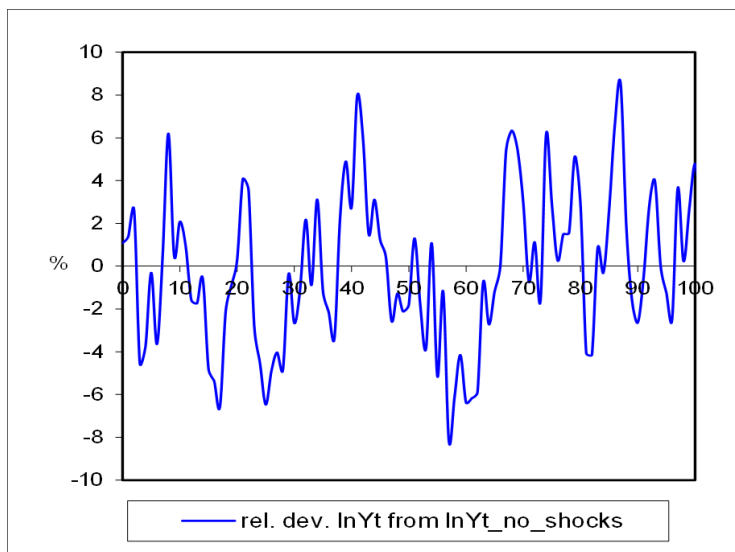
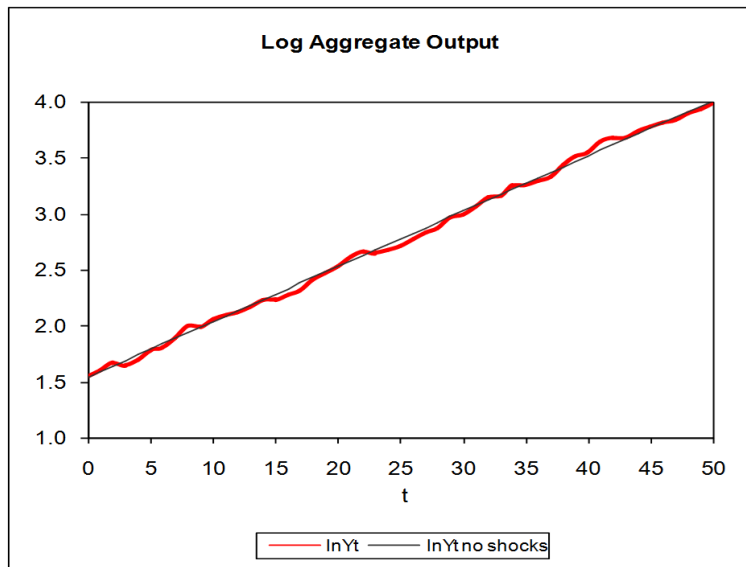
Hence, $\ln y_t$ is normally distributed, and we can calculate its mean and variance analogously.

Plots

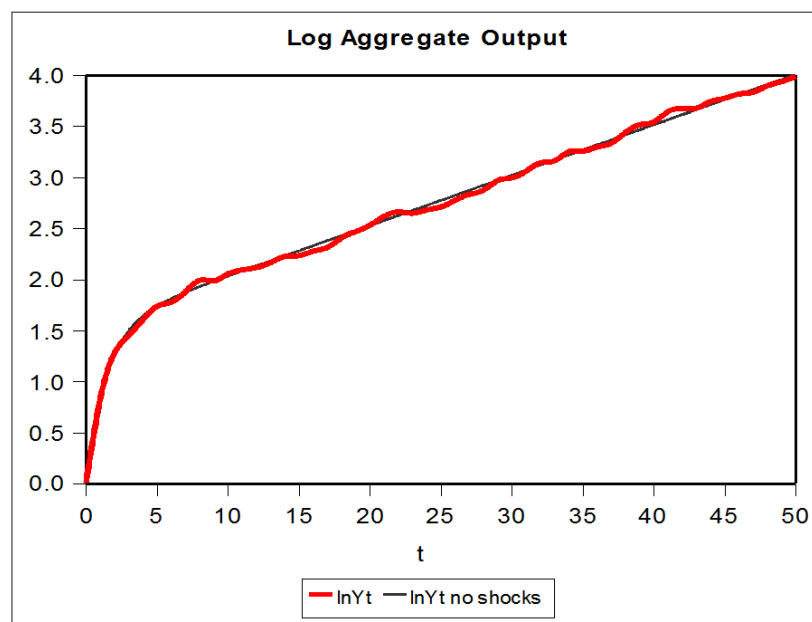
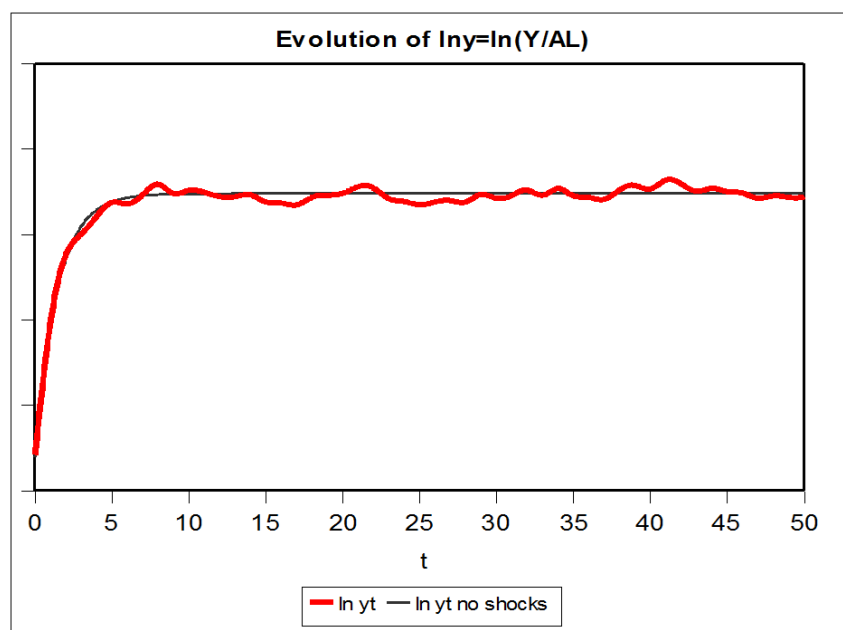
Assume $k_0 = k_{nss}$. Plot the evolution of $\ln y_t$ and $\ln Y_t$, and the empirical and limiting distributions of $\ln y_t$.

We assume $\alpha = 0.5$, $\rho = 0.05$, $n = 0.03$, and $g = 0.02$. These values imply: $k_{nss} = 0.218$ and $y_{nss} = 0.467$.



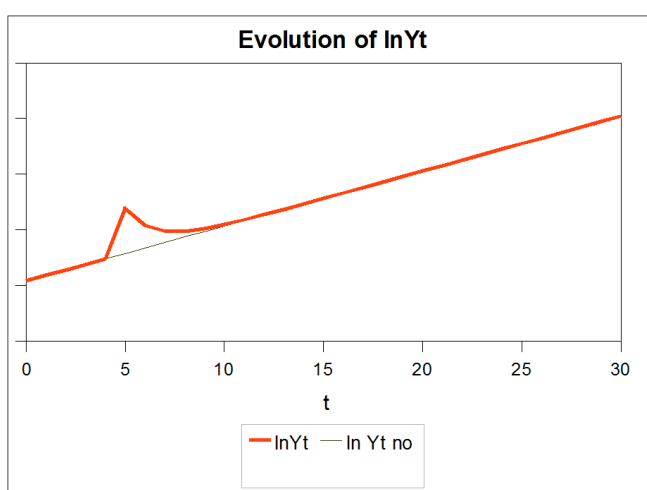
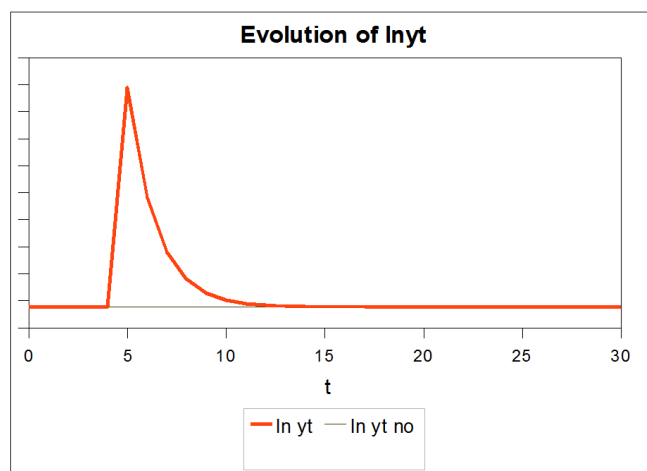


Now assume $k_0 < k_{nss}$. We get:



Impulse-Response Analysis

Set $\theta_t = 1$ ($\ln \theta_t = 0$) and assume the economy evolves along the balanced-growth path (nonstochastic steady state). At some period τ , θ_τ suddenly jumps up, and then returns to 1 from $\tau + 1$ onward.



Appendix

Phase Diagram for the Deterministic Model

From the Euler equation: $c_{t+1} - c_t = \left[\frac{\alpha\beta}{(1+n)(1+g)} k_{t+1}^{\alpha-1} - 1 \right] c_t$.

Using the resource constraint to eliminate k_{t+1} we get:

$$c_{t+1} - c_t = \left[\frac{\alpha\beta}{(1+n)(1+g)} \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)^{\alpha-1} - 1 \right] c_t.$$

The resource const. gives: $k_{t+1} - k_t = \frac{k_t^\alpha - c_t}{(1+n)(1+g)} - k_t$.

Hence, we have the following system of two difference equations in k and c :

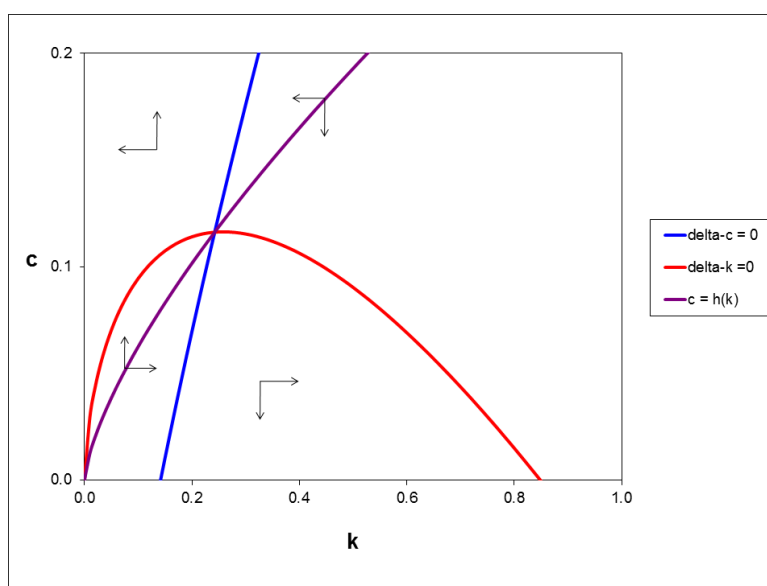
$$\begin{cases} \Delta c_{t+1} = \left[\frac{\alpha\beta}{(1+n)(1+g)} \left(\frac{k_t^\alpha - c_t}{(1+n)(1+g)} \right)^{-(1-\alpha)} - 1 \right] c_t \\ \Delta k_{t+1} = \frac{k_t^\alpha - c_t}{(1+n)(1+g)} - k_t \end{cases}$$

The $\Delta c_{t+1} = 0$ and $\Delta k_{t+1} = 0$ schedules are:

$$\left\{ \begin{array}{l} \Delta c_{t+1}=0: \quad c_t = k_t^\alpha - (1+n)(1+g) \underbrace{\left[\frac{\alpha\beta}{(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}}}_{k_{ss}} \\ \Delta k_{t+1}=0: \quad c_t = k_t^\alpha - (1+n)(1+g)k_t \end{array} \right.$$

Now we plot the schedules given above together with the policy function:

$$c_t = (1 - \alpha\beta) k_t^\alpha$$



Remark 1. For the plot we set: $\alpha = 0.7$, $\rho = 0.05$, $n = 0.03$, and $g = 0.02$. These values give: $k_{ss} = 0.242$, $y_{ss} = 0.371$, and $c_{ss} = 0.116$.

Remark 2. The three lines cross at $k = k_{ss}$.

Remark 3. The arrows indicate the qualitative dynamics of the system.

Remark 4. Notice that the policy function is the stable arm of the system.

Remark 5. In the continuous-time version of this model, $\Delta c_{t+1} = 0$ would produce a vertical line at $k = k_{ss}$.

Euler Equation and Transversality Condition**I. A Finite-Horizon Problem**

Consider the following problem:

$$\begin{aligned}
 & \max \sum_{t=0}^T \beta^t u(c_t) \\
 & \text{s.t. } c_t + k_{t+1} - (1 - \delta)k_t = f(k_t) \quad \forall t \in \{0, 1, \dots, T\} \\
 & k_{T+1} \geq 0 \\
 & k_0 > 0 \text{ given}
 \end{aligned}$$

where:

$$u' > 0, \quad u'' < 0, \quad \lim_{c \rightarrow 0^+} u'(c) = \infty$$

$$f(0) = 0, \quad f' > 0, \quad f'' < 0, \quad \lim_{k \rightarrow 0^+} f'(k) = \infty, \quad \lim_{k \rightarrow \infty} f'(k) = 0$$

Remark. In principle, we should include a set of non-negativity constraints, $c_t \geq 0$ and $k_{t+1} \geq 0$. However, the Inada conditions $\lim_{c \rightarrow 0^+} u'(c) = \infty$ and $\lim_{k \rightarrow 0^+} f'(k) = \infty$ ensure that $c_t > 0$ and $k_{t+1} > 0$ at the optimum (except for k_{T+1} , that's why we kept the constraint $k_{T+1} \geq 0$). We should also write $c_t + k_{t+1} - (1 - \delta)k_t \leq f(k_t)$. But $u' > 0$ ensures this constraint will always hold with equality at the optimum.

The Lagrangian for the problem is:

$$\mathcal{L} = \sum_{t=0}^T \beta^t u(c_t) + \sum_{t=0}^T \lambda_t [f(k_t) + (1 - \delta)k_t - k_{t+1} - c_t] + \mu k_{T+1}$$

The Kuhn-Tucker first-order necessary conditions (FONC) are:

- (1) $\frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow \quad \beta^t u'(c_t) = \lambda_t \quad \forall t \in \{0, 1, \dots, T\}$
- (2) $\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \Rightarrow \quad \lambda_t = \lambda_{t+1} [f'(k_{t+1}) + (1 - \delta)] \quad \forall t \in \{0, 1, \dots, T-1\}$
- (3) $\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow \quad \lambda_T = \mu$
- (4) $\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow \quad c_t = f(k_t) + (1 - \delta)k_t - k_{t+1} \quad \forall t \in \{0, 1, \dots, T\}$
- (5) $\frac{\partial \mathcal{L}}{\partial \mu} \geq 0 \Rightarrow \quad k_{T+1} \geq 0$
- (6) $[CS] \quad \mu k_{T+1} = 0, \quad \mu \geq 0$

where $[CS]$ is the *complementary-slackness* condition.

Euler Equation

Combining (1) and (2) we get: $\beta^t u'(c_t) = \beta^{t+1} u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)] \Rightarrow$

$$(7) \quad u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)]$$

The interpretation of this expression is simple. Consider a (small) reduction of period- t consumption: $dc_t = -\Delta$, with $\Delta > 0$. This reduces utility by $u'(c_t)dc_t = -u'(c_t)\Delta$. The Δ units of capital accumulated through the reduction of current consumption increase future consumption by $dc_{t+1} = [f'(k_{t+1}) + (1 - \delta)]\Delta$. The increase in utility induced by this additional consumption is $\beta u'(c_{t+1})dc_{t+1} = \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)]\Delta$. The expression above says that, at the optimum, reshuffling consumption accross consecutive periods cannot increase utility: $dU = u'(c_t)dc_t + \beta u'(c_{t+1})dc_{t+1} = -u'(c_t)\Delta + \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)]\Delta = [-u'(c_t) + \beta u'(c_{t+1})(f'(k_{t+1}) + (1 - \delta))]\Delta = 0$, where the last equality follows from (7).

Another way of looking at this is by rewriting the expression above as follows: $\frac{u'(c_t)}{\beta u'(c_{t+1})} = f'(k_{t+1}) + (1 - \delta)$. The left-hand side is the marginal rate of substitution between consumption in periods t and $t + 1$; the right-hand side is the marginal rate of transformation (*i.e.*, the rate at which current consumption can be transformed into future consumption through the available technology). At the optimum, the rate at which the consumer is *willing* to substitute consumption must be equal to the rate at which the consumer *can* substitute consumption.

Using (4) we can rewrite (7) as follows:

$$(8) \quad u'(f(k_t) + (1 - \delta)k_t - k_{t+1}) = \beta u'(f(k_{t+1}) + (1 - \delta)k_{t+1} - k_{t+2}) [f'(k_{t+1}) + (1 - \delta)]$$

This is the *Euler Equation*. It's a second-order difference equation in k . Solving this difference equation requires two boundary conditions. We already have one, the initial condition $k_0 > 0$. We'll use the complementary-slackness condition to get the second one.

Transversality Condition (TVC)

From (3) and (6) we get:

$$(9) \quad \lambda_T k_{T+1} = 0$$

From (1) we get:

$$(10) \quad \lambda_T = \beta^T u'(c_T)$$

Combining (9) and (10):

$$(11) \quad \beta^T u'(c_T) k_{T+1} = 0$$

Since $\beta > 0$ and $u'(c_T) > 0$, (11) implies:

$$(12) \quad k_{T+1} = 0$$

This is our second boundary condition, and is usually called the *transversality condition* (because it is a description of how the optimal path crosses the terminal line; to "transverse" means to "go across").

II. An Infinite-Horizon Problem

Now suppose the problem is:

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & c_t + k_{t+1} - (1 - \delta)k_t = f(k_t) \quad \forall t \in \{0, 1, \dots\} \\ & k_0 > 0 \text{ given} \end{aligned}$$

The Lagrangian is:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t [f(k_t) + (1 - \delta)k_t - k_{t+1} - c_t]$$

Then:

$$(13) \quad \frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow \quad \beta^t u'(c_t) = \lambda_t \quad \forall t \in \{0, 1, \dots\}$$

$$(14) \quad \frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \Rightarrow \quad \lambda_t = \lambda_{t+1} [f'(k_{t+1}) + (1 - \delta)] \quad \forall t \in \{0, 1, \dots\}$$

$$(15) \quad \frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow \quad c_t = f(k_t) + (1 - \delta)k_t - k_{t+1} \quad \forall t \in \{0, 1, \dots\}$$

Combining (13)-(15) we obtain the same Euler Equation as before:

$$(16) \quad u'(f(k_t) + (1 - \delta)k_t - k_{t+1}) = \beta u'(f(k_{t+1}) + (1 - \delta)k_{t+1} - k_{t+2}) [f'(k_{t+1}) + (1 - \delta)]$$

Again, we need two boundary conditions. We have one, the initial value of the capital stock, k_0 . But, how about the second? We don't have the complementary slackness condition to derive it (because there's no such thing as the "final" capital stock in an infinite-horizon setup). Fortunately, it is possible to derive a transversality condition by taking the limit for $T \rightarrow \infty$ of the TVC for the finite-horizon case. We have to be careful, however, because the relevant limit is not $\lim_{T \rightarrow \infty} k_{T+1}$. This doesn't make sense, since we've shown in (12)

that, for the finite-horizon case, $k_{T+1} = 0$. Instead, we have to go back one step and take the limit of expression (11), to get:

$$(17) \quad \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0$$

Now we can use (15) to write:

$$(18) \quad \lim_{T \rightarrow \infty} \beta^T u'(f(k_T) + (1 - \delta)k_T - k_{T+1}) k_{T+1} = 0$$

which is our TVC (and gives us the second boundary condition needed to solve the Euler Equation).

III. Some Remarks

Our informal argument for the derivation of the transversality condition for infinite-horizon models does not make clear whether the TVC is actually a *necessary* condition for an optimum. This has been debated quite a bit in the literature. There are papers showing conditions under which the TVC is necessary for an optimum. One of them is Kamihigashi (2002). For extensions to the stochastic case, see Kamihigashi (2003) and (2005). You can also check Karatzas and Sudderth (2010).

Stokey and Lucas (1989) [Theorem 4.15, page 98] give conditions under which the Euler Equation together with the TVC are *sufficient* conditions for an optimum.

It is common in the literature to eliminate c_t by substituting the constraint into the objective function. In this case we can define $F(k_t, k_{t+1}) \equiv u(f(k_t) + (1 - \delta)k_t - k_{t+1})$ and rewrite the TVC as follows:

$$(19) \quad \lim_{T \rightarrow \infty} \beta^T [-F_2(k_T, k_{T+1})] k_{T+1} = 0$$

From the TVC we know: $\lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0$. Equivalently, we can write: $\lim_{T \rightarrow \infty} \beta^{T-1} u'(c_{T-1}) k_T = 0$. But we know from the Euler Equation that: $u'(c_{T-1}) = \beta u'(c_T) [f'(k_T) + (1 - \delta)]$. Then, multiplying by k_T : $u'(c_{T-1}) k_T = \beta u'(c_T) [f'(k_T) + (1 - \delta)] k_T$. Substituting into the TVC: $\lim_{T \rightarrow \infty} \beta^{T-1} \beta u'(c_T) [f'(k_T) + (1 - \delta)] k_T = 0$. Then:

$$(20) \quad \lim_{T \rightarrow \infty} \beta^T u'(c_T) [f'(k_T) + (1 - \delta)] k_T = 0$$

which is an alternative form of the TVC. Notice, however, that this form of the TVC, unlike the original, was derived imposing the Euler Equation. If we define $F(k_t, k_{t+1}) \equiv u(f(k_t) + (1 - \delta)k_t - k_{t+1})$ as our objective function, (20) becomes:

$$(21) \quad \lim_{T \rightarrow \infty} \beta^T F_1(k_T, k_{T+1}) k_T = 0$$

which is the form of the TVC given in Stokey and Lucas (1989), page 98.

Sometimes the TVC is written using the Lagrange multiplier. From (9) we get:

$$(21) \quad \lim_{T \rightarrow \infty} \lambda_T k_{T+1} = 0$$

The TVC can be interpreted as saying that the present discounted value of wealth (the capital stock) at infinity must be zero; in other words, k_{T+1} should not grow too fast compared to its marginal value (shadow price) $\lambda_T = \beta^T u'(c_T)$. Hence, the TVC rules out overaccumulation of wealth. The idea is that if one saves too much and spends too little, then one is not behaving optimally.

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Real Business Cycles

Technology

$$Y_t = \theta_t F(K_t, A_t L_t)$$

where F displays standard neoclassical properties, and

$$A_t = (1 + g)^t A_0$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

where $A_0 > 0$, $g \geq 0$, $\theta > 0$, $\rho \in (0, 1)$, and $\{\varepsilon_{t+1}\}_{t=0}^{\infty}$ is a sequence of i.i.d random variables with distribution $\varepsilon_{t+1} \sim N(0, \sigma_{\varepsilon}^2)$. Notice that the evolution of $\ln \theta_{t+1}$ can be written as follows: $\ln(\frac{\theta_{t+1}}{\theta}) = \rho \ln(\frac{\theta_t}{\theta}) + \varepsilon_{t+1}$.

Capital evolves according to:

$$K_{t+1} = I_t + (1 - \delta)K_t$$

The resource constraint is:

$$C_t + I_t \leq Y_t$$

Population evolves according to:

$$N_t = (1 + n)^t N_0, \quad n \geq 0$$

The utility fn. of the representative household is:

$$\tilde{U} = \sum_{t=0}^{\infty} \tilde{\beta}^t u(\tilde{c}_t, 1 - \tilde{l}_t) \frac{N_t}{H}$$

where H is the fixed number of households (so $\frac{N_t}{H}$ gives the number of members of the household), $\tilde{c}_t = \frac{C_t}{N_t}$ is consumption for the representative member of each household (*i.e.*, consumption per capita), $1 - \tilde{l}_t$ is leisure per capita (total time is normalized to one and $\tilde{l}_t = \frac{L_t}{N_t}$ is the time spent working in per capita terms); $u(\cdot)$ is the per-period utility function, and $\tilde{\beta} \in (0, 1)$ is the discount factor.

For any variable X , define:

$$x_t \equiv \frac{X_t}{A_t N_t}$$

Recall that, in this model, $N_t \neq L_t$.

By CRS:

$$\begin{aligned} y_t &\equiv \frac{Y_t}{A_t N_t} = \frac{\theta_t F(K_t, A_t L_t)}{A_t N_t} \\ &= \theta_t F\left(\frac{K_t}{A_t N_t}, \frac{A_t L_t}{A_t N_t}\right) = \theta_t F\left(k_t, \frac{L_t}{N_t}\right) \\ &= \theta_t F(k_t, \tilde{l}_t) \end{aligned}$$

Remark. We can also write: $y_t = \theta_t F\left(\frac{K_t}{A_t N_t}, \frac{A_t L_t}{A_t N_t}\right) = \theta_t F(k_t, A_t l_t) = \theta_t F(k_t, A_0(1+g)^t l_t)$.

We can rewrite the resource constraint as follows:

$$C_t + K_{t+1} - (1 - \delta)K_t \leq Y_t$$

Dividing by $A_t N_t$:

$$\frac{C_t}{A_t N_t} + \frac{K_{t+1}}{A_{t+1} N_{t+1}} \frac{A_{t+1} N_{t+1}}{A_t N_t} - (1 - \delta) \frac{K_t}{A_t N_t} \leq \frac{Y_t}{A_t N_t}$$

Then:

$$c_t + k_{t+1}(1+g)(1+n) - (1-\delta)k_t \leq \theta_t F(k_t, \tilde{l}_t)$$

We can rewrite the utility function as follows:

$$\begin{aligned}
\tilde{U} &= \sum_{t=0}^{\infty} \tilde{\beta}^t u(\tilde{c}_t, 1 - \tilde{l}_t) \frac{N_t}{H} \\
&= \sum_{t=0}^{\infty} \tilde{\beta}^t u(A_t c_t, 1 - \tilde{l}_t) \frac{N_t}{H} \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \left(\frac{1+n}{1+\tilde{\rho}} \right)^t u(A_t c_t, 1 - \tilde{l}_t) \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t u(A_t c_t, 1 - \tilde{l}_t)
\end{aligned}$$

where $\beta \equiv \frac{1+n}{1+\tilde{\rho}} = \tilde{\beta}(1+n)$, with $\tilde{\rho} > n$, so $\beta \in (0, 1)$.

**Particular Case: Log Utility, Cobb-Douglas
Technology, 100% Depreciation**

Assume:

$$u(\tilde{c}_t, 1 - \tilde{l}_t) = \ln \tilde{c}_t + \gamma \ln(1 - \tilde{l}_t), \quad \gamma > 0$$

$$Y_t = \theta_t K_t^\alpha (A_t L_t)^{1-\alpha}, \quad \alpha \in (0, 1)$$

$$K_{t+1} = I_t \quad (\text{i.e., } \delta = 1)$$

Then:

$$\begin{aligned} y_t &= \theta_t F(k_t, \tilde{l}_t) \\ &= \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} \end{aligned}$$

The constraint becomes:

$$c_t + k_{t+1}(1 + g)(1 + n) \leq \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$$

The utility function becomes

$$\begin{aligned}
\tilde{U} &= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln(\tilde{c}_t) + \gamma \ln(1 - \tilde{l}_t) \right] = \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln(A_t c_t) + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln A_t + \ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \ln[A_0(1+g)^t] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t [\ln A_0 + t \ln(1+g)] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \left[\ln A_0 \sum_{t=0}^{\infty} \beta^t + \ln(1+g) \sum_{t=0}^{\infty} \beta^t t \right] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \left[\ln A_0 \frac{1}{1-\beta} + \ln(1+g) \frac{\beta}{(1-\beta)^2} \right] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \text{constant}_1 + \text{positive constant}_2 \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right]
\end{aligned}$$

Then we can work with:

$$U \equiv \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right]$$

Planner's Problem

The planner chooses contingent plans for consumption (c_t), labor ($\tilde{l}_t$), and capital (k_{t+1}), in order to solve:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right]$$

$$\text{s.t. } c_t + (1 + g)(1 + n)k_{t+1} \leq \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_0 > 0, \theta_0 > 0 \text{ given}$$

Remark 1. When choosing c_t , $\tilde{l}_t$ and k_{t+1} , the planner knows the value of θ_t (the past values too).

Remark 2. $\mathbb{E}_0$ denotes the conditional expectation as of time 0.

Remark 3. It's clear that the constraint will be binding at the optimum.

Remark 4. Our assumptions ensure that consumption and capital will be strictly positive at the optimum, and $\tilde{l}_t \in (0, 1)$.

Bellman Equation (BE)

For any period t :

$$V(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1})$$

$$\text{s.t.} \quad (1+g)(1+n)k_{t+1} = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t$$

$$\ln \theta_{t+1} = (1-\rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

Strategy

First we'll use the FOC and the Envelope Condition to obtain some intuition. Then we'll solve the problem in closed form using the Guess-Verify method.

To facilitate the interpretation of the FOC we generalize notation and write $u(c_t)$ instead of $\ln c_t$ and $v(1 - \tilde{l}_t)$

instead of $\gamma \ln(1 - \tilde{l}_t)$. Assume $u' > 0$, $v' > 0$, $u'' < 0$ and $v'' < 0$. Also, use $\theta_t F(k_t, \tilde{l}_t)$ instead of $\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$. Finally, allow for a depreciation rate smaller than one. Then, BE becomes:

$$V(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} u(c_t) + v(1 - \tilde{l}_t) + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1})$$

$$\text{s.t.} \quad (1+g)(1+n)k_{t+1} = \theta_t F(k_t, \tilde{l}_t) + (1-\delta)k_t - c_t$$

$$\ln \theta_{t+1} = (1-\rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

FOC (use the constraint to eliminate c_t and get the partial derivatives w.r.t. k_{t+1} and $\tilde{l}_t$).

$$(1) \quad u'(c_t)(1+g)(1+n) = \beta \mathbb{E}_t V_1(k_{t+1}, \theta_{t+1})$$

$$(2) \quad v'(1 - \tilde{l}_t) = u'(c_t) \theta_t F_2(k_t, \tilde{l}_t)$$

$$(3) \quad c_t = \theta_t F(k_t, \tilde{l}_t) + (1-\delta)k_t - (1+g)(1+n)k_{t+1}$$

Envelope

$$(4) \quad V_1(k_t, \theta_t) = u'(c_t) [\theta_t F_1(k_t, \tilde{l}_t) + (1 - \delta)]$$

Shifting (4) one period ahead and substituting into (1):

$$(5) \quad u'(c_t)(1+g)(1+n) = \beta \mathbb{E}_t \left\{ u'(c_{t+1}) [\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)] \right\}$$

Summarizing:

$$(6) \quad v'(1 - \tilde{l}_t) = u'(c_t) \theta_t F_2(k_t, \tilde{l}_t)$$

$$(7) \quad u'(c_t) = \mathbb{E}_t \left\{ \beta u'(c_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1+g)(1+n)} \right\}$$

$$(8) \quad c_t + (1 + g)(1 + n)k_{t+1} = \theta_t F(k_t, \tilde{l}_t) + (1 - \delta)k_t$$

Hence, we've obtained an intratemporal condition between consumption and leisure, (6), an intertemporal condition for consumption, (7), and the budget constraint, (8). We have three equations (per period) to solve for the evolution of three variables: c_t , $\tilde{l}_t$, and k_{t+1} .

Combining expressions (6) and (7) we can obtain three additional conditions (which are redundant from the point of view of the model's solution).

Substituting (6) into both sides of (7) we get an intertemporal condition for labor/leisure:

$$(9) \quad v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \begin{aligned} & \beta v'(1 - \tilde{l}_{t+1}) \times \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \\ & \times \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})} \end{aligned} \right\}$$

Substituting (6) into the left-hand side of (7) we get an intertemporal condition between l_t and c_{t+1} :

$$(10) \quad v'(1 - \tilde{l}_t) = \beta \mathbb{E}_t \left\{ u'(c_{t+1}) \left[\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta) \right] \theta_t F_2(k_t, \tilde{l}_t) \right\}$$

Finally, substituting (6) into the right-hand side of (7) we get an intertemporal condition between c_t and l_{t+1} :

$$(11) \quad u'(c_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n) \theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})} \right\}$$

Interpretation of (6).

Consider a small increase in current labor, $d\tilde{l}_t = \Delta > 0$. Hence, current leisure goes down by $d(1 - \tilde{l}_t) = -\Delta$. This induces a change in utility equal to: $v'(1 - \tilde{l}_t)d(1 - \tilde{l}_t) = -v'(1 - \tilde{l}_t)\Delta < 0$.

The extra amount of labor increases current output by: $dy_t = \theta_t F_2(k_t, \tilde{l}_t)d\tilde{l}_t = \theta_t F_2(k_t, \tilde{l}_t)\Delta$ (k_t is given). Imagine you use all this extra output to increase current consumption. Then: $dc_t = dy_t = \theta_t F_2(k_t, \tilde{l}_t)\Delta$. This increases utility by $u'(c_t)dc_t = u'(c_t)\theta_t F_2(k_t, \tilde{l}_t)\Delta$.

Equation (6) says that, at the optimum, the reduction in utility induced by a reduction in leisure (increase in labor) must be exactly compensated by the increase in utility induced by the extra consumption (obtained through the increase in labor). [Notice that Δ is in both terms, so it cancels out]

Another way: from (6) we get: $\frac{v'(1-\tilde{l}_t)}{u'(c_t)} = \theta_t F_2(k_t, \tilde{l}_t)$. This says that, at the optimum, the marginal rate of substitution between leisure and consumption, $\frac{v'(1-\tilde{l}_t)}{u'(c_t)}$, must be equal to the marginal rate of transformation between them, which equals the marginal product of labor, $\theta_t F_2(k_t, \tilde{l}_t)$.

Interpretation of (7) [Euler Equation for Consumption]

Consider a small reduction in cons. at t , $dc_t = -\Delta < 0$. This changes utility by $u'(c_t)dc_t = -u'(c_t)\Delta < 0$.

Now use the reduction in consumption to increase the capital stock. From the resource constraint we get (keeping output constant): $dc_t + (1+g)(1+n)dk_{t+1} = 0$. Then: $dk_{t+1} = \frac{\Delta}{(1+g)(1+n)}$.

The increase in capital increases next-period output by: $dy_{t+1} = \theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1})dk_{t+1} = \frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1})\Delta}{(1+g)(1+n)}$. In addition, you get the undepreciated capital: $(1-\delta)dk_{t+1} = \frac{(1-\delta)\Delta}{(1+g)(1+n)}$. Imagine you use all these extra resources to increase next-period consumption. Then: $dc_{t+1} = \frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}\Delta$. The increase in expected utility (recall θ_{t+1} is unknown at t) induced by the extra consumption is:

$$\mathbb{E}_t\{\beta u'(c_{t+1})dc_{t+1}\} = \mathbb{E}_t\left\{\beta u'(c_{t+1})\frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}\right\}\Delta.$$

Hence, (7) says that, at the optimum, the reduction in utility induced by a reduction in current consumption must be exactly compensated by the increase in expected utility induced by next period's additional consumption.

Think about the case with no uncertainty. Then, (8) becomes: $u'(c_t) = \beta u'(c_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$. We can rewrite this as follows: $\frac{u'(c_t)}{\beta u'(c_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$. This says that, at the optimum, the marginal rate of substitution between current and next-period consumption, $\frac{u'(c_t)}{\beta u'(c_{t+1})}$, must equal the marginal rate of transformation between current and next-period consumption, $\frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$.

Interpretation of (9) [Euler Equation for Labor/Leisure]

Consider a small reduction in current leisure, $d(1 - \tilde{l}_t) = -\Delta$. This changes utility by $v'(1 - \tilde{l}_t)d(1 - \tilde{l}_t) = -v'(1 - \tilde{l}_t)\Delta < 0$.

The reduction in leisure increases labor by: $d\tilde{l}_t = \Delta$. This increases current output by: $dy_t = \theta_t F_2(k_t, \tilde{l}_t)d\tilde{l}_t = \theta_t F_2(k_t, \tilde{l}_t)\Delta$.

Imagine you use this extra output to increase the capital stock. From the resource constraint we get (keeping c_t constant): $(1 + g)(1 + n)dk_{t+1} = dy_t$. Then: $dk_{t+1} = \frac{\theta_t F_2(k_t, \tilde{l}_t)\Delta}{(1+g)(1+n)}$.

For a given level of next-period labor, the increase in capital would increase next-period output by: $dy_{t+1} = \theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1})dk_{t+1}$.

In addition, you would get the undepreciated capital: $(1 - \delta)dk_{t+1}$.

Hence, total additional resources would be $dy_{t+1} + (1 - \delta)dk_{t+1} = [\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)] dk_{t+1}$.

Imagine you use these extra resources to reduce the amount of next-period work. How much would you be able to reduce $\tilde{l}_{t+1}$? From the prod. fn. we know that $dy_{t+1} = \theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1})dk_{t+1} + \theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})d\tilde{l}_{t+1}$. Then $dy_{t+1} + (1-\delta)dk_{t+1} = \left[\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta) \right] dk_{t+1} + \theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})d\tilde{l}_{t+1}$.

Setting $dy_{t+1} + (1-\delta)dk_{t+1} = 0$ we get:

$$\begin{aligned} d\tilde{l}_{t+1} &= -\frac{\left[\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta) \right] dk_{t+1}}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})} \\ &= -\frac{\left[\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta) \right] \frac{\theta_t F_2(k_t, \tilde{l}_t) \Delta}{(1+g)(1+n)}}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})} \\ &= -\frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t, \tilde{l}_t) \Delta}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})}. \end{aligned}$$

Then, next-period leisure increases by:

$$d(1 - \tilde{l}_{t+1}) = \frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t, \tilde{l}_t) \Delta}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})}.$$

This increases exp. ut. by: $\mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) d(1 - \tilde{l}_{t+1}) \right\} = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})} \right\} \Delta$.

Hence, (9) says that, at the optimum, the reduction in utility induced by a reduction in current leisure must be exactly compensated by the increase in expected utility induced by next period's additional leisure.

Think about the case with no uncertainty. Then: $v'(1 - \tilde{l}_t) = \beta v'(1 - \tilde{l}_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})}$.

We can rewrite this as follows:

$$\frac{v'(1 - \tilde{l}_t)}{\beta v'(1 - \tilde{l}_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})}.$$

This says that, at the optimum, the marginal rate of substitution between current and next-period leisure must equal the marginal rate of transformation between current and next-period leisure.

The interpretation of (10) and (11) can be done analogously.

Wealth and Substitution Effects

Consider the deterministic case, for intuition.

Wealth Effect

An increase in θ_t has a positive wealth effect, since $y_t = \theta_t F(k_t, \tilde{l}_t)$ goes up for given values of k_t and $\tilde{l}_t$.

If consumption and leisure are normal goods, then c_t and $1 - \tilde{l}_t$ will go up (and $\tilde{l}_t$ will go down) for every t .

If the shock is temporary (i.e., θ_t increases but $\theta_{t+1}, \theta_{t+2}, \dots$ remain constant) then c_t will increase only a little, $\tilde{l}_t$ will go down only a little, and k_{t+1} will go up a lot in order to increase future output possibilities and then future consumption and leisure.

If the shock is permanent (i.e., θ_t increases for all t) the increase in c_t and the reduction in $\tilde{l}_t$ will be big, and the effect on k_{t+1} will be basically zero, since there's no incentive to shift resources towards the future.

Substitution Effects

The analysis of the wealth effects did not consider the impact of θ_t on marginal products.

An increase of θ_t increases the current marginal product of labor. This has an intratemporal substitution effect: reduce current leisure (increase labor) and increase current consumption. This can be seen from the intratemporal FOC: $\frac{v'(1-\tilde{l}_t)}{u'(c_t)} = \theta_t F_2(k_t, \tilde{l}_t)$.

Remark. Strictly speaking, the expression above does not imply that consumption and labor have to go up after the positive technology shock. However, to think about the substitution effect we need to eliminate the wealth effect. If there's no wealth effect, consumption and leisure have to go in opposite directions (with no extra wealth, to consume more you have to work more). The same applies to the comments below.

An increase of θ_t relative to θ_{t+1} increases the current marginal product of labor relative to tomorrow's. For a given marginal product of capital, this has an intertemporal substitution effect on labor/leisure: work more in t , work less in $t + 1$. From (9):

$$\frac{v'(1-\tilde{l}_t)}{\beta v'(1-\tilde{l}_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})}.$$

The latter effect requires an increase in investment in t , and then k_{t+1} increases.

In period t , an expected increase in θ_{t+1} increases the expected marginal product of capital. This has an intertemporal substitution effect on consumption: reduce c_t and increase c_{t+1} . From (7): $\frac{u'(c_t)}{\beta u'(c_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$.

In period t , an expected increase in θ_{t+1} increases the expected marginal product of capital. For given relative marginal products of labor, this has an intert. subst. effect on labor: increase $\tilde{l}_t$ and reduce $\tilde{l}_{t+1}$. From (9):

$$\frac{v'(1-\tilde{l}_t)}{\beta v'(1-\tilde{l}_{t+1})} = \frac{\theta_{t+1}F_1(k_{t+1},\tilde{l}_{t+1})+(1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t,\tilde{l}_t)}{\theta_{t+1}F_2(k_{t+1},\tilde{l}_{t+1})}.$$

The two effects above imply that the increase in the expected marginal product of capital increases investment in period t , and hence increases k_{t+1} .

Define:

$$R_t \equiv \theta_t F_1(k_t, \tilde{l}_t) \quad MPK_t$$

$$w_t \equiv \theta_t F_2(k_t, \tilde{l}_t) A_t \quad MPL_t$$

Remark: $MPK_t = \frac{\partial Y_t}{\partial K_t} = \theta_t F_1(K_t, A_t L_t) = \theta_t F_1(k_t, \tilde{l}_t)$, where the last equality follows from the homogeneity of degree zero of F_1 in K and L (dividing by $A_t N_t$). Analogously, $MPL_t = \frac{\partial Y_t}{\partial L_t} = \theta_t \frac{\partial F(K_t, A_t L_t)}{\partial L_t} = \theta_t F_2(K_t, A_t L_t) A_t = \theta_t F_2(k_t, \tilde{l}_t) A_t$, where the last equality follows from the homogeneity of degree zero of F_2 in K and L (dividing by $A_t N_t$).

Since F is homogeneous of degree one in K and L , Euler's Theorem implies $Y_t = \theta_t \frac{\partial F}{\partial L_t} L_t + \theta_t \frac{\partial F}{\partial K_t} K_t \Rightarrow Y_t = \theta_t F_2 A_t L_t + \theta_t F_1 K_t \Rightarrow Y_t = w_t L_t + R_t K_t$. Dividing by $A_t N_t$:

$$y_t = w_t l_t + R_t k_t$$

$$= \frac{w_t}{A_t} \tilde{l}_t + R_t k_t$$

Using the expressions above, we can rewrite the FOCs as follows:

$$v'(1 - \tilde{l}_t) = u'(c_t) \frac{w_t}{A_t}$$

$$u'(c_t) = \mathbb{E}_t \left\{ \beta u'(c_{t+1}) \frac{R_{t+1} + (1 - \delta)}{(1 + g)(1 + n)} \right\}$$

$$c_t + (1 + g)(1 + n)k_{t+1} = \frac{w_t}{A_t} \tilde{l}_t + R_t k_t + (1 - \delta)k_t$$

The Euler equation for labor/leisure becomes:

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{R_{t+1} + (1 - \delta)}{(1 + g)(1 + n)} \frac{\frac{w_t}{A_t}}{\frac{w_{t+1}}{A_{t+1}}} \right\}$$

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{R_{t+1} + (1 - \delta)}{(1 + g)(1 + n)} \frac{w_t}{w_{t+1}} \frac{A_{t+1}}{A_t} \right\}$$

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{R_{t+1} + (1 - \delta)}{1 + n} \frac{w_t}{w_{t+1}} \right\}$$

where the last line uses $\frac{A_{t+1}}{A_t} = 1 + g$.

Notice that the expressions above are exactly the ones we would get in the decentralized version of the model. In this case, w_t would be the wage rate at t , R_t would be the rental rate of capital at t , and $y_t = w_t l_t + R_t k_t$ is the zero-profit condition. We can also reinterpret the wealth and substitution effects (e.g., the wealth and substitution effects of a temporary change in wages).

If we add a credit market to the decentralized version, the Euler equations for consumption and labor can be written as follows:

$$u'(c_t) = \mathbb{E}_t \left\{ \beta u'(c_{t+1}) \frac{1+r_t}{(1+g)(1+n)} \right\}$$

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{1+r_t}{1+n} \frac{w_t}{w_{t+1}} \right\}$$

where r_t is the bond's real interest rate (determined in period t , paid in $t+1$). In this case, $1+r_t$ can be moved outside the expectations operator (because r_t is known in period t).

A Closed-Form Solution for our Special Case

The Bellman Equation is:

$$V(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) \\ + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1}) \end{array} \right\}$$

$$\text{s.t.} \quad (1 + g)(1 + n)k_{t+1} = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

Guess:

$$V^G(k_t, \theta_t) = D + E \ln k_t + F \ln \theta_t$$

Then:

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) \\ + \beta \mathbb{E}_t \{ D + E \ln k_{t+1} + F \ln \theta_{t+1} \} \end{array} \right\}$$

$$\text{s.t.} \quad (1 + g)(1 + n)k_{t+1} = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

Then

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \\ \beta \mathbb{E}_t \left\{ \begin{array}{l} D + E \ln \left(\frac{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}{(1+g)(1+n)} \right) \\ + F[(1-\rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}] \end{array} \right\} \end{array} \right\}$$

Then:

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \beta D \\ + \beta E \ln \left(\frac{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}{(1+g)(1+n)} \right) \\ + \beta F(1 - \rho) \ln \theta \\ + \beta F \rho \ln \theta_t + \beta F \mathbb{E}_t \varepsilon_{t+1} \end{array} \right\}$$

Then:

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \beta D \\ + \beta E \ln \left(\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t \right) \\ - \beta E \ln[(1+g)(1+n)] \\ + \beta F(1 - \rho) \ln \theta + \beta F \rho \ln \theta_t \end{array} \right\}$$

FOC:

$$c_t : \quad \frac{1}{c_t} = \beta E \frac{1}{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}$$

$$\tilde{l}_t : \quad \frac{\gamma}{1-\tilde{l}_t} = \beta E \frac{(1-\alpha)\theta_t k_t^\alpha \tilde{l}_t^{-\alpha}}{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}$$

From the FOC for c_t we get:

$$c_t = \frac{1}{1 + \beta E} \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$$

Substituting the expression above into the FOC for $\tilde{l}_t$ we get:

$$\tilde{l}_t = \frac{(1 - \alpha)(1 + \beta E)}{(1 - \alpha)(1 + \beta E) + \gamma} \equiv \tilde{l}(E)$$

Remark. $\tilde{l}$ is constant over time.

Substituting $\tilde{l}_t = \tilde{l}(E)$ into the expression for c_t :

$$c_t = \frac{1}{1 + \beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha}$$

Substituting the previous expressions into the resource constraint:

$$k_{t+1} = \frac{1}{(1 + g)(1 + n)} \frac{\beta E}{1 + \beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha}$$

Substituting into the objective function:

$$\begin{aligned}\tilde{V}(k_t, \theta_t) &= \left\{ \begin{aligned} &\ln \left(\frac{1}{1+\beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha} \right) + \gamma \ln(1 - \tilde{l}(E)) \\ &+ \beta D + \beta E \ln \left(\frac{1}{(1+g)(1+n)} \frac{\beta E}{1+\beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha} \right) \\ &+ \beta F[(1-\rho) \ln \theta + \rho \ln \theta_t] \end{aligned} \right\} \\ \tilde{V}(k_t, \theta_t) &= \left\{ \begin{aligned} &\ln \left(\frac{1}{1+\beta E} [\tilde{l}(E)]^{1-\alpha} \right) + \ln \theta_t + \alpha \ln k_t + \gamma \ln(1 - \tilde{l}(E)) \\ &+ \beta D + \beta E \ln \left(\frac{1}{(1+g)(1+n)} \frac{\beta E}{1+\beta E} [\tilde{l}(E)]^{1-\alpha} \right) \\ &+ \beta E \ln \theta_t + \beta E \alpha \ln k_t + \beta F(1-\rho) \ln \theta + \beta F \rho \ln \theta_t \end{aligned} \right\} \\ \tilde{V}(k_t, \theta_t) &= \left\{ \begin{aligned} &\left[\ln \left(\frac{1}{1+\beta E} [\tilde{l}(E)]^{1-\alpha} \right) + \gamma \ln(1 - \tilde{l}(E)) + \beta D \right. \\ &\quad \left. + \beta E \ln \left(\frac{[\tilde{l}(E)]^{1-\alpha}}{(1+g)(1+n)} \frac{\beta E}{1+\beta E} \right) + \beta F(1-\rho) \ln \theta \right] \\ &+ (1 + \beta E + \rho \beta F) \ln \theta_t + \alpha(1 + \beta E) \ln k_t \end{aligned} \right\}\end{aligned}$$

Then:

$$\tilde{V}(k_t, \theta_t) = \Omega(D, E, F) + \alpha(1 + \beta E) \ln k_t + (1 + \beta E + \rho\beta F) \ln \theta_t$$

For $\tilde{V}(k_t, \theta_t)$ to equal $V^G(k_t, \theta_t) \forall (k_t, \theta_t)$ we need

$$\begin{aligned} D &= \Omega(D, E, F) \\ E &= \alpha(1 + \beta E) \\ F &= 1 + \beta E + \rho\beta F \end{aligned}$$

From the second equation we obtain:

$$E = \frac{\alpha}{1 - \alpha\beta}$$

Substituting into the third equation:

$$F = \frac{1}{(1 - \alpha\beta)(1 - \rho\beta)}$$

Substituting the values of E and F into the first equation we can obtain the value of the additive constant, D .

Using the results above we conclude:

$$V(k_t, \theta_t) = D + \frac{\alpha}{1 - \alpha\beta} \ln k_t + \frac{1}{(1 - \alpha\beta)(1 - \rho\beta)} \ln \theta_t$$

and

$$\tilde{l} = \frac{1 - \alpha}{(1 - \alpha) + \gamma(1 - \alpha\beta)} \in (0, 1)$$

$$c_t = (1 - \alpha\beta) \theta_t k_t^\alpha \tilde{l}^{1-\alpha}$$

$$k_{t+1} = \frac{\alpha\beta}{(1 + g)(1 + n)} \theta_t k_t^\alpha \tilde{l}^{1-\alpha}$$

Remark 1. Notice that $c_t = (1 - \alpha\beta) y_t$ and $k_{t+1} = \frac{\alpha\beta}{(1+g)(1+n)} y_t$. Hence, constant saving rate, $s = \alpha\beta$.

Remark 2. Labor and leisure are constant. Therefore, wealth and substitution effects cancel each other.

Nonstochastic Steady State (Balanced Growth)

Since $\tilde{l}$ is constant, it's clear that:

$$\tilde{l}_{nss} = \tilde{l} = \frac{1 - \alpha}{(1 - \alpha) + \gamma(1 - \alpha\beta)}$$

Recall $\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$ with $\varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$ and $\rho \in (0, 1)$. Setting $\varepsilon_{t+1} = \mathbb{E}\varepsilon_{t+1} = 0$ and $\theta_{t+1} = \theta_t = \theta_{nss}$ we get:

$$\theta_{nss} = \theta$$

Setting $\theta_t = \theta$ and $k_{t+1} = k_t = k_{nss}$ in $k_{t+1} = \frac{\alpha\beta}{(1+g)(1+n)} \theta_t k_t^\alpha \tilde{l}^{1-\alpha}$ we get: $k_{nss} = \frac{\alpha\beta}{(1+g)(1+n)} \theta k_{nss}^\alpha \tilde{l}^{1-\alpha}$.
Then:

$$k_{nss} = \left[\frac{\alpha\beta\theta}{(1+g)(1+n)} \right]^{\frac{1}{1-\alpha}} \tilde{l}$$

From $y_t = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$ we get:

$$y_{nss} = \theta k_{nss}^\alpha \tilde{l}_{nss}^{1-\alpha}$$

From $c_t = (1 - \alpha\beta) y_t$ we get:

$$c_{nss} = (1 - \alpha\beta) y_{nss}$$

Recall that: $x_t \equiv \frac{X_t}{A_t N_t} = \frac{\tilde{x}_t}{A_t}$. Therefore, if x is constant in the nonstochastic steady state, the aggregate variable X grows at (gross) rate $(1 + g)(1 + n)$, and the per-capita variable $\tilde{x}_t$ grows at (gross) rate $1 + g$.

Calibration

Strategy: Choose parameter values so that the balanced-growth path of the model economy matches certain long-term features of the measured economy.

Remark. Go back to the case with $\delta < 1$ (because we want to calibrate its value).

Calibration of n

The rate of population growth, n , can be measured directly, as the average population growth over a long period.

Calibration of g

In the nonstochastic steady state of the model, g equals the growth rate of per capita GDP. Hence, g can be measured directly as the average growth of per capita GDP over a long period.

Remark. Later we'll provide another way of calibrating g .

Calibration of α

From $Y_t = \theta_t K_t^\alpha (A_t L_t)^{1-\alpha}$ we get: $\frac{\partial Y_t}{\partial L_t} = (1 - \alpha) \frac{Y_t}{L_t}$.

Now think about the behavior of firms in the decentralized version of the model.

Real profits are: $Y_t - w_t L_t - R_t K_t$.

Profit maximization with respect to L_t : $\frac{\partial Y_t}{\partial L_t} = w_t$.

Then: $(1 - \alpha) \frac{Y_t}{L_t} = w_t$.

Then: $1 - \alpha = \frac{w_t L_t}{Y_t}$.

Therefore, $1 - \alpha$ can be measured as the average share of labor income over a long period.

Calibration of ρ and σ_ε^2

The production function is: $Y_t = \theta_t K_t^\alpha (A_t L_t)^{1-\alpha}$

Then: $Y_t = (\theta_t A_t^{1-\alpha}) K_t^\alpha L_t^{1-\alpha} = z_t K_t^\alpha L_t^{1-\alpha}$.

Then: $\ln Y_t = \ln z_t + \alpha \ln K_t + (1 - \alpha) \ln L_t$.

Then: $\ln z_t = \ln Y_t - \alpha \ln K_t - (1 - \alpha) \ln L_t$.

Given our value of α , the r-h-s can be estimated from data. This gives an estimate of $\ln z_t$ (Solow Residual).

We know $\ln z_t = \ln(\theta_t A_t^{1-\alpha})$.

Then: $\ln z_t = (1 - \alpha) \ln A_t + \ln \theta_t$.

Recall $A_t = (1 + g)^t A_0$.

Then: $\ln A_t = \ln A_0 + \ln(1 + g) \times t$.

Then: $\ln z_t = (1 - \alpha)[\ln A_0 + \ln(1 + g) \times t] + \ln \theta_t$.

Then: $\ln z_t = (1 - \alpha) \ln A_0 + (1 - \alpha) \ln(1 + g) \times t + \ln \theta_t$.

Then: $\ln z_t = \beta_0 + \beta_1 t + u_t$.

Then, fitting a linear trend to $\ln z_t$, and given our calibrated value for α , we can estimate:

(i) A_0 from $\beta_0 = (1 - \alpha) \ln A_0$ (this is not very interesting for our purposes);

(ii) g , from $\beta_1 = (1 - \alpha) \ln(1 + g)$;

(iii) the noise $u_t = \ln \theta_t$ for each t (from the regression residuals)

With the estimated time series for $\ln \theta_t$ we can run a regression of the form

$$\ln \theta_t = \vartheta + \rho \ln \theta_{t-1} + \varepsilon_t.$$

This gives:

- (i) an estimate of ρ ;
- (ii) an estimate of θ , from $\vartheta = (1 - \rho) \ln \theta$ (again, not very interesting for our purposes).
- (iii) an estimate of ε_t for each t (from the regression residuals).

With these residuals we can calibrate σ_ε^2 .

Calibration of δ

We know: $K_{t+1} = I_t + (1 - \delta)K_t$.

Then: $\frac{K_{t+1}}{A_{t+1}N_{t+1}} \frac{A_{t+1}N_{t+1}}{A_t N_t} = \frac{I_t}{A_t N_t} + (1 - \delta) \frac{K_t}{A_t N_t}$.

Then: $k_{t+1}(1 + g)(1 + n) = i_t + (1 - \delta)k_t$.

In the nonstochastic steady-state:

$$k_{nss}(1 + g)(1 + n) = i_{nss} + (1 - \delta)k_{nss}.$$

Then:

$$\delta = \frac{i_{nss}}{k_{nss}} + 1 - (1 + g)(1 + n).$$

Hence, with a long time series of the investment/capital ratio $\frac{I_t}{K_t}$ ($= \frac{i_t}{k_t}$) we can measure the depreciation rate (recall we have already calibrated g and n).

Calibration of β

Evaluating the Euler Equation for consumption at the nss:

$$u'(c_{nss}) = \beta u'(c_{nss}) \frac{\alpha \theta k_{nss}^{\alpha-1} \tilde{l}^{1-\alpha} + (1-\delta)}{(1+g)(1+n)}.$$

$$\text{Then: } (1+g)(1+n) = \beta \left[\alpha \frac{\theta k_{nss}^{\alpha} \tilde{l}^{1-\alpha}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } (1+g)(1+n) = \beta \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } (1+g)(1+n) = \tilde{\beta}(1+n) \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } (1+g) = \tilde{\beta} \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } \tilde{\beta} = (1+g) \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right]^{-1}.$$

Therefore, with a long time series of the output/capital ratio $\frac{Y_t}{K_t}$ ($= \frac{y_t}{k_t}$) we can measure the discount factor $\tilde{\beta}$ (recall we have already calibrated α , δ , and g).

Calibration of γ

From the intratemporal FOC:

$$v'(1 - \tilde{l}_{nss}) = u'(c_{nss})\theta F_2(k_{nss}, \tilde{l}_{nss}).$$

For logarithmic preferences and Cobb-Douglas technology we get:

$$\frac{\gamma}{1 - \tilde{l}_{nss}} = \frac{1}{c_{nss}}(1 - \alpha)\theta k_{nss}^{\alpha} \tilde{l}_{nss}^{-\alpha}.$$

$$\text{Then: } \frac{\gamma}{1 - \tilde{l}_{nss}} = \frac{1}{c_{nss}}(1 - \alpha)\frac{y_{nss}}{\tilde{l}_{nss}}.$$

$$\text{Then: } \gamma = (1 - \alpha)\frac{y_{nss}}{c_{nss}}\frac{1 - \tilde{l}_{nss}}{\tilde{l}_{nss}}.$$

Therefore, given an estimate of the long-run fraction of time devoted to market activities $\tilde{l}_{nss}$, and a long time series of the output/consumption ratio $\frac{Y_t}{C_t}$ ($= \frac{y_t}{c_t}$), we can measure the preference parameter γ (recall we have already calibrated α).

Example

For quarterly US data

α	δ	ρ	σ_ε	g	$\tilde{\beta}$	γ	n
0.4	0.012	0.95	0.007	0.0156/4	0.987	1.78	0.012/4

Notice that: $\beta = \tilde{\beta}(1 + n) = 0.9988 < 1$.

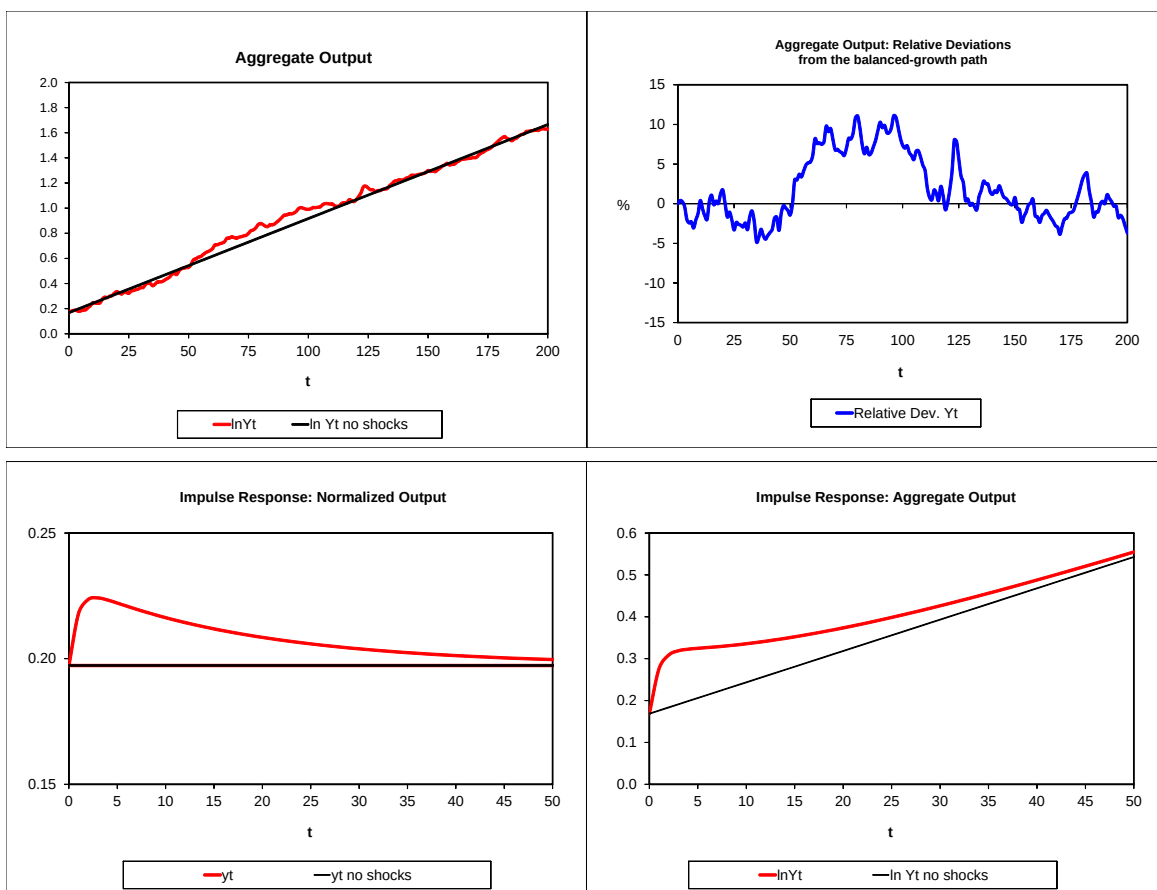
Adapted from

Cooley, T. and Prescott, E., "Economic Growth and Business Cycles", Chapter 1 in *Frontiers of Business Cycle Research*, edited by Thomas Cooley, Princeton University Press, 1995.

Plots ($\delta = 1$, parameters not calibrated, just an example)

Start at the nonstochastic steady state (no transition).

Set $A_0 = 6$, $L_0 = 1$, $\theta = 1$, $\alpha = 0.3$, $\tilde{\beta} = 0.99$, $n = 0.0025$, $g = 0.005$, $\gamma = 2$, $\rho = 0.95$, $\sigma_\varepsilon = 0.01$.
Then: $\tilde{l} = 0.33$, $k_{nss} = 0.058$, $y_{nss} = 0.197$.



Hodrick-Prescott (HP) Filter

Let $\{y_t\}$ be the time series to be decomposed into the trend component, $\{\tau_t\}$, and the cyclical component $\{c_t\}$, in such a way that $y_t = \tau_t + c_t$ (e.g., y_t may be the natural log of GDP).

The HP filter chooses the $\{\tau_t\}$ that solves:

$$\min \sum_{t=1}^T (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2$$

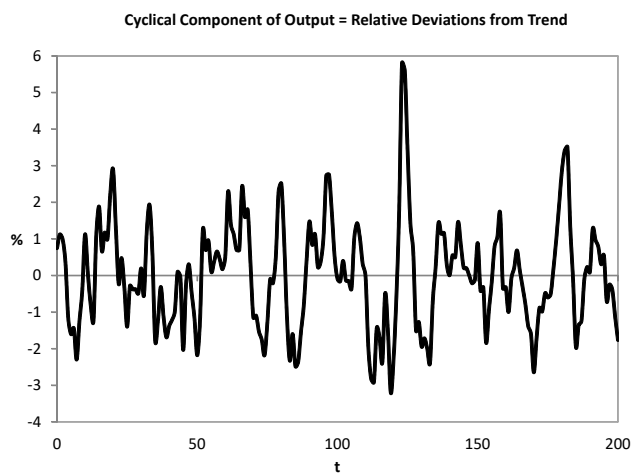
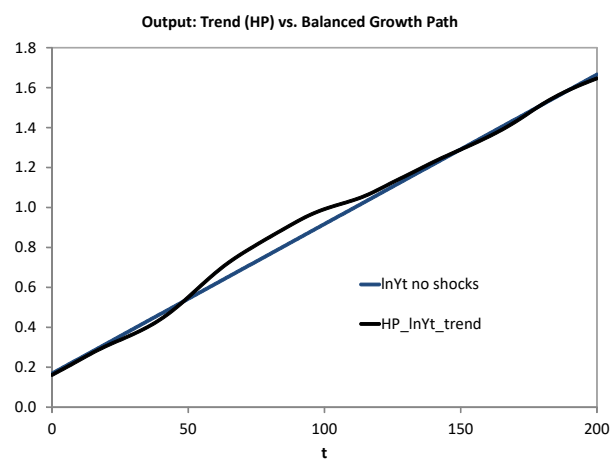
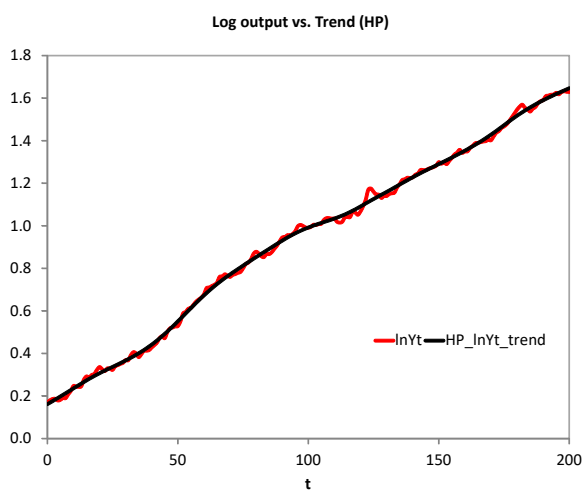
where λ is the penalty attached to the volatility of the trend.

We have: $\lambda = 0 \Rightarrow \tau_t = y_t$, and $\lambda \rightarrow \infty \Rightarrow \tau_t = \tau_0 + at$ (i.e., linear trend)

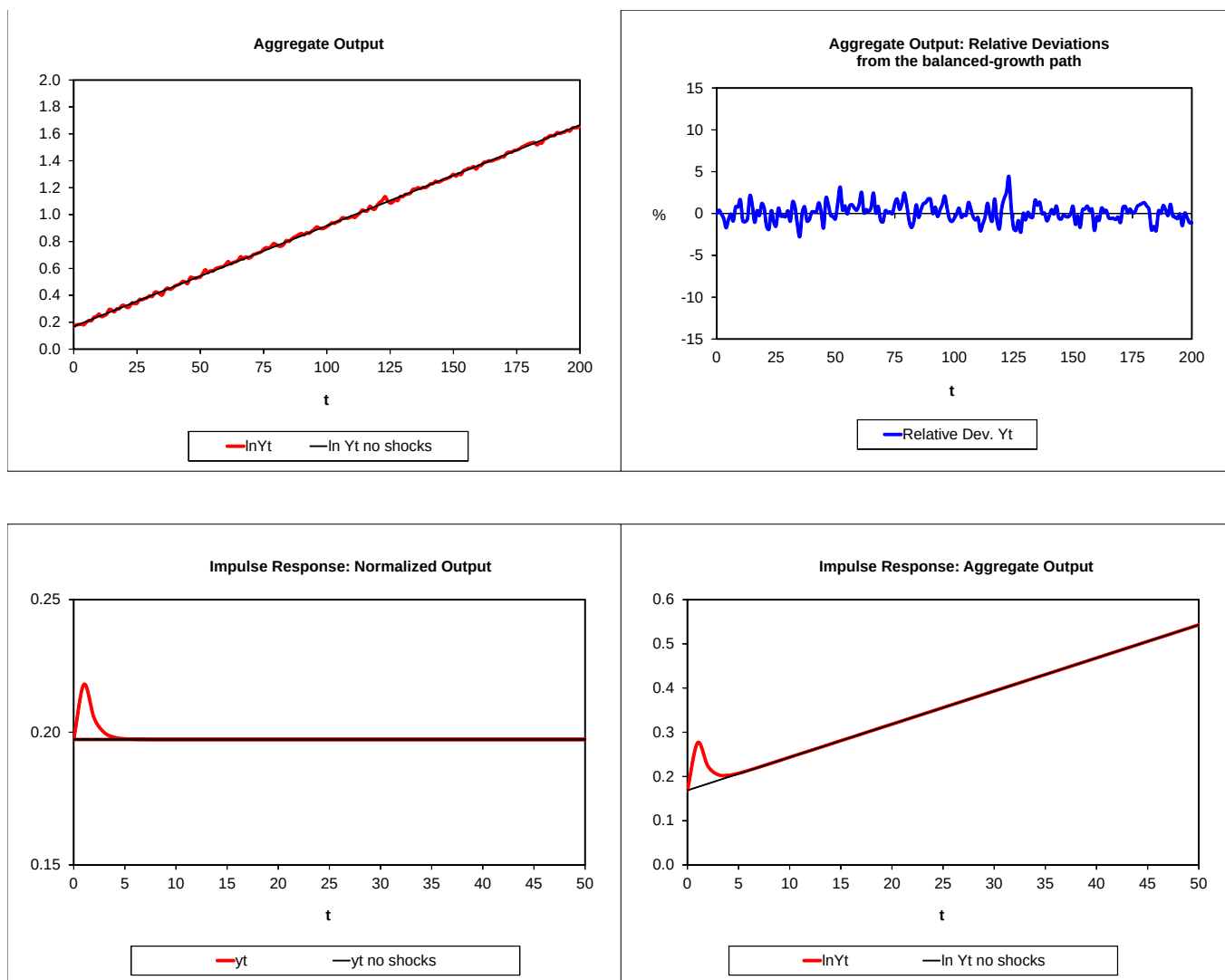
The standard value for quarterly data is $\lambda = 1600$. For annual data: $\lambda = 100$ (Ravn and Uhlig suggest $\lambda = 6.25$). For monthly data: $\lambda = 14400$ (Ravn and Uhlig suggest $\lambda = 129600$).

Remark. The cyclical component obtained from the HP filter is invariant to the presence of a linear trend.

Use the HP filter to detrend the series.



Reduce ρ to 0.1



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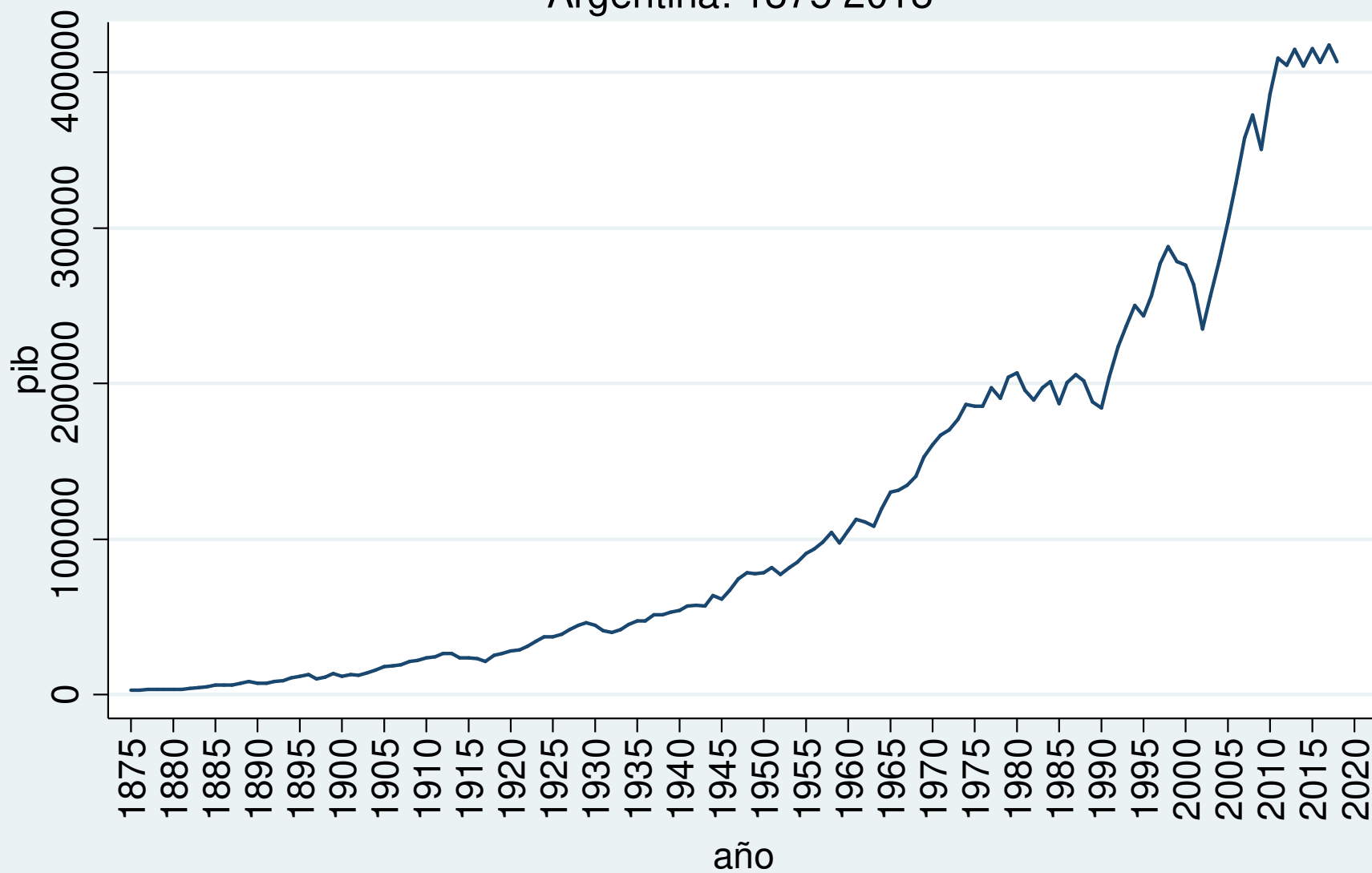
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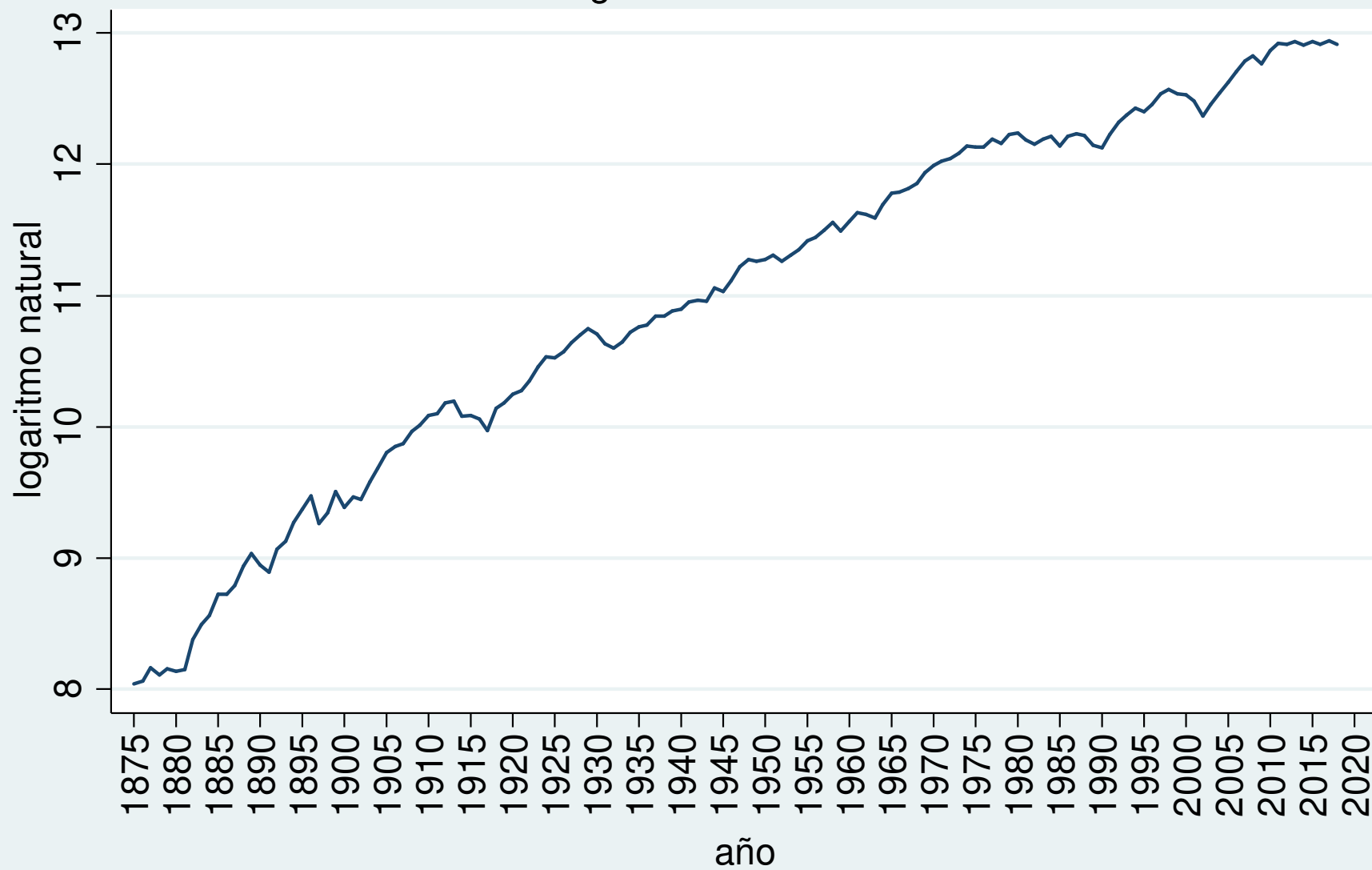
PIB real

Argentina: 1875 2018



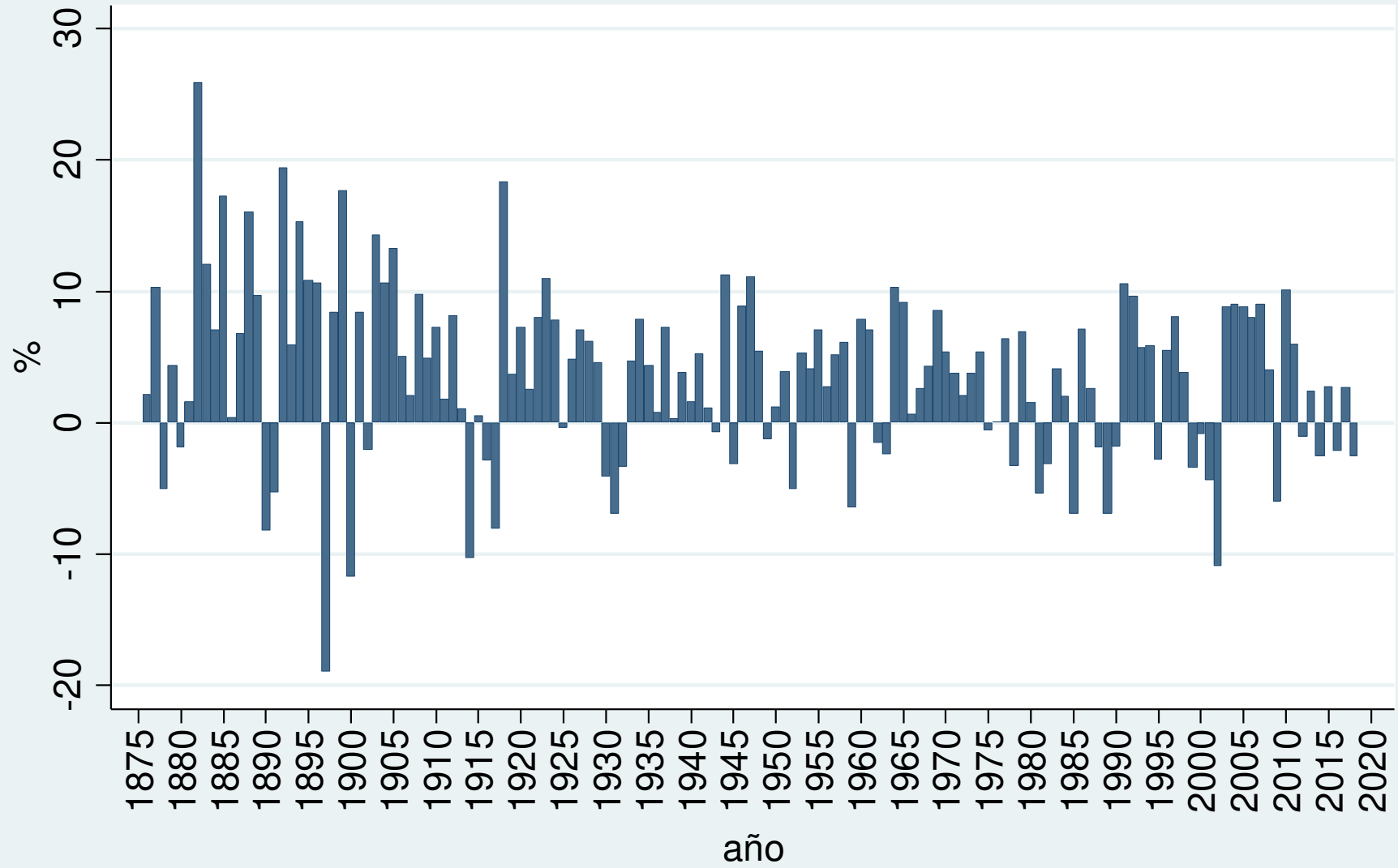
PIB real

Argentina: 1875 2018



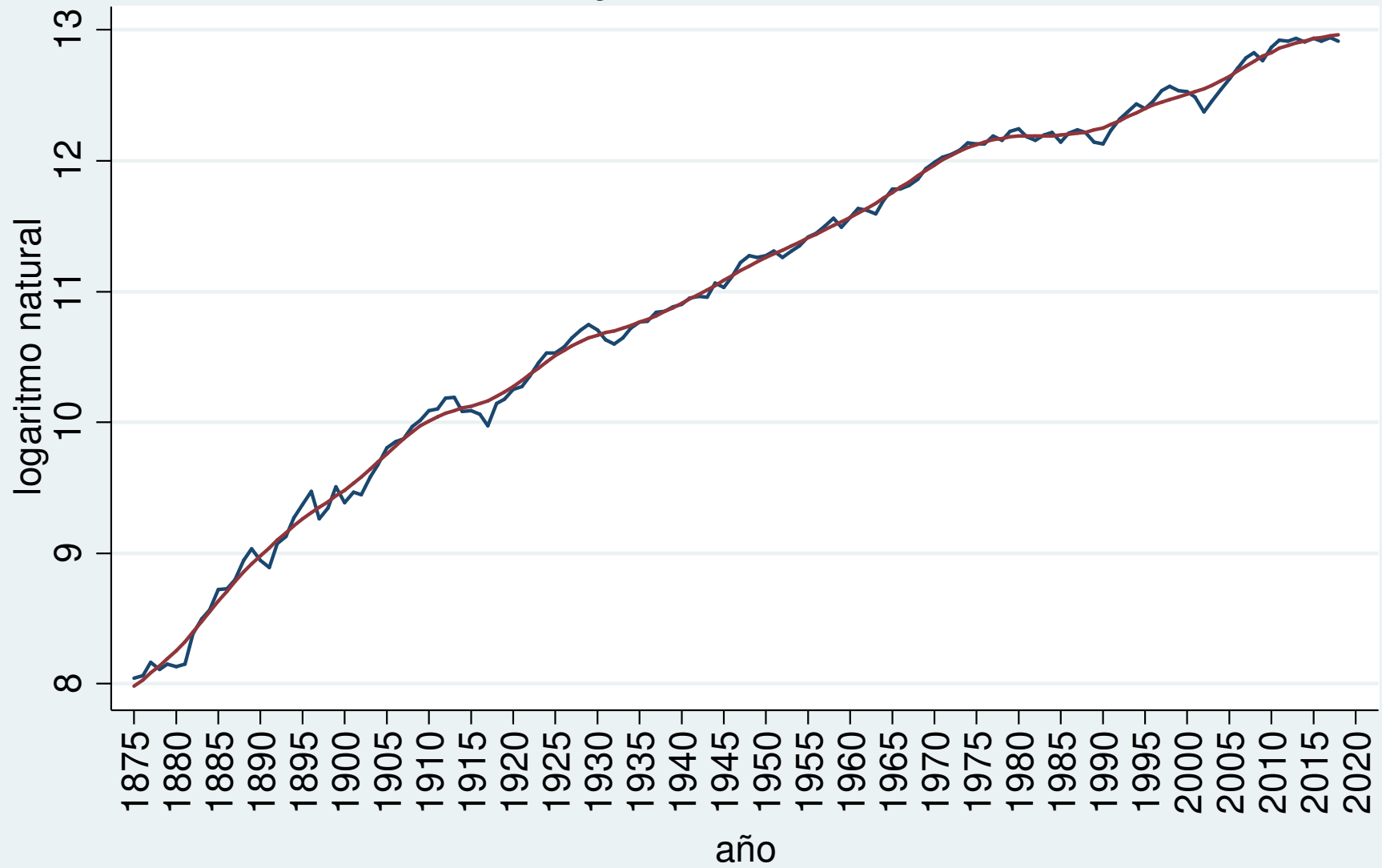
PIB real: Tasa de Crecimiento

Argentina: 1875 2018



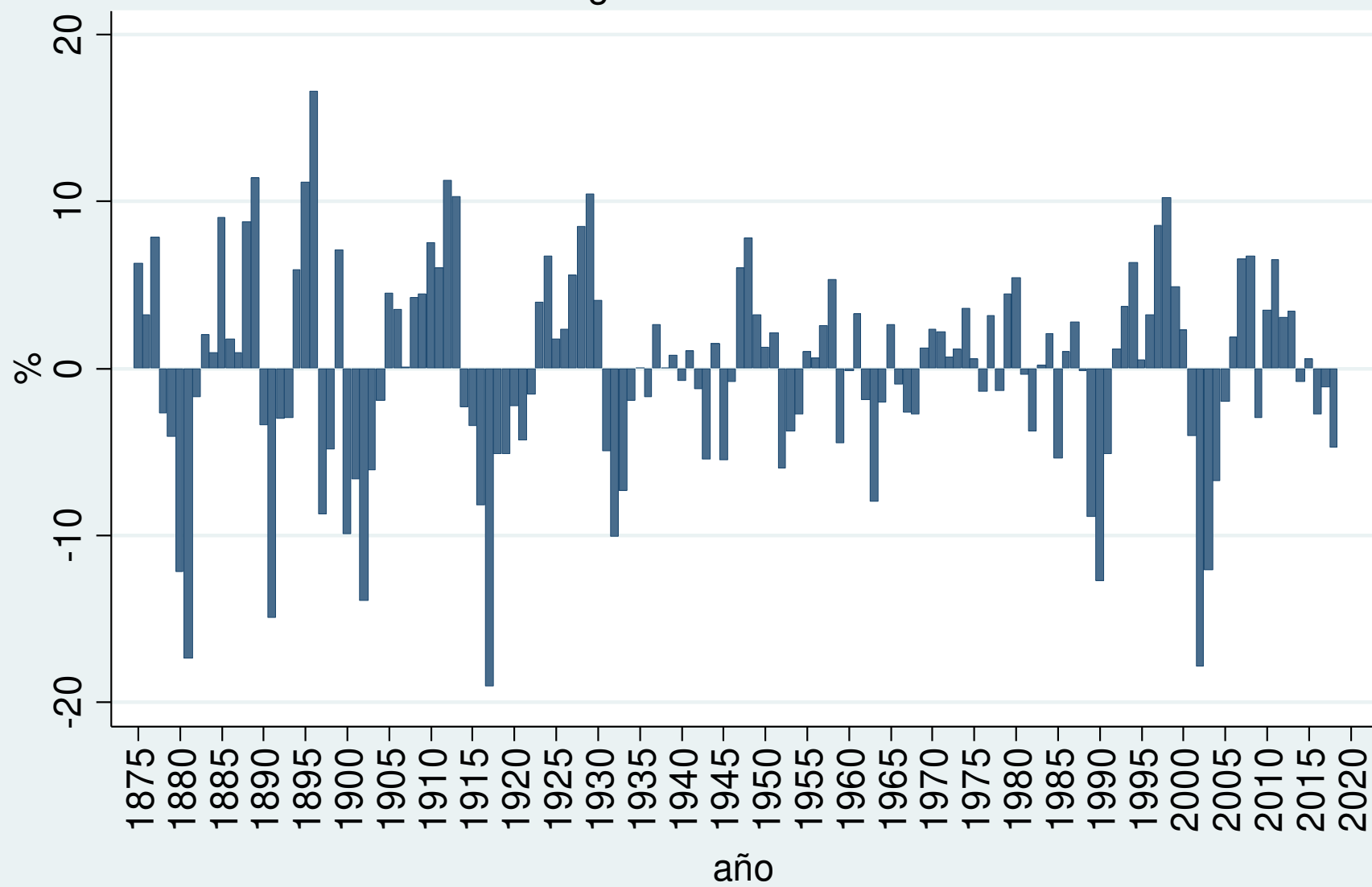
PIB real y Tendencia

Argentina: 1875 2018



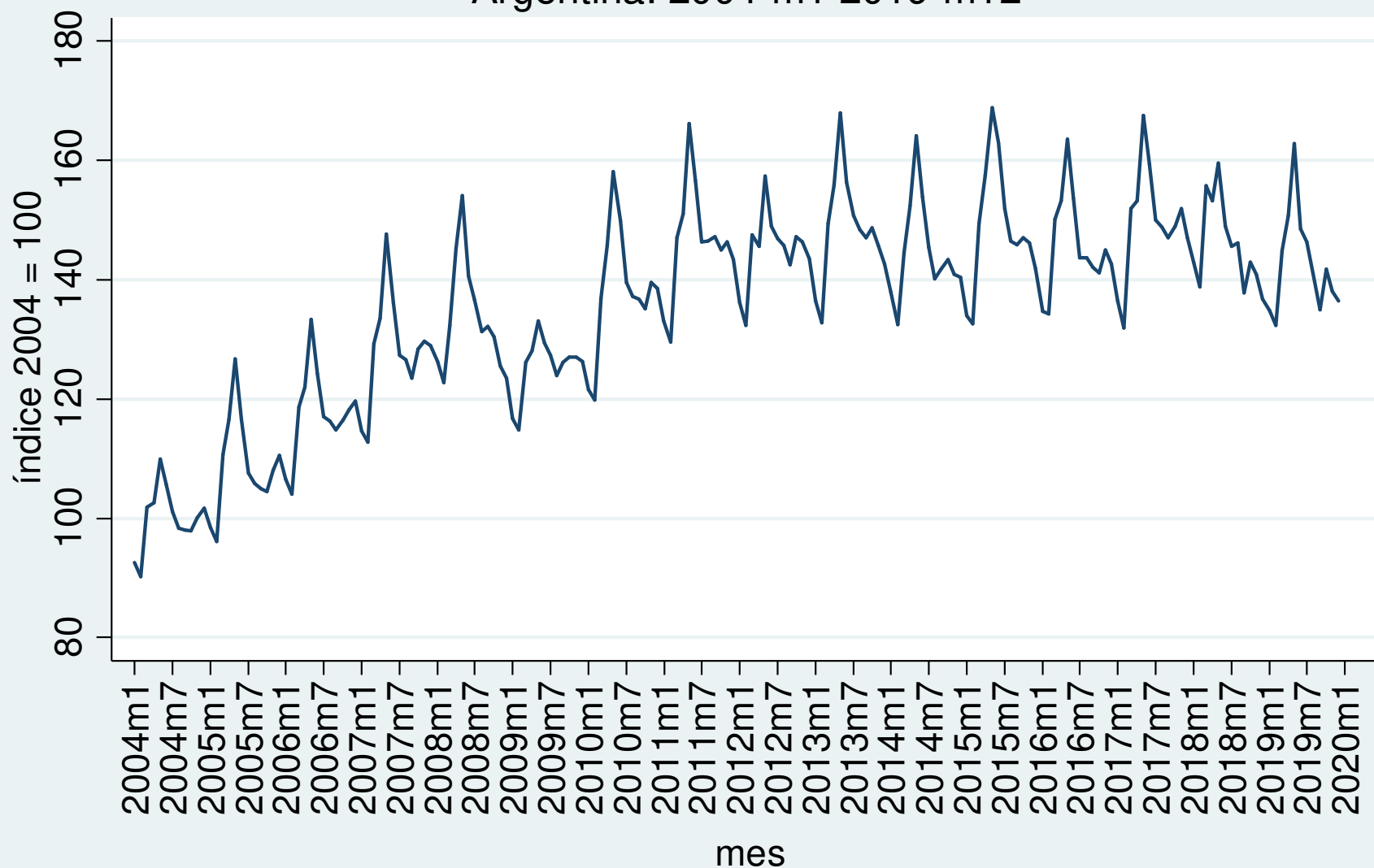
PIB: ciclo

Argentina: 1875 2018



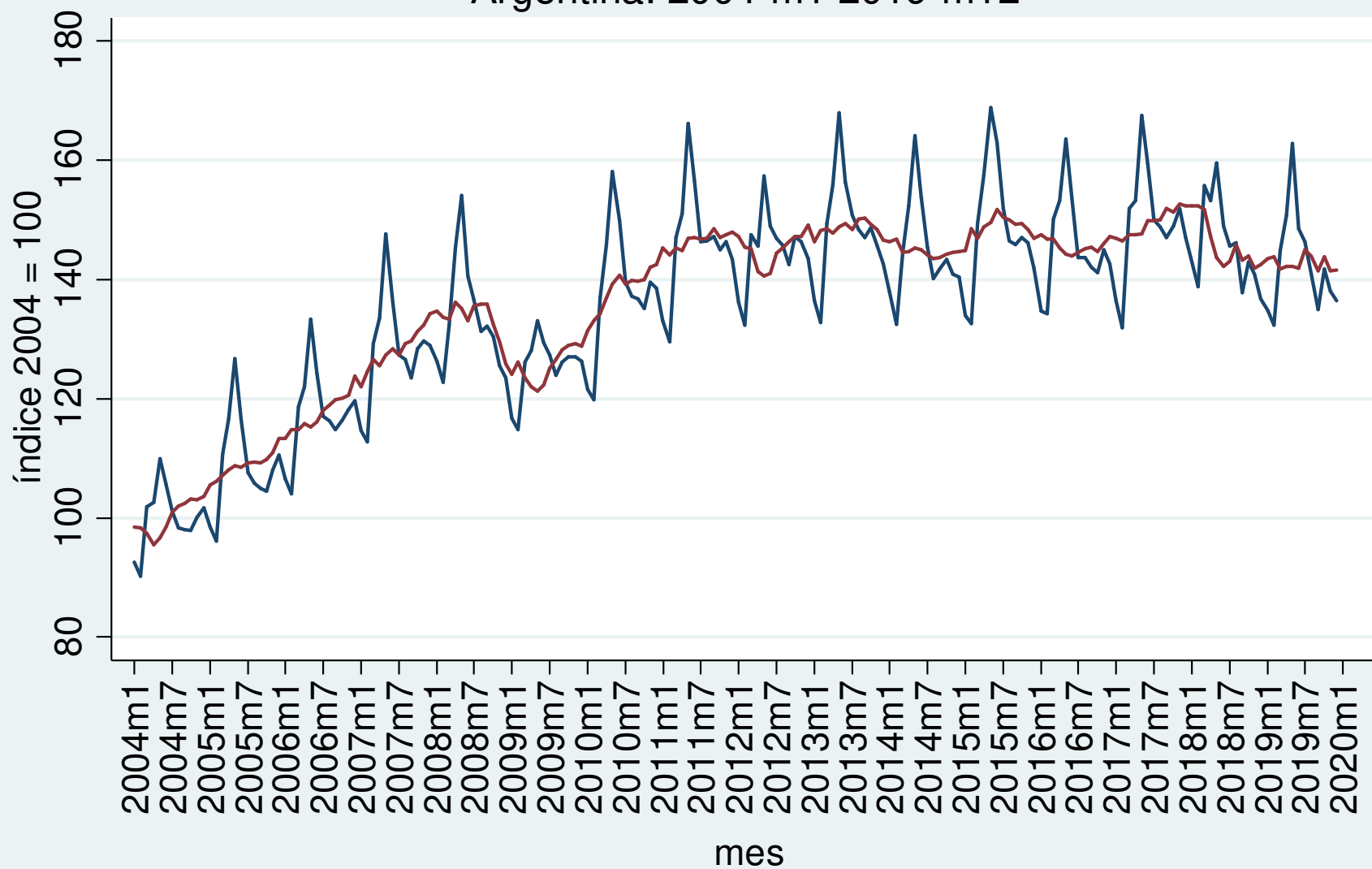
EMAE: Estimador Mensual de Actividad Económica

Argentina: 2004-m1 2019-m12



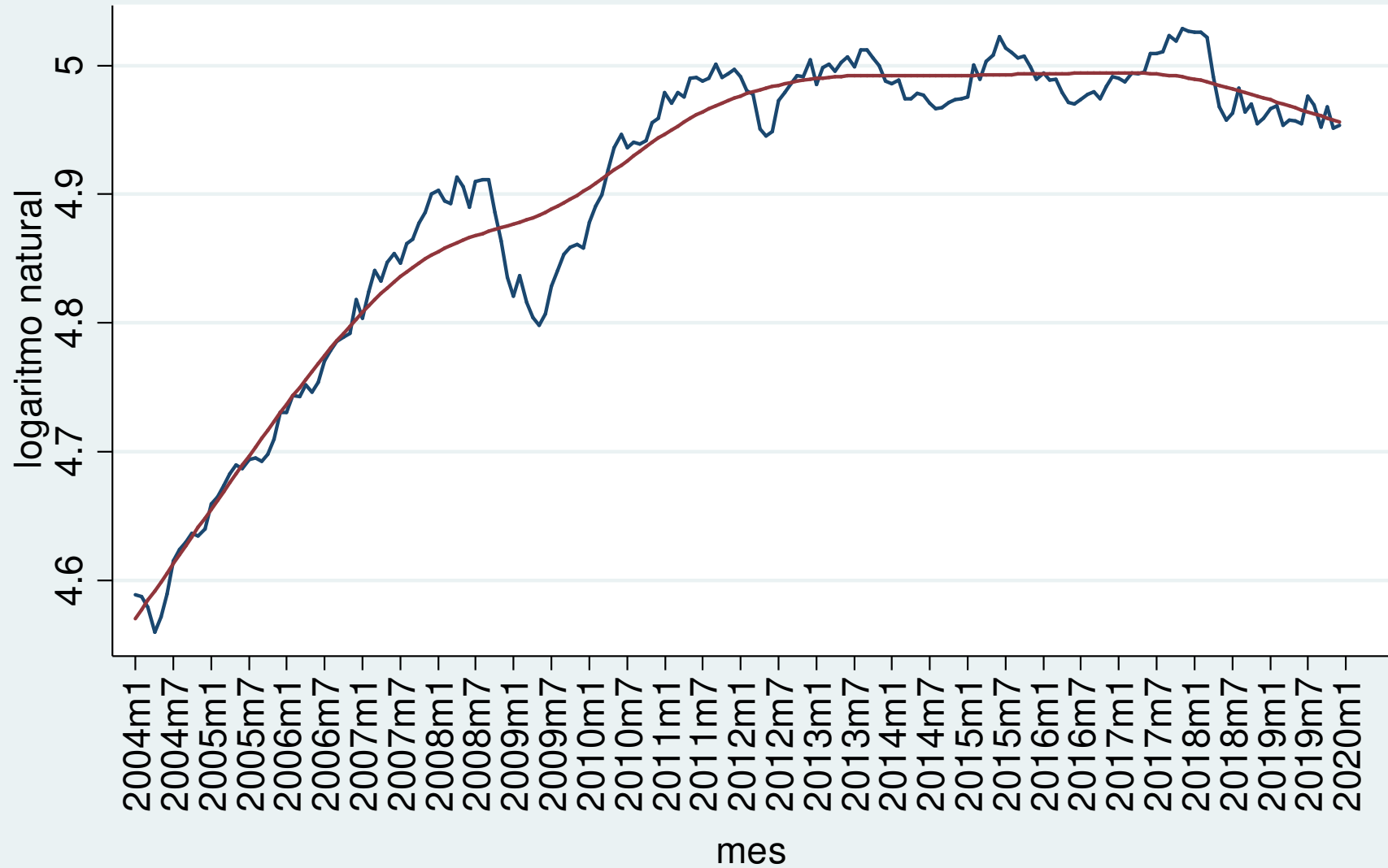
EMAE: nivel y serie desestacionalizada

Argentina: 2004-m1 2019-m12



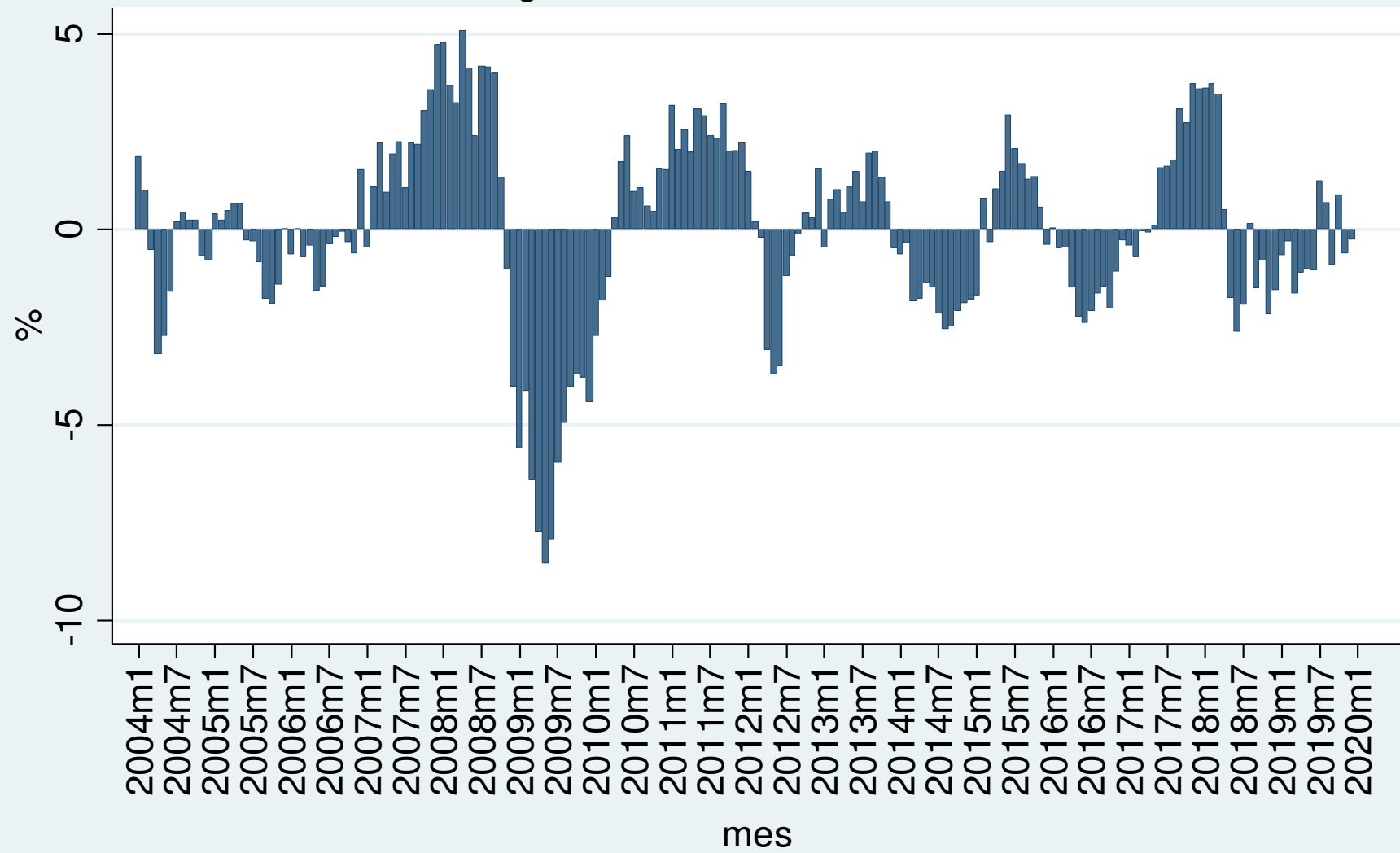
EMAE: serie desestacionalizada y tendencia

Argentina: 2004-m1 2019-m12



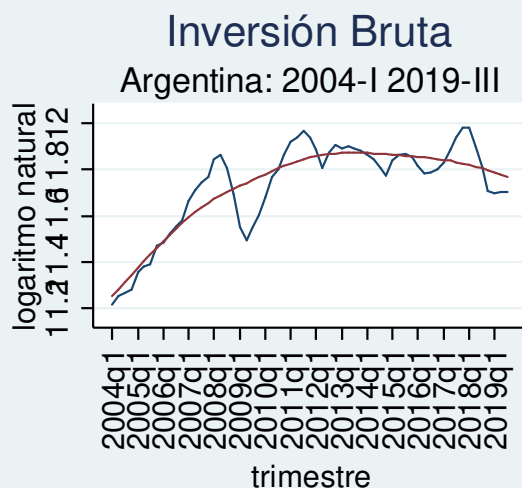
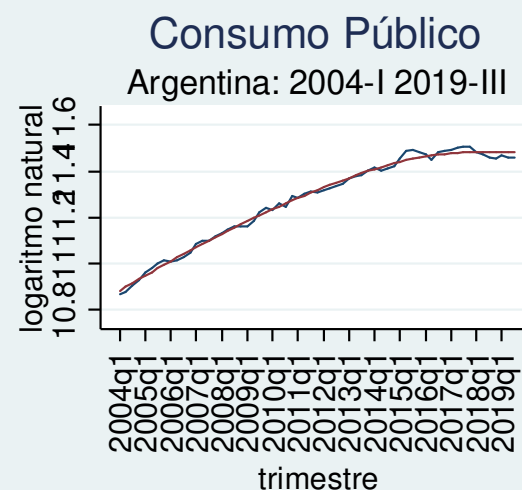
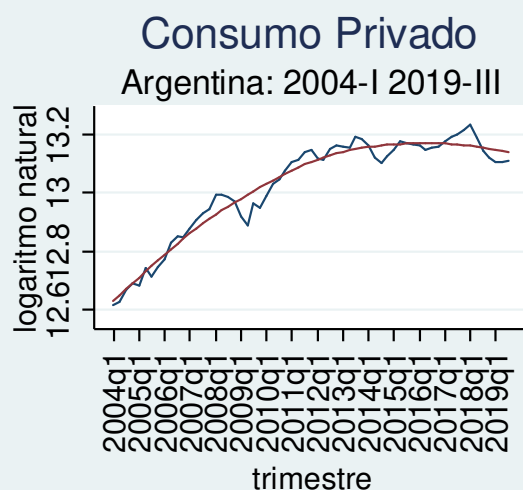
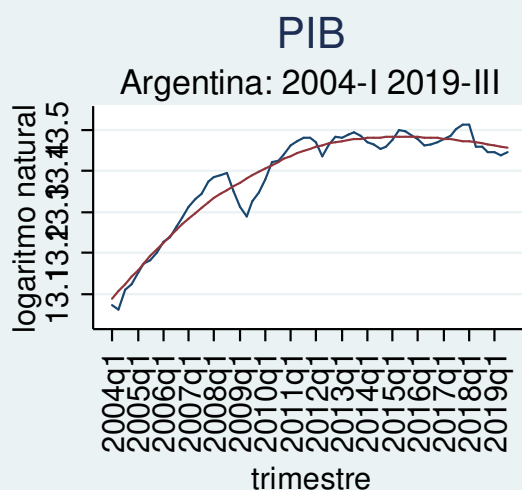
EMAE: ciclo

Argentina: 2004-m1 2019-m12



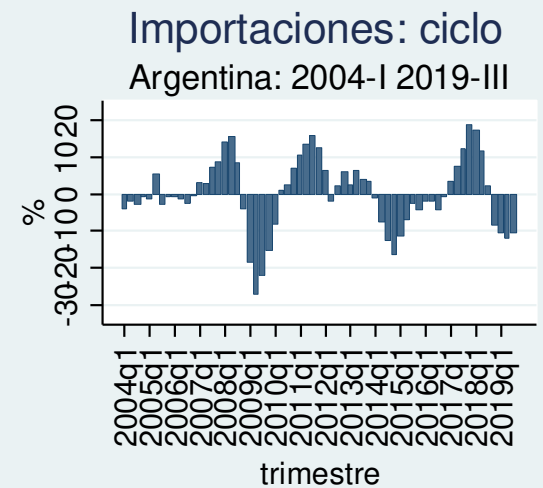
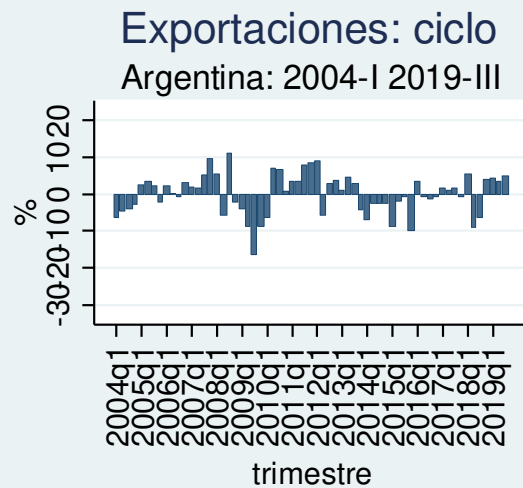
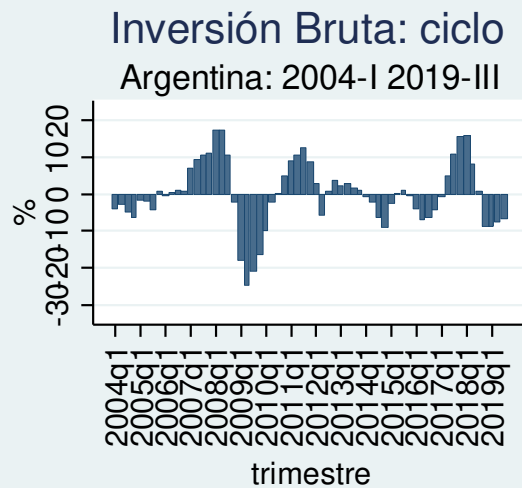
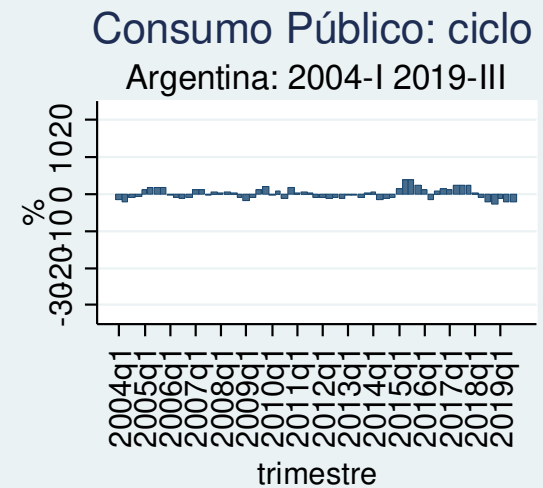
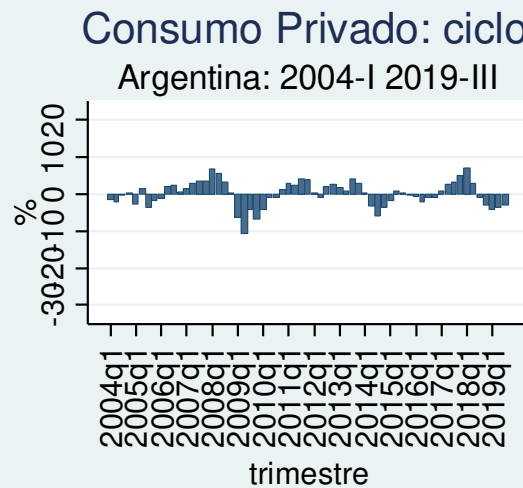
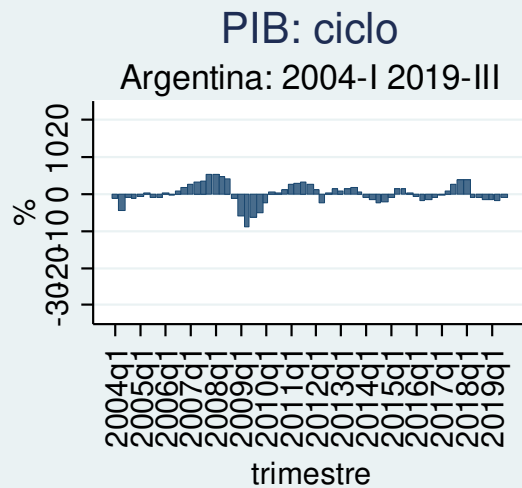
PIB y Componentes del Gasto: Nivel y Tendencia

Argentina: 2004-I 2019-III



PIB y Componentes del Gasto: Ciclo

Argentina: 2004-I 2019-III



Desvío Estándar Componente Cíclico

sd_pib	2.8
sd_con	3.0
sd_conp	3.4
sd_cong	1.6
sd_inv	8.7
sd_expo	5.5
sd_impo	9.6
sd_nxpib	2.0

Desvío Estándar del Componente Cíclico con relación al del PIB

sd_con / sd_pib	1.07
sd_conp / sd_pib	1.22
sd_cong / sd_pib	0.57
sd_inv / sd_pib	3.12
sd_expo / sd_pib	1.97
sd_impo / sd_pib	3.43
sd_nxpib / sd_pib	0.71

Correlaciones de los componentes cíclicos con el componente cíclico del PIB

	pib
pib	1.00
con	0.89
conp	0.89
cong	0.32
inv	0.94
expo	0.63
impo	0.86
nxpib	-0.55

Schmitt-Grohé and Uribe

Table 1.3: HP-Filtered Business Cycles

Statistic	All Countries	Poor Countries	Emerging Countries	Rich Countries
<u>Standard Deviations</u>				
σ_y	3.79	4.12	3.98	2.07
σ_c/σ_y	1.08	1.09	1.23	0.87
σ_g/σ_y	2.29	2.53	2.29	1.23
σ_i/σ_y	3.77	3.80	3.79	3.62
σ_x/σ_y	3.50	3.47	3.67	3.42
σ_m/σ_y	3.65	3.70	3.52	3.63
σ_{tb}/y	1.79	1.64	2.92	0.89
σ_{ca}/y	1.78	1.71	2.63	1.02
<u>Correlations with y</u>				
y	1.00	1.00	1.00	1.00
c	0.60	0.53	0.68	0.82
g/y	-0.08	0.02	-0.06	-0.56
i	0.69	0.65	0.71	0.86
x	0.19	0.18	0.13	0.30
m	0.32	0.23	0.46	0.58
tb/y	-0.18	-0.08	-0.34	-0.37
tb	-0.20	-0.11	-0.36	-0.36
ca/y	-0.32	-0.29	-0.39	-0.38
ca	-0.33	-0.29	-0.41	-0.37

Table 1.6: Duration and Amplitude of Business Cycles in Emerging and Developed Economies

Group of Countries	Duration		Amplitude	
	Contraction	Expansion	Contraction	Expansion
Latin America	3.5	16.0	6.2	21.3
OECD	3.6	23.8	2.2	20.2

Source: Calderón and Fuentes (2010).

Note: The data is quarterly real GDP from 1980:1 to 2006:4. The countries included in the Latin America group are: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Ecuador, Mexico, Paraguay, Peru, Uruguay, and Venezuela. The countries included in the OECD group are Australia, Canada, France, Germany, Italy, Japan, New Zealand, Portugal, Spain, Sweden, United Kingdom, and the United States.

Government Expenditure in a Real Business-Cycle Model

Technology:

$$Y_t = \theta_t K_t^\alpha L_t^{1-\alpha} \quad \alpha \in (0, 1)$$

with

$$\ln \theta_{t+1} = (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1}$$

where $\psi \in (0, 1)$, $\theta > 0$, and $\{\varepsilon_{t+1}\}_{t=0}^{\infty}$ is a sequence of i.i.d. random variables with $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.

Government expenditure evolves according to

$$\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \eta_{t+1}$$

where $\phi \in (0, 1)$, $G > 0$, and $\{\eta_{t+1}\}_{t=0}^{\infty}$ is a sequence of i.i.d. random variables with $\eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$.

The shocks to technology and government expenditure are independent from each other.

The government's budget constraint in period t is

$$G_t = T_t$$

where T_t denotes revenues from lump-sum taxes.

The utility function of the representative agent is

$$U = \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - L_t)$$

with

$$u(C_t, 1 - L_t) = \ln C_t + \gamma \ln(1 - L_t)$$

where $\gamma > 0$ and $\beta \equiv \frac{1}{1+\rho}$ with $\rho > 0$.

The budget constraint of the representative agent is

$$C_t + I_t = w_t L_t + r_t K_t + \Pi_t - T_t$$

where

$$I_t = K_{t+1} - (1 - \delta)K_t$$

and Π_t are profits (that will be zero in equilibrium).

Profit Maximization

$$\Pi_t \equiv \max_{K_t, L_t} \theta_t K_t^\alpha L_t^{1-\alpha} - w_t L_t - r_t K_t$$

FOC

$$K_t : r_t = \alpha \theta_t K_t^{\alpha-1} L_t^{1-\alpha} = \alpha \frac{\theta_t K_t^\alpha L_t^{1-\alpha}}{K_t}$$

$$L_t : w_t = (1 - \alpha) \theta_t K_t^\alpha L_t^{-\alpha} = (1 - \alpha) \frac{\theta_t K_t^\alpha L_t^{1-\alpha}}{L_t}$$

Then (d stands for demand):

$$r_t = \alpha \theta_t \left(\frac{K_t}{L_t} \right)^{\alpha-1} = \alpha \frac{Y_t}{K_t^d}$$

$$w_t = (1 - \alpha) \theta_t \left(\frac{K_t}{L_t} \right)^\alpha = (1 - \alpha) \frac{Y_t}{L_t^d}$$

and

$$\Pi_t = 0$$

Notice that the real wage equals the marginal product of labor, which is proportional to the average productivity of labor $\frac{Y_t}{L_t}$.

Utility Maximization

$$\max_{\{C_t, L_t, K_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\ln C_t + \gamma \ln(1 - L_t)]$$

$$\text{s.t.} \quad C_t + K_{t+1} - (1 - \delta)K_t = w_t L_t + r_t K_t - T_t$$

$$K_0 > 0 \text{ given}$$

Lagrangian

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \left\{ \begin{array}{l} \beta^t [\ln C_t + \gamma \ln(1 - L_t)] + \\ \lambda_t [w_t L_t + r_t K_t + (1 - \delta)K_t - T_t - C_t - K_{t+1}] \end{array} \right\}$$

FOC (s stands for supply):

$$C_t : \quad \frac{\beta^t}{C_t} = \lambda_t$$

$$L_t : \quad \frac{\beta^t \gamma}{1 - L_t^s} = \lambda_t w_t$$

$$K_{t+1} : \quad \lambda_t = \mathbb{E}_t \{ \lambda_{t+1} (r_{t+1} + 1 - \delta) \}$$

$$\lambda_t : \quad C_t + K_{t+1}^s - (1 - \delta)K_t = w_t L_t^s + r_t K_t - T_t$$

Combining the first-order conditions w.r.t. C_t and L_t we get:

$$\frac{\gamma C_t}{1 - L_t^s} = w_t$$

Combining the first-order conditions w.r.t. C_t and K_{t+1} we get the Euler Equation for consumption:

$$1 = \mathbb{E}_t \left\{ \beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

Equilibrium requires:

$$\left. \begin{aligned} K_t^d &= K_t^s = K_t \\ L_t^d &= L_t^s = L_t \end{aligned} \right\} \text{factor markets}$$

$$\left. \begin{aligned} Y_t &= C_t + I_t + G_t \\ Y_t &= \theta_t K_t^\alpha L_t^{1-\alpha} \\ I_t &= K_{t+1} - (1 - \delta)K_t \end{aligned} \right\} \text{output market}$$

$$\left. \begin{aligned} r_t &= \alpha \frac{Y_t}{K_t} \\ w_t &= (1 - \alpha) \frac{Y_t}{L_t} \end{aligned} \right\} \text{profit max}$$

$$\left. \begin{aligned} \frac{\gamma C_t}{1 - L_t} &= w_t \\ 1 &= \mathbb{E}_t \left\{ \beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\} \end{aligned} \right\} \text{utility max}$$

$$T_t = G_t \} \text{ gov. budget constraint}$$

Remark 1. Equilibrium in the market for output together with the zero-profit condition and the government's budget constraint ensure that the budget constraint of the household is satisfied.

Remark 2. We don't actually need to solve for T_t , since it's always equal to G_t .

Hence, we would like to solve the following system of nonlinear stochastic dynamic equations:

$$Y_t = \theta_t K_t^\alpha L_t^{1-\alpha}$$

$$w_t = (1 - \alpha) \frac{Y_t}{L_t}$$

$$r_t = \alpha \frac{Y_t}{K_t}$$

$$Y_t = C_t + I_t + G_t$$

$$I_t = K_{t+1} - (1 - \delta)K_t$$

$$\frac{\gamma C_t}{1 - L_t} = w_t$$

$$1 = \mathbb{E}_t \left\{ \beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

$$\ln \theta_{t+1} = (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1}$$

$$\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \eta_{t+1}$$

We have 9 equations per period to solve for the evolution of 9 variables: 7 endog. variables ($K_t, Y_t, C_t, I_t, L_t, r_t, w_t$) and 2 exogenous variables (θ_t, G_t). Of the 7 endogenous variables, one is a state variable (K_t) and six are jump variables ($Y_t, C_t, I_t, L_t, r_t, w_t$).

Nonstochastic Steady State

We use upper bars to denote nonstochastic steady-state values.

Setting $\varepsilon_{t+1} = 0$ and $\theta_{t+1} = \theta_t$ into $\ln \theta_{t+1} = (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1}$ we get

$$\bar{\theta} = \theta$$

Setting $\eta_{t+1} = 0$ and $G_{t+1} = G_t$ into $\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \eta_{t+1}$ we get

$$\bar{G} = G$$

Setting $C_t = C_{t+1}$ and $r_{t+1} = \bar{r}$ into the stochastic Euler equation we get:

$$\begin{aligned}\bar{r} &= \frac{1}{\beta} - 1 + \delta \\ &= \rho + \delta\end{aligned}$$

We know $r_t = \alpha \frac{Y_t}{K_t} = \alpha \theta_t \left(\frac{K_t}{L_t} \right)^{\alpha-1}$. Setting $r_t = \bar{r}$ and $\theta_t = \bar{\theta}$ we get

$$\overline{\left(\frac{K}{L} \right)} = \left(\frac{\alpha \bar{\theta}}{\bar{r}} \right)^{\frac{1}{1-\alpha}}$$

We know $w_t = (1 - \alpha) \frac{Y_t}{L_t} = (1 - \alpha) \theta_t \left(\frac{K_t}{L_t} \right)^{\alpha}$. Then:

$$\begin{aligned} \bar{w} &= (1 - \alpha) \bar{\theta} \overline{\left(\frac{K}{L} \right)}^{\alpha} \\ &= (1 - \alpha) \bar{\theta}^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \end{aligned}$$

We finish in two steps. First we express the rest of the variables in terms of $\bar{K}$ (and other known variables/parameters). Then we solve for $\bar{K}$.

Setting $K_{t+1} = K_t$ into $I_t = K_{t+1} - (1 - \delta)K_t$ we get:

$$\bar{I} = \delta \bar{K}$$

From $\overline{\left(\frac{K}{L}\right)} = \left(\frac{\alpha\bar{\theta}}{\bar{r}}\right)^{\frac{1}{1-\alpha}}$ we get:

$$\bar{L} = \left(\frac{\bar{r}}{\alpha\bar{\theta}}\right)^{\frac{1}{1-\alpha}} \bar{K}$$

From the production function: $\bar{Y} = \bar{\theta}\bar{K}^\alpha\bar{L}^{1-\alpha}$. Using the expression above: we get:

$$\bar{Y} = \frac{\bar{r}}{\alpha}\bar{K}$$

From $Y_t = C_t + I_t + G_t$ we get: $\bar{C} = \bar{Y} - \bar{I} - \bar{G}$.

Using the previous results: $\bar{C} = \frac{\bar{r}}{\alpha}\bar{K} - \delta\bar{K} - \bar{G}$.

Then:

$$\bar{C} = \left(\frac{\bar{r}}{\alpha} - \delta\right)\bar{K} - \bar{G}$$

From $\frac{\gamma C_t}{1-L_t} = w_t$ we get: $\gamma \bar{C} = \bar{w}(1 - \bar{L})$.

Substituting our previous results in the expression above:

$$\gamma \left[\left(\frac{\bar{r}}{\alpha} - \delta \right) \bar{K} - \bar{G} \right] = \bar{w} \left[1 - \left(\frac{\bar{r}}{\alpha \bar{\theta}} \right)^{\frac{1}{1-\alpha}} \bar{K} \right]$$

Solving for $\bar{K}$ we get

$$\bar{K} = \alpha \frac{\bar{w} + \gamma \bar{G}}{(\gamma + 1 - \alpha) \bar{r} - \gamma \alpha \delta}$$

where $\bar{G}$, $\bar{r}$ and $\bar{w}$ are already known.

Notice that the denominator is positive, so $\bar{K} > 0$.

Now we can substitute the value of $\bar{K}$ into our previous expressions to obtain $\bar{I}$, $\bar{L}$, $\bar{Y}$, and $\bar{C}$.

Solution Strategy

1. Log-linearize the equations characterizing the equilibrium of the system around the nonstochastic steady state. This gives a new system of linear stochastic difference equations whose solution approximates the one of the original nonlinear system (as long as we are close enough to the nonstochastic steady state).
2. Write the linear system in an appropriate matrix form.
3. Calibrate the model.
4. Solve for the recursive equilibrium law of motion using the method of undetermined coefficients.
5. Analyze the solution via impulse-response analysis and second-order properties (variances, covariances, etc.). If needed, detrend the variables using an appropriate filter (e.g., the Hodrick-Prescott filter).

Log-linearization

We follow Uhlig's approach to log-linearization (Uhlig, 1999). See the Appendix for a different approach.

For any variable X define the period- t log-deviation from its nonstochastic steady-state value:

$$\widetilde{X}_t \equiv \ln X_t - \ln \overline{X}$$

Then:

$$X_t = \overline{X} e^{\widetilde{X}_t}$$

We establish a few approximations that will be useful later on. Recall that a first-order Taylor approximation of $f(x, y)$ around $(x, y) = (0, 0)$ gives:

$$\begin{aligned} f(x, y) &\cong f(0, 0) + f_x(0, 0)(x - 0) + f_y(0, 0)(y - 0) \\ &= f(0, 0) + f_x(0, 0)x + f_y(0, 0)y \end{aligned}$$

Claim 1 :

$$e^{\widetilde{X}_t + a\widetilde{Y}_t} \cong 1 + \widetilde{X}_t + a\widetilde{Y}_t$$

Proof. A first-order Taylor approximation of $e^{\widetilde{X}_t + a\widetilde{Y}_t}$ around $(\widetilde{X}_t, \widetilde{Y}_t) = (0, 0)$ gives:

$$\begin{aligned} e^{\widetilde{X}_t + a\widetilde{Y}_t} &\cong e^{0+a0} + e^{0+a0}(\widetilde{X}_t - 0) + ae^{0+a0}(\widetilde{Y}_t - 0) \\ &= 1 + \widetilde{X}_t + a\widetilde{Y}_t \end{aligned}$$

■

Corollary 1 :

$$e^{\widetilde{X}_t} \cong 1 + \widetilde{X}_t$$

Claim 2 :

$$\widetilde{X}_t \widetilde{Y}_t \cong 0$$

Proof. A first-order Taylor approximation of $\widetilde{X}_t \widetilde{Y}_t$ around $(\widetilde{X}_t, \widetilde{Y}_t) = (0, 0)$ gives:

$$\begin{aligned} \widetilde{X}_t \widetilde{Y}_t &\cong 0 + 0(\widetilde{X}_t - 0) + 0(\widetilde{Y}_t - 0) \\ &= 0 \end{aligned}$$

■

Claim 3 :

$$\mathbb{E}_t\{ae^{\widetilde{X}_{t+1}}\} \cong a + a\mathbb{E}_t\{\widetilde{X}_{t+1}\}$$

Proof. Using $e^{\widetilde{X}_{t+1}} \cong 1 + \widetilde{X}_{t+1}$ inside the curly brackets:

$$\begin{aligned} \mathbb{E}_t\{ae^{\widetilde{X}_{t+1}}\} &\cong \mathbb{E}_t\{a(1 + \widetilde{X}_{t+1})\} \\ &= a + a\mathbb{E}_t\{\widetilde{X}_{t+1}\} \end{aligned}$$

■

Now we log-linearize the 9 equations that characterize the equilibrium of our model.

For the technology shock we get:

$$\begin{aligned}\tilde{\theta}_{t+1} &\equiv \ln \theta_{t+1} - \ln \bar{\theta} \\ &= (1 - \psi) \ln \theta + \psi \ln \theta_t + \varepsilon_{t+1} - \ln \bar{\theta} \\ &= (1 - \psi) \ln \bar{\theta} + \psi \ln \theta_t + \varepsilon_{t+1} - \ln \bar{\theta} \\ &= \psi (\ln \theta_t - \ln \bar{\theta}) + \varepsilon_{t+1} \\ &= \psi \tilde{\theta}_t + \varepsilon_{t+1}\end{aligned}$$

Analogously, for government expenditure we get:

$$\tilde{G}_{t+1} = \phi \tilde{G}_t + \eta_{t+1}$$

From the production function we get:

$$\begin{aligned}
\tilde{Y}_t &\equiv \ln Y_t - \ln \bar{Y} \\
&= \ln \left(\theta_t K_t^\alpha L_t^{1-\alpha} \right) - \ln \left(\bar{\theta} \bar{K}^\alpha \bar{L}^{1-\alpha} \right) \\
&= \ln \theta_t + \alpha \ln K_t + (1 - \alpha) \ln L_t \\
&\quad - \ln \bar{\theta} - \alpha \ln \bar{K} - (1 - \alpha) \ln \bar{L} \\
&= \left(\ln \theta_t - \ln \bar{\theta} \right) + \alpha \left(\ln K_t - \ln \bar{K} \right) \\
&\quad + (1 - \alpha) \left(\ln L_t - \ln \bar{L} \right) \\
&= \tilde{\theta}_t + \alpha \tilde{K}_t + (1 - \alpha) \tilde{L}_t
\end{aligned}$$

From $w_t = (1 - \alpha) \frac{Y_t}{L_t}$ we get:

$$\begin{aligned}
\tilde{w}_t &\equiv \ln w_t - \ln \bar{w} \\
&= \ln \left((1 - \alpha) \frac{Y_t}{L_t} \right) - \ln \left((1 - \alpha) \frac{\bar{Y}}{\bar{L}} \right) \\
&= \ln(1 - \alpha) + \ln Y_t - \ln L_t \\
&\quad - \ln(1 - \alpha) - \ln \bar{Y} + \ln \bar{L} \\
&= \left(\ln Y_t - \ln \bar{Y} \right) - \left(\ln L_t - \ln \bar{L} \right) \\
&= \tilde{Y}_t - \tilde{L}_t
\end{aligned}$$

Analogously, from $r_t = \alpha \frac{Y_t}{K_t}$ we get:

$$\tilde{r}_t = \tilde{Y}_t - \tilde{K}_t$$

Notice that the previous expressions are exact relations. No approximations were needed to derive them.

The following log-linearizations make use of the approximations derived previously (we keep using the $=$ sign instead of $\cong$ just for simplicity).

From $Y_t = C_t + I_t + G_t$ we get:

$$\overline{Y}e^{\tilde{Y}_t} = \overline{C}e^{\tilde{C}_t} + \overline{I}e^{\tilde{I}_t} + \overline{G}e^{\tilde{G}_t}$$

$$\overline{Y}(1 + \tilde{Y}_t) = \overline{C}(1 + \tilde{C}_t) + \overline{I}(1 + \tilde{I}_t) + \overline{G}(1 + \tilde{G}_t)$$

$$\overline{Y} + \overline{Y}\tilde{Y}_t = \overline{C} + \overline{C}\tilde{C}_t + \overline{I} + \overline{I}\tilde{I}_t + \overline{G} + \overline{G}\tilde{G}_t$$

$$\overline{Y} + \overline{Y}\tilde{Y}_t = (\overline{C} + \overline{I} + \overline{G}) + \overline{C}\tilde{C}_t + \overline{I}\tilde{I}_t + \overline{G}\tilde{G}_t$$

$$\overline{Y} + \overline{Y}\tilde{Y}_t = \overline{Y} + \overline{C}\tilde{C}_t + \overline{I}\tilde{I}_t + \overline{G}\tilde{G}_t$$

Then:

$$\overline{Y}\tilde{Y}_t = \overline{C}\tilde{C}_t + \overline{I}\tilde{I}_t + \overline{G}\tilde{G}_t$$

From $I_t = K_{t+1} - (1 - \delta)K_t$ we get:

$$\bar{I}e^{\tilde{I}_t} = \bar{K}e^{\tilde{K}_{t+1} - (1-\delta)\tilde{K}_t}$$

$$\bar{I}(1 + \tilde{I}_t) = \bar{K}(1 + \tilde{K}_{t+1}) - (1 - \delta)\bar{K}(1 + \tilde{K}_t)$$

$$\bar{I} + \bar{I}\tilde{I}_t = \bar{K} + \bar{K}\tilde{K}_{t+1} - (1 - \delta)\bar{K} - (1 - \delta)\bar{K}\tilde{K}_t$$

$$\bar{I} + \bar{I}\tilde{I}_t = \delta\bar{K} + \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

$$\bar{I} + \bar{I}\tilde{I}_t = \bar{I} + \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

$$\bar{I}\tilde{I}_t = \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

$$\delta\bar{K}\tilde{I}_t = \bar{K}\left[\tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t\right]$$

Then:

$$\delta\tilde{I}_t = \tilde{K}_{t+1} - (1 - \delta)\tilde{K}_t$$

From $\frac{\gamma C_t}{1-\bar{L}_t} = w_t$ we get:

$$\gamma C_t = w_t(1 - \bar{L}_t)$$

$$\gamma \bar{C} e^{\tilde{C}_t} = \bar{w} e^{\tilde{w}_t} (1 - \bar{L} e^{\tilde{L}_t})$$

$$\gamma \bar{C} (1 + \tilde{C}_t) = \bar{w} (1 + \tilde{w}_t) [1 - \bar{L} (1 + \tilde{L}_t)]$$

$$\gamma \bar{C} + \gamma \bar{C} \tilde{C}_t = (\bar{w} + \bar{w} \tilde{w}_t) [(1 - \bar{L}) - \bar{L} \tilde{L}_t]$$

$$\gamma \bar{C} + \gamma \bar{C} \tilde{C}_t = \bar{w} (1 - \bar{L}) - \bar{w} \bar{L} \tilde{L}_t + \bar{w} \tilde{w}_t (1 - \bar{L}) - \bar{w} \tilde{w}_t \bar{L} \tilde{L}_t$$

$$\bar{w} (1 - \bar{L}) + \bar{w} (1 - \bar{L}) \tilde{C}_t = \bar{w} (1 - \bar{L}) - \bar{w} \bar{L} \tilde{L}_t + \bar{w} (1 - \bar{L}) \tilde{w}_t - \bar{w} \bar{L} \tilde{w}_t \tilde{L}_t$$

$$\bar{w} (1 - \bar{L}) \tilde{C}_t = -\bar{w} \bar{L} \tilde{L}_t + \bar{w} (1 - \bar{L}) \tilde{w}_t$$

$$\tilde{C}_t = -\frac{\bar{w} \bar{L}}{\bar{w} (1 - \bar{L})} \tilde{L}_t + \frac{\bar{w} (1 - \bar{L})}{\bar{w} (1 - \bar{L})} \tilde{w}_t$$

Then:

$$\tilde{C}_t = \tilde{w}_t - \frac{\bar{L}}{1 - \bar{L}} \tilde{L}_t$$

From $1 = \mathbb{E}_t\{\beta \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta)\}$ we get:

$$1 = \mathbb{E}_t \left\{ \beta \frac{\bar{C}_e \tilde{C}_t}{\bar{C}_e \tilde{C}_{t+1}} \left(\bar{r} e^{\tilde{r}_{t+1}} + 1 - \delta \right) \right\}$$

$$1 = \mathbb{E}_t \{ \beta e^{\tilde{C}_t - \tilde{C}_{t+1}} [\bar{r}(1 + \tilde{r}_{t+1}) + 1 - \delta] \}$$

$$1 = \mathbb{E}_t \{ \beta (1 + \tilde{C}_t - \tilde{C}_{t+1}) [\bar{r} + \bar{r} \tilde{r}_{t+1} + 1 - \delta] \}$$

$$1 = \mathbb{E}_t \{ \beta (1 + \tilde{C}_t - \tilde{C}_{t+1}) [(\bar{r} + 1 - \delta) + \bar{r} \tilde{r}_{t+1}] \}$$

$$1 = \mathbb{E}_t \{ (\beta + \beta \tilde{C}_t - \beta \tilde{C}_{t+1}) (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) \}$$

$$1 = \mathbb{E}_t \{ \beta (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) + \beta \tilde{C}_t (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) - \beta \tilde{C}_{t+1} (\frac{1}{\beta} + \bar{r} \tilde{r}_{t+1}) \}$$

$$1 = \mathbb{E}_t \{ 1 + \beta \bar{r} \tilde{r}_{t+1} + \tilde{C}_t + \beta \bar{r} \tilde{C}_t \tilde{r}_{t+1} - \tilde{C}_{t+1} - \beta \bar{r} \tilde{C}_{t+1} \tilde{r}_{t+1} \}$$

$$1 = \mathbb{E}_t \{ 1 + \beta \bar{r} \tilde{r}_{t+1} + \tilde{C}_t - \tilde{C}_{t+1} \}$$

$$0 = \mathbb{E}_t \{ \beta \bar{r} \tilde{r}_{t+1} + \tilde{C}_t - \tilde{C}_{t+1} \}$$

Then:

$$0 = \mathbb{E}_t \{ \tilde{C}_{t+1} - \beta \bar{r} \tilde{r}_{t+1} - \tilde{C}_t \}$$

Summarizing

$$\tilde{Y}_t = \tilde{\theta}_t + \alpha \tilde{K}_t + (1 - \alpha) \tilde{L}_t$$

$$\tilde{w}_t = \tilde{Y}_t - \tilde{L}_t$$

$$\tilde{r}_t = \tilde{Y}_t - \tilde{K}_t$$

$$\bar{Y} \tilde{Y}_t = \bar{C} \tilde{C}_t + \bar{I} \tilde{I}_t + \bar{G} \tilde{G}_t$$

$$\delta \tilde{I}_t = \tilde{K}_{t+1} - (1 - \delta) \tilde{K}_t$$

$$\tilde{C}_t = \tilde{w}_t - \frac{\bar{L}}{1 - \bar{L}} \tilde{L}_t$$

$$0 = \mathbb{E}_t \left\{ \tilde{C}_{t+1} - \beta \bar{r} \tilde{r}_{t+1} - \tilde{C}_t \right\}$$

$$\tilde{\theta}_{t+1} = \psi \tilde{\theta}_t + \varepsilon_{t+1}$$

$$\tilde{G}_{t+1} = \phi \tilde{G}_t + \eta_{t+1}$$

where $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and $\eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$ (independent from each other).

Matrix Form

Define the following column vectors:

$$x_t \equiv [\widetilde{K}_{t+1}] , \quad y_t \equiv \begin{bmatrix} \widetilde{Y}_t \\ \widetilde{C}_t \\ \widetilde{I}_t \\ \widetilde{L}_t \\ \widetilde{r}_t \\ \widetilde{w}_t \end{bmatrix} , \quad z_t \equiv \begin{bmatrix} \widetilde{\theta}_t \\ \widetilde{G}_t \end{bmatrix} , \quad \epsilon_{t+1} \equiv \begin{bmatrix} \varepsilon_{t+1} \\ \eta_{t+1} \end{bmatrix}$$

Then, we can rewrite our log-linearized system in matrix form, as follows:

$$\mathbf{0} = Ax_t + Bx_{t-1} + Cy_t + Dz_t$$

$$\mathbf{0} = \mathbb{E}_t \left\{ \begin{array}{l} Fx_{t+1} + Gx_t + Hx_{t-1} + \\ Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t \end{array} \right\}$$

$$z_{t+1} = Nz_t + \epsilon_{t+1} \quad \text{with} \quad \mathbb{E}_t\{\epsilon_{t+1}\} = \mathbf{0}$$

where

$$A = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad B = \begin{bmatrix} -\alpha \\ 0 \\ -1 \\ 0 \\ -(1-\delta) \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & -(1-\alpha) & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & 0 \\ \bar{Y} & -\bar{C} & -\bar{I} & 0 & 0 & 0 \\ 0 & 0 & -\delta & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{\bar{L}}{1-\bar{L}} & 0 & -1 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -\overline{G} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$F = [0], \quad G = [0], \quad H = [0]$$

$$J = \begin{bmatrix} 0 & 1 & 0 & 0 & -\beta \bar{r} & 0 \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} 0 & 0 \end{bmatrix}, \quad M = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$N = \begin{bmatrix} \psi & 0 \\ 0 & \phi \end{bmatrix}$$

Calibration

For quarterly data (similar to Hansen & Wright, 1992):

$$\alpha = 0.36$$

$$\beta = 0.99$$

$$\gamma = 2.5$$

$$\delta = 0.025$$

$$\psi = 0.95$$

$$\phi = 0.96$$

$$\bar{\theta} = 1$$

$$\bar{G} = 0.27$$

$$\sigma_{\varepsilon} = 0.007$$

$$\sigma_{\eta} = 0.021$$

Then:

$$\bar{r} = 0.035$$

$$\bar{w} = 2.37$$

$$\overline{K} = 12.5$$

$$\overline{L} = 0.329$$

$$\overline{Y} = 1.219$$

$$\overline{I} = 0.312$$

$$\overline{C} = 0.636$$

$$\frac{\overline{G}}{\overline{\overline{Y}}} = 0.22$$

$$\frac{\overline{I}}{\overline{\overline{Y}}} = 0.26$$

$$\frac{\overline{K}}{4\overline{\overline{Y}}} = 2.56$$

Solution (Uhlig, 1999)

We look for recursive law of motions

$$x_t = Px_{t-1} + Qz_t$$

$$y_t = Rx_{t-1} + Sz_t$$

with the property that the equilibrium described by these rules is *stable*.

Solution strategy: solve for P , Q , R , and S using the method of undetermined coefficients.

We assume that the matrix C in the block of nonexpectational equations is invertible. For the more general case see Uhlig (1999).

We start by substituting $y_t = Rx_{t-1} + Sz_t$ into the first block $0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t$. Then:

$$0 = Ax_t + Bx_{t-1} + C(Rx_{t-1} + Sz_t) + Dz_t$$

$$0 = Ax_t + Bx_{t-1} + CRx_{t-1} + CSz_t + Dz_t$$

$$0 = Ax_t + (B + CR)x_{t-1} + (D + CS)z_t$$

Then:

$$Ax_t = -(B + CR)x_{t-1} - (D + CS)z_t$$

From $x_t = Px_{t-1} + Qz_t$ we get:

$$Ax_t = APx_{t-1} + AQz_t$$

For the right-hand sides of the two expressions above to coincide *for every pair* (x_{t-1}, z_t) , we need:

$$AP = -(B + CR)$$

$$AQ = -(D + CS)$$

From the first expression we get

$$R = -C^{-1}(AP + B)$$

and from the second

$$S = -C^{-1}(AQ + D)$$

Hence, we've managed to express R in terms of P , and S in terms of Q . Now we have to find P and Q .

Substituting $x_t = Px_{t-1} + Qz_t$, $y_t = Rx_{t-1} + Sz_t$, and $z_{t+1} = Nz_t + \epsilon_{t+1}$ into $\mathbf{0} = \mathbb{E}_t\{Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t\}$ we get:

$$\mathbf{0} = \mathbb{E}_t \left\{ \begin{array}{l} F(Px_t + Qz_{t+1}) + G(Px_{t-1} + Qz_t) + Hx_{t-1} \\ + J(Rx_t + Sz_{t+1}) + Ky_t + Lz_{t+1} + Mz_t \end{array} \right\}$$

$$\mathbf{0} = \mathbb{E}_t \left\{ \begin{array}{l} FPx_t + FQz_{t+1} + GPx_{t-1} + GQz_t + Hx_{t-1} \\ + JRx_t + JSz_{t+1} + Ky_t + Lz_{t+1} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQ\mathbb{E}_t\{z_{t+1}\} + GPx_{t-1} + GQz_t + Hx_{t-1} + \\ JRx_t + JS\mathbb{E}_t\{z_{t+1}\} + Ky_t + L\mathbb{E}_t\{z_{t+1}\} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQ\mathbb{E}_t\{Nz_t + \epsilon_{t+1}\} + GPx_{t-1} + \\ GQz_t + Hx_{t-1} + JRx_t + JS\mathbb{E}_t\{Nz_t + \epsilon_{t+1}\} + \\ Ky_t + L\mathbb{E}_t\{Nz_t + \epsilon_{t+1}\} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQNz_t + FQ\mathbb{E}_t\{\epsilon_{t+1}\} + GPx_{t-1} + \\ GQz_t + Hx_{t-1} + JRx_t + JSNz_t + JS\mathbb{E}_t\{\epsilon_{t+1}\} + \\ Ky_t + LNz_t + L\mathbb{E}_t\{\epsilon_{t+1}\} + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} FPx_t + FQNz_t + GPx_{t-1} + \\ GQz_t + Hx_{t-1} + JRx_t + JSNz_t + \\ Ky_t + LNz_t + Mz_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP + JR)x_t + \\ (FQN + GQ + JSN + M + LN)z_t + \\ (GP + H)x_{t-1} + Ky_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP + JR)(Px_{t-1} + Qz_t) + \\ (FQN + GQ + JSN + M + LN)z_t + \\ (GP + H)x_{t-1} + Ky_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP + JR)Px_{t-1} + (FP + JR)Qz_t + \\ (FQN + GQ + JSN + M + LN)z_t + \\ (GP + H)x_{t-1} + Ky_t \end{array} \right\}$$

$$0 = \left\{ \begin{array}{l} (FP^2 + JRP + GP + H)x_{t-1} + \\ [(FP + JR)Q + FQN + GQ + JSN + M + LN]z_t + \\ Ky_t \end{array} \right\}$$

Then:

$$\begin{aligned} Ky_t = & -[FP^2 + (JR + G)P + H]x_{t-1} \\ & -[(FP + JR + G)Q + FQN + JSN + LN + M]z_t \end{aligned}$$

From $y_t = Rx_{t-1} + Sz_t$ we get:

$$Ky_t = KRx_{t-1} + KSz_t$$

Comparing the previous two expressions we conclude:

$$KR = -[FP^2 + (JR + G)P + H]$$

$$KS = -[(FP + JR + G)Q + FQN + JSN + LN + M]$$

From the first of the two expressions above we obtain:

$KR = -FP^2 - JRP - GP - H$. Then we can use $R = -C^{-1}(AP + B)$ to eliminate R . We get:

$$-KC^{-1}(AP+B) = -FP^2 + JC^{-1}(AP+B)P - GP - H$$

$$-KC^{-1}AP - KC^{-1}B = -FP^2 + JC^{-1}AP^2 + JC^{-1}BP - GP - H$$

$$0 = -(F - JC^{-1}A)P^2 + (JC^{-1}B - G + KC^{-1}A)P - H + KC^{-1}B$$

Then:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P + (H - KC^{-1}B) = 0$$

The previous expression is a matrix quadratic equation in P . Once the model is calibrated, this equation can be solved with a computer. There are different solution methods available (see the Appendix).

Stability requires that every eigenvalue of P has modulus smaller than one. See the Appendix and Uhlig (1999) for a discussion.

Notice that, once we have found P , we can obtain R from $R = -C^{-1}(AP + B)$.

Now we need to solve for Q using the fact that

$$KS = -[(FP + JR + G)Q + FQN + JSN + LN + M]$$

Notice that we have already found P and R .

We start by rewriting the expression above as follows:

$$KS = -(FP + JR + G)Q - FQN - JSN - LN - M$$

Using $S = -C^{-1}(AQ + D)$ to eliminate S we get:

$$-KC^{-1}(AQ + D) = -(FP + JR + G)Q - FQN \\ + JC^{-1}(AQ + D)N - LN - M$$

$$-KC^{-1}AQ - KC^{-1}D = -(FP + JR + G)Q \\ - FQN + JC^{-1}AQN + JC^{-1}DN - LN - M$$

$$[FP + JR + G - KC^{-1}A]Q - KC^{-1}D = \\ -(F - JC^{-1}A)QN + JC^{-1}DN - LN - M$$

Then:

$$(F - JC^{-1}A)QN + [FP + JR + G - KC^{-1}A]Q \\ = JC^{-1}DN - LN - M + KC^{-1}D$$

Notice that the different ways in which Q enters in the left-hand side of the expression above prevents us from solving for Q directly. We can, however, solve for Q using the **vec operator** (column-wise vectorization).

Aside on the vec operator

Let A be the 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Then:

$$\text{vec}(A) = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{bmatrix}$$

The previous definition generalizes in the obvious way to any matrix.

Theorem 1. $\text{vec}(A + B) = \text{vec}(A) + \text{vec}(B)$

Theorem 2. $\text{vec}(ABC) = (C' \otimes A)\text{vec}(B)$

Corollary: Setting $C = I$ (identity matrix) in Theorem 2 we get: $\text{vec}(AB) = (I \otimes A)\text{vec}(B)$

Reminder. The symbol $\otimes$ denotes de Kronecker product between two matrices.

For example, if

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

and

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

we get:

$$\begin{aligned} A \otimes B &= \begin{bmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{bmatrix} \\ &= \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{bmatrix} \end{aligned}$$

Recall we have:

$$\begin{aligned} & (F - JC^{-1}A)QN + [FP + JR + G - KC^{-1}A]Q \\ &= JC^{-1}DN - LN - M + KC^{-1}D \end{aligned}$$

Applying the *vec* operator to both sides of the expression above we get:

$$\begin{aligned} & \text{vec}\{(F - JC^{-1}A)QN + [FP + JR + G - KC^{-1}A]Q\} \\ &= \text{vec}(JC^{-1}DN - LN - M + KC^{-1}D) \end{aligned}$$

$$\begin{aligned} & \text{vec}\{(F - JC^{-1}A)QN\} + \text{vec}\{[FP + JR + G - KC^{-1}A]Q\} \\ &= \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \quad \text{by Th. 1} \end{aligned}$$

$$\begin{aligned} & [N' \otimes (F - JC^{-1}A)]\text{vec}(Q) + [I \otimes (FP + JR + G - KC^{-1}A)]\text{vec}(Q) \\ &= \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \quad \text{by Th.2 and its Corollary} \end{aligned}$$

$$\begin{aligned} & [N' \otimes (F - JC^{-1}A) + I \otimes (FP + JR + G - KC^{-1}A)]\text{vec}(Q) \\ &= \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \end{aligned}$$

Then:

$$\begin{aligned} \text{vec}(Q) = & [N' \otimes (F - JC^{-1}A) + I \otimes (FP + JR + G - KC^{-1}A)]^{-1} \\ & \times \text{vec}[(JC^{-1}D - L)N - M + KC^{-1}D] \end{aligned}$$

provided the inverse exists.

From $\text{vec}(Q)$ we can recover Q .

Having found Q we can obtain S from

$$S = -C^{-1}(AQ + D)$$

Hence, we've managed to solve for P , Q , R , and S .

Now we are ready to solve our calibrated model using Uhlig's Matlab code (usually called Uhlig's Toolkit).

Notational convention. To use Uhlig's toolkit we have to change the time convention for the capital stock. What we have called K_{t+1} ($\widetilde{K}_{t+1}$) has to be renamed as K_t ($\widetilde{K}_t$). Analogously, K_t ($\widetilde{K}_t$) has to be changed to K_{t-1} ($\widetilde{K}_{t-1}$).

After running the code we get:

$$P = [0.9561]$$

$$Q = [0.1051 \quad -0.0032]$$

$$R = \begin{bmatrix} 0.1437 \\ 0.6474 \\ -0.7577 \\ -0.3380 \\ -0.8563 \\ 0.4817 \end{bmatrix}, \quad S = \begin{bmatrix} 1.3543 & 0.1111 \\ 0.5293 & -0.1477 \\ 4.2041 & -0.1299 \\ 0.5535 & 0.1737 \\ 1.3543 & 0.1111 \\ 0.8007 & -0.0625 \end{bmatrix}$$

Then, $x_t = Px_{t-1} + Qz_t$ becomes:

$$\widetilde{K}_{t+1} = 0.9561\widetilde{K}_t + 0.1051\widetilde{\theta}_t - 0.0032\widetilde{G}_t$$

Also, $y_t = Rx_{t-1} + Sz_t$ gives:

$$\widetilde{Y}_t = 0.1437\widetilde{K}_t + 1.3543\widetilde{\theta}_t + 0.1111\widetilde{G}_t$$

$$\widetilde{C}_t = 0.6474\widetilde{K}_t + 0.5293\widetilde{\theta}_t - 0.1477\widetilde{G}_t$$

$$\widetilde{I}_t = -0.7577\widetilde{K}_t + 4.2041\widetilde{\theta}_t - 0.1299\widetilde{G}_t$$

$$\widetilde{L}_t = -0.3380\widetilde{K}_t + 0.5535\widetilde{\theta}_t + 0.1737\widetilde{G}_t$$

$$\widetilde{r}_t = -0.8563\widetilde{K}_t + 1.3543\widetilde{\theta}_t + 0.1111\widetilde{G}_t$$

$$\widetilde{w}_t = 0.4817\widetilde{K}_t + 0.8007\widetilde{\theta}_t - 0.0625\widetilde{G}_t$$

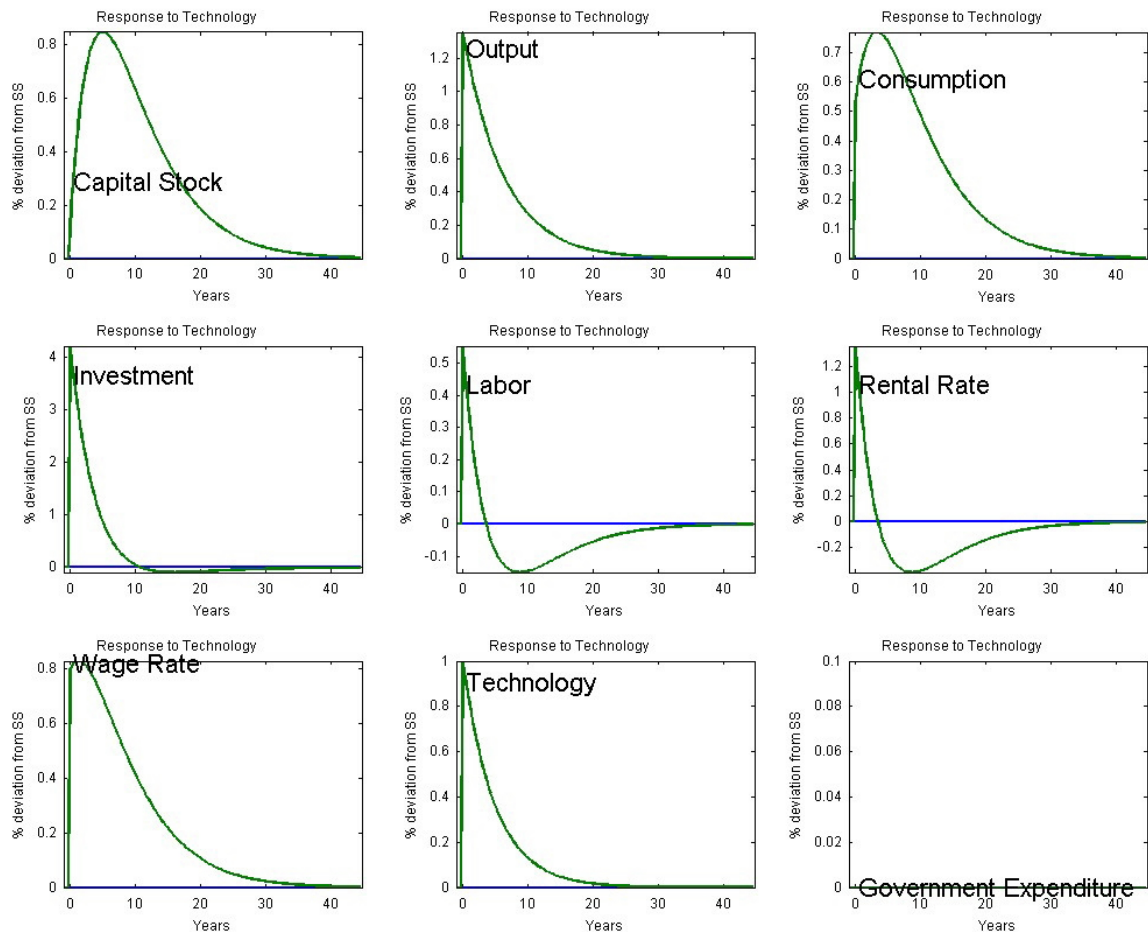
Finally, recall $z_{t+1} = Nz_t + \epsilon_{t+1}$ gives:

$$\widetilde{\theta}_{t+1} = 0.95\widetilde{\theta}_t + \varepsilon_{t+1} \quad \text{with } \varepsilon_{t+1} \sim \mathcal{N}(0, 0.007^2)$$

$$\widetilde{G}_{t+1} = 0.96\widetilde{G}_t + \eta_{t+1} \quad \text{with } \eta_{t+1} \sim \mathcal{N}(0, 0.021^2)$$

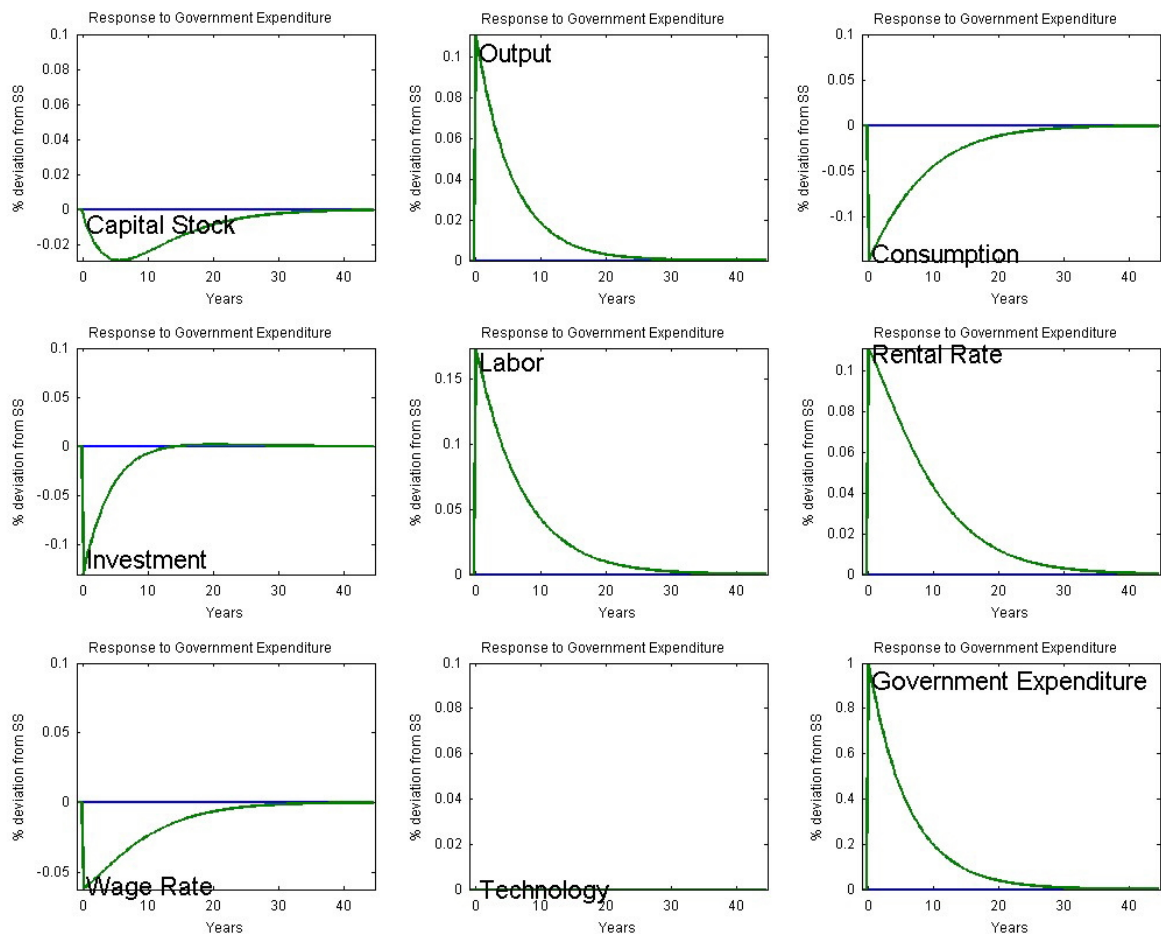
Impulse Responses

Response to a 1% Shock to Technology

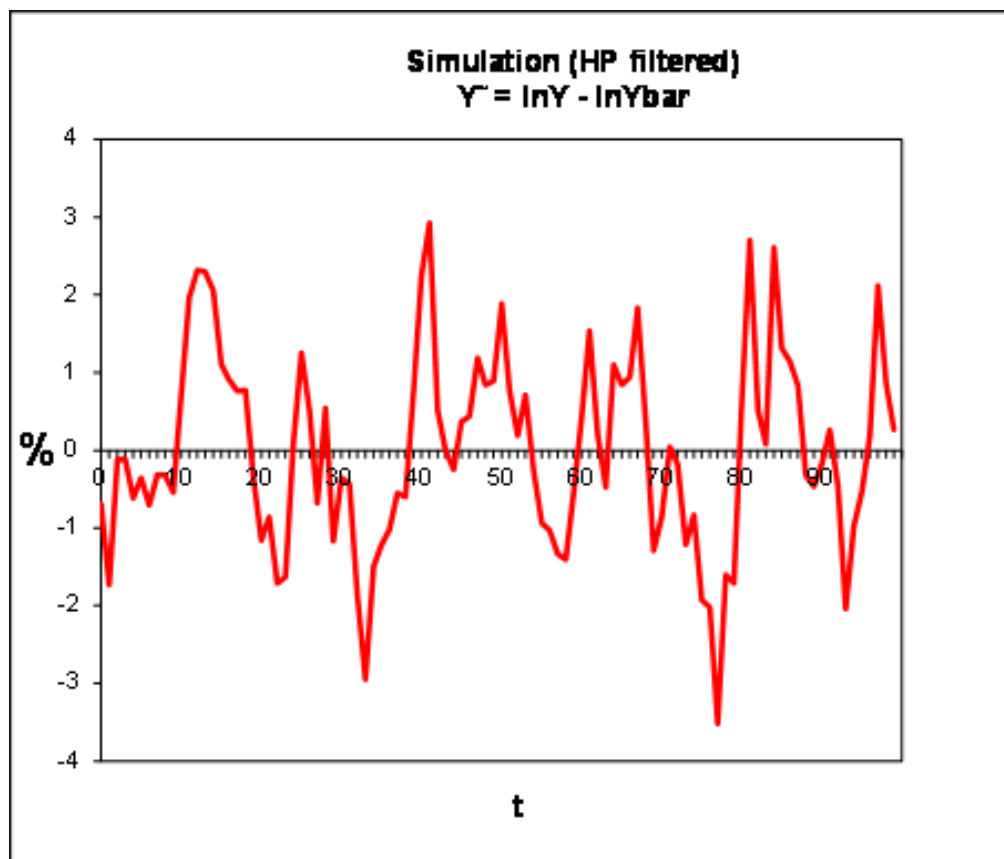


Impulse Responses

Response to a 1% Shock to Government Expenditure



Simulation for the cyclical component of output



Cyclical Properties of Time Series

Standard Deviations (%)

		Data	Model
Output	Y	1.92	1.28
Consumption	C	0.86	0.66
Investment	I	5.33	3.88
Labor (hours)	L	1.50	0.69
Wage	w	1.10	0.77

Note. The column "Data" reports figures for the U.S. over the period 1947:1-1991:3 (Hansen & Wright 1992, Table 2). The column "Model" reports the sample means of statistics computed for each of 100 simulations of our calibrated model. The length of each simulation is 179 periods (quarters), which coincides with the length of the sample in Hansen & Wright (1992). Similar figures are reported by Hansen & Wright (1992) in Table 3. Both, U.S. data and model data were detrended using the HP filter with $\lambda = 1600$.

Standard Deviations Relative to Output

		Data	Model
Output	Y	1.00	1.00
Consumption	C	0.45	0.52
Investment	I	2.78	3.04
Labor (hours)	L	0.78	0.54
Wage	w	0.57	0.61

Contemporaneous Correlations with Output

		Data	Model
Output	Y	1.00	1.00
Consumption	C	0.71	0.59
Investment	I	0.73	0.94
Labor (hours)	L	0.82	0.85
Wage	w	0.63	0.88

Hours vs. Productivity

	Data	Model
σ_L/σ_w	1.37	0.90
$Corr(L, w)$	0.07	0.50

Recall that w is proportional to $\frac{Y}{L}$ (average productivity of labor). Hence $\tilde{w} = \widetilde{\left(\frac{Y}{L}\right)} = \tilde{Y} - \tilde{L}$.

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Appendix

Log-linearization

Consider the expression $y_t = f(x_t)$, where f is a differentiable function, and let $\bar{y} = f(\bar{x})$ be its steady-state value. We want to approximate the log-deviation $\ln y_t - \ln \bar{y}$ with a linear function of the log-deviation $\ln x_t - \ln \bar{x}$, for values of x_t close to $\bar{x}$.

From $y_t = f(x_t)$ we get: $\ln y_t = \ln f(x_t)$. We can rewrite this as follows: $\ln y_t = \ln f(e^{\ln x_t}) \equiv g(\ln x_t)$. Now we take a first-order Taylor approximation of $g(\ln x_t)$ around $\ln \bar{x}$:¹

$$g(\ln x_t) \cong g(\ln \bar{x}) + g'(\ln \bar{x})(\ln x_t - \ln \bar{x})$$

Using $g(\ln x_t) \equiv \ln f(e^{\ln x_t})$ we get:

$$\ln f(e^{\ln x_t}) \cong \ln f(e^{\ln \bar{x}}) + \frac{1}{f(e^{\ln \bar{x}})} f'(e^{\ln \bar{x}}) e^{\ln \bar{x}} (\ln x_t - \ln \bar{x})$$

We can rewrite the expression above as follows:

$$\begin{aligned} \ln f(x_t) &\cong \ln f(\bar{x}) + \frac{\bar{x}}{f(\bar{x})} f'(\bar{x}) (\ln x_t - \ln \bar{x}) \\ &= \ln f(\bar{x}) + e_f(\bar{x}) (\ln x_t - \ln \bar{x}) \end{aligned}$$

where $e_f(\bar{x}) \equiv \frac{\bar{x}}{f(\bar{x})} f'(\bar{x})$ is the elasticity of the function f evaluated at $\bar{x}$. Recalling that $y_t = f(x_t)$ we get the expression we were looking for:

$$\ln y_t - \ln \bar{y} \cong e_f(\bar{x}) (\ln x_t - \ln \bar{x})$$

Defining $\hat{y}_t \equiv \ln y_t - \ln \bar{y}$ and $\hat{x}_t \equiv \ln x_t - \ln \bar{x}$ we get:

$$\hat{y}_t \cong e_f(\bar{x}) \hat{x}_t$$

Notice that we can write: $\hat{x}_t \equiv \ln x_t - \ln \bar{x} = \ln \left(\frac{x_t}{\bar{x}} \right) = \ln \left(1 + \frac{x_t - \bar{x}}{\bar{x}} \right)$. For values of x_t sufficiently close to $\bar{x}$ we know that $\ln \left(1 + \frac{x_t - \bar{x}}{\bar{x}} \right) \cong \frac{x_t - \bar{x}}{\bar{x}}$ (this follows from a first-order Taylor approximation of $m(x_t) \equiv \ln \left(1 + \frac{x_t - \bar{x}}{\bar{x}} \right)$ around $\bar{x}$). Then: $\hat{x}_t \cong \frac{x_t - \bar{x}}{\bar{x}}$. Therefore, we can write:

$$\frac{y_t - \bar{y}}{\bar{y}} \cong e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}}$$

¹ An alternative approach to log-linearization is presented in: Uhlig, Harald (1999). "A Toolkit for Analysing Nonlinear Dynamic Stochastic Models Easily," in Ramon Marimon and Andrew Scott (eds.), *Computational Methods for the Study of Dynamic Economies*, Oxford: Oxford University Press, pp. 30-61.

In fact, we can derive the previous expression directly. First, write $y_t = f(x_t) = e^{\ln f(x_t)}$. Then: $\frac{y_t}{\bar{y}} = \frac{e^{\ln f(x_t)}}{e^{\ln f(\bar{x})}} = e^{\ln f(x_t) - \ln f(\bar{x})} \equiv h(x_t)$. A first-order Taylor approximation of $h(x_t)$ around $\bar{x}$ gives:

$$h(x_t) \cong h(\bar{x}) + h'(\bar{x})(x_t - \bar{x})$$

Using $h(x_t) \equiv e^{\ln f(x_t) - \ln f(\bar{x})}$ we get:

$$\begin{aligned} e^{\ln f(x_t) - \ln f(\bar{x})} &\cong e^0 + e^0 \frac{1}{f(\bar{x})} f'(\bar{x})(x_t - \bar{x}) \\ &= 1 + \frac{\bar{x}}{f(\bar{x})} f'(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \\ &= 1 + e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \end{aligned}$$

Finally, using $e^{\ln f(x_t) - \ln f(\bar{x})} = \frac{y_t}{\bar{y}}$ we obtain: $\frac{y_t}{\bar{y}} \cong 1 + e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \Rightarrow \frac{y_t}{\bar{y}} - 1 \cong e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}} \Rightarrow$

$$\frac{y_t - \bar{y}}{\bar{y}} \cong e_f(\bar{x}) \frac{x_t - \bar{x}}{\bar{x}}$$

which is what we wanted to show.

The previous expression suggests a simple way of log-linearizing $y_t = f(x_t)$ through differentiation. Start by taking the natural logarithm: $\ln y_t = \ln f(x_t)$. Then totally differentiate, evaluating all derivatives at their steady-state levels: $\frac{dy_t}{y} = \frac{f'(\bar{x})}{f(\bar{x})} dx_t \Rightarrow \frac{dy_t}{y} = \frac{\bar{x}}{f(\bar{x})} f'(\bar{x}) \frac{dx_t}{\bar{x}} \Rightarrow \frac{dy_t}{y} = e_f(\bar{x}) \frac{dx_t}{\bar{x}}$. Finally, interpret dx_t as $x_t - \bar{x}$ and dy_t as $y_t - \bar{y}$. Similarly, we can directly differentiate $y_t = f(x_t)$ to get $dy_t = f'(\bar{x}) dx_t$, and then divide by $\bar{y} = f(\bar{x})$ to obtain: $\frac{dy_t}{y} = \frac{f'(\bar{x})}{f(\bar{x})} dx_t$. Multiplying and dividing the right-hand side by $\bar{x}$ we get: $\frac{dy_t}{y} = \frac{\bar{x} f'(\bar{x})}{f(\bar{x})} \frac{dx_t}{\bar{x}} \Rightarrow \frac{dy_t}{y} = e_f(\bar{x}) \frac{dx_t}{\bar{x}}$.

A similar procedure works for multivariate functions, $y_t = f(x_{1t}, x_{2t}, \dots, x_{nt})$. In this case we get:

$$\hat{y}_t \cong e_{f,1}(\bar{x}) \hat{x}_{1t} + e_{f,2}(\bar{x}) \hat{x}_{2t} + \dots + e_{f,n}(\bar{x}) \hat{x}_{nt}$$

where $e_{f,i}(\bar{x})$ is the partial elasticity of f with respect to its i -th argument, evaluated at $\bar{x}$.

In stochastic models, we usually have expressions of the form

$$y_t = \mathbb{E}_t\{f(x_{t+1})\}$$

Taking natural logarithms we get: $\ln y_t = \ln \mathbb{E}_t\{f(x_{t+1})\}$. Now we make the approximation $\ln \mathbb{E}_t\{f(x_{t+1})\} \cong \mathbb{E}_t\{\ln f(x_{t+1})\}$ to get:²

$$\ln y_t \cong \mathbb{E}_t\{\ln f(x_{t+1})\}$$

²Since $\ln(\cdot)$ is a strictly concave function, we know from Jensen's Inequality that $\ln \mathbb{E}_t\{f(x_{t+1})\} > \mathbb{E}_t\{\ln f(x_{t+1})\}$.

Since $\ln \bar{y} = \ln f(\bar{x})$, the expression above gives: $\ln y_t - \ln \bar{y} \cong \mathbb{E}_t\{\ln f(x_{t+1})\} - \ln f(\bar{x}) \Rightarrow$

$$\ln y_t - \ln \bar{y} \cong \mathbb{E}_t\{\ln f(x_{t+1}) - \ln f(\bar{x})\}$$

Using $\ln f(x_{t+1}) \cong \ln f(\bar{x}) + e_f(\bar{x})(\ln x_{t+1} - \ln \bar{x})$ we can rewrite the expression above as follows: $\ln y_t - \ln \bar{y} \cong \mathbb{E}_t\{e_f(\bar{x})(\ln x_{t+1} - \ln \bar{x})\} \Rightarrow$

$$\hat{y}_t \cong e_f(\bar{x})\mathbb{E}_t\{\hat{x}_{t+1}\}$$

Log-linearizing a standard RBC model

The following equations describe a standard RBC model:³

$\ln \lambda_{t+1} = (1 - \gamma) \ln \lambda + \gamma \ln \lambda_t + \epsilon_{t+1}$	Evolution of TFP
$K_{t+1} = I_t + (1 - \delta)K_t$	Law of motion for capital
$Y_t = \lambda_t K_t^\theta H_t^{1-\theta}$	Cobb-Douglas technology
$w_t = (1 - \theta) \frac{Y_t}{H_t}$	FOC for labor (firm)
$r_t = \theta \frac{Y_t}{K_t}$	FOC for capital (firm)
$Y_t = C_t + I_t$	Resource constraint
$\frac{aC_t}{1-H_t} = w_t$	Intratemporal optimality condition (household)
$\frac{1}{C_t} = \mathbb{E}_t \left\{ \frac{\beta}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$	Euler Equation (household)

From $\ln \lambda_{t+1} = (1 - \gamma) \ln \lambda + \gamma \ln \lambda_t + \epsilon_{t+1}$, and $\bar{\lambda} = \lambda$ we get:

$$\begin{aligned} \hat{\lambda}_{t+1} &\equiv \ln \lambda_{t+1} - \ln \bar{\lambda} \\ &= (1 - \gamma) \ln \lambda + \gamma \ln \lambda_t + \epsilon_{t+1} - \ln \lambda \\ &= \gamma (\ln \lambda_t - \ln \lambda) + \epsilon_{t+1} \\ &= \gamma \hat{\lambda}_t + \epsilon_{t+1} \end{aligned}$$

Notice that this case does not require any approximation. The same is true for the production function and the two first-order conditions for the firm: $Y_t = \lambda_t K_t^\theta H_t^{1-\theta}$, $w_t = (1 - \theta) \frac{Y_t}{H_t}$, and $r_t = \theta \frac{Y_t}{K_t}$. Taking natural logs and subtracting the log of the steady-state values we get:

$$\begin{aligned} \hat{Y}_t &= \hat{\lambda}_t + \theta \hat{K}_t + (1 - \theta) \hat{H}_t \\ \hat{w}_t &= \hat{Y}_t - \hat{H}_t \\ \hat{r}_t &= \hat{Y}_t - \hat{K}_t \end{aligned}$$

³The per-period utility function is: $u(C_t, H_t) = \ln C_t + a \ln(1 - H_t)$.

From $K_{t+1} = I_t + (1 - \delta)K_t$ we get $dK_{t+1} = dI_t + (1 - \delta)dK_t$. Dividing by $\bar{K} = \frac{\bar{I}}{\delta}$ we get: $\frac{dK_{t+1}}{\bar{K}} = \delta \frac{dI_t}{\bar{I}} + (1 - \delta) \frac{dK_t}{\bar{K}} \Rightarrow$

$$\hat{K}_{t+1} = \delta \hat{I}_t + (1 - \delta) \hat{K}_t$$

From $Y_t = C_t + I_t$ we get $dY_t = dC_t + dI_t$. Dividing by $\bar{Y}$ we get: $\frac{dY_t}{\bar{Y}} = \frac{1}{\bar{Y}} dC_t + \frac{1}{\bar{Y}} dI_t$. Then: $\frac{dY_t}{\bar{Y}} = \frac{\bar{C}}{\bar{Y}} \frac{dC_t}{\bar{C}} + \frac{\bar{I}}{\bar{Y}} \frac{dI_t}{\bar{I}} \Rightarrow \hat{Y}_t = \frac{\bar{C}}{\bar{Y}} \hat{C}_t + \frac{\bar{I}}{\bar{Y}} \hat{I}_t \Rightarrow$

$$\bar{Y} \hat{Y}_t = \bar{C} \hat{C}_t + \bar{I} \hat{I}_t$$

From $\frac{aC_t}{1-H_t} = w_t$ we get: $\ln a + \ln C_t - \ln(1 - H_t) = \ln w_t$. Differentiating we get: $\frac{dC_t}{C_t} + \frac{1}{1-H_t} dH_t = \frac{dw_t}{w_t} \Rightarrow \frac{dC_t}{C_t} + \frac{\bar{H}}{1-\bar{H}} \frac{dH_t}{\bar{H}} = \frac{dw_t}{w_t} \Rightarrow$

$$\hat{C}_t + \frac{\bar{H}}{1-\bar{H}} \hat{H}_t = \hat{w}_t$$

From $\frac{1}{C_t} = \mathbb{E}_t \left\{ \frac{\beta}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$ we get: $-\ln C_t = \ln \mathbb{E}_t \left\{ \frac{\beta}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$. Approximating the log of the expectation with the expectation of the log we get: $-\ln C_t \cong \mathbb{E}_t \{ \ln \beta - \ln C_{t+1} + \ln(r_{t+1} + 1 - \delta) \}$. Differentiating (and dropping the approximation sign): $-\frac{dC_t}{C_t} = \mathbb{E}_t \left\{ -\frac{dC_{t+1}}{C_{t+1}} + \frac{1}{\bar{r}+1-\delta} dr_{t+1} \right\}$. Since $\frac{1}{\bar{r}+1-\delta} = \beta$ we get: $-\frac{dC_t}{C_t} = \mathbb{E}_t \left\{ -\frac{dC_{t+1}}{C_{t+1}} + \beta dr_{t+1} \right\} \Rightarrow -\frac{dC_t}{C_t} = \mathbb{E}_t \left\{ -\frac{dC_{t+1}}{C_{t+1}} + \beta \bar{r} \frac{dr_{t+1}}{\bar{r}} \right\} \Rightarrow$

$$-\hat{C}_t = \mathbb{E}_t \{ -\hat{C}_{t+1} + \beta \bar{r} \hat{r}_{t+1} \} \Rightarrow$$

$$\mathbb{E}_t \hat{C}_{t+1} - \beta \bar{r} \mathbb{E}_t \hat{r}_{t+1} = \hat{C}_t$$

Uhlig's Method

Solving for P using Generalized Eigenvalues and Eigenvectors

We want to find the matrix P that solves:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P + (H - KC^{-1}B) = \mathbf{0}$$

Define:

$$\Psi \equiv F - JC^{-1}A$$

$$\Gamma \equiv JC^{-1}B - G + KC^{-1}A$$

$$\Theta \equiv H - KC^{-1}B$$

Then, we can rewrite our original matrix equation as follows:

$$\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$$

Remark. P, Ψ, Γ, Θ are $m \times m$ matrices, where m is the number of endogenous state variables (*i.e.*, the number of elements in vector x_t).

It will be useful to define the following two matrices:

$$\Xi \equiv \begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix}$$

$$\Delta \equiv \begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix}$$

Remark. In the expressions above, I denotes the $m \times m$ identity matrix, and $\mathbf{0}$ denotes an $m \times m$ matrix of zeros. Hence, Ξ and Δ are $2m \times 2m$ matrices.

Definition: Generalized Eigenvalue Problem.

The generalized eigenvalue problem for a pair of (possibly complex) $n \times n$ matrices (Ξ, Δ) is the problem of finding the (possibly complex) scalar λ and the (possibly complex) $n \times 1$ vectors $v \neq \mathbf{0}$ that satisfy $\Xi v = \lambda \Delta v$. The values of λ that satisfy the equation are the *generalized eigenvalues* (of Ξ with respect to Δ), and the corresponding vectors $v \neq \mathbf{0}$ are the *generalized (right) eigenvectors*.

Remark 1. When Δ is nonsingular we can reduce the generalized eigenvalue problem to a standard eigenvalue problem. Premultiplying $\Xi x = \lambda \Delta x$ by Δ^{-1} we get $\Delta^{-1}\Xi x = \lambda x$. Hence, the generalized eigenvalues of Ξ with respect to Δ coincide with the standard eigenvalues of $\Delta^{-1}\Xi$.

Remark 2. In our particular case, $n = 2m$. Hence, there are $2m$ generalized eigenvalues (but some may be repeated).

Claim 1 (Theorem 3.4, Part 1, in Uhlig (1999))

Let v be a generalized eigenvector of Ξ with respect to Δ , and λ its associated generalized eigenvalue. Then, v can be written in the following way:

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda x \\ x \end{bmatrix} \text{ for some } x \text{ of dimension } m \times 1.$$

Proof

From $\Xi v = \lambda \Delta v$ we get:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Then:

$$(1) \quad \Gamma v_1 + \Theta v_2 = \lambda \Psi v_1$$

$$(2) \quad v_1 = \lambda v_2$$

From (2):

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda v_2 \\ v_2 \end{bmatrix}$$

Finally, setting $x = v_2$:

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda x \\ x \end{bmatrix}$$

■

Claim 2 (Theorem 3.4, Part 2, in Uhlig (1999))

Suppose there are m generalized eigenvalues of Ξ w.r.t. Δ - $\lambda_1, \dots, \lambda_m$ - such that their associated generalized eigenvectors - $v_1, \dots, v_m$ - can be written as $v_1 = \begin{bmatrix} \lambda_1 x_1 \\ x_1 \end{bmatrix}, \dots, v_m = \begin{bmatrix} \lambda_m x_m \\ x_m \end{bmatrix}$ with $x_1, \dots, x_m$ linearly independent. Then

$$P = \Omega \Lambda \Omega^{-1}$$

is a solution to $\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$, where $\Omega \equiv [x_1, x_2, \dots, x_m]$ and $\Lambda \equiv \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$.

Proof.

First notice that we can always write $v_i = \begin{bmatrix} \lambda_i x_i \\ x_i \end{bmatrix}$ because of Claim 1. Notice also that Ω and Λ are $m \times m$ matrices. Moreover, Ω is invertible because of the assumption that $x_1, \dots, x_m$ are linearly independent.

From (1) above we get: $\Gamma v_1 + \Theta v_2 = \lambda \Psi v_1$. Since $v = \begin{bmatrix} \lambda x \\ x \end{bmatrix}$ we get: $\Gamma \lambda x + \Theta x = \lambda \Psi \lambda x$. Then:

$$\Psi x \lambda^2 - \Gamma x \lambda - \Theta x = \mathbf{0}$$

Applying the expression above to our m eigenvalues/eigenvectors we get:

$$\begin{aligned}\Psi x_1 \lambda_1^2 - \Gamma x_1 \lambda_1 - \Theta x_1 &= \mathbf{0} \\ \Psi x_m \lambda_m^2 - \Gamma x_m \lambda_m - \Theta x_m &= \mathbf{0}\end{aligned}$$

Recall that the left-hand and right-hand sides of the expressions above are $m \times 1$ matrices. We can put them all together in the following $m \times m$ matrices:

$$[\Psi x_1 \lambda_1^2 - \Gamma x_1 \lambda_1 - \Theta x_1, \dots, \Psi x_m \lambda_m^2 - \Gamma x_m \lambda_m - \Theta x_m] = [\mathbf{0}, \dots, \mathbf{0}]$$

Now we can write:

$$[\Psi x_1 \lambda_1^2, \dots, \Psi x_m \lambda_m^2] - [\Gamma x_1 \lambda_1, \dots, \Gamma x_m \lambda_m] - [\Theta x_1, \dots, \Theta x_m] = \mathbf{0}$$

where the right-hand side is an $m \times m$ null matrix.

Then:

$$\Psi [x_1 \lambda_1^2, \dots, x_m \lambda_m^2] - \Gamma [x_1 \lambda_1, \dots, x_m \lambda_m] - \Theta [x_1, \dots, x_m] = \mathbf{0}$$

Then:

$$\Psi [x_1, x_2, \dots, x_m] \begin{bmatrix} \lambda_1^2 & 0 & \dots & 0 \\ 0 & \lambda_2^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_m^2 \end{bmatrix} - \Gamma [x_1, x_2, \dots, x_m] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_m \end{bmatrix} - \Theta [x_1, x_2, \dots, x_m] = \mathbf{0}$$

Then:

$$\Psi \Omega \Lambda^2 - \Gamma \Omega \Lambda - \Theta \Omega = \mathbf{0}$$

Since Ω is invertible we can post-multiply the expression above by Ω^{-1} . We get: $\Psi \Omega \Lambda^2 \Omega^{-1} - \Gamma \Omega \Lambda \Omega^{-1} - \Theta \Omega \Omega^{-1} = \mathbf{0} \Omega^{-1} \Rightarrow$

$$(3) \quad \Psi \Omega \Lambda^2 \Omega^{-1} - \Gamma \Omega \Lambda \Omega^{-1} - \Theta = \mathbf{0}$$

Now define

$$P \equiv \Omega \Lambda \Omega^{-1}$$

and notice that $P^2 = \Omega \Lambda \Omega^{-1} \Omega \Lambda \Omega^{-1} = \Omega \Lambda \Omega \Lambda \Omega^{-1} \Rightarrow$

$$P^2 = \Omega \Lambda^2 \Omega^{-1}$$

Then, we can rewrite (3) as follows:

$$\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$$

which is what we wanted to show.

■

Stability

Consider a process of the form:

$$x_t = \rho x_{t-1} + \epsilon_t$$

where x_t and ϵ_t are scalar random variables. Assume ϵ_t is stationary. It is well known that the stability condition for such a process is $|\rho| < 1$.¹

Now consider the following generalization of the expression above:

$$x_t = Px_{t-1} + Qz_t$$

where x_t and z_t are vectors, P and Q are matrices, and z_t is stationary. In this case, stability requires that each eigenvalue of P has modulus less than one ("it lies within the unit circle"). The matrix Q does not play any role because z_t has been assumed to be stationary.

Now recall from Claim 2 that P can be written as follows: $P = \Omega\Lambda\Omega^{-1}$. From this, it follows that the eigenvalues of P coincide with the eigenvalues of Λ : $Px = \lambda x \Rightarrow \Omega\Lambda\Omega^{-1}x = \lambda x \Rightarrow \Omega^{-1}\Omega\Lambda\Omega^{-1}x = \lambda\Omega^{-1}x \Rightarrow \Lambda\Omega^{-1}x = \lambda\Omega^{-1}x \Rightarrow \Lambda v = \lambda v$, where $v = \Omega^{-1}x$ is an eigenvector of Λ associated to λ . Since Λ is diagonal, its eigenvalues coincide with its diagonal elements. Now, according to Claim 2, the diagonal elements of Λ are m of the generalized eigenvalues of Ξ w.r.t. Δ . Therefore, we conclude that our recursive solution $x_t = Px_{t-1} + Qz_t$ will be stable as long as we can find m generalized eigenvalues of Ξ w.r.t. Δ with modulus lower than one.

If there are exactly m generalized eigenvalues with modulus less than one, there will be a unique stable solution (there may well be additional unstable solutions). Notice that m is the dimension of the vector x_t . Therefore, the stable solution will be unique when the number of eigenvalues with modulus less than one coincides with the number of endogenous state variables.

If there are more than m eigenvalues with modulus less than one, then there will be indeterminacy, *i.e.*, multiple stable solutions (we can construct multiple P s with eigenvalues lower than one in modulus).

If there are less than m eigenvalues with modulus less than one there will be no stable solution.

¹Loosely speaking, stability means that x_t does not explode over time. Iterating over $x_t = \rho x_{t-1} + \epsilon_t$ we get $x_t = \rho^{n+1}x_{t-(n+1)} + \rho^n\epsilon_{t-n} + \dots + \rho\epsilon_{t-1} + \epsilon_t = \rho^{n+1}y_{t-(n+1)} + \sum_{j=0}^n \rho^j \epsilon_{t-j}$. When $|\rho| < 1$, $\lim_{n \rightarrow \infty} \rho^{n+1} = 0$. Moreover, for a stationary solution we have: (i) $\lim_{n \rightarrow \infty} \rho^{n+1}x_{t-(n+1)} = 0$ (in an appropriate sense); and (ii) $\lim_{n \rightarrow \infty} \sum_{j=0}^n \rho^j \epsilon_{t-j} = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-j}$ is well defined. Therefore, we get: $x_t = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-j}$.

Uhlig's Method

Solving for P using the Generalized Schur Decomposition

We want to find the matrix P that solves:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P + (H - KC^{-1}B) = \mathbf{0}$$

Define:

$$\Psi \equiv F - JC^{-1}A$$

$$\Gamma \equiv JC^{-1}B - G + KC^{-1}A$$

$$\Theta \equiv H - KC^{-1}B$$

Then, we can rewrite our original matrix equation as follows:

$$\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$$

Remark. P, Ψ, Γ, Θ are $m \times m$ matrices, where m is the number of endogenous state variables (*i.e.*, the number of elements in vector x_t).

It will be useful to define the following two matrices:

$$\Xi \equiv \begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix}$$
$$\Delta \equiv \begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix}$$

Remark. In the expressions above, I denotes the $m \times m$ identity matrix, and $\mathbf{0}$ denotes an $m \times m$ matrix of zeros. Hence, Ξ and Δ are $2m \times 2m$ matrices.

Now we are going to apply the Generalized Schur Decomposition to the matrix pair (Ξ, Δ) . We start by recalling the notion of a generalized eigenvalue of a matrix pair, and then we state the Generalized Schur Decomposition Theorem.

Definition: Generalized Eigenvalue Problem.

The generalized eigenvalue problem for a pair of (possibly complex) $n \times n$ matrices (Ξ, Δ) is the problem of finding the (possibly complex) scalar λ and the (possibly complex) $n \times 1$ vectors $v \neq \mathbf{0}$ that satisfy $\Xi v = \lambda \Delta v$. The values of λ that satisfy the equation are the *generalized eigenvalues* (of Ξ with respect to Δ), and the corresponding vectors $v \neq \mathbf{0}$ are the *generalized (right) eigenvectors*.

Remark. When Δ is nonsingular we can reduce the generalized eigenvalue problem to a standard eigenvalue problem. Premultiplying $\Xi x = \lambda \Delta x$ by Δ^{-1} we get $\Delta^{-1}\Xi x = \lambda x$. Hence, the generalized eigenvalues of Ξ with respect to Δ coincide with the standard eigenvalues of $\Delta^{-1}\Xi$.

Theorem: Generalized Schur Decomposition.

Let Ξ and Δ be (possibly) complex $n \times n$ matrices. Then, there exist (possibly) complex, unitary $n \times n$ matrices Q and Z such that

$$\begin{aligned}\Xi &= QTZ \\ \Delta &= QSZ\end{aligned}$$

where S and T are (possibly) complex, upper triangular $n \times n$ matrices. The diagonal elements of T divided by the diagonal elements of S , $\frac{t_{ii}}{s_{ii}}$, are the generalized eigenvalues of the matrix pair (Ξ, Δ) . If $s_{ii} \neq 0$ and $t_{ii} \neq 0$, the generalized eigenvalue is finite; if $s_{ii} = 0$ and $t_{ii} \neq 0$, the generalized eigenvalue is infinite (by convention); if $s_{ii} = 0$ and $t_{ii} = 0$, the matrix pair (Ξ, Δ) is singular (*i.e.*, $|\Xi - \lambda\Delta| = 0$) and there is an infinity of generalized eigenvalues. Furthermore, the pairs (s_{ii}, t_{ii}) can be arranged in any order.

Remark 1. In our particular case, $n = 2m$.

Remark 2. A matrix Z is unitary if and only if $Z'Z = I$, where Z' denotes the conjugate transpose of Z , and I is the identity matrix. The conjugate transpose of Z is formed by transposing Z and replacing each element with its complex conjugate.

Remark 3. A maxtrix S is upper triangular when each element below the diagonal is equal to zero.

Remark 4. The decomposition above is usually called the QZ decomposition, in reference to the two unitary matrices mentioned in the theorem.

Remark 5. A QZ decomposition always exists, but it may not be unique.

Remark 6. Do not confuse matrices Q and S in the theorem above with the matrices Q and S that we used in our recursive representation of the solution ($x_t = Px_{t-1} + Qz_t$, $y_t = Rx_{t-1} + Sz_t$).

Now we partition Q , Z , S , and T comformably with Ξ and Δ . We get (recall that S and T are upper triangular):

$$\begin{aligned}Q &= \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix}, Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \\ S &= \begin{bmatrix} S_{11} & S_{12} \\ \mathbf{0} & S_{22} \end{bmatrix}, T = \begin{bmatrix} T_{11} & T_{12} \\ \mathbf{0} & T_{22} \end{bmatrix}\end{aligned}$$

where each block is an $m \times m$ matrix.

Claim (Theorem 3.5, Part 1, in Uhlig (1999))

Suppose Z_{21} and Q_{21} are nonsingular (*i.e.*, invertible). Then

$$P = -Z_{21}^{-1}Z_{22}$$

is a solution to $\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$.

Proof

To show that $P = -Z_{21}^{-1}Z_{22}$ solves $\Psi P^2 - \Gamma P - \Theta = \mathbf{0}$ we can show that

$$P = -Z_{21}^{-1}Z_{22} \quad \Rightarrow \quad (\Psi P^2 - \Gamma P - \Theta)x = \mathbf{0} \quad \forall x \in \mathbb{R}^m$$

From $(\Psi P^2 - \Gamma P - \Theta)x = \mathbf{0}$ we get:

$$\Psi P^2 x = \Gamma P x + \Theta x$$

We also have (trivially):

$$Px = Px$$

Combining the expressions above we get:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2 x \\ Px \end{bmatrix} = \begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix}$$

More compactly, we can write the expression above as follows:

$$\Delta \begin{bmatrix} P^2 x \\ Px \end{bmatrix} = \Xi \begin{bmatrix} Px \\ x \end{bmatrix}$$

Therefore, we want to show that

$$P = -Z_{21}^{-1}Z_{22} \quad \Rightarrow \quad \Delta \begin{bmatrix} P^2 x \\ Px \end{bmatrix} = \Xi \begin{bmatrix} Px \\ x \end{bmatrix} \quad \forall x \in \mathbb{R}^m$$

Notice that the second block in the system above must hold for sure (since it is just $Px = Px$). On the other hand, the first block of the system is what we really need to show (*i.e.*, that $\Psi P^2 x = \Gamma P x + \Theta x$). To prove this we'll do the following: (i) Fix an arbitrary $x \in \mathbb{R}^m$; (ii) rewrite the left-hand side of the system above using $\Delta = QSZ$, partition conformably, and assume $P = -Z_{21}^{-1}Z_{22}$; (iii) rewrite the right-hand side using $\Xi = QTZ$, partition conformably, and assume $P = -Z_{21}^{-1}Z_{22}$; (iv) impose equality between the second block of the left-hand side and the second block of the right-hand side, and show that this implies equality between the first block in the left-hand side and the first block in the right-hand side.

For any $x \in \mathbb{R}^m$, we can use the QZ decomposition to rewrite the left-hand side as follows:

$$\Delta \begin{bmatrix} P^2x \\ Px \end{bmatrix} = QSZ \begin{bmatrix} P^2x \\ Px \end{bmatrix}$$

Partitioning the matrices above we get:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ \mathbf{0} & S_{22} \end{bmatrix} \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11} & Q_{11}S_{12} + Q_{12}S_{22} \\ Q_{21}S_{11} & Q_{21}S_{12} + Q_{22}S_{22} \end{bmatrix} \begin{bmatrix} Z_{11}P^2x + Z_{12}Px \\ Z_{21}P^2x + Z_{22}Px \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11} & Q_{11}S_{12} + Q_{12}S_{22} \\ Q_{21}S_{11} & Q_{21}S_{12} + Q_{22}S_{22} \end{bmatrix} \begin{bmatrix} Z_{11}P^2x + Z_{12}Px \\ (Z_{21}P + Z_{22})Px \end{bmatrix}$$

If $P = -Z_{21}^{-1}Z_{22}$ we get: $Z_{21}P + Z_{22} = -Z_{21}Z_{21}^{-1}Z_{22} + Z_{22} = -Z_{22} + Z_{22} = \mathbf{0}$.

Then:

$$\begin{bmatrix} \Psi & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11} & Q_{11}S_{12} + Q_{12}S_{22} \\ Q_{21}S_{11} & Q_{21}S_{12} + Q_{22}S_{22} \end{bmatrix} \begin{bmatrix} Z_{11}P^2x + Z_{12}Px \\ \mathbf{0} \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Psi P^2x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}S_{11}(Z_{11}P^2x + Z_{12}Px) \\ Q_{21}S_{11}(Z_{11}P^2x + Z_{12}Px) \end{bmatrix}$$

Then:

$$(1) \quad \Psi P^2x = Q_{11}S_{11}(Z_{11}P^2x + Z_{12}Px)$$

$$(2) \quad Px = Q_{21}S_{11}(Z_{11}P^2x + Z_{12}Px)$$

Analogously, for any $x \in \mathbb{R}^m$, we can use the QZ decomposition to rewrite the right-hand side as follows:

$$\Xi \begin{bmatrix} Px \\ x \end{bmatrix} = QTZ \begin{bmatrix} Px \\ x \end{bmatrix}$$

Partitioning the matrices above we get:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} \begin{bmatrix} T_{11} & T_{12} \\ \mathbf{0} & T_{22} \end{bmatrix} \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11} & Q_{11}T_{12} + Q_{12}T_{22} \\ Q_{21}T_{11} & Q_{21}T_{12} + Q_{22}T_{22} \end{bmatrix} \begin{bmatrix} Z_{11}Px + Z_{12}x \\ Z_{21}Px + Z_{22}x \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11} & Q_{11}T_{12} + Q_{12}T_{22} \\ Q_{21}T_{11} & Q_{21}T_{12} + Q_{22}T_{22} \end{bmatrix} \begin{bmatrix} Z_{11}Px + Z_{12}x \\ (Z_{21}P + Z_{22})x \end{bmatrix}$$

If $P = -Z_{21}^{-1}Z_{22}$ we get: $Z_{21}P + Z_{22} = -Z_{21}Z_{21}^{-1}Z_{22} + Z_{22} = -Z_{22} + Z_{22} = \mathbf{0}$.

Then:

$$\begin{bmatrix} \Gamma & \Theta \\ I & \mathbf{0} \end{bmatrix} \begin{bmatrix} Px \\ x \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11} & Q_{11}T_{12} + Q_{12}T_{22} \\ Q_{21}T_{11} & Q_{21}T_{12} + Q_{22}T_{22} \end{bmatrix} \begin{bmatrix} Z_{11}Px + Z_{12}x \\ \mathbf{0} \end{bmatrix}$$

Then:

$$\begin{bmatrix} \Gamma Px + \Theta x \\ Px \end{bmatrix} = \begin{bmatrix} Q_{11}T_{11}(Z_{11}Px + Z_{12}x) \\ Q_{21}T_{11}(Z_{11}Px + Z_{12}x) \end{bmatrix}$$

Then:

$$(3) \quad \Gamma Px + \Theta x = Q_{11}T_{11}(Z_{11}Px + Z_{12}x)$$

$$(4) \quad Px = Q_{21}T_{11}(Z_{11}Px + Z_{12}x)$$

We know (2) and (4) must be equal to each other. Hence, $Q_{21}S_{11}(Z_{11}P^2x + Z_{12}Px) = Q_{21}T_{11}(Z_{11}Px + Z_{12}x)$. Since, by assumption, Q_{21} is invertible, we can premultiply by Q_{21}^{-1} to get:

$$(5) \quad S_{11}(Z_{11}P^2x + Z_{12}Px) = T_{11}(Z_{11}Px + Z_{12}x)$$

Substituting (5) into (3) we get:

$$(6) \quad \Gamma Px + \Theta x = Q_{11}S_{11}(Z_{11}P^2x + Z_{12}Px)$$

But the right-hand side of (6) coincides with the right hand side of (1). Therefore, the left-hand sides must coincide too. Hence, we have shown that:

$$(7) \quad P = -Z_{21}^{-1}Z_{22} \Rightarrow \Psi P^2x = \Gamma Px + \Theta x$$

or, equivalently,

$$(8) \quad P = -Z_{21}^{-1}Z_{22} \Rightarrow (\Psi P^2 - \Gamma P - \Theta)x = \mathbf{0}$$

Since $x \in \mathbb{R}^m$ was chosen arbitrarily, we conclude that the implication above holds $\forall x \in \mathbb{R}^m$, which is what we wanted to show. ■

Money in the Utility Function

Government

$$\begin{aligned} T_t &= M_t - M_{t-1} \\ &= \mu_t M_{t-1} - M_{t-1} \\ &= (\mu_t - 1) M_{t-1} \end{aligned}$$

$$\ln \mu_{t+1} = (1 - \alpha) \ln \mu + \alpha \ln \mu_t + \eta_{t+1}$$

where $t \in \{0, 1, \dots\}$, T_t are lump-sum transfers during period t , M_t is the money supply at the end of t , the initial stock $M_{-1} > 0$ is given, μ_t is the gross growth rate of the money supply, $\alpha \in (0, 1)$, $\mu > 0$, and the η_t 's are *iid* with zero mean and variance σ_η^2 .

Technology

$$Y_t = Z_t K_t^\theta H_t^{1-\theta}, \quad \theta \in (0, 1)$$

$$\ln Z_{t+1} = (1 - \gamma) \ln Z + \gamma \ln Z_t + \varepsilon_{t+1}$$

with $\gamma \in (0, 1)$, $Z > 0$, and the ε_t 's are *iid* with zero mean and variance σ_ε^2 .

We assume that the η_t 's and ε_t 's are independent from each other at all leads and lags.

Remark on Notation

Capital letters are used to distinguish per capita variables that a competitive household takes as parametric from individual-specific variables that are chosen by the household (in equilibrium these will be the same). Aggregate variables and per capita variables coincide because there's a mass-one continuum of (identical) households.

Representative Household

Expected Utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u \left(c_t, h_t, \frac{m_t}{p_t} \right)$$

where

$$u \left(c_t, h_t, \frac{m_t}{p_t} \right) = \ln c_t - B h_t + a \ln \left(\frac{m_t}{p_t} \right)$$

with $\beta \in (0, 1)$, $B > 0$, $a > 0$. We assume $\beta < \mu$.

Budget Constraint

$$c_t + k_{t+1} + \frac{b_t}{p_t} + \frac{m_t}{p_t} = w_t h_t + r_t k_t + (1 - \delta) k_t \\ + \frac{(1 + i_{t-1}) b_{t-1}}{p_t} + \frac{m_{t-1}}{p_t} + \frac{T_t}{p_t}$$

where we have imposed zero profits (because technology displays CRS), and $m_{-1} = M_{-1} > 0$, $b_{-1} = 0$ and $k_0 > 0$ are given.

Profit Maximization

$$\Pi_t \equiv \max_{K_t, H_t} Z_t K_t^\theta H_t^{1-\theta} - w_t H_t - r_t K_t$$

FOC give:

$$w_t = (1 - \theta) Z_t \left(\frac{K_t}{H_t} \right)^\theta = (1 - \theta) \frac{Y_t}{H_t}$$

$$r_t = \theta Z_t \left(\frac{K_t}{H_t} \right)^{-(1-\theta)} = \theta \frac{Y_t}{K_t}$$

$$\Pi_t = Y_t - w_t H_t - r_t K_t = 0 \quad (\text{zero profits})$$

Utility Maximization

Lagrangian

$$\mathcal{L} = \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t u(c_t, h_t, m_t/p_t) + \sum_{t=0}^{\infty} \beta^t \lambda_t [w_t h_t + r_t k_t + (1-\delta)k_t + \frac{(1+i_{t-1})b_{t-1}}{p_t} + \frac{m_{t-1}}{p_t} + \frac{T_t}{p_t} - c_t - k_{t+1} - \frac{b_t}{p_t} - \frac{m_t}{p_t}] \right\}$$

First-Order Conditions (FOC)

$$c_t : \quad u_c(t) = \lambda_t$$

$$h_t : \quad -u_h(t) = \lambda_t w_t$$

$$m_t : \quad u_m(t) \frac{1}{p_t} - \lambda_t \frac{1}{p_t} + \beta \mathbb{E}_t \left\{ \lambda_{t+1} \frac{1}{p_{t+1}} \right\} = 0$$

$$k_{t+1} : \quad \lambda_t = \beta \mathbb{E}_t \{ \lambda_{t+1} (r_{t+1} + 1 - \delta) \}$$

$$b_t : \quad \lambda_t \frac{1}{p_t} = \beta \mathbb{E}_t \left\{ \lambda_{t+1} \frac{1+i_t}{p_{t+1}} \right\}$$

plus the budget constraint and appropriate transversality conditions. Notice that we use u_m to denote the partial derivative of u with respect to its third argument, $\frac{m}{p}$.

From the FOCs w.r.t. c_t and h_t we get:

$$\frac{-u_h(t)}{u_c(t)} = w_t$$

From the FOCs w.r.t. c_t and k_{t+1} we get:

$$u_c(t) = \beta \mathbb{E}_t \{u_c(t+1)(r_{t+1} + 1 - \delta)\}$$

From the FOCs w.r.t. c_t and b_t we get:

$$\begin{aligned} u_c(t) &= \beta \mathbb{E}_t \{u_c(t+1) (1 + i_t) \frac{p_t}{p_{t+1}}\} \\ &= \beta (1 + i_t) \mathbb{E}_t \{u_c(t+1) \frac{p_t}{p_{t+1}}\} \end{aligned}$$

From the FOCs w.r.t. c_t and m_t we get:

$$u_c(t) = u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\}$$

Combining the previous two expressions we get:

$$\begin{aligned} u_c(t) &= u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \\ &= u_m(t) + \frac{u_c(t)}{1 + i_t} \end{aligned}$$

Then:

$$\frac{u_m(t)}{u_c(t)} = \frac{i_t}{1 + i_t}$$

Exercise: Interpret.

For $u(c_t, h_t, m_t/p_t) = \ln c_t - Bh_t + a \ln(m_t/p_t)$ we have:

$$u_c = \frac{1}{c}$$

$$u_h = -B$$

$$u_m = \frac{a}{\frac{m}{p}}$$

Define:

$$m_t^r \equiv \frac{m_t}{p_t} \quad (\text{real balances})$$

$$\pi_{t+1} \equiv \frac{p_{t+1}}{p_t} \quad (\text{gross inflation rate})$$

Then, $\frac{-u_h(t)}{u_c(t)} = w_t$ becomes:

$$Bc_t = w_t$$

From $u_c(t) = \beta \mathbb{E}_t \{u_c(t+1)(r_{t+1} + 1 - \delta)\}$ we get:

$$\frac{1}{\beta} = \mathbb{E}_t \left\{ \frac{c_t}{c_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

From $u_c(t) = u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\}$ we get:

$$\frac{1}{c_t} = \frac{a}{m_t^r} + \beta \mathbb{E}_t \left\{ \frac{1}{c_{t+1}} \frac{1}{\pi_{t+1}} \right\}$$

From $\frac{u_m(t)}{u_c(t)} = \frac{i_t}{1+i_t}$ we get:

$$m_t^r = a \frac{1+i_t}{i_t} c_t$$

Equilibrium requires:

$$c_t = C_t$$

$$k_t = K_t$$

$$h_t = H_t$$

$$m_t = M_t$$

$$b_t = B_t = 0$$

Using the conditions above and the government's budget constraint into the budget constraint of the household we get:

$$C_t + K_{t+1} + \frac{B_t}{p_t} + \frac{M_t}{p_t} = w_t H_t + r_t K_t + (1 - \delta) K_t + \frac{(1 + i_{t-1}) B_{t-1}}{p_t} + \frac{M_{t-1}}{p_t} + \frac{(\mu_t - 1) M_{t-1}}{p_t}$$

Since $I_t = K_{t+1} - (1 - \delta) K_t$ (gross investment), $\mu_t M_{t-1} = M_t$, and $w_t H_t + r_t K_t = Y_t$ (zero profits), we get:

$$C_t + I_t = Y_t$$

In addition, from $M_t = \mu_t M_{t-1}$ we get: $\frac{M_t}{p_t} = \mu_t \frac{M_{t-1}}{p_t}$.

Then: $\frac{M_t}{p_t} = \mu_t \frac{M_{t-1}}{p_{t-1}} \frac{p_{t-1}}{p_t} \Rightarrow$

$$M_t^r = \frac{\mu_t}{\pi_t} M_{t-1}^r$$

Summarizing (12 eqs., 12 var.):

$$Y_t = Z_t K_t^\theta H_t^{1-\theta}$$

$$w_t = (1 - \theta) \frac{Y_t}{H_t}$$

$$r_t = \theta \frac{Y_t}{K_t}$$

$$I_t = K_{t+1} - (1 - \delta)K_t$$

$$Y_t = C_t + I_t$$

$$M_t^r = a^{\frac{1+i_t}{i_t}} C_t$$

$$BC_t = w_t$$

$$M_t^r = \frac{\mu_t}{\pi_t} M_{t-1}^r$$

$$\frac{1}{\beta} = \mathbb{E}_t \left\{ \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$$

$$\frac{1}{C_t} = \frac{a}{M_t^r} + \beta \mathbb{E}_t \left\{ \frac{1}{C_{t+1}} \frac{1}{\pi_{t+1}} \right\}$$

$$\ln Z_{t+1} = (1 - \gamma) \ln Z + \gamma \ln Z_t + \varepsilon_{t+1}$$

$$\ln \mu_{t+1} = (1 - \alpha) \ln \mu + \alpha \ln \mu_t + \eta_{t+1}$$

Nonstochastic Steady State

We have:

$$\bar{\mu} = \mu$$

$$\bar{Z} = Z$$

From $M_t^r = \frac{\mu_t}{\pi_t} M_{t-1}^r$ we get: $\bar{M}^r = \frac{\bar{\mu}}{\bar{\pi}} \bar{M}^r$. Then:

$$\begin{aligned}\bar{\pi} &= \bar{\mu} \\ &= \mu\end{aligned}$$

From $\frac{1}{\beta} = \mathbb{E}_t \left\{ \frac{C_t}{C_{t+1}} (r_{t+1} + 1 - \delta) \right\}$, $C_t = C_{t+1}$, and $r_{t+1} = \bar{r}$ we get:

$$\bar{r} = \frac{1}{\beta} - 1 + \delta$$

From $r_t = \theta \frac{Y_t}{K_t} = \theta Z_t \left(\frac{K_t}{H_t} \right)^{-(1-\theta)}$ we get: $\bar{r} = \theta \bar{Z} \left(\frac{\bar{K}}{\bar{H}} \right)^{-(1-\theta)}$.

Then:

$$\frac{\bar{K}}{\bar{H}} = \left(\frac{\theta \bar{Z}}{\bar{r}} \right)^{\frac{1}{1-\theta}}$$

From $w_t = (1 - \theta) \frac{Y_t}{H_t} = (1 - \theta) Z_t \left(\frac{K_t}{H_t} \right)^\theta$ we get:

$$\bar{w} = (1 - \theta) \bar{Z} \left(\frac{\bar{K}}{\bar{H}} \right)^\theta$$

From $BC_t = w_t$ we get:

$$\bar{C} = \frac{\bar{w}}{B}$$

From $\frac{1}{C_t} = \frac{a}{M_t^r} + \beta \mathbb{E}_t \left\{ \frac{1}{C_{t+1}} \frac{1}{\pi_{t+1}} \right\}$ we get: $\frac{1}{C} = \frac{a}{\bar{M}^r} + \beta \frac{1}{C} \frac{1}{\bar{\pi}}$. Then, using $\bar{\pi} = \mu$, we get:

$$\bar{M}^r = a \bar{C} \frac{\mu}{\mu - \beta}$$

Recall we have assumed $\mu > \beta$. Then, $\bar{M}^r > 0$.

From $M_t^r = a \frac{1+i_t}{i_t} C_t$ we get: $\overline{M}^r = a \frac{1+\bar{i}}{\bar{i}} \overline{C}$. Using the previous expression to eliminate $\overline{M}^r$ we get: $a \overline{C} \frac{\mu}{\mu-\beta} = a \frac{1+\bar{i}}{\bar{i}} \overline{C}$. Then: $\frac{\mu}{\mu-\beta} = \frac{1+\bar{i}}{\bar{i}}$. Then:

$$\bar{i} = \frac{\mu - \beta}{\beta}$$

Since we have assumed $\mu > \beta$, we get $\bar{i} > 0$.

From the expression above we get:

$$\begin{aligned} 1 + \bar{i} &= \frac{1}{\beta} \mu \\ &= \frac{1}{\beta} \pi \end{aligned}$$

This shows that the gross real interest rate in the nss is $\frac{1}{\beta}$, which is consistent with the expression we found earlier for the rental rate of capital: $\bar{r} - \delta = \frac{1}{\beta} - 1$.

Now we express the rest of the variables in terms of $\bar{K}$ and then solve for $\bar{K}$.

From $I_t = K_{t+1} - (1 - \delta)K_t$ we get:

$$\bar{I} = \delta \bar{K}$$

From $\frac{\bar{K}}{\bar{H}} = \left(\frac{\theta \bar{Z}}{\bar{r}}\right)^{\frac{1}{1-\theta}}$ we get:

$$\bar{H} = \left(\frac{\theta \bar{Z}}{\bar{r}}\right)^{\frac{-1}{1-\theta}} \bar{K}$$

From $Y_t = Z_t K_t^\theta H_t^{1-\theta}$ we get: $\bar{Y} = \bar{Z} \bar{K}^\theta \bar{H}^{1-\theta}$. Using the expression above: $\bar{Y} = \bar{Z} \bar{K}^\theta \left(\frac{\theta \bar{Z}}{\bar{r}}\right)^{-1} \bar{K}^{1-\theta}$. Then:

$$\bar{Y} = \frac{\bar{r}}{\theta} \bar{K}$$

From $Y_t = C_t + I_t$ we get: $\bar{Y} = \bar{C} + \bar{I}$. Using the previous expressions: $\frac{\bar{r}}{\theta} \bar{K} = \frac{\bar{w}}{B} + \delta \bar{K}$. Then:

$$\bar{K} = \frac{\bar{w}}{\left(\frac{\bar{r}}{\theta} - \delta\right) B}$$

where $\bar{K} > 0$ because $\bar{r} > \delta$ and $\theta \in (0, 1)$.

Notice that the only real variable that depends on μ in the nss is $\bar{M}^r$. This is because the nominal interest rate $\bar{i}$ increases with $\bar{\pi} = \mu$.

Log-linearization

For any variable X define:

$$\widetilde{X}_t \equiv \ln X_t - \ln \bar{X}$$

Using the standard method to log-linearize our system, we obtain:

$$\widetilde{Y}_t = \widetilde{Z}_t + \theta \widetilde{K}_t + (1 - \theta) \widetilde{H}_t$$

$$\widetilde{w}_t = \widetilde{Y}_t - \widetilde{H}_t$$

$$\widetilde{r}_t = \widetilde{Y}_t - \widetilde{K}_t$$

$$\delta \widetilde{I}_t = \widetilde{K}_{t+1} - (1 - \delta) \widetilde{K}_t$$

$$\bar{Y} \widetilde{Y}_t = \bar{C} \widetilde{C}_t + \bar{I} \widetilde{I}_t$$

$$\widetilde{M}_t^r = \widetilde{C}_t - \frac{\beta}{\mu} \widetilde{i}_t$$

$$\tilde{C}_t = \tilde{w}_t$$

$$\tilde{M}_t^r = \tilde{M}_{t-1}^r + \tilde{\mu}_t - \tilde{\pi}_t$$

$$0 = \mathbb{E}_t \left\{ \tilde{C}_{t+1} - \tilde{C}_t - \beta \bar{r} \tilde{r}_{t+1} \right\}$$

$$0 = \mathbb{E}_t \left\{ \beta \tilde{C}_{t+1} + \beta \tilde{\pi}_{t+1} - \mu \tilde{C}_t + (\mu - \beta) \tilde{M}_t^r \right\}$$

$$\tilde{Z}_{t+1} = \gamma \tilde{Z}_t + \varepsilon_{t+1} \quad \text{with} \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$$\tilde{\mu}_{t+1} = \alpha \tilde{\mu}_t + \eta_{t+1} \quad \text{with} \quad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$$

Remark. To obtain the second expectational equation use the fact that $\frac{a\bar{C}}{\bar{M}^r} = \frac{\mu-\beta}{\mu}$.

Price Level

From $\pi_{t+1} \equiv \frac{p_{t+1}}{p_t}$ we get: $p_{t+1} = \pi_{t+1}p_t \Rightarrow$

$$\ln p_{t+1} = \ln \pi_{t+1} + \ln p_t.$$

From $\tilde{\pi}_{t+1} = \ln \pi_{t+1} - \ln \bar{\pi}$ and $\bar{\pi} = \mu$ we get:

$$\ln \pi_{t+1} = \ln \mu + \tilde{\pi}_{t+1}.$$

Then:

$$\ln p_{t+1} = \ln \mu + \tilde{\pi}_{t+1} + \ln p_t$$

At $t = 0$, p_0 is determined from the money-market equilibrium condition: $M_0^r = a \frac{1+i_0}{i_0} C_0 \Rightarrow \frac{M_0}{p_0} = a \frac{1+i_0}{i_0} C_0 \Rightarrow$

$$p_0 = \frac{1}{aC_0} \frac{i_0}{1+i_0} M_0$$

Notice that, from the solution of the model, we know $\tilde{C}_0$ and $\tilde{i}_0$. Therefore, we know $C_0 = \bar{C}e^{\tilde{C}_0}$ and $i_0 = \bar{i}e^{\tilde{i}_0}$. Hence, given M_0 , the expression above determines p_0 .

Matrix Form

Define:

$$x_t \equiv \begin{bmatrix} \widetilde{K}_{t+1} \\ \widetilde{M}_t^r \end{bmatrix}, \quad y_t \equiv \begin{bmatrix} \widetilde{Y}_t \\ \widetilde{C}_t \\ \widetilde{I}_t \\ \widetilde{H}_t \\ \widetilde{r}_t \\ \widetilde{w}_t \\ \widetilde{i}_t \\ \widetilde{\pi}_t \end{bmatrix}$$

$$z_t \equiv \begin{bmatrix} \widetilde{Z}_t \\ \widetilde{\mu}_t \end{bmatrix}, \quad \epsilon_t \equiv \begin{bmatrix} \varepsilon_t \\ \eta_t \end{bmatrix}$$

Then, we can rewrite our log-linearized system in matrix form, as follows:

$$\mathbf{0} = Ax_t + Bx_{t-1} + Cy_t + Dz_t$$

$$\mathbf{0} = \mathbb{E}_t\{Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t \\ + Lz_{t+1} + Mz_t\}$$

$$z_{t+1} = Nz_t + \epsilon_{t+1} \quad \text{with} \quad \mathbb{E}_t\{\epsilon_{t+1}\} = \mathbf{0}$$

where

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -\theta & 0 \\ 0 & 0 \\ -1 & 0 \\ -(1-\delta) & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & -(1-\theta) & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -\delta & 0 & 0 & 0 & 0 & 0 \\ \overline{Y} & -\overline{C} & -\overline{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & \frac{\beta}{\mu} & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$F = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 \\ 0 & \mu - \beta \end{bmatrix}, \quad H = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$J = \begin{bmatrix} 0 & 1 & 0 & 0 & -\beta\bar{r} & 0 & 0 & 0 \\ 0 & \beta & 0 & 0 & 0 & 0 & 0 & \beta \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\mu & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$N = \begin{bmatrix} \gamma & 0 \\ 0 & \alpha \end{bmatrix}$$

Calibration

For quarterly data (similar the the CIA model of Cooley & Hansen 1989):

$$\overline{Z} = 1$$

$$\theta = 0.36$$

$$\beta = 0.99$$

$$B = 2.86$$

$$\delta = 0.025$$

$$\gamma = 0.95$$

$$\alpha = 0.48$$

$$\sigma_{\varepsilon} = 0.00721$$

$$\sigma_{\eta} = 0.009$$

$$a = 0.01$$

Remark. The value of a is chosen so that, when $\mu = 1$, the value of $\overline{M}^r$ is similar to the one implied by the CIA model of Cooley & Hansen (1989). This is the value chosen by McCandless (2008), Chapter 9.

We solve the model for different values of μ :

$$\mu \in \{1.015, 1.15\}$$

This calibration implies:

	$\mu = 1.015$	$\mu = 1.15$
$\bar{r}$	0.0351	0.0351
$\bar{w}$	2.3706	2.3706
$\bar{C}$	0.8288	0.8288
$\bar{K}$	11.4324	11.4324
$\bar{H}$	0.3009	0.3009
$\bar{Y}$	1.1147	1.1147
$\bar{I}$	0.2858	0.2858
$\bar{M}^r$	0.3365	0.0596
$\bar{i}$	0.0253	0.1616
$\bar{\pi}$	1.015	1.15

Solution

$$x_t = Px_{t-1} + Qz_t$$

$$y_t = Rx_{t-1} + Sz_t$$

For $\mu = 1.015$ we get:

$$P = \begin{bmatrix} 0.9418 & 0 \\ 0.5316 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.1552 & 0 \\ 0.4703 & -0.8803 \end{bmatrix}$$

$$R = \begin{bmatrix} 0.0550 & 0 \\ 0.5316 & 0 \\ -1.3273 & 0 \\ -0.4766 & 0 \\ -0.9450 & 0 \\ 0.5316 & 0 \\ 0 & 0 \\ -0.5316 & 1 \end{bmatrix}, \quad S = \begin{bmatrix} 1.9417 & 0 \\ 0.4703 & 0 \\ 6.2091 & 0 \\ 1.4715 & 0 \\ 1.9417 & 0 \\ 0.4703 & 0 \\ 0 & 0.9026 \\ -0.4703 & 1.8803 \end{bmatrix}$$

Then:

$$\widetilde{K}_{t+1} = 0.9418\widetilde{K}_t + 0.1552\widetilde{Z}_t$$

$$\widetilde{M}_t^r = 0.5316\widetilde{K}_t + 0.4703\widetilde{Z}_t - 0.8803\widetilde{\mu}_t$$

$$\widetilde{Y}_t = 0.0550\widetilde{K}_t + 1.9417\widetilde{Z}_t$$

$$\widetilde{C}_t = 0.5316\widetilde{K}_t + 0.4703\widetilde{Z}_t$$

$$\widetilde{I}_t = -1.3273\widetilde{K}_t + 6.2091\widetilde{Z}_t$$

$$\widetilde{H}_t = -0.4766\widetilde{K}_t + 1.4715\widetilde{Z}_t$$

$$\widetilde{r}_t = -0.9450\widetilde{K}_t + 1.9417\widetilde{Z}_t$$

$$\widetilde{w}_t = 0.5316\widetilde{K}_t + 0.4703\widetilde{Z}_t$$

$$\widetilde{i}_t = 0.9026\widetilde{\mu}_t$$

$$\widetilde{\pi}_t = -0.5316\widetilde{K}_t + \widetilde{M}_{t-1}^r - 0.4703\widetilde{Z}_t + 1.8803\widetilde{\mu}_t$$

Notice that the solution for $\widetilde{K}_{t+1}$, $\widetilde{Y}_t$, $\widetilde{C}_t$, $\widetilde{I}_t$, $\widetilde{H}_t$, $\widetilde{r}_t$, and $\widetilde{w}_t$, coincides exactly with the solution of the real (non-monetary) model in Hansen (1985). Upon reflection this is not surprising since the previous system of equations can be broken down in two blocks, as follows:

Block 1

$$\widetilde{Y}_t = \widetilde{Z}_t + \theta \widetilde{K}_t + (1 - \theta) \widetilde{H}_t$$

$$\widetilde{w}_t = \widetilde{Y}_t - \widetilde{H}_t$$

$$\widetilde{r}_t = \widetilde{Y}_t - \widetilde{K}_t$$

$$\delta \widetilde{I}_t = \widetilde{K}_{t+1} - (1 - \delta) \widetilde{K}_t$$

$$\overline{Y} \widetilde{Y}_t = \overline{C} \widetilde{C}_t + \overline{I} \widetilde{I}_t$$

$$\widetilde{C}_t = \widetilde{w}_t$$

$$0 = \mathbb{E}_t \left\{ \widetilde{C}_{t+1} - \widetilde{C}_t - \beta \overline{r} \widetilde{r}_{t+1} \right\}$$

$$\widetilde{Z}_{t+1} = \gamma \widetilde{Z}_t + \varepsilon_{t+1} \quad \text{with} \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

Block 2

$$\widetilde{M}_t^r = \widetilde{C}_t - \frac{\beta}{\mu} \widetilde{i}_t$$

$$\widetilde{M}_t^r = \widetilde{M}_{t-1}^r + \widetilde{\mu}_t - \widetilde{\pi}_t$$

$$0 = \mathbb{E}_t \left\{ \beta \widetilde{C}_{t+1} + \beta \widetilde{\pi}_{t+1} - \mu \widetilde{C}_t + (\mu - \beta) \widetilde{M}_t^r \right\}$$

$$\widetilde{\mu}_{t+1} = \alpha \widetilde{\mu}_t + \eta_{t+1} \quad \text{with} \quad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2)$$

Notice that Block 1 is exactly the same as the one in Hansen's model. This block of 8 equations can be solved for $\widetilde{K}_{t+1}$, $\widetilde{Y}_t$, $\widetilde{C}_t$, $\widetilde{I}_t$, $\widetilde{H}_t$, $\widetilde{r}_t$, $\widetilde{w}_t$, and $\widetilde{Z}_t$ (8 variables). Given the solution for $\widetilde{C}_t$ obtained from Block 1 we can solve Block 2 (4 eq.) for $\widetilde{M}_t^r$, $\widetilde{i}_t$, $\widetilde{\pi}_t$, and $\widetilde{\mu}_t$ (4 variables). Remark. Separability of real balances in the utility function is important for this result.

Notice that the only real variable that is affected by money growth is M^r (real balances). When this happens we say that money is superneutral. It is natural to expect that M^r is affected by money growth since we know that money growth affects the nominal interest rate, which is a determinant of the demand for real balances.

Percent Standard Deviations

Real Money	0.9587	(0.088)	0.53
Output	1.8029	(0.232)	1.00
Consumption	0.5182	(0.082)	0.29
Investment	5.7645	(0.731)	3.20
Labor	1.3733	(0.174)	0.74
Rental Rate	1.8373	(0.232)	1.02
Wage	0.5182	(0.082)	0.29
Nominal Interest Rate	0.831	(0.064)	0.46
Gross Inflation	1.6503	(0.119)	0.92

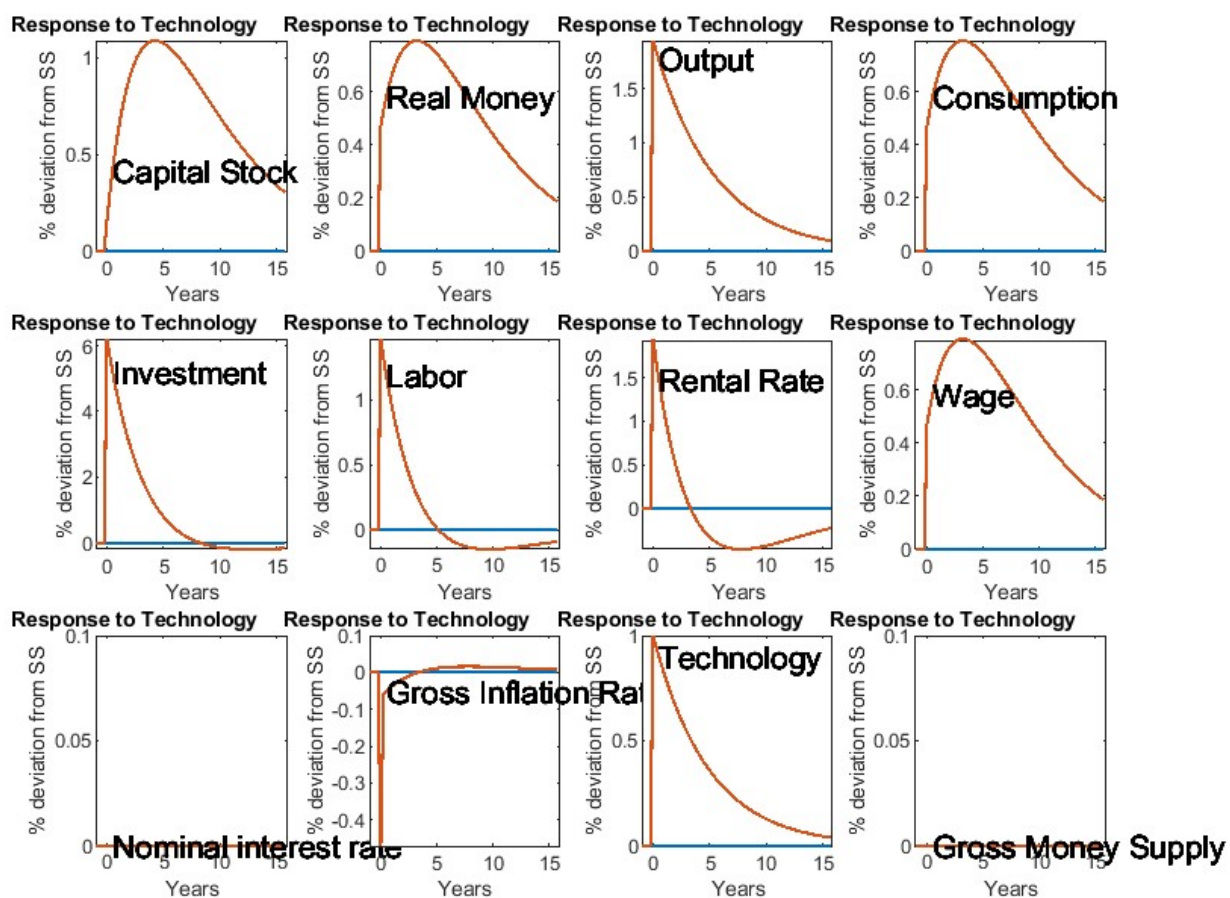
Standard errors in parenthesis; 100 simulations, 115 periods each. Third column: s.d. relative to output.

Contemporaneous Correlations with Output

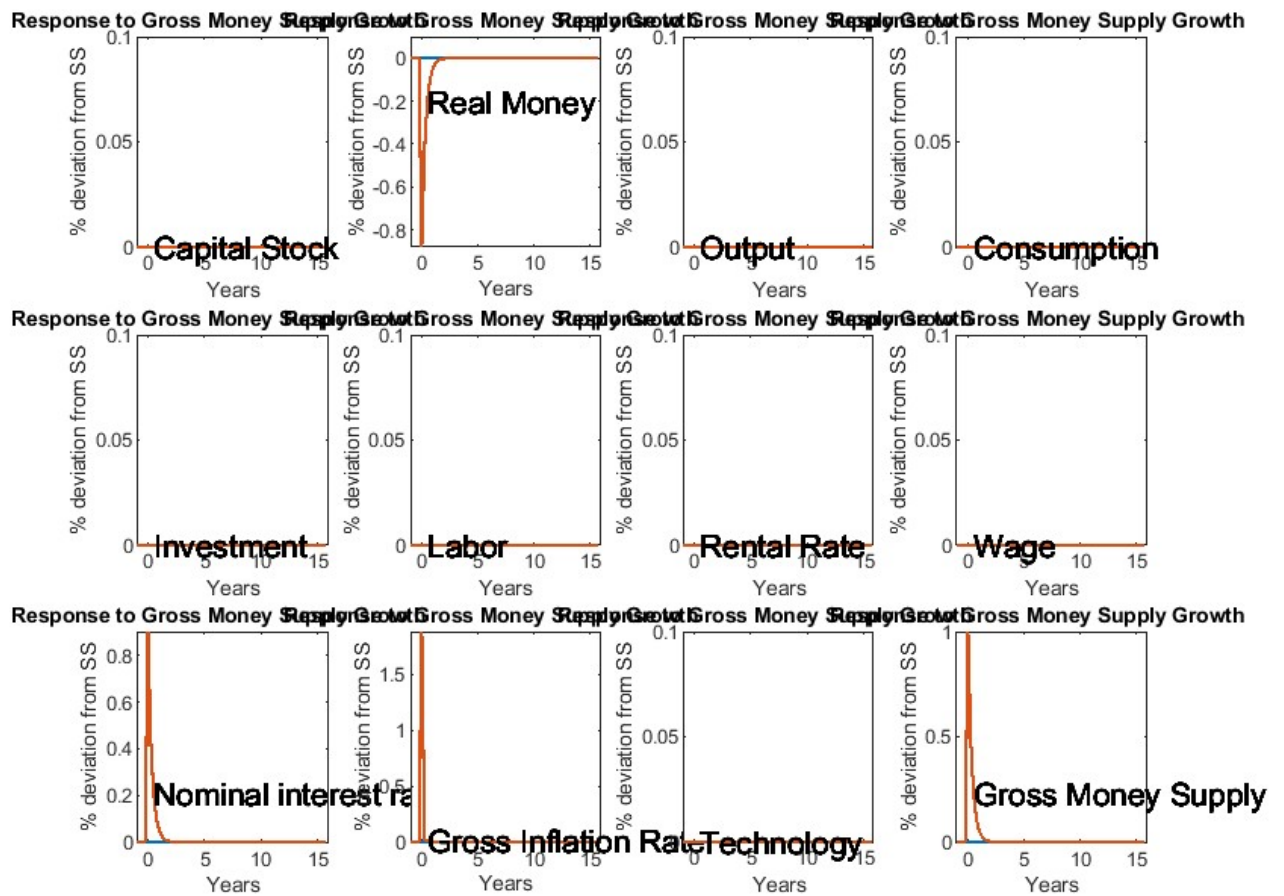
Real Money	0.46	(0.11)
Output	1.00	(0.00)
Consumption	0.88	(0.02)
Investment	0.99	(0.00)
Labor	0.98	(0.00)
Rental Rate	0.96	(0.01)
Wage	0.88	(0.02)
Nominal Interest Rate	0.01	(0.14)
Gross Inflation	−0.1	(0.1)

Standard errors in parenthesis; 100 simulations, 115 periods each.

Impulse Responses: Technology Shock



Impulse Responses: Money Supply Shock



A different version of the model

Government

$$G_t = \frac{M_t - M_{t-1}}{p_t}$$

$$\ln G_{t+1} = (1 - \phi) \ln G + \phi \ln G_t + \xi_{t+1}$$

Remark 1. Interpret G_t as the fiscal deficit at t , and assume all the deficit is monetized.

Remark 2. Drop the lump-sum transfers from the budget constraint of the household.

Remark 3. We assume G_t is exogenous and M_t is endogenous (fiscal dominance)

Now the resource constraint becomes:

$$Y_t = C_t + I_t + G_t$$

In steady state, seigniorage ($S_t \equiv \frac{M_t - M_{t-1}}{p_t}$) is:

$$\begin{aligned}\bar{S} &= \frac{\bar{\mu} - 1}{\bar{\mu}} \bar{M}^r = \frac{\bar{\mu} - 1}{\bar{\mu}} \frac{a\bar{C}}{1 - \frac{\beta}{\bar{\mu}}} \\ &= \frac{\bar{\mu} - 1}{\bar{\mu} - \beta} a\bar{C}\end{aligned}$$

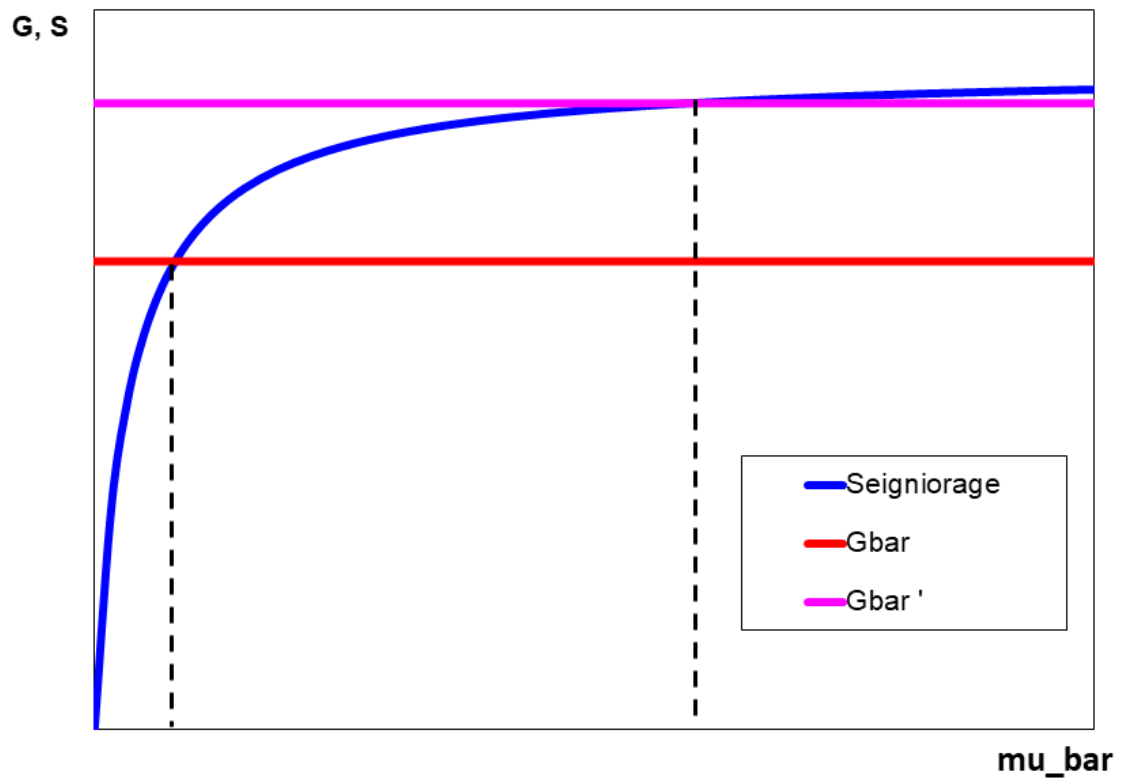
The steady-state value of G_t is:

$$\bar{G} = G$$

The equilibrium value of $\bar{\mu}$ solves: $\bar{G} = \bar{S}$. Then:

$$G = \frac{\bar{\mu} - 1}{\bar{\mu} - \beta} a\bar{C}$$

Remark. In this model, $\bar{C}$ is independent of $\bar{\mu}$ and G .



Appendix
Money in the Utility Function
when the nominal interest rate is zero

From the FOCs w.r.t. c_t and m_t we get:

$$u_c(t) = u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \quad (1)$$

From the FOCs w.r.t. c_t and b_t we get:

$$u_c(t) = \beta (1 + i_t) \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \quad (2)$$

Suppose $i_t = 0$. Then, (2) becomes:

$$u_c(t) = \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} \quad (3)$$

Substituting (1) into (3) we get:

$$u_m(t) + \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\} = \beta \mathbb{E}_t \left\{ u_c(t+1) \frac{p_t}{p_{t+1}} \right\}$$

Then:

$$u_m(t) = 0$$

The expression above implies that, when $i_t = 0$, the household would demand real balances until satiation (zero marginal utility).

Notice, however, that the expression above cannot hold if the utility function always displays $u_m > 0$, like in our logarithmic case. What's happening? Well, with $i_t = 0$ and $u_m > 0$, the household could borrow like crazy ($b_t/p_t \rightarrow -\infty$) and use the proceeds to increase real balances ($m_t/p_t \rightarrow +\infty$), and this operation would always increase utility ($u_m > 0$). Hence, the maximization problem of the household would have no solution.

References

McCandless, George (2008). *The ABCs of RBCs. An Introduction to Dynamic Macroeconomic Models*, Harvard University Press. Chapters 8 & 9.

Cooley, T. and Hansen, G. (1989). "The Inflation Tax in a Real Business Cycle Model," *The American Economic Review*, Vol. 79, No. 4 (September), pp. 733-748.

Money, Government Deficits, and Inflation

Money Demand:

$$\frac{M_t^d}{P_t} = \Phi(\bar{i}_t, Y_t)$$

where

$$1 + i_t = (1 + r_t) \frac{\mathbb{E}_t P_{t+1}}{P_t}$$

For simplicity, assume:

$$\begin{aligned} Y_t &= Y \quad \forall t \\ r_t &= r \quad \forall t \end{aligned}$$

Then:

$$\frac{M_t^d}{P_t} = \Psi\left(\frac{\mathbb{E}_t \bar{P}_{t+1}}{P_t}\right)$$

$$\text{where } \Psi\left(\frac{\mathbb{E}_t P_{t+1}}{P_t}\right) \equiv \Phi\left((1 + r) \frac{\mathbb{E}_t P_{t+1}}{P_t} - 1, Y\right).$$

For simplicity, we assume that the function Ψ is linear:

$$\frac{M_t^d}{P_t} = c - b \frac{\mathbb{E}_t P_{t+1}}{P_t}$$

with $c > 0$ and $b > 0$.

Equilibrium requires:

$$M_t^d = M_t$$

where M_t is the money supply (in nominal terms).

Then:

$$\frac{M_t}{P_t} = c - b \frac{\mathbb{E}_t P_{t+1}}{P_t}$$

From the previous expression we get:

$$P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$$

where $\alpha \equiv \frac{b}{c} > 0$.

The expression above is a first-order stochastic linear difference equation in the price level.

Assume:

$$\alpha < 1$$

Then, we can solve $P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$ forward:

$$\begin{aligned} P_t &= \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1} \\ &= \frac{M_t}{c} + \alpha \mathbb{E}_t \left\{ \frac{M_{t+1}}{c} + \alpha \mathbb{E}_{t+1} P_{t+2} \right\} \\ &= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \mathbb{E}_{t+1} P_{t+2} \\ &= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t P_{t+2} \quad \text{by LIE} \\ &= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \left\{ \frac{M_{t+2}}{c} + \alpha \mathbb{E}_{t+2} P_{t+3} \right\} \\ &= \frac{1}{c} M_t + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \frac{\alpha^2}{c} \mathbb{E}_t M_{t+2} + \alpha^3 \mathbb{E}_t \mathbb{E}_{t+2} P_{t+3} \\ &= \frac{1}{c} (M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2}) + \alpha^3 \mathbb{E}_t P_{t+3} \end{aligned}$$

If we keep iterating forward we get:

$$P_t = \frac{1}{c} (M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2} + \dots + \alpha^n \mathbb{E}_t M_{t+n}) + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Then:

$$P_t = \frac{1}{c} \sum_{i=0}^n \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Letting $n \rightarrow \infty$:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} + \lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

assuming that $\sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \equiv \lim_{n \rightarrow \infty} \sum_{i=0}^n \alpha^i \mathbb{E}_t M_{t+i}$ is finite.

Fundamental Solution

Notation: we use a $*$ to denote this particular solution.

Suppose the price level satisfies:

$$\lim_{n \rightarrow \infty} \alpha^n \mathbb{E}_t P_{t+n} = 0$$

Then, the previous expression for P_t becomes:

$$P_t^* = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \quad (1)$$

The expression above shows that the price level at t depends on the money supply expected to prevail from t into the indefinite future.

According to previous equation, if government deficits are to influence the price level path, it can only be through their effect on the expected path of base money. However, the government deficit and path of base money are

not rigidly linked in any immutable way. The reason is that the government can, at least to a point, borrow by issuing interest-bearing government debt, and so need not necessarily issue base money to cover its deficit. To see this more precisely, we need to write down the budget constraint of the consolidated government:

$$G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1}) + \frac{M_t - M_{t-1}}{P_t} \quad (2)$$

Under the system formed by equations (1) and (2), the inflationary consequences of government deficits depend sensitively on the government's strategy for servicing the debt that it issues.

We consider first a regime in which government deficits have no effects on the rate of inflation. In this regime, the government always finances its entire deficit or surplus by issuing or retiring interest-bearing government debt. Then:

$$M_t - M_{t-1} = 0 \quad \forall t$$

and

$$G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1}) \quad \forall t$$

From $M_t - M_{t-1} = 0 \ \forall t$ we get:

$$M_t = M \quad \forall t$$

Then, (1) becomes:

$$\begin{aligned} P_t^* &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \\ &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M \\ &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i M \\ &= \frac{M}{c} \sum_{i=0}^{\infty} \alpha^i \\ &= \frac{M}{c} [1 + \alpha + \alpha^2 + \dots] \\ &= \frac{M}{c} \frac{1}{1 - \alpha} \quad \text{since } \alpha \in (0, 1) \end{aligned}$$

That is:

$$P_t^* = \frac{1}{(1 - \alpha)c} M$$

Hence, P_t^* is constant over time and proportional to M .

Also, from $G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1})$ we get:

$$D_{t-1} = \frac{T_t - G_t}{1 + r_{t-1}} + \frac{D_t}{1 + r_{t-1}}$$

Shifting the expression above one period ahead:

$$D_t = \frac{T_{t+1} - G_{t+1}}{1 + r_t} + \frac{D_{t+1}}{1 + r_t}$$

Since $\mathbb{E}_t D_t = D_t$, the expression above gives:

$$D_t = \mathbb{E}_t \left\{ \frac{T_{t+1} - G_{t+1}}{1 + r_t} + \frac{D_{t+1}}{1 + r_t} \right\}$$

Iterating forward on the previous expression we get:

$$\begin{aligned} D_t &= \mathbb{E}_t \left\{ \frac{T_{t+1} - G_{t+1}}{1 + r_t} + \frac{1}{1 + r_t} \left(\frac{T_{t+2} - G_{t+2}}{1 + r_{t+1}} + \frac{D_{t+2}}{1 + r_{t+1}} \right) \right\} \\ &= \mathbb{E}_t \left\{ \frac{T_{t+1} - G_{t+1}}{1 + r_t} + \frac{T_{t+2} - G_{t+2}}{(1 + r_t)(1 + r_{t+1})} + \frac{D_{t+2}}{(1 + r_t)(1 + r_{t+1})} \right\} \end{aligned}$$

Then:

$$\begin{aligned}
D_t &= \mathbb{E}_t \left\{ \begin{aligned} &\frac{T_{t+1}-G_{t+1}}{1+r_t} + \frac{T_{t+2}-G_{t+2}}{(1+r_t)(1+r_{t+1})} \\ &+ \frac{1}{(1+r_t)(1+r_{t+1})} \left(\frac{T_{t+3}-G_{t+3}}{1+r_{t+2}} + \frac{D_{t+3}}{1+r_{t+2}} \right) \end{aligned} \right\} \\
&= \mathbb{E}_t \left\{ \begin{aligned} &\frac{T_{t+1}-G_{t+1}}{1+r_t} + \frac{T_{t+2}-G_{t+2}}{(1+r_t)(1+r_{t+1})} + \frac{T_{t+3}-G_{t+3}}{(1+r_t)(1+r_{t+1})(1+r_{t+2})} \\ &+ \frac{D_{t+3}}{(1+r_t)(1+r_{t+1})(1+r_{t+2})} \end{aligned} \right\} \\
&= \mathbb{E}_t \left\{ \begin{aligned} &\frac{T_{t+1}-G_{t+1}}{1+r_t} + \frac{T_{t+2}-G_{t+2}}{(1+r_t)(1+r_{t+1})} + \dots + \frac{T_{t+n+1}-G_{t+n+1}}{(1+r_t)(1+r_{t+1})\dots(1+r_{t+n})} \\ &+ \frac{D_{t+n+1}}{(1+r_t)(1+r_{t+1})\dots(1+r_{t+n})} \end{aligned} \right\}
\end{aligned}$$

Then:

$$D_t = \mathbb{E}_t \sum_{j=0}^n \frac{1}{R_{tj}} (T_{t+j+1} - G_{t+j+1}) + \mathbb{E}_t \left\{ \frac{1}{R_{tn}} D_{t+n+1} \right\}$$

where $R_{tj} \equiv (1+r_t)(1+r_{t+1})\dots(1+r_{t+j})$.

Remark: with r_t constant we get $R_{tj} = (1+r)^{j+1}$.

Now let $n \rightarrow \infty$ and set $\lim_{n \rightarrow \infty} E_t \left\{ \frac{1}{R_{tn}} D_{t+n+1} \right\} = 0$. Then:

$$D_t = \mathbb{E}_t \sum_{j=0}^{\infty} \frac{T_{t+j+1} - G_{t+j+1}}{R_{tj}}$$

In this regime, a positive value of interest-bearing government debt signals a stream of future government budgets that is in surplus in the present value sense indicated by the previous equation.

Substituting the expression above into $D_{t-1} = \frac{T_t - G_t}{1+r_{t-1}} + \frac{D_t}{1+r_{t-1}}$ we get:

$$(1 + r_{t-1})D_{t-1} = (T_t - G_t) + \mathbb{E}_t \sum_{j=0}^{\infty} \frac{T_{t+j+1} - G_{t+j+1}}{R_{tj}}$$

In this regime, government deficits have no effects on the price path because they are permitted to have no effects on the path of base money. For the path of base money to be unaffected by government deficits, it is necessary that government (primary) deficits be temporary and be accompanied by exactly offsetting future government surpluses.

There are alternative debt-servicing strategies under which government deficits *are* inflationary. To take an example at the opposite pole from the previous regime, consider the following rule:

$$D_t = 0 \quad \forall t$$

Then, equation (2) becomes

$$G_t - T_t = \frac{M_t - M_{t-1}}{P_t}$$

According to this rule, deficits are always to be financed entirely by issuing additional base money, with interest-bearing government debt never being issued. In this regime, the time path of government deficits affects the time path of the price level in a rigid and immediate way that is described by the previous budget constraint and the equation for P_t^* . Under this regime it is possible for the government budget to be persistently in deficit, within limits imposed by the budget constraint and the money-demand equation. Deficits need not be temporary.

As a simple example, suppose that the primary deficit is constant:

$$G_t - T_t = \Delta \quad \forall t$$

This requires a constant seigniorage:

$$\frac{M_t - M_{t-1}}{P_t} = \Delta \quad \forall t$$

We guess that, in this case, there is an equilibrium with a constant growth of the money supply:

$$M_{t+1} = (1 + \mu)M_t \quad \forall t$$

Then:

$$M_{t+i} = (1 + \mu)^i M_t$$

Substituting into the expression for P_t^* :

$$\begin{aligned} P_t^* &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \\ &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t \left\{ (1 + \mu)^i M_t \right\} \\ &= \frac{M_t}{c} \sum_{i=0}^{\infty} [\alpha(1 + \mu)]^i \end{aligned}$$

Then, assuming $\alpha(1 + \mu) < 1$:

$$P_t^* = \frac{M_t}{c[1 - \alpha(1 + \mu)]}$$

Since P_t^* is proportional to M_t , we get:

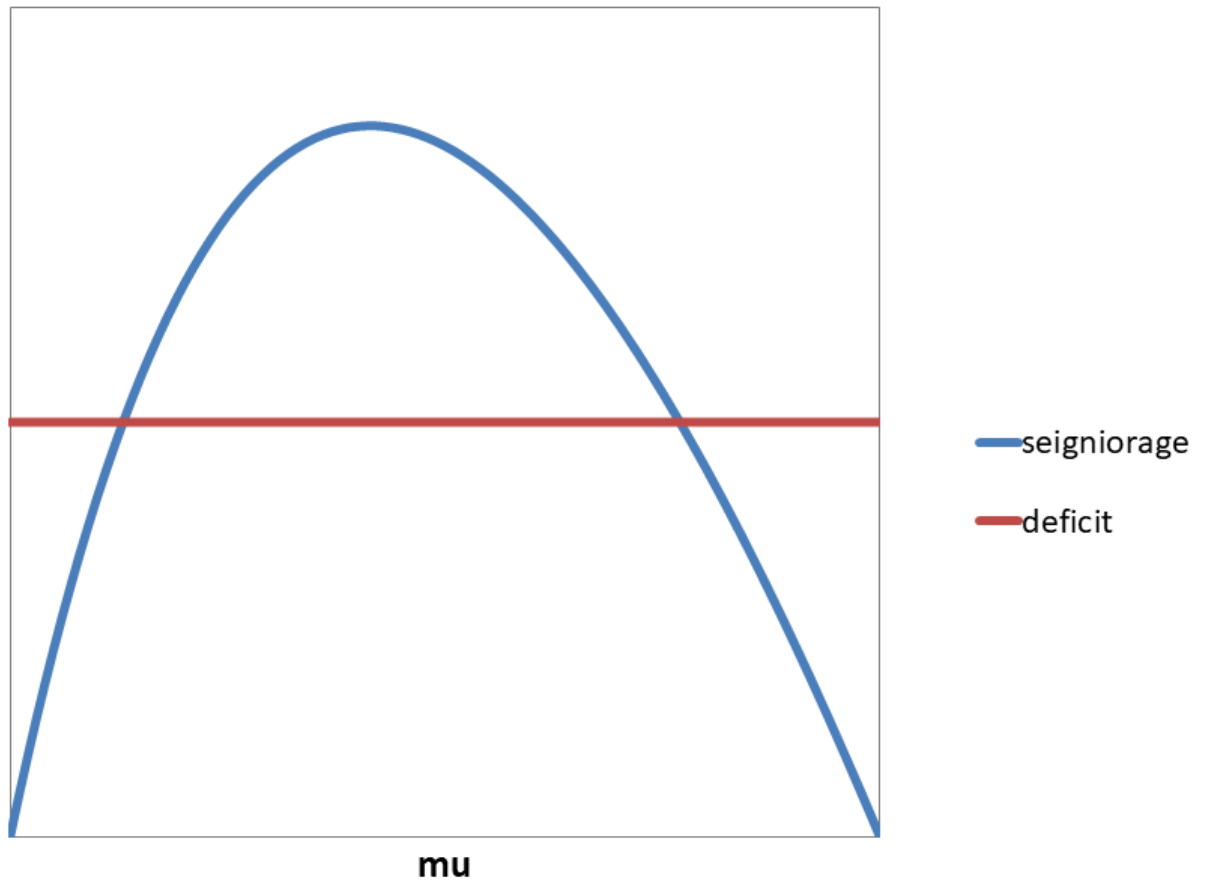
$$\pi_t = \mu \quad \forall t$$

where $\pi_t \equiv \frac{P_t - P_{t-1}}{P_{t-1}}$ is the inflation rate at t .

To determine the equilibrium value of μ we need to go back to the budget constraint:

$$\begin{aligned} \Delta &= \frac{M_t - M_{t-1}}{P_t} \\ &= \frac{M_t - M_{t-1}}{M_{t-1}} \frac{M_{t-1}}{P_t} \\ &= \mu \frac{M_{t-1}}{M_t} \frac{M_t}{P_t} \\ &= \frac{\mu}{1 + \mu} c[1 - \alpha(1 + \mu)] \end{aligned}$$

From the expression above we can solve for μ . In this particular case we get a quadratic equation than can be solved in closed form, and we obtain multiple equilibria (assuming the equilibria exist). See the figure below.



Appendix I

Budget Constraint of the Consolidated Government

Treasury (T)

Budget constraint:

$$G_t^T + r_{t-1}D_{t-1}^T = T_t + (D_t^T - D_{t-1}^T) + V_t$$

Central Bank (CB)

Balance Sheet:

$D_t^{T,CB}$	M_t/P_t
	D_t^{CB}
	NW_t

Budget constraint:

$$G_t^{CB} + r_{t-1}D_{t-1}^{CB} + V_t + (D_t^{T,CB} - D_{t-1}^{T,CB}) = r_{t-1}D_{t-1}^{T,CB} + (D_t^{CB} - D_{t-1}^{CB}) + \frac{M_t - M_{t-1}}{P_t}$$

Consolidated Government

Combining the budget constraints of T and CB:

$$\begin{aligned} & G_t^T + r_{t-1}D_{t-1}^T + G_t^{CB} + r_{t-1}D_{t-1}^{CB} + V_t + (D_t^{T,CB} - D_{t-1}^{T,CB}) \\ = & T_t + (D_t^T - D_{t-1}^T) + V_t + r_{t-1}D_{t-1}^{T,CB} + (D_t^{CB} - D_{t-1}^{CB}) + \frac{M_t - M_{t-1}}{P_t} \end{aligned}$$

Then:

$$\begin{aligned} & (G_t^T + G_t^{CB}) + r_{t-1}(D_{t-1}^T - D_{t-1}^{T,CB} + D_{t-1}^{CB}) \\ = & T_t + (D_t^T - D_t^{T,CB} + D_t^{CB}) - (D_{t-1}^T - D_{t-1}^{T,CB} + D_{t-1}^{CB}) + \frac{M_t - M_{t-1}}{P_t} \end{aligned}$$

Then:

$$G_t + r_{t-1}D_{t-1} = T_t + D_t - D_{t-1} + \frac{M_t - M_{t-1}}{P_t}$$

where

$$G_t \equiv G_t^T + G_t^{CB}$$

$$D_t \equiv D_t^T - D_t^{T,CB} + D_t^{CB}$$

Appendix II:

Solution with Bubbles

The equation we want to solve is

$$P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$$

with $\alpha \in (0, 1)$.

We have already shown that:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} + \lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Now we want a solution with

$$\lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1} \neq 0$$

We propose the following candidate:

$$P_t = P_t^* + B_t$$

where P_t^* is the fundamental solution and B_t is an unspecified random variable (a bubble).

Now we check under what conditions (if any) the expression above is a solution.

For $P_t = P_t^* + B_t$ to be a solution it must satisfy our original equation, $P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$. Hence, we need:

$$P_t^* + B_t = \frac{M_t}{c} + \alpha \mathbb{E}_t \{P_{t+1}^* + B_{t+1}\}$$

Then:

$$P_t^* + B_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}^* + \alpha \mathbb{E}_t B_{t+1}$$

But we know the fundamental solution satisfies the original equation:

$$P_t^* = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}^*$$

Combining the previous two expressions we get:

$$B_t = \alpha \mathbb{E}_t B_{t+1}$$

Hence, $P_t = P_t^* + B_t$ will be a solution as long as the random variable B_t (the bubble) satisfies the condition above.

From the previous expression we get:

$$\mathbb{E}_t B_{t+1} = \frac{1}{\alpha} B_t$$

Notice that $\frac{1}{\alpha} > 1$, since $\alpha \in (0, 1)$.

We have:

$$\mathbb{E}_{t+1} B_{t+2} = \frac{1}{\alpha} B_{t+1}$$

Then:

$$\mathbb{E}_t \mathbb{E}_{t+1} B_{t+2} = \mathbb{E}_t \left\{ \frac{1}{\alpha} B_{t+1} \right\}$$

$$\begin{aligned} \mathbb{E}_t B_{t+2} &= \frac{1}{\alpha} \mathbb{E}_t B_{t+1} \quad \text{by LIE} \\ &= \frac{1}{\alpha} \frac{1}{\alpha} B_t \\ &= \frac{1}{\alpha^2} B_t \end{aligned}$$

And:

$$\begin{aligned} \mathbb{E}_{t+2} B_{t+3} &= \frac{1}{\alpha} B_{t+2} \\ \mathbb{E}_t \mathbb{E}_{t+2} B_{t+3} &= \frac{1}{\alpha} \mathbb{E}_t B_{t+2} \\ \mathbb{E}_t B_{t+3} &= \frac{1}{\alpha} \mathbb{E}_t B_{t+2} \quad \text{by LIE} \\ \mathbb{E}_t B_{t+3} &= \frac{1}{\alpha} \frac{1}{\alpha^2} B_t \\ \mathbb{E}_t B_{t+3} &= \frac{1}{\alpha^3} B_t \end{aligned}$$

Keep going to get:

$$\mathbb{E}_t B_{t+n} = \frac{1}{\alpha^n} B_t$$

Then:

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{E}_t B_{t+n} &= \lim_{n \rightarrow \infty} \frac{1}{\alpha^n} B_t \\ &= B_t \lim_{n \rightarrow \infty} \frac{1}{\alpha^n} \\ &= \begin{cases} +\infty & \text{if } B_t > 0 \\ -\infty & \text{if } B_t < 0 \end{cases} \end{aligned}$$

Hence, the bubble is explosive.

Just to be sure, we check that $\lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1} \neq 0$ whenever there is a bubble ($B_t \neq 0$).

Using $P_t = P_t^* + B_t$ we get:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1} &= \lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t \{P_{t+n+1}^* + B_{t+n+1}\} \\
&= \underbrace{\lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1}^*}_{= 0} \\
&\quad + \lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t B_{t+n+1} \\
&= \lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t B_{t+n+1} \\
&= \lim_{n \rightarrow \infty} \alpha^{n+1} \frac{1}{\alpha^{n+1}} B_t \\
&= \lim_{n \rightarrow \infty} B_t \\
&= B_t \\
&\neq 0 \text{ whenever } B_t \neq 0
\end{aligned}$$

Example

Suppose:

$$M_t = M$$

and $\{B_t\}$ satisfies

$$B_{t+1} = \frac{1}{\alpha}B_t + \eta_{t+1}$$

where $\{\eta_{t+1}\}$ is a zero-mean iid sequence of shocks.

Notice that

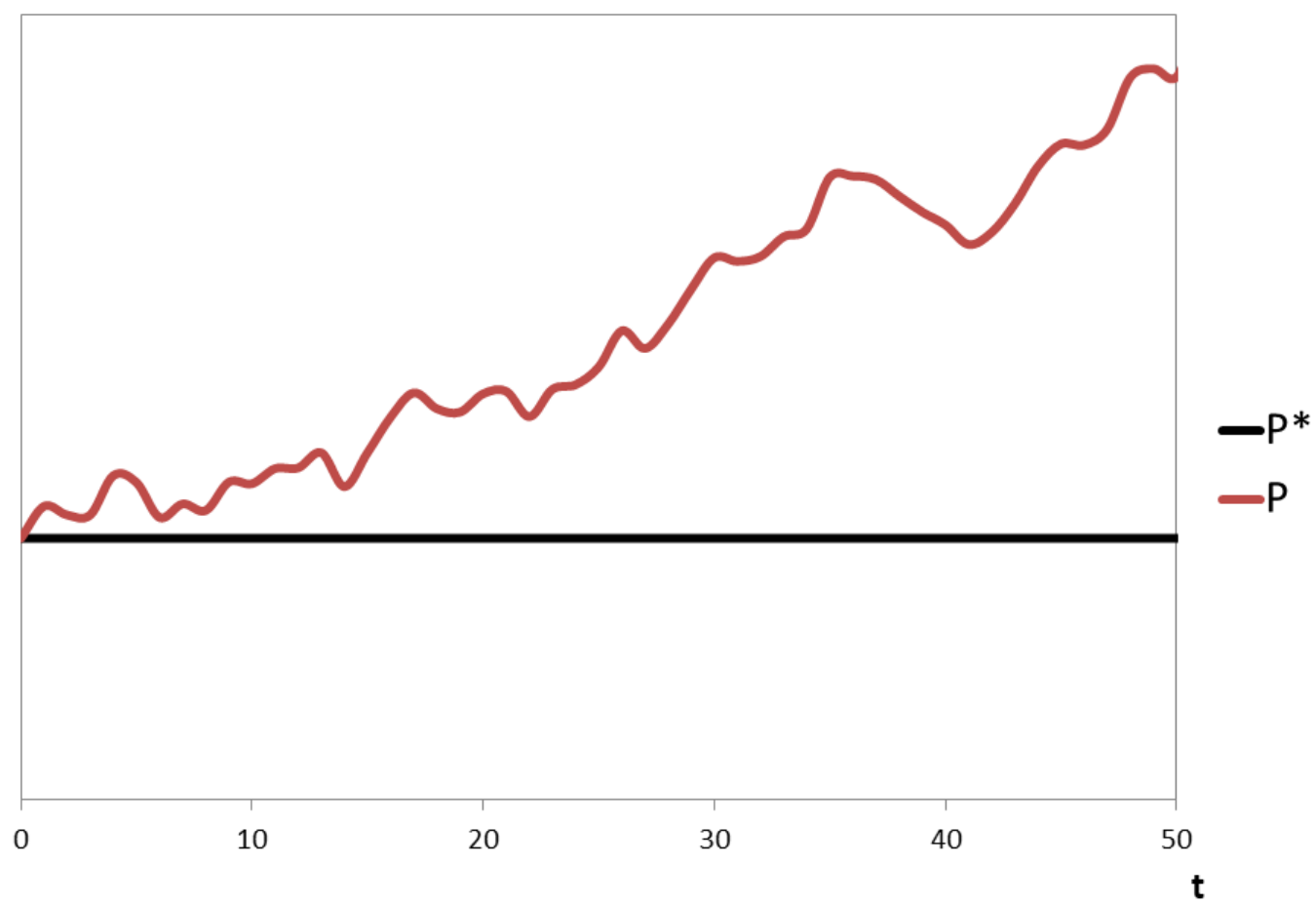
$$\begin{aligned}\mathbb{E}_t B_{t+1} &= \frac{1}{\alpha}B_t + \mathbb{E}_t \eta_{t+1} \\ &= \frac{1}{\alpha}B_t\end{aligned}$$

and then

$$P_t = P_t^* + B_t$$

is a solution with $P_t^* = \frac{1}{(1-\alpha)c}M$ and $B_t = \frac{1}{\alpha}B_{t-1} + \eta_t$.

Figure



Bursting Bubbles

Now suppose:

$$B_{t+1} = \begin{cases} \frac{\alpha}{p}B_t + \eta_{t+1} & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

where $p \in (0, 1)$ and $\{\eta_{t+1}\}$ is a zero-mean iid sequence of shocks.

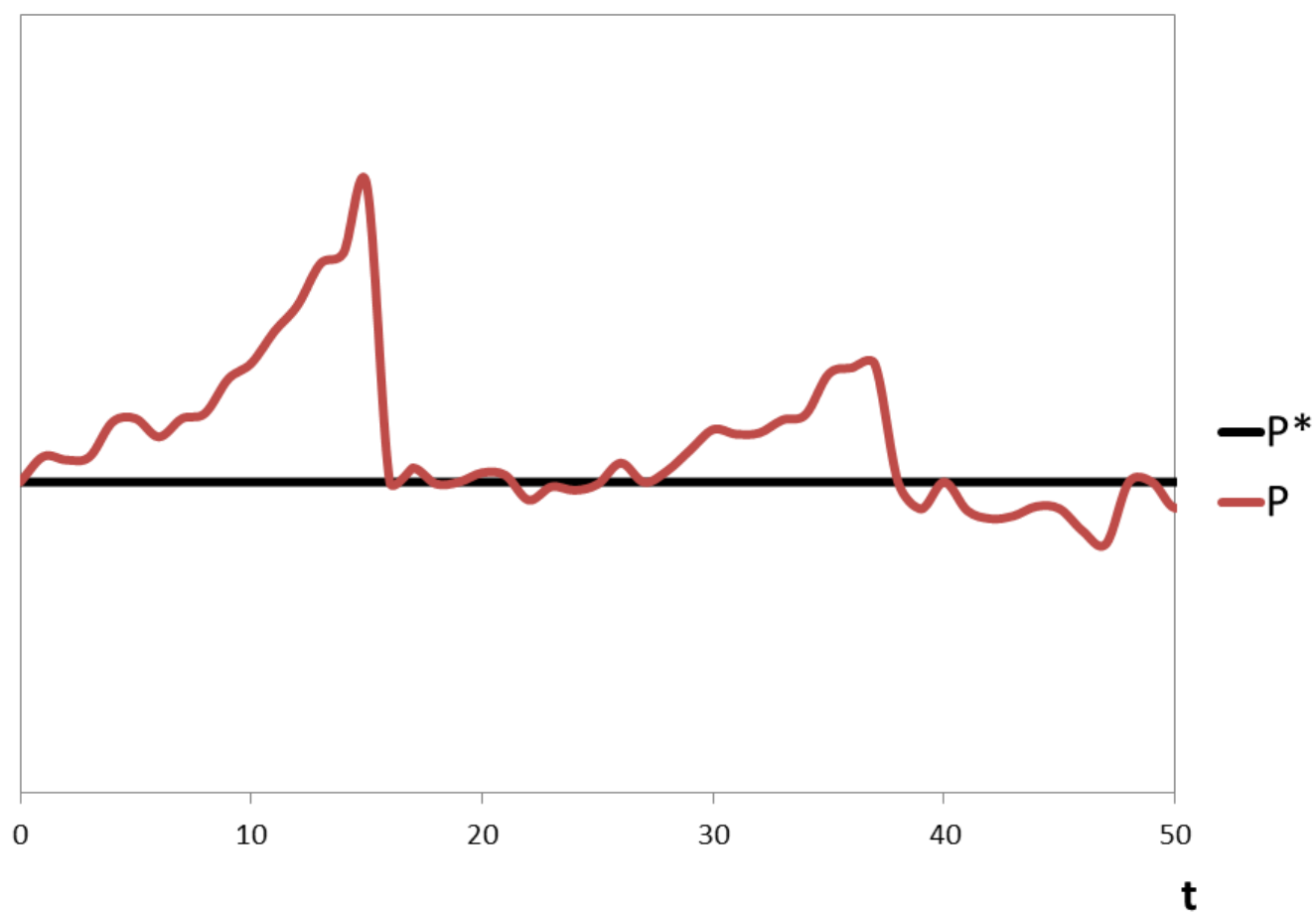
From the expression above we get:

$$\begin{aligned} \mathbb{E}_t B_{t+1} &= p\mathbb{E}_t\left\{\frac{\alpha}{p}B_t + \eta_{t+1}\right\} + (1 - p)\mathbb{E}_t\{0\} \\ &= p\frac{\alpha}{p}B_t + \mathbb{E}_t\eta_{t+1} \\ &= \alpha B_t \end{aligned}$$

Hence, the random variable B_t qualifies as a bubble.

Example of a bursting bubble and fundamental solution

$$P_t^* = \frac{1}{(1-\alpha)c}M.$$



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Unpleasant Monetarist Arithmetic

Discrete time: $t \in \{1, 2, \dots\}$.

Money Demand:

$$\frac{M_t^d}{P_t} = \Phi(\bar{i}_t, \bar{Y}_t)$$

where

$$1 + i_t = (1 + r_t) \frac{\mathbb{E}_t P_{t+1}}{P_t}$$

For simplicity, assume:

$$\begin{aligned} Y_t &= Y \quad \forall t \geq 0 \\ r_t &= r \quad \forall t \geq 0 \end{aligned}$$

Then:

$$\frac{M_t^d}{P_t} = \Psi\left(\frac{\mathbb{E}_t \bar{P}_{t+1}}{P_t}\right)$$

where $\Psi\left(\frac{\mathbb{E}_t P_{t+1}}{P_t}\right) \equiv \Phi\left((1 + r) \frac{\mathbb{E}_t P_{t+1}}{P_t} - 1, Y\right)$.

For simplicity, we assume that the function Ψ is linear:

$$\frac{M_t^d}{P_t} = c - b \frac{\mathbb{E}_t P_{t+1}}{P_t}$$

with $c > 0$ and $b > 0$.

Equilibrium requires:

$$M_t^d = M_t$$

where M_t is the money supply (in nominal terms).

Then:

$$\frac{M_t}{P_t} = c - b \frac{\mathbb{E}_t P_{t+1}}{P_t}$$

From the previous expression we get:

$$P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$$

where $\alpha \equiv \frac{b}{c} > 0$.

The expression above is a first-order stochastic linear difference equation in the price level.

Assume $c > b$, which implies

$$\alpha < 1$$

Then, we can solve $P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$ forward:

$$\begin{aligned} P_t &= \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1} \\ &= \frac{M_t}{c} + \alpha \mathbb{E}_t \left\{ \frac{M_{t+1}}{c} + \alpha \mathbb{E}_{t+1} P_{t+2} \right\} \\ &= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \mathbb{E}_{t+1} P_{t+2} \\ &= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t P_{t+2} \quad \text{by LIE} \\ &= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \left\{ \frac{M_{t+2}}{c} + \alpha \mathbb{E}_{t+2} P_{t+3} \right\} \\ &= \frac{1}{c} M_t + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \frac{\alpha^2}{c} \mathbb{E}_t M_{t+2} + \alpha^3 \mathbb{E}_t \mathbb{E}_{t+2} P_{t+3} \\ &= \frac{1}{c} (M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2}) + \alpha^3 \mathbb{E}_t P_{t+3} \end{aligned}$$

If we keep iterating forward we get:

$$P_t = \frac{1}{c}(M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2} + \dots + \alpha^n \mathbb{E}_t M_{t+n}) + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Then:

$$P_t = \frac{1}{c} \sum_{i=0}^n \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Letting $n \rightarrow \infty$:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} + \lim_{n \rightarrow \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

assuming that $\sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \equiv \lim_{n \rightarrow \infty} \sum_{i=0}^n \alpha^i \mathbb{E}_t M_{t+i}$ is finite.

Suppose the price level satisfies:

$$\lim_{n \rightarrow \infty} \alpha^n \mathbb{E}_t P_{t+n} = 0$$

Then, the previous expression for P_t becomes:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \quad (1)$$

which is the so-called fundamental solution (no bubbles).

The expression above shows that the price level at t depends on the money supply expected to prevail from t into the indefinite future.

According to previous equation, if government deficits are to influence the price level path, it can only be through their effect on the expected path of base money. However, the government deficit and path of base money are not rigidly linked in any immutable way. The reason is that the government can, at least to a point, borrow by issuing interest-bearing government debt, and so need not necessarily issue base money to cover its deficit. To see

this more precisely, we need to write down the budget constraint of the consolidated government:

$$G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1}) + \frac{M_t - M_{t-1}}{P_t} \quad (2)$$

with $M_0 > 0$, $D_0 > 0$ and $r_0 = r > 0$ given.

Under the system formed by equations (1) and (2), the inflationary consequences of government deficits depend sensitively on the government's strategy for servicing the debt that it issues.

Consider the following fiscal-monetary policy combination:

$$\text{a) } G_t - T_t = \begin{cases} \overline{\Delta} & \text{if } t \in \{1, 2, \dots, T\} \\ \underline{\Delta} & \text{if } t \in \{T + 1, T + 2, \dots\} \end{cases}$$

b) For $t \in \{1, 2, \dots, T\}$, M_t grows at the constant exogenous rate θ and D_t evolves endogenously to satisfy the budget constraint of the government;

c) For $t \in \{T + 1, T + 2, \dots\}$, $D_t = D_T$ and M_t evolves endogenously to satisfy the budget constraint of the government (if possible).

To find an equilibrium we work backwards in time.

We know that $r_{t-1} = r$, $G_t - T_t = \underline{\Delta}$ and $D_t = D_T$ $\forall t > T$.

Then, $G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1}) + \frac{M_t - M_{t-1}}{P_t}$ becomes

$$\begin{aligned} G_t - T_t + rD_T &= (D_T - D_T) + \frac{M_t - M_{t-1}}{P_t} \quad \forall t > T \\ \underline{\Delta} + rD_T &= \frac{M_t - M_{t-1}}{P_t} \quad \forall t > T \end{aligned} \quad (3)$$

We guess that there is an equilibrium with a constant growth of the money supply after T :

$$M_{t+1} = (1 + \mu)M_t \quad \forall t \geq T$$

where M_T is predetermined.

Then:

$$M_{t+i} = (1 + \mu)^i M_t \quad \forall t \geq T \quad (4)$$

with $i \in \{0, 1, 2, \dots\}$.

Substituting (4) into (1) we get:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t\{(1 + \mu)^i M_t\} \quad \forall t \geq T$$

$$P_t = \frac{M_t}{c} \sum_{i=0}^{\infty} [\alpha(1 + \mu)]^i$$

$$P_t = \frac{M_t}{c} \frac{1}{1 - \alpha(1 + \mu)}$$

where we have assumed $\alpha(1 + \mu) < 1$.

Then:

$$P_t = \frac{1}{c [1 - \alpha(1 + \mu)]} M_t \quad \forall t \geq T \quad (5)$$

Remark 1 $\frac{M_t}{P_t} = c [1 - \alpha(1 + \mu)] \quad \forall t \geq T$

Remark 2. $\pi_t = \mu \quad \forall t > T$

Now we consider $t \leq T$.

From $P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i}$ we get:

$$P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \frac{1}{c} \sum_{i=T-t+1}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \quad \forall t \leq T$$

$$P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \frac{1}{c} \alpha^{T+1-t} \sum_{i=T+1-t}^{\infty} \alpha^{i-(T+1-t)} \mathbb{E}_t M_{t+i}$$

$$P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \frac{1}{c} \sum_{i=T+1-t}^{\infty} \alpha^{i-(T+1-t)} \mathbb{E}_t \{ \mathbb{E}_{T+1} M_{t+i} \} \quad \text{LIE}$$

$$P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \frac{1}{c} \sum_{j=0}^{\infty} \alpha^j \mathbb{E}_t \{ \mathbb{E}_{T+1} M_{T+1+j} \}$$

$$P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \mathbb{E}_t \left\{ \frac{1}{c} \sum_{j=0}^{\infty} \alpha^j \mathbb{E}_{T+1} M_{T+1+j} \right\}$$

Using (1):

$$P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \mathbb{E}_t P_{T+1} \quad \forall t \leq T \quad (6)$$

We know $M_{t+1} = (1 + \theta)M_t \quad \forall t \in \{1, \dots, T\}$, with $M_1 = (1 + \theta)M_0 > 0$ given. Then:

$$M_{t+i} = (1 + \theta)^i M_t \quad \forall t \in \{1, \dots, T\} \quad (7)$$

where $i \in \{0, 1, \dots, T - t\}$.

Substituting (5) and (7) into (6):

$$P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t \{ (1 + \theta)^i M_t \} + \alpha^{T+1-t} \mathbb{E}_t \left\{ \frac{M_{T+1}}{c[1 - \alpha(1 + \mu)]} \right\}$$

$$P_t = \frac{M_t}{c} \sum_{i=0}^{T-t} [\alpha(1 + \theta)]^i + \frac{\alpha^{T+1-t} M_{T+1}}{c[1 - \alpha(1 + \mu)]}$$

$$P_t = \frac{M_t}{c} \sum_{i=0}^{T-t} [\alpha(1 + \theta)]^i + \frac{\alpha^{T+1-t}(1 + \mu)M_T}{c[1 - \alpha(1 + \mu)]}$$

$$P_t = \frac{M_t}{c} \frac{1 - [\alpha(1 + \theta)]^{T+1-t}}{1 - \alpha(1 + \theta)} + \frac{\alpha^{T+1-t}(1 + \mu)M_T}{c[1 - \alpha(1 + \mu)]} \quad \forall t \leq T$$

But $M_T = (1 + \theta)^{T-t} M_t$. Then:

$$\begin{aligned}
P_t &= \frac{M_t}{c} \frac{1 - [\alpha(1 + \theta)]^{T+1-t}}{1 - \alpha(1 + \theta)} + \frac{\alpha^{T+1-t}(1 + \mu)(1 + \theta)^{T-t} M_t}{c [1 - \alpha(1 + \mu)]} \quad \forall t \leq T \\
P_t &= \left[\frac{1 - [\alpha(1 + \theta)]^{T+1-t}}{1 - \alpha(1 + \theta)} + \frac{\alpha^{T+1-t}(1 + \mu)(1 + \theta)^{T-t}}{1 - \alpha(1 + \mu)} \right] \frac{M_t}{c} \\
P_t &= \frac{[1 - \alpha(1 + \mu)] \left[1 - [\alpha(1 + \theta)]^{T+1-t} \right] + [1 - \alpha(1 + \theta)] \left[\alpha^{T+1-t}(1 + \mu)(1 + \theta)^{T-t} \right]}{[1 - \alpha(1 + \theta)] [1 - \alpha(1 + \mu)]} \frac{M_t}{c} \\
P_t &= \frac{1 - [\alpha(1 + \theta)]^{T+1-t} - \alpha(1 + \mu) + \alpha(1 + \mu) [\alpha(1 + \theta)]^{T+1-t} + \alpha^{T+1-t}(1 + \mu)(1 + \theta)^{T-t} - \alpha(1 + \theta) \alpha^{T+1-t}(1 + \mu)(1 + \theta)^{T-t}}{[1 - \alpha(1 + \theta)] [1 - \alpha(1 + \mu)]} \frac{M_t}{c} \\
P_t &= \frac{1 - \alpha^{T+1-t}(1 + \theta)^{T+1-t} - \alpha(1 + \mu) + (1 + \mu) \alpha^{T+2-t}(1 + \theta)^{T+1-t} + \alpha^{T+1-t}(1 + \mu)(1 + \theta)^{T-t} - (1 + \mu) \alpha^{T+2-t}(1 + \theta)^{T+1-t}}{[1 - \alpha(1 + \theta)] [1 - \alpha(1 + \mu)]} \frac{M_t}{c} \\
P_t &= \frac{1 - \alpha^{T+1-t}(1 + \theta)^{T+1-t} - \alpha(1 + \mu) + \alpha^{T+1-t}(1 + \mu)(1 + \theta)^{T-t}}{[1 - \alpha(1 + \theta)] [1 - \alpha(1 + \mu)]} \frac{M_t}{c} \\
P_t &= \frac{1 - \alpha(1 + \mu) + [-(1 + \theta) + (1 + \mu)] \alpha^{T+1-t}(1 + \theta)^{T-t}}{[1 - \alpha(1 + \theta)] [1 - \alpha(1 + \mu)]} \frac{M_t}{c}
\end{aligned}$$

Then:

$$P_t = \frac{1 - \alpha(1 + \mu) + (\mu - \theta) \alpha^{T+1-t}(1 + \theta)^{T-t}}{[1 - \alpha(1 + \theta)] [1 - \alpha(1 + \mu)]} \frac{M_t}{c} \quad \forall t \leq T \quad (8)$$

Remark: we still need to find the equilibrium value of μ .

Finding μ

Seigniorage:

$$\begin{aligned} S_t &\equiv \frac{M_t - M_{t-1}}{P_t} \\ &= \frac{M_t - M_{t-1}}{M_{t-1}} \frac{M_{t-1}}{P_t} \\ &= \frac{M_t - M_{t-1}}{M_{t-1}} \frac{M_{t-1}}{M_t} \frac{M_t}{P_t} \\ &= \frac{\mu_t}{1 + \mu_t} \frac{M_t}{P_t} \end{aligned}$$

where $\mu_t \equiv \frac{M_t - M_{t-1}}{M_{t-1}}$ is the growth rate of the nominal money supply.

For $t > T$ we have $\mu_t = \mu$ and $\frac{M_t}{P_t} = c[1 - \alpha(1 + \mu)]$.
Then:

$$S_t = \frac{\mu}{1 + \mu} c[1 - \alpha(1 + \mu)] \quad \forall t > T \quad (9)$$

Substituting (9) into (3):

$$\underline{\Delta} + rD_T = \frac{\mu}{1+\mu}c[1 - \alpha(1 + \mu)] \quad \forall t > T \quad (10)$$

To find μ from the equation above we first need to find the value of D_T , which will depend on μ (as we'll see below).

Finding D_T

From (2) we get: $G_t - T_t + r_{t-1}D_{t-1} = (D_t - D_{t-1}) + S_t \Rightarrow$

$$D_t = (1 + r)D_{t-1} + \bar{\Delta} - S_t \quad \forall t \leq T$$

But we know $S_t \equiv \frac{M_t - M_{t-1}}{P_t} = \frac{\mu_t}{1 + \mu_t} \frac{M_t}{P_t} \Rightarrow$

$$\begin{aligned} S_t &= \frac{\theta}{1 + \theta} \frac{M_t}{P_t} \quad \forall t \leq T \\ &= \frac{\theta}{1 + \theta} \frac{c[1 - \alpha(1 + \theta)][1 - \alpha(1 + \mu)]}{1 - \alpha(1 + \mu) + (\mu - \theta)\alpha^{T+1-t}(1 + \theta)^{T-t}} \quad \forall t \leq T \end{aligned}$$

where we used (8) in the second line.

Combining the previous two expressions we get:

$$D_t = (1+r)D_{t-1} + \bar{\Delta} - \frac{\theta}{1+\theta} \frac{c[1-\alpha(1+\theta)][1-\alpha(1+\mu)]}{1-\alpha(1+\mu) + (\mu-\theta)\alpha^{T+1-t}(1+\theta)^{T-t}} \quad (11)$$

which holds $\forall t \leq T$.

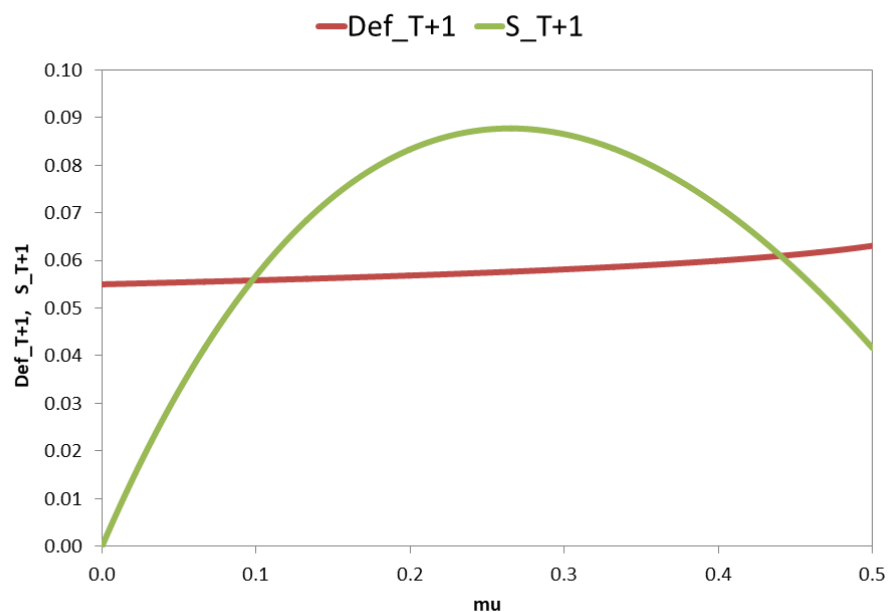
For any given μ , the expression above recursively determines the evolution of D_t from $t = 1$ until $t = T$, starting with D_0 . For $t = T$ we get

$$D_T(\mu)$$

Substituting the expression above into (10) we get:

$$\underline{\Delta} + rD_T(\mu) = \frac{\mu}{1+\mu} c[1-\alpha(1+\mu)] \quad \forall t > T \quad (12)$$

Graphically



Remark 1. Multiple equilibria. We'll assume we are in the upward-sloping side of the seigniorage Laffer curve.

Remark 2. For a high-enough deficit, equilibrium will not exist.

Solution strategy:

(i) Choose a value for μ .

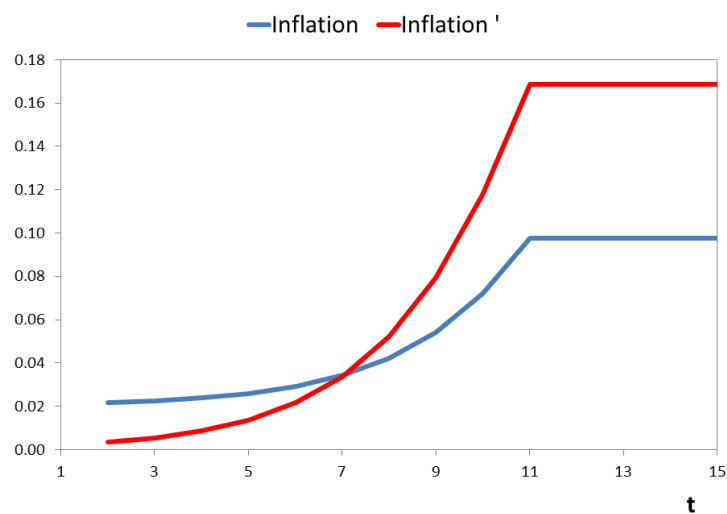
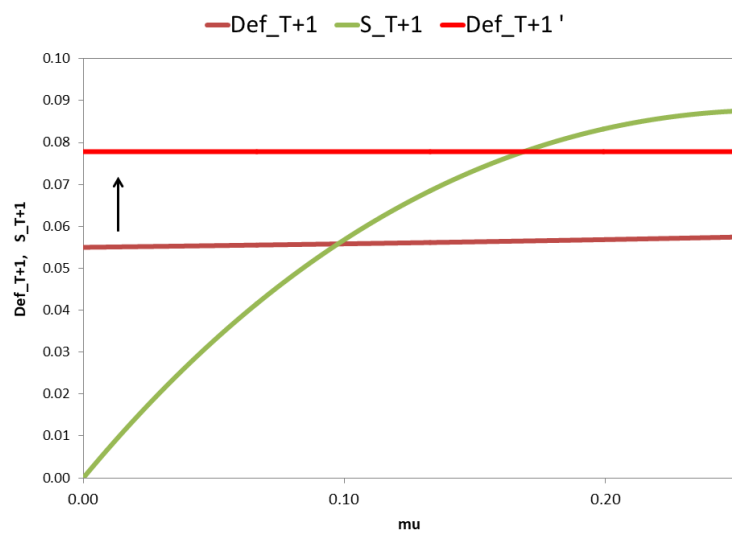
(ii) Iterate on (11) to find $D_T(\mu)$.

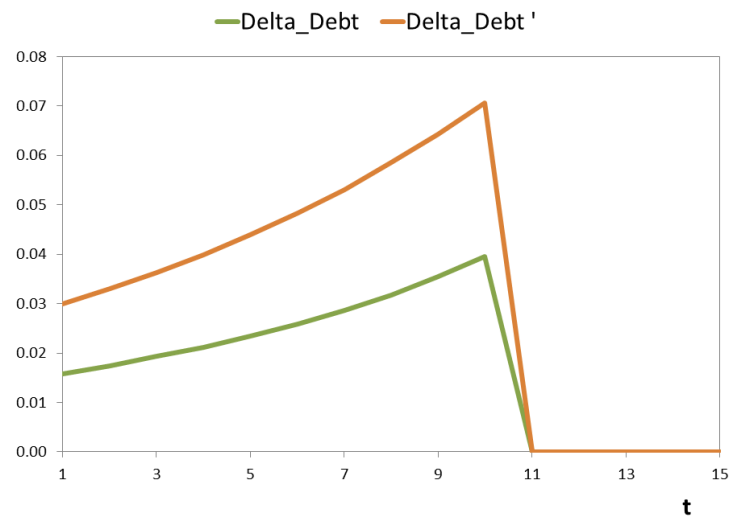
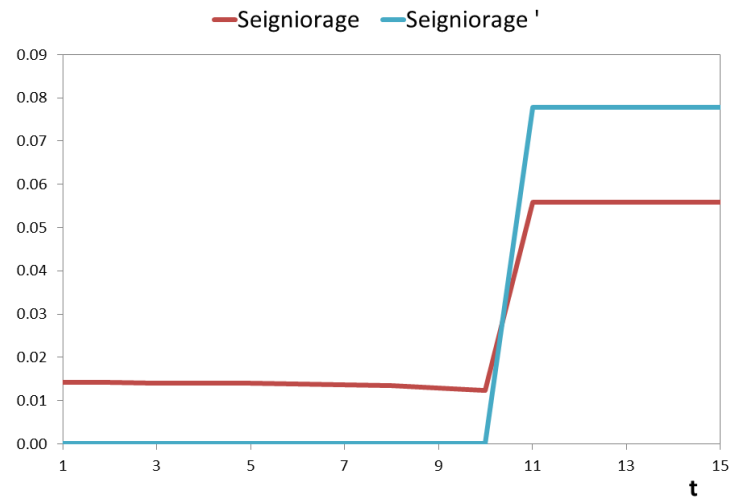
(iii) Compute $\underline{\Delta} + rD_T(\mu) - \frac{\mu}{1+\mu}c[1 - \alpha(1 + \mu)]$

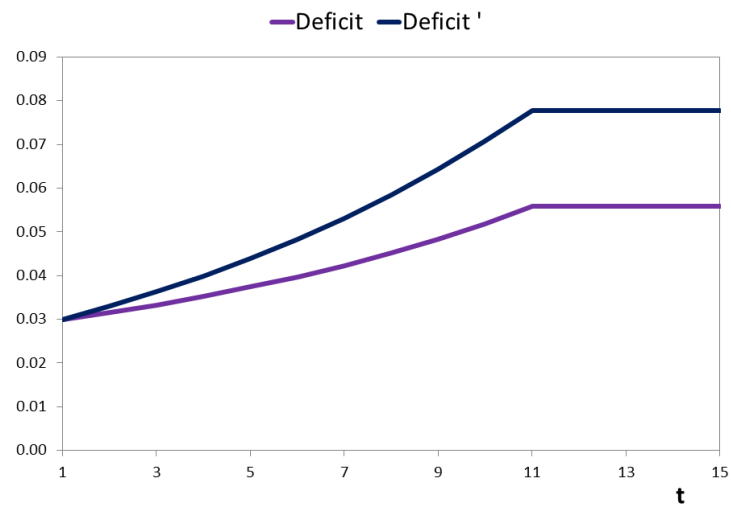
(iv) Change μ until the expression above equals 0, so (12) holds.

Unpleasant Monetarist Arithmetic

$$\downarrow \theta \Rightarrow \uparrow \mu$$







Appendix I

From (6) we get:

$$\frac{P_t}{P_{t-1}} = \frac{\frac{1-\alpha(1+\mu)+(\mu-\theta)\alpha^{T+1-(t+1)}(1+\theta)^{T-(t+1)} M_{t+1}}{[1-\alpha(1+\theta)][1-\alpha(1+\mu)]} \frac{c}{c}}{\frac{1-\alpha(1+\mu)+(\mu-\theta)\alpha^{T+1-t}(1+\theta)^{T-t} M_t}{[1-\alpha(1+\theta)][1-\alpha(1+\mu)]} \frac{c}{c}} \quad \forall t \in \{1, \dots, T-1\}$$

Then, $\forall t \in \{1, \dots, T-1\}$:

$$\begin{aligned} \frac{P_t}{P_{t-1}} &= \frac{1 - \alpha(1 + \mu) + (\mu - \theta) \alpha^{T+1-(t+1)}(1 + \theta)^{T-(t+1)} M_{t+1}}{1 - \alpha(1 + \mu) + (\mu - \theta) \alpha^{T+1-t}(1 + \theta)^{T-t} M_t} \\ &= \frac{1 - \alpha(1 + \mu) + (\mu - \theta) \alpha^{T-t}(1 + \theta)^{T-t-1}}{1 - \alpha(1 + \mu) + (\mu - \theta) \alpha^{T+1-t}(1 + \theta)^{T-t}} (1 + \theta) \\ &= \frac{1 - \alpha(1 + \mu) + (\mu - \theta) \alpha^{T-t}(1 + \theta)^{T-t}}{1 - \alpha(1 + \mu) + (\mu - \theta) \alpha^{T+1-t}(1 + \theta)^{T-t}} \end{aligned}$$

Differentiating the expression above with respect to t we get:

$$\frac{\partial \left(\frac{P_t}{P_{t-1}} \right)}{\partial t} \begin{cases} > 0 & \text{if } \mu > \theta \\ = 0 & \text{if } \mu = \theta \\ < 0 & \text{if } \mu < \theta \end{cases}$$

Appendix II

From $G_t + rD_{t-1} = T_t + (D_t - D_{t-1}) + S_t$ we get:

$$(1 + r)D_{t-1} = T_t - G_t + S_t + D_t$$

Then:

$$D_{t-1} = \frac{X_t}{1 + r} + \frac{D_t}{1 + r}$$

where $X_t \equiv T_t - G_t + S_t$.

Iterating forward:

$$\begin{aligned} D_0 &= \frac{X_1}{1 + r} + \frac{D_1}{1 + r} \\ &= \frac{X_1}{1 + r} + \frac{1}{1 + r} \left(\frac{X_2}{1 + r} + \frac{D_2}{1 + r} \right) \\ &= \frac{X_1}{1 + r} + \frac{X_2}{(1 + r)^2} + \frac{D_2}{(1 + r)^2} \\ &= \frac{X_1}{1 + r} + \frac{X_2}{(1 + r)^2} + \frac{1}{(1 + r)^2} \left(\frac{X_3}{1 + r} + \frac{D_3}{1 + r} \right) \\ &= \frac{X_1}{1 + r} + \frac{X_2}{(1 + r)^2} + \frac{X_3}{(1 + r)^3} + \frac{D_3}{(1 + r)^3} \end{aligned}$$

Then:

$$D_0 = \frac{X_1}{1+r} + \frac{X_2}{(1+r)^2} + \dots + \frac{X_T}{(1+r)^T} + \frac{D_T}{(1+r)^T}$$

Then:

$$(1+r)D_0 = X_1 + \frac{X_2}{1+r} + \dots + \frac{X_T}{(1+r)^{T-1}} + \frac{D_T}{(1+r)^{T-1}}$$

Then:

$$(1+r)D_0 = \sum_{t=1}^T \frac{X_t}{(1+r)^{t-1}} + \frac{D_T}{(1+r)^{T-1}}$$

Letting $T \rightarrow \infty$:

$$(1+r)D_0 = \sum_{t=1}^{\infty} \frac{X_t}{(1+r)^{t-1}} + \lim_{T \rightarrow \infty} \frac{D_T}{(1+r)^{T-1}}$$

Imposing $\lim_{T \rightarrow \infty} \frac{D_T}{(1+r)^{T-1}} = 0$ (TVC) and assuming $\sum_{t=1}^{\infty} \frac{X_t}{(1+r)^{t-1}}$ is finite:

$$(1+r)D_0 = \sum_{t=1}^{\infty} \frac{X_t}{(1+r)^{t-1}}$$

Then:

$$(1 + r)D_0 = \sum_{t=1}^{\infty} \frac{T_t - G_t + S_t}{(1 + r)^{t-1}}$$

Then:

$$\sum_{t=1}^{\infty} \frac{G_t - T_t}{(1 + r)^{t-1}} + (1 + r)D_0 = \sum_{t=1}^{\infty} \frac{S_t}{(1 + r)^{t-1}}$$

Notice that, for a given left-hand side, a reduction of S_t in some period requires an increase in S_t in some other period.

References

Sargent, Thomas (1980). "Rational Expectations and the Reconstruction of Macroeconomics," Federal Reserve Bank of Minneapolis *Quarterly Review*, Vol. 4, No. 3 (Summer), pp. 15-19.

Sargent, T. & Wallace, N. (1981). "Some Unpleasant Monetarist Arithmetic," Federal Reserve Bank of Minneapolis *Quarterly Review*, Vol. 5, No. 3 (Fall), pp. 1-17.

Basic New Keynesian Model

Reference

Galí, Jordi (2008). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework*, Princeton University Press.

◆ Dynamic Stochastic General Equilibrium Model

◆ Monopolistic Competition

◆ Price Rigidity

◆ Short-run Non-neutrality of Monetary Policy

Government

Budget constraint

$$T_t = M_t - M_{t-1}$$

where T_t are lump-sum transfers and M_t is the quantity of money at the end of period t .

Policy Rule

1. Money Growth Rule
2. Taylor Rule

We'll specify these policy rules later.

Firms

There's a continuum of firms indexed by $i \in [0, 1]$. Each firm produces a differentiated product, but they all use an identical technology, represented by the production function

$$Y_t(i) = A_t N_t(i)^{1-\alpha} \quad \alpha \in (0, 1)$$

where A_t evolves exogenously according to

$$\ln A_{t+1} = (1 - \rho_a) \ln A + \rho_a \ln A_t + \varepsilon_{t+1}^A$$

with $A > 0$, $\rho_a \in (0, 1)$, and $\mathbb{E}\{\varepsilon_{t+1}^A\} = 0$.

The cost function associated to the production function above is:

$$\Psi_t(Y_t(i)) = W_t \underbrace{A_t^{-\frac{1}{1-\alpha}} Y_t(i)^{\frac{1}{1-\alpha}}}_{N_t}$$

Marginal cost is:

$$\psi_t(i) \equiv \Psi'_t(Y_t(i)) = \frac{1}{1-\alpha} W_t A_t^{-\frac{1}{1-\alpha}} Y_t(i)^{\frac{\alpha}{1-\alpha}}$$

Monopolistic Competition: Each firm operates in a monopolistically competitive market and faces a demand schedule to be derived below.

Price Rigidity [Calvo (1983), Yun (1996)]: In any period t , each firm may reset its price with probability $1 - \theta$, independently of the time elapsed since the last adjustment. Thus, in each period, a measure $1 - \theta$ of producers reset their prices, while a fraction θ keep their prices unchanged. With this formulation, the expected duration of a price is $\frac{1}{1-\theta}$ (see below). Therefore, θ becomes a natural index of price stickiness ($\theta \rightarrow 0$ full flexibility).

The probability that a price is fixed for $k \in \{1, 2, \dots\}$ periods is

$$\Pr(k) = \theta^{k-1} (1 - \theta).$$

Notice that $\sum_{k=1}^{\infty} \Pr(k) = 1$.

Then, expected duration (ED) is:

$$\begin{aligned}
 ED &= \sum_{k=1}^{\infty} \text{Pr}(k)k \\
 &= \sum_{k=1}^{\infty} \theta^{k-1} (1 - \theta) k \\
 &= (1 - \theta) \sum_{k=1}^{\infty} k \theta^{k-1}
 \end{aligned}$$

Then:

$$\begin{aligned}
 ED &= (1 - \theta) [1 + 2\theta + 3\theta^2 + 4\theta^3 + \dots] \\
 &= (1 - \theta) [1 + \theta + \theta + \theta^2 + \theta^2 + \theta^2 + \theta^3 + \theta^3 + \theta^3 + \theta^3 + \dots] \\
 &= (1 - \theta) [(1 + \theta + \theta^2 + \theta^3 + \dots) + (\theta + \theta^2 + \theta^3 + \dots) + (\theta^2 + \theta^3 + \theta^4 + \dots) + \dots] \\
 &= (1 - \theta) [(1 + \theta + \theta^2 + \theta^3 + \dots) + \theta(1 + \theta + \theta^2 + \dots) + \theta^2(1 + \theta + \theta^2 + \dots) + \dots] \\
 &= (1 - \theta) \left[\frac{1}{1 - \theta} + \theta \frac{1}{1 - \theta} + \theta^2 \frac{1}{1 - \theta} + \dots \right] \\
 &= (1 - \theta) \frac{1}{1 - \theta} [1 + \theta + \theta^2 + \dots] \\
 &= [1 + \theta + \theta^2 + \dots] \\
 &= \frac{1}{1 - \theta}
 \end{aligned}$$

Representative Household

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U \left(C_t, N_t, \frac{M_t}{P_t} \right)$$

$$U \left(C_t, N_t, \frac{M_t}{P_t} \right) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} + \frac{\chi}{1-\nu} \left(\frac{M_t}{P_t} \right)^{1-\nu}$$

$$C_t \equiv \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

with $\sigma > 0$, $\varphi > 0$, $\nu > 0$, $\epsilon > 1$, $\chi > 0$, and

$$\int_0^1 P_t(i) C_t(i) di + B_t + M_t = W_t N_t + (1+i_{t-1}) B_{t-1} + M_{t-1} + \Omega_t + T_t$$

where Ω_t = nominal profits at t , and T_t = lump-sum transfers at t .

Remark. Setting $\chi = 0$ we get a model without money in the utility function. When $\sigma = 1$ replace $\frac{C_t^{1-\sigma}}{1-\sigma}$ with $\ln C$. Analogously for $\nu = 1$.

Utility Maximization

We split the problem in two: an intratemporal problem and an intertemporal problem.

Intratemporal Problem

For any given level of expenditure in consumption goods, I , maximize the value of the consumption index C_t :

$$\begin{aligned} \max_{C_t(\cdot)} \quad & \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} \\ \text{s.t.} \quad & \int_0^1 P_t(i) C_t(i) di = I \end{aligned}$$

Lagrangian

$$\mathcal{L} = \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} + \lambda \left[I - \int_0^1 P_t(i) C_t(i) di \right]$$

FOC (see the Appendix). For each $h \in [0, 1]$ we have:

$$\frac{\epsilon}{\epsilon-1} \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}-1} \frac{\epsilon-1}{\epsilon} C_t(h)^{\frac{\epsilon-1}{\epsilon}-1} - \lambda P_t(h) = 0$$

Then:

$$\left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C_t(h)^{\frac{-1}{\epsilon}} = \lambda P_t(h) \quad \forall h \in [0, 1]$$

Then, for each pair of goods, j and h , we get:

$$\left(\frac{C_t(h)}{C_t(j)} \right)^{-\frac{1}{\epsilon}} = \frac{P_t(h)}{P_t(j)}$$

Notice that ϵ measures the elasticity of substitution between goods: $-\frac{d \ln(\frac{C_t(h)}{C_t(j)})}{d \ln(\frac{P_t(h)}{P_t(j)})} = \epsilon$.

From the expression above we get (changing index from h to i):

$$C_t(i) = \left(\frac{P_t(i)}{P_t(j)} \right)^{-\epsilon} C_t(j) \quad \forall i \in [0, 1]$$

Substituting into the constraint:

$$I = \int_0^1 P_t(i) C_t(i) di = \int_0^1 P_t(i) \left(\frac{P_t(i)}{P_t(j)} \right)^{-\epsilon} C_t(j) di = \frac{C_t(j)}{P_t(j)^{-\epsilon}} \int_0^1 P_t(i)^{1-\epsilon} di$$

Then:

$$C_t(j) = P_t(j)^{-\epsilon} \frac{I}{\int_0^1 P_t(i)^{1-\epsilon} di}$$

Now let the price level be:

$$P_t = \left[\int_0^1 P_t(i)^{1-\epsilon} di \right]^{\frac{1}{1-\epsilon}}$$

Remark. P_t is defined by:

$$P_t \equiv \min_{C_t(\cdot)} \int_0^1 P_t(i) C_t(i) di$$

$$\text{s.t.} \quad C_t \equiv \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} = 1$$

Then, our previous expression for $C_t(j)$ becomes: $C_t(j) = P_t(j)^{-\epsilon} \frac{I}{P_t^{1-\epsilon}}$. Rearranging, and changing the index to i , we can write:

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} \frac{I}{P_t}$$

Substituting the expression above into the objective we get:

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} = \left(\int_0^1 \left[\left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} \frac{I}{P_t} \right]^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

$$C_t = \frac{I}{P_t} \frac{1}{P_t^{-\epsilon}} \left(\int_0^1 P_t(i)^{1-\epsilon} di \right)^{-\frac{\epsilon}{1-\epsilon}} = \frac{I}{P_t} \frac{1}{P_t^{-\epsilon}} \left(P_t^{1-\epsilon} \right)^{-\frac{\epsilon}{1-\epsilon}} = \frac{I}{P_t} \frac{1}{P_t^{-\epsilon}} P_t^{-\epsilon}$$

Then:

$$C_t = \frac{I}{P_t}$$

Substituting the expression above into the one for $C_t(i)$ we get:

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} C_t \quad \forall i \in [0, 1]$$

Also, notice that $C_t = \frac{I}{P_t}$ implies $P_t C_t = I$, and then:

$$P_t C_t = \int_0^1 P_t(i) C_t(i) di$$

Intertemporal Problem

Using our last result we can write the intertemporal problem as follows:

$$\begin{aligned} \max \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U\left(C_t, N_t, \frac{M_t}{P_t}\right) \\ \text{s.t.} \quad & P_t C_t + B_t + M_t = W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} + \Omega_t + T_t \end{aligned}$$

Lagrangian

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t U\left(C_t, N_t, \frac{M_t}{P_t}\right) + \right. \\ & \left. \sum_{t=0}^{\infty} \beta^t \lambda_t \left[W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} \right. \right. \\ & \left. \left. + \Omega_t + T_t - P_t C_t - B_t - M_t \right] \right\} \end{aligned}$$

FOC

$$C_t : \quad U_{c,t} = \lambda_t P_t$$

$$N_t : \quad -U_{n,t} = \lambda_t W_t$$

$$M_t : \quad \lambda_t = U_{m,t} \frac{1}{P_t} + \beta \mathbb{E}_t \{ \lambda_{t+1} \}$$

$$B_t : \quad \lambda_t = \beta (1 + i_t) \mathbb{E}_t \{ \lambda_{t+1} \}$$

Remark. $U_m \equiv \frac{\partial U}{\partial (\frac{M}{P})}$.

Combining the FOCs w.r.t. C_t and N_t we get:

$$\frac{-U_{n,t}}{U_{c,t}} = \frac{W_t}{P_t}$$

Combining the FOCs w.r.t. C_t and B_t we get:

$$U_{c,t} = (1 + i_t) \beta \mathbb{E}_t \left\{ U_{c,t+1} \frac{P_t}{P_{t+1}} \right\}$$

Combining the FOCs w.r.t. C_t and M_t we get:

$$U_{c,t} = U_{m,t} + \beta \mathbb{E}_t \left\{ U_{c,t+1} \frac{P_t}{P_{t+1}} \right\}$$

Combining the previous two expressions we get:

$$\frac{U_{m,t}}{U_{c,t}} = \frac{i_t}{1 + i_t}$$

For our particular utility function we have:

$$\begin{aligned} U_{c,t} &= C_t^{-\sigma} \\ U_{n,t} &= -N_t^\varphi \\ U_{m,t} &= \chi \left(\frac{M_t}{P_t} \right)^{-\nu} \end{aligned}$$

Then, our previous expressions become ($d = \text{demand}$):

$$C_t^\sigma N_t^\varphi = \frac{W_t}{P_t}$$

$$\frac{M_t^d}{P_t} = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{i_t}{1 + i_t} \right)^{-\frac{1}{\nu}}$$

$$1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} (1 + i_t) \right\}$$

Stochastic Discount Factor

The stochastic discount factor for one-period nominal payoffs is:

$$Q_{t,t+1} = \beta \frac{U_{c,t+1}}{U_{c,t}} \frac{P_t}{P_{t+1}} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}}$$

Analogously, for k -period nominal payoffs we have:

$$Q_{t,t+k} = \beta^k \frac{U_{c,t+k}}{U_{c,t}} \frac{P_t}{P_{t+k}} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}}$$

Intuition: Think of buying, in period t , S_t nominal units of an asset that will pay the (possibly random) amount D_{t+k} per unit in period $t+k$. Let q_t be the per-unit price of this asset. Then, you spend $q_t S_t$ in period t in order to get $S_t D_{t+k}$ (possibly random) nominal units in period $t+k$. Then, combining the FOC w.r.t. C_t and S_t we can show that q_t will satisfy $q_t = \mathbb{E}_t \left\{ \beta^k \frac{U_{c,t+k}}{U_{c,t}} \frac{P_t}{P_{t+k}} D_{t+k} \right\}$. That is: $q_t = \mathbb{E}_t \left\{ Q_{t,t+k} D_{t+k} \right\}$.

Optimal Price Setting by Firms

If firm $i \in [0, 1]$ gets to reoptimize in period t , it will choose the price $P_t^*(i)$ that maximizes the current market value of the profits generated while that price remains effective. The probability that the price set at t remains fixed at $t + k$, $P_{t+k}(i) = P_t^*(i)$, is θ^k for $k \in \{0, 1, 2, \dots\}$. Let $Y_{t+k|t}(i)$ denote output in period $t + k$ for a firm that last reset its price in period t . Then, nominal profits in $t + k$ will be: $\Omega_{t+k|t}(i) = P_t^*(i)Y_{t+k|t}(i) - \Psi_{t+k}(Y_{t+k|t}(i))$. Since the firm is owned by the household, and nominal profits are stochastic, the appropriate discount factor is $Q_{t,t+k}$. Hence, the optimal $P_t^*(i)$ solves:

$$\begin{aligned} \max \quad & \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[P_t^*(i) Y_{t+k|t}(i) - \Psi_{t+k}(Y_{t+k|t}(i)) \right] \right\} \\ \text{s.t.} \quad & Y_{t+k|t}(i) = \left(\frac{P_t^*(i)}{P_{t+k}} \right)^{-\epsilon} C_{t+k} \end{aligned}$$

where the processes for P_{t+k} , C_{t+k} , and then $Q_{t,t+k}$, are taken as given.

Notice that the monopolistically competitive producer is constrained by the demand of its product that was derived from utility maximization.

From the structure of the problem above we see that all firms setting their prices at t will choose the same price. Hence, from now on we drop the i index.

Substituting the constraint into the objective we get:

$$\max_{P_t^*} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[\frac{P_t^{*1-\epsilon}}{P_{t+k}^{-\epsilon}} C_{t+k} - \Psi_{t+k} \left(\frac{P_t^{*- \epsilon}}{P_{t+k}^{-\epsilon}} C_{t+k} \right) \right] \right\}$$

FOC

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[(1-\epsilon) \frac{P_t^{*- \epsilon}}{P_{t+k}^{-\epsilon}} C_{t+k} - \Psi'_{t+k|t} (-\epsilon) \frac{P_t^{*- \epsilon - 1}}{P_{t+k}^{-\epsilon}} C_{t+k} \right] \right\} = 0$$

where $\Psi'_{t+k|t}$ denotes the derivative of Ψ_{t+k} evaluated at $Y_{t+k|t}$.

Then:

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[(1-\epsilon) Y_{t+k|t} + \psi_{t+k|t} \epsilon P_t^{*-1} Y_{t+k|t} \right] \right\} = 0$$

where $\psi_{t+k|t} \equiv \Psi'_{t+k|t}$.

Then:

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} (1-\epsilon) P_t^{*-1} \left[P_t^* + \psi_{t+k|t} \frac{\epsilon}{1-\epsilon} \right] \right\} = 0$$

Then:

$$(1-\epsilon) P_t^{*-1} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \left[P_t^* - \frac{\epsilon}{\epsilon-1} \psi_{t+k|t} \right] \right\} = 0$$

Finally:

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \left[P_t^* - \frac{\epsilon}{\epsilon-1} \psi_{t+k|t} \right] \right\} = 0$$

From the expression above we get (recall $\epsilon > 1$):

$$P_t^* = \frac{\epsilon}{\epsilon - 1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \right\}}$$

Multiplying and dividing the expression above by $\psi_{t|t}$ we get:

$$P_t^* = \frac{\epsilon}{\epsilon - 1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}} \psi_{t|t}$$

That is:

$$P_t^* = \mathcal{M}_t \psi_{t|t}$$

with

$$\mathcal{M}_t \equiv \frac{\epsilon}{\epsilon - 1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}}$$

Remark. $\mathcal{M}_t$ is a markup over current marginal cost. Notice that the markup is endogenous, and depends not only on the current values of certain variables but also on the expected value of future variables.

The markup can be rewritten as follows:

$$\mathcal{M}_t = \frac{\epsilon}{\epsilon - 1} \frac{Q_{t,t} Y_{t|t} \psi_{t|t} + \sum_{k=1}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{Q_{t,t} Y_{t|t} \psi_{t|t} + \sum_{k=1}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}}$$

When prices are fully flexible ($\theta = 0$) we get:

$$\mathcal{M}_t^{flex} = \frac{\epsilon}{\epsilon - 1}$$

Then:

$$P_t^{*,flex} = \frac{\epsilon}{\epsilon - 1} \psi_{t|t}$$

which is the standard expression for the optimal price as a markup over marginal cost when prices are flexible and the elasticity of demand is constant.

Price-Level Dynamics

Let ω_k denote the probability that a firm adjusted its price k periods ago. That is, for any period t , ω_k is the probability that a firm adjusted its price for the last time in $t - k$. Then:

$$\omega_k = (1 - \theta) \theta^k$$

Remark: $\sum_{k=0}^{\infty} \omega_k = \sum_{k=0}^{\infty} (1 - \theta) \theta^k = (1 - \theta) \sum_{k=0}^{\infty} \theta^k = (1 - \theta) \frac{1}{1 - \theta} = 1$.

Since there's a continuum of producers uniformly distributed over $[0, 1]$, ω_k gives the fraction of producers that adjusted prices k periods ago. Also, recall that all firms that adjust their prices in a given period set the same price. Then (ordering the firms appropriately), we have:

$$\begin{aligned} P_t &= \left[\int_0^1 P_t(i)^{1-\epsilon} di \right]^{\frac{1}{1-\epsilon}} \\ &= \left[\int_0^{\omega_0} P_t^{*1-\epsilon} di + \int_{\omega_0}^{\omega_0+\omega_1} P_{t-1}^{*1-\epsilon} di + \int_{\omega_0+\omega_1}^{\omega_0+\omega_1+\omega_2} P_{t-2}^{*1-\epsilon} di + \dots \right]^{\frac{1}{1-\epsilon}} \\ &= \left[\omega_0 P_t^{*1-\epsilon} + \omega_1 P_{t-1}^{*1-\epsilon} + \omega_2 P_{t-2}^{*1-\epsilon} + \dots \right]^{\frac{1}{1-\epsilon}} \\ &= \left[\sum_{k=0}^{\infty} \omega_k P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \end{aligned}$$

$$\begin{aligned}
P_t &= \left[\sum_{k=0}^{\infty} (1-\theta) \theta^k P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \sum_{k=1}^{\infty} (1-\theta) \theta^k P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \theta \sum_{k=1}^{\infty} (1-\theta) \theta^{k-1} P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \theta \sum_{k=0}^{\infty} (1-\theta) \theta^k P_{t-1-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \theta P_{t-1}^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}
\end{aligned}$$

Then:

$$P_t = \left[\theta P_{t-1}^{1-\epsilon} + (1-\theta) P_t^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}}$$

Notice that $\theta = 0$ (flex. prices) implies $P_t^{flex} = P_t^{*,flex}$.

For future reference, define:

$$\Pi_t \equiv \frac{P_t}{P_{t-1}} \quad (\text{gross inflation rate})$$

$$\pi_t \equiv \ln \Pi_t = \ln P_t - \ln P_{t-1} \quad (\text{inflation rate})$$

Market Clearing

$$N_t = N_t^d \quad \text{where } N_t^d \equiv \int_0^1 N_t(i) di$$

$$M_t = M_t^d$$

$$B_t = 0$$

Remark: $Y_t(i) = C_t(i) \forall i \in [0, 1]$ has already been imposed in the profit-max problem.

Define:

$$Y_t \equiv \left(\int_0^1 Y_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

Then, $Y_t(i) = C_t(i) \forall i$ implies

$$Y_t = C_t$$

In addition, since $P_t C_t = \int_0^1 P_t(i) C_t(i) di$ we get:

$$P_t Y_t = \int_0^1 P_t(i) Y_t(i) di$$

Notice that the expression above holds even though $Y_t \neq \int_0^1 Y_t(i) di$.

Aggregate Output

From $Y_t(i) = A_t N_t(i)^{1-\alpha}$ we get: $N_t(i) = \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}}$.

Then:

$$\begin{aligned} N_t &= \int_0^1 N_t(i) di \\ &= \int_0^1 \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}} di \\ &= A_t^{\frac{-1}{1-\alpha}} \int_0^1 Y_t(i)^{\frac{1}{1-\alpha}} di \end{aligned}$$

Using $Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} Y_t$ we get:

$$\begin{aligned} N_t &= A_t^{\frac{-1}{1-\alpha}} \int_0^1 \left[\left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} Y_t \right]^{\frac{1}{1-\alpha}} di \\ &= A_t^{\frac{-1}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \end{aligned}$$

Then:

$$Y_t = A_t N_t^{1-\alpha} \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{-(1-\alpha)}$$

Then:

$$Y_t = A_t N_t^{1-\alpha} D_t^{-1}$$

where

$$D_t \equiv \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{(1-\alpha)}$$

captures price dispersion across goods/firms.

Walras Law

Aggregate profits are:

$$\begin{aligned}\Omega_t &\equiv \int_0^1 \Omega_t(i) di = \int_0^1 [P_t(i)Y_t(i) - W_t N_t(i)] di \\ &= \int_0^1 P_t(i)Y_t(i) di - W_t \int_0^1 N_t(i) di \\ &= P_t Y_t - W_t N_t^d\end{aligned}$$

Substituting the expression above and $T_t = M_t - M_{t-1}$ into the household's budget constraint we get:

$$\begin{aligned}P_t C_t + B_t + M_t^d &= W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} + \Omega_t + T_t \\ &= W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} + P_t Y_t - W_t N_t^d \\ &\quad + M_t - M_{t-1} \\ &= W_t N_t + P_t Y_t - W_t N_t^d + M_t \quad \text{imposing } B_{t-1} = 0\end{aligned}$$

Then:

$$P_t(C_t - Y_t) + B_t + (M_t^d - M_t) + W_t(N_t^d - N_t) = 0$$

Nonstochastic Steady State

We look for a *symmetric* nonstochastic steady state with constant money supply and no inflation.

$$M_t = M_{t+1} = \overline{M}$$

$$P_t = P_{t+1} = \overline{P}$$

Symmetry implies:

$$\overline{P}(i) = \overline{P}(j) \quad \forall i, j \in [0, 1]$$

Substituting into the definition of P_t we get:

$$\overline{P} = \overline{P}(i)$$

Then: $\overline{\Pi} = \frac{\overline{P}}{\overline{P}}$ gives:

$$\overline{\Pi} = 1$$

and $\bar{\pi} = \ln \bar{P} - \ln \bar{P}$ yields:

$$\bar{\pi} = 0$$

Substituting into $C_t(i) = \left(\frac{P_t(i)}{P_t(j)} \right)^{-\epsilon} C_t(j)$ we get:

$$\bar{C}(i) = \bar{C}(j)$$

Substituting into the definition of C_t we get:

$$\bar{C} = \bar{C}(i)$$

Since $Y_t(i) = C_t(i)$ we get:

$$\bar{Y}(i) = \bar{C}(i)$$

$$\bar{Y}(i) = \bar{Y}(j)$$

$$\bar{Y} = \bar{Y}(i)$$

$$\bar{Y} = \bar{C}$$

From $Y_t(i) = A_t N_t(i)^{1-\alpha}$ and $\bar{Y} = \bar{Y}(i)$ we get:

$$\bar{N}(i) = \bar{N}(j)$$

and then $N_t = \int_0^1 N_t(i) di$ gives

$$\bar{N} = \bar{N}(i)$$

From $\ln A_{t+1} = (1 - \rho_a) \ln A + \rho_a \ln A_t + \varepsilon_{t+1}^A$ we get:

$$\bar{A} = A$$

From $D_t \equiv \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{(1-\alpha)}$ we get:

$$\bar{D} = 1$$

Then, $Y_t = A_t N_t^{1-\alpha} D_t^{-1}$ gives:

$$\bar{Y} = A \bar{N}^{1-\alpha}$$

From $\bar{P} = \left[\theta \bar{P}^{1-\epsilon} + (1-\theta) \bar{P}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}}$ we get:

$$\bar{P}^* = \bar{P}$$

From $1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} (1 + i_t) \right\}$ we get:

$$1 + \bar{i} = \frac{1}{\beta}$$

For future reference, define:

$$\bar{i}_t \equiv \ln(1 + i_t)$$

Then:

$$\begin{aligned} \bar{i} &= \ln(1 + \bar{i}) \\ &= \ln\left(\frac{1}{\beta}\right) = -\ln \beta \equiv \rho \end{aligned}$$

From $\frac{M_t}{P_t} = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{i_t}{1+i_t} \right)^{-\frac{1}{\nu}}$ we get:

$$\frac{\bar{M}}{\bar{P}} = \chi^{\frac{1}{\nu}} \bar{C}^{\frac{\sigma}{\nu}} (1 - \beta)^{-\frac{1}{\nu}}$$

From $C_t^\sigma N_t^\varphi = \frac{W_t}{P_t}$ we get:

$$\frac{\overline{W}}{\overline{P}} = \overline{C}^\sigma \overline{N}^\varphi$$

The expression above and $\overline{C} = \overline{Y} = A\overline{N}^{1-\alpha}$ give:

$$\frac{\overline{W}}{\overline{P}} = A^\sigma \overline{N}^{(1-\alpha)\sigma + \varphi}$$

From $\psi_t = \frac{1}{1-\alpha} W_t A_t^{-\frac{1}{1-\alpha}} Y_t(i)^{\frac{\alpha}{1-\alpha}}$ and $Y_t(i) = A_t N_t(i)^{1-\alpha}$ we get $\psi_t = \frac{1}{1-\alpha} W_t A_t^{-1} N_t(i)^\alpha$. Then:

$$\overline{\psi} = \frac{1}{1-\alpha} \frac{\overline{W} \overline{N}^\alpha}{A}$$

From $\mathcal{M}_t \equiv \frac{\epsilon}{\epsilon-1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}}$ we get:

$$\overline{\mathcal{M}} = \frac{\epsilon}{\epsilon-1}$$

From $P_t^* = \mathcal{M}_t \psi_{t|t}$, $\bar{P}^* = \bar{P}$, and $\bar{\mathcal{M}} = \frac{\epsilon}{\epsilon-1}$ we get:

$$\bar{P} = \frac{\epsilon}{\epsilon-1} \bar{\psi}$$

Using $\bar{\psi} = \frac{1}{1-\alpha} \frac{\bar{W} \bar{N}^\alpha}{A}$ into the previous expression gives:

$$\frac{\bar{W}}{\bar{P}} = \frac{\epsilon-1}{\epsilon} (1-\alpha) A \bar{N}^{-\alpha}$$

Combining the expression above with $\frac{\bar{W}}{\bar{P}} = A^\sigma \bar{N}^{(1-\alpha)\sigma+\varphi}$ we get: $A^\sigma \bar{N}^{(1-\alpha)\sigma+\varphi} = \frac{\epsilon-1}{\epsilon} (1-\alpha) A \bar{N}^{-\alpha}$. Then:

$$\bar{N} = \left[\frac{\epsilon-1}{\epsilon} (1-\alpha) A^{1-\sigma} \right]^{\frac{1}{\alpha+\sigma(1-\alpha)+\varphi}}$$

Having found $\bar{N}$ we can use our previous expressions to find the values of the other variables in the nonstochastic steady state.

Notice that $\overline{N}$ is independent of $\overline{M}$.

Therefore, all real variables are independent of $\overline{M}$ (e.g., $\overline{C} = \overline{Y} = A\overline{N}^{1-\alpha}$).

From $\frac{\overline{M}}{\overline{P}} = \chi^{\frac{1}{\nu}} \overline{C}^{\frac{\sigma}{\nu}} (1 - \beta)^{-\frac{1}{\nu}}$ we get:

$$\overline{P} = \frac{1}{\chi^{\frac{1}{\nu}} \overline{C}^{\frac{\sigma}{\nu}} (1 - \beta)^{-\frac{1}{\nu}}} \overline{M}$$

Since $\overline{C}$ is independent of $\overline{M}$, the expression above shows that $\overline{P}$ is proportional to $\overline{M}$.

Hence, money is neutral in the long run.

Log-linearization

For any variable X define the period- t log-deviation from the nonstochastic steady state as follows:

$$\widetilde{X}_t \equiv \ln X_t - \ln \bar{X}$$

Then:

$$\begin{aligned} X_t &= \bar{X} e^{\widetilde{X}_t} \\ &\cong \bar{X} (1 + \widetilde{X}_t) \end{aligned}$$

For $i_t \equiv \ln(1 + i_t)$ define:

$$\widetilde{i}_t \equiv i_t - \bar{i}$$

Notice that: $\widetilde{i}_t \equiv i_t - \bar{i} = \ln(1 + i_t) - \ln(1 + \bar{i}) \cong i_t - \bar{i}$.

Then:

$$\widetilde{i}_t \cong i_t - \bar{i}$$

From $\ln A_{t+1} = (1 - \rho_a) \ln A + \rho_a \ln A_t + \varepsilon_{t+1}^A$ we get:

$$\widetilde{A}_{t+1} = \rho_a \widetilde{A}_t + \varepsilon_{t+1}^A$$

From $\Pi_t \equiv \frac{P_t}{P_{t-1}}$ we get: $\ln \Pi_t = \ln P_t - \ln P_{t-1}$ and $\ln \bar{\Pi} = \ln \bar{P} - \ln \bar{P}$. Then: $\tilde{\Pi}_t \equiv \ln \Pi_t - \ln \bar{\Pi} = (\ln P_t - \ln \bar{P}) - (\ln P_{t-1} - \ln \bar{P})$. Then:

$$\tilde{\Pi}_t = \tilde{P}_t - \tilde{P}_{t-1}$$

Notice that $\tilde{P}_t - \tilde{P}_{t-1} = (\ln P_t - \ln \bar{P}) - (\ln P_{t-1} - \ln \bar{P}) = \ln P_t - \ln P_{t-1} = \ln \left(\frac{P_t}{P_{t-1}} \right) = \ln \Pi_t \equiv \pi_t$. Then:

$$\begin{aligned} \tilde{\Pi}_t &= \ln \Pi_t \\ &= \pi_t \end{aligned}$$

Hence:

$$\pi_t = \tilde{P}_t - \tilde{P}_{t-1}$$

From $C_t = Y_t$ we get:

$$\tilde{C}_t = \tilde{Y}_t$$

Recall $Y_t = A_t N_t^{1-\alpha} D_t^{-1}$ with $D_t \equiv \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{(1-\alpha)}$.

Then: $\ln Y_t = \ln A_t + (1-\alpha) \ln N_t - \ln D_t$ and $\ln \bar{Y} = \ln \bar{A} + (1-\alpha) \ln \bar{N} - \ln \bar{D}$. Subtracting the second from the first we get:

$$\tilde{Y}_t = \tilde{A}_t + (1-\alpha)\tilde{N}_t - \tilde{D}_t$$

where

$$\begin{aligned} \tilde{D}_t &\equiv \ln D_t - \ln \bar{D} \\ &= \ln D_t \quad \text{since } \bar{D} = 1 \\ &= (1-\alpha) \ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \\ &\equiv d_t \end{aligned}$$

A first-order Taylor approximation of d_t around the symmetric nss with zero inflation gives (see the Appendix):

$$\tilde{D}_t = d_t \cong 0$$

Then, the expression for aggregate output becomes:

$$\tilde{Y}_t = \tilde{A}_t + (1-\alpha)\tilde{N}_t$$

From $P_t = [\theta P_{t-1}^{1-\epsilon} + (1-\theta) P_t^*]^{1-\epsilon}$ and $\bar{P} = \bar{P}^*$ we get:

$$P_t^{1-\epsilon} = \theta P_{t-1}^{1-\epsilon} + (1-\theta) P_t^{*1-\epsilon}$$

$$\left(\bar{P} e^{\tilde{P}_t}\right)^{1-\epsilon} = \theta \left(\bar{P} e^{\tilde{P}_{t-1}}\right)^{1-\epsilon} + (1-\theta) \left(\bar{P}^* e^{\tilde{P}_t^*}\right)^{1-\epsilon}$$

$$\bar{P}^{1-\epsilon} e^{(1-\epsilon)\tilde{P}_t} = \theta \bar{P}^{1-\epsilon} e^{(1-\epsilon)\tilde{P}_{t-1}} + (1-\theta) \bar{P}^{1-\epsilon} e^{(1-\epsilon)\tilde{P}_t^*} \quad \text{using } \bar{P}^* = \bar{P}$$

$$e^{(1-\epsilon)\tilde{P}_t} = \theta e^{(1-\epsilon)\tilde{P}_{t-1}} + (1-\theta) e^{(1-\epsilon)\tilde{P}_t^*}$$

$$1 + (1-\epsilon)\tilde{P}_t = \theta [1 + (1-\epsilon)\tilde{P}_{t-1}] + (1-\theta) [1 + (1-\epsilon)\tilde{P}_t^*]$$

$$(1-\epsilon)\tilde{P}_t = \theta (1-\epsilon)\tilde{P}_{t-1} + (1-\theta)(1-\epsilon)\tilde{P}_t^*$$

Then:

$$\tilde{P}_t = \theta \tilde{P}_{t-1} + (1-\theta)\tilde{P}_t^*$$

From $C_t^\sigma N_t^\varphi = \frac{W_t}{P_t}$ we get $\sigma \ln C_t + \varphi \ln N_t = \ln W_t - \ln P_t$ and $\sigma \ln \bar{C} + \varphi \ln \bar{N} = \ln \bar{W} - \ln \bar{P}$. Subtracting the second from the first:

$$\sigma \tilde{C}_t + \varphi \tilde{N}_t = \tilde{W}_t - \tilde{P}_t$$

From $1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} (1 + i_t) \right\}$, $i_t \equiv \ln(1 + i_t)$ and $\Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}$ we get:

$$1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{e^{i_t}}{\Pi_{t+1}} \right\}$$

The term inside curly brackets gives: $\ln \beta - \sigma \ln C_{t+1} + \sigma \ln C_t - \ln \Pi_{t+1} + i_t$, and $\ln \beta - \sigma \ln \bar{C} + \sigma \ln \bar{C} - \ln \bar{\Pi} + \bar{i}$. Subtracting the second from the first: $-\sigma \tilde{C}_{t+1} + \sigma \tilde{C}_t - \tilde{\Pi}_{t+1} + \tilde{i}_t$. Then, using $\tilde{\Pi}_{t+1} = \pi_{t+1}$, we get:

$$0 = \mathbb{E}_t \left\{ -\sigma \tilde{C}_{t+1} + \sigma \tilde{C}_t + \tilde{i}_t - \pi_{t+1} \right\}$$

From $\frac{M_t}{P_t} = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{i_t}{1+i_t} \right)^{-\frac{1}{\nu}}$, $M_t^r \equiv \frac{M_t}{P_t}$, and $i_t \equiv \ln(1 + i_t)$, we get: $M_t^r = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{e^{i_t} - 1}{e^{i_t}} \right)^{-\frac{1}{\nu}}$. Then:

$$M_t^r = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(1 - e^{-i_t} \right)^{-\frac{1}{\nu}}$$

From the expression above we get: $\ln M_t^r = \frac{1}{\nu} \ln \chi + \frac{\sigma}{\nu} \ln C_t - \frac{1}{\nu} \ln \left(1 - e^{-i_t} \right)$ and $\ln \bar{M}^r = \frac{1}{\nu} \ln \chi + \frac{\sigma}{\nu} \ln \bar{C} - \frac{1}{\nu} \ln \left(1 - e^{-\bar{i}} \right)$. Subtracting the second from the first: $\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \frac{1}{\nu} \left[\ln \left(1 - e^{-i_t} \right) - \ln \left(1 - e^{-\bar{i}} \right) \right]$. A first-order Taylor approximation of $\ln \left(1 - e^{-i_t} \right)$ around $i_t = \bar{i}$ gives: $\ln \left(1 - e^{-i_t} \right) \cong \ln \left(1 - e^{-\bar{i}} \right) + \frac{e^{-\bar{i}}}{1 - e^{-\bar{i}}} \left(i_t - \bar{i} \right)$. Then: $\ln \left(1 - e^{-i_t} \right) - \ln \left(1 - e^{-\bar{i}} \right) \cong \frac{e^{\ln \beta}}{1 - e^{\ln \beta}} \widetilde{i}_t = \frac{\beta}{1 - \beta} \widetilde{i}_t$. Then: $\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \frac{\beta}{\nu(1 - \beta)} \widetilde{i}_t$. Then:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \eta \widetilde{i}_t \quad \text{with} \quad \eta \equiv \frac{\beta}{\nu(1 - \beta)}$$

$$\text{From } P_t^* = \frac{\epsilon}{\epsilon-1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \right\}} \text{ and}$$

$$Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} \text{ we get:}$$

$$\begin{aligned} P_t^* \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \right\} \\ = \frac{\epsilon}{\epsilon-1} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \psi_{t+k|t} \right\} \end{aligned}$$

Log-linearizing the left-hand side of the expression above we get:

$$\begin{aligned} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k P_t^* \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \right\} \\ = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \bar{P}^* e^{\tilde{P}_t^*} \left(\frac{\bar{C}_e \tilde{C}_{t+k}}{\bar{C}_e \tilde{C}_t} \right)^{-\sigma} \frac{\bar{P}_e \tilde{P}_t}{\bar{P}_e \tilde{P}_{t+k}} \bar{Y}_e \tilde{Y}_{t+k|t} \right\} \end{aligned}$$

$$\begin{aligned}
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{\tilde{P}_t^*} \left(e^{\tilde{C}_{t+k} - \tilde{C}_t} \right)^{-\sigma} e^{\tilde{P}_t - \tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{\tilde{P}_t^*} e^{-\sigma(\tilde{C}_{t+k} - \tilde{C}_t)} e^{\tilde{P}_t - \tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{\tilde{P}_t^* - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t}} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 + \tilde{P}_t^* - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} \right\}
\end{aligned}$$

Log-linearizing the right-hand side of the expression above we get:

$$\begin{aligned}
&\frac{\epsilon}{\epsilon-1} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \psi_{t+k|t} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \left(\frac{\bar{C}_e \tilde{C}_{t+k}}{\bar{C}_e \tilde{C}_t} \right)^{-\sigma} \frac{\bar{P}_e \tilde{P}_t}{\bar{P}_e \tilde{P}_{t+k}} \bar{Y}_e \tilde{Y}_{t+k|t} \bar{\psi}_e \tilde{\psi}_{t+k|t} \right\}
\end{aligned}$$

$$\begin{aligned}
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \left(e^{\tilde{C}_{t+k}-\tilde{C}_t} \right)^{-\sigma} e^{\tilde{P}_t-\tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} e^{\tilde{\psi}_{t+k|t}} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{-\sigma(\tilde{C}_{t+k}-\tilde{C}_t)} e^{\tilde{P}_t-\tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} e^{\tilde{\psi}_{t+k|t}} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{-\sigma\tilde{C}_{t+k}+\sigma\tilde{C}_t+\tilde{P}_t-\tilde{P}_{t+k}+\tilde{Y}_{t+k|t}+\tilde{\psi}_{t+k|t}} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \right\}
\end{aligned}$$

Equating both sides, and using $\overline{P} = \frac{\epsilon}{\epsilon-1} \overline{\psi}$:

$$\begin{aligned}
&\overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 + \tilde{P}_t^* - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \right\}
\end{aligned}$$

$$\begin{aligned}
& \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 + \tilde{P}_t^* - \sigma \tilde{C}_{t+k} + \sigma \tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} \right\} \\
&= \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 - \sigma \tilde{C}_{t+k} + \sigma \tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \right\} \\
&= \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} 1 - \sigma \tilde{C}_{t+k} + \sigma \tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \\ -1 - \tilde{P}_t^* + \sigma \tilde{C}_{t+k} - \sigma \tilde{C}_t - \tilde{P}_t + \tilde{P}_{t+k} - \tilde{Y}_{t+k|t} \end{array} \right\} \\
&= \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} - \tilde{P}_t^* \right\} \\
& \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{P}_t^* \right\} = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\} \\
& \tilde{P}_t^* \sum_{k=0}^{\infty} (\theta\beta)^k = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\} \\
& \tilde{P}_t^* \frac{1}{1-\theta\beta} = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\}
\end{aligned}$$

Then:

$$\tilde{P}_t^* = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\}$$

Define the individual marginal cost in real terms:

$$MC_{t+k|t} \equiv \frac{\psi_{t+k|t}}{P_{t+k}}$$

Then:

$$\widetilde{MC}_{t+k|t} = \tilde{\psi}_{t+k|t} - \tilde{P}_{t+k}$$

Then, we can write:

$$\tilde{P}_t^* = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k|t} + \tilde{P}_{t+k} \right\}$$

Subtracting $\tilde{P}_{t-1}$ from both sides:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k|t} + \tilde{P}_{t+k} \right\} - \tilde{P}_{t-1}$$

Using $\tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t\{\tilde{P}_{t-1}\}$ we get:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t\left\{\widetilde{MC}_{t+k|t} + \tilde{P}_{t+k} - \tilde{P}_{t-1}\right\}$$

Now we are going to eliminate the individual real marginal cost, $\widetilde{MC}_{t+k|t}$, from the expression above and replace it by the economy's average real marginal cost $\widetilde{MC}_{t+k}$, to be defined below.

From $\psi_{t+k|t} = \frac{1}{1-\alpha} W_{t+k} A_{t+k}^{-\frac{1}{1-\alpha}} Y_{t+k|t}^{\frac{\alpha}{1-\alpha}}$ we get:

$$\tilde{\psi}_{t+k|t} = \tilde{W}_{t+k} - \frac{1}{1-\alpha} \tilde{A}_{t+k} + \frac{\alpha}{1-\alpha} \tilde{Y}_{t+k|t}$$

Then, $\widetilde{MC}_{t+k|t} = \tilde{\psi}_{t+k|t} - \tilde{P}_{t+k}$ gives:

$$\widetilde{MC}_{t+k|t} = \tilde{W}_{t+k} - \frac{1}{1-\alpha} \tilde{A}_{t+k} + \frac{\alpha}{1-\alpha} \tilde{Y}_{t+k|t} - \tilde{P}_{t+k}$$

From $Y_t = A_t N_t^{1-\alpha} D_t^{-1}$ we get: $N_t = A_t^{-\frac{1}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}} D_t^{\frac{1}{1-\alpha}}$.

Then: $W_t N_t = W_t A_t^{-\frac{1}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}} D_t^{\frac{1}{1-\alpha}}$. Then, mar-

ginal cost is: $\psi_t = \frac{1}{1-\alpha} W_t A_t^{-\frac{1}{1-\alpha}} Y_t^{\frac{\alpha}{1-\alpha}} D_t^{\frac{1}{1-\alpha}}$. Then:

$\tilde{\psi}_t = \tilde{W}_t - \frac{1}{1-\alpha} \tilde{A}_t + \frac{\alpha}{1-\alpha} \tilde{Y}_t + \frac{1}{1-\alpha} \tilde{D}_t$. Since $\tilde{D}_t \cong 0$, we get:

$$\tilde{\psi}_t = \tilde{W}_t - \frac{1}{1-\alpha} \tilde{A}_t + \frac{\alpha}{1-\alpha} \tilde{Y}_t$$

Defining $MC_t \equiv \frac{\psi_t}{P_t}$ we get, for period t :

$$\widetilde{MC}_t = \tilde{\psi}_t - \tilde{P}_t$$

Combining the previous two expressions we get, for period $t+k$:

$$\widetilde{MC}_{t+k} = \tilde{W}_{t+k} - \frac{1}{1-\alpha} \tilde{A}_{t+k} + \frac{\alpha}{1-\alpha} \tilde{Y}_{t+k} - \tilde{P}_{t+k}$$

Comparing the expressions for $\widetilde{MC}_{t+k|t}$ and $\widetilde{MC}_{t+k}$ we get:

$$\widetilde{MC}_{t+k|t} = \widetilde{MC}_{t+k} + \frac{\alpha}{1-\alpha} (\tilde{Y}_{t+k|t} - \tilde{Y}_{t+k})$$

But $Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}}\right)^{-\epsilon} Y_{t+k}$ gives:

$$\tilde{Y}_{t+k|t} = -\epsilon \left(\tilde{P}_t^* - \tilde{P}_{t+k} \right) + \tilde{Y}_{t+k}$$

Combining the previous two expressions we get:

$$\widetilde{MC}_{t+k|t} = \widetilde{MC}_{t+k} - \frac{\alpha\epsilon}{1-\alpha} \left(\tilde{P}_t^* - \tilde{P}_{t+k} \right)$$

Going back to the expression for $\tilde{P}_t^* - \tilde{P}_{t-1}$:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} - \frac{\alpha\epsilon}{1-\alpha} \left(\tilde{P}_t^* - \tilde{P}_{t+k} \right) \\ + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} \\ - \frac{\alpha\epsilon}{1-\alpha} \left(\tilde{P}_t^* - \tilde{P}_{t+k} + \tilde{P}_{t-1} - \tilde{P}_{t-1} \right) \\ + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} \\ -\frac{\alpha\epsilon}{1-\alpha} \left[(\tilde{P}_t^* - \tilde{P}_{t-1}) - (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right] \\ + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} - \frac{\alpha\epsilon}{1-\alpha} (\tilde{P}_t^* - \tilde{P}_{t-1}) \\ + \frac{\alpha\epsilon}{1-\alpha} (\tilde{P}_{t+k} - \tilde{P}_{t-1}) + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \frac{-\alpha\epsilon}{1-\alpha} (\tilde{P}_t^* - \tilde{P}_{t-1}) \sum_{k=0}^{\infty} (\theta\beta)^k$$

$$+ (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \left(1 + \frac{\alpha\epsilon}{1-\alpha} \right) (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \frac{-\alpha\epsilon}{1-\alpha} (\tilde{P}_t^* - \tilde{P}_{t-1}) \frac{1}{1 - \theta\beta}$$

$$+ (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1 - \alpha + \alpha\epsilon}{1-\alpha} (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = \frac{-\alpha\epsilon}{1-\alpha}(\tilde{P}_t^* - \tilde{P}_{t-1})$$

$$+ (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\left(\tilde{P}_t^* - \tilde{P}_{t-1} \right) \left(1 + \frac{\alpha\epsilon}{1-\alpha} \right) =$$

$$= (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\left(\tilde{P}_t^* - \tilde{P}_{t-1} \right) \frac{1-\alpha+\alpha\epsilon}{1-\alpha} =$$

$$= (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = \frac{1-\alpha}{1-\alpha+\alpha\epsilon} (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \frac{1-\alpha}{1-\alpha+\alpha\epsilon} \widetilde{MC}_{t+k} + (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

Then:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \Theta \widetilde{MC}_{t+k} + (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

where

$$\Theta \equiv \frac{1 - \alpha}{1 - \alpha + \alpha\epsilon}$$

Then:

$$\begin{aligned} \tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta) \Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\} \end{aligned}$$

But (see the Appendix):

$$\sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t-1}) = \frac{1}{1 - \theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1})$$

Then:

$$\begin{aligned}\tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + (1 - \theta\beta) \mathbb{E}_t \left\{ \frac{1}{1 - \theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) \right\}\end{aligned}$$

$$\begin{aligned}\tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) \right\}\end{aligned}$$

$$\begin{aligned}\tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ (\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) \right\}\end{aligned}$$

Using $\tilde{P}_{t+k} - \tilde{P}_{t+k-1} = \pi_{t+k}$ we get:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \pi_{t+k} \right\}$$

From the expression above we get:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ (1 - \theta\beta)\Theta \sum_{k=1}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} + \sum_{k=1}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \pi_{t+k} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ (1 - \theta\beta)\Theta \theta\beta \sum_{k=1}^{\infty} (\theta\beta)^{k-1} \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} + \theta\beta \sum_{k=1}^{\infty} (\theta\beta)^{k-1} \mathbb{E}_t \left\{ \pi_{t+k} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ \theta\beta \left[(1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+1+k} \right\} + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \pi_{t+1+k} \right\} \right]$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ \theta\beta \mathbb{E}_t \left\{ (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \widetilde{MC}_{t+1+k} + \sum_{k=0}^{\infty} (\theta\beta)^k \pi_{t+1+k} \right\}$$

Using the Law of Iterated Expectations ($\mathbb{E}_t \mathbb{E}_{t+1} X_{t+k} = \mathbb{E}_t X_{t+k}$) we get:

$$\begin{aligned} \tilde{P}_t^* - \tilde{P}_{t-1} = & (1 - \theta\beta) \Theta \widetilde{MC}_t + \pi_t + \\ & + \theta\beta \mathbb{E}_t \left\{ \begin{aligned} & (1 - \theta\beta) \Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_{t+1} \{ \widetilde{MC}_{t+1+k} \} \\ & + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_{t+1} \{ \pi_{t+1+k} \} \end{aligned} \right\} \end{aligned}$$

Then:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \Theta \widetilde{MC}_t + \pi_t + \theta\beta \mathbb{E}_t \{ \tilde{P}_{t+1}^* - \tilde{P}_t \}$$

Now we want to eliminate $\tilde{P}_t^*$ and $\tilde{P}_{t+1}^*$ from the last expression. From $\tilde{P}_t = \theta \tilde{P}_{t-1} + (1 - \theta) \tilde{P}_t^*$ we get:

$$\tilde{P}_t - \tilde{P}_{t-1} = \theta \tilde{P}_{t-1} + (1 - \theta) \tilde{P}_t^* - \tilde{P}_{t-1}$$

$$\tilde{P}_t - \tilde{P}_{t-1} = (1 - \theta) \tilde{P}_t^* - (1 - \theta) \tilde{P}_{t-1}$$

$$\tilde{P}_t - \tilde{P}_{t-1} = (1 - \theta)(\tilde{P}_t^* - \tilde{P}_{t-1})$$

Then:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = \frac{1}{1 - \theta}(\tilde{P}_t - \tilde{P}_{t-1})$$

Substituting into the previous expression we get:

$$\frac{1}{1 - \theta}(\tilde{P}_t - \tilde{P}_{t-1}) = (1 - \theta\beta)\Theta\widetilde{MC}_t + \pi_t + \theta\beta\mathbb{E}_t\left\{\frac{1}{1 - \theta}(\tilde{P}_{t+1} - \tilde{P}_t)\right\}$$

$$(\tilde{P}_t - \tilde{P}_{t-1}) = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + (1 - \theta)\pi_t + \theta\beta\mathbb{E}_t\left\{\tilde{P}_{t+1} - \tilde{P}_t\right\}$$

$$\pi_t = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + (1 - \theta)\pi_t + \theta\beta\mathbb{E}_t\{\pi_{t+1}\}$$

$$[1 - (1 - \theta)]\pi_t = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + \theta\beta\mathbb{E}_t\{\pi_{t+1}\}$$

Then:

$$\theta\pi_t = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + \theta\beta\mathbb{E}_t\{\pi_{t+1}\}$$

Notice that $\theta = 0 \Rightarrow \widetilde{MC}_t^{flex} = 0$.

Dividing by $\theta > 0$ we get:

$$\pi_t = \lambda \widetilde{MC}_t + \beta \mathbb{E}_t \{ \pi_{t+1} \}$$

where

$$\lambda \equiv \frac{(1-\theta)(1-\theta\beta)\Theta}{\theta}$$

Remark. Iterating forward on the expression above, and imposing $\lim_{n \rightarrow \infty} \beta^n \mathbb{E}_t \pi_{t+n} = 0$, we get:

$$\pi_t = \lambda \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t \{ \widetilde{MC}_{t+k} \}$$

The next step is to show that $\widetilde{MC}_t$ is proportional to a measure of the *output gap* to be defined below.

Remark. This step is not really necessary to solve the model, but it is standard practice to write the model in this way. You should keep in mind, however, that not every model can be written in this form.

From $\sigma\tilde{C}_t + \varphi\tilde{N}_t = \tilde{W}_t - \tilde{P}_t$ and $\tilde{C}_t = \tilde{Y}_t$ we get:
 $\tilde{W}_t - \tilde{P}_t = \sigma\tilde{Y}_t + \varphi\tilde{N}_t$

From $\tilde{Y}_t = \tilde{A}_t + (1 - \alpha)\tilde{N}_t$ we get: $\tilde{N}_t = \frac{\tilde{Y}_t - \tilde{A}_t}{1 - \alpha}$

Then: $\tilde{W}_t - \tilde{P}_t = \sigma\tilde{Y}_t + \varphi\frac{\tilde{Y}_t - \tilde{A}_t}{1 - \alpha}$

Then:

$$\tilde{W}_t - \tilde{P}_t = \left(\sigma + \frac{\varphi}{1 - \alpha}\right)\tilde{Y}_t - \frac{\varphi}{1 - \alpha}\tilde{A}_t$$

Recall that: $\widetilde{MC}_t = \tilde{W}_t - \tilde{P}_t - \frac{1}{1 - \alpha}\tilde{A}_t + \frac{\alpha}{1 - \alpha}\tilde{Y}_t$.

Then, using the expression above we get:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi}{1 - \alpha}\right)\tilde{Y}_t - \frac{\varphi}{1 - \alpha}\tilde{A}_t - \frac{1}{1 - \alpha}\tilde{A}_t + \frac{\alpha}{1 - \alpha}\tilde{Y}_t.$$

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi}{1 - \alpha} + \frac{\alpha}{1 - \alpha}\right)\tilde{Y}_t - \left(\frac{\varphi}{1 - \alpha} + \frac{1}{1 - \alpha}\right)\tilde{A}_t$$

Then:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha}\right)\tilde{Y}_t - \frac{1 + \varphi}{1 - \alpha}\tilde{A}_t$$

Let Y_t^n denote the *natural level of output*, that is, the output level that would prevail under flexible prices ($\theta = 0$). The expression above gives:

$$\widetilde{MC}_t^n = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n - \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t$$

We know from our study of the price-setting problem that, under full flexibility, the optimal price satisfies: $P_t^{*,flex} = \frac{\epsilon}{\epsilon-1} \psi_{t|t}$, with $P_t^{*,flex} = P_t^{flex}$ and $\psi_{t|t}^{flex} = \psi_t^{flex}$.

Then: $P_t^{flex} = \frac{\epsilon}{\epsilon-1} \psi_t^{flex}$. Hence: $\frac{\psi_t^{flex}}{P_t^{flex}} = \frac{\epsilon-1}{\epsilon}$. Since

$MC_t^n \equiv \frac{\psi_t^{flex}}{P_t^{flex}}$, we get:

$$MC_t^n = \frac{\epsilon - 1}{\epsilon}$$

Then:

$$\widetilde{MC}_t^n = 0$$

Then, our previous expression becomes:

$$0 = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n - \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t.$$

Then:

$$\left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n = \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t$$

$$\frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 - \alpha} \tilde{Y}_t^n = \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t$$

$$\tilde{Y}_t^n = \frac{1 + \varphi}{\sigma(1 - \alpha) + \varphi + \alpha} \tilde{A}_t$$

Then:

$$\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$$

where

$$\psi_y^n = \frac{1 + \varphi}{\sigma(1 - \alpha) + \varphi + \alpha}$$

The expression above gives: $\tilde{A}_t = \frac{\sigma(1-\alpha)+\varphi+\alpha}{1+\varphi} \tilde{Y}_t^n$.

Substituting into our last expression for $\widetilde{MC}_t$ we get:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t - \frac{1 + \varphi}{1 - \alpha} \frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 + \varphi} \tilde{Y}_t^n$$

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t - \frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 - \alpha} \tilde{Y}_t^n$$

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t - \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n$$

Then:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) (\tilde{Y}_t - \tilde{Y}_t^n)$$

Define the **output gap** as the log-deviation of output from its natural level:

$$\tilde{y}_t \equiv \ln Y_t - \ln Y_t^n$$

That is, the output gap measures the percent deviation of current output from its hypothetical flexible-price level.

From our analysis of the nss we know $\bar{Y} = A\bar{N}^{1-\alpha}$ with $\bar{N} = \left[\frac{\epsilon-1}{\epsilon}(1-\alpha)A^{1-\sigma} \right]^{\frac{1}{\alpha+\sigma(1-\alpha)+\varphi}}$. Notice that $\bar{Y}$ is independent of θ . Hence, $\bar{Y} = \bar{Y}^n$. This implies:

$$\tilde{y}_t = \ln Y_t - \ln Y_t^n$$

$$\tilde{y}_t = \ln Y_t - \ln Y_t^n + \ln \bar{Y} - \ln \bar{Y}$$

$$\tilde{y}_t = \ln Y_t - \ln Y_t^n + \ln \bar{Y}^n - \ln \bar{Y} \quad \text{using } \bar{Y} = \bar{Y}^n$$

$$\tilde{y}_t = (\ln Y_t - \ln \bar{Y}) - (\ln Y_t^n - \ln \bar{Y}^n)$$

Then:

$$\tilde{y}_t = \tilde{Y}_t - \tilde{Y}_t^n$$

Substituting the latter into $\widetilde{MC}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right) (\tilde{Y}_t - \tilde{Y}_t^n)$ we get:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right) \tilde{y}_t$$

Substituting the expression above into $\pi_t = \lambda \widetilde{MC}_t + \beta \mathbb{E}_t \{\pi_{t+1}\}$ we get :

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \{\pi_{t+1}\}$$

where

$$\begin{aligned} \kappa &\equiv \lambda \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \\ &= \frac{(1 - \theta)(1 - \theta\beta)\Theta}{\theta} \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \\ &= \frac{(1 - \theta)(1 - \theta\beta)(1 - \alpha)}{\theta(1 - \alpha + \alpha\epsilon)} \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \\ &= \frac{(1 - \theta)(1 - \theta\beta)}{\theta} \frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 - \alpha + \alpha\epsilon} \end{aligned}$$

The expression above is often referred to as the **New Keynesian Phillips Curve (NKPC)**.

Remark. Iterating forward on the expression above, and imposing $\lim_{n \rightarrow \infty} \beta^n \mathbb{E}_t \pi_{t+n} = 0$, we get:

$$\pi_t = \kappa \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t \{\tilde{y}_{t+k}\}$$

Now we'll derive a dynamic version of the IS curve.

Combining the Euler equation $0 = \mathbb{E}_t \left\{ -\sigma \tilde{C}_{t+1} + \sigma \tilde{C}_t + \tilde{i}_t - \pi_{t+1} \right\}$

with $\tilde{C}_t = \tilde{Y}_t$ we get:

$$0 = \mathbb{E}_t \left\{ -\sigma \tilde{Y}_{t+1} + \sigma \tilde{Y}_t + \tilde{i}_t - \pi_{t+1} \right\}$$

$$\sigma \tilde{Y}_t = \sigma \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \tilde{i}_t + \mathbb{E}_t \left\{ \pi_{t+1} \right\}$$

Then:

$$\tilde{Y}_t = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \frac{1}{\sigma} \left(\tilde{i}_t - \mathbb{E}_t \left\{ \pi_{t+1} \right\} \right)$$

Since $\tilde{i}_t = i_t - \bar{i}$ we can rewrite the expression above as follows:

$$\tilde{Y}_t = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \frac{1}{\sigma} \left[(i_t - \mathbb{E}_t \left\{ \pi_{t+1} \right\}) - \bar{i} \right]$$

Define the *expected real interest rate* as follows:

$$r_t \equiv i_t - \mathbb{E}_t \{ \pi_{t+1} \}$$

Since there's no inflation in our nss, we have:

$$\begin{aligned} \bar{r} &= \bar{i} \\ &= \rho \end{aligned}$$

Then, we can write:

$$\begin{aligned} \tilde{Y}_t &= \mathbb{E}_t \{ \tilde{Y}_{t+1} \} - \frac{1}{\sigma} (r_t - \bar{r}) \\ &= \mathbb{E}_t \{ \tilde{Y}_{t+1} \} - \frac{1}{\sigma} \tilde{r}_t \end{aligned}$$

where

$$\begin{aligned} \tilde{r}_t &\equiv r_t - \bar{r} \\ &= i_t - \mathbb{E}_t \{ \pi_{t+1} \} - \bar{i} \\ &= \tilde{i}_t - \mathbb{E}_t \{ \pi_{t+1} \} \end{aligned}$$

When prices are flexible, the expression above becomes:

$$\begin{aligned}\tilde{Y}_t^n &= \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} (r_t^n - \bar{r}^n) \\ &= \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} \tilde{r}_t^n \quad \text{where } \tilde{r}_t^n \equiv r_t^n - \bar{r}^n\end{aligned}$$

Notice that:

$$\bar{r}^n = \bar{r} = \bar{i} = \rho$$

The expression above implies:

$$\begin{aligned}\tilde{r}_t - \tilde{r}_t^n &= (r_t - \bar{r}) - (r_t^n - \bar{r}^n) \\ &= (r_t - \bar{r}) - (r_t^n - \bar{r}) \\ &= r_t - r_t^n\end{aligned}$$

Now we can write:

$$\tilde{Y}_t - \tilde{Y}_t^n = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \frac{1}{\sigma} \tilde{r}_t - \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} + \frac{1}{\sigma} \tilde{r}_t^n$$

$$\tilde{Y}_t - \tilde{Y}_t^n = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} - \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

Using $\tilde{y}_t = \tilde{Y}_t - \tilde{Y}_t^n$:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

which is the *Dynamic IS curve* (DIS).

Using $\tilde{r}_t - \tilde{r}_t^n = r_t - r_t^n$ we can rewrite the DIS curve as follows:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (r_t - r_t^n)$$

Remark. Iterating forward on the expression above, and imposing $\lim_{n \rightarrow \infty} \mathbb{E}_t \tilde{y}_{t+n} = 0$, we get:

$$\tilde{y}_t = -\frac{1}{\sigma} \sum_{k=0}^{\infty} \mathbb{E}_t \{ r_{t+k} - r_{t+k}^n \}$$

Now we find an expression for $\tilde{r}_t^n$.

From $\tilde{Y}_t^n = \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} \tilde{r}_t^n$ we get: $\tilde{r}_t^n = \sigma \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n - \tilde{Y}_t^n \right\}$.

But $\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$ gives: $\tilde{Y}_{t+1}^n - \tilde{Y}_t^n = \psi_y^n (\tilde{A}_{t+1} - \tilde{A}_t)$.

Then:

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \left\{ \tilde{A}_{t+1} - \tilde{A}_t \right\}$$

Using $\tilde{r}_t^n = r_t^n - \bar{r}^n = r_t^n - \rho$:

$$r_t^n = \rho + \sigma \psi_y^n \mathbb{E}_t \left\{ \tilde{A}_{t+1} - \tilde{A}_t \right\}$$

Remark. $\tilde{A}_{t+1} - \tilde{A}_t = (\ln A_{t+1} - \ln A) - (\ln A_t - \ln A) = \ln A_{t+1} - \ln A_t = a_{t+1} - a_t = \Delta a_{t+1}$ (using Galí's notation).

From $\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \widetilde{\eta}_t^i$ and $\widetilde{C}_t = \widetilde{Y}_t$ we get:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{Y}_t - \widetilde{\eta}_t^i$$

Using $\widetilde{y}_t = \widetilde{Y}_t - \widetilde{Y}_t^n$ we get:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} (\widetilde{y}_t + \widetilde{Y}_t^n) - \widetilde{\eta}_t^i$$

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{y}_t + \frac{\sigma}{\nu} \widetilde{Y}_t^n - \widetilde{\eta}_t^i$$

Using $\widetilde{Y}_t^n = \psi_y^n \widetilde{A}_t$ we get:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{y}_t + \frac{\sigma}{\nu} \psi_y^n \widetilde{A}_t - \widetilde{\eta}_t^i$$

which is our LM curve.

Before making the policy rules explicit, we derive equations for the real wage and the markup over marginal cost.

Define:

$$w_t \equiv \frac{W_t}{P_t}$$

Then:

$$\bar{w} = \frac{\bar{W}}{\bar{P}}$$

and

$$\tilde{w}_t = \tilde{W}_t - \tilde{P}_t$$

Using $\tilde{W}_t - \tilde{P}_t = (\sigma + \frac{\varphi}{1-\alpha})\tilde{Y}_t - \frac{\varphi}{1-\alpha}\tilde{A}_t$ we get:

$$\tilde{w}_t = (\sigma + \frac{\varphi}{1-\alpha})\tilde{Y}_t - \frac{\varphi}{1-\alpha}\tilde{A}_t$$

To log-linearize the markup over marginal cost, recall that $\overline{\mathcal{M}} = \frac{\epsilon}{\epsilon-1}$ and $P_t^* = \mathcal{M}_t \psi_{t|t}$. Then: $\frac{P_t^*}{P_t} = \mathcal{M}_t \frac{\psi_{t|t}}{P_t}$.
Then: $\frac{P_t^*}{P_t} = \mathcal{M}_t MC_{t|t}$. Then:

$$\mathcal{M}_t = \frac{P_t^*}{P_t} \frac{1}{MC_{t|t}}$$

Then:

$$\widetilde{\mathcal{M}}_t = \widetilde{P}_t^* - \widetilde{P}_t - \widetilde{MC}_{t|t}$$

Using $\widetilde{MC}_{t|t} = \widetilde{MC}_t - \frac{\alpha\epsilon}{1-\alpha}(\widetilde{P}_t^* - \widetilde{P}_t)$ we get: $\widetilde{\mathcal{M}}_t = \widetilde{P}_t^* - \widetilde{P}_t - \widetilde{MC}_t + \frac{\alpha\epsilon}{1-\alpha}(\widetilde{P}_t^* - \widetilde{P}_t)$. Then:

$$\widetilde{\mathcal{M}}_t = \left(1 + \frac{\alpha\epsilon}{1-\alpha}\right)(\widetilde{P}_t^* - \widetilde{P}_t) - \widetilde{MC}_t$$

We know $\widetilde{MC}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right) \widetilde{y}_t$. Then:

$$\widetilde{\mathcal{M}}_t = \left(1 + \frac{\alpha\epsilon}{1-\alpha}\right)(\widetilde{P}_t^* - \widetilde{P}_t) - \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right) \widetilde{y}_t$$

From $\tilde{P}_t = \theta \tilde{P}_{t-1} + (1 - \theta) \tilde{P}_t^*$ we get $\tilde{P}_t^* - \tilde{P}_t = \frac{\theta}{1-\theta}(\tilde{P}_t - \tilde{P}_{t-1})$. Using $\pi_t = \tilde{P}_t - \tilde{P}_{t-1}$ we get: $\tilde{P}_t^* - \tilde{P}_t = \frac{\theta}{1-\theta}\pi_t$. Then:

$$\widetilde{\mathcal{M}}_t = \left(1 + \frac{\alpha\epsilon}{1-\alpha}\right) \frac{\theta}{1-\theta} \pi_t - \left(\sigma + \frac{\varphi + \alpha}{1-\alpha}\right) \tilde{y}_t$$

Policy Rules

Money-Growth Rule

$$M_t = \mu_t M_{t-1}$$

with

$$\ln \mu_{t+1} = \rho_\mu \ln \mu_t + \varepsilon_{t+1}^\mu, \quad \varepsilon_{t+1}^\mu \sim N(0, \sigma_{\varepsilon^\mu}^2)$$

Notice that the rule above implies that money growth in the nss is zero:

$$\bar{\mu} = 1$$

Also:

$$\tilde{\mu}_{t+1} = \rho_{\mu} \tilde{\mu}_t + \varepsilon_{t+1}^{\mu}$$

where $\tilde{\mu}_t \equiv \ln \mu_t - \ln \bar{\mu} = \ln \mu_t$.

From $M_t = \mu_t M_{t-1}$ we get: $\frac{M_t}{P_t} = \mu_t \frac{M_{t-1}}{\Pi_t P_{t-1}}$. Then:

$$M_t^r = \frac{\mu_t}{\Pi_t} M_{t-1}^r$$

Then: $\tilde{M}_t^r = \tilde{\mu}_t - \tilde{\Pi}_t + \tilde{M}_{t-1}^r$. Then:

$$\tilde{M}_t^r = \tilde{\mu}_t - \pi_t + \tilde{M}_{t-1}^r$$

Taylor Rule

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

with

$$v_{t+1} = \rho_v v_t + \varepsilon_{t+1}^v, \quad \varepsilon_{t+1}^v \sim N(0, \sigma_{\varepsilon^v}^2)$$

Notice that the nss value of v is zero:

$$\bar{v} = 0$$

Using $\tilde{i}_t \equiv i_t - \bar{i} = i_t - \rho$ we can rewrite the Taylor Rule as follows:

$$\tilde{i}_t = \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

Remark. Taylor (1993) writes the Taylor Rule, for annual data, as follows:

$$i_t = \pi_t + \frac{1}{2} \tilde{y}_t + \frac{1}{2} (\pi_t - \pi^*) + \rho$$

with $\pi^* = 2\%$ and $\rho = 2\%$.

Then (in percentage points): $i_t = \frac{3}{2}\pi_t + \frac{1}{2}\tilde{y}_t + (\rho - \frac{1}{2}\pi^*) = \frac{3}{2}\pi_t + \frac{1}{2}\tilde{y}_t + (2 - \frac{1}{2}2)$. Then:

$$i_t = 1.5\pi_t + \frac{1}{2}\tilde{y}_t + 1$$

If we set $\phi_\pi = 1.5$, $\phi_y = 0.5$ and $\rho = 2\%$ our rule is similar to Taylor's, except that we have $\pi^* = 0$, and we have added a shock v_t . Another difference is that Taylor measures the output gap as the percentage deviation of output from trend (and not as the deviation from the flexible-price output, which is *unobservable*).

Summarizing

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \{ \pi_{t+1} \}$$

NKPC

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

DIS

$$\tilde{M}_t^r = \frac{\sigma}{\nu} \tilde{y}_t + \frac{\sigma}{\nu} \psi_y^n \tilde{A}_t - \eta \dot{\tilde{t}}_t$$

LM

$$\tilde{r}_t = \tilde{i}_t - \mathbb{E}_t \{ \pi_{t+1} \}$$

Fisher Equation

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \{ \tilde{A}_{t+1} - \tilde{A}_t \}$$

Natural Int. Rate

$$\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$$

Natural Output

$$\tilde{y}_t = \tilde{Y}_t - \tilde{Y}_t^n$$

Output Gap

$$\tilde{Y}_t = \tilde{A}_t + (1 - \alpha) \tilde{N}_t$$

Output

$$\tilde{w}_t = (\sigma + \frac{\varphi}{1-\alpha}) \tilde{Y}_t - \frac{\varphi}{1-\alpha} \tilde{A}_t$$

Real Wage

$$\tilde{\mathcal{M}}_t = (1 + \frac{\alpha \epsilon}{1-\alpha}) \frac{\theta}{1-\theta} \pi_t - \left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right) \tilde{y}_t$$

Markup

$$\tilde{A}_{t+1} = \rho_a \tilde{A}_t + \varepsilon_{t+1}^A$$

TFP Shock

Money-Growth Rule

$$\widetilde{M}_t^r = \widetilde{\mu}_t - \pi_t + \widetilde{M}_{t-1}^r$$

$$\widetilde{\mu}_{t+1} = \rho_\mu \widetilde{\mu}_t + \varepsilon_{t+1}^\mu$$

Taylor Rule

$$\widetilde{i}_t = \phi_\pi \pi_t + \phi_y \widetilde{y}_t + v_t$$

$$v_{t+1} = \rho_v v_t + \varepsilon_{t+1}^v$$

Under both policy rules we have a dynamic stochastic system of 13 equations in 13 variables: $\pi_t, \tilde{y}_t, \tilde{i}_t, \tilde{r}_t, \tilde{r}_t^n, \tilde{M}_t^r, \tilde{Y}_t^n, \tilde{Y}_t, \tilde{N}_t, \tilde{w}_t, \tilde{\mathcal{M}}_t, \tilde{A}_t$ plus $\tilde{\mu}_t$ (money-growth rule) or v_t (Taylor rule).

We can solve the system by Uhlig's method. We use "brute force" and declare all endogenous variables to be state variables. Define:

$$\begin{aligned} x_t &\equiv \left(\tilde{M}_t^r, \tilde{y}_t, \pi_t, \tilde{i}_t, \tilde{r}_t, \tilde{r}_t^n, \tilde{Y}_t^n, \tilde{Y}_t, \tilde{N}_t, \tilde{w}_t, \tilde{\mathcal{M}}_t \right)' \\ z_t &\equiv \left(\tilde{A}_t, \tilde{\mu}_t \right)', \quad \varepsilon_{t+1} \equiv \left(\varepsilon_{t+1}^A, \varepsilon_{t+1}^\mu \right)' \quad \text{money-growth rule} \\ z_t &\equiv \left(\tilde{A}_t, v_t \right)', \quad \varepsilon_{t+1} \equiv \left(\varepsilon_{t+1}^A, \varepsilon_{t+1}^v \right)' \quad \text{Taylor rule} \end{aligned}$$

Then, we can write the system in the following form:

$$\begin{aligned} \mathbf{0} &= \mathbb{E}_t \{ Fx_{t+1} + Gx_t + Hx_{t-1} + Lz_{t+1} + Mz_t \} \\ z_{t+1} &= Nz_t + \varepsilon_{t+1} \end{aligned}$$

The solution will be of the form:

$$x_t = Px_{t-1} + Qz_t$$

Money-Growth Rule

$$N_{2 \times 2} = \begin{bmatrix} \rho_a & 0 \\ 0 & \rho_\mu \end{bmatrix}$$

$$F_{11 \times 11} = \begin{bmatrix} 0 & 0 & \beta & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$${}_{11 \times 11} G = \begin{bmatrix} 0 & \kappa & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & -\frac{1}{\sigma} & \frac{1}{\sigma} & 0 & 0 & 0 & 0 & 0 \\ 1 & -\frac{\sigma}{\nu} & 0 & \eta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & G_{8,9} & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{10,8} & 0 & 1 & 0 \\ 1 & G_{11,2} & G_{11,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where

$$G_{8,9} = -(1 - \alpha)$$

$$G_{10,8} = -\left(\sigma + \frac{\varphi}{1 - \alpha}\right)$$

$$G_{11,2} = \sigma + \frac{\varphi + \alpha}{1 - \alpha}$$

$$G_{11,3} = -\left(1 + \frac{\alpha \epsilon}{1 - \alpha}\right) \frac{\theta}{1 - \theta}$$

$$\begin{aligned}
{}_{11 \times 11} H &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
{}_{11 \times 2} L &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\sigma\psi_y^n & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad {}_{11 \times 2} M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -\frac{\sigma}{\nu}\psi_y^n & 0 \\ 0 & 0 \\ \sigma\psi_y^n & 0 \\ -\psi_y^n & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \\ \frac{\phi}{1-\alpha} & 0 \\ 0 & 0 \end{bmatrix}
\end{aligned}$$

Taylor Rule

$$N_{2 \times 2} = \begin{bmatrix} \rho_a & 0 \\ 0 & \rho_v \end{bmatrix}$$

$$F_{11 \times 11} = \begin{bmatrix} 0 & 0 & \beta & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{11 \times 11} = \begin{bmatrix} 0 & \kappa & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & -\frac{1}{\sigma} & \frac{1}{\sigma} & 0 & 0 & 0 & 0 & 0 \\ 1 & -\frac{\sigma}{\nu} & 0 & \eta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & G_{8,9} & 0 & 0 \\ 0 & -\phi_y & -\phi_\pi & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{10,8} & 0 & 1 & 0 \\ 1 & G_{11,2} & G_{11,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where

$$G_{8,9} = -(1 - \alpha)$$

$$G_{10,8} = -(\sigma + \frac{\varphi}{1 - \alpha})$$

$$G_{11,2} = \sigma + \frac{\varphi + \alpha}{1 - \alpha}$$

$$G_{11,3} = -(1 + \frac{\alpha \epsilon}{1 - \alpha}) \frac{\theta}{1 - \theta}$$

$${}_{11 \times 11} H = [\mathbf{0}] \quad (\text{all zeros})$$

$${}_{11 \times 2} L = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\sigma\psi_y^n & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad {}_{11 \times 2} M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -\frac{\sigma}{\nu}\psi_y^n & 0 \\ 0 & 0 \\ \sigma\psi_y^n & 0 \\ -\psi_y^n & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \\ \frac{\phi}{1-\alpha} & 0 \\ 0 & 0 \end{bmatrix}$$

Calibration (quarterly data)

$$\theta = \frac{2}{3}, \alpha = \frac{1}{3}, \beta = 0.99, \sigma = 1, \nu = 1,$$

$$\eta = 4, \varphi = 1, \phi_\pi = 1.5, \phi_y = \frac{0.5}{4}, \chi = 1$$

$$\epsilon = 6, A = 1, \rho_a = 0.9, \rho_\mu = \rho_v = 0.5$$

These values give:

$$\Theta = 0.25, \lambda = 0.0425, \kappa = 0.1275, \psi_y^n = 1$$

Remark. We follow Galí's calibration and set $\eta = 4$ even though this does not satisfy $\eta = \frac{\beta}{\nu(1-\beta)}$ for our values of β and ν .

Solution under Money-growth Rule

$$P = \begin{bmatrix} 0.7039 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.7039 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.2961 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.2084 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.7039 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1.0559 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1.7598 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.2569 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0.2407 & 0.4537 \\ -0.7593 & 1.1204 \\ -0.2407 & 0.5463 \\ 0.0000 & 0.1667 \\ 0.1454 & -0.2408 \\ -0.1000 & 0.0000 \\ 1.0000 & 0.0000 \\ 0.2407 & 1.1204 \\ -1.1389 & 1.6806 \\ -0.8982 & 2.8010 \\ 0.3519 & 1.0088 \end{bmatrix}$$

Then:

$$\widetilde{M}_t^r = 0.7039 \widetilde{M}_{t-1}^r + 0.2407 \widetilde{A}_t + 0.4537 \widetilde{\mu}_t$$

$$\widetilde{y}_t = 0.7039 \widetilde{M}_{t-1}^r - 0.7593 \widetilde{A}_t + 1.1204 \widetilde{\mu}_t$$

$$\pi_t = 0.2961 \widetilde{M}_{t-1}^r - 0.2407 \widetilde{A}_t + 0.5463 \widetilde{\mu}_t$$

$$\widetilde{i}_t = 0.1667 \widetilde{\mu}_t$$

$$\widetilde{r}_t = -0.20846 \widetilde{M}_{t-1}^r + 0.1454 \widetilde{A}_t - 0.2408 \widetilde{\mu}_t$$

$$\widetilde{r}_t^n = -0.1 \widetilde{A}_t$$

$$\widetilde{Y}_t^n = \widetilde{A}_t$$

$$\widetilde{Y}_t = 0.7039 \widetilde{M}_{t-1}^r + 0.2407 \widetilde{A}_t + 1.1204 \widetilde{\mu}_t$$

$$\widetilde{N}_t = 1.0559 \widetilde{M}_{t-1}^r - 1.1389 \widetilde{A}_t + 1.6806 \widetilde{\mu}_t$$

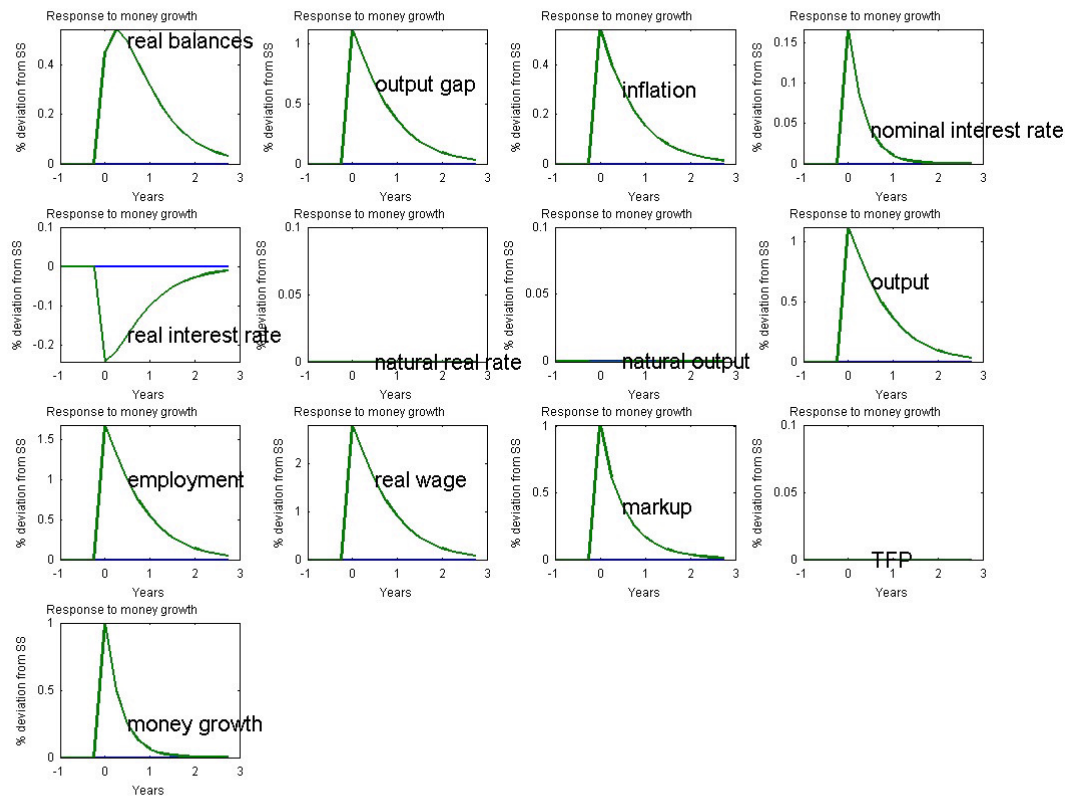
$$\widetilde{w}_t = 1.7598 \widetilde{M}_{t-1}^r - 0.8982 \widetilde{A}_t + 2.8010 \widetilde{\mu}_t$$

$$\widetilde{\mathcal{M}}_t = 0.2569 \widetilde{M}_{t-1}^r + 0.3519 \widetilde{A}_t + 1.0088 \widetilde{\mu}_t$$

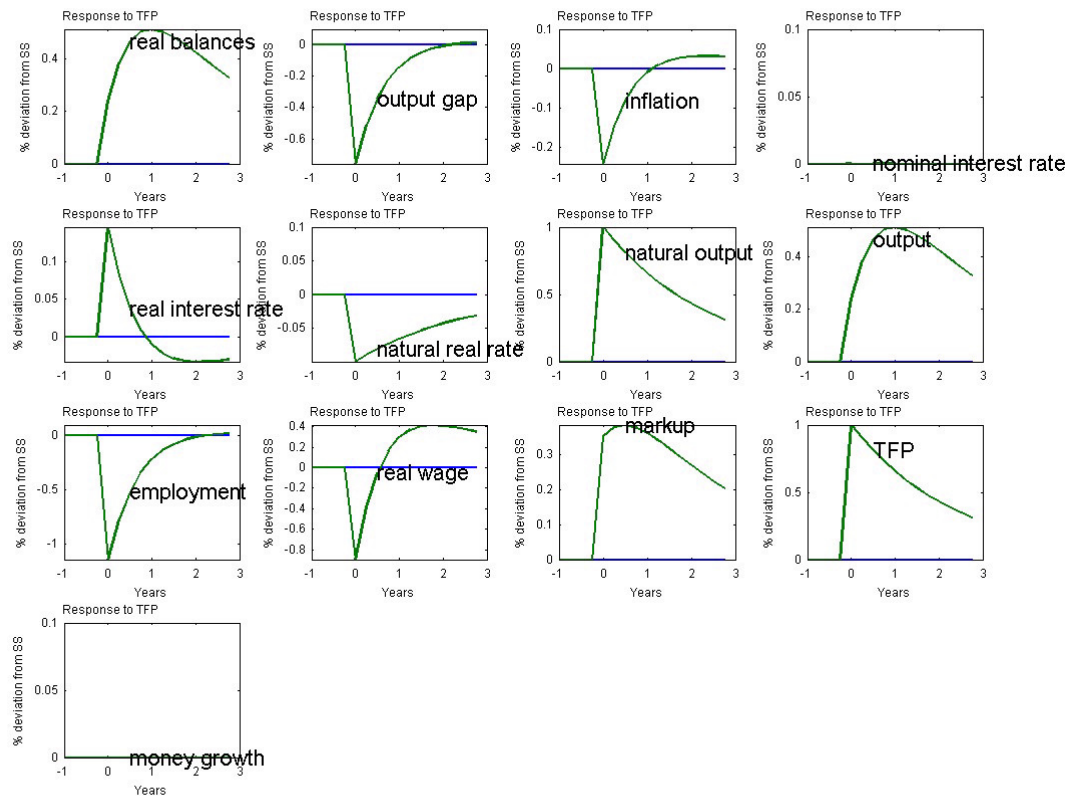
$$\widetilde{A}_{t+1} = 0.90 \widetilde{A}_t + \varepsilon_{t+1}^A$$

$$\widetilde{\mu}_{t+1} = 0.50 \widetilde{\mu}_t + \varepsilon_{t+1}^\mu$$

Impulse Responses to a Money Shock ($\theta = \frac{2}{3}$)



Impulse Responses to a TFP shock ($\theta = \frac{2}{3}$)



Taylor Rule

The solution is:

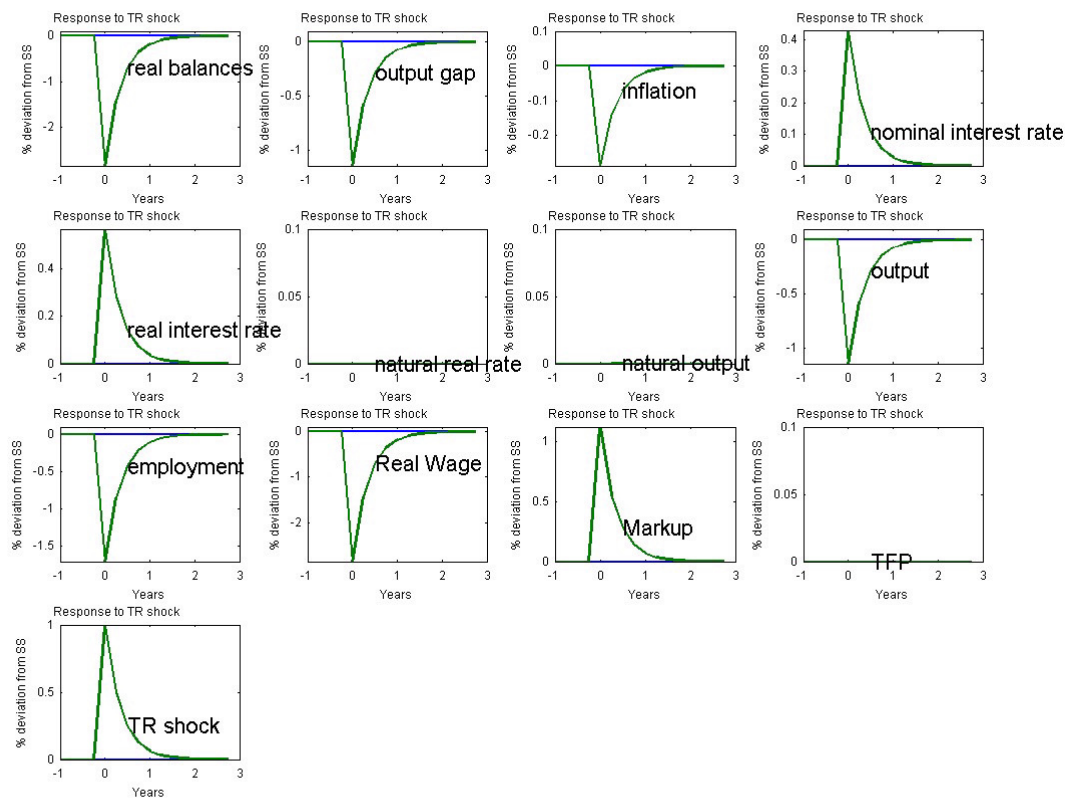
$$P = [0] \quad (\text{all zeros})$$

$$Q = \begin{bmatrix} 1.7033 & -2.8434 \\ -0.1079 & -1.1396 \\ -0.1262 & -0.2877 \\ -0.2028 & 0.4260 \\ -0.0892 & 0.5698 \\ -0.1000 & 0.0000 \\ 1.0000 & 0.0000 \\ 0.8921 & -1.1396 \\ -0.1618 & -1.7094 \\ 0.7303 & -2.8491 \\ -0.6860 & 1.1171 \end{bmatrix}$$

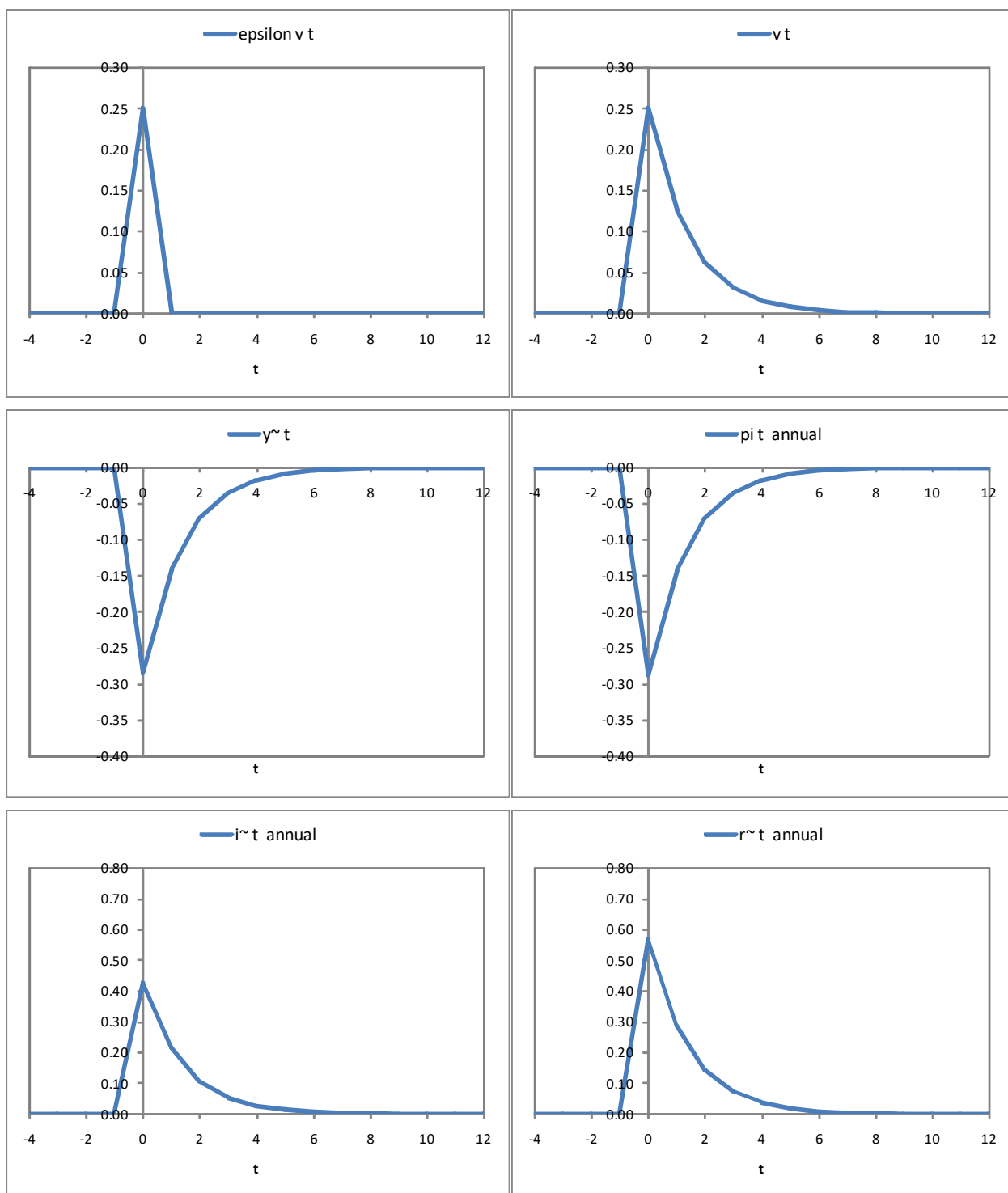
Then:

$$\begin{aligned}\widetilde{M}_t^r &= 1.7033 \widetilde{A}_t - 2.8434 v_t \\ \widetilde{y}_t &= -0.1079 \widetilde{A}_t - 1.1396 v_t \\ \pi_t &= -0.1262 \widetilde{A}_t - 0.2877 v_t \\ \widetilde{i}_t &= -0.2028 \widetilde{A}_t + 0.4260 v_t \\ \widetilde{r}_t &= -0.0892 \widetilde{A}_t + 0.5698 v_t \\ \widetilde{r}_t^n &= -0.1 \widetilde{A}_t \\ \widetilde{Y}_t^n &= \widetilde{A}_t \\ \widetilde{Y}_t &= 0.8921 \widetilde{A}_t - 1.1396 v_t \\ \widetilde{N}_t &= -0.1618 \widetilde{A}_t - 1.7094 v_t \\ \widetilde{w}_t &= 0.7303 \widetilde{A}_t - 2.8491 v_t \\ \widetilde{\mathcal{M}}_t &= -0.6860 \widetilde{A}_t + 1.1171 v_t \\ \widetilde{A}_{t+1} &= 0.90 \widetilde{A}_t + \varepsilon_{t+1}^A \\ v_{t+1} &= 0.50 v_t + \varepsilon_{t+1}^v\end{aligned}$$

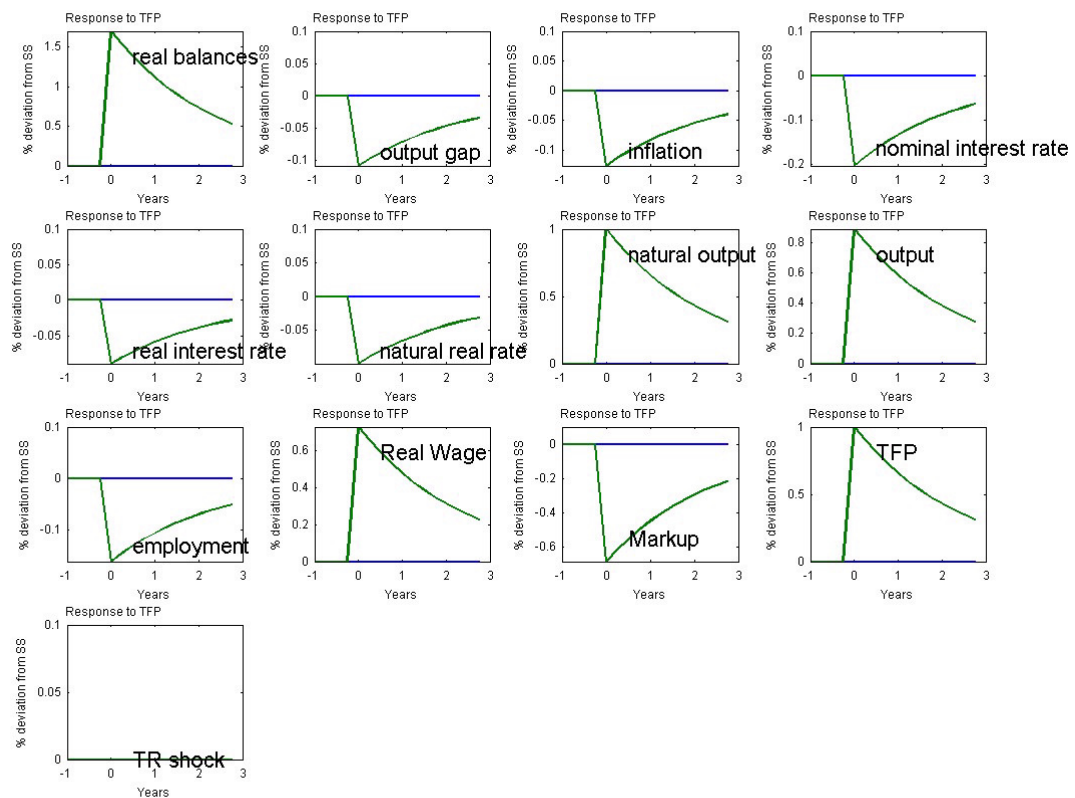
Impulse Responses to a Taylor Rule Shock ($\theta = \frac{2}{3}$)



Galí's Figure 3.1



Impulse Responses to a TFP shock ($\theta = \frac{2}{3}$)



Closed-form Solution and Graphical Representation

We work with the Taylor-Rule version of the model. Analogous results can be obtained using the Money-Growth version.

Strategy: Find a closed-form solution for $\tilde{y}$ and π , and represent the equilibrium with two plots; (i) DIS-MP in $\tilde{y}$ - $\tilde{i}$ space; (ii) DAD-DAS in $\tilde{y}$ - π space

where

DIS: Dynamic IS Curve

MP: Monetary Policy (Taylor Rule) Curve

DAD: Dynamic Aggregate Demand

DAS: Dynamic Aggregate Supply

Closed-Form Solution

From the Taylor Rule we get:

$$\tilde{i}_t = [\phi_\pi \pi_t + v_t] + \phi_y \tilde{y}_t \quad \text{MP}$$

which shows that, for given values of π_t and v_t , there's a positive relationship between $\tilde{i}_t$ and $\tilde{y}_t$.

Combining the original DIS curve and the Fisher Equation we get:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{i}_t - \mathbb{E}_t \{ \pi_{t+1} \} - \tilde{r}_t^n)$$

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} + \frac{1}{\sigma} \tilde{r}_t^n - \frac{1}{\sigma} \tilde{i}_t$$

But:

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \{ \tilde{A}_{t+1} - \tilde{A}_t \}$$

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \{ \rho_a \tilde{A}_t + \varepsilon_{t+1}^A - \tilde{A}_t \}$$

$$\tilde{r}_t^n = \sigma \psi_y^n [(\rho_a - 1) \tilde{A}_t + \mathbb{E}_t \varepsilon_{t+1}^A]$$

$$\tilde{r}_t^n = -\sigma \psi_y^n (1 - \rho_a) \tilde{A}_t$$

Then:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} + \frac{1}{\sigma} [-\sigma \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} - \psi_y^n (1 - \rho_a) \tilde{A}_t - \frac{1}{\sigma} \tilde{i}_t$$

Then:

$$\tilde{y}_t = [\mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t \quad \text{DIS}$$

which shows that, for given values of $\mathbb{E}_t \tilde{y}_{t+1}$, $\mathbb{E}_t \pi_{t+1}$ and $\tilde{A}_t$, there's a negative relationship between $\tilde{i}_t$ and $\tilde{y}_t$.

Now we can use MP to eliminate $\tilde{i}_t$ from DIS. We get:

$$\tilde{y}_t = [\mathbb{E}_t\{\tilde{y}_{t+1} + \frac{1}{\sigma}\pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t] - \frac{1}{\sigma}\tilde{i}_t$$

$$\tilde{y}_t = [\mathbb{E}_t\{\tilde{y}_{t+1} + \frac{1}{\sigma}\pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t] - \frac{1}{\sigma}[\phi_\pi\pi_t + v_t + \phi_y\tilde{y}_t]$$

$$\tilde{y}_t(1 + \frac{\phi_y}{\sigma}) = [\mathbb{E}_t\{\tilde{y}_{t+1} + \frac{1}{\sigma}\pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t - \frac{1}{\sigma}v_t] - \frac{\phi_\pi}{\sigma}\pi_t$$

$$\tilde{y}_t \frac{\sigma + \phi_y}{\sigma} = [\frac{1}{\sigma}\mathbb{E}_t\{\sigma\tilde{y}_{t+1} + \pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t - \frac{1}{\sigma}v_t] - \frac{\phi_\pi}{\sigma}\pi_t$$

$$\tilde{y}_t = \frac{\sigma}{\sigma + \phi_y} [\frac{1}{\sigma}\mathbb{E}_t\{\sigma\tilde{y}_{t+1} + \pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t - \frac{1}{\sigma}v_t] - \frac{\sigma}{\sigma + \phi_y} \frac{\phi_\pi}{\sigma} \pi_t$$

Then:

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\mathbb{E}_t\{\sigma\tilde{y}_{t+1} + \pi_{t+1}\} - \sigma\psi_y^n(1-\rho_a)\tilde{A}_t - v_t] - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t \quad \text{DAD}$$

which shows that, for given values of $\mathbb{E}_t\tilde{y}_{t+1}$, $\mathbb{E}_t\pi_{t+1}$, $\tilde{A}_t$ and v_t , there's a negative relationship between $\tilde{y}_t$ and π_t

From the NKPC we get:

$$\pi_t = \beta \mathbb{E}_t \{\pi_{t+1}\} + \kappa \tilde{y}_t \quad \text{DAS}$$

which shows that, given $\mathbb{E}_t \pi_{t+1}$, there's a positive relationship between $\tilde{y}_t$ and π_t .

Now we can use DAD and DAS to solve for the equilibrium values of π_t and $\tilde{y}_t$. We'll guess the form of the solution and then use the method of undetermined coefficients to obtain the values of the parameters.

Guess:

$$\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$$

$$\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$$

where ψ_{ya} , ψ_{yv} , $\psi_{\pi a}$ and $\psi_{\pi v}$ are coefficients to be determined.

From $\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$ we get:

$$\tilde{y}_{t+1} = \psi_{ya} \tilde{A}_{t+1} + \psi_{yv} v_{t+1}$$

$$\mathbb{E}_t \tilde{y}_{t+1} = \psi_{ya} \mathbb{E}_t \tilde{A}_{t+1} + \psi_{yv} \mathbb{E}_t v_{t+1} \Rightarrow$$

$$\mathbb{E}_t \tilde{y}_{t+1} = \psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t$$

From $\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$ we get:

$$\pi_{t+1} = \psi_{\pi a} \tilde{A}_{t+1} + \psi_{\pi v} v_{t+1}$$

$$\mathbb{E}_t \pi_{t+1} = \psi_{\pi a} \mathbb{E}_t \tilde{A}_{t+1} + \psi_{\pi v} \mathbb{E}_t v_{t+1} \Rightarrow$$

$$\mathbb{E}_t \pi_{t+1} = \psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t$$

Substituting the expressions for $\tilde{y}_t$ and $\mathbb{E}_t\pi_{t+1}$ into DAS we get:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t$$

$$\pi_t = \beta(\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) + \kappa(\psi_{y a} \tilde{A}_t + \psi_{y v} v_t) \Rightarrow$$

$$\pi_t = (\beta \psi_{\pi a} \rho_a + \kappa \psi_{y a}) \tilde{A}_t + (\beta \psi_{\pi v} \rho_v + \kappa \psi_{y v}) v_t$$

Comparing the expression above with $\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$ we get:

$$\psi_{\pi a} = \beta \psi_{\pi a} \rho_a + \kappa \psi_{y a}$$

$$\psi_{\pi v} = \beta \psi_{\pi v} \rho_v + \kappa \psi_{y v}$$

Substituting the expressions for π_t , $\mathbb{E}_t \tilde{y}_{t+1}$, and $\mathbb{E}_t \pi_{t+1}$ into DAD we get:

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\sigma \mathbb{E}_t \tilde{y}_{t+1} + \mathbb{E}_t \pi_{t+1} - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t] - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} \left[\begin{aligned} &\sigma (\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t) + (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) \\ &- \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t \end{aligned} \right]$$

$$- \frac{\phi_\pi}{\sigma + \phi_y} (\psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t)$$

$$\tilde{y}_t = \frac{\sigma \psi_{ya} \rho_a + \psi_{\pi a} \rho_a - \sigma \psi_y^n (1 - \rho_a) - \phi_\pi \psi_{\pi a}}{\sigma + \phi_y} \tilde{A}_t + \frac{\sigma \psi_{yv} \rho_v + \psi_{\pi v} \rho_v - 1 - \phi_\pi \psi_{\pi v}}{\sigma + \phi_y} v_t \Rightarrow$$

$$\tilde{y}_t = \frac{\sigma \psi_{ya} \rho_a + (\rho_a - \phi_\pi) \psi_{\pi a} - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y} \tilde{A}_t + \frac{\sigma \psi_{yv} \rho_v + (\rho_v - \phi_\pi) \psi_{\pi v} - 1}{\sigma + \phi_y} v_t$$

Comparing the expression above with $\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$ we get:

$$\psi_{ya} = \frac{\sigma \psi_{ya} \rho_a + (\rho_a - \phi_\pi) \psi_{\pi a} - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y}$$

$$\psi_{yv} = \frac{\sigma \psi_{yv} \rho_v + (\rho_v - \phi_\pi) \psi_{\pi v} - 1}{\sigma + \phi_y}$$

Hence, we need to solve the following system of four equations in the four unknowns $\psi_{\pi a}$, $\psi_{\pi v}$, ψ_{ya} , ψ_{yv} :

$$\psi_{\pi a} = \beta \psi_{\pi a} \rho_a + \kappa \psi_{ya} \quad (1)$$

$$\psi_{\pi v} = \beta \psi_{\pi v} \rho_v + \kappa \psi_{yv} \quad (2)$$

$$\psi_{ya} = \frac{\sigma \psi_{ya} \rho_a + (\rho_a - \phi_\pi) \psi_{\pi a} - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y} \quad (3)$$

$$\psi_{yv} = \frac{\sigma \psi_{yv} \rho_v + (\rho_v - \phi_\pi) \psi_{\pi v} - 1}{\sigma + \phi_y} \quad (4)$$

From (1) we get: $\psi_{\pi a}(1 - \beta\rho_A) = \kappa\psi_{ya}$. Then:

$$\psi_{\pi a} = \frac{\kappa}{1-\beta\rho_A}\psi_{ya} \quad (5)$$

From (2) we get: $\psi_{\pi v}(1 - \beta\rho_v) = \kappa\psi_{yv}$. Then:

$$\psi_{\pi v} = \frac{\kappa}{1-\beta\rho_v}\psi_{yv} \quad (6)$$

From (3) we get:

$$(\sigma + \phi_y)\psi_{ya} = \sigma\rho_a\psi_{ya} - \sigma\psi_y^n(1 - \rho_a) + (\rho_a - \phi_\pi)\psi_{\pi a}$$

$$(\sigma + \phi_y - \sigma\rho_a)\psi_{ya} = -\sigma\psi_y^n(1 - \rho_a) + (\rho_a - \phi_\pi)\psi_{\pi a}$$

Using (5):

$$(\sigma + \phi_y - \sigma\rho_a)\psi_{ya} = -\sigma\psi_y^n(1 - \rho_a) + (\rho_a - \phi_\pi)\frac{\kappa}{1-\beta\rho_a}\psi_{ya}$$

$$(\sigma + \phi_y - \sigma\rho_a - \frac{\kappa(\rho_a - \phi_\pi)}{1-\beta\rho_a})\psi_{ya} = -\sigma\psi_y^n(1 - \rho_a)$$

$$\frac{(1-\beta\rho_a)(\sigma+\phi_y-\sigma\rho_a)-\kappa(\rho_a-\phi_\pi)}{1-\beta\rho_a}\psi_{ya}=-\sigma\psi_y^n(1-\rho_a)$$

$$\frac{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)}{1-\beta\rho_a}\psi_{ya}=-\sigma\psi_y^n(1-\rho_a)\Rightarrow$$

$$\psi_{ya} = \frac{-\sigma\psi_y^n(1-\rho_a)(1-\beta\rho_a)}{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)} \quad (7)$$

From (4) we get:

$$(\sigma+\phi_y)\psi_{yv}=\sigma\psi_{yv}\rho_v+(\rho_v-\phi_\pi)\psi_{\pi v}-1$$

$$(\sigma+\phi_y-\sigma\rho_v)\psi_{yv}=(\rho_v-\phi_\pi)\psi_{\pi v}-1$$

Using (6):

$$(\sigma+\phi_y-\sigma\rho_v)\psi_{yv}=(\rho_v-\phi_\pi)\frac{\kappa}{1-\beta\rho_v}\psi_{yv}-1$$

$$(\sigma+\phi_y-\sigma\rho_v-\frac{\kappa(\rho_v-\phi_\pi)}{1-\beta\rho_v})\psi_{yv}=-1$$

$$\frac{(1-\beta\rho_v)(\sigma+\phi_y-\sigma\rho_v)-\kappa(\rho_v-\phi_\pi)}{1-\beta\rho_v}\psi_{yv}=-1$$

$$\frac{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)}{1-\beta\rho_v}\psi_{yv}=-1\Rightarrow$$

$$\psi_{yv} = \frac{-(1-\beta\rho_v)}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)} \quad (8)$$

Substituting (7) back into (5) we get:

$$\psi_{\pi a} = \frac{\kappa}{1-\beta\rho_a} \frac{-\sigma\psi_y^n(1-\rho_a)(1-\beta\rho_a)}{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)} \Rightarrow$$

$$\psi_{\pi a} = \frac{-\sigma\psi_y^n(1-\rho_a)\kappa}{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)} \quad (9)$$

Finally, substituting (8) back into (6) we get:

$$\psi_{\pi v} = \frac{\kappa}{1-\beta\rho_v} \frac{-(1-\beta\rho_v)}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)} \Rightarrow$$

$$\psi_{\pi v} = \frac{-\kappa}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)} \quad (10)$$

Summarizing, the solution is:

$$\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$$

$$\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$$

with

$$\psi_{ya} = -\sigma \psi_y^n (1 - \rho_a)(1 - \beta \rho_a) \Lambda_a$$

$$\psi_{\pi a} = -\sigma \psi_y^n (1 - \rho_a) \kappa \Lambda_a$$

$$\psi_{yv} = -(1 - \beta \rho_v) \Lambda_v$$

$$\psi_{\pi v} = -\kappa \Lambda_v$$

where:

$$\Lambda_a \equiv \frac{1}{(1 - \beta \rho_a)[\sigma(1 - \rho_a) + \phi_y] + \kappa(\phi_\pi - \rho_a)}$$

$$\Lambda_v \equiv \frac{1}{(1 - \beta \rho_v)[\sigma(1 - \rho_v) + \phi_y] + \kappa(\phi_\pi - \rho_v)}$$

Our calibration gives:

$$\psi_{ya} = -0.1079, \psi_{\pi a} = -0.1262, \psi_{yv} = -1.1396, \psi_{\pi v} = -0.2877$$

Once we've found the solution for π_t and $\tilde{y}_t$ we can use the Taylor Rule to obtain the equilibrium value of the nominal interest rate:

$$\tilde{i}_t = \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

Then we can get the real interest rate:

$$\tilde{r}_t = \tilde{i}_t - \mathbb{E}_t \{\pi_{t+1}\}$$

Aggregate output is (recall $\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$):

$$\tilde{Y}_t = \tilde{Y}_t^n + \tilde{y}_t$$

From the production function, $\tilde{Y}_t = \tilde{A}_t + (1 - \alpha)\tilde{N}_t$, we get:

$$\tilde{N}_t = \frac{\tilde{Y}_t - \tilde{A}_t}{1 - \alpha}$$

Finally, we can use the LM equation to get real money balances:

$$M_t^r = \frac{\sigma}{\nu} \tilde{y}_t + \frac{\sigma}{\nu} \psi_y^n \tilde{A}_t - \eta \tilde{i}_t$$

Plots

We know:

$$\tilde{i}_t = (\phi_\pi \pi_t + v_t) + \phi_y \tilde{y}_t \quad \text{MP}$$

$$\tilde{y}_t = [\mathbb{E}_t \tilde{y}_{t+1} + \frac{1}{\sigma} \mathbb{E}_t \pi_{t+1} - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t \quad \text{DIS}$$

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\sigma \mathbb{E}_t \tilde{y}_{t+1} + \mathbb{E}_t \pi_{t+1} - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t] - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t \quad \text{DAD}$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \quad \text{DAS}$$

From our solution we know:

$$\begin{aligned} \mathbb{E}_t \tilde{y}_{t+1} &= \psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t \\ \mathbb{E}_t \pi_{t+1} &= \psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t \end{aligned}$$

Then, we can eliminate the expected values of future variables from DIS, DAD and DAS.

From DIS we get:

$$\tilde{y}_t = [\mathbb{E}_t \tilde{y}_{t+1} + \frac{1}{\sigma} \mathbb{E}_t \pi_{t+1} - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = [\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t + \frac{1}{\sigma} (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = [\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t + \frac{1}{\sigma} \psi_{\pi a} \rho_a \tilde{A}_t + \frac{1}{\sigma} \psi_{\pi v} \rho_v v_t - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = [(\psi_{ya} + \frac{1}{\sigma} \psi_{\pi a}) \rho_a - \psi_y^n (1 - \rho_a)] \tilde{A}_t + (\psi_{yv} + \frac{1}{\sigma} \psi_{\pi v}) \rho_v v_t - \frac{1}{\sigma} \tilde{i}_t$$

From DAD we get:

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\sigma \mathbb{E}_t \tilde{y}_{t+1} + \mathbb{E}_t \pi_{t+1} - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t] - \frac{\phi \pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{\sigma (\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t) + (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t}{\sigma + \phi_y}$$

$$- \frac{\phi \pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{\sigma \psi_{ya} \rho_a \tilde{A}_t + \sigma \psi_{yv} \rho_v v_t + \psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t}{\sigma + \phi_y}$$

$$- \frac{\phi \pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{[\sigma\psi_{ya}\rho_a + \psi_{\pi a}\rho_a - \sigma\psi_y^n(1-\rho_a)]\tilde{A}_t + (\sigma\psi_{yv}\rho_v + \psi_{\pi v}\rho_v - 1)v_t}{\sigma + \phi_y} - \frac{\phi_\pi}{\sigma + \phi_y}\pi_t$$

$$\tilde{y}_t = \frac{(\sigma\psi_{ya} + \psi_{\pi a})\rho_a - \sigma\psi_y^n(1-\rho_a)}{\sigma + \phi_y}\tilde{A}_t + \frac{(\sigma\psi_{yv} + \psi_{\pi v})\rho_v - 1}{\sigma + \phi_y}v_t - \frac{\phi_\pi}{\sigma + \phi_y}\pi_t$$

From DAS we get:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t$$

$$\pi_t = \beta(\psi_{\pi a}\rho_a\tilde{A}_t + \psi_{\pi v}\rho_v v_t) + \kappa \tilde{y}_t$$

Hence, we've arrived to a system that we can plot:

$$\tilde{i}_t = (\phi_\pi \pi_t + v_t) + \phi_y \tilde{y}_t \quad \text{MP}$$

$$\begin{aligned} \tilde{y}_t = & [(\psi_{ya} + \frac{1}{\sigma} \psi_{\pi a}) \rho_a - \psi_y^n (1 - \rho_a)] \tilde{A}_t \\ & + (\psi_{yv} + \frac{1}{\sigma} \psi_{\pi v}) \rho_v v_t - \frac{1}{\sigma} \tilde{i}_t \end{aligned} \quad \text{DIS}$$

$$\begin{aligned} \tilde{y}_t = & \frac{(\sigma \psi_{ya} + \psi_{\pi a}) \rho_a - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y} \tilde{A}_t \\ & + \frac{(\sigma \psi_{yv} + \psi_{\pi v}) \rho_v - 1}{\sigma + \phi_y} v_t - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t \end{aligned} \quad \text{DAD}$$

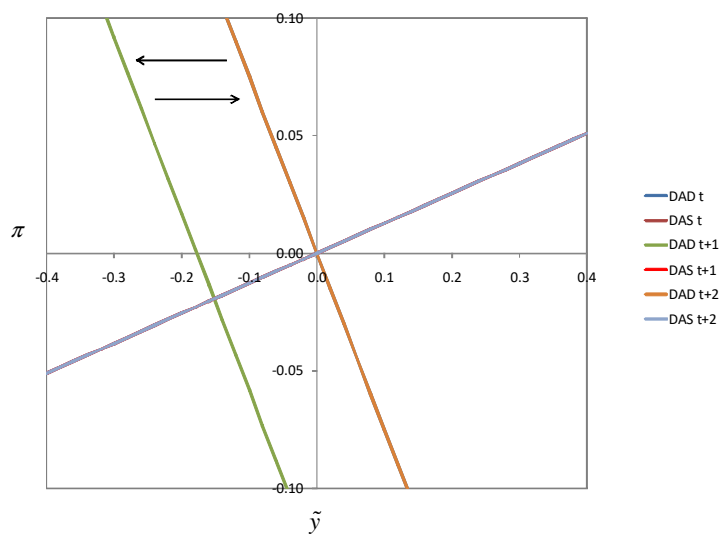
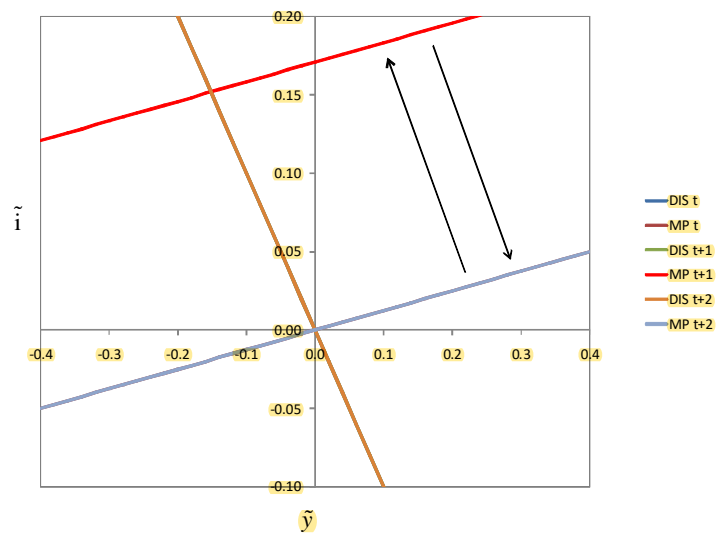
$$\pi_t = \beta (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) + \kappa \tilde{y}_t \quad \text{DAS}$$

The DAD-DAS plot determines $\tilde{y}_t$ and π_t

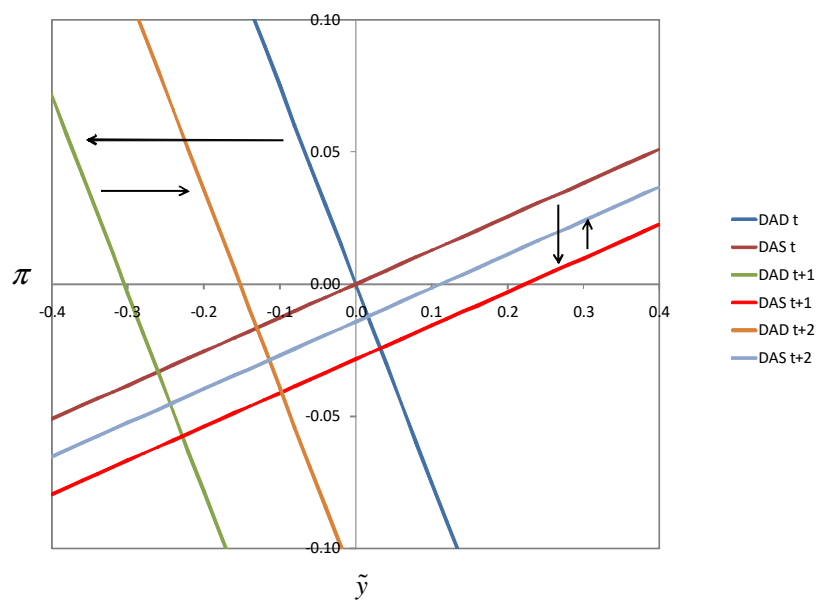
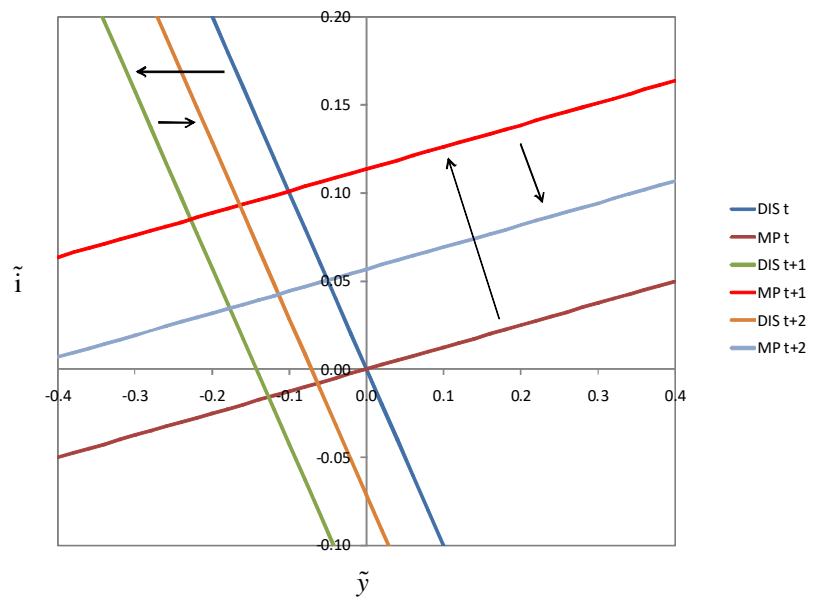
Then π_t determines the position of MP (we can actually eliminate π_t from MP using $\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$)

Then the DIS-MP plot determines $\tilde{i}_t$

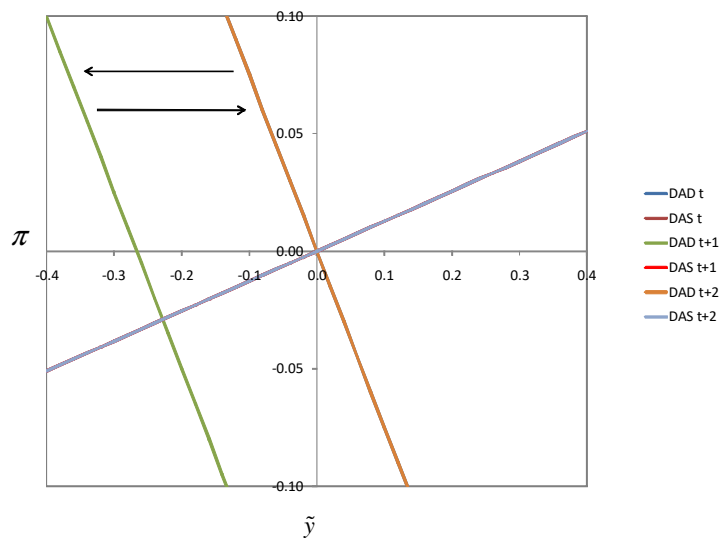
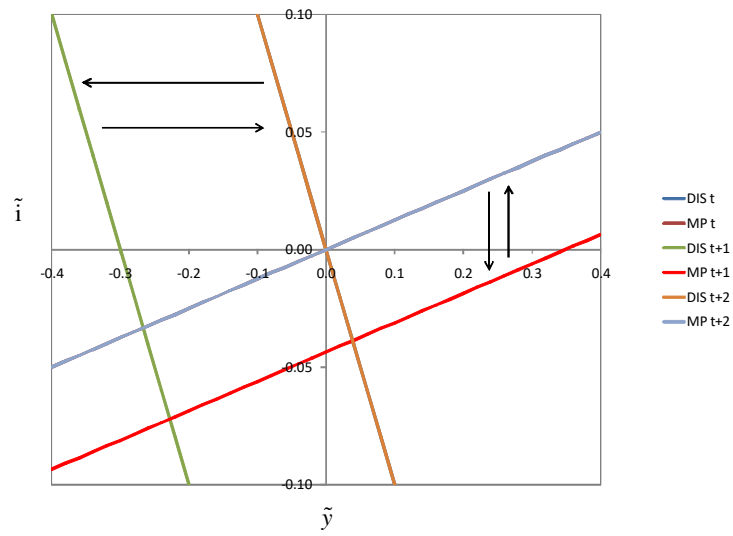
A temporary shock to the Taylor Rule ($\varepsilon_t^v > 0$, $\rho_v = 0$)



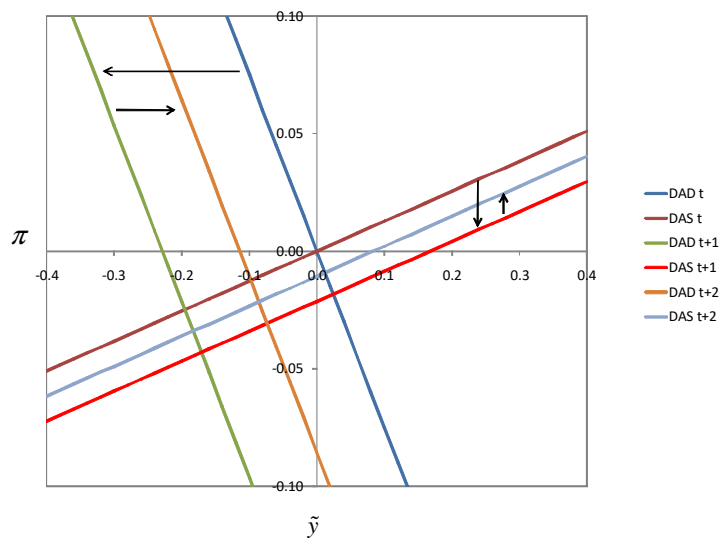
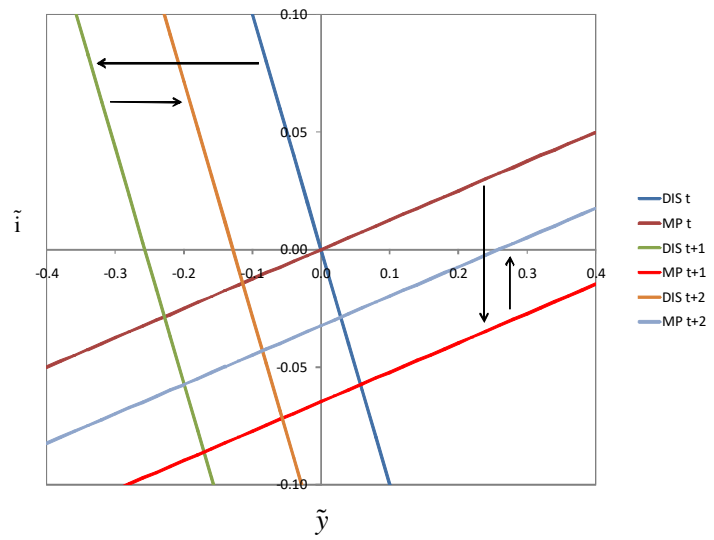
A persistent shock to the Taylor Rule ($\varepsilon_t^v > 0$, $\rho_v > 0$)



A temporary shock to technology ($\varepsilon_t^A > 0, \rho_a = 0$)



A persistent shock to technology ($\varepsilon_t^A > 0, \rho_a > 0$)



Appendix

First-order conditions for the intratemporal problem

Consider the following problem:

$$\begin{aligned} \max \quad & \left(\int_0^1 C(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} \\ \text{s.t.} \quad & \int_0^1 P(i)C(i)di = I \end{aligned}$$

Form the Lagrangian:

$$\mathcal{L} = \left(\int_0^1 C(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} + \lambda \left[I - \int_0^1 P(i)C(i)di \right]$$

To derive the first-order conditions we cannot take the pointwise derivative with respect to $C(i)$ for any fixed i , because such derivative is simply zero. Instead, we'll use a variational argument. First notice that what we are choosing is a real function $C(\bullet)$ over the domain $[0, 1]$.

Let $C^*(\bullet)$ be the solution of our problem, and define a new function $C(\bullet, \alpha)$ by perturbing $C^*(\bullet)$, as follows:

$$C(i, \alpha) = C^*(i) + \alpha F(i)$$

where $\alpha \in \mathbb{R}$ and $F(\bullet)$ is an arbitrary (integrable) real function over $[0, 1]$. Notice that, for a given function $F(\bullet)$, each value of α defines a new function. Also, when $\alpha = 0$ we have $C(i, 0) = C^*(i)$.

Now we can write the Lagrangian as a function of α , as follows:

$$\mathcal{L}(\alpha) = \left(\int_0^1 C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} + \lambda \left[I - \int_0^1 P(i) C(i, \alpha) di \right]$$

Since our objective is maximized at $\alpha = 0$, which is obviously an interior point of $\mathbb{R}$, the FOC of our problem is $\mathcal{L}'(0) = 0$. From the expression above, and using Leibniz's rule for the differentiation of integrals, we get:

$$\begin{aligned} \mathcal{L}'(\alpha) = & \frac{\epsilon}{\epsilon-1} \left(\int_0^1 C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}-1} \int_0^1 \left[\frac{\epsilon-1}{\epsilon} C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}-1} \frac{\partial C(i, \alpha)}{\partial \alpha} \right] di \\ & - \lambda \int_0^1 P(i) \frac{\partial C(i, \alpha)}{\partial \alpha} di \end{aligned}$$

Using $\frac{\partial C(i, \alpha)}{\partial \alpha} = F(i)$ we get:

$$\mathcal{L}'(\alpha) = \left(\int_0^1 C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} \int_0^1 C(i, \alpha)^{\frac{-1}{\epsilon}} F(i) di - \lambda \int_0^1 P(i) F(i) di$$

Evaluating at $\alpha = 0$ we get (recall $C(i, 0) = C^*(i)$):

$$\mathcal{L}'(0) = \left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} \int_0^1 C^*(i)^{\frac{-1}{\epsilon}} F(i) di - \lambda \int_0^1 P(i) F(i) di$$

We can rewrite the expression above as follows:

$$\begin{aligned} \mathcal{L}'(0) &= \int_0^1 \left[\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} F(i) \right] di - \int_0^1 \lambda P(i) F(i) di \\ &= \int_0^1 \left[\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} - \lambda P(i) \right] F(i) di \end{aligned}$$

We want to equate the expression above to zero:

$$\int_0^1 \left[\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} - \lambda P(i) \right] F(i) di = 0$$

Since $F(\bullet)$ can be chosen arbitrarily, the only way to satisfy the expression above for any such $F(\bullet)$ is to set the term in brackets equal to zero for each $i \in [0, 1]$. To see this, notice that if the term in brackets is different from zero for some $i \in [0, 1]$, say > 0 , then it will be > 0 over some interval (because of continuity). Hence, choosing an $F(\bullet)$ that is positive over such interval, the product $[\bullet] F(\bullet)$ will be positive over that interval, and then the whole integral will be different from zero. Therefore, we conclude:

$$\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} - \lambda P(i) = 0 \quad \forall i \in [0, 1]$$

Price dispersion

Claim 1. Up to first order, $\int_0^1 [\ln P_t(i) - \ln P_t] di \cong 0$.

Proof.

From $P_t = \left[\int_0^1 P_t(i)^{1-\epsilon} di \right]^{\frac{1}{1-\epsilon}}$ we get:

$$P_t^{1-\epsilon} = \int_0^1 P_t(i)^{1-\epsilon} di$$

Dividing by $P_t^{1-\epsilon}$ we get:

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} di = 1$$

Notice that $\left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} = e^{\ln \left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon}} = e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)}$.

A first-order Taylor approximation of $e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)}$ around $\ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) = \ln(1) = 0$ gives:

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^{(1-\epsilon) \ln(1)} + e^{(1-\epsilon) \ln(1)} (1-\epsilon) \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) \right]$$

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^0 + e^0 (1-\epsilon) \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln(1) \right]$$

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 + (1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)$$

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 + (1-\epsilon) [\ln P_t(i) - \ln P_t]$$

Then:

$$\left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} \cong 1 + (1-\epsilon) [\ln P_t(i) - \ln P_t]$$

Then:

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} di \cong \int_0^1 [1 + (1-\epsilon) (\ln P_t(i) - \ln P_t)] di$$

$$1 \cong \int_0^1 [1 + (1-\epsilon) (\ln P_t(i) - \ln P_t)] di$$

$$1 \cong 1 + (1 - \epsilon) \int_0^1 [\ln P_t(i) - \ln P_t] di$$

$$0 \cong (1 - \epsilon) \int_0^1 [\ln P_t(i) - \ln P_t] di$$

Then:

$$\int_0^1 [\ln P_t(i) - \ln P_t] di \cong 0$$



Claim 2. Up to first order we have $d_t \cong 0$, where $d_t \equiv (1 - \alpha) \ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]$.

Proof. Notice that $\left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} = e^{\ln \left(\frac{P_t(i)}{P_t} \right) \frac{-\epsilon}{1-\alpha}}$. Then $\left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} = e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)}$.

A first-order Taylor approximation of $e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)}$ around $\ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) = \ln(1) = 0$ gives:

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^{-\frac{\epsilon}{1-\alpha} \ln(1)} + e^{-\frac{\epsilon}{1-\alpha} \ln(1)} \frac{-\epsilon}{1-\alpha} \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) \right]$$

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^0 + e^0 \frac{-\epsilon}{1-\alpha} \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln(1) \right]$$

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 - \frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)$$

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 - \frac{\epsilon}{1-\alpha} [\ln P_t(i) - \ln P_t]$$

Then:

$$\left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} \cong 1 - \frac{\epsilon}{1-\alpha} [\ln P_t(i) - \ln P_t]$$

Then:

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \cong \int_0^1 \left[1 - \frac{\epsilon}{1-\alpha} [\ln P_t(i) - \ln P_t] \right] di$$

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \cong 1 - \frac{\epsilon}{1-\alpha} \int_0^1 [\ln P_t(i) - \ln P_t] di$$

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \cong 1 \quad \text{by Claim 1.}$$

Then:

$$\ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \cong \ln(1)$$

Then:

$$\ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \cong 0$$

Then:

$$d_t \equiv (1-\alpha) \ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \cong (1-\alpha) 0$$

Then:

$$d_t \cong 0.$$



Intermediate step in the derivation of the NKPC

Claim 3. The following equality holds:

$$\sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t-1}) = \frac{1}{1-\theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1})$$

Proof. We have:

$$\begin{aligned} & \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t-1}) = \\ &= \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1} + \tilde{P}_{t+k-1} - \dots - \tilde{P}_{t+1} + \tilde{P}_{t+1} - \tilde{P}_t + \tilde{P}_t - \tilde{P}_{t-1}) \\ &= \sum_{k=0}^{\infty} (\theta\beta)^k \left[(\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) + (\tilde{P}_{t+k-1} - \tilde{P}_{t+k-2}) + \dots + (\tilde{P}_t - \tilde{P}_{t-1}) \right] \\ &= (\tilde{P}_t - \tilde{P}_{t-1}) + (\theta\beta) \left[(\tilde{P}_{t+1} - \tilde{P}_t) + (\tilde{P}_t - \tilde{P}_{t-1}) \right] + \\ & \quad + (\theta\beta)^2 \left[(\tilde{P}_{t+2} - \tilde{P}_{t+1}) + (\tilde{P}_{t+1} - \tilde{P}_t) + (\tilde{P}_t - \tilde{P}_{t-1}) \right] + \\ & \quad + (\theta\beta)^3 \left[(\tilde{P}_{t+3} - \tilde{P}_{t+2}) + (\tilde{P}_{t+2} - \tilde{P}_{t+1}) + (\tilde{P}_{t+1} - \tilde{P}_t) + (\tilde{P}_t - \tilde{P}_{t-1}) \right] + \dots \end{aligned}$$

$$\begin{aligned}
&= \left(\tilde{P}_t - \tilde{P}_{t-1} \right) \left[1 + \theta\beta + (\theta\beta)^2 + (\theta\beta)^3 + \dots \right] + \\
&\quad + \left(\tilde{P}_{t+1} - \tilde{P}_t \right) \theta\beta \left[1 + \theta\beta + (\theta\beta)^2 + (\theta\beta)^3 + \dots \right] + \\
&\quad + \left(\tilde{P}_{t+2} - \tilde{P}_{t+1} \right) (\theta\beta)^2 \left[1 + \theta\beta + (\theta\beta)^2 + (\theta\beta)^3 + \dots \right] + \dots \\
&= \left(\tilde{P}_t - \tilde{P}_{t-1} \right) \frac{1}{1-\theta\beta} + \left(\tilde{P}_{t+1} - \tilde{P}_t \right) \frac{\theta\beta}{1-\theta\beta} + \left(\tilde{P}_{t+2} - \tilde{P}_{t+1} \right) \frac{(\theta\beta)^2}{1-\theta\beta} + \dots \\
&= \frac{1}{1-\theta\beta} \left[\left(\tilde{P}_t - \tilde{P}_{t-1} \right) + \theta\beta \left(\tilde{P}_{t+1} - \tilde{P}_t \right) + (\theta\beta)^2 \left(\tilde{P}_{t+2} - \tilde{P}_{t+1} \right) + \dots \right] \\
&= \frac{1}{1-\theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1})
\end{aligned}$$

■

Solving the Basic New Keynesian Model using the Blanchard-Kahn Method

The three-equation model is:¹

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \quad \text{DAS}$$

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma} (\tilde{i}_t - \mathbb{E}_t \pi_{t+1} - \tilde{r}_t^n) \quad \text{DIS}$$

$$\tilde{i}_t = \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t \quad \text{TR}$$

where $\tilde{A}_{t+1} = \rho_a \tilde{A}_t + \varepsilon_{t+1}^A$, $v_{t+1} = \rho_v v_t + \varepsilon_{t+1}^v$ and $\tilde{r}_t^n = -\sigma \psi_y^n (1 - \rho_a) \tilde{A}_t$.²

Substituting TR into DIS and rearranging we obtain the following two-equation model:

$$\beta \mathbb{E}_t \pi_{t+1} = \pi_t - \kappa \tilde{y}_t \quad \text{DAS}$$

$$\mathbb{E}_t \pi_{t+1} + \sigma \mathbb{E}_t \tilde{y}_{t+1} = \phi_\pi \pi_t + (\sigma + \phi_y) \tilde{y}_t - \tilde{r}_t^n + v_t \quad \text{DAD}$$

Then:

$$\begin{bmatrix} \beta & 0 \\ 1 & \sigma \end{bmatrix} \begin{bmatrix} \mathbb{E}_t \pi_{t+1} \\ \mathbb{E}_t \tilde{y}_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -\kappa \\ \phi_\pi & \sigma + \phi_y \end{bmatrix} \begin{bmatrix} \pi_t \\ \tilde{y}_t \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \tilde{r}_t^n \\ v_t \end{bmatrix}$$

or:

$$A \mathbb{E}_t w_{t+1} = B w_t + C \eta_t$$

where

$$A \equiv \begin{bmatrix} \beta & 0 \\ 1 & \sigma \end{bmatrix}, \quad B \equiv \begin{bmatrix} 1 & -\kappa \\ \phi_\pi & \sigma + \phi_y \end{bmatrix}, \quad C \equiv \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix}, \quad w_t \equiv \begin{bmatrix} \pi_t \\ \tilde{y}_t \end{bmatrix}, \quad \eta_t \equiv \begin{bmatrix} \tilde{r}_t^n \\ v_t \end{bmatrix}$$

Since A is nonsingular, we can rewrite the expression above in the form of Blanchard & Kahn (1980). Premultiplying both sides by A^{-1} we get: $\mathbb{E}_t w_{t+1} = A^{-1} B w_t + A^{-1} C \eta_t \Rightarrow$

$$\mathbb{E}_t w_{t+1} = \Omega w_t + \Theta \eta_t$$

where

$$\begin{aligned} \Omega &\equiv A^{-1} B = \begin{bmatrix} \frac{1}{\beta} & 0 \\ -\frac{1}{\sigma\beta} & \frac{1}{\sigma} \end{bmatrix} \begin{bmatrix} 1 & -\kappa \\ \phi_\pi & \sigma + \phi_y \end{bmatrix} = \begin{bmatrix} \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\sigma\beta} & \frac{\beta(\sigma + \phi_y) + \kappa}{\sigma\beta} \end{bmatrix} \\ \Theta &\equiv A^{-1} C = \begin{bmatrix} \frac{1}{\beta} & 0 \\ -\frac{1}{\sigma\beta} & \frac{1}{\sigma} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -\frac{1}{\sigma} & \frac{1}{\sigma} \end{bmatrix} \end{aligned}$$

Now we want to diagonalize Ω . That is, we want to write

$$\Omega = \Gamma \Lambda \Gamma^{-1}$$

where Λ is a diagonal matrix with the eigenvalues of Ω in its diagonal, and Γ is the matrix having as columns the right-eigenvectors of Ω (see the Appendix).³

Using the expression above into $\mathbb{E}_t w_{t+1} = \Omega w_t + \Theta \eta_t$ we get:

$$\mathbb{E}_t w_{t+1} = \Gamma \Lambda \Gamma^{-1} w_t + \Theta \eta_t$$

Premultiplying by Γ^{-1} :

$$\Gamma^{-1} \mathbb{E}_t w_{t+1} = \Lambda \Gamma^{-1} w_t + \Gamma^{-1} \Theta \eta_t$$

Then:

$$\mathbb{E}_t (\Gamma^{-1} w_{t+1}) = \Lambda \Gamma^{-1} w_t + \Gamma^{-1} \Theta \eta_t$$

¹ See Chapter 3 in Galí (2008).

² Notice that $\tilde{r}_{t+1}^n = -\sigma \psi_y^n (1 - \rho_a) \tilde{A}_{t+1} \Rightarrow \tilde{r}_{t+1}^n = -\sigma \psi_y^n (1 - \rho_a) (\rho_a \tilde{A}_t + \varepsilon_{t+1}^A) \Rightarrow$

$\tilde{r}_{t+1}^n = -\sigma \psi_y^n (1 - \rho_a) \rho_a \tilde{A}_t - \sigma \psi_y^n (1 - \rho_a) \varepsilon_{t+1}^A \Rightarrow \tilde{r}_{t+1}^n = \rho_a \tilde{r}_t^n + \varepsilon_{t+1}^r$, where $\varepsilon_{t+1}^r \equiv -\sigma \psi_y^n (1 - \rho_a) \varepsilon_{t+1}^A$.

³ Instead of the standard diagonalization method, Blanchard & Kahn use the more general Jordan (canonical) decomposition (which applies also to non-diagonalizable matrices). See Chapter 21 in Sydsæter, Strøm & Berck (2005) for a concise exposition.

Then:

$$\mathbb{E}_t z_{t+1} = \Lambda z_t + \Phi \eta_t$$

where $z_t \equiv \Gamma^{-1} w_t$ and $\Phi \equiv \Gamma^{-1} \Theta$. Notice that each element of z_t is a linear combination of the elements of w_t (π_t and $\tilde{y}_t$).

The expression above gives:

$$\begin{bmatrix} \mathbb{E}_t z_{1,t+1} \\ \mathbb{E}_t z_{2,t+1} \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} z_{1,t} \\ z_{2,t} \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} \tilde{r}_t^n \\ v_t \end{bmatrix}$$

Then:

$$\mathbb{E}_t z_{1,t+1} = \lambda_1 z_{1,t} + \phi_{11} \tilde{r}_t^n + \phi_{12} v_t \quad (1)$$

$$\mathbb{E}_t z_{2,t+1} = \lambda_2 z_{2,t} + \phi_{21} \tilde{r}_t^n + \phi_{22} v_t \quad (2)$$

Notice that the diagonalization of Ω has allowed us to transform the original system of equations into two separate equations that can be solved individually.

Now we focus on (1).⁴ We first notice that we don't have an initial (boundary) condition for $z_{1,t}$.⁵ Instead, our boundary condition will be a no-explosion condition to be specified below. If $\lambda_1 > 1$, there will be a unique non-explosive solution that can be found by *forward* iteration. To find this solution we start by rewriting (1) as follows:⁶

$$z_{1,t} = -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) + \frac{1}{\lambda_1} \mathbb{E}_t z_{1,t+1}$$

Then:

$$\begin{aligned} z_{1,t} &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) + \frac{1}{\lambda_1} \mathbb{E}_t z_{1,t+1} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) + \frac{1}{\lambda_1} \mathbb{E}_t \left\{ -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_{t+1}^n + \phi_{12} v_{t+1}) + \frac{1}{\lambda_1} \mathbb{E}_{t+1} z_{1,t+2} \right\} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t \{ \phi_{11} \tilde{r}_{t+1}^n + \phi_{12} v_{t+1} \} + \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t \{ \mathbb{E}_{t+1} z_{1,t+2} \} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) + \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t z_{1,t+2} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) + \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t \left\{ -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_{t+2}^n + \phi_{12} v_{t+2}) + \frac{1}{\lambda_1} \mathbb{E}_{t+2} z_{1,t+3} \right\} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) - \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t \{ \phi_{11} \tilde{r}_{t+2}^n + \phi_{12} v_{t+2} \} + \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t \{ \mathbb{E}_{t+2} z_{1,t+3} \} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) - \left(\frac{1}{\lambda_1} \right)^3 (\phi_{11} \rho_a^2 \tilde{r}_t^n + \phi_{12} \rho_v^2 v_t) + \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t z_{1,t+3} \\ &= -\frac{\phi_{11}}{\lambda_1} \left[1 + \frac{\rho_a}{\lambda_1} + \left(\frac{\rho_a}{\lambda_1} \right)^2 \right] \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \left[1 + \frac{\rho_v}{\lambda_1} + \left(\frac{\rho_v}{\lambda_1} \right)^2 \right] v_t + \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t z_{1,t+3} \\ &\dots \\ &= -\frac{\phi_{11}}{\lambda_1} \left[1 + \frac{\rho_a}{\lambda_1} + \left(\frac{\rho_a}{\lambda_1} \right)^2 + \dots + \left(\frac{\rho_a}{\lambda_1} \right)^{j-1} \right] \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \left[1 + \frac{\rho_v}{\lambda_1} + \left(\frac{\rho_v}{\lambda_1} \right)^2 + \dots + \left(\frac{\rho_v}{\lambda_1} \right)^{j-1} \right] v_t + \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j} \end{aligned}$$

Then:

$$z_{1,t} = -\frac{\phi_{11}}{\lambda_1} \frac{1 - \left(\frac{\rho_a}{\lambda_1} \right)^j}{1 - \frac{\rho_a}{\lambda_1}} \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \frac{1 - \left(\frac{\rho_v}{\lambda_1} \right)^j}{1 - \frac{\rho_v}{\lambda_1}} v_t + \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j}$$

For $j \rightarrow \infty$ we get:

$$z_{1,t} = -\frac{\phi_{11}}{\lambda_1} \frac{1}{1 - \frac{\rho_a}{\lambda_1}} \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \frac{1}{1 - \frac{\rho_v}{\lambda_1}} v_t + \lim_{j \rightarrow \infty} \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j}$$

⁴The analysis for (2) is completely analogous.

⁵We know $z_{1,t}$ is a linear combination of π_t and $\tilde{y}_t$. But π_t and $\tilde{y}_t$ are nonpredetermined (jump) variables for which we don't have initial conditions.

⁶Notice that $\lambda_1 > 1 \Rightarrow \frac{1}{\lambda_1} < 1$.

Now we impose the following no-explosion condition:

$$\lim_{j \rightarrow \infty} \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j} = 0$$

Then:

$$z_{1,t} = -\frac{\phi_{11}}{\lambda_1} \frac{\lambda_1}{\lambda_1 - \rho_a} \hat{r}_t^n - \frac{\phi_{12}}{\lambda_1} \frac{\lambda_1}{\lambda_1 - \rho_v} v_t$$

Finally:

$$z_{1,t} = \frac{\phi_{11}}{\rho_a - \lambda_1} \hat{r}_t^n + \frac{\phi_{12}}{\rho_v - \lambda_1} v_t \quad (3)$$

which is the *unique* non-explosive solution for $z_{1,t}$.

Analogously, the solution for $z_{2,t}$ is:

$$z_{2,t} = \frac{\phi_{21}}{\rho_a - \lambda_2} \hat{r}_t^n + \frac{\phi_{22}}{\rho_v - \lambda_2} v_t \quad (4)$$

Now we want to uncover the solutions for the original variables. From $z_t \equiv \Gamma^{-1} w_t$ we get $w_t = \Gamma z_t$. Then:

$$\begin{bmatrix} \pi_t \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} z_{1,t} \\ z_{2,t} \end{bmatrix}$$

where the elements of Γ have been denoted by γ_{ij} .

Then:

$$\pi_t = \gamma_{11} z_{1,t} + \gamma_{12} z_{2,t} \quad (5)$$

$$\tilde{y}_t = \gamma_{21} z_{1,t} + \gamma_{22} z_{2,t} \quad (6)$$

Substituting (3) and (4) into (5) and (6) we get:

$$\begin{aligned} \pi_t &= \gamma_{11} \left(\frac{\phi_{11}}{\rho_a - \lambda_1} \hat{r}_t^n + \frac{\phi_{12}}{\rho_v - \lambda_1} v_t \right) + \gamma_{12} \left(\frac{\phi_{21}}{\rho_a - \lambda_2} \hat{r}_t^n + \frac{\phi_{22}}{\rho_v - \lambda_2} v_t \right) \\ \tilde{y}_t &= \gamma_{21} \left(\frac{\phi_{11}}{\rho_a - \lambda_1} \hat{r}_t^n + \frac{\phi_{12}}{\rho_v - \lambda_1} v_t \right) + \gamma_{22} \left(\frac{\phi_{21}}{\rho_a - \lambda_2} \hat{r}_t^n + \frac{\phi_{22}}{\rho_v - \lambda_2} v_t \right) \end{aligned}$$

Then:

$$\begin{aligned} \pi_t &= \left(\frac{\gamma_{11}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{12}\phi_{21}}{\rho_a - \lambda_2} \right) \hat{r}_t^n + \left(\frac{\gamma_{11}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{12}\phi_{22}}{\rho_v - \lambda_2} \right) v_t \\ \tilde{y}_t &= \left(\frac{\gamma_{21}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{22}\phi_{21}}{\rho_a - \lambda_2} \right) \hat{r}_t^n + \left(\frac{\gamma_{21}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{22}\phi_{22}}{\rho_v - \lambda_2} \right) v_t \end{aligned}$$

Then:

$$\pi_t = \psi_{\pi r} \hat{r}_t^n + \psi_{\pi v} v_t \quad (7)$$

$$\tilde{y}_t = \psi_{y r} \hat{r}_t^n + \psi_{y v} v_t \quad (8)$$

where

$$\begin{aligned} \psi_{\pi r} &\equiv \frac{\gamma_{11}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{12}\phi_{21}}{\rho_a - \lambda_2} \\ \psi_{\pi v} &\equiv \frac{\gamma_{11}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{12}\phi_{22}}{\rho_v - \lambda_2} \\ \psi_{y r} &\equiv \frac{\gamma_{21}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{22}\phi_{21}}{\rho_a - \lambda_2} \\ \psi_{y v} &\equiv \frac{\gamma_{21}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{22}\phi_{22}}{\rho_v - \lambda_2} \end{aligned}$$

Using the fact that $\tilde{r}_t^n = -\sigma\psi_y^n(1 - \rho_a)\tilde{A}_t$ we can rewrite (7) and (8) as follows:

$$\pi_t = \psi_{\pi a}\tilde{A}_t + \psi_{\pi v}v_t \quad (9)$$

$$\tilde{y}_t = \psi_{ya}\tilde{A}_t + \psi_{yv}v_t \quad (10)$$

where

$$\psi_{\pi a} \equiv -\psi_{\pi r}\sigma\psi_y^n(1 - \rho_a)$$

$$\psi_{ya} \equiv -\psi_{yr}\sigma\psi_y^n(1 - \rho_a)$$

Using the Matlab code `nlm_bk.m` you can check that this solution coincides with the one we found in class using the method of undetermined coefficients.

Appendix

Recall that an eigenvalue of a square matrix Ω is a (possibly complex) number λ for which there is a (possibly complex) non-zero vector $\mathbf{v}$ such that $\Omega\mathbf{v} = \lambda\mathbf{v}$.⁷ The vector $\mathbf{v}$ is called a right eigenvector of Ω . From $\Omega\mathbf{v} = \lambda\mathbf{v}$ we get $(\Omega - \lambda I)\mathbf{v} = \mathbf{0}$. Therefore, to compute the eigenvalues of Ω we need to solve the equation $\det(\Omega - \lambda I) = 0$.

For $\Omega = \begin{bmatrix} \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\beta\sigma} & \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \end{bmatrix}$, we need to solve:

$$\det \left(\begin{bmatrix} \frac{1}{\beta} - \lambda & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\beta\sigma} & \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} - \lambda \end{bmatrix} \right) = 0$$

Then:

$$\lambda^2 - \left(\frac{1}{\beta} + \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \right) \lambda + \left(\frac{\beta(\sigma + \phi_y) + \kappa}{\beta^2\sigma} + \frac{\kappa(\beta\phi_\pi - 1)}{\beta^2\sigma} \right) = 0$$

$$\lambda^2 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \lambda + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} = 0$$

where the left-hand-side is the characteristic polynomial of Ω .

Then, the roots of the characteristic polynomial are:

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \pm \sqrt{\left(\frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \right)^2 - 4 \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma}} \right)$$

Then:

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta\sigma + \beta\phi_y + \kappa}{\beta\sigma} \pm \sqrt{\frac{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2}{\beta^2\sigma^2} - 4 \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma}} \right)$$

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta\sigma + \beta\phi_y + \kappa}{\beta\sigma} \pm \sqrt{\frac{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4(\sigma + \phi_y + \kappa\phi_\pi)\beta\sigma}{\beta^2\sigma^2}} \right)$$

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta\sigma + \beta\phi_y + \kappa}{\beta\sigma} \pm \frac{1}{\beta\sigma} \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

$$\lambda_{1,2} = \frac{1}{2\beta\sigma} \left(\sigma + \beta\sigma + \beta\phi_y + \kappa \pm \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

Then:

$$\lambda_1 = \frac{1}{2\beta\sigma} \left(\sigma + \beta\sigma + \beta\phi_y + \kappa - \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

$$\lambda_2 = \frac{1}{2\beta\sigma} \left(\sigma + \beta\sigma + \beta\phi_y + \kappa + \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

With the eigenvalues we can find the eigenvectors. Evaluating $(\Omega - \lambda I)\mathbf{v} = \mathbf{0}$ at $\lambda = \lambda_1$ we get the right-eigenvector associated to λ_1 , denoted by $\mathbf{v}^1$. We have:

$$\begin{bmatrix} \frac{1}{\beta} - \lambda_1 & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\beta\sigma} & \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} - \lambda_1 \end{bmatrix} \begin{bmatrix} v_1^1 \\ v_2^1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Then:

$$\left(\frac{1}{\beta} - \lambda_1 \right) v_1^1 - \frac{\kappa}{\beta} v_2^1 = 0$$

$$\frac{\beta\phi_\pi - 1}{\beta\sigma} v_1^1 + \left(\frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} - \lambda_1 \right) v_2^1 = 0$$

⁷See Chapter 21 in Sydsæter, Strøm & Berck (2005) for a concise exposition.

By linear dependence (recall $\det(\Omega - \lambda I) = 0$) we can use any one of the two equations above. From the first one we get: $(\frac{1}{\beta} - \lambda_1)v_1^1 = \frac{\kappa}{\beta}v_2^1 \Rightarrow v_2^1 = \frac{\beta}{\kappa}(\frac{1}{\beta} - \lambda_1)v_1^1 \Rightarrow$

$$\mathbf{v}^1 = \begin{bmatrix} v_1^1 \\ \frac{\beta}{\kappa}(\frac{1}{\beta} - \lambda_1)v_1^1 \end{bmatrix}$$

Normalizing $v_1^1 = 1$ we get:

$$\mathbf{v}^1 = \begin{bmatrix} 1 \\ \frac{1}{\kappa}(1 - \beta\lambda_1) \end{bmatrix}$$

Analogously, for λ_2 we get:

$$\mathbf{v}^2 = \begin{bmatrix} 1 \\ \frac{1}{\kappa}(1 - \beta\lambda_2) \end{bmatrix}$$

Now we can construct the matrix of eigenvectors as follows:

$$\Gamma = \begin{bmatrix} 1 & 1 \\ \frac{1}{\kappa}(1 - \beta\lambda_1) & \frac{1}{\kappa}(1 - \beta\lambda_2) \end{bmatrix}$$

It is not difficult, but tedious, to show that $\Gamma\Lambda\Gamma^{-1} = \Omega$.⁸

Finally, we want to say something about the sign and size of the eigenvalues. We first notice that the characteristic polynomial can be written as follows:

$$\begin{aligned} P(\lambda) &= \lambda^2 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma}\lambda + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} \\ &= \lambda^2 - \text{tr}(\Omega)\lambda + \det(\Omega) \\ &= \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2 \\ &= (\lambda - \lambda_1)(\lambda - \lambda_2) \end{aligned}$$

where we have used $\lambda_1 + \lambda_2 = \text{tr}(\Omega)$ and $\lambda_1\lambda_2 = \det(\Omega)$.

Now we notice that

$$\begin{aligned} \lambda_1 + \lambda_2 &= \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} > 1 \\ \lambda_1\lambda_2 &= \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} > 1 \end{aligned}$$

because in both cases the numerator is higher than σ while the denominator is strictly smaller than σ .

From $\lambda_1\lambda_2 > 1 > 0$ we learn that both eigenvalues must have the same sign. Then, $\lambda_1 + \lambda_2 > 1$ implies that both eigenvalues must be positive. Then, $\lambda_1\lambda_2 > 1$ implies that either $\lambda_1 > 1$ or $\lambda_2 > 1$ (or both). Now we evaluate $P(\lambda)$ at $\lambda = 1$ to get: $P(1) = (1 - \lambda_1)(1 - \lambda_2)$. Since either $\lambda_1 > 1$ or $\lambda_2 > 1$, it follows that $P(1) > 0 \Leftrightarrow \lambda_1 > 1$ and $\lambda_2 > 1$. But $P(1) = 1 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma}$. Then, $P(1) > 0 \Leftrightarrow 1 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} > 0$. Then: $P(1) > 0 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$. Finally, we conclude:

$$\lambda_1 > 1 \text{ and } \lambda_2 > 1 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$$

which gives a necessary and sufficient condition for the existence of a unique non-explosive solution. Notice that a sufficient (but not necessary) condition is $\phi_\pi > 1$.

We can give some intuition for the expression above. Consider a situation with constant inflation and output gap. From the Taylor Rule we get: $\frac{\partial i}{\partial \pi} = \phi_\pi + \phi_y \frac{\partial \tilde{y}}{\partial \pi}$. But the Phillips Curve gives: $\pi = \beta\pi + \kappa\tilde{y} \Rightarrow \tilde{y} = \frac{1-\beta}{\kappa}\pi \Rightarrow \frac{\partial \tilde{y}}{\partial \pi} = \frac{1-\beta}{\kappa}$. Substituting into the expression for $\frac{\partial i}{\partial \pi}$ we get: $\frac{\partial i}{\partial \pi} = \phi_\pi + \phi_y \frac{1-\beta}{\kappa}$. Then, $\frac{\partial i}{\partial \pi} > 1 \Leftrightarrow \phi_\pi + \phi_y \frac{1-\beta}{\kappa} > 1$. But $\phi_\pi + \phi_y \frac{1-\beta}{\kappa} > 1 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$. Then:

$$\frac{\partial i}{\partial \pi} > 1 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$$

Hence, the nominal interest rate must increase more than one-for-one with the inflation rate. In other words, the *real* interest rate must increase in response to inflation.

⁸ Notice that $\det(\Gamma) = \frac{\beta}{\kappa}(\lambda_1 - \lambda_2)$. Hence, for Γ to be invertible we need $\lambda_1 \neq \lambda_2$.

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Search Theory and Unemployment

The Model

Partial Equilibrium

Infinite Horizon

Continuous time

Two states:

e = employed

u = unemployed (not working and searching for a job)

Preferences:

$$U = \mathbb{E}_0 \int_0^{\infty} e^{-\rho t} y(t) dt$$

where

$$y(t) = \begin{cases} w(t) & \text{if employed at } t \\ z & \text{if unemployed at } t \end{cases}$$

Discrete-time version: period of length h (small).

We'll let $h \rightarrow 0$ to get the continuous-time version

$\lambda h =$ prob. of receiving a job offer while unemployed

$1 - \lambda h =$ prob. of receiving no offer

$w \in [\underline{w}, \overline{w}] =$ wage attached to the job offer

$F(w) = \Pr\{W \leq w\}$ cdf wages ($f(w) \equiv F'(w)$ pdf)

$wh =$ wage income per period if employed at wage w

$\alpha h =$ prob. of being fired while employed

$1 - \alpha h =$ prob. keeping the job

The wage is constant as long as the job is kept

$zh = (b - c)h =$ net income per period if unemployed

Remark

More precisely, employment offers follow a time-homogeneous Poisson process with arrival rate λ . Hence, the probability of receiving n offers during time-interval h is:

$$q(n, h) = \frac{(\lambda h)^n}{n!} e^{-\lambda h}$$

Remark. $\mathbb{E}\{n\} = \lambda h$, $Var\{n\} = \lambda h$.

Hence:

$$q(0, h) = e^{-\lambda h}$$

$$q(1, h) = \lambda h e^{-\lambda h}$$

$$q(n \geq 2, h) = 1 - (1 + \lambda h) e^{-\lambda h}$$

A first-order Taylor approximation of $q(0, h) = e^{-\lambda h}$ around $h = 0$ gives:

$$q(0, h) \cong q(0, 0) + q_h(0, 0)(h - 0)$$

$$q(0, h) \cong e^{-\lambda 0} - \lambda e^{-\lambda 0}(h - 0) \Rightarrow$$

Then:

$$q(0, h) \cong 1 - \lambda h$$

A first-order Taylor approximation of $q(1, h) = \lambda h e^{-\lambda h}$ around $h = 0$ gives:

$$q(1, h) \cong q(1, 0) + q_h(1, 0)(h - 0)$$

$$q(1, h) \cong \lambda 0 e^{-\lambda 0} + (\lambda e^{-\lambda 0} - \lambda 0 e^{-\lambda 0} \lambda)(h - 0) \Rightarrow$$

$$q(1, h) \cong \lambda h$$

It follows that $q(n \geq 2, h) = 1 - q(0, h) - q(1, h) \cong 1 - (1 - \lambda h) - \lambda h \Rightarrow$

$$q(n \geq 2, h) \cong 0$$

Hence, during a sufficiently short period of time, an unemployed agent essentially receives either zero offers or just one offer.

During time-interval h , an employed worker is fired with probability $p(h) = 1 - e^{-\alpha h}$, and remains employed with probability $1 - p(h) = e^{-\alpha h}$.

A first-order Taylor approximation of $p(h)$ around $h = 0$ gives:

$$p(h) \cong p(0) + p'(0)(h - 0)$$

$$p(h) \cong 1 - e^{-\alpha 0} + \alpha e^{-\alpha 0}(h - 0)$$

$$p(h) \cong 1 - 1 + \alpha h \Rightarrow$$

$$p(h) \cong \alpha h$$

Then:

$$1 - p(h) \cong 1 - \alpha h$$

Discount Factor:

$$e^{-\rho h} = \frac{1}{e^{\rho h}}$$

A first-order Taylor approximation of $e^{\rho h}$ around $h = 0$ gives:

$$e^{\rho h} \cong e^{\rho 0} + e^{\rho 0} \rho (h - 0)$$

$$e^{\rho h} \cong 1 + \rho h$$

Then:

$$e^{-\rho h} \cong \frac{1}{1 + \rho h}$$

Back to the Model

Notice that the only decision to be made by an agent is to decide whether to accept an offer while unemployed (and stop looking for a job) or reject it (and keep looking for another opportunity). The agent is not allowed to search for a new job while employed (on-the-job search is ruled out by assumption).

We'll show that the optimal decision rule is to set a reservation wage and accept any offer with an attached wage higher or equal than the reservation wage.

Define:

$V_e(w, t, h)$ = value of employment at time t , with wage w , for a period of length h

$V_u(t, h)$ = value of unemployment at time t , for a period of length h .

Then:

$$V_e(w, t, h) = wh + \frac{1}{1 + \rho h} [\alpha h V_u(t + h, h) + (1 - \alpha h) V_e(w, t + h, h)]$$

Since the problem is stationary, $V_e(w, t, h)$ and $V_u(t, h)$ are the same for all t . Therefore, we can simply write $V_e(w, h)$ and $V_u(h)$. Then:

$$V_e(w, h) = wh + \frac{1}{1 + \rho h} [\alpha h V_u(h) + (1 - \alpha h) V_e(w, h)]$$

The expression above gives:

$$V_e(w, h) (1 - \frac{1 - \alpha h}{1 + \rho h}) = wh + \frac{\alpha h}{1 + \rho h} V_u(h)$$

$$V_e(w, h) \frac{(\rho + \alpha)h}{1 + \rho h} = wh + \frac{\alpha h}{1 + \rho h} V_u(h) \Rightarrow$$

$$V_e(w, h) = \frac{1 + \rho h}{\alpha + \rho} w + \frac{\alpha}{\alpha + \rho} V_u(h)$$

Continuous time: $h \rightarrow 0$

$$V_e(w) = \frac{w}{\alpha + \rho} + \frac{\alpha}{\alpha + \rho} V_u$$

where $V_e(w) \equiv V_e(w, 0)$ and $V_u \equiv V_u(0)$.

The expression above can be rewritten as follows:

$$\rho V_e(w) = w + \alpha[V_u - V_e(w)]$$

Asset-Pricing Interpretation

ρ = risk-free rate

$V_e(w)$ = current price

w = dividend

$\alpha[V_u - V_e(w)]$ = expected capital gain/loss

Risk-neutral investor $\Rightarrow \rho = \frac{w + \alpha(V_u - V_e)}{V_e}$

Or: $\rho = \frac{w + [\alpha V_u + (1 - \alpha)V_e] - V_e}{V_e}$

$V_u(h)$ must satisfy:

$$V_u(h) = zh + \frac{1}{1+\rho h} [(1-\lambda h)V_u(h) + \lambda h \mathbb{E} \max\{V_e(W, h), V_u(h)\}]$$

Then:

$$V_u(h)(1 - \frac{1-\lambda h}{1+\rho h}) = zh + \frac{\lambda h}{1+\rho h} \mathbb{E} \max\{V_e(W, h), V_u(h)\}$$

$$V_u(h) \frac{(\rho+\lambda)h}{1+\rho h} = zh + \frac{\lambda h}{1+\rho h} \mathbb{E} \max\{V_e(W, h), V_u(h)\}$$

$$V_u(h) = \frac{1+\rho h}{(\rho+\lambda)h} zh + \frac{1+\rho h}{(\rho+\lambda)h} \frac{\lambda h}{1+\rho h} \mathbb{E} \max\{V_e(W, h), V_u(h)\} \Rightarrow$$

$$V_u(h) = \frac{1+\rho h}{\rho+\lambda} z + \frac{\lambda}{\rho+\lambda} \mathbb{E} \max\{V_e(W, h), V_u(h)\}$$

Continuous time: $h \rightarrow 0$

$$V_u = \frac{z}{\rho + \lambda} + \frac{\lambda}{\rho + \lambda} \mathbb{E} \max\{V_e(W), V_u\}$$

From the expression above we get:

$$(\rho + \lambda)V_u = z + \lambda \mathbb{E} \max\{V_e(W), V_u\}$$

$$\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u$$

$$\rho V_u = z + \lambda [\mathbb{E} \max\{V_e(W), V_u\} - V_u]$$

$$\rho V_u = z + \lambda \mathbb{E} \{\max\{V_e(W), V_u\} - V_u\} \Rightarrow$$

$$\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W) - V_u, 0\}$$

Or:

$$\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u$$

$$\rho = \frac{z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u}{V_u}$$

$$\rho = \frac{z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u + V_u - V_u}{V_u} \Rightarrow$$

$$\rho = \frac{z + [\lambda \mathbb{E} \max\{V_e(W), V_u\} + (1 - \lambda)V_u] - V_u}{V_u}$$

Remark. Analogous asset-pricing interpretation.

Define:

$$\omega \equiv \rho V_u$$

From $V_e(w) = \frac{w}{\alpha+\rho} + \frac{\alpha}{\alpha+\rho}V_u$ we get:

$$V_e(w) - V_u = \frac{w}{\alpha+\rho} + \frac{\alpha}{\alpha+\rho}V_u - V_u$$

$$V_e(w) - V_u = \frac{w}{\alpha+\rho} - \frac{\rho V_u}{\alpha+\rho}$$

$$V_e(w) - V_u = \frac{w}{\alpha+\rho} - \frac{\omega}{\alpha+\rho} \Rightarrow$$

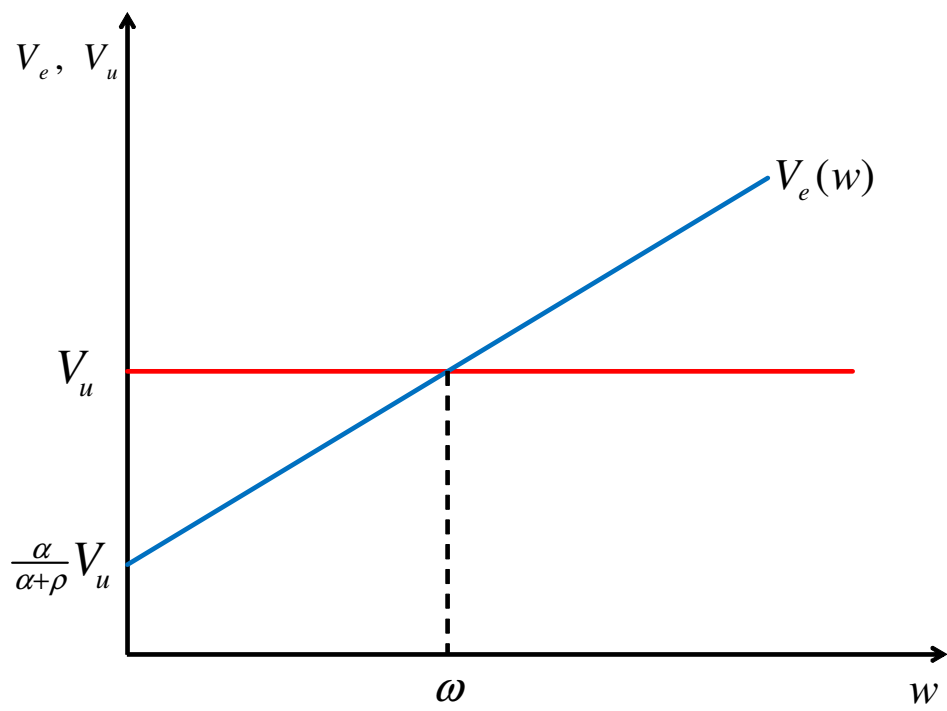
$$V_e(w) - V_u = \frac{w - \omega}{\rho + \alpha}$$

Therefore:

$$V_e(w) > V_u \text{ if and only if } w > \omega$$

Hence, it is optimal to accept an employment offer and become employed (stop searching) when the wage attached to the offer is above ω . That's why ω is called the *reservation wage*.

Plot



Using the definition of ω we can rewrite $\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W) - V_u, 0\}$ as follows:

$$\omega - z = \lambda \mathbb{E} \max\{V_e(W) - V_u, 0\}$$

Using $V_e(w) - V_u = \frac{w-\omega}{\rho+\alpha}$ we get:

$$\omega - z = \lambda \mathbb{E} \max\{\frac{W-\omega}{\rho+\alpha}, 0\} \Rightarrow$$

$$\omega - z = \frac{\lambda}{\rho + \alpha} \mathbb{E} \max\{W - \omega, 0\}$$

The expression above determines the value of ω (and then the value of V_u , since $V_u = \frac{\omega}{\rho}$)

Interpretation:

l-h-s: marginal cost of search when wage offer is ω

r-h-s: marginal return to search when wage offer is ω

Notice that:

$$\mathbb{E} \max\{W - \omega, 0\} = \int_{\underline{w}}^{\bar{w}} \max\{w - \omega, 0\} f(w) dw$$

$$\mathbb{E} \max\{W - \omega, 0\} = \int_{\underline{w}}^{\omega} 0 f(w) dw + \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

$$\mathbb{E} \max\{W - \omega, 0\} = \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

Or:

$$\omega - z = \frac{\lambda[1 - F(\omega)]}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) \frac{f(w)}{1 - F(\omega)} dw$$

$$\omega - z = \frac{\lambda[1 - F(\omega)]}{\rho + \alpha} \left[\int_{\omega}^{\bar{w}} w \frac{f(w)}{1 - F(\omega)} dw - \omega \int_{\omega}^{\bar{w}} \frac{f(w)}{1 - F(\omega)} dw \right] \Rightarrow$$

$$\omega - z = \frac{\lambda[1 - F(\omega)]}{\rho + \alpha} [\mathbb{E}\{w \mid w \geq \omega\} - \omega]$$

Integrating $\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$ by parts we get:

$$u = w - \omega, \quad v = F(w)$$

$$du = dw, \quad dv = F'(w) dw = f(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = (w - \omega) F(w) \Big|_{\omega}^{\bar{w}} - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = (\bar{w} - \omega) F(\bar{w}) - (\omega - \omega) F(\omega) - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = (\bar{w} - \omega) - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = \int_{\omega}^{\bar{w}} dw - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = \int_{\omega}^{\bar{w}} [1 - F(w)] dw$$

Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} [1 - F(w)] dw$$

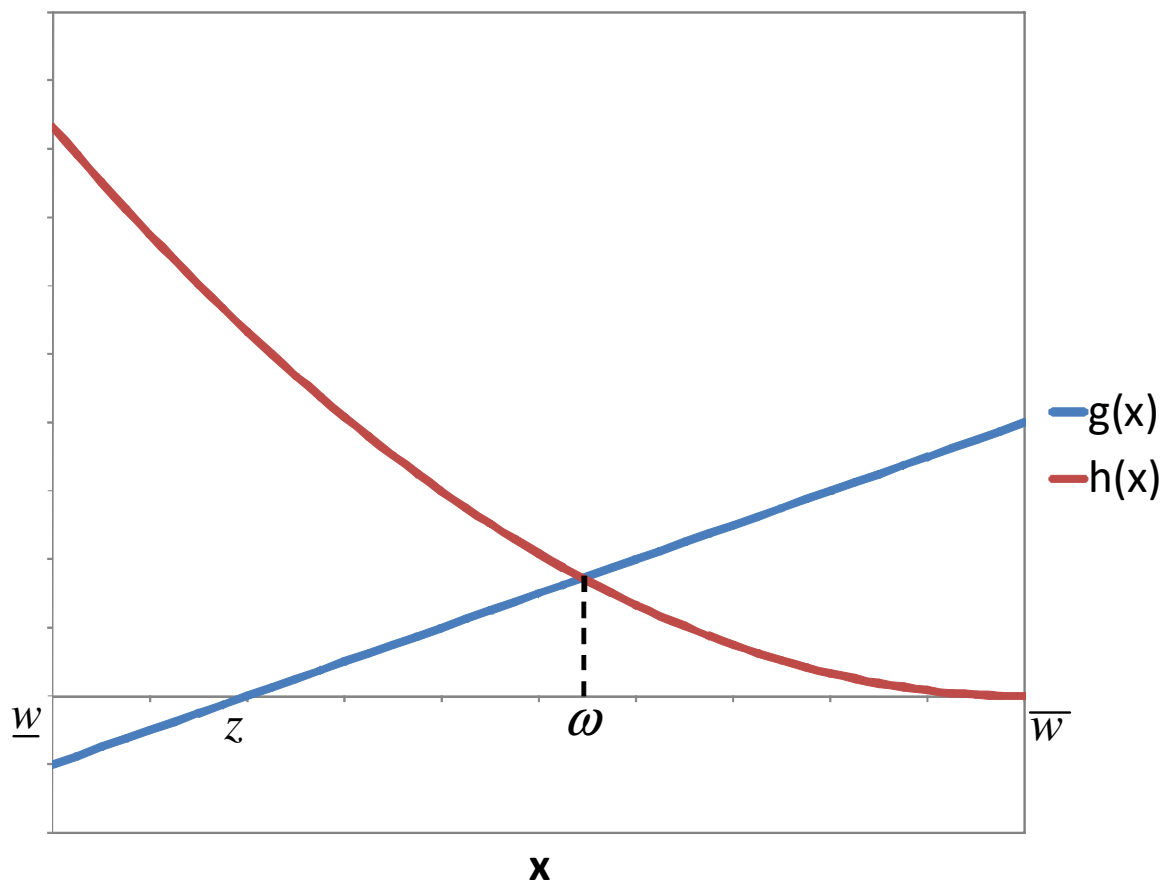
Define:

$$g(x) = x - z \quad \forall x \in [\underline{w}, \bar{w}]$$

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\bar{w}} [1 - F(w)] dw \quad \forall x \in [\underline{w}, \bar{w}]$$

$$= \frac{\lambda}{\rho + \alpha} \int_x^{\bar{w}} (w - x) f(w) dw \quad \forall x \in [\underline{w}, \bar{w}]$$

Then:



Remark:

$$h(\underline{w}) = \frac{\lambda}{\alpha + \rho}(\mu - \underline{w}) > 0 \quad \text{where } \mu \equiv \mathbb{E}W$$

$$h(\overline{w}) = 0$$

$$h'(x) = \frac{-\lambda}{\alpha + \rho}[1 - F(x)] \quad \begin{cases} < 0 & \forall x \in [\underline{w}, \overline{w}) \\ = 0 & \text{at } x = \overline{w} \end{cases}$$

$$h''(x) = \frac{\lambda}{\alpha + \rho}f(x) > 0 \quad \forall x \in [\underline{w}, \overline{w}]$$

Unemployment

Define:

$$\theta_u \equiv \lambda[1 - F(\omega)]$$

$$\theta_e \equiv \alpha$$

Then:

$$\theta_u h = \lambda h[1 - F(\omega)]$$

$$\theta_e h = \alpha h$$

We have:

$$\lambda h = \text{prob. offer}$$

$$1 - F(\omega) = \text{prob. accepting an offer } (w > \omega) \\ \text{conditional on receiving one}$$

$$\theta_u h = \text{prob. leaving unemployment}$$

$$\theta_e h = \alpha h = \text{prob. leaving employment}$$

Consider an economy populated by a continuum of mass L of *ex ante* identical agents.

Define:

$$\begin{aligned} E(t) &= \text{employment at } t \\ U(t) &= \text{unemployment at } t \\ L &= \text{labor force (fixed)} \end{aligned}$$

Then:

$$E(t) + U(t) = L \text{ for all } t$$

Then:

$$U(t + h) - U(t) = \theta_e h E(t) - \theta_u h U(t)$$

Dividing by h :

$$\frac{U(t + h) - U(t)}{h} = \theta_e E(t) - \theta_u U(t)$$

Letting $h \rightarrow 0$:

$$\dot{U}(t) = \theta_e E(t) - \theta_u U(t)$$

In steady-state: $E(t) = E$, $U(t) = U$, and $\dot{U}(t) = 0$.

Define:

$$u \equiv \frac{U}{L} = \text{steady-state unemployment rate}$$

Setting $\dot{U}(t) = 0$ and using $E = L - U$, we obtain:

$$0 = \theta_e E - \theta_u U$$

$$0 = \theta_e (L - U) - \theta_u U$$

$$0 = \theta_e L - (\theta_e + \theta_u)U$$

$$(\theta_e + \theta_u)U = \theta_e L$$

$$\frac{U}{L} = \frac{\theta_e}{\theta_e + \theta_u} \Rightarrow$$

$$\begin{aligned} u &= \frac{\theta_e}{\theta_e + \theta_u} \\ &= \frac{\alpha}{\alpha + \lambda[1 - F(\omega)]} \end{aligned}$$

Expected duration of employment and unemployment spells

Let T_u denote the (random) duration of a completed unemployment spell. Then (see the Appendix):

$$\begin{aligned} D_u(t) &\equiv \Pr\{T_u \leq t\} \\ &= 1 - e^{-\theta_u t} \end{aligned}$$

The associated density function, $d_u(t) = D'_u(t)$, is:

$$d_u(t) = \theta_u e^{-\theta_u t}$$

From the expression above we can calculate the expected duration of completed unemployment spells. We get:

$$\begin{aligned}\mathbb{E}T_u &= \int_0^\infty t\theta_u e^{-\theta_u t} dt \\ &= \frac{1}{\theta_u}\end{aligned}$$

Remark. Integrate by parts using $u = t$ and $v = -e^{-\theta_u t}$.

Analogously, denoting by T_e the duration of completed employment spells, we get:

$$\mathbb{E}T_e = \frac{1}{\theta_e}$$

Substituting $\mathbb{E}T_u = \frac{1}{\theta_u}$ and $\mathbb{E}T_e = \frac{1}{\theta_e}$ into $u = \frac{\theta_e}{\theta_e + \theta_u}$ we get:

$$\begin{aligned}u &= \frac{\frac{1}{\mathbb{E}T_e}}{\frac{1}{\mathbb{E}T_e} + \frac{1}{\mathbb{E}T_u}} = \frac{\frac{1}{\mathbb{E}T_e}}{\frac{\mathbb{E}T_u + \mathbb{E}T_e}{\mathbb{E}T_e \mathbb{E}T_u}} \Rightarrow \\ u &= \frac{\mathbb{E}T_u}{\mathbb{E}T_u + \mathbb{E}T_e}\end{aligned}$$

Comparative Statics

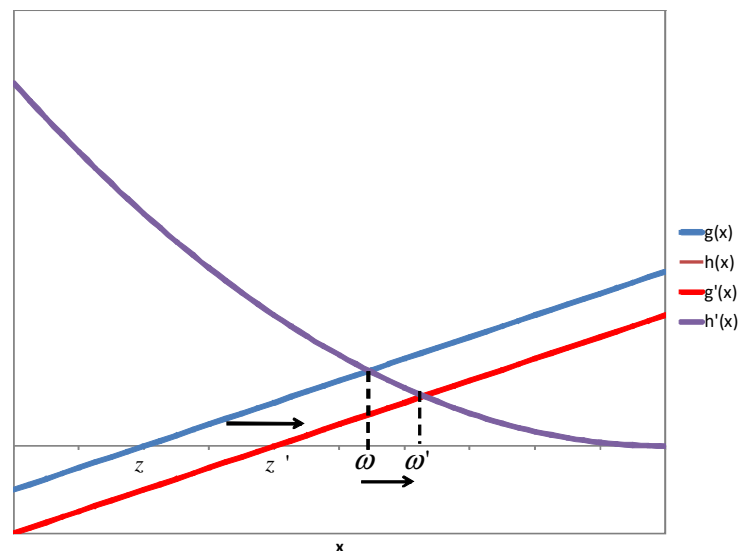
Recall that:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} [1 - F(w)] dw$$

Change in z

$$\frac{\partial \omega}{\partial z} = \frac{\rho + \alpha}{\rho + \alpha + \theta_u} \in (0, 1)$$

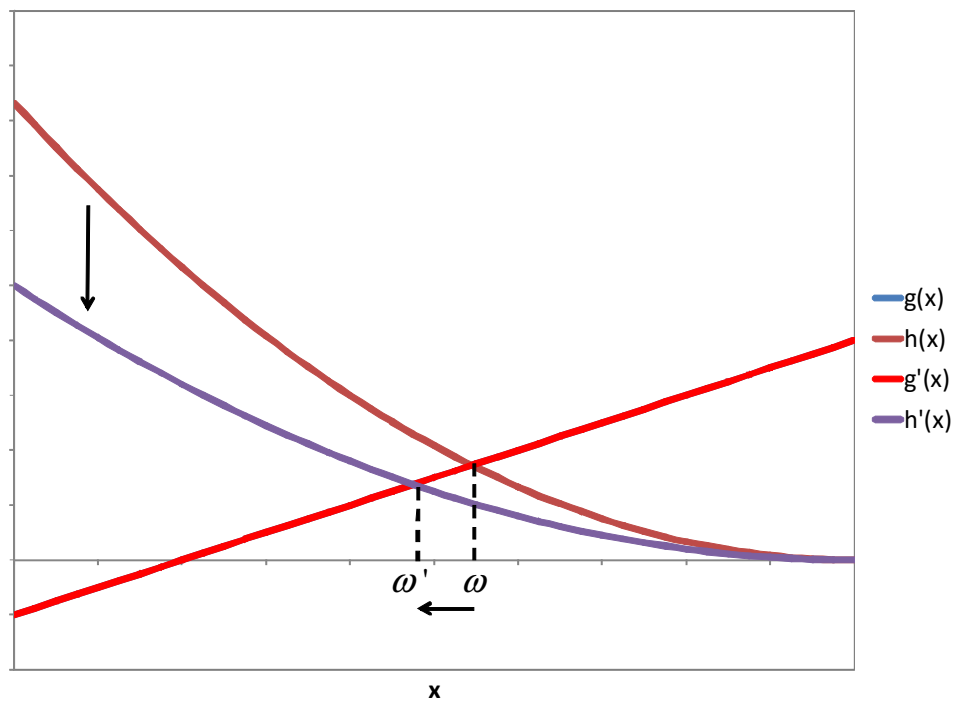
$$\begin{aligned} \uparrow z = b - c &\Rightarrow \uparrow \omega \Rightarrow \uparrow F(\omega) \Rightarrow \downarrow 1 - F(\omega) \Rightarrow \\ \downarrow \theta_u = \lambda[1 - F(\omega)] &\Rightarrow \uparrow u = \frac{\theta_e}{\theta_e + \theta_u} \end{aligned}$$



Change in ρ

$$\frac{\partial \omega}{\partial \rho} = -\frac{\omega - z}{\rho + \alpha + \theta_u} < 0$$

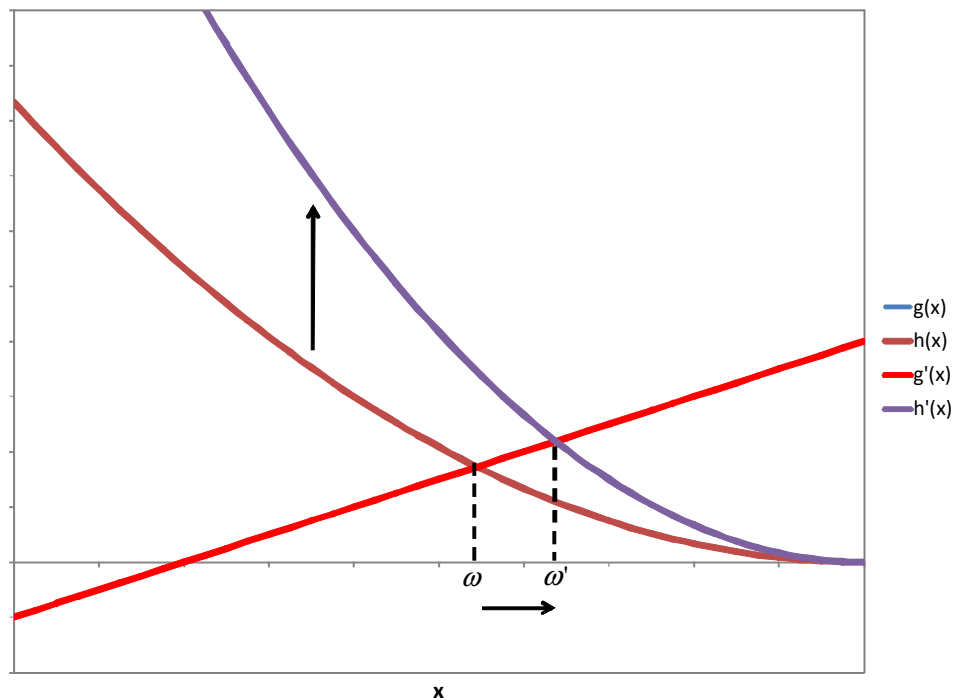
$$\uparrow \rho \Rightarrow \downarrow \omega \Rightarrow \downarrow F(\omega) \Rightarrow \uparrow 1 - F(\omega) \Rightarrow \uparrow \theta_u = \lambda[1 - F(\omega)] \Rightarrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$$



Change in λ

$$\begin{aligned}\frac{\partial \omega}{\partial \lambda} &= \frac{1}{\rho + \alpha + \theta_u} \int_{\omega}^{\bar{w}} [1 - F(w)] dw > 0 \\ &= \frac{\rho + \alpha}{\rho + \alpha + \theta_u} \frac{\omega - z}{\lambda} > 0\end{aligned}$$

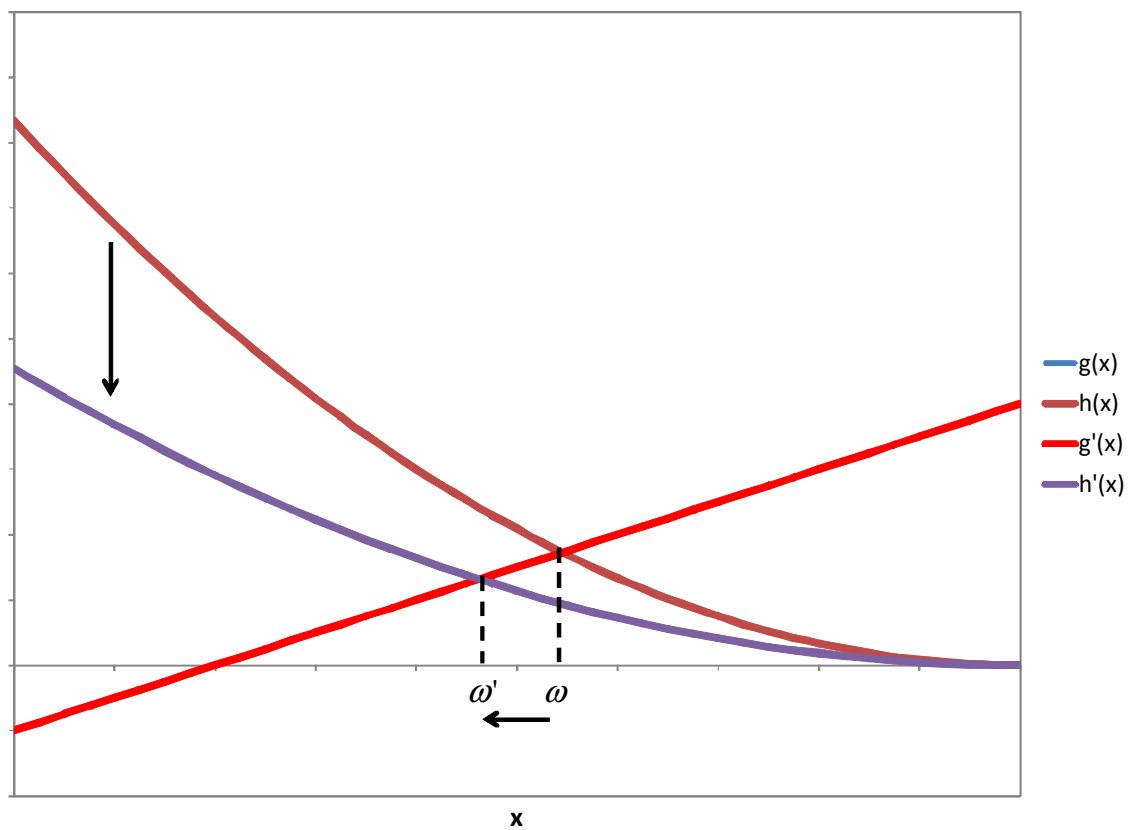
$$\uparrow \lambda \Rightarrow \uparrow \omega \Rightarrow \uparrow F(\omega) \Rightarrow \downarrow 1 - F(\omega) \Rightarrow \downarrow \uparrow \theta_u = \lambda[1 - F(\omega)] \Rightarrow \uparrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$$



Change in α

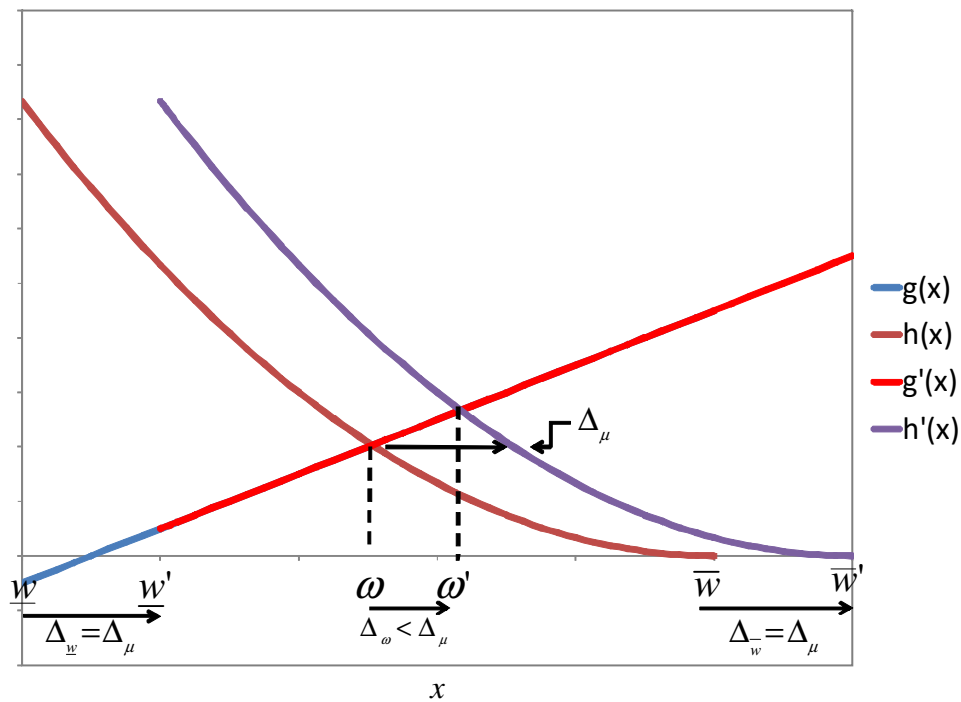
$$\frac{\partial \omega}{\partial \alpha} = -\frac{\omega - z}{\rho + \alpha + \theta_u} < 0$$

$$\left. \begin{array}{l} \uparrow \alpha \Rightarrow \downarrow \omega \Rightarrow \downarrow F(\omega) \Rightarrow \uparrow 1 - F(\omega) \Rightarrow \uparrow \theta_u = \lambda[1 - F(\omega)] \\ \uparrow \alpha \Rightarrow \uparrow \theta_e \end{array} \right\} \Rightarrow \uparrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$$



A shift to the right of the wage distribution

The mean $\mu \equiv \mathbb{E}W$ increases and the support shifts too.



Then: $\uparrow \mu \Rightarrow \uparrow \omega$

Then: $\begin{cases} \uparrow \mu \Rightarrow \downarrow F(\omega; \mu) \text{ for given } \omega \\ \uparrow \omega \Rightarrow \uparrow F(\omega; \mu) \text{ for given } \mu \end{cases}$

Since $\uparrow \omega < \uparrow \mu$ we know $\downarrow F(\omega; \mu) \Rightarrow \uparrow 1 - F(\omega; \mu) \Rightarrow \uparrow \theta_u = \lambda[1 - F(\omega; \mu)] \Rightarrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$

Appendix

Expected Duration of Employment and Unemployment Spells

Let T_u denote the (random) duration of a completed unemployment spell, and let D_u denote its cumulative distribution function: $D_u(t) \equiv \Pr\{T_u \leq t\}$. Consider the following question: Given that the unemployment spell has already lasted t units of time, what is the probability that it will end in the next interval $h > 0$? In other words, we are interested in

$$\Pr\{T_u \in [t, t+h] \mid T_u \geq t\}.$$

Using the law of conditional probabilities we can rewrite the expression above as follows:

$$\begin{aligned} \Pr\{T_u \in [t, t+h] \mid T_u \geq t\} &= \frac{\Pr\{T_u \in [t, t+h] \text{ and } T_u \geq t\}}{\Pr\{T_u \geq t\}} \\ &= \frac{\Pr\{T_u \in [t, t+h]\}}{\Pr\{T_u \geq t\}} \\ &= \frac{\Pr\{T_u \leq t+h\} - \Pr\{T_u \leq t\}}{1 - \Pr\{T_u \leq t\}} \quad 1 \\ &= \frac{D_u(t+h) - D_u(t)}{1 - D_u(t)} \end{aligned}$$

It is standard to characterize this aspect of the distribution using the *hazard function*

$$\theta_u(t) \equiv \lim_{h \rightarrow 0} \frac{\Pr\{T_u \in [t, t+h] \mid T_u \geq t\}}{h}$$

Using our previous result we can write:

$$\begin{aligned} \theta_u(t) &= \lim_{h \rightarrow 0} \frac{D_u(t+h) - D_u(t)}{[1 - D_u(t)]h} \\ &= \frac{1}{1 - D_u(t)} \lim_{h \rightarrow 0} \frac{D_u(t+h) - D_u(t)}{h} \\ &= \frac{D'_u(t)}{1 - D_u(t)} \\ &= \frac{d_u(t)}{S_u(t)} \end{aligned}$$

where $d_u(t) \equiv D'_u(t)$ is the density function and $S_u(t) \equiv 1 - D_u(t) = \Pr\{T_u > t\}$ is the so-called *survival function*.²

¹With no mass points, $\Pr\{T_u = t\} = 0$. Then: $\Pr\{T_u \leq t\} = \Pr\{T_u < t\}$.

²The terms *hazard* and *survival* come from biostatistics, where it is standard to ask for the probability that an individual dies during interval $[t, t+h]$ given that it has already survived up to t . When talking about unemployment spells, this vocabulary is a bit counterintuitive because a higher probability of survival means a higher probability of remaining unemployed.

Notice that: $\frac{d \ln S_u(\tau)}{d\tau} = \frac{1}{S_u(\tau)} S'_u(\tau) = \frac{-D'_u(\tau)}{S_u(\tau)} = -\frac{d_u(\tau)}{S_u(\tau)} = -\theta_u(\tau)$. Then:

$$\theta_u(\tau) = -\frac{d \ln S_u(\tau)}{d\tau}$$

Integrating both sides of the expression above from 0 to t we get:

$$S_u(t) = e^{-\int_0^t \theta_u(\tau) d\tau}$$

The expression above, together with $d_u(t) = \theta_u(t)S_u(t)$, gives:

$$d_u(t) = \theta_u(t)e^{-\int_0^t \theta_u(\tau) d\tau}$$

The hazard function evaluated at t gives the *rate* at which unemployment spells are terminated, given that they have already lasted until t . In our search model, the rate at which a worker escapes unemployment equals the rate at which employment offers arrive, multiplied by the probability that the offer is accepted (conditional on arrival). Since offers arrive at rate λ and the probability of acceptance is $1 - F(\omega)$, we get:

$$\theta_u = \lambda[1 - F(\omega)].$$

Notice that the reservation wage is stationary, F does not change over time, and λ is constant. Hence, the hazard rate is independent of the elapsed duration of the unemployment spell. In this case we have $\int_0^t \theta_u(\tau) d\tau = \int_0^t \theta_u d\tau = \theta_u t$. Hence, $d_u(t) = \theta_u(t)e^{-\int_0^t \theta_u(\tau) d\tau}$ becomes:

$$d_u(t) = \theta_u e^{-\theta_u t}$$

which shows that the distribution of completed unemployment spells is exponential, with parameter θ_u . The associated survival function is

$$S_u(t) = e^{-\theta_u t}.$$

Once we've found the distribution of completed unemployment spells, we can calculate their expected duration:

$$\begin{aligned} \mathbb{E}T_u &= \int_0^\infty t d_u(t) dt \\ &= \int_0^\infty t \theta_u e^{-\theta_u t} dt. \end{aligned}$$

Integrating by parts, using $u = t$ and $v = -e^{-\theta_u t}$, we obtain:

$$\mathbb{E}T_u = \frac{1}{\theta_u}.$$

It can also be shown that:

$$VarT_u = \frac{1}{\theta_u^2}.$$

We can repeat the calculations for employment spells. Letting T_e denote the duration of completed employment spells, and using analogous notation, we obtain:

$$\theta_e = \alpha$$

$$d_e(t) = \alpha e^{-\alpha t}$$

$$D_e(t) = 1 - e^{-\alpha t}$$

$$S_e(t) = e^{-\alpha t}$$

$$\mathbb{E}T_e = \frac{1}{\alpha}$$

and

$$VarT_e = \frac{1}{\alpha^2}.$$

Determination of the reservation wage: graphical representation

In class we defined:

$$g(x) = x - z \quad \forall x \in [\underline{w}, \overline{w}]$$

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\overline{w}} (w - x) f(w) dw \quad \forall x \in [\underline{w}, \overline{w}]$$

More generally, we can define the functions above over all the real numbers, as follows:

$$g(x) = x - z \quad \forall x \in \mathbb{R}$$

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\infty} (w - x) f(w) dw \quad \forall x \in \mathbb{R}$$

The function g is obviously a straight line with slope equal to 1 and intercept (at $x = 0$) equal to $-z$.

For h we get three different parts.

Part 1. $x < \underline{w}$

In this case, we have:

$$\begin{aligned} h(x) &= \frac{\lambda}{\rho + \alpha} \int_x^{\infty} (w - x) f(w) dw \\ h(x) &= \frac{\lambda}{\rho + \alpha} \left[\underbrace{\int_x^{\underline{w}} (w - x) f(w) dw}_{=0} + \int_{\underline{w}}^{\overline{w}} (w - x) f(w) dw + \underbrace{\int_{\overline{w}}^{\infty} (w - x) f(w) dw}_{=0} \right] \\ h(x) &= \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\overline{w}} (w - x) f(w) dw \\ h(x) &= \frac{\lambda}{\rho + \alpha} \left[\int_{\underline{w}}^{\overline{w}} w f(w) dw - x \int_{\underline{w}}^{\overline{w}} f(w) dw \right] \\ h(x) &= \frac{\lambda}{\rho + \alpha} (\mathbb{E}W - x) \end{aligned}$$

Then:

$$h(x) = \frac{\lambda}{\rho + \alpha} (\mu - x) \quad \forall x \in (-\infty, \underline{w})$$

The expression above is a straight line with slope equal to $-\frac{\lambda}{\rho + \alpha}$ and intercept (at $x = 0$) equal to $\frac{\lambda}{\rho + \alpha} \mu$.

Part 2. $x \in [\underline{w}, \overline{w}]$

In this case, we have:

$$\begin{aligned} h(x) &= \frac{\lambda}{\rho + \alpha} \int_x^{\infty} (w - x) f(w) dw \\ h(x) &= \frac{\lambda}{\rho + \alpha} \int_x^{\overline{w}} (w - x) f(w) dw + \underbrace{\frac{\lambda}{\rho + \alpha} \int_{\overline{w}}^{\infty} (w - x) f(w) dw}_{=0} \end{aligned}$$

Then:

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\overline{w}} (w - x) f(w) dw \quad \forall x \in [\underline{w}, \overline{w}]$$

This is the part we analyzed in class

Part 3. $x > \overline{w}$

In this case, we have:

$$h(x) = \frac{\lambda}{\rho+\alpha} \int_x^\infty (w-x) \underbrace{f(w)}_{=0} dw$$

Then:

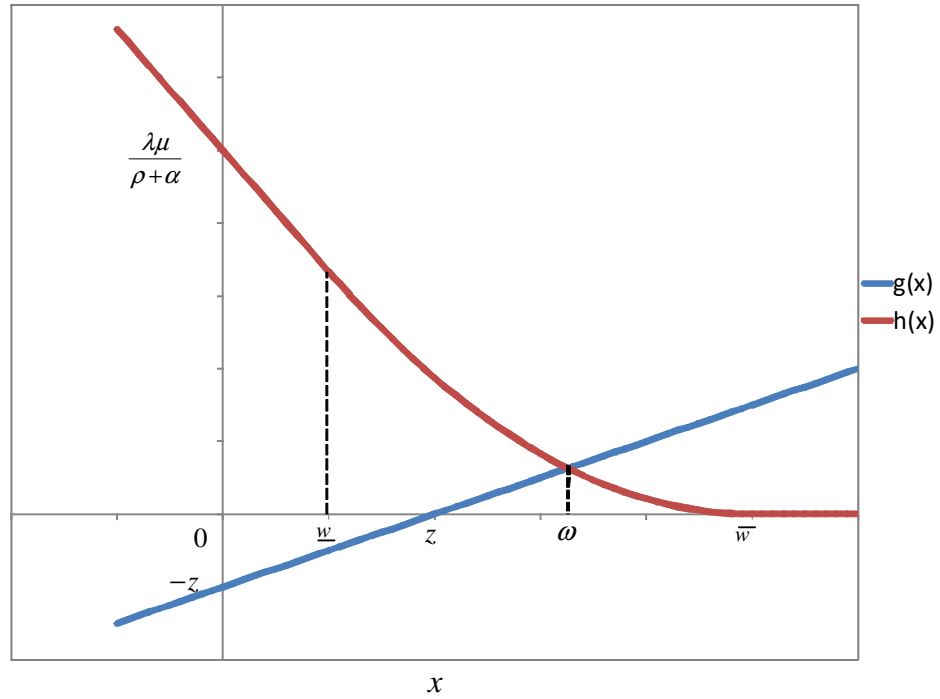
$$h(x) = 0 \quad \forall x > \overline{w}$$

Summarizing

$$g(x) = x - z \quad \forall x \in \mathbb{R}$$

$$h(x) = \begin{cases} \frac{\lambda}{\rho+\alpha}(\mu - x) & \forall x < \underline{w} \\ \frac{\lambda}{\rho+\alpha} \int_x^{\overline{w}} (w-x)f(w)dw & \forall x \in [\underline{w}, \overline{w}] \\ 0 & \forall x > \overline{w} \end{cases}$$

Plot



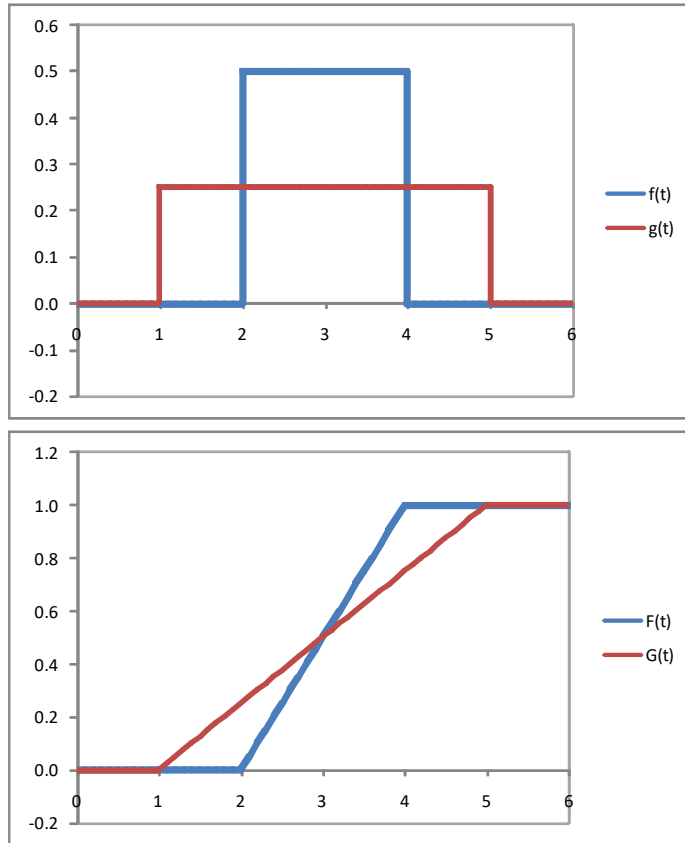
Mean-Preserving Spreads and the Reservation Wage

In what follows, we are going to consider non-negative random variables with cumulative distribution functions (cdf's) of the form $F : \mathbb{R}_+ \rightarrow [0, 1]$, with positive density $f(x) \equiv F'(x) > 0$ on a closed interval $[\underline{x}, \bar{x}]$, where $\bar{x} > \underline{x} \geq 0$, and $f(x) = 0 \ \forall x \notin [\underline{x}, \bar{x}]$. Notice that $F(x) = 0 \ \forall x \leq \underline{x}$ and $F(x) = 1 \ \forall x \geq \bar{x}$.

Definition (Informal): Simple Mean-Preserving Spread (MPS)

A distribution G is obtained by a simple mean-preserving spread of F if it is derived from F by shifting part of the mass of the distribution toward the tails, keeping the mean constant.

The figure below provides an example where a uniform distribution (G) is obtained from another uniform distribution (F) through a simple MPS. We plot the probability density functions (pdf's) and the cdf's for both random variables.



Theorem: Rothschild and Stiglitz (1970).

Consider two random variables X and Y with distributions F and G , respectively. Assume that X and Y have the same mean. Then, the following statements are equivalent:

- (a) G can be obtained from F by a sequence of simple MPS.
- (b) $\int_0^x G(t)dt \geq \int_0^x F(t)dt \quad \forall x \in \mathbb{R}_+$.
- (c) There is a random variable Z with $\mathbb{E}\{Z|X\} = 0$ such that $Y \stackrel{d}{=} X + Z$.
- (d) $\int_0^\infty u(x)f(x)dx \geq \int_0^\infty u(x)g(x)dx$ for any bounded concave fn. $u : \mathbb{R}_+ \rightarrow \mathbb{R}$.

Proof. See:

Rothschild, M. and Stiglitz, J. (1970). "Increasing Risk I: A Definition", *Journal of Economic Theory*, Vol. 2, No. 3 (September), pp. 225-243.

Rothschild, M. and Stiglitz, J. (1971). "Increasing Risk II: Its Economic Consequences", *Journal of Economic Theory*, Vol. 3, No. 1 (March), pp. 66-84.

Rothschild, M. and Stiglitz, J. (1972). "Addendum to 'Increasing Risk: I. A definition'", *Journal of Economic Theory*, Vol. 5, No. 2 (October), p. 206.

Rasmusen, E. and Petrakis, E. (1992). "Defining the Mean-Preserving Spread: 3-pt versus 4-pt," in *Decision Making Under Risk and Uncertainty: New Models and Empirical Findings*, John Geweke (editor), Amsterdam: Kluwer.

Machina, M. and Pratt, J. (1997). "Increasing Risk, Some Direct Constructions," *Journal of Risk and Uncertainty*, Vol. 14, pp. 103-127.

Leshno, M., Levy, H. and Spector, Y. (1997). "A Comment on Rothschild and Stiglitz's 'Increasing Risk: I. A Definition'," *Journal of Economic Theory*, Vol. 77, No. 1 (November), pp. 223-228.

Müller, Alfred, (1998). "Comparing Risks with Unbounded Distributions," *Journal of Mathematical Economics*, Vol. 30, No. 2 (September), pp. 229-239.

■

Remark 1. The symbol $\stackrel{d}{=}$ indicates equality *in distribution*. That is, Y is a random variable that takes the same values with the same probabilities as the random variable $X + Z$. Equality in distribution is a much weaker condition than equality: two random variables are equal if they take on the same value in every state, which is sufficient but not necessary for equality in distribution. For example $X = \begin{cases} 0 & \text{if } s = 1 \\ 1 & \text{if } s = 2 \end{cases}$ and $Y = \begin{cases} 1 & \text{if } s = 1 \\ 0 & \text{if } s = 2 \end{cases}$ are different random variables, but they are equal in distribution when $\Pr(s = 1) = \Pr(s = 2) = \frac{1}{2}$.

Remark 2. Mean independence ($\mathbb{E}\{Z|X\} = 0$) is weaker than independence but stronger than uncorrelatedness. Independence implies mean independence, and mean independence implies uncorrelatedness, but the converse statements are not true in general.

Remark 3. Condition $\int_0^\infty u(x)f(x)dx \geq \int_0^\infty u(y)g(y)dy$ means that F second-order stochastically dominates G . It shows that any risk-averse, expected-utility maximizer will prefer lottery F to lottery G . When this happens we say that F is less risky than G or, equivalently, that G is riskier than F .

Remark 4. The cdf's F and G may cross more than once.

The part of the previous theorem that is useful for us is (b). To see the intuition for it we need first a couple of results.

Claim 1. Suppose that X is a random variable with the properties described at the beginning of this note. Let b denote a real number such that $b \geq \bar{x}$. Then, $\mathbb{E}\{X\} = b - \int_0^b F(x)dx = \int_0^b [1 - F(x)]dx$.

Proof.

We know

$$\mathbb{E}\{X\} = \int_{\underline{x}}^{\bar{x}} xf(x)dx$$

The expression above can be rewritten as follows:

$$\mathbb{E}\{X\} = \underbrace{\int_0^{\underline{x}} xf(x)dx}_{=0} + \int_{\underline{x}}^{\bar{x}} xf(x)dx + \underbrace{\int_{\bar{x}}^b xf(x)dx}_{=0}$$

Then:

$$\mathbb{E}\{X\} = \int_0^b xf(x)dx$$

Integrating by parts we get (with $u = x$ and $dv = f(x)dx$):

$$\mathbb{E}\{X\} = xF(x)\Big|_0^b - \int_0^b F(x)dx$$

$$\mathbb{E}\{X\} = [bF(b) - 0F(0)] - \int_0^b F(x)dx$$

Then:

$$\mathbb{E}\{X\} = b - \int_0^b F(x)dx$$

where we have used $F(b) = 1$, since $b \geq \bar{x}$.

The expression above can be rewritten as follows:

$$\mathbb{E}\{X\} = \int_0^b dx - \int_0^b F(x)dx$$

Then:

$$\mathbb{E}\{X\} = \int_0^b [1 - F(x)]dx$$

■

Claim 2. Consider two non-negative random variables, X and Y , with supports $[\underline{x}, \bar{x}]$ and $[\underline{y}, \bar{y}]$, and let $b > \bar{x}$ and $b > \bar{y}$. Let F and G denote the cdfs of X and Y , respectively. Assume both random variables have the same mean: $\mathbb{E}\{X\} = \mathbb{E}\{Y\}$. Then: $\int_0^b F(x)dx = \int_0^b G(x)dx$. Moreover, $\int_0^\infty F(x)dx = \int_0^\infty G(x)dx$.
Proof.

From the proof of our previous claim we know:

$$\mathbb{E}\{X\} = b - \int_0^b F(x)dx$$

$$\mathbb{E}\{Y\} = b - \int_0^b G(x)dx$$

Then, the equality of means implies:

$$\begin{aligned} b - \int_0^b F(x)dx &= b - \int_0^b G(x)dx \\ - \int_0^b F(x)dx &= - \int_0^b G(x)dx \Rightarrow \end{aligned}$$

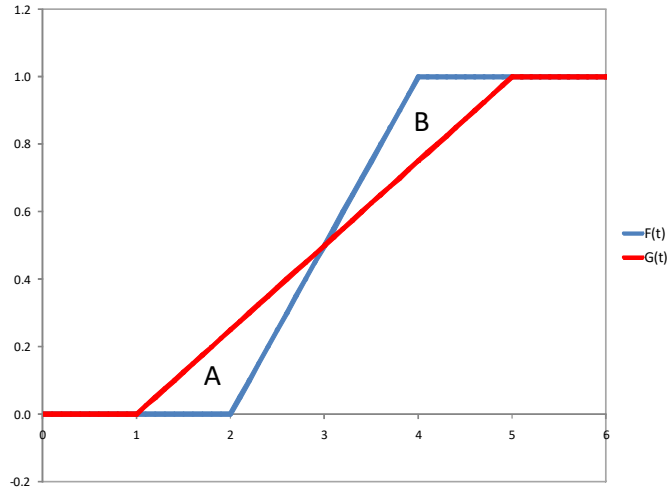
$$\int_0^b F(x)dx = \int_0^b G(x)dx$$

Since $F(x) = G(x) = 1 \ \forall x > b$, the expression above implies:

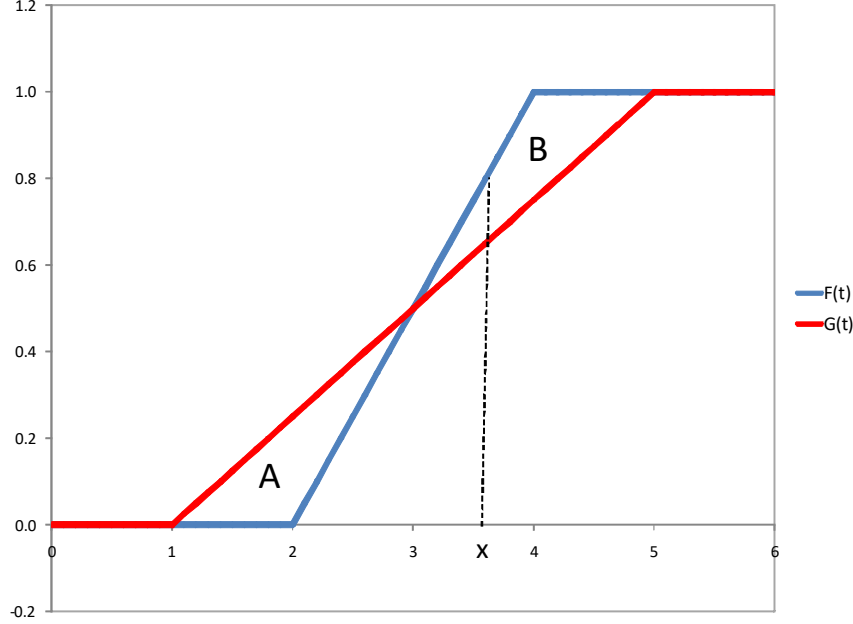
$$\int_0^\infty F(x)dx = \int_0^\infty G(x)dx$$

■

Claim 2 shows that the areas denoted by A and B in the following figure must be equal.



But then the next figure shows that the area below G from 0 to x must be bigger than the area below F . And this is the result established in part (b) of the theorem above: $\int_0^x G(t)dt \geq \int_0^x F(t)dt$.



Part (b) of Rothschild and Stiglitz's theorem can be used to show the effect of a mean-preserving spread of the wage distribution on the reservation wage. Recall that the reservation wage, ω , is implicitly defined by the following expression:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

It will be convenient to rewrite the expression above in a different way. Adding and subtracting $\frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw$ from the right-hand side we get:

$$\begin{aligned} \omega - z &= \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw + \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\bar{w}} (w - \omega) f(w) dw - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} \left(\int_{\underline{w}}^{\bar{w}} w f(w) dw - \omega \int_{\underline{w}}^{\bar{w}} f(w) dw \right) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} (\mathbb{E}W - \omega) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} (\mu - \omega + z - z) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ (\omega - z) \left(1 + \frac{\lambda}{\rho + \alpha} \right) &= \frac{\lambda}{\rho + \alpha} (\mu - z) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ (\omega - z) \frac{\rho + \alpha + \lambda}{\rho + \alpha} &= \frac{\lambda}{\rho + \alpha} (\mu - z) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \end{aligned}$$

Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha + \lambda}(\mu - z) - \frac{\lambda}{\rho + \alpha + \lambda} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw$$

Integrating the second term of the right-hand side by parts, with $u = w - \omega$ and $dv = f(w)dw$, we get:

$$\begin{aligned} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw &= (w - \omega) F(w) \Big|_{\underline{w}}^{\omega} - \int_{\underline{w}}^{\omega} F(w) dw \\ \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw &= (\omega - \omega) F(\omega) - (\underline{w} - \omega) F(\underline{w}) - \int_{\underline{w}}^{\omega} F(w) dw \\ \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw &= 0 \times F(\omega) - (\underline{w} - \omega) \times 0 - \int_{\underline{w}}^{\omega} F(w) dw \end{aligned}$$

Then:

$$\int_{\underline{w}}^{\omega} (w - \omega) f(w) dw = - \int_{\underline{w}}^{\omega} F(w) dw$$

Combining the previous two expressions we get:

$$\omega - z = \frac{\lambda}{\rho + \alpha + \lambda}(\mu - z) + \frac{\lambda}{\rho + \alpha + \lambda} \int_{\underline{w}}^{\omega} F(w) dw$$

But $\int_{\underline{w}}^{\omega} F(w) dw = \int_0^{\omega} F(w) dw$, since $F(w) = 0 \ \forall w \leq \underline{w}$. Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha + \lambda}(\mu - z) + \frac{\lambda}{\rho + \alpha + \lambda} \int_0^{\omega} F(w) dw$$

Suppose now that the wage distribution changes from F to G , where G is obtained from F through a mean-preserving spread (in other words, the wage distribution becomes riskier). Then μ remains unchanged, and the term $\int_0^{\omega} F(w) dw$ in the right-hand side of the expression above changes to $\int_0^{\omega} G(w) dw$. But we know from Rothschild and Stiglitz's theorem that $\int_0^{\omega} G(w) dw \geq \int_0^{\omega} F(w) dw \ \forall \omega \geq 0$. Hence, the right-hand side of the expression above increases. But this requires an increase in the left-hand side. Hence, ω must increase. We conclude that a riskier wage distribution induces an increase in the reservation wage. The intuition for this result is that the higher incidence of very good wage offers induced by the MPS increases the value of searching for a job, while the higher incidence of very bad offers is less detrimental because these offers will be rejected anyway (the option to work will not be exercised at such low wages).

A Matching Model of Unemployment

Continuous Time. Infinite Horizon.

$$\text{Workers} \begin{cases} E = \text{employed} \\ U = \text{unemployed} \end{cases}$$

$$\text{Jobs} \begin{cases} F = \text{filled} \\ V = \text{vacant} \end{cases}$$

Each job can have at most one worker. Then: $E = F$.

The labor force, L , is fixed. Then: $E(t) + U(t) = L$ at every t .

We will consider only steady states (E and U constant)

The number of jobs is endogenous. Vacancies can be created and eliminated freely.

When the job is vacant the firm is actively engaged in hiring at a fixed cost c per unit of time.

An employed worker produces output at rate y per unit of time, and is paid a wage w per unit of time.

Output (y) is exogenous; w is determined endogenously through *bargaining* between the worker and the firm.

Worker's utility, per unit time: $u_W = \begin{cases} w & \text{if employed} \\ z & \text{if unemployed} \end{cases}$

We assume $y > z$.

Firm's profits from a job, per unit of time:

$$u_J = \begin{cases} y - w & \text{if the job is filled} \\ -c & \text{if the job is vacant} \end{cases}$$

Workers' objective function is the expected present discounted value of utility.

Firms' objective function is the expected present discounted value of profits.

The discount rate, $r > 0$, is exogenous and constant (the same for workers and firms).

Matching

When unemployed workers and vacant jobs meet, new matches (*i.e.*, new filled jobs = new employed workers) are produced at rate M per unit of time, according to a so-called *matching function* (technology):

$$M = m(U, V)$$

We assume $m(\bullet)$ is increasing in both arguments, concave, and homogeneous of degree one. Also: $m(0, V) = m(U, 0) = 0$.

Example: $M = AU^\gamma V^{1-\gamma}$ with $A > 0$ and $\gamma \in (0, 1)$.

Define:

$$u \equiv \frac{U}{L} \quad \text{unemp. rate}$$

$$v \equiv \frac{V}{L} \quad \text{vacancy rate}$$

$$m \equiv \frac{M}{L}$$

Remark: we abuse notation and use m to denote both $m \equiv \frac{M}{L}$ and the matching function, $m(\bullet)$.

Since $m(\bullet)$ is h-d-one, we can write:

$$m = m(u, v)$$

Define:

$$p \equiv \frac{m(U, V)}{U} = \frac{m(u, v)}{u} = \text{job-finding rate}$$

$$q \equiv \frac{m(U, V)}{V} = \frac{m(u, v)}{v} = \text{job-filling rate}$$

$$\theta \equiv \frac{V}{U} = \frac{v}{u} = \text{labor-market tightness}$$

Since $m(\bullet)$ is h-d-one, we get: $p = \frac{m(u, v)}{u} = m(1, \frac{v}{u}) = m(1, \theta)$ and $q = \frac{m(u, v)}{v} = m(\frac{u}{v}, 1) = m(\frac{1}{\theta}, 1)$. Now we can write (abusing notation):

$$p = p(\theta) \equiv m(1, \theta)$$

$$q = q(\theta) \equiv m(\frac{1}{\theta}, 1)$$

Notice that $p' > 0$ and $q' < 0$. Also, the concavity of $m(\bullet)$ implies $p'' \leq 0$. Finally, notice that $p(\theta) = \theta q(\theta)$.

We assume jobs are destroyed at the exogenous rate α per unit of time. Hence, during a time interval of length h we have: $E(t+h) - E(t) = Mh - \alpha h E$. Dividing by h and letting $h \rightarrow 0$ we get: $\dot{E} = M - \alpha E$. In steady state $\dot{E} = 0$, then $M = \alpha E$. Then:

$$m(U, V) = \alpha E$$

Dividing by L , using h-d-one, and letting $e \equiv \frac{E}{L}$ (employment rate) we get:

$$m(u, v) = \alpha e$$

Hence, in s-s, job creation = job destruction.

From $E + U = L$ we get:

$$e + u = 1$$

From $m(U, V) = \alpha E$ and $p \equiv \frac{m(U, V)}{U}$ we can find the steady-state unemployment rate:

$$pU = \alpha E$$

$$pu = \alpha e \quad (\text{dividing by } L)$$

$$pu = \alpha(1 - u)$$

$$(p + \alpha)u = \alpha \Rightarrow$$

$$u = \frac{\alpha}{\alpha + p}$$

Recalling that p is a function of θ we can write:

$$u(\theta) = \frac{\alpha}{\alpha + p(\theta)}$$

Stationary Value Functions

For a period of length h , the value of employment is:

$$V_E(h) = wh + \frac{1}{1 + rh} [\alpha h V_U(h) + (1 - \alpha h) V_E(h)]$$

The expression above gives:

$$V_E(h)(1 - \frac{1 - \alpha h}{1 + rh}) = wh + \frac{\alpha h}{1 + rh} V_U(h)$$

$$V_E(h) \frac{(r + \alpha)h}{1 + rh} = wh + \frac{\alpha h}{1 + rh} V_U(h) \Rightarrow$$

$$V_E(h) = \frac{1 + rh}{\alpha + r} w + \frac{\alpha}{\alpha + r} V_U(h)$$

Continuous time: $h \rightarrow 0$

$$V_E = \frac{w}{\alpha + r} + \frac{\alpha}{\alpha + r} V_U$$

where $V_E \equiv V_E(0)$ and $V_U \equiv V_U(0)$.

The expression above can be rewritten as follows:

$$rV_E = w + \alpha(V_U - V_E)$$

The value of unemployment must satisfy (writing p instead of $p(\theta)$, for simplicity):

$$V_U(h) = zh + \frac{1}{1 + rh} [phV_E(h) + (1 - ph)V_U(h)]$$

The expression above gives:

$$V_U(h)(1 - \frac{1-ph}{1+rh}) = zh + \frac{ph}{1+rh} V_E(h)$$

$$V_U(h) \frac{(r+p)h}{1+rh} = zh + \frac{ph}{1+rh} V_E(h) \Rightarrow$$

$$V_U(h) = \frac{1 + rh}{p + r} z + \frac{p}{p + r} V_E(h)$$

Continuous time: $h \rightarrow 0$

$$V_U = \frac{z}{p + r} + \frac{p}{p + r} V_E$$

The expression above can be rewritten as follows:

$$rV_U = z + p(V_E - V_U)$$

The value of a filled job must satisfy:

$$V_F(h) = (y - w)h + \frac{1}{1 + rh} [\alpha h V_V(h) + (1 - \alpha h) V_F(h)]$$

The expression above gives:

$$V_F(h) \left(1 - \frac{1 - \alpha h}{1 + rh}\right) = (y - w)h + \frac{\alpha h}{1 + rh} V_V(h)$$

$$V_F(h) \frac{(r + \alpha)h}{1 + rh} = (y - w)h + \frac{\alpha h}{1 + rh} V_V(h)$$

Then:

$$V_F(h) = \frac{1 + rh}{\alpha + r} (y - w) + \frac{\alpha}{\alpha + r} V_V(h)$$

Continuous time: $h \rightarrow 0$

$$V_F = \frac{y - w}{\alpha + r} + \frac{\alpha}{\alpha + r} V_V$$

The expression above can be rewritten as follows:

$$rV_F = (y - w) + \alpha(V_V - V_F)$$

The value of a vacant job must satisfy (writing q instead of $q(\theta)$, for simplicity):

$$V_V(h) = -ch + \frac{1}{1+rh}[qhV_F(h) + (1-qh)V_V(h)]$$

The expression above gives:

$$V_V(h)(1 - \frac{1-qh}{1+rh}) = -ch + \frac{qh}{1+rh}V_F(h)$$

$$V_V(h)\frac{(r+q)h}{1+rh} = -ch + \frac{qh}{1+rh}V_F(h)$$

Then:

$$V_V(h) = -\frac{1+rh}{q+r}c + \frac{q}{q+r}V_F(h)$$

Continuous time: $h \rightarrow 0$

$$V_V = \frac{-c}{q+r} + \frac{q}{q+r}V_F$$

The expression above can be rewritten as follows:

$$rV_V = -c + q(V_F - V_V)$$

Wage Determination (Nash Bargaining)

Once they meet, a firm and a worker bargain over the wage, w_i (i indexes this particular firm-worker pair).

The joint payoff from disagreement (no match) is:

$$\mathcal{J}_d = V_U + V_V$$

Remark. The values of V_U and V_V do not depend on w_i , the wage negotiated by this particular firm-worker pair. They are determined by conditions in the rest of the market (not influenced by a particular firm or worker).

The joint payoff from agreement (a match is formed) is:

$$\begin{aligned}\mathcal{J}_a &= V_E(w_i) + V_F(w_i) \\ &= \frac{w_i}{\alpha + r} + \frac{\alpha}{\alpha + r} V_U + \frac{y - w_i}{\alpha + r} + \frac{\alpha}{\alpha + r} V_V \\ &= \frac{y + \alpha(V_U + V_V)}{\alpha + r}\end{aligned}$$

Remark. In our case, the joint payoff from agreement is independent from the negotiated wage (transferable utility).

Surplus from agreement:

$$\begin{aligned} S &= \mathcal{J}_a - \mathcal{J}_d \\ &= \frac{y + \alpha(V_U + V_V)}{\alpha + r} - (V_U + V_V) \\ &= \frac{y - r(V_U + V_V)}{\alpha + r} \end{aligned}$$

Remark 1. Notice that $S = V_E + V_F - (V_U + V_V) = (V_E - V_U) + (V_F - V_V)$.

Remark 2. $S > 0$ requires $y > r(V_U + V_V)$. If $S < 0$ players won't agree.

Nash's Bargaining Solution (NBS):

$$V_E = V_U + \beta S$$

$$V_F = V_V + (1 - \beta)S$$

where $\beta \in (0, 1)$ is a parameter that can be interpreted as the bargaining power of the worker.

From the expressions above we get:

$$\beta(V_F - V_V) = (1 - \beta)(V_E - V_U)$$

Remark 1. Writing $V_E(w_i)$ and $V_F(w_i)$ makes clear that, for given values of V_V and V_U , the expression above determines the negotiated wage.

Remark 2. The NBS can be obtained from:

$$\max_{w_i} [V_E(w_i) - V_U]^\beta [V_F(w_i) - V_V]^{1-\beta}$$

Recall that $V'_E(w_i) = -V'_F(w_i)$.

To complete the model, recall that vacancies can be created and destroyed costlessly. Hence, in equilibrium:

$$V_V = 0$$

Finally, recall that in a symmetric equilibrium we have

$$w_i = w$$

Summarizing:

$$m = m(u, v)$$

$$m = \alpha e$$

$$\theta = \frac{v}{u}$$

$$u = \frac{\alpha}{\alpha + p(\theta)}$$

$$rV_E = w + \alpha(V_U - V_E)$$

$$rV_U = z + p(\theta)(V_E - V_U)$$

$$rV_F = y - w + \alpha(V_V - V_F)$$

$$rV_V = -c + q(\theta)(V_F - V_V)$$

$$\beta(V_F - V_V) = (1 - \beta)(V_E - V_U)$$

$$V_V = 0$$

Remark. 10 eq. in 10 var.: $m, u, v, e, \theta, w, V_E, V_U, V_F, V_V$.

Solving the Model

We show how to find the equilibrium values of w , θ , u and v . The equilibrium levels of the rest of the variables follow easily from the equations summarized above.

We start by using the optimization conditions of the firm to find a relationship between w and θ .

From $rV_F = y - w + \alpha(V_V - V_F)$ we get:

$$V_F - V_V = \frac{(y - rV_V) - w}{r + \alpha}$$

Notice that $V_F - V_V > 0$ requires $w < y - rV_V$. Hence, $y - rV_V$ is like a reservation wage for the firm.

Substituting $V_V = 0$ into the expression above we get:

$$V_F = \frac{y - w}{r + \alpha}$$

Hence, the reservation wage for the firm is equal to y (i.e., the firm won't fill a job if it has to pay $w > y$).

Substituting $V_V = 0$ into $rV_V = -c + q(\theta)(V_F - V_V)$ we get:

$$V_F = \frac{c}{q(\theta)}$$

Combining the two expressions above we get: $\frac{y-w}{r+\alpha} = \frac{c}{q(\theta)}$. Then:

$$y - w = \frac{r + \alpha}{q(\theta)}c$$

Notice that the r-h-s is positive whenever $c > 0$. Hence, the l-h-s implies $y > w$ (i.e., the wage rate is lower than the firm's reservation wage).

The expression above gives:

$$w = y - \frac{r + \alpha}{q(\theta)}c$$

This is a negative relationship between w and θ (recall q is decreasing in θ). We call this relationship the *job creation condition* (JC).

Now we combine the opt. conditions of a worker with the NBS to find another relationship between w and θ .

From $rV_E = w + \alpha(V_U - V_E)$ we get:

$$V_E - V_U = \frac{w - rV_U}{r + \alpha}$$

Remark. We can think of rV_U as the reservation wage of the worker. Then, $V_E > V_U$ if and only if w is higher than the worker's reservation wage (otherwise, the worker will reject the agreement).

From the bargaining condition we know: $V_E - V_U = \frac{\beta}{1-\beta}(V_F - V_V)$. But we also know $V_F = \frac{y-w}{r+\alpha}$ and $V_V = 0$. Then:

$$V_E - V_U = \frac{\beta}{1-\beta} \frac{y-w}{r+\alpha}$$

Since $y > w$, this expression shows that $V_E - V_U > 0$, and then $w > rV_U$. Therefore: $rV_U < w < y$.

Equating the previous two expressions we get $\frac{w-rV_U}{r+\alpha} = \frac{\beta}{1-\beta} \frac{y-w}{r+\alpha}$. Solving for w we obtain:

$$w = \beta y + (1 - \beta)rV_U$$

Notice that w is a weighted average between the two reservation wages, and the weights depend on the bargaining powers.

We can eliminate rV_U from the previous expression by writing it as a function of θ . First use $rV_U = z + p(\theta)(V_E - V_U)$ and the bargaining condition to get: $rV_U = z + p(\theta)\frac{\beta}{1-\beta}(V_F - V_V)$. Now recall $V_F = \frac{c}{q(\theta)}$ and $V_V = 0$ to get: $rV_U = z + p(\theta)\frac{\beta}{1-\beta}\frac{c}{q(\theta)}$. Using $p(\theta) = \theta q(\theta)$ we get: $rV_U = z + \theta q(\theta)\frac{\beta}{1-\beta}\frac{c}{q(\theta)} \Rightarrow$

$$rV_U = z + \frac{\beta c}{1 - \beta}\theta$$

Substituting the expression above into the previous one, $w = \beta y + (1 - \beta)rV_U$, we get:

$$w = \beta y + (1 - \beta)z + \beta c\theta$$

The expression above gives a positive (linear) relation between labor-market tightness and the equilibrium wage. We call this relation the **wage curve (WC)**.

Combining the job creation condition with the wage curve we can find the equilibrium values of the wage rate, w^* , and labor-market tightness, θ^* .

Equating JC and WC we get:

$$y - \frac{r+\alpha}{q(\theta)}c = \beta y + (1 - \beta)z + \beta c\theta$$

$$(1 - \beta)y = (1 - \beta)z + \beta c\theta + \frac{r+\alpha}{q(\theta)}c$$

$$(1 - \beta)(y - z) = c \left[\beta\theta + \frac{r+\alpha}{q(\theta)} \right]$$

Then:

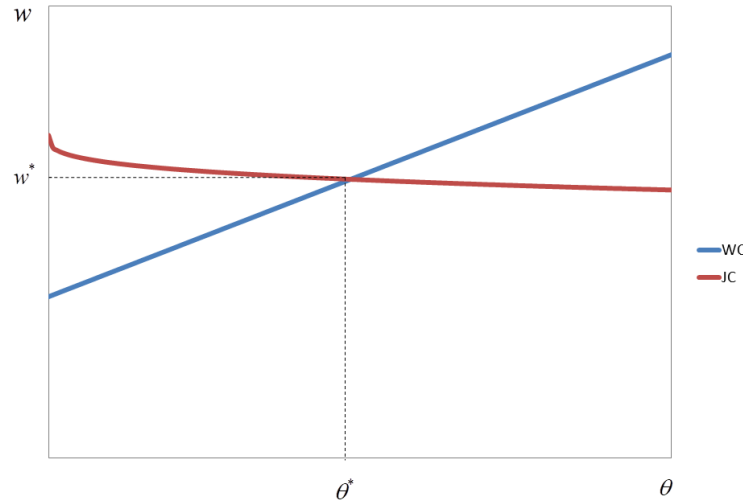
$$\frac{r + \alpha}{q(\theta)} + \beta\theta = (1 - \beta)\frac{y - z}{c}$$

The expression above defines the equilibrium value of labor-market tightness, θ^* . Typically, the solution will have to be found numerically.

Then we can go back to WC to get the equilibrium wage:

$$w^* = z + \beta(y - z) + \beta c\theta^*$$

We can represent the determination of θ^* and w^* graphically, by plotting the JC and WC together.



$$\text{JC: } w = y - \frac{r+\alpha}{q(\theta)}c \quad \text{WC: } w = \beta y + (1 - \beta)z + \beta c\theta$$

Existence

Under standard assumptions on the matching function, $q(\theta) \rightarrow \infty$ as $\theta \rightarrow 0$. Hence, JC will cut the vertical axis at y . On the other hand, WC cuts the vertical axis at $\beta y + (1 - \beta)z$. Hence, if $y > z$, the intercept of JC will be above the intercept of WC. Since JC is continuous and decreasing, and WC is continuous and increasing, they will cross at a unique positive value for θ . Hence, a unique equilibrium value θ^* will exist.

Once we've found θ^* we can obtain the equilibrium unemployment rate from:

$$u^* = \frac{\alpha}{\alpha + p(\theta^*)}$$

Recalling that $\theta = \frac{v}{u}$, we can obtain the equilibrium vacancy rate from:

$$v^* = \theta^* u^*$$

We can represent the determination of u^* and v^* graphically.

Combining $u = \frac{\alpha}{\alpha + p(\theta)}$ with $\theta = \frac{v}{u}$ we get:

$$u = \frac{\alpha}{\alpha + p(\frac{v}{u})}$$

The expression above implicitly defines v in terms of u . This relation is usually called the *Beveridge Curve* (BC).

Recall that $p = \frac{m(u,v)}{u}$. Then, we can rewrite the expression above as follows:

$$\alpha u + m(u, v) = \alpha$$

It is now easy to show that BC has negative slope (recall that $m(\bullet)$ is increasing in both arguments). Moreover, the concavity of $m(\bullet)$ implies that BC is convex.

Now we find another relation between v and u .

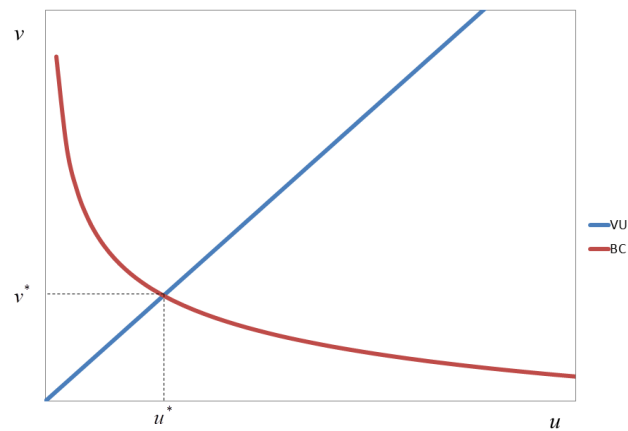
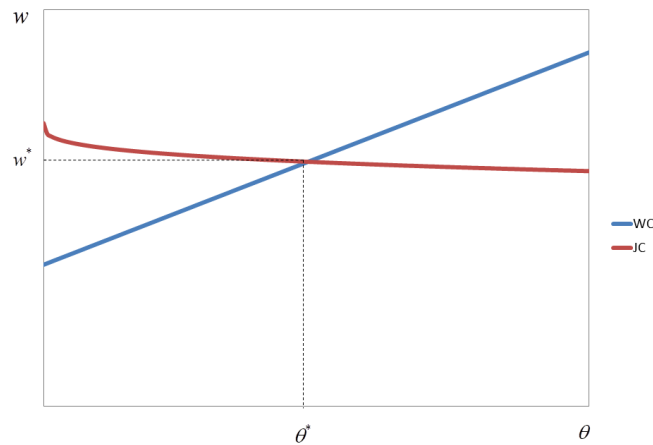
Since $\theta \equiv \frac{v}{u}$, we know the equilibrium levels of the vacancy rate and the unemployment rate must satisfy:

$$v = \theta^* u$$

We call the expression above the VU curve (recall θ^* has already been determined).

Now we can combine the BC and VU curves in the same plot in order to represent the determination of u^* and v^* .

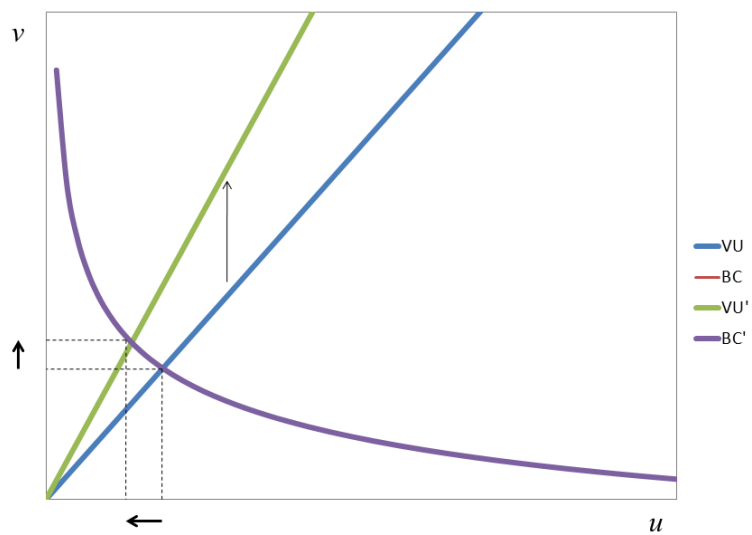
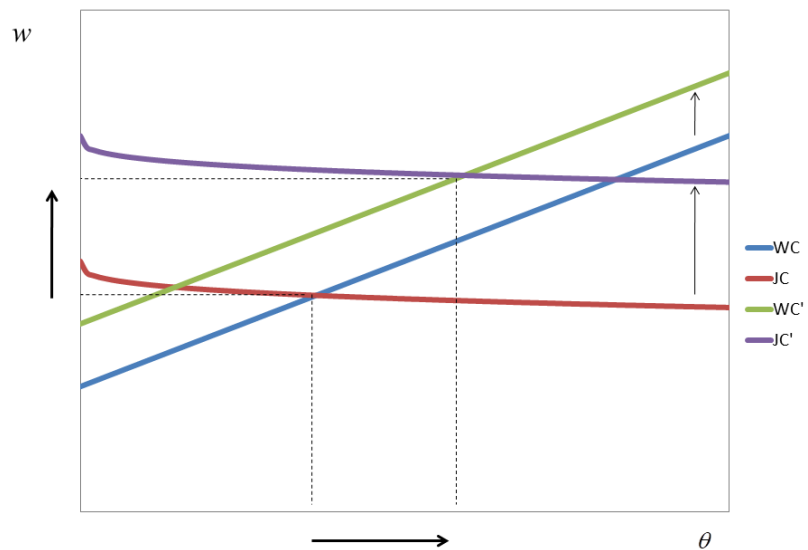
Hence, we can characterize the full equilibrium using the following two plots:



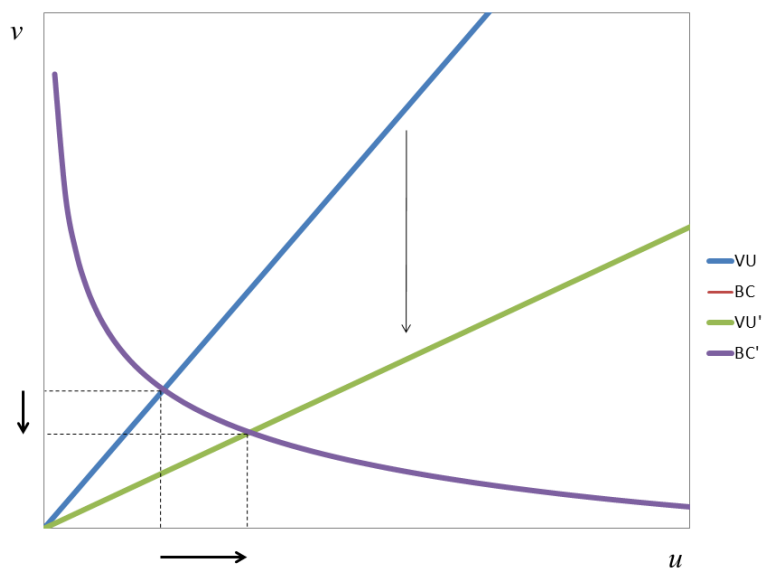
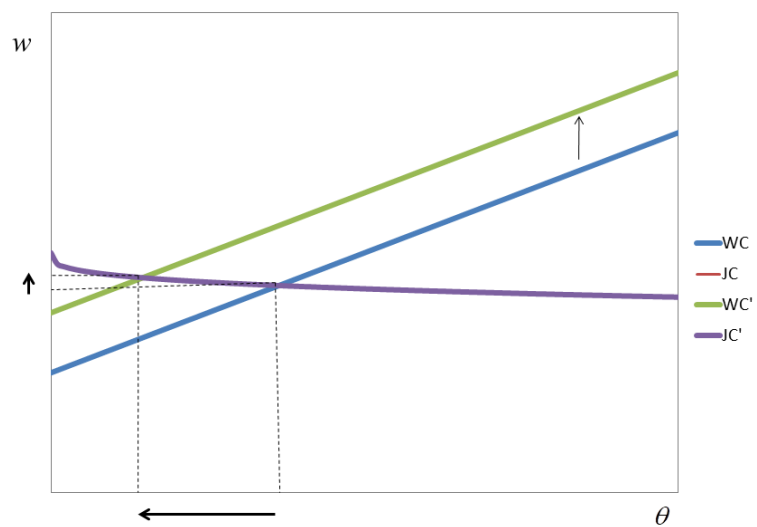
The equilibrium levels of the other variables can be recovered easily from the rest of the equations of the model. For example, $m^* = m(u^*, v^*)$, $p^* = p(\theta^*)$, $q^* = q(\theta^*)$, and so on.

Comparative Statics

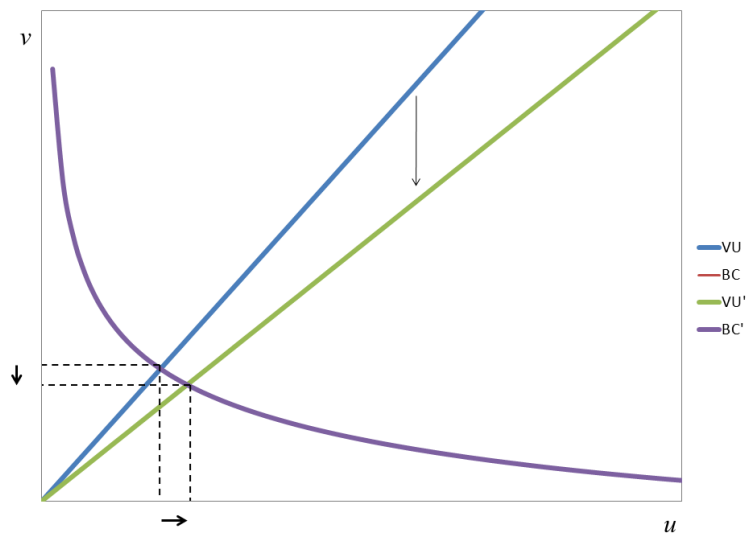
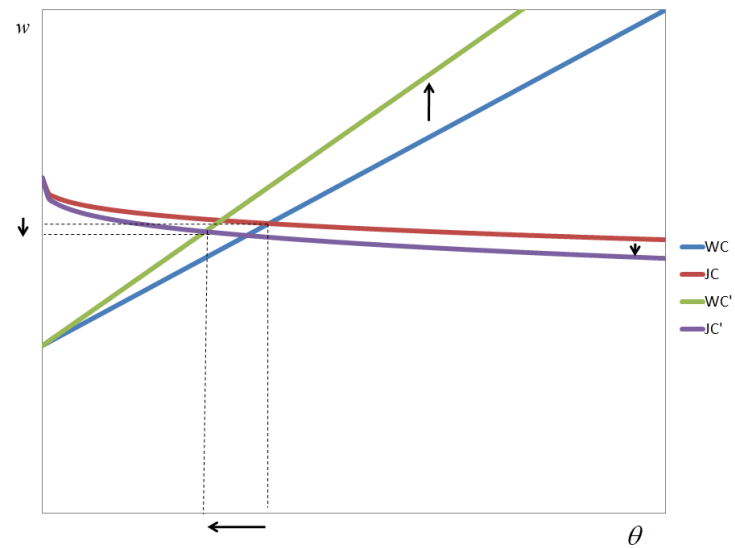
Increase in y



Increase in z

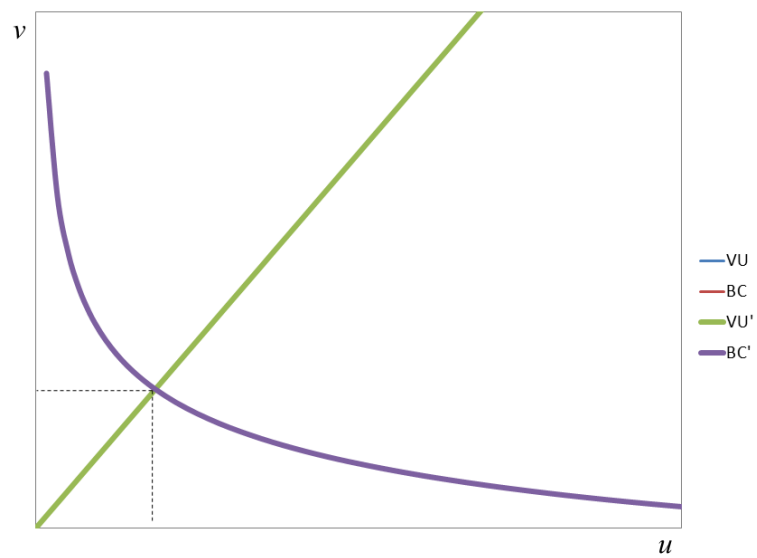
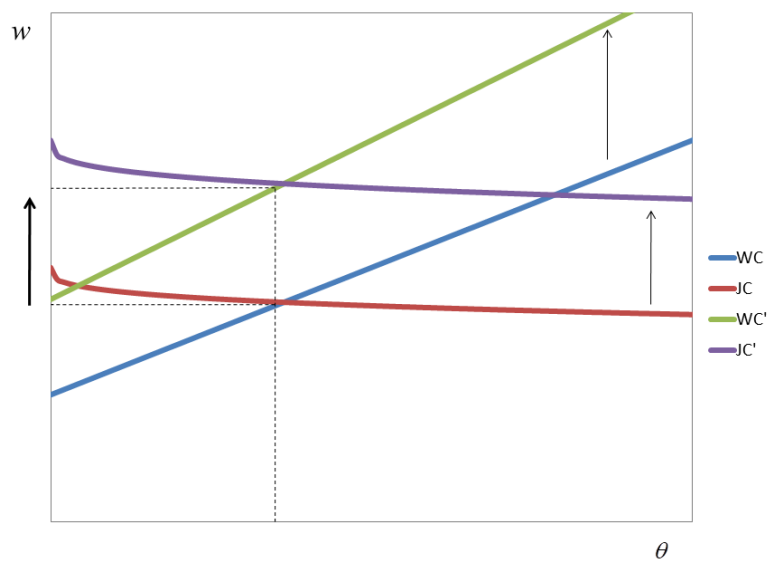


Increase in c

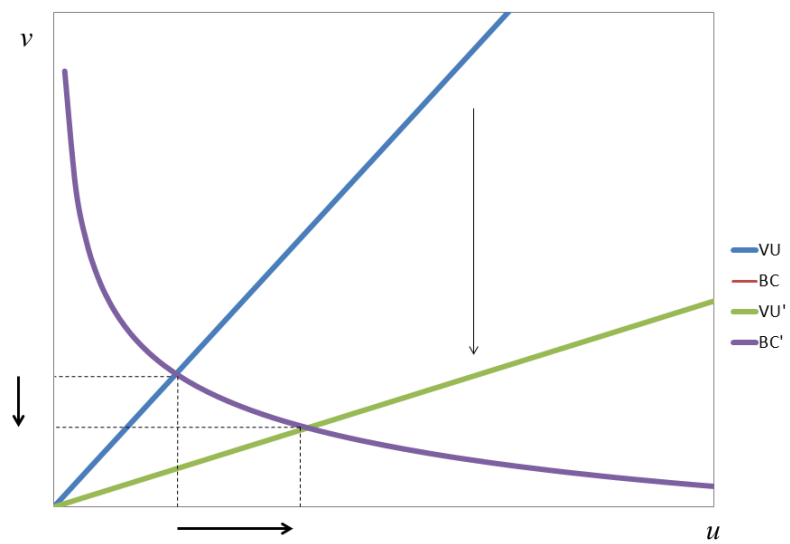
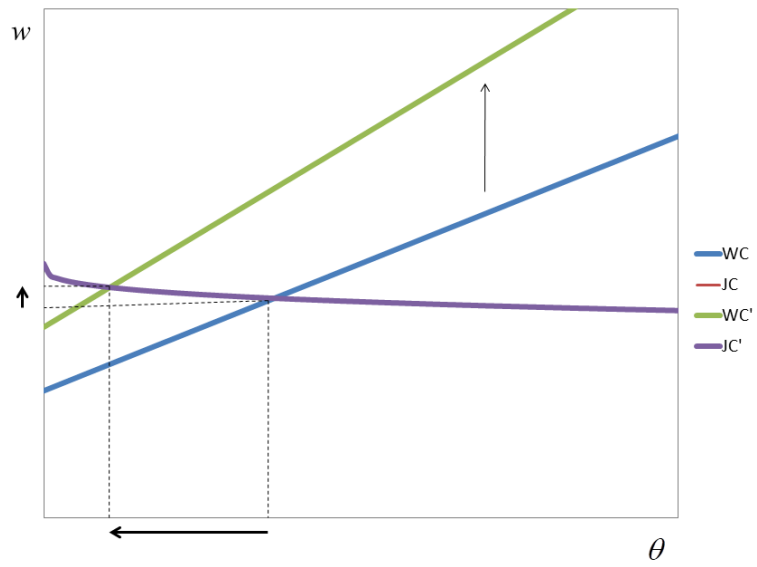


The effect on w is ambiguous.

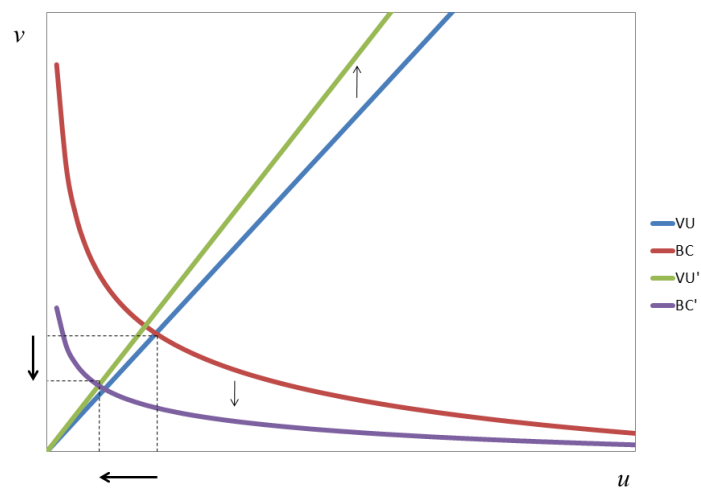
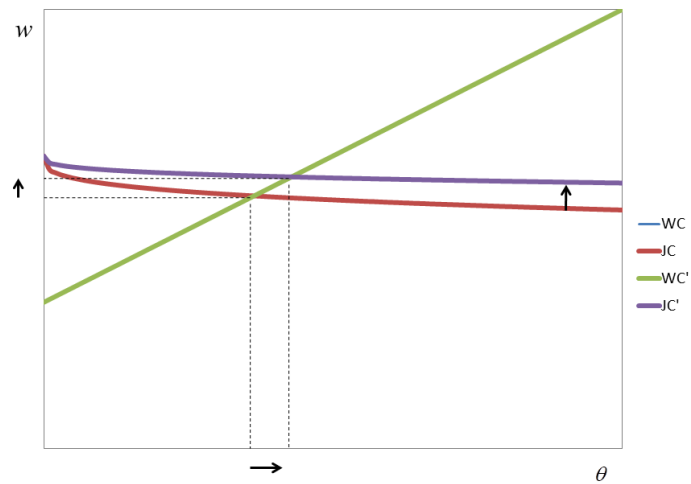
Proportional increase in y , c , and z



Increase in β

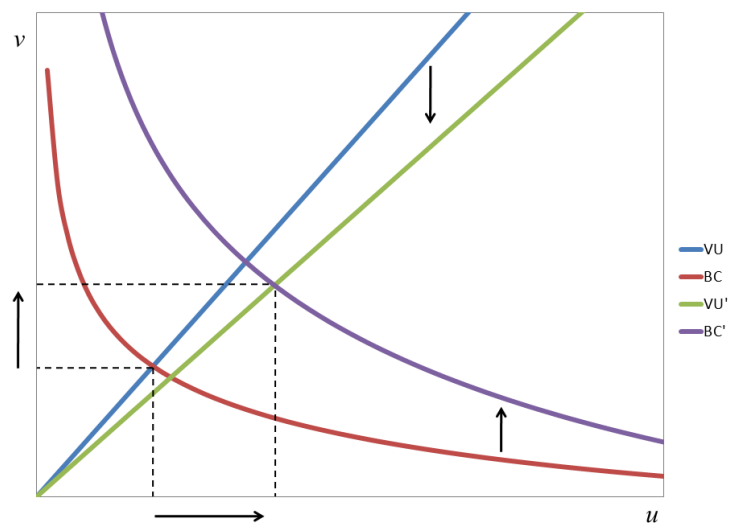
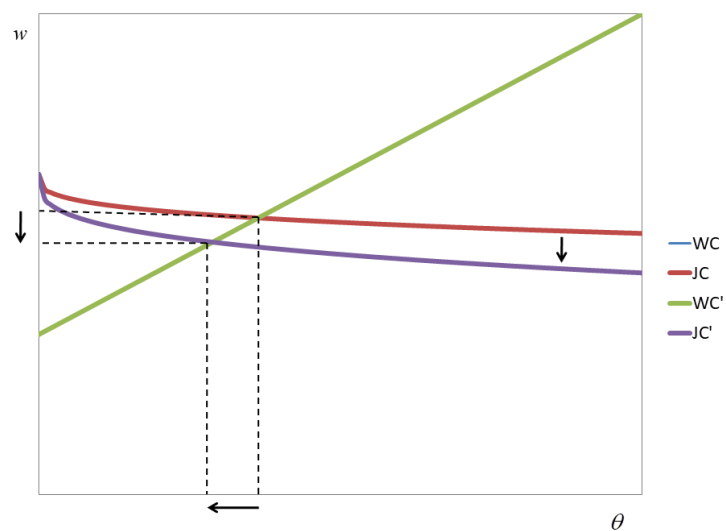


Increase in A (matching efficiency)



The effect on v is ambiguous.

Increase in α



The effect on v is ambiguous.

Matching and Efficiency

Consider the matching model studied in class. Let say that two paths $\theta(\bullet)$ and $u(\bullet)$ are efficient if they solve the following problem:

$$\begin{aligned} \max \quad & \int_0^\infty e^{-rt} [(1-u)y + uz - c\theta u] dt \\ \text{s.t.} \quad & \dot{u} = \alpha(1-u) - p(\theta)u \\ & \theta \in \mathbb{R}_{++} \\ & u(0) = u_0 \\ & \lim_{t \rightarrow \infty} u(t) \geq 0 \end{aligned}$$

where $r > 0$, $y > 0$, $z > 0$, $c > 0$, $\alpha > 0$, and $u_0 \in [0, 1]$ is given.

Remark. For simplicity, we have dropped the dependence of u and θ on t .

The objective functional is the present discounted value of social output/utility: a fraction $e = 1 - u$ of employed agents produce y units of output/utility, while a fraction u of unemployed agents get z units of utility/output; in addition, each vacant job $v = \theta u$ costs c units of output/utility. The planner is subject to the same matching frictions faced by private agents. Hence, the change in the unemployment rate, $\dot{u}$, is equal to the difference between the new unemployed agents generated by the process of job destruction, $\alpha(1-u)$, and the new employed agents generated by the matching process, $p(\theta)u$. In principle, we should include restrictions of the form $u \in [0, 1]$ and $\theta \in \mathbb{R}_+ \forall t$, but we'll concentrate on interior solutions, where $u \in (0, 1)$ and $\theta \in \mathbb{R}_{++}$. It's clear from the statement of the problem that u is the state variable and θ is the control variable.

We can use optimal control techniques to obtain necessary conditions for a maximum. The current-value Hamiltonian is:

$$H(t, u, \theta, \lambda) = (1-u)y + uz - c\theta u + \lambda[\alpha(1-u) - p(\theta)u]$$

where λ is the co-state variable.

The Maximum Principle establishes that the control variable must maximize the Hamiltonian, and that the co-state variable must satisfy $\dot{\lambda} - r\lambda = -H_u$.

A necessary condition for the maximization of H w.r.t. θ is $H_\theta = 0$. Then:

$$-cu - \lambda p'(\theta)u = 0 \quad (1)$$

From $\dot{\lambda} - r\lambda = -H_u$ we get: $\dot{\lambda} - r\lambda = -[-y + z - c\theta - \lambda\alpha - \lambda p(\theta)]$. Then:

$$\dot{\lambda} = (y - z + c\theta) + \lambda[r + \alpha + p(\theta)] \quad (2)$$

From (1) we get: $\lambda p'(\theta)u = -cu \Rightarrow \lambda p'(\theta) = -c \Rightarrow$

$$\lambda = \frac{-c}{p'(\theta)} \quad (3)$$

We know that $p(\theta) = \theta q(\theta)$. Then: $p'(\theta) = q(\theta) + \theta q'(\theta)$. Then: $p'(\theta) = q(\theta)[1 + \frac{\theta}{q(\theta)}q'(\theta)] \Rightarrow p'(\theta) = q(\theta)[1 - \frac{-\theta}{q(\theta)}q'(\theta)] \Rightarrow$

$$p'(\theta) = q(\theta)[1 - \eta(\theta)] \quad (4)$$

where

$$\eta(\theta) \equiv \frac{-\theta}{q(\theta)}q'(\theta)$$

Substituting (4) into (3) we get:

$$\lambda = \frac{-c}{q(\theta)[1 - \eta(\theta)]} \quad (5)$$

We know that θ is constant in steady state. Hence, (5) implies that λ is constant in steady state, too. Therefore:

$$\dot{\lambda} = 0 \quad (6)$$

Substituting (5) and (6) into (2) gives:

$$\begin{aligned} \dot{\lambda} &= (y - z + c\theta) + \lambda[r + \alpha + p(\theta)] \\ 0 &= (y - z + c\theta) + \frac{-c}{q(\theta)[1 - \eta(\theta)]}[r + \alpha + p(\theta)] \\ 0 &= y - z + c \left[\theta - \frac{r + \alpha + p(\theta)}{q(\theta)[1 - \eta(\theta)]} \right] \\ 0 &= \frac{y - z}{c} + \theta - \frac{r + \alpha + p(\theta)}{q(\theta)[1 - \eta(\theta)]} \quad \text{dividing by } c \\ 0 &= \frac{y - z}{c} + \theta - \frac{r + \alpha + \theta q(\theta)}{q(\theta)[1 - \eta(\theta)]} \quad \text{using } p(\theta) = \theta q(\theta) \\ \frac{r + \alpha + \theta q(\theta)}{q(\theta)[1 - \eta(\theta)]} - \theta &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta) - \theta q(\theta)[1 - \eta(\theta)]}{q(\theta)[1 - \eta(\theta)]} &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta) - \theta q(\theta) + \theta q(\theta)\eta(\theta)}{q(\theta)[1 - \eta(\theta)]} &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta)\eta(\theta)}{q(\theta)[1 - \eta(\theta)]} &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta)\eta(\theta)}{q(\theta)} &= [1 - \eta(\theta)] \frac{y - z}{c} \\ \frac{r + \alpha}{q(\theta)} + \frac{\theta q(\theta)\eta(\theta)}{q(\theta)} &= [1 - \eta(\theta)] \frac{y - z}{c} \end{aligned}$$

Then:

$$\frac{r + \alpha}{q(\theta)} + \eta(\theta)\theta = [1 - \eta(\theta)] \frac{y - z}{c} \quad (7)$$

But we know from our lecture notes that, in steady-state, the equilibrium value of θ must satisfy:

$$\frac{r + \alpha}{q(\theta)} + \beta\theta = (1 - \beta) \frac{y - z}{c} \quad (8)$$

Comparing (7) and (8) we conclude that the steady-state equilibrium will be efficient whenever:

$$\beta = \eta(\theta) \quad (9)$$

The expression above is usually called the *Hosios condition*.¹

Remarks.

Notice that $\eta(\theta)$, which is the elasticity of $q(\theta)$ with respect to θ , is also the elasticity of the matching function with respect to unemployment. To see this, recall that $q(\theta) \equiv m(\frac{1}{\theta}, 1) = \frac{m(u, v)}{v}$. Then: $\eta(\theta) \equiv -\frac{\theta}{q(\theta)}q'(\theta) = -\frac{\frac{v}{m(u, v)}m_1(\frac{1}{\theta}, 1)\frac{-1}{\theta^2}}{\frac{v}{m(u, v)}} = \frac{v}{u} \frac{v}{m(u, v)} m_1(\frac{u}{v}, 1) \frac{1}{(\frac{v}{u})^2} = \frac{v}{m(u, v)} m_1(\frac{u}{v}, 1) \frac{1}{\frac{v}{u}} = \frac{u}{m(u, v)} m_1(\frac{u}{v}, 1)$.

Notice also, that $1 - \eta(\theta)$ is the elasticity of $p(\theta)$ with respect to θ . To see this, recall that $p(\theta) = \theta q(\theta)$. Then: $\frac{\theta}{p(\theta)}p'(\theta) = \frac{\theta}{\theta q(\theta)}[q'(\theta)\theta + q(\theta)] = \frac{1}{q(\theta)}[q'(\theta)\theta + q(\theta)] = \frac{q'(\theta)\theta}{q(\theta)} + 1 = -\frac{q'(\theta)\theta}{q(\theta)} + 1 = 1 - \eta(\theta)$.

We know $p'(\theta) > 0$, then $\frac{\theta}{p(\theta)}p'(\theta) = 1 - \eta(\theta) > 0$. Then $\eta(\theta) < 1$. We also know $\eta(\theta) > 0$. Hence: $\eta(\theta) \in (0, 1)$.

We know unemployment increases with β (the bargaining power of the worker). Hence, when $\beta > \eta(\theta)$ unemployment is above the efficient level. Conversely, when $\beta < \eta(\theta)$ unemployment is below the efficient level. This is consistent with the fact that both firms and workers are causing congestion for each other. One more hiring firm makes searching workers better off, but it makes other hiring firms worse off. Analogously, one more searching worker makes hiring firms better off, but it makes other searching workers worse off. Hence, equilibrium unemployment can be above or below the efficient level.

Since vacancies are filled at Poisson rate $q(\theta)$, the expected duration of a vacant job is $\varphi(\theta) \equiv \frac{1}{q(\theta)}$. Since unemployed workers move into employment at a Poisson rate $p(\theta)$, the expected duration of unemployment spells is $\psi(\theta) \equiv \frac{1}{p(\theta)}$. To provide some rationale for condition (9), notice that the elasticity of φ with respect to vacancies is $\varepsilon_{\varphi}(\frac{v}{u}) \equiv \frac{\frac{v}{\varphi(\frac{v}{u})} \frac{\partial \varphi(\frac{v}{u})}{\partial v}}{\frac{v}{\varphi(\frac{v}{u})}} = \frac{\frac{v}{\varphi(\frac{v}{u})} \varphi'(\frac{v}{u}) \frac{1}{u}}{\frac{v}{\varphi(\frac{v}{u})}} = \frac{\varphi'(\frac{v}{u})}{\varphi(\frac{v}{u})} \varphi(\frac{v}{u}) = \frac{\theta}{\varphi(\theta)} \frac{d}{d\theta}(\frac{1}{q(\theta)}) = \theta q(\theta) \frac{-q'(\theta)}{[q(\theta)]^2} = -\frac{\theta}{q(\theta)} q'(\theta) = \eta(\theta)$. Also, the elasticity of ψ with respect to unemployment is $\varepsilon_{\psi}(\frac{v}{u}) \equiv \frac{\frac{u}{\psi(\frac{v}{u})} \frac{\partial \psi(\frac{v}{u})}{\partial u}}{\frac{u}{\psi(\frac{v}{u})}} = \frac{\frac{u}{\psi(\frac{v}{u})} \psi'(\frac{v}{u}) \frac{-v}{u^2}}{\frac{u}{\psi(\frac{v}{u})}} = -\frac{\frac{v}{\psi(\frac{v}{u})} \psi'(\frac{v}{u})}{\psi(\frac{v}{u})} = -\frac{\theta}{\psi(\theta)} \psi'(\theta) = -\frac{\theta}{\frac{1}{p(\theta)}} \frac{d}{d\theta}(\frac{1}{p(\theta)}) = -\theta p(\theta) \frac{-p'(\theta)}{[p(\theta)]^2} = \frac{\theta}{p(\theta)} p'(\theta) = 1 - \eta(\theta)$. Hence, if $\eta(\theta)$ is high (and then $1 - \eta(\theta)$ is low), it is an indication that at the margin firms are causing more congestion to other firms than the congestion that workers are causing to other workers. In this case, the social planner "taxes" firms by giving workers a higher share in the wage bargain (a higher β).

¹Hosios, Arthur J. (1990). "On the Efficiency of Matching and Related Models of Search and Unemployment," *Review of Economic Studies*, Vol. 57, No. 2 (April), pp. 279-298.

Macroeconomía Dinámica

SIR Models

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Basic SIR Model

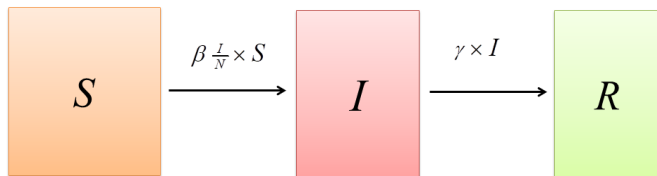
- In every period, each individual is in one of three states:

S = susceptible

I = infectious

R = removed (recovered or dead)

- Graphically:



- Notice that the removed cannot become susceptible or infectious again (permanent immunity).
- Remark: infectious $\neq$ infected; an infected person may not transmit the disease to others.

Basic SIR Model

- Constant population N (including the deceased):

$$S_t + I_t + R_t = N$$

- Initial conditions:

$$S_0 = (1 - \epsilon)N \cong N, \quad I_0 = \epsilon N > 0, \quad R_0 = 0$$

- Laws of motion (discrete-time version):

$$S_{t+1} - S_t = -\beta \left(\frac{I_t}{N} \right) S_t$$

$$I_{t+1} - I_t = \beta \left(\frac{I_t}{N} \right) S_t - \gamma I_t$$

$$R_{t+1} - R_t = \gamma I_t$$

- Notice that these equations are consistent with a constant N .
- Time is usually measured in days.

Basic SIR Model

- Interpretation of $\beta \left(\frac{I_t}{N} \right) S_t$.
 - β is the average number of contacts (sufficient for transmission), per unit time (per day), of one individual.
 - $\beta \left(\frac{I_t}{N} \right)$ is the average number of contacts with infectious people, per unit time (per day), of one susceptible person.
 - $\beta \left(\frac{I_t}{N} \right) S_t$ is the number of new infectious cases, per unit time (per day), due to S_t susceptibles.
- Suppose $\beta = \frac{1}{5} = 0.2$, $\frac{I_t}{N} = \frac{10}{40} = \frac{1}{4} = 0.25$ and $S_t = 20$ million. Then:
 - $\beta = \frac{1}{5}$ means that, on average, there's one contact every five days.
 - $\beta \left(\frac{I_t}{N} \right) = \frac{1}{5} \frac{1}{4} = \frac{1}{20}$ means that, on average, there's one contact with an infectious person every twenty days.
 - $\beta \left(\frac{I_t}{N} \right) S_t = \frac{1}{20} 20 = 1$ means that, on average, one million people become infectious each day (and stop being susceptible).

Basic SIR Model

- Alternatively:

- $\beta \left(\frac{S_t}{N} \right)$ is the average number of susceptibles that get the disease (and become infectious) from one infectious individual, per unit time (per day).
- $\beta \left(\frac{S_t}{N} \right) I_t$ is the number of new cases, per unit time (per day), due to I_t infectious individuals.

- Example: $\beta = \frac{1}{5} = 0.2$, $\frac{S_t}{N} = \frac{20}{40} = \frac{1}{2}$ and $I_t = 10$ million. Then:

- $\beta \left(\frac{S_t}{N} \right) = \frac{1}{5} \frac{1}{2} = \frac{1}{10}$ means that, on average, an infectious individual transmits the disease to one susceptible person (that becomes infectious) every ten days.
- $\beta \left(\frac{S_t}{N} \right) I_t = \frac{1}{10} 10 = 1$ means that, on average, one million people become infectious each day (and stop being susceptible).

- Interpretation of γI_t .
 - γ is the average number of infectious people that become removed (through recovery or death), per unit time (per day).
 - γI_t is the number of infectious people that become removed (by recovery or death), per unit time (per day).
- Suppose $\gamma = \frac{1}{10} = 0.1$ and $I_t = 10$ million. Then:
 - $\gamma = \frac{1}{10}$ means that, on average, one infectious person becomes removed every ten days. In other words: on average, the infectious period lasts ten days ($1/\gamma$).
 - $\gamma I_t = \frac{1}{10} 10 = 1$ means that, on average, a million of the infectious become removed each day.

Basic SIR Model

- Continuous time:

$$\dot{S}(t) = -\beta \left(\frac{I(t)}{N} \right) S(t)$$

$$\dot{I}(t) = \beta \left(\frac{I(t)}{N} \right) S(t) - \gamma I(t)$$

$$\dot{R}(t) = \gamma I(t)$$

with $S(t) + I(t) + R(t) = N$, $S(0) = (1 - \epsilon)N \cong N$, $I(0) = \epsilon N > 0$ and $R(0) = 0$.

- For any variable X we define $\dot{X}(t) \equiv \frac{dX}{dt}(t)$.

- Interpretation

- γ is the removal rate per unit of time; hence, during a short time interval h , a fraction γh of the infectious becomes removed.
 - When h is small: $R(t+h) - R(t) \cong \dot{R}(t) h = \gamma h I(t)$.
- β is the contact rate per unit of time; hence, during a short time interval h , an individual has βh contacts (sufficient for transmission).
 - When h is small: $S(t+h) - S(t) \cong \dot{S}(t) h = -\beta h \left(\frac{I(t)}{N} \right) S(t)$.

Basic SIR Model

- We can simplify notation by dropping the time index:

$$\dot{S} = -\beta \left(\frac{I}{N} \right) S$$

$$\dot{I} = \beta \left(\frac{I}{N} \right) S - \gamma I$$

$$\dot{R} = \gamma I$$

- We could simplify notation further by normalizing population to one: $N = 1$. In this case, S , I and R can be interpreted as fractions of the population (including the deceased). Since at $t = 0$ there are no deceased, S , I and R can be interpreted as fractions of the initial population (of living individuals).
- From now on we'll work in continuous time, but everything can be done in discrete time.

Basic SIR Model

- From $\dot{I}(t) = \beta \left(\frac{I(t)}{N} \right) S(t) - \gamma I(t)$ we can compute the growth rate of infectious cases at t :

$$\frac{\dot{I}(t)}{I(t)} = \beta \frac{S(t)}{N} - \gamma$$

- This growth rate will be positive (increasing number of infectious) when $\beta \frac{S(t)}{N} > \gamma$, that is, when

$$\mathcal{R}_t^e \equiv \frac{\beta S(t)}{\gamma N} > 1$$

where $\mathcal{R}_t^e$ is the so-called **effective reproduction number** at t .

- For an epidemic to start we need a positive growth rate at $t = 0$. That is:

$$\mathcal{R}_0^e \equiv \frac{\beta S(0)}{\gamma N} > 1$$

Basic SIR Model

- Since $\frac{S(0)}{N} \cong 1$, we have $\mathcal{R}_0^e \cong \frac{\beta}{\gamma}$. Hence, it is standard to define the so-called **basic reproduction number** as follows:

$$\mathcal{R}_0 \equiv \frac{\beta}{\gamma}$$

- Interpretation:
 - β is the average number of contacts of an (infectious) individual, per unit time.
 - $1/\gamma$ is the average duration of the infectious period.
 - $\mathcal{R}_0 = \beta \times \frac{1}{\gamma}$ is the expected number of new infectious individuals per sick person.
 - Example: if an individual has one contact (with transmission) every five days ($\beta = \frac{1}{5}$) and is infectious during 10 days ($\gamma = 1/10$), it will transmit the disease to $\beta \times \frac{1}{\gamma} = \frac{1}{5} \times 10 = 2$ individuals (that will become infectious).
- Remark: $\mathcal{R}_0^e \equiv \frac{\beta}{\gamma} \frac{S(0)}{N}$ and $\mathcal{R}_0 \equiv \frac{\beta}{\gamma} \Rightarrow \mathcal{R}_0^e = \mathcal{R}_0 \frac{S(0)}{N}$.

- Phase Diagram

- From $\dot{I} = \beta \left(\frac{I}{N} \right) S - \gamma I$ we can get the $\dot{I} = 0$ schedule:
 $\dot{I} = 0 \Rightarrow \beta \left(\frac{I}{N} \right) S - \gamma I = 0 \Rightarrow \beta \frac{S}{N} = \gamma$ (for $I > 0$). Then:

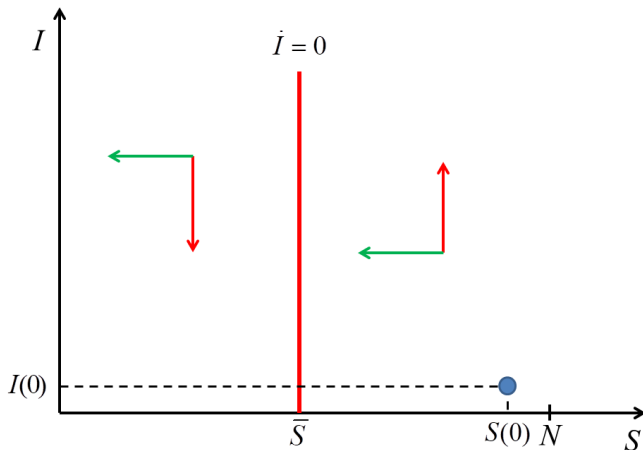
$$\begin{aligned}\bar{S} &= \frac{\gamma}{\beta} N \\ &= \frac{N}{\mathcal{R}_0} \\ &= \frac{S(0)}{\mathcal{R}_0^e}\end{aligned}$$

- If $S > \bar{S}$ we get $\dot{I} > 0$, so the number of infectious is increasing over time. The opposite happens if $S < \bar{S}$.
- From $\dot{S} = -\beta \left(\frac{I}{N} \right) S$ we see that S is always falling (as long as S and I are positive).
- Finally, recall the initial conditions: $S(0) = (1 - \epsilon)N \cong N$,
 $I(0) = \epsilon N > 0$.

Basic SIR Model

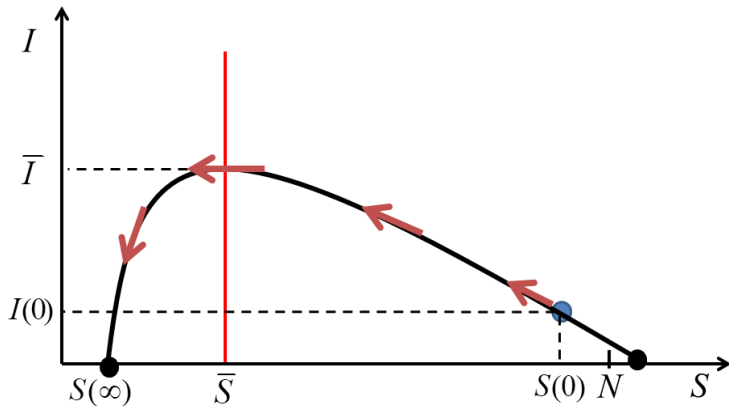
- Phase Diagram (cont.)

- Assuming $S(0) > \bar{S} = \frac{S(0)}{\mathcal{R}_0^e}$ (i.e., $\mathcal{R}_0^e > 1$):



Basic SIR Model

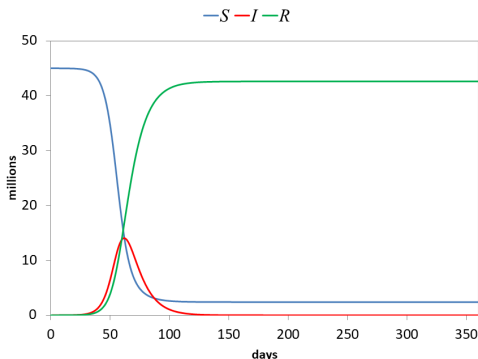
- Phase Diagram (cont.)



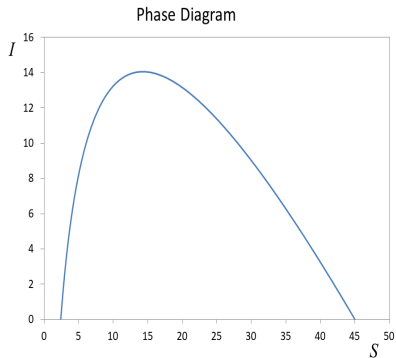
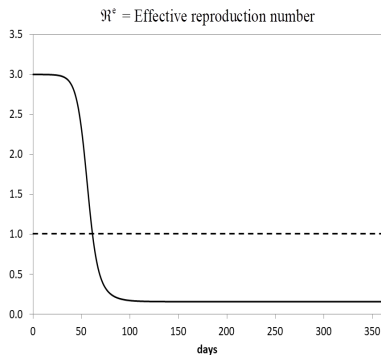
- Remarks on the previous figure
 - The equation of the black curve can be derived in closed form (see the Appendix).
 - $\bar{S}$ is the **herd immunity** threshold.
 - Once $S < \bar{S}$, the epidemic starts to disappear: I starts falling from its maximum $\bar{I}$ towards zero.
 - It can be shown that $\bar{I} = N - \bar{S} + \bar{S} \ln \left(\frac{S(0)}{\bar{S}} \right)$ (see the Appendix).
 - $S(\infty)$ denotes the long-run number of susceptibles. It is implicitly defined by $S(\infty) - \bar{S} \ln S(\infty) = N - \bar{S} \ln S(0)$ (see the Appendix).
 - $I(\infty) = 0$: no infectious individuals in the long run.
 - $N - S(\infty) - I(\infty) = N - S(\infty) = R(\infty)$ is the long-run number of removed individuals (either recovered or dead).
 - Notice that $R(\infty) = N - S(\infty)$ gives the number of persons that were sick at some time during the epidemic.

Basic SIR Model

- Given a set of parameter values, we can simulate the evolution of S , I and R .
 - We show simulations for the discrete-time version of the model, than can be easily obtained with a standard spreadsheet (e.g., Excel).
- Suppose: $\gamma = 1/10$, $\beta = 0.3$, $\mathcal{R}_0 = \frac{\beta}{\gamma} = 3$, $N = 45,000,000$, $I_0 = 1,000$, $R_0 = 0$, $S_0 = N - I_0$.

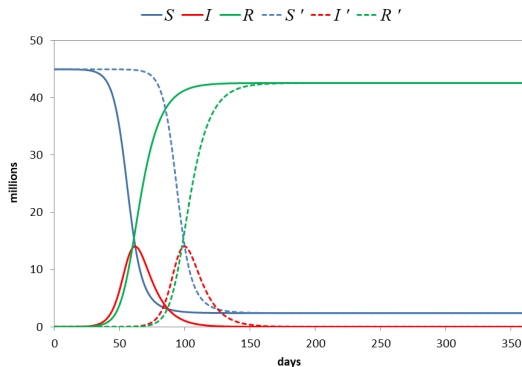


Basic SIR Model



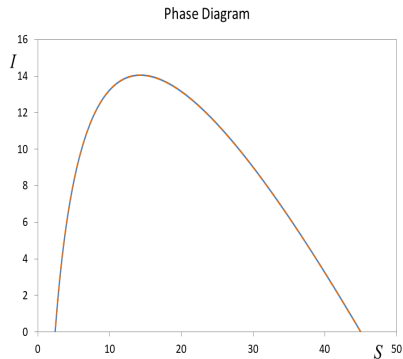
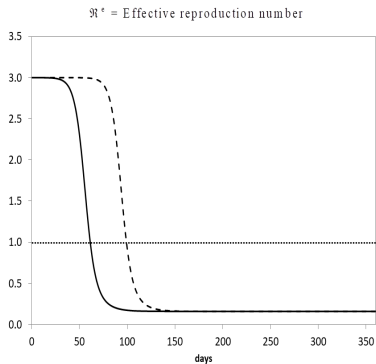
Basic SIR Model

- Reduce I_0 from 1,000 to 1, keeping the other parameters constant.



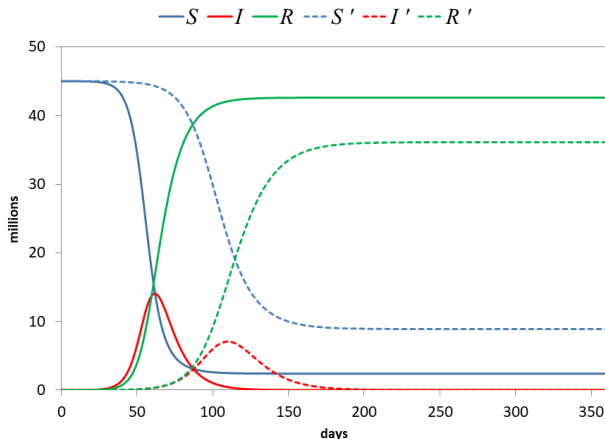
The solid lines (dashed lines) show the results for the original (new) parametrization.

Basic SIR Model

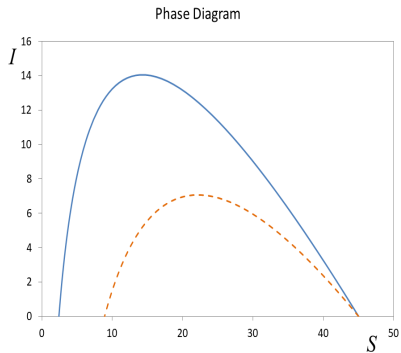
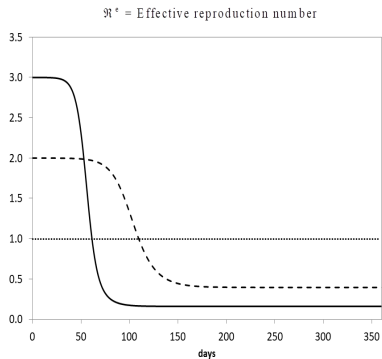


Basic SIR Model

- Reduce β from 0.3 to 0.2, keeping the other parameters constant. This reduces $\mathcal{R}_0$ from 3 to 2.

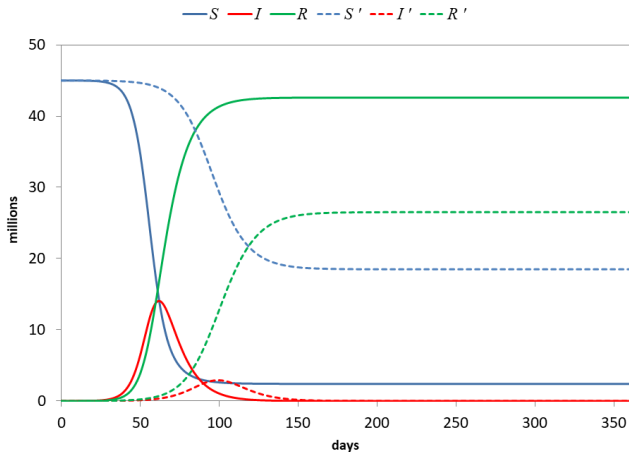


Basic SIR Model

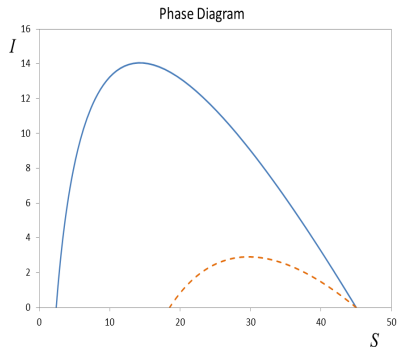
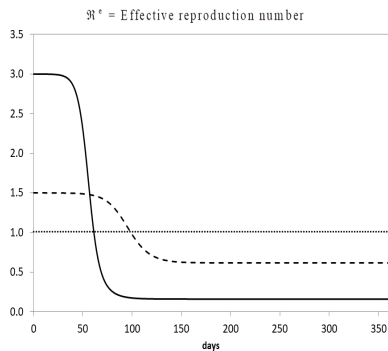


Basic SIR Model

- Increase γ from $\frac{1}{10}$ to $\frac{1}{5}$, keeping the other parameters constant. This reduces $\mathcal{R}_0$ from 3 to 1.5.

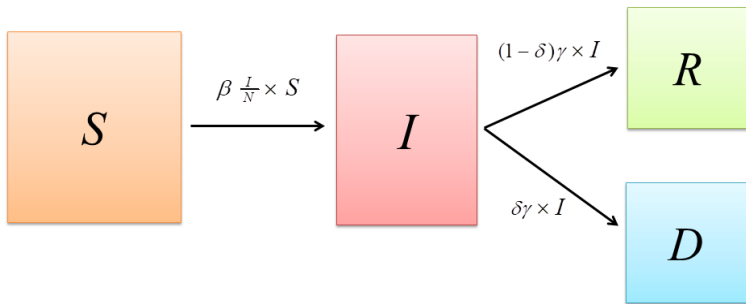


Basic SIR Model



SIRD Model

- Split the Removed into Recovered or Dead.
 - R = recovered
 - D = dead
 - $R + D$ = removed
- Graphically



SIRD Model

- Total population, including the deceased, is constant:

$$N = S(t) + I(t) + R(t) + D(t)$$

- We use $P(t)$ to denote the number of persons that are alive at t :

$$P(t) = S(t) + I(t) + R(t)$$

- The laws of motion are the following (dropping the time index):

$$\dot{S} = -\beta \left(\frac{I}{N} \right) S$$

$$\dot{I} = \beta \left(\frac{I}{N} \right) S - \gamma I$$

$$\dot{R} = (1 - \delta) \gamma I$$

$$\dot{D} = \delta \gamma I$$

$$\dot{P} = -\delta \gamma I$$

- Initial conditions:

$$S_0 = (1 - \epsilon)N \cong N, \quad I_0 = \epsilon N > 0, \quad R_0 = 0, \quad D_0 = 0$$

- The parameter δ is the **Case Fatality Rate (CFR)**.

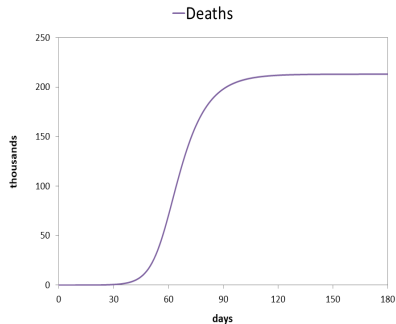
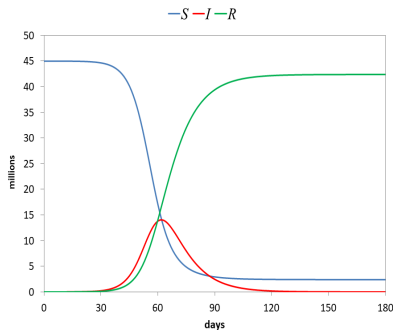
- From $\dot{R} + \dot{D} = \gamma I$ we get $(\dot{R} + \dot{D})h = \gamma hI$. Then, in a short time interval, h , approximately $\gamma h \times I$ infectious individuals become removed. A fraction δ of those die, $\dot{D}h = \delta \gamma hI$, and a fraction $1 - \delta$ recover, $\dot{R}h = (1 - \delta) \gamma hI$.
- Combining $\dot{D} = \delta \gamma I$ with $\dot{R} = (1 - \delta) \gamma I$ we get $\dot{D} = \frac{\delta}{1 - \delta} \dot{R}$. Integrating from 0 to t , and imposing $D(0) = R(0) = 0$, we get $D(t) = \frac{\delta}{1 - \delta} R(t)$. Then:
$$\delta = \frac{D(t)}{D(t) + R(t)}.$$
- We can think of $\delta \gamma$ as the case fatality rate per unit of time (per day). If $\delta = 0.01$ (1%) and $\gamma = \frac{1}{10}$, $\delta \gamma = 0.001$ means that 0.1% of the infectious die each day.
- Integrating $\dot{D} = \delta \gamma I$ from 0 to t , and imposing $D(0) = 0$ we get $D(t) = \delta \gamma \int_0^t I(x) dx$. Then:
$$\delta \gamma = \frac{D(t)}{\int_0^t I(x) dx}.$$
- The CFR is not the same as the **mortality rate**, which is usually defined as the flow of deaths over a period of time divided by the total population (at the beginning of the period). If we take $t = 0$ as the initial time we get:
$$\text{mortality rate}_t = \frac{D(t) - D(0)}{P(0)} = \frac{D(t)}{N}.$$

SIRD Model

- In reality, the CFR may not be constant. One possibility is that there are congestion effects in the health care system. This can be modeled by assuming that the CFR is an increasing function of the number of infectious people (or a fraction of the infectious people, reflecting those that may need hospitalization).
 - Examples used in the literature: $\delta(t) = \varphi + \kappa I(t)$; $\delta(t) = \varphi + \kappa I(t)^2$.
- In the SIRD model, the effective reproduction number ($\mathcal{R}_t^e$) and the basic reproduction number ($\mathcal{R}_0$) are the same as those in the basic SIR model.
- The structure of the phase diagram in (S, I) space is also the same.
- The dynamics for S , I , and the number of removed individuals are exactly the same as those induced by the basic SIR model. But now we have the extra information about the number of dead people (D).
- In the equations for $\dot{S}$ and $\dot{I}$, I is divided by N (which includes the deceased). This is standard in the literature, but it would be better to divide by P . Usually this is not problematic, since the number of deaths is relatively small. As mentioned earlier, N is sometimes normalized to 1, and variables are interpreted as fractions of the initial population (of living individuals).

SIRD Model

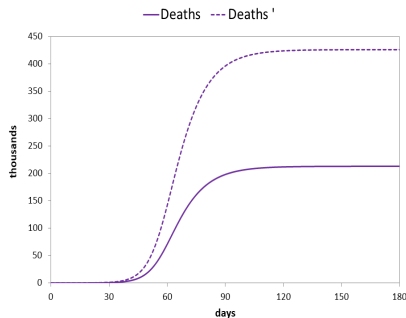
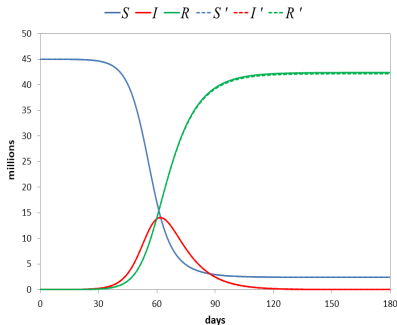
- Suppose, again, that $\gamma = 1/10$, $\beta = 0.3$, $\mathcal{R}_0 = \frac{\beta}{\gamma} = 3$, $N = 45,000,000$ and $I_0 = 1,000$. Also: $R_0 = D_0 = 0$ and $S_0 = N - I_0$. Assume $\delta = 0.005$.



- Even with a small fatality rate (0.5%) the number of deaths is high (213,000 in a population of 45 million).

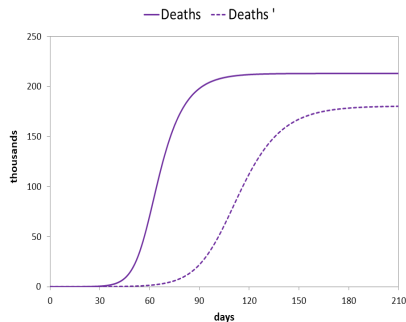
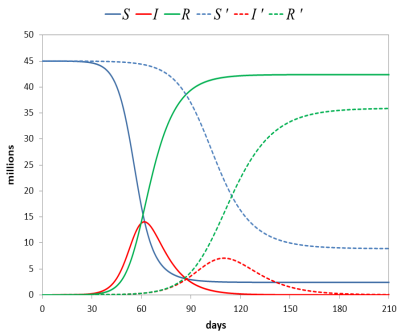
SIRD Model

- A higher fatality rate increases the number of deaths and reduces the number of recovered, without affecting the number of removed individuals. Also, S and I remain unaffected.
- $\delta = 0.01$ (1%)



SIRD Model

- Reduce β from 0.3 to 0.2, keeping the other parameters constant. This reduces $\mathcal{R}_0$ from 3 to 2.



- The number of deaths falls from 213,000 to 180,500.

SIRD Model with Lockdown

- Suppose the government responds to the epidemic by mandating a uniform lockdown.
- Let $L(t)$ denote the fraction of $P(t)$ that is in lockdown at t .
- Assume $L(t) \in [0, \bar{L}]$, with $\bar{L} < 1$.
 - Full lockdown is not feasible: some goods are essential, etc.
- Let $\theta(t) \in [0, 1]$ denote the effectiveness of the lockdown at t .
 - $\theta = \frac{3}{4}$ means that 25% of the individuals do not obey the lockdown.
- The effective lockdown at t is $\theta(t)L(t)$.
- The number of susceptibles that can become infectious is now $[1 - \theta(t)L(t)]S(t)$.
- The number of infectious that can transmit the disease is now $[1 - \theta(t)L(t)]I(t)$.

SIRD Model with Lockdown

- Normalizing $N = 1$, and dropping the time index, we get the following equations:

$$\dot{S} = -\beta (1 - \theta L) I (1 - \theta L) S$$

$$\dot{I} = \beta (1 - \theta L) I (1 - \theta L) S - \gamma I$$

$$\dot{R} = (1 - \delta) \gamma I$$

$$\dot{D} = \delta \gamma I$$

$$\dot{P} = -\delta \gamma I$$

SIRD Model with Lockdown

- We can rewrite the equations for $\dot{S}$ and $\dot{I}$ as follows:

$$\dot{S} = -\beta(1 - \theta L)^2 IS$$

$$\dot{I} = \beta(1 - \theta L)^2 IS - \gamma I$$

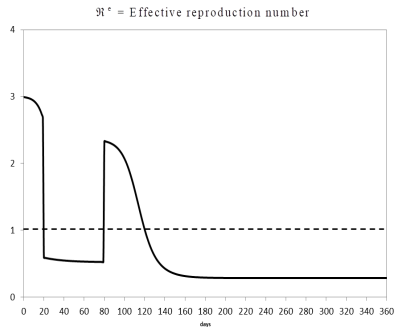
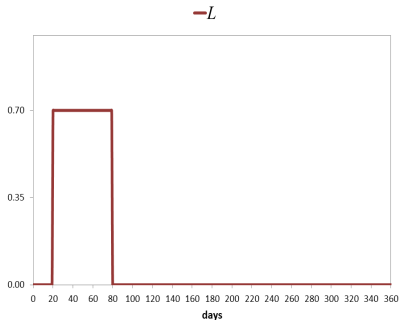
- Notice that the lockdown works like a reduction in the contact rate.
 - If we define the effective contact rate as $\beta^e(t) \equiv \beta[1 - \theta(t)L(t)]^2$ we get (dropping the time index): $\dot{S} = -\beta^e IS$ and $\dot{I} = \beta^e IS - \gamma I$.
- The effective reproduction number is now (recall we have normalized $N = 1$):

$$\mathcal{R}_t^e \equiv \frac{\beta}{\gamma}[1 - \theta(t)L(t)]^2 S(t) = \frac{\beta^e(t)}{\gamma} S(t)$$

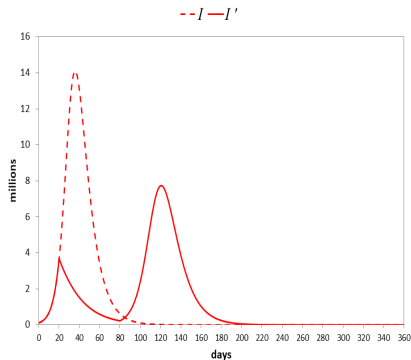
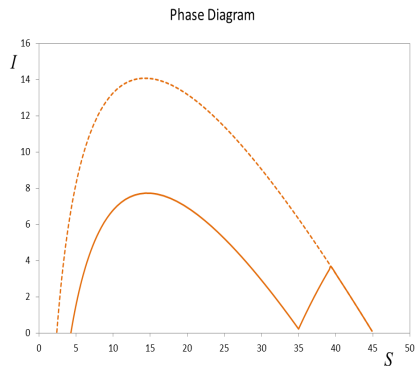
- The basic reproduction number is always calculated in the absence of any policy. Hence, it is still given by $\mathcal{R}_0 \equiv \frac{\beta}{\gamma}$.

SIRD Model with Lockdown

- Suppose $\gamma = 1/10$, $\delta = 0.005$, $\beta = 0.3$, $\mathcal{R}_0 = \frac{\beta}{\gamma} = 3$, $N = 45,000,000$ and $I_0 = 112,500$, $R_0 = D_0 = 0$ and $S_0 = N - I_0$.
- In addition, assume $L(t) = 0.7 \ \forall t \in [20, 80)$ and $L(t) = 0$ otherwise, with constant effectiveness $\theta = 0.75$.



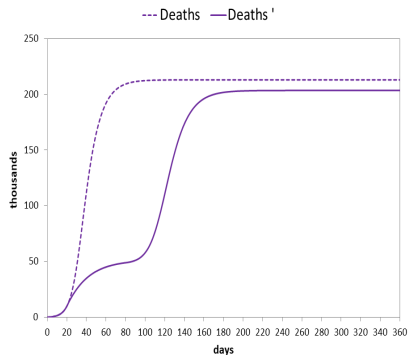
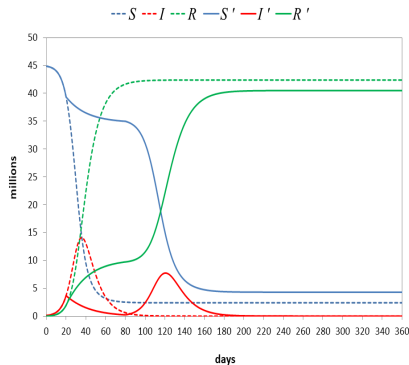
SIRD Model with Lockdown



For comparison, the dashed lines show the case with no lockdown.

Notice the emergence of a second wave of infections when the lockdown is lifted.

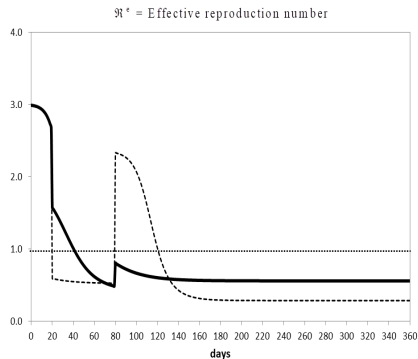
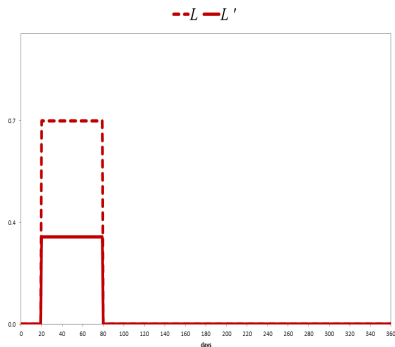
SIRD Model with Lockdown



Notice that the final number of deaths is somewhat smaller but still high.

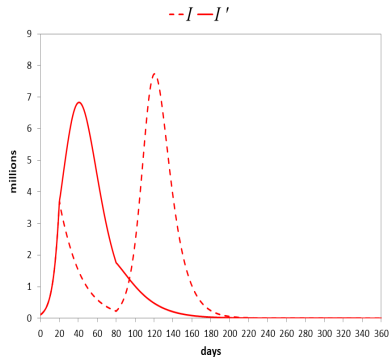
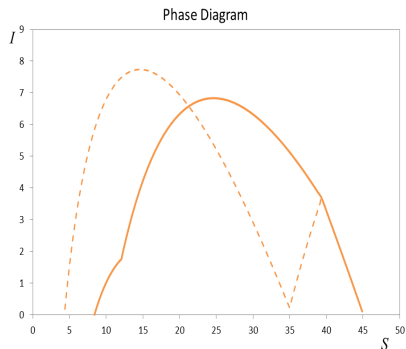
SIRD Model with Lockdown

- A milder lockdown



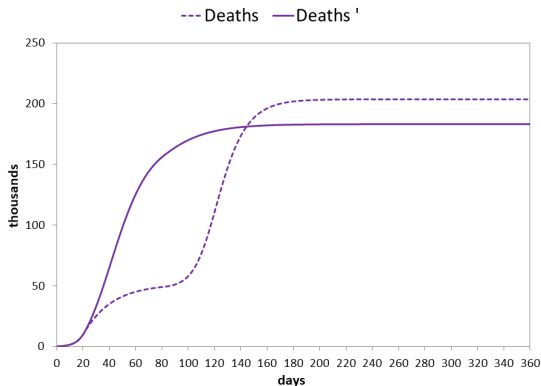
For comparison, the dashed lines show the case with a tighter lockdown.

SIRD Model with Lockdown



No second wave with the milder lockdown. Herd immunity is reached earlier.

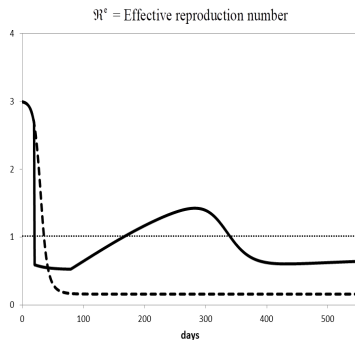
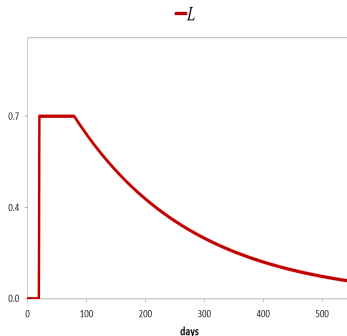
SIRD Model with Lockdown



Notice that, in the end, the total number of deaths is smaller with the milder lockdown.

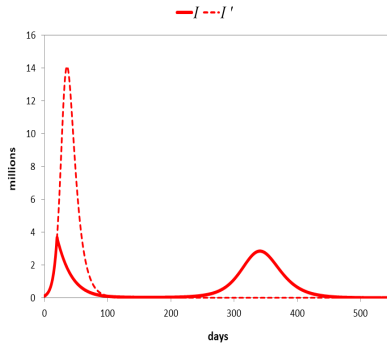
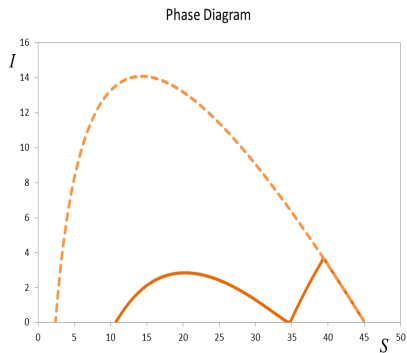
SIRD Model with Lockdown

- A tight lockdown that is lifted gradually.
 - The rest of the parameters remain unchanged.

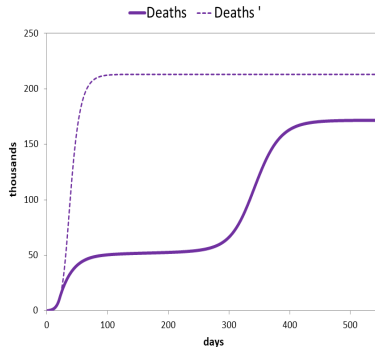
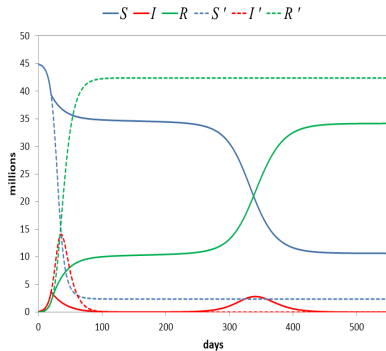


For comparison, the dashed lines show the case with no lockdown.

SIRD Model with Lockdown



SIRD Model with Lockdown



SIRD Model: Testing and Tracing

- Consider the SIRD model with lockdown.
- We want to introduce a testing-tracing-quarantine (TTQ) policy.
- Suppose that, at time t , the government can identify (trace), test as infectious, and subsequently quarantine, a fraction $\psi(t)$ of $I(t)$.
- Assume quarantined individuals don't transmit the virus. Hence, the relevant number of infectious in the laws of motion for $S(t)$ and $I(t)$ is $[1 - \psi(t)]I(t)$.
- Then, dropping the time index and normalizing $N = 1$, we get:

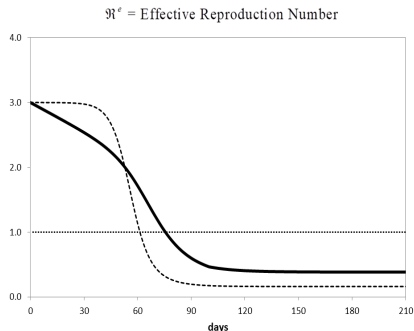
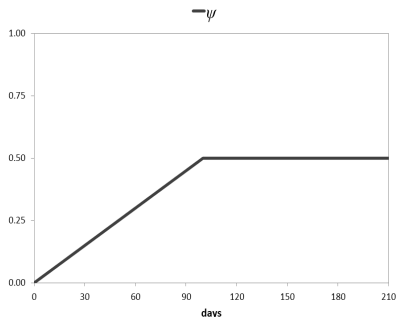
$$\begin{aligned}\dot{S} &= -\beta (1 - \theta L)^2 (1 - \psi) IS \\ \dot{I} &= \beta (1 - \theta L)^2 (1 - \psi) IS - \gamma I\end{aligned}$$

- The rest of the equations remain unchanged.
- The effective reproduction number is now

$$\begin{aligned}\mathcal{R}_t^e &\equiv \frac{\beta}{\gamma} [1 - \theta(t)L(t)]^2 [1 - \psi(t)] S(t) \\ &= \frac{\beta^e(t)}{\gamma} S(t)\end{aligned}$$

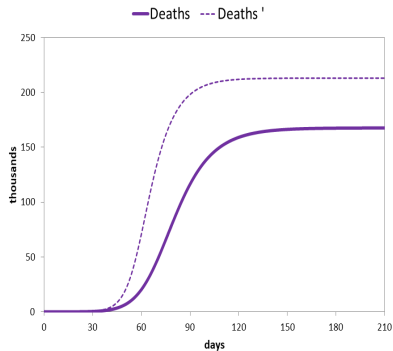
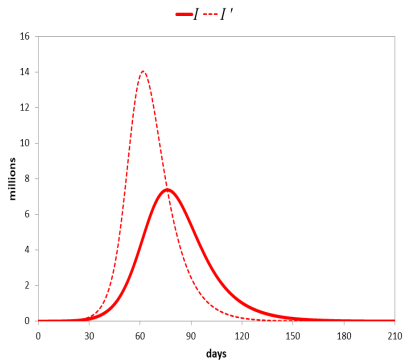
SIRD Model: Testing and Tracing

- Suppose $\gamma = 1/10$, $\delta = 0.005$, $N = 45,000,000$ and $I_0 = 1000$, $R_0 = D_0 = 0$ and $S_0 = N - I_0$.
- Let $\psi(t)$ increase linearly from 0 to 0.5 during the first one hundred days and remain there afterwards. There's no lockdown.



The dashed lines show the case of no testing.

SIRD Model: Testing and Tracing



SIRD Model: Social Distancing in reduced form

- We've seen that, in the model with lockdown and TTQ, the equations for $\dot{S}$ and $\dot{I}$ can be rewritten as $\dot{S}(t) = -\beta^e(t)I(t)S(t)$ and $\dot{I}(t) = \beta^e(t)I(t)S(t) - \gamma I(t)$, with $\beta^e(t) \equiv \beta[1 - \theta(t)L(t)]^2[1 - \psi(t)]$.
- This suggests that we could model social-distancing measures, including the endogenous response of individuals, through a time-varying contact rate, $\beta(t)$.
- In this case, the equations for $\dot{S}$ and $\dot{I}$ become

$$\begin{aligned}\dot{S}(t) &= -\beta(t)I(t)S(t) \\ \dot{I}(t) &= \beta(t)I(t)S(t) - \gamma I(t)\end{aligned}$$

and we need a description of the evolution of $\beta(t)$ over time.

- The rest of the equations of the model remain unaffected.

SIRD Model: Social Distancing in reduced form

- Suppose

$$\beta(t) = \beta_0 e^{-\lambda t} + \beta_*(1 - e^{-\lambda t})$$

where $\beta_0 > 0$, $\beta_* > 0$ and $\lambda > 0$.

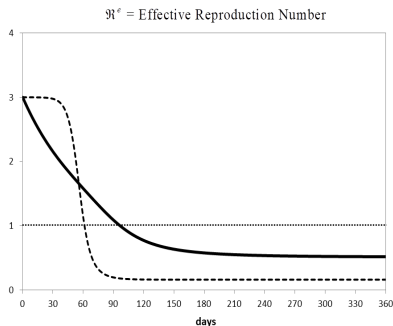
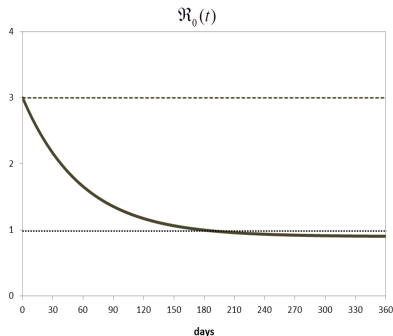
- With this functional form, $\beta(t)$ starts at β_0 and converges monotonically to β_* at a rate determined by λ .
 - The standard case with $\beta(t)$ constant is obtained by setting $\beta_0 = \beta_*$.
- If we define $\mathcal{R}_0(t) \equiv \frac{\beta(t)}{\gamma}$, the expression above is equivalent to

$$\mathcal{R}_0(t) = \mathcal{R}_0 e^{-\lambda t} + \mathcal{R}_*(1 - e^{-\lambda t})$$

where $\mathcal{R}_0 \equiv \frac{\beta_0}{\gamma}$ and $\mathcal{R}_* \equiv \frac{\beta_*}{\gamma}$.

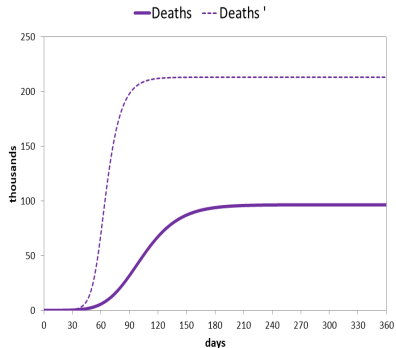
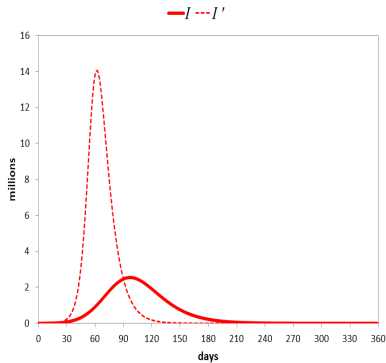
SIRD Model: Social Distancing in reduced form

- Suppose $\gamma = 1/10$, $\delta = 0.005$, $N = 45,000,000$ and $I_0 = 1000$, $R_0 = D_0 = 0$ and $S_0 = N - I_0$.
- Also, set $\beta_0 = 0.3$ ($\mathcal{R}_0 = 3$), $\beta_* = 0.09$ ($\mathcal{R}_* = 0.9$), and $\lambda = 0.017$.



The dashed lines show the case of constant β .

SIRD Model: Social Distancing in reduced form



SIRD Model: Vaccine

- Consider a standard SIRD model with lockdown.
- Suppose a vaccine is discovered at $t = \tau$.
- Assume, for simplicity, that all susceptible individuals can be instantaneously vaccinated at $t = \tau$.
 - Suppose infectious individuals are all symptomatic and can be identified as such. In this case, there's no need to vaccinate them. Recovered individuals need not be vaccinated either.
- Before τ , the variables evolve according to the equations given earlier for the SIRD model with lockdown.
- At $t = \tau$ all susceptibles become recovered. Then:

$$S(\tau) = 0$$

$$R(\tau) = R(\tau^-) + S(\tau^-)$$

- For $t > \tau$ we have:

$$S(t) = 0 \quad (\Rightarrow \dot{S}(t) = 0)$$

$$\dot{I}(t) = -\gamma I(t)$$

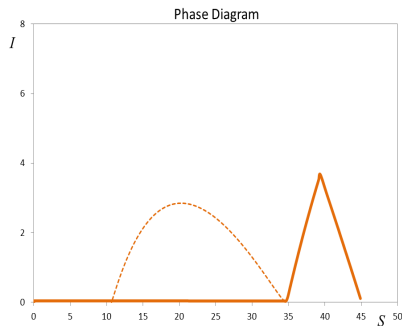
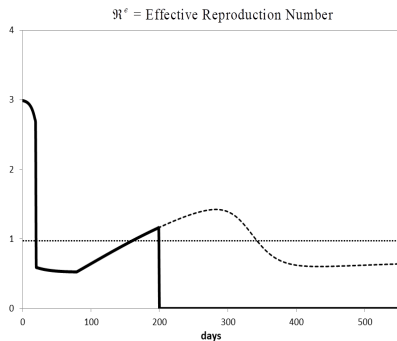
$$\dot{R}(t) = (1 - \delta) \gamma I(t)$$

$$\dot{D}(t) = \delta \gamma I(t)$$

$$\dot{P}(t) = -\delta \gamma I(t)$$

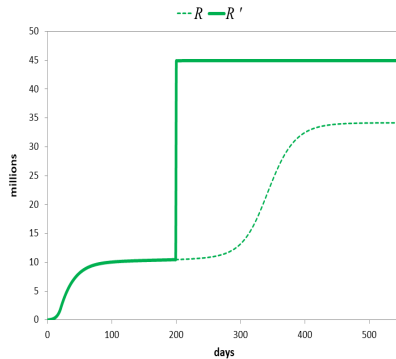
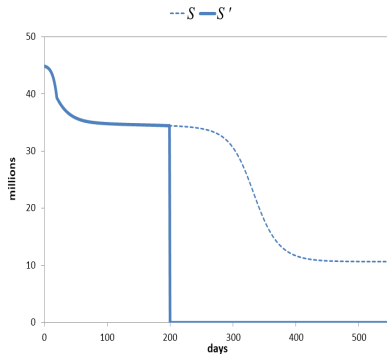
SIRD Model: Vaccine

- Consider our previous simulation with a lockdown that is lifted gradually.
- Assume a vaccine becomes available at $t = 200$.

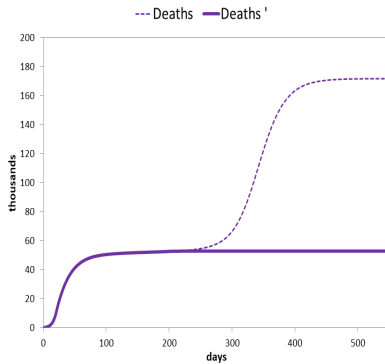
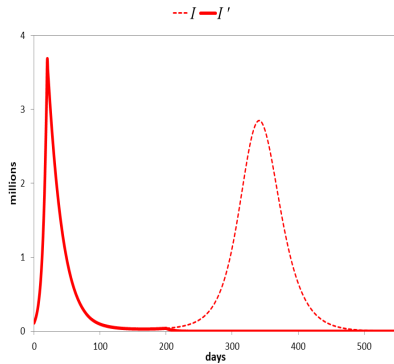


The dashed lines show the case without the arrival of the vaccine.

SIRD Model: Vaccine



SIRD Model: Vaccine



SIRD Model: Treatment

- Consider a standard SIRD model.
- Suppose a treatment that cures the disease becomes available at $t = \tau$.
- Before τ , the variables evolve according to the equations given earlier for the SIRD model with lockdown.
- Assume that, once it becomes available, the treatment is provided to all infectious individuals at every $t \geq \tau$, who are instantly transformed into recovered individuals.
- As a result, the number of new deaths from the disease goes to zero (the stock of deceased stabilizes).
- Since there are no more infectious individuals, the number of susceptibles stabilizes.
- Then:

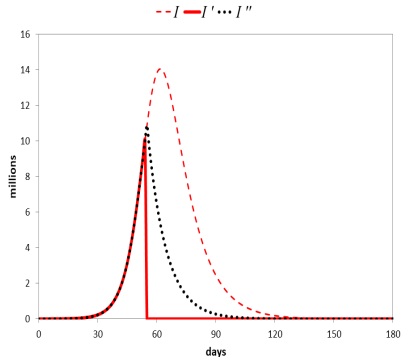
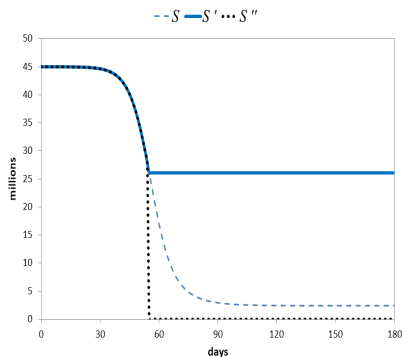
$$I(t) = 0 \quad \forall t \geq \tau$$

$$S(t) = S(\tau^-) \quad \forall t \geq \tau$$

$$D(t) = D(\tau^-) \quad \forall t \geq \tau$$

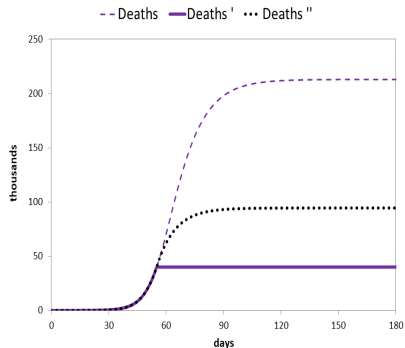
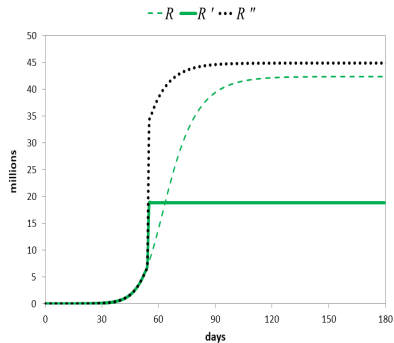
$$R(t) = N - D(\tau^-) - S(\tau^-) \quad \forall t \geq \tau$$

SIRD Model: Treatment



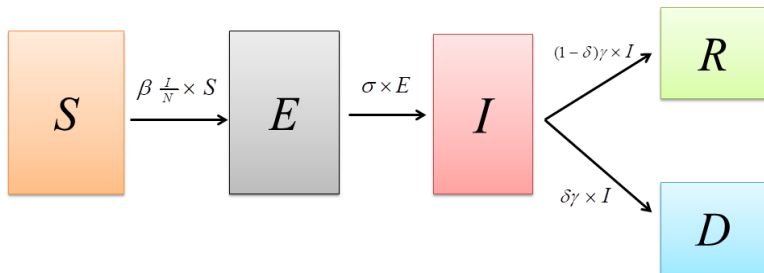
Dashed lines show the case with no treatment and no vaccine; dotted lines show the case with a vaccine.

SIRD Model: Treatment



Extensions: SEIRD Model

- New state: $E = \text{exposed}$
 - Individuals that are infected but not yet infectious (incubation period).
- $N = S + E + I + R + D$ (constant).
- $P = S + E + I + R$.



Extensions: SEIRD Model

- Normalizing $N = 1$:

$$\dot{S} = -\beta IS$$

$$\dot{E} = \beta IS - \sigma E$$

$$\dot{I} = \sigma E - \gamma I$$

$$\dot{R} = (1 - \delta) \gamma I$$

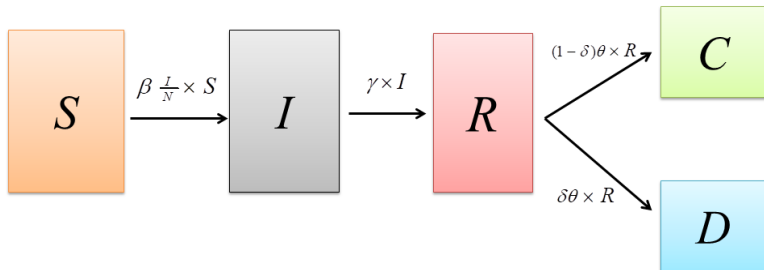
$$\dot{D} = \delta \gamma I$$

$$\dot{P} = -\delta \gamma I$$

- $\sigma = \frac{1}{5}$ means that, on average, the incubation period lasts five days.

Extensions: SIRCD Model

- Now $R = \text{resolving}$, and $C = \text{reCovered}$.
- $N = S + I + R + C + D$ (constant).
- $P = S + I + R + C$.



Extensions: SIRCD Model

- Normalizing $N = 1$:

$$\dot{S} = -\beta IS$$

$$\dot{I} = \beta IS - \gamma I$$

$$\dot{R} = \gamma I - \theta R$$

$$\dot{C} = (1 - \delta) \theta R$$

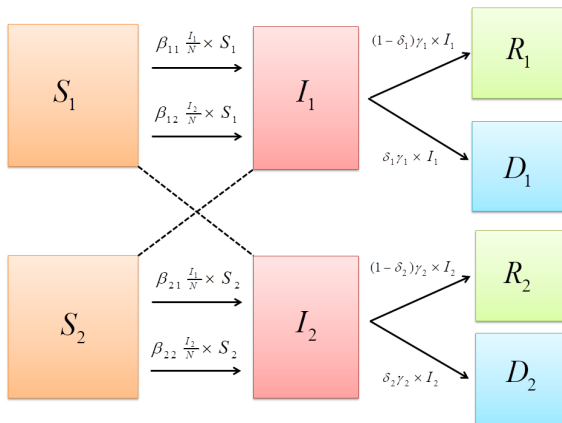
$$\dot{D} = \delta \theta R$$

$$\dot{P} = -\delta \theta R$$

- Now $\gamma = \frac{1}{5}$ means that, on average, a person with the virus is infectious for five days, while $\theta = \frac{1}{10}$ means that, on average, an infected person that stopped being infectious will be ill for another 10 days.

Extensions: Multi-group SIRD Model with Lockdown

- Two groups (for simplicity): $g \in \{1, 2\}$.
- Useful to model heterogeneity: different fatality rates by age, targeted lockdowns, etc.
- Figure (without lockdown):



Extensions: Multi-group SIRD Model with Lockdown

- We have:

$$N_g = S_g + I_g + R_g + D_g \quad (\text{constant})$$

$$P_g = S_g + I_g + R_g$$

$$N = N_1 + N_2$$

$$P = P_1 + P_2$$

$$S = S_1 + S_2$$

$$I = I_1 + I_2$$

$$R = R_1 + R_2$$

$$D = D_1 + D_2$$

Extensions: Multi-group SIRD Model with Lockdown

- Laws of motion, normalizing $N = 1$:

$$\dot{S}_1 = -\beta_{11} (1 - \theta_1 L_1) I_1 (1 - \theta_1 L_1) S_1 - \beta_{12} (1 - \theta_2 L_2) I_2 (1 - \theta_1 L_1) S_1$$

$$\dot{S}_2 = -\beta_{21} (1 - \theta_1 L_1) I_1 (1 - \theta_2 L_2) S_2 - \beta_{22} (1 - \theta_2 L_2) I_2 (1 - \theta_2 L_2) S_2$$

$$\dot{I}_1 = \beta_{11} (1 - \theta_1 L_1) I_1 (1 - \theta_1 L_1) S_1 + \beta_{12} (1 - \theta_2 L_2) I_2 (1 - \theta_1 L_1) S_1 - \gamma_1 I_1$$

$$\dot{I}_2 = \beta_{21} (1 - \theta_1 L_1) I_1 (1 - \theta_2 L_2) S_2 + \beta_{22} (1 - \theta_2 L_2) I_2 (1 - \theta_2 L_2) S_2 - \gamma_2 I_2$$

$$\dot{R}_1 = (1 - \delta_1) \gamma_1 I_1$$

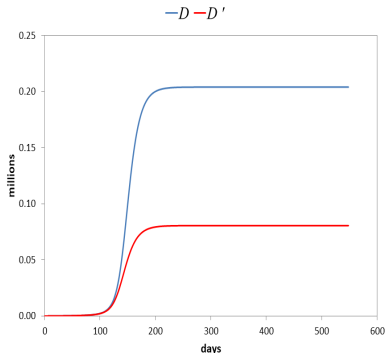
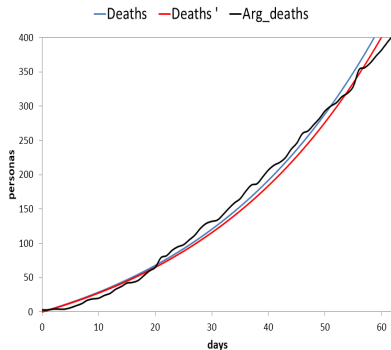
$$\dot{R}_2 = (1 - \delta_2) \gamma_2 I_2$$

$$\dot{D}_1 = \delta_1 \gamma_1 I_1 = -\dot{P}_1$$

$$\dot{D}_2 = \delta_2 \gamma_2 I_2 = -\dot{P}_2$$

Estimation Difficulties

- Identification problem: δ and $I(0)$.



Appendix

- We are going to derive a closed-form solution for the evolution of the epidemic in (S, I) space.
- From $\dot{I}(t) = \beta \left(\frac{I(t)}{N} \right) S(t) - \gamma I(t)$ and $\dot{S}(t) = -\beta \left(\frac{I(t)}{N} \right) S(t)$ we get:

$$\frac{\dot{I}(t)}{\dot{S}(t)} = \frac{\beta \left(\frac{I(t)}{N} \right) S(t) - \gamma I(t)}{-\beta \left(\frac{I(t)}{N} \right) S(t)}$$

$$\frac{\dot{I}(t)}{\dot{S}(t)} = -1 + \frac{\gamma}{\beta} \frac{N}{S(t)}$$

- Then:

$$\dot{I}(t) = -\dot{S}(t) + \bar{S} \frac{\dot{S}(t)}{S(t)}$$

- Integrating both sides of the previous expression from 0 to t :

$$\int_0^t \dot{I}(x) dx = - \int_0^t \dot{S}(x) dx + \bar{S} \int_0^t \frac{\dot{S}(x)}{S(x)} dx$$

- Then:

$$I(x)]_0^t = -[S(x)]_0^t + \bar{S} [\ln S(x)]_0^t$$

$$I(t) - I(0) = -[S(t) - S(0)] + \bar{S} [\ln S(t) - \ln S(0)]$$

- Then:

$$I(t) = [I(0) + S(0) - \bar{S} \ln S(0)] + \bar{S} \ln S(t) - S(t)$$

- Since $I(0) + S(0) = N$:

$$I(t) = [N - \bar{S} \ln S(0)] + \bar{S} \ln S(t) - S(t)$$

- Dropping the time index from the previous expression we get:

$$I = [N - \bar{S} \ln S(0)] + \bar{S} \ln S - S$$

which gives the equation for the black line in the phase diagram.

- Now we can solve for the value of S that maximizes the number of infectious, I .
- Differentiating the expression above with respect to S and equating it to zero we get: $\bar{S} \frac{1}{S} - 1 = 0$. Then:

$$S = \bar{S}$$

which coincides with the value given in the phase diagram.

- Evaluating the objective function at $S = \bar{S}$ we get:

$$\bar{I} = N - \bar{S} \ln S(0) + \bar{S} \ln \bar{S} - \bar{S} \Rightarrow \bar{I} = N - \bar{S} - \bar{S} [\ln S(0) - \ln \bar{S}] \Rightarrow$$

$$\bar{I} = N - \bar{S} + \bar{S} \ln \left(\frac{S(0)}{\bar{S}} \right)$$

- We can compute the long-run number of susceptibles, $S(\infty)$, by setting $I = 0$ in $I = [N - \bar{S} \ln S(0)] + \bar{S} \ln S - S$.
- We get: $0 = N - \bar{S} \ln S(0) + \bar{S} \ln S(\infty) - S(\infty) \Rightarrow$

$$S(\infty) - \bar{S} \ln S(\infty) = N - \bar{S} \ln S(0)$$

- The expression above implicitly defines the value of $S(\infty)$.
 - This equation has a solution with $S(\infty) > N$, which is not the one we're looking for (we can see this from the phase diagram). We are looking for a solution with $S(\infty) < \bar{S} < N$.

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